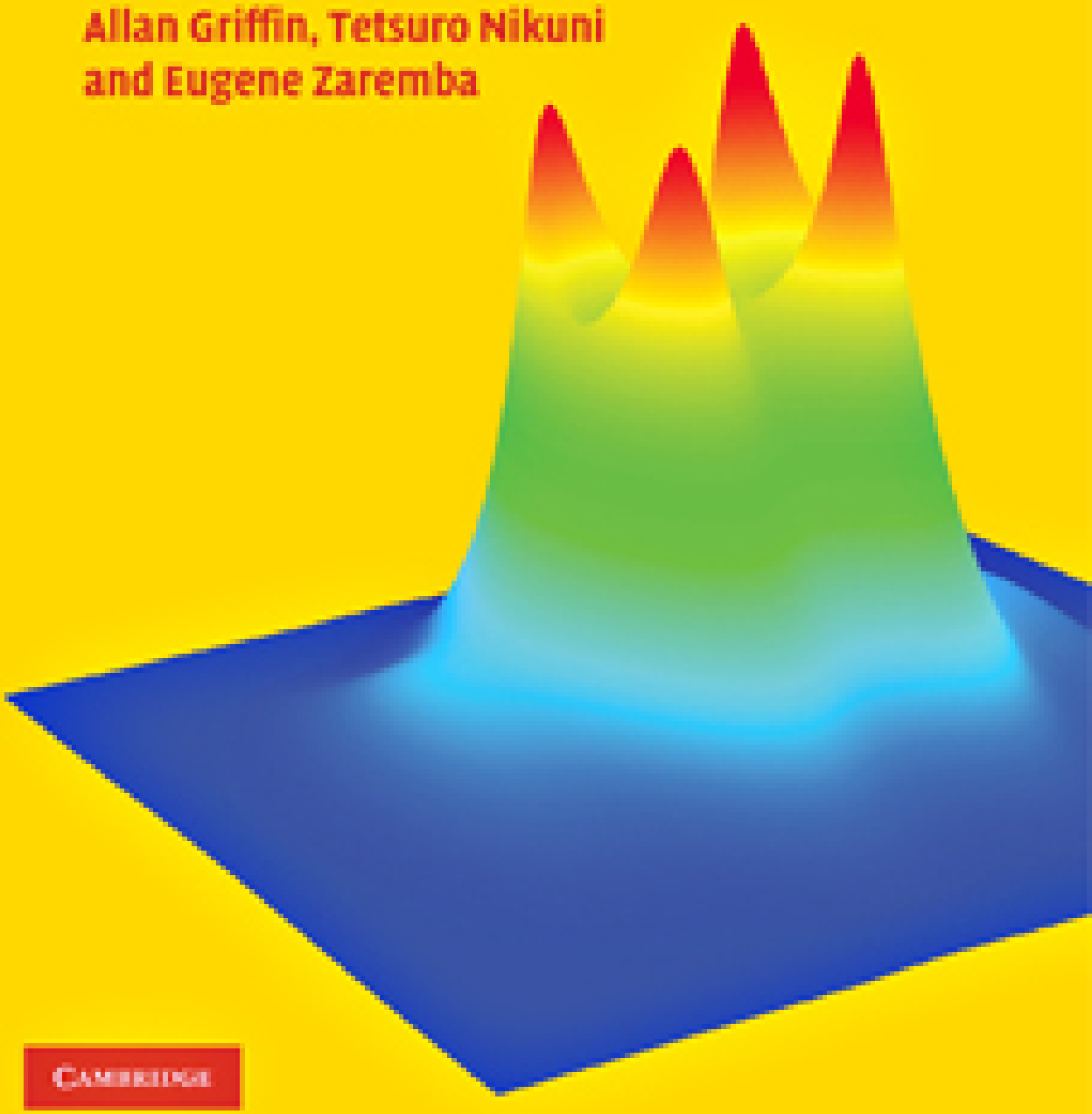
Bose-Condensed Gases at Finite Temperatures

Allan Griffin, Tetsuro Nikuni
and Eugene Zaremba



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BOSE-CONDENSED GASES AT FINITE TEMPERATURES

The discovery of Bose–Einstein condensation (BEC) in trapped ultracold atomic gases in 1995 has led to an explosion of theoretical and experimental research on the properties of Bose-condensed dilute gases.

The first treatment of BEC at finite temperatures, this book presents a thorough account of the theory of two-component dynamics and non-equilibrium behaviour in superfluid Bose gases. It uses a simplified microscopic model to give a clear, explicit account of collective modes in both the collisionless and collision-dominated regions. Major topics such as kinetic equations, local equilibrium and two-fluid hydrodynamics are introduced at an elementary level, before more detailed treatments and microscopic derivations of the underlying equations are given. Explicit predictions are worked out and linked to experiments. Providing a platform for future experimental and theoretical studies on the finite temperature dynamics of trapped Bose gases, this book is ideal for researchers and graduate students in ultracold atom physics, atomic, molecular and optical physics, and condensed matter physics.

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Preface

Since the creation of Bose–Einstein condensation (BEC) in trapped atomic gases in 1995, there has been an enormous amount of research on ultracold quantum gases. However, most theoretical studies have ignored the dynamical effect of the thermally excited atoms. In this book, we try to give a clear development of the key ideas and theoretical techniques needed to deal with the dynamics and nonequilibrium behaviour of trapped Bose gases at finite temperatures. By limiting ourselves from the beginning to a relatively simple microscopic model, we can concentrate on the new physics which arises when dealing with the correlated motions of both the condensate and noncondensate degrees of freedom. This book also sets the stage for the future generalizations that will be needed to understand the coupled dynamics of the superfluid and normal fluid components in strongly interacting Bose gases, where there is significant depletion of the condensate even at $T = 0$.

The core of this book is based on a long paper published by the authors (Zaremba, Nikuni and Griffin, 1999). In the last decade, together with our coworkers, we have extended and applied this work in many additional papers. The starting point of our approach is not original, in that it consists of combining the Gross–Pitaevskii equation for the condensate with a Boltzmann equation for the noncondensate atoms. The kinetic equation for trapped superfluid Bose gases we use was first developed and studied in a pioneering series of papers by Kirkpatrick and Dorfman in 1985 on a uniform Bose gas at finite temperatures. In the initial phase of writing this book, we intended to refer to our coupled equations for the condensate and the thermal cloud by some name other than ZNG, but nothing we came up with was particularly descriptive or natural and we finally gave up the attempt. Therefore, we simply refer to them as the ZNG coupled equations, following the custom in the current literature of the last few years.

In the introductory chapter we discuss in more detail how our starting

equations and general approach for dealing with a Bose-condensed gas at finite temperature are built on the work of previous theorists over the last five decades. On a longer time scale, our book makes use of methods for solving kinetic equations for Bose gases that were originally developed for classical gases by Boltzmann, Enskog and Chapman over a century ago.

This book should be of interest to all graduate students and researchers working in the area of ultracold dilute quantum gases and BEC. We hope that it will be particularly useful to experimental researchers, who may find the original theoretical literature difficult to understand, not to mention widely dispersed among many separate papers. It should also appeal to those in the wider theoretical and mathematical physics community who are interested in nonequilibrium problems in general, in quantum transport theory, in many body theory or in quantum field theory at finite temperatures. We have made no attempt to repeat the material on BEC in trapped atomic gases discussed in the two excellent books by Pethick and Smith (2008) and by Pitaevskii and Stringari (2003), and we hope these books will be referred to as needed by the reader.

The book includes a detailed treatment of the collision-dominated two-fluid region. In a graduate course on ultracold atoms, an introduction to the two-fluid hydrodynamics in trapped superfluid gases could be based on Chapters 2, 3, 14 and 15 as well as Section 8.1 of Chapter 8.

Originally our plan was to discuss the connections with other theoretical approaches used to deal with trapped Bose gases at finite temperatures. However, once we found that a clear exposition and development of our approach and the physics underlying it already filled 450 pages, our plan to review other formalisms in any depth was not feasible. We give references to other approaches at various places in the book. In addition, excellent reviews given at a recent workshop are available online (see footnote 2 in Chapter 1).

Allan Griffin would like to acknowledge the work of his students on the theory of interacting Bose gases, both before and after the discovery of BEC in 1995. In particular, Edward Taylor has been helpful in editing Chapters 14 and 16. Tetsuro Nikuni thanks Satoru Konabe and Takashi Inoue for valuable discussions and useful suggestions.

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1

Overview and introduction

Since the dramatic discovery of Bose–Einstein condensation (BEC) in trapped atomic gases in 1995 (Anderson *et al.*, 1995), there has been an explosion of theoretical and experimental research on the properties of Bose-condensed dilute gases. The first phase of this research was discussed in the influential review article by Dalfovo *et al.* (1999) and in the proceedings of the 1998 Varenna Summer School on BEC (Inguscio *et al.*, 1999). More recently, this research has been well documented in two monographs, by Pethick and Smith (2008, second edition)[1] and by Pitaevskii and Stringari (2003). Most of this research, both experimental and theoretical, has concentrated on the case of low temperatures (well below the BEC transition temperature, $T_{\rm BEC}$), where one is effectively dealing with a *pure* Bose condensate. The total fraction of noncondensate atoms in such experiments can be as small as 10% of the total number of atoms and, equally importantly, this low-density cloud of thermally excited atoms is spread over a much larger spatial region compared with the high-density condensate, which is localized at the centre of the trapping potential. Thus most studies of Bose-condensed gases at low temperatures have concentrated entirely on the condensate degree of freedom and its response to various perturbations. This region is well described by the famous Gross–Pitaevskii (GP) equation of motion for the condensate order parameter $\Phi(\mathbf{r}, t)$. As shown by research since 1995, this pure condensate domain is very rich in physics.

The main goal of the present book, in contrast, is to describe the dynamics of dilute trapped atomic gases at finite temperatures such that the noncondensate atoms also play an important role. This means that we shall be concerned with a trapped Bose gas composed of two distinct components,

[1] The first edition of the Pethick and Smith book was published in 2002. We give page references to the expanded second edition, published in 2008. The first 13 chapters in both editions cover similar material.

the condensate and the noncondensate. These two components satisfy quite different equations of motion but can be strongly coupled to each other and hence can modify each other significantly. The coupled dynamics of a two-component superfluid Bose gas brings in a whole new class of phenomena. At a theoretical level, one clearly has to deal with a "generalized" GP equation for $\Phi(\mathbf{r}, t)$ which now includes the effect of the mean fields and collisions associated with the noncondensate atoms. Broadly speaking, the thermal cloud atoms will be described by a kinetic equation for a normal gas of atoms, such as the well-known Boltzmann equation for a classical gas. The major difference is that in a trapped Bose-condensed gas the thermal atoms are coupled to the condensate component via mean fields and collisions.

The theory of Bose-condensed gases has been an active research topic since the ground-breaking work of Bogoliubov in 1947. The present book is built on the rich body of research carried out in the period 1957–67 by Lee and Yang, Beliaev, Pitaevskii, Hugenholtz and Pines, Hohenberg and Martin, Gavoret and Nozières, Kane and Kadanoff and many others. More specifically, what we shall call the Zaremba–Nikuni–Griffin (ZNG) approximation (see the preface and Chapter 3) is very much an extension for trapped gases of the pioneering studies by Kirkpatrick and Dorfman (1983, 1985a,b). These authors derived a kinetic equation for the thermal atoms in a uniform Bose-condensed gas and used it to give an explicit derivation of the Landau two-fluid equations that account for transport coefficients. However, when their papers were published in 1985, research interest in BEC in gases was very low and their work had little impact.

Looking back, it is perhaps surprising that in the early work on Bose-condensed gases there was almost no explicit discussion of a time-dependent equation of motion for the condensate. It was only after the discovery of BEC in trapped ultracold atomic gases in 1995 that the time-dependent Gross–Pitaevskii equation and its extensions became central in theoretical discussions, even though this equation had been developed in 1961. The ZNG theory "stitches" together a generalized GP equation for the Bose condensate (which includes the coupling to the thermal cloud atoms) and a kinetic equation for the thermal cloud atoms. The ZNG approximation is thus the offspring of a "civil union" between Gross–Pitaevskii and Bogoliubov on the one hand and Boltzmann on the other.

So far, the study of the dynamics of a trapped Bose-condensed gas at finite temperatures has not been a topic of systematic experimental studies. Part of the reason for this, we believe, has been the implicit belief that the presence of a thermal cloud of noncondensate atoms just complicates the behaviour of a pure $T = 0$ condensate and is not the source of any interesting

"new physics". A theme of this book is that this attitude is not justified. The coupling of the condensate and noncondensate degrees of freedom at finite temperatures leads to a two-component system in which *both* components can exhibit coherent collective behaviour, resulting in many new phenomena. Indeed, under certain conditions, a trapped Bose gas at finite temperatures can exhibit two-fluid phenomena that are precisely analogous to the well-known macroscopic quantum behaviour in superfluid ^{4}He (For details, see Chapters 15 and 17).

The present book is devoted first (Chapters 3–7) to deriving coupled equations for a two-component superfluid Bose gas within a simple but realistic microscopic approximation for each component. The second major goal (Chapters 8–19) is to solve these approximate equations for the dynamics of a trapped Bose gas at finite temperatures in two different regions, the collisionless (or mean-field dominated) domain and the hydrodynamic (or collision-dominated) domain. In the collisionless region, for which there are considerable experimental data, our coupled equations give results that are in quantitative agreement with the observed temperature-dependent frequency and damping of the many kinds of collective modes that can be excited in ultracold Bose gases.

In the quite different hydrodynamic region, collisions bring the system into a state of local thermodynamic equilibrium. We prove that our approximate model equations lead to the well-known two-fluid hydrodynamical description first derived (Landau, 1941; Khalatnikov, 1965) for superfluid ^{4}He. This connection allows one to make a very detailed comparison between the hydrodynamics of a trapped Bose-condensed gas and that of liquid ^{4}He and emphasizes the key role of the Bose broken symmetry (Hohenberg and Martin, 1965; Anderson, 1966, 1994; Bogoliubov, 1970). Strangely enough, the hydrodynamics of a dilute Bose-condensed gas can be even more complex than that of superfluid ^{4}He. The reason, as we discuss in Chapter 15, is that in a trapped gas the condensate and noncondensate can be out of *diffusive* local equilibrium with each other.

Discussions that start with the dynamics of a pure condensate at $T = 0$ can give the impression that a trapped Bose-condensed gas is some completely new phase of matter, unconnected with other interacting many body systems. The point of view of the present book is quite different, in that we start with the normal phase. That is, a Bose superfluid (gas or liquid) is viewed as a normal fluid in which, as a result of a second-order phase transition, an extra new degree of freedom emerges, namely, a condensate described by the macroscopic wavefunction $\Phi(\mathbf{r},t)$. A crucial question is how this new "superfluid" degree of freedom couples into and modifies the "nor-

mal fluid" degrees of freedom. Answering this question leads, in our opinion, to a deeper insight into the dynamics of a two-component Bose superfluid. This approach allows one to put the two extreme limits, a pure condensate ($T \ll T_{\rm BEC}$) and a pure thermal cloud ($T > T_{\rm BEC}$) into a broader context. It also sets the stage for developing a two-fluid description in trapped gases similar to that used to describe the low-frequency hydrodynamics of superfluid ^{4}He.

The theory of interacting Bose-condensed fluids is most usefully discussed using quantum field operators. This approach was initiated by Bogoliubov (1947) in a simple model calculation, formalized in a systematic way by Beliaev (1958a), and then developed by Gavoret and Nozières (1964), Hohenberg and Martin (1965), Bogoliubov (1970) and many others in the early 1960s. This many body formalism is discussed in detail in the well-known texts Abrikosov *et al.* (1963) and Fetter and Walecka (1971). We recall that

$$\begin{array}{l} \text{the operator } \hat{\psi}^\dagger(\mathbf{r}) \text{ creates an atom at } \mathbf{r}; \\ \text{the operator } \hat{\psi}(\mathbf{r}) \text{ destroys an atom at } \mathbf{r}. \end{array} \tag{1.1}$$

These quantum field operators satisfy the usual Bose commutation relation

$$\left[\hat{\psi}(\mathbf{r}), \hat{\psi}^\dagger(\mathbf{r}')\right] = \delta(\mathbf{r} - \mathbf{r}'). \tag{1.2}$$

All observables can be written in terms of these operators; for example, the density $\hat{n}(\mathbf{r}) = \hat{\psi}^\dagger(\mathbf{r})\hat{\psi}(\mathbf{r})$ and the interaction energy is given by

$$\hat{V} = \tfrac{1}{2} \int d\mathbf{r} \int d\mathbf{r}'\, \hat{\psi}^\dagger(\mathbf{r}')\hat{\psi}^\dagger(\mathbf{r})v(\mathbf{r} - \mathbf{r}')\hat{\psi}(\mathbf{r}')\hat{\psi}(\mathbf{r}), \tag{1.3}$$

where $v(\mathbf{r})$ is the interatomic potential.

The crucial idea, due to Bogoliubov (1947) and later generalized by Beliaev (1958a,b), is to separate out the condensate component of the quantum field operators, setting

$$\hat{\psi}(\mathbf{r}) = \langle\hat{\psi}(\mathbf{r})\rangle + \tilde{\psi}(\mathbf{r}), \tag{1.4}$$

where

$$\langle\hat{\psi}(\mathbf{r})\rangle \equiv \Phi(\mathbf{r}) \tag{1.5}$$

is the Bose macroscopic wavefunction. This quantity plays the role of the "order parameter" for the Bose superfluid phase transition:

$$\Phi(\mathbf{r}) = \begin{cases} 0, & \text{if } T > T_{\rm BEC}, \\ \neq 0, & \text{if } T < T_{\rm BEC}. \end{cases} \tag{1.6}$$

We note that $\Phi(\mathbf{r}) = \sqrt{n_{\rm c}}e^{i\theta}$ is a two-component order parameter in that it

has both amplitude and phase. Clearly, $\Phi(\mathbf{r})$ is not simply related to many-particle wavefunctions $\Psi(\mathbf{r}_1, \mathbf{r}_2, \ldots, \mathbf{r}_N)$. The thermal average in $\langle\hat{\psi}(\mathbf{r})\rangle$ involves introducing a small *symmetry-breaking* perturbation that allows $\Phi(\mathbf{r})$ to be finite,

$$\hat{H}_{\rm SB} = \lim_{\eta\to 0} \int d\mathbf{r} \left[\eta(\mathbf{r})\hat{\psi}^\dagger(\mathbf{r}) + \eta^*(\mathbf{r})\hat{\psi}(\mathbf{r})\right]. \tag{1.7}$$

The philosophy behind the concept of symmetry-breaking and its use in a variety of condensed matter systems was extensively discussed by Bogoliubov (1970) in a beautiful and convincing article which is highly recommended to all readers. Since the early 1960s, the concept of a broken-symmetry order parameter has increasingly become the basis of all modern treatments of different phases of matter in all branches of physics. A lucid systematic account of this approach as a basis for statistical mechanics is developed in recent monographs by Mazenko (2000, 2003).

The Beliaev (1958a) decomposition (1.4) of the quantum field operator implies that the single-particle density matrix has the property

$$\lim_{|\mathbf{r}-\mathbf{r}'|\to\infty} \rho_1(\mathbf{r}, \mathbf{r}') \equiv \langle\hat{\psi}(\mathbf{r})\hat{\psi}^\dagger(\mathbf{r}')\rangle = \Phi(\mathbf{r})\Phi^*(\mathbf{r}'). \tag{1.8}$$

Penrose (1951) and Penrose and Onsager (1956) first gave a formal definition of what BEC is in an interacting Bose gas in terms of the asymptotic property given in (1.8). Specifically, they defined the wavefunction of the Bose condensate as the eigenstate of $\rho_1(\mathbf{r}, \mathbf{r}')$ which is macroscopically occupied. The Penrose–Onsager approach is nicely summarized in Section 2.1 of Pitaevskii and Stringari (2003) and Section 2.1 of Leggett (2006).

The later, independent, formulation of Beliaev (1958a), based on separating the condensate part of the quantum field operator $\hat{\psi}(\mathbf{r})$ as in (1.4), extended the powerful field theoretic techniques for dealing with the dynamics of many body systems to include Bose condensation. The Beliaev Green's function approach allows one to avoid working directly with many body wavefunctions (as in the original approach of Penrose and Onsager, 1956) and instead work with equations of motion for Green's functions. The Beliaev formulation involves defining the Bose condensate as a broken-symmetry order parameter (1.5), which can be time dependent. This extends the original Penrose–Onsager formulation, which was based on number conserving eigenstates. The usefulness of working with number nonconserving states and anomalous Green's functions was clarified by Anderson in a classic article written in 1965 (reprinted in Anderson, 1994, p. 229). See also Section 2.2 of Pitaevskii and Stringari (2003).

To avoid any misunderstanding about the use of number nonconserving

states, we emphasize that all conservation laws (including the continuity equation for density fluctuations) can be satisfied in theories based on a broken-symmetry order parameter. As is well understood (see for example Griffin, 1993), conservation laws play a crucial role in Bose superfluids.

We note that Beliaev's name is attached to two papers on interacting Bose systems which have quite different goals. In the first paper, Beliaev (1958a) set up a *general* formalism to deal with an interacting Bose-condensed system using diagrammatic methods incorporating the presence of a Bose order parameter $\Phi(\mathbf{r})$. In a companion paper, Beliaev (1958b) used this formalism to calculate the excitation spectrum of the single-particle Green's functions $\tilde{G}_{\alpha\beta}(\mathbf{q},\omega)$ to second order in the interaction. This is the famous Beliaev second-order approximation, generalizing the first-order Bogoliubov approximation.

The condensate wavefunction $\Phi(\mathbf{r},t)$ is a *coherent* state, with a "clamped" value of the phase, rather than a Fock state of fixed N with no well-defined phase. The order parameter $\Phi(\mathbf{r},t)$ acts like a *classical* field, since quantum fluctuations are negligible when the number of atoms $N_{\rm c}$ in the single-particle condensate wavefunction is large. As noted above, Anderson (1966, 1994) deserves great credit for understanding (in the period 1958–1963) the new physics involved in working with a broken-symmetry state $\Phi(\mathbf{r},t)$ with a well-defined phase, both in BCS superconductors and in superfluid ^{4}He. This broken-symmetry state nicely captures the physics of the Josephson effect and superfluidity. The external symmetry-breaking perturbation (1.7) allows the system to set up off-diagonal symmetry-breaking fields internally, which persist even when the external perturbation is set to zero at the end of the calculation ($\eta \to 0$).

The same sort of physics is the basis of the well-known Bardeen–Cooper–Schrieffer (BCS) theory of superconductors, based on the formation of bound pairs of fermions called Cooper pairs, which are bosons and hence can Bose-condense. Indeed, it is interesting to recall that before the BCS theory appeared in 1957, Bogoliubov's pioneering paper, published in 1947, was largely unknown or ignored. After the physics of the BCS theory was reformulated in a simple fashion involving symmetry-breaking mean fields related to a Cooper-pair condensate (Gor'kov, 1958), theorists quickly realized that the Bogoliubov theory of Bose-condensed gases involved a similar kind of "off-diagonal" mean field. Since the 1960s, in condensed matter physics our understanding of superconductors has gone hand-in-hand with our understanding of superfluid ^{4}He. In particular, the BCS theory has played an important role as an example of how a broken-symmetry theory captures

the physics of superfluid motion, and it has the advantage of *not* working with a number-conserving approximation.

One of the stunning developments in ultracold gases is the experimental realization of the BCS–BEC crossover in a two-component Fermi gas (for an introduction, see Chapter 17 of Pethick and Smith, 2008). Using a Feshbach resonance to tune the magnitude and sign of the s-wave scattering length a between Fermi atoms in different hyperfine states, one can go in a smooth fashion from a BCS phase with Cooper pairs immersed in a gas of fermion BCS quasiparticles to a Bose condensate phase in which all the fermion excitations have paired up to form Bose molecules. This BEC of molecules is a very promising new "Bose gas", since molecules made up of two fermions are very stable against three-body decay in a two-component Fermi gas, because of the Pauli exclusion principle. Moreover, the molecular scattering length is proportional to a and hence can be very large (Petrov *et al.*, 2004, 2005) near the Feshbach resonance (where $|a| \to \infty$). As a final bonus, the creation of bosonic molecules via the destruction of two fermions is a concrete illustration of a physical process that does *not* conserve the total number of bosons and thus can be thought of as a symmetry-breaking perturbation of the type (1.7).

One can formulate the Gross–Pitaevskii and Bogoliubov approximations directly in terms of variational many-particle wavefunctions. However, such formulations are usually limited to simple mean-field approximations at $T = 0$. The explicit introduction of the broken-symmetry order parameter $\Phi(\mathbf{r}, t)$ gives a more systematic way (Beliaev, 1958a; Hohenberg and Martin, 1965; Bogoliubov, 1970; Nozières and Pines, 1990) of isolating the role of the Bose condensate within a general treatment of an interacting Bose-condensed fluid at finite temperatures. This approach was developed in order to understand the characteristic properties of a Bose superfluid such as liquid ^{4}He, in spite of the fact that one could not do quantitative calculations on such a strongly interacting system. A major goal of this book is to show that the resulting formalism allows one to treat, in an easy and natural manner, questions related to damping as well as superfluidity at finite temperatures in both the collisionless and hydrodynamic regions.

As already noted, this book deals with the Bose condensate by means of the approach formalized and developed by Beliaev (1958a,b), which is based on separating out the Bose-condensate degree of freedom as an order parameter related to a broken symmetry. The finite value of the condensate order parameter leads to new correlations in space and time between the noncondensate atoms. As a result, besides using the ordinary single-particle Green's functions, it is natural to introduce anomalous ("off-

diagonal") single-particle Green's functions to describe the new condensate-induced correlations between the atoms outside the condensate. In the early 1960s, it was realized that one could give a compact version of many body perturbation theory for Bose superfluids by working with a 2×2 matrix single-particle Green's function $\tilde{G}_{\alpha\beta}$ describing the noncondensate atoms. A similar scheme for BCS superconductors grew out of the related work of Gor'kov (1958). For over five decades, these Green's function techniques have been used with great success in dealing with both Bose and Fermi superfluids.

By the 1960s, it was realized that there are three paradigms for quantum fluids:

(1) Bose superfluids (associated with a Bose condensate wavefunction Φ);
(2) normal Fermi fluids (associated with the key role of a Fermi surface);
(3) Fermi superfluids (associated with Cooper pairs that form a Bose condensate).

The first two kinds of quantum fluid were magnificently described in two books by Nozières and Pines written in the early 1960s, although the book on superfluid Bose liquids (Nozières and Pines, 1990) was only published decades later. One of the clearest discussions of the connection between the order parameter $\Phi(\mathbf{r}, t)$ and superfluidity in Bose fluids is given in Chapters 4 and 5 of Nozières and Pines (1990). The classic account formulating the various levels of theory for Bose superfluids is the monumental paper by Hohenberg and Martin (1965). This paper shows the central unifying role of the broken-symmetry order parameter $\Phi(\mathbf{r}, t)$, summarizes the physics involved in both the collisionless and hydrodynamic domains and gives criteria for developing and judging various microscopic approximation schemes for correlation functions, using thermal Green's function techniques. The general philosophy and approach of the present book has been strongly influenced by Hohenberg and Martin's seminal paper.

The introductory review article by Leggett (2001) and his recent book on quantum liquids (Leggett, 2006) give a thoughtful account of many basic assumptions used in current theories on ultracold gases. However, we do not share Leggett's reservations about the usefulness of the concept of a Bose order parameter arising from a broken symmetry. Treating the condensate as a new degree of freedom becomes especially convenient when one is attempting to deal with the dynamics of a trapped Bose-condensed gas at finite temperatures, as we hope to illustrate in the present book.

We have emphasized that our approach to nonequilibrium problems in Bose gases involves separating out the condensate right at the beginning,

following the classic Bogoliubov–Beliaev approach. This means that the condensate and noncondensate dynamics are treated in quite different ways, $\Phi(\mathbf{r}, t)$ playing the role of the order parameter. There is an alternative approach used in the current literature in which the equations of motion for all low-energy bosonic modes (for which the occupation numbers $N_i \gg 1$) are treated classically. This "classical field" approach had its origins in the theory of lasers and quantum optics. It leads to what are called stochastic GP equations, in which an analogue of $\Phi(\mathbf{r}, t)$ describes both the condensate mode and low-energy excitations on an equal basis. We note that such stochastic GP equations have a basis completely different from that of the condensate generalized GP equation we introduce in Chapter 3. For reviews, applications and further references to such equations, see Sinatra, Lobo and Castin (2001); Davis, Morgan and Burnett (2001); Gardiner, Anglin and Fudge (2002); Bradley, Blakie and Gardiner (2004); and Brewczyk, Gajda, and Rzazewski (2007). While having advantages in certain problems, the classical field approach has not yet been extensively implemented in the study of collective modes in trapped Bose gases. Indeed, it is not clear how this approach could be used to discuss the collisional hydrodynamic region.

For a comparison of various formalisms for dealing with trapped Bose-condensed gases at finite temperatures, we refer to the proceedings of a recent workshop on this topic[2].

1.1 Historical overview of Bose superfluids

To put the coupled equations for the condensate and thermal cloud into context, we now briefly review some features of the theory of superfluidity in liquid ^{4}He. In later chapters, we often make connections between the properties of superfluid Bose gases at finite temperatures and superfluid ^{4}He.

The original discovery of superfluidity in liquid ^{4}He was announced in the famous papers by Kapitza (1938) working in Moscow and by Allen and Misener (1938) based in Cambridge. These and subsequent experiments in the following years (for a review see Wilks, 1967) showed that, in comparison with classical fluids, superfluid ^{4}He could exhibit very bizarre behaviour. The attempt to understand this behaviour led to the development of a two-fluid theory of the hydrodynamic behaviour of liquid ^{4}He by Landau (1941). An earlier but less complete two-fluid theory based on a dilute Bose gas

[2] *Proc. Workshop on Nonequilibrium Behaviour in Superfluid Gases at Finite Temperatures*, Sandbjerg, Denmark June 10–13, 2007 (http://www.phys.au.dk/nonequilibrium/Home.html). See also the long tutorial review by Proukakis and Jackson (2008).

was developed by Tisza in the period 1938–40. For further discussion of the history of BEC and superfluids, see Griffin (1999a) and Balibar (2007).

In this early work, superfluidity (the term was coined in 1938 by Kapitza) was entirely associated with the *relative* motion of the normal and superfluid components under a variety of conditions (Khalatnikov, 1965; Wilks, 1967). The main conclusion of this early research was that while the normal fluid exhibited the finite viscosity and thermal conductivity typical of an ordinary fluid, the superfluid component (which exhibited only irrotational flow) did not. In more recent times, an aspect of superfluidity that has been emphasized as most central (for example, see Chapter 4 of Nozières and Pines, 1990; Leggett, 2001, 2006) is that the superfluid velocity is associated with the gradient of the phase of the macroscopic wavefunction $\Phi(\mathbf{r}, t)$. However, an equally important property to understand is why superfluidity persists *even* in the presence of a dissipative normal fluid. This question can best be addressed by studying the local equilibrium region induced by strong collisions, a region described by the two-fluid hydrodynamic equations.

In essence, Landau developed his generic two-fluid hydrodynamics by generalizing the standard theory of classical hydrodynamics (see, for example, Huang, 1987) to include the equations of motion for a new "superfluid" degree of freedom. We recall that classical fluid dynamics was developed well before the existence of atoms had been demonstrated. Since the work of Maxwell and Boltzmann in the 1880s, it has been known that a "coarse-grained" hydrodynamic description of a fluid, in terms of just a few quantities such as the local density $n(\mathbf{r}, t)$ and the local velocity $\mathbf{v}(\mathbf{r}, t)$, is only valid when the collisions between atoms are strong enough to produce "local equilibrium". As a result, hydrodynamics describes only low-frequency phenomena, for which the fluid is in local equilibrium. This is defined by the condition $\omega\tau \ll 1$, where ω is the frequency of the collective mode and τ is the collisional relaxation time to reach local equilibrium.

The description of a fluid in terms of a few hydrodynamic local variables was developed by Bernoulli, Euler and others in the eighteenth century. In work which led to a microscopic basis for these hydrodynamic theories, Boltzmann introduced the key concept of a kinetic equation to describe the nonequilibrium behaviour of atoms in a dilute classical gas. He developed the idea that such a gas would approach thermal equilibrium in several distinct stages. Initially the dynamical behaviour is very complex. However, the system eventually reaches the so-called "kinetic" stage, which can be described by a single-particle distribution function $f(\mathbf{p}, \mathbf{r}, t)$. The latter is given by the solution of a kinetic equation, the structure of which (even when we generalize it to deal with a Bose-condensed gas) is usefully written

in the form

$$\hat{\mathcal{L}} f(\mathbf{p}, \mathbf{r}, t) = C[f(\mathbf{p}, \mathbf{r}, t)]. \tag{1.9}$$

The differential operator $\hat{\mathcal{L}}$ describes the dynamical changes in $f(\mathbf{p}, \mathbf{r}, t)$, as time develops, under the influence of external (and self-induced internal) fields. The r.h.s. involves a collision integral $C[f]$, which is a functional of the distribution function $f(\mathbf{p}, \mathbf{r}, t)$.

The so-called *collisionless* limit describes the case when the collision terms on the r.h.s. of (1.9) can be neglected in a first approximation, with f the solution of

$$\hat{\mathcal{L}} f = 0. \tag{1.10}$$

In the *opposite* limit, the form of $f(\mathbf{p}, \mathbf{r}, t)$ is determined by the collisions, i.e. by the condition that $f = \tilde{f}$ where

$$C[\tilde{f}] = 0. \tag{1.11}$$

Satisfying this condition is quite different from neglecting the collision term in (1.9) to obtain (1.10). In (1.11), the collision integral is in general finite and only vanishes if the distribution function f takes on a *special* form denoted by $\tilde{f}$. The solution of (1.11) determines the unique local-equilibrium solution $f(\mathbf{p}, \mathbf{r}, t)$ enforced by rapid collisions. This special solution $\tilde{f}$ is identical to the Bose distribution describing thermal equilibrium, except that thermodynamic parameters such as temperature, pressure and chemical potential now depend on position and time. These local variables characterize "local hydrodynamic equilibrium", which is the domain reached by the system before complete thermal equilibrium, the so-called *hydrodynamic* stage. The differential equations of fluid dynamics, which determine these local variables, can be derived from the more fundamental kinetic equation for $f(\mathbf{p}, \mathbf{r}, t)$ by taking moments.

In his seminal 1941 paper, Landau did not connect the superfluid component with the motion of a "Bose condensate". Indeed, he rejected the efforts by Tisza (1938, 1940) and London (1938a, b) to use a Bose-condensed gas as a model for superfluid ^{4}He. However, since the period 1957–65, Landau's superfluid degree of freedom has been understood microscopically in terms of the complex order parameter $\Phi(\mathbf{r}, t)$. In this microscopic theory, the superfluid velocity field $\mathbf{v}_{\rm s}(\mathbf{r}, t)$ is always related to the gradient of the phase of $\Phi(\mathbf{r}, t)$, as we shall discuss in Chapters 2 and 3.

Landau's pioneering work on superfluid ^{4}He has two separate aspects, which are logically distinct and should not be confused with each other:

(a) *The first aspect is the two-fluid equations describing hydrodynamic*

behaviour. These equations are generic and apply (under certain conditions) to trapped Bose gases as well as to superfluid ^{4}He. In the period 1947–50, Landau and Khalatnikov extended these equations to include hydrodynamic damping of the normal fluid (described by various kinds of viscosities, thermal conductivity, etc.). This work is described in the classic monograph by Khalatnikov (1965).

(b) *The second aspect is a "microscopic" theory of the elementary excitations describing the normal fluid in the superfluid phase of liquid* ^{4}He. This is the famous phonon–roton quasiparticle spectrum. Within this picture of a weakly interacting gas of *quasiparticles*, Landau and coworkers could calculate the thermodynamic and transport properties of superfluid ^{4}He. These quantities enter the two-fluid equations of Landau and Khalatnikov and hence determine the velocity and damping of the first sound and second sound modes.

As we have noted, Landau's original formulation of his two-fluid hydrodynamic equations was phenomenological, in that the superfluid component was not given an explicit microscopic basis. Indeed, Landau (1941) does not even mention atoms in his famous paper. The first explicit derivation of the Landau two-fluid hydrodynamic equations starting from the existence of the macroscopic broken-symmetry order parameter $\Phi(\mathbf{r}, t)$ was given by Bogoliubov in a 1963 preprint, but only published some years later (Bogoliubov, 1970). This proof was built on Bogoliubov's pioneering work on deriving hydrodynamic equations for classical liquids.

In Fig. 1.1, we show some of the classic developments on the nonequilibrium behaviour of interacting gases; these form the conceptual background of the present book. The arrows in this figure indicate the connections between the transport and hydrodynamic equations, first developed for classical gases and later extended to deal with Bose-condensed fluids.

1.2 Summary of chapters

Recent research on trapped Bose gases and Bose–Einstein condensation is authoritatively reviewed in the books by Pethick and Smith (2008) and Pitaevskii and Stringari (2003). These reflect increasing interest in the field of ultracold atoms as well as the rapid progress made since 1995. However, as we have noted, both books concentrate on the Gross–Pitaevskii limit of a pure condensate (i.e. very low temperatures); the effects associated with the dynamics of the thermal cloud are only briefly mentioned. In the present book, in contrast, we focus on the dynamics of trapped Bose gases

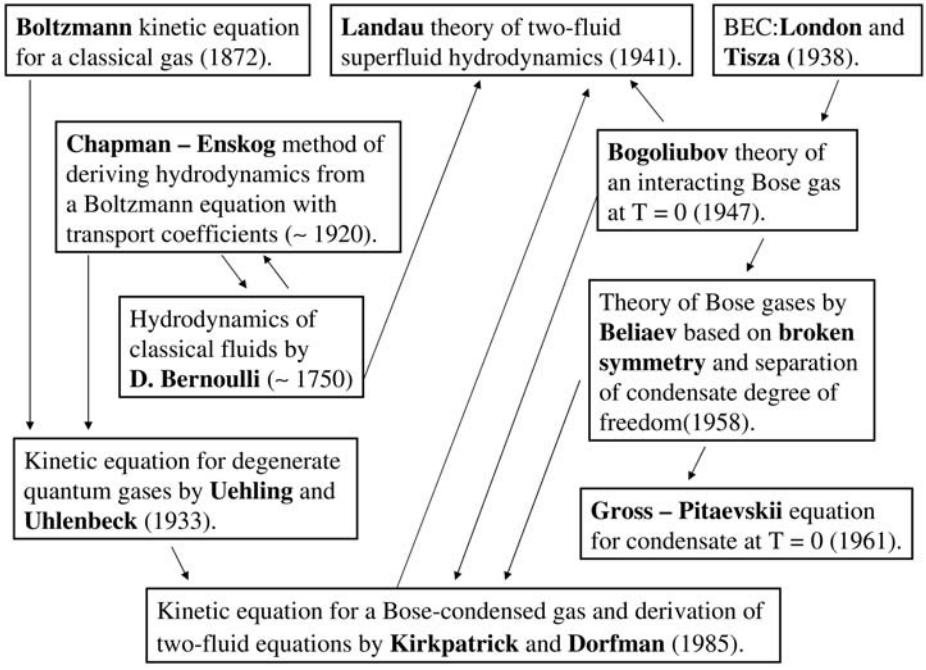


Fig. 1.1. This flow diagram shows some of the key theoretical developments that form the basis and approach of the present book.

at finite temperatures and explore the consequences of having two coupled components (the condensate and noncondensate atoms) which act quite differently. Clearly the finite-temperature domain includes the important limit in which the Bose condensate vanishes and we are left with a normal Bose gas. In addition, the equations of two-fluid hydrodynamics, used to describe superfluid ^{4}He, must emerge as the description of trapped Bose gases when collisions are strong enough to produce local equilibrium.

With our coworkers, we have written a series of interconnected papers on the nonequilibrium behaviour of trapped Bose gases at finite temperatures. In particular, we have derived and developed the simplest nontrivial approximation for the coupled dynamics of the condensate and thermal atoms, which we shall call the Zaremba–Nikuni–Griffin or ZNG equations. The ZNG equations have been solved numerically using Monte Carlo techniques by Jackson and Zaremba and the results used to explain the frequency and damping of collective modes in dilute Bose gases at finite temperatures, in quantitative detail. In the other extreme, the collisional hydrodynamic

limit, the ZNG equations have been shown to lead to two-fluid hydrodynamic equations, including all transport processes; these are analogous to the Landau–Khalatnikov equations, well known in superfluid ^{4}He. The fact that the ZNG approximation, simplified as it is, describes both the collisionless and hydrodynamic regions is strong evidence that it contains the essential physics for understanding trapped Bose superfluid gases.

However, there is a deficiency in our treatment of the collisional hydrodynamic (two-fluid) region at finite temperatures, starting from the ZNG coupled equations: namely, when the interactions are strong enough to produce local equilibrium, it is expected that there will be significant "quantum depletion" of the Bose condensate. That is, even at $T = 0$ the fraction of atoms outside the condensate should be significant (as in superfluid ^{4}He). Such depletion effects are completely neglected in the ZNG approximation, where the noncondensate component only arises due to thermal excitation of atoms out of the Bose condensate at finite temperature. As a result, in the two-fluid hydrodynamic equations that we derive in Chapters 15 and 17, the superfluid density is equal to the condensate density and consequently the normal fluid density is equal to the thermal cloud density. Moreover, the thermodynamic functions and transport coefficients in these equations are based on Hartree–Fock excitations.

The justification for basing our analysis in this book on the ZNG coupled equations is a combination of their simplicity coupled to the fact that they do lead to the Landau two-fluid equations which describe local collisional hydrodynamics. The ZNG equations allow us to derive (see Chapters 15 and 17) the Landau two-fluid equations from a well-defined microscopic model. In general, we expect that, even for strong interactions, our microscopic expressions for the coefficients in the two-fluid hydrodynamic equations will be good approximations at high temperatures in trapped gases, where the thermal depletion of the condensate dominates. An attempt to derive the Landau two-fluid equations with the inclusion of quantum depletion would require a much more complicated analysis, starting from the kind of generalized kinetic equations derived in Chapter 7.

In the course of setting up the theory leading to the coupled ZNG equations, we discuss the Bose order parameter, correlation functions, Green's functions, response functions, conservation laws, the reasons why the condensate and noncondensate atoms are governed by such different laws of motion, the meaning of the Boltzmann equations and their region of validity and, finally, what the collisional hydrodynamic domain really represents. Most of these key concepts originate from the kinetic theory of classical gases and condensed matter physics. We hope that this book will show the useful-

ness of these powerful ideas in understanding the nonequilibrium collective behaviour of Bose-condensed gases.

At several places in the book, we will work out the theory for a uniform Bose gas, for simplicity. Needless to say, if a fundamental problem or inconsistency arises (for example, conservation laws are not obeyed) in the case of a uniform Bose gas, an analogous problem will occur in dealing with a trapped gas.

A brief outline of the content of the subsequent chapters may be useful.

In Chapter 2, we review the dynamics of a pure condensate at $T = 0$ using the linearized GP equation. This $T = 0$ limit is especially appealing since one can ignore all the complications that arise from the presence of noncondensate atoms (the thermal cloud). We discuss the normal mode solutions of the $T = 0$ GP equation of motion using the "quantum hydrodynamic" formalism, which is expressed in terms of the local condensate density $n_c(\mathbf{r}, t) = |\Phi(\mathbf{r}, t)|^2$ and the superfluid velocity $\mathbf{v}_c(\mathbf{r}, t)$. Within the Thomas–Fermi approximation, Stringari (1996b) first showed that the equations of motion for these two variables can be combined to give a wave equation for oscillations of the condensate at $T = 0$.

In Chapter 3, we discuss the dynamics of the coupled condensate and noncondensate components, starting from the finite-T generalized GP equation for $\Phi(\mathbf{r}, t)$ and a kinetic equation for the single-particle distribution function $f(\mathbf{p}, \mathbf{r}, t)$ describing the noncondensate atoms. The coupled equations of motion summarized in Chapter 3 form the basis of the rest of this book. We should note that the simplified microscopic model which we use for the coupled condensate and thermal cloud components is not adequate to deal with the region of very low temperatures, or the region very close to the superfluid transition temperature where fluctuation effects play an important role.

In the kinetic equation describing the thermal atoms in a Bose-condensed gas at finite temperature, the two-body collision terms are of two distinct types (Kirkpatrick and Dorfman, 1983, 1985a). Collisions only involving noncondensate atoms are referred to as C_{22} processes. Collisions which involve one condensate atom are referred to as C_{12} processes. The C_{12} collision integral in the kinetic equation only arises in the Bose-condensed phase and plays a crucial role in the dynamics of the condensate at finite temperatures. Indeed, the effect of the C_{12} collisions is a major theme in every chapter of this book.

Chapters 4–7 form a major subsection of the book, giving a review of the microscopic theory of interacting Bose-condensed systems. The theory is built on the idea of a broken symmetry and formulated in terms of

correlation functions. In Chapter 4, we discuss the Green's function approach describing time-dependent phenomena in a superfluid Bose fluid, a formalism developed in the decade 1956–67. Following Hohenberg and Martin (1965), we classify various methods of computing response functions based on particular choices of the single-particle self-energies. In Chapter 5, we discuss the excitation spectrum given by the famous second-order Beliaev approximation for the single-particle self-energies. We also show how a *time-dependent* theory generated by the first-order self-energy approximation (Hartree–Fock–Bogoliubov theory) can generate a density response function which exhibits the Beliaev excitation spectrum. In Chapters 6 and 7 we develop the powerful Kadanoff–Baym approach (Kane and Kadanoff, 1965; Imamović-Tomasović and Griffin, 2001), based on nonequilibrium Green's functions and use it to derive the ZNG equations introduced in Chapter 3 as the simplest realization of this more general theory. Chapters 4–7 are fairly theoretical, and the reader who is mainly interested in final results and applications can go immediately to Chapter 8.

Chapters 8–13 are devoted to the ZNG theory predictions for Bose gas dynamics in what is traditionally called the "collisionless region", the region most studied in current BEC experiments. In the collisionless region the coupled dynamics is dominated by the condensate mean field, collisions playing an important subsidiary role. More precisely, the collision time is much larger than the period of the collective mode. In Chapters 8 and 9, we apply the ZNG equations to several problems in the simplest approximation, namely one in which the thermal cloud is always in *thermal equilibrium*. We explain how this limitation to a static thermal cloud still gives rise to the damping or growth of condensate collective modes, owing to collisions between atoms in the time-dependent condensate and the static thermal cloud.

In Chapter 8, we also use the ZNG coupled equations to give a microscopic basis for various phenomenological GP equations used in the literature which include dissipation. In Chapter 9 on vortices, we give a detailed treatment of the kinetic equation and the collision integrals in the presence of rotating anisotropic trap potentials. These results (which have not been published before) can be used to treat the effect of C_{12} collisions with the thermal cloud atoms on the formation of vortex lattices in a condensate at finite temperatures. This is illustrated within the static thermal cloud approximation.

In Chapter 10, we derive a simple set of coupled "moment equations", starting from our ZNG equations. This set of equations can be mathematically "closed" by approximating the collision term, which allows one to

solve them analytically. These analytic results are very useful, since they apply formally to both the collisionless and hydrodynamic dynamical regions. While this moment method does not include Landau damping, the results complement those obtained in other chapters.

In Chapters 11–13, we discuss the numerical solutions of the ZNG coupled equations of motion in the collisionless region, this time including the full dynamics of the thermal cloud. In Chapter 11, we discuss the Monte Carlo numerical techniques developed by Jackson and Zaremba (2002a) to solve the coupled ZNG equations of motion. This chapter is intended to be used as a "primer" on how to carry out such calculations for oneself. These calculations can explain a variety of experiments on the frequency and damping of collective modes at finite temperature in quantitative detail, as reviewed in Chapter 12. At finite temperatures, the damping of the collective modes is mainly due to Landau damping, as discussed in detail in Chapter 13.

The collision-dominated "hydrodynamic" domain is the subject of the last part of the book, in Chapters 14–19. In this domain the noncondensate is in *local* hydrodynamic equilibrium induced by frequent collisions between the atoms in the thermal cloud. As a result, the noncondensate can be completely described in terms of hydrodynamic variables such as the noncondensate local density $\tilde{n}(\mathbf{r}, t)$, local velocity $\mathbf{v}_{\rm n}(\mathbf{r}, t)$ and local pressure $\tilde{P}(\mathbf{r}, t)$. The condensate and noncondensate exhibit coupled coherent oscillations at the *same* frequency. In a uniform Bose superfluid, these oscillations correspond to first and second sound. This region in trapped atomic Bose gases has not been much explored experimentally. However, as noted earlier, it seems accessible using the very stable Bose molecules that arise in the BEC limit of the BCS–BEC crossover in two-component superfluid Fermi gases with a Feshbach resonance.

In Chapter 14, we review the famous two-fluid hydrodynamic equations first derived by Landau in 1941, which form the basis of our understanding of the hydrodynamic behaviour of superfluid ^{4}He (Khalatnikov, 1965; Wilks, 1967). For uniform systems, we discuss first sound and second sound, with emphasis on the difference between Bose gases and superfluid ^{4}He. Chapter 15 gives a very detailed discussion of how the Landau two-fluid description emerges naturally from the ZNG equations when collisions are strong enough to produce *local equilibrium*. Of particular interest is the key role of the relaxation time describing the rate at which the condensate and noncondensate atoms come into "diffusive" equilibrium with each other. This process arises from the C_{12} collisions between the two components. In Chapter 16, we develop a new variational approach to solve the two-fluid

equations in a trapped Bose gas, where the thermodynamic properties are spatially dependent.

In Chapter 17, we show how the coupled ZNG equations for the condensate and thermal cloud atoms lead to a closed set of hydrodynamic equations that are precisely of the form first discussed by Khalatnikov and Landau for superfluid ^{4}He (for a classic account see Khalatnikov, 1965). The derivations in Chapters 15 and 17 are of general interest at a conceptual level for two reasons: (a) they prove that the ZNG equations have all the correct physics and (b) they allow one to understand in a very explicit manner (for perhaps the first time) how the Landau two-fluid description of a superfluid Bose system has its basis in the existence of a Bose broken symmetry. As we have noted, of course, the Landau two-fluid hydrodynamic equations are much more general than the particular microscopic model used in our derivation. The two-fluid equations allow one to see the precise relation between the dynamics of trapped Bose-condensed gases and the dynamics of superfluid ^{4}He in the collision-dominated hydrodynamic region.

Chapter 18 gives an explicit calculation of all transport coefficients (the thermal conductivity, shear viscosity and the four coefficients of second viscosity), using the equations derived in Chapter 17. The associated relaxation times are of great importance since they define in a precise fashion the "crossover" region between the collisionless and collisional hydrodynamic domains. Some more technical details of the derivation of the transport coefficients are given in Appendices B and C. We derive explicit expressions for calculating the frequency (Chapter 16) and damping (Chapter 19) of hydrodynamic modes in trapped gases in the two-fluid region. These chapters are based on a variational formulation of the two-fluid equations describing a trapped gas.

The experimental achievement of the conditions for local equilibrium, and hence for collisional hydrodynamics, in trapped Bose gases must be viewed as one of the great challenges in future work on ultracold atom physics. We hope that this book will both stimulate and guide such investigations.

2

Condensate dynamics at $T = 0$

Trapped Bose-condensed atomic gases are remarkable because, in spite of the fact that these are very dilute systems, they exhibit robust coherent dynamic behaviour when perturbed. These quantum "wisps of matter" are a new phase of highly coherent matter. While binary collisions are very infrequent, the large coherent mean field associated with the Bose condensate ensures that interactions play a crucial role in determining the *collective* response of these trapped superfluid gases.

In our discussion of the theory of collective oscillations of atomic condensates, the macroscopic Bose wavefunction $\Phi(\mathbf{r}, t)$ plays a central role. This wavefunction is the BEC order parameter. As discussed in Chapter 1, the initial attempts at defining this order parameter began with the pioneering work of London (1938a), were further developed by Bogoliubov (1947) and finally extended to deal with any Bose superfluid using the general quantum field theoretic formalism developed by Beliaev (1958a). Almost all this early theoretical work was limited to $T = 0$ where, in a dilute weakly interacting Bose gas, all the atoms are in the condensate. The first extension of these ideas to nonuniform Bose condensates was by Pitaevskii (1961) and, independently, by Gross (1961), which led to the now famous Gross–Pitaevskii (GP) equation of motion for $\Phi(\mathbf{r}, t)$. Before the discovery of BEC in trapped gases, the time-dependent GP equation was mainly used to study vortices in Bose superfluids, which involve a spatially nonuniform ground state. Apart from this application, the GP equation was largely unknown. The situation changed overnight in 1995 with the creation of trapped nonuniform Bose condensates in atomic gases.

This chapter is a review of the ground state solution of the $T = 0$ GP equation and of small-amplitude fluctuations (collective oscillations) about this equilibrium state. This review is needed as the starting point for generalizations in the following chapters, which deal with finite temperatures.

For much more detailed accounts of the GP equation at $T = 0$, we refer to the review by Fetter (1999) as well as the texts by Pethick and Smith (2008) and Pitaevskii and Stringari (2003).

2.1 Gross–Pitaevskii (GP) equation

At $T = 0$, all the atoms in a dilute Bose gas are described by a macroscopic wavefunction $\Phi(\mathbf{r}, t)$, which is the solution of the time-dependent Hartree GP equation

$$i\hbar \frac{\partial \Phi(\mathbf{r}, t)}{\partial t} = \left[-\frac{\hbar^2 \nabla^2}{2m} + V_{\rm trap}(\mathbf{r}) + V_{\rm H}(\mathbf{r}, t) \right] \Phi(\mathbf{r}, t). \tag{2.1}$$

The trapping potential will always be of the form

$$V_{\rm trap}(\mathbf{r}) = \tfrac{1}{2}\, m(\omega_x^2 x^2 + \omega_y^2 y^2 + \omega_z^2 z^2), \tag{2.2}$$

which describes an anisotropic harmonic potential in terms of the three trap frequencies ω_i. The two most important special cases are a spherical trap ($\omega_x = \omega_y = \omega_z \equiv \omega_0$) and an axially symmetric trap ($\omega_x = \omega_y \equiv \omega_\perp \neq \omega_z$). The self-consistent condensate Hartree potential is

$$V_{\rm H}(\mathbf{r}, t) = \int d\mathbf{r}' v(\mathbf{r} - \mathbf{r}') n_{\rm c}(\mathbf{r}, t) = g n_{\rm c}(\mathbf{r}, t). \tag{2.3}$$

At the ultracold temperatures of interest in BEC experiments, we are dealing with extremely low energy atoms. As a result, we can use the s-wave approximation and describe the interatomic potential by a pseudopotential

$$\begin{aligned} v(\mathbf{r} - \mathbf{r}') &= \frac{4\pi a \hbar^2}{m} \delta(\mathbf{r} - \mathbf{r}') \\ &\equiv g\delta(\mathbf{r} - \mathbf{r}'). \end{aligned} \tag{2.4}$$

Here a is the s-wave scattering length for the real potential. The condensate density is given by $n_{\rm c}(\mathbf{r}, t) = |\Phi(\mathbf{r}, t)|^2$, and hence (2.1) reduces to a nonlinear Schrodinger equation (NLSE) for $\Phi(\mathbf{r}, t)$, namely

$$i\hbar \frac{\partial \Phi(\mathbf{r}, t)}{\partial t} = \left[-\frac{\hbar^2 \nabla^2}{2m} + V_{\rm trap}(\mathbf{r}) + g|\Phi(\mathbf{r}, t)|^2 \right] \Phi(\mathbf{r}, t). \tag{2.5}$$

Since all atoms are in an identical quantum state, there is a Hartree mean field but no Fock (or exchange) mean field in (2.5).

The $T = 0$ GP equation (2.5) has been the subject of thousands of papers since the discovery of BEC in laser-cooled trapped atomic gases. As reviewed at length in the literature, it gives a quantitative description of

both the static and dynamic (linear and nonlinear) behaviour in trapped Bose gases at $T \lesssim 0.5\, T_{\rm BEC}$. For typical atomic Bose gases, this interaction-induced depletion of the condensate is estimated to be of the order of a few per cent (see, for example, Hutchinson *et al.*, 1997). This $T = 0$ GP equation has been extended to deal with two-component Bose gases, involving two different atomic hyperfine states. The effect of external perturbations related to laser and radio frequency fields (used to manipulate the Bose condensates) is easily incorporated.

Our main purpose in this section is to illustrate the physics involved in the GP equation (2.5) by using it to discuss the collective oscillations of a pure condensate. In Chapter 3, we discuss the extension of this equation to finite temperature to deal with the effect of noncondensate atoms. First we briefly review the static equilibrium solution $\Phi_0(\mathbf{r})$ of the GP equation, given by

$$\langle \hat{\psi}(\mathbf{r}, t) \rangle = \Phi(\mathbf{r}, t) = \Phi_0(\mathbf{r}) e^{-i\mu t/\hbar}, \tag{2.6}$$

where μ is the chemical potential of the condensate. The physics behind this can be seen

$$\begin{aligned} \langle N-1|\hat{\psi}(\mathbf{r}, t)|N\rangle &= e^{iE_{N-1}t/\hbar}\langle N-1|\hat{\psi}(\mathbf{r})|N\rangle e^{-iE_N t/\hbar} \\ &= \langle N-1|\sqrt{N}|N-1\rangle e^{-i(E_N - E_{N-1})t/\hbar} \\ &= \sqrt{N} e^{-i\mu t/\hbar}, \end{aligned} \tag{2.7}$$

where N is the number of atoms. Using (2.6) in (2.5) gives an equation for the static order parameter $\Phi_0(\mathbf{r})$:

$$\mu_{\rm c0}\Phi_0(\mathbf{r}) = \left[-\frac{\hbar^2\nabla^2}{2m} + V_{\rm trap}(\mathbf{r}) + g|\Phi_0(\mathbf{r})|^2 \right] \Phi_0(\mathbf{r}). \tag{2.8}$$

Assuming a vortex-free ground state $\Phi_0(\mathbf{r}) = \sqrt{n_{\rm c0}(\mathbf{r})}$, this GP equation for the static condensate wavefunction leads to the following expression for the equilibrium condensate chemical potential,

$$\mu_{\rm c0} = -\frac{\hbar^2\nabla^2\sqrt{n_{\rm c0}(\mathbf{r})}}{2m\sqrt{n_{\rm c0}(\mathbf{r})}} + V_{\rm trap}(\mathbf{r}) + g n_{\rm c0}(\mathbf{r}). \tag{2.9}$$

This condensate chemical potential may be viewed as the eigenvalue of the time-independent GP equation (2.8).

A standard first approximation in solving (2.9) is to ignore the kinetic energy associated with the condensate amplitude $\sqrt{n_{\rm c}(\mathbf{r})}$, i.e. to neglect the $-\hbar^2\nabla^2/2m$ term. In this "Thomas–Fermi" (TF) approximation, the static

GP equation for $\Phi_0(\mathbf{r})$ reduces to (Dalfovo *et al.*, 1999)

$$\left(V_{\rm trap}(\mathbf{r}) + g|\Phi_0(\mathbf{r})|^2\right) = \mu_{\rm c0}, \tag{2.10}$$

which is easily inverted to give the condensate density profile

$$\begin{aligned} n_{\rm c0}(\mathbf{r}) &= \frac{1}{g}[\mu_{\rm c0} - V_{\rm trap}(\mathbf{r})] \\ &= \frac{1}{g}\left[\mu_{\rm c0} - \tfrac{1}{2}m\omega_0^2 r^2\right] \geq 0. \end{aligned} \tag{2.11}$$

Clearly in the TF approximation for an *isotropic* harmonic trap ($\omega_{0x} = \omega_{0y} = \omega_{0z} = \omega_0$ in (2.2)), the size of the condensate is $R_{\rm TF}$, given by

$$\mu_{\rm c0} = \tfrac{1}{2}m\omega_0^2 R_{\rm TF}^2. \tag{2.12}$$

Using (2.11), one finds $\mu_{\rm c0}$ from the condition that $\int d\mathbf{r}\, n_{\rm c0}(\mathbf{r}) = N_{\rm c} = N$, which gives

$$\mu_{\rm c0} = \frac{\hbar\omega_0}{2}\left(\frac{15N_{\rm c}a}{a_{\rm ho}}\right)^{2/5}, \quad a_{\rm ho} \equiv \left(\frac{\hbar}{m\omega_0}\right)^{1/2}. \tag{2.13}$$

The oscillator length $a_{\rm ho}$ is the size of the ground state Gaussian wavefunction of an atom in a parabolic potential well. Combining (2.12) and (2.13) gives

$$\begin{aligned} R_{\rm TF} &= a_{\rm ho}\left(\frac{15N_{\rm c}a}{a_{\rm ho}}\right)^{1/5} \\ &\gg a_{\rm ho}, \quad \text{if } \frac{N_{\rm c}a}{a_{\rm ho}} \gg 1. \end{aligned} \tag{2.14}$$

The TF approximation (2.11) for $n_{\rm c0}(\mathbf{r})$ is very good for large $N_{\rm c}$, except for a small region near the edge of the condensate ($r \simeq R_{\rm TF}$). In this approximation, $n_{\rm c0}(\mathbf{r})$ abruptly vanishes at $r = R_{\rm TF}$, while in reality there is a more gradual decrease (the condensate wavefunction tends to, but never reaches, zero). In the first BEC experiments after 1995, the typical values of $a_{\rm ho}$ and a ensured that the TF approximation was very good if the total number of condensate atoms $N_{\rm c}$ was $\gtrsim 10^4$ atoms (Edwards *et al.*, 1996).

To discuss small amplitude condensate fluctuations around the static equilibrium value of $\Phi_0(\mathbf{r})$, we first reformulate the time-dependent GP equation in terms of the condensate density and phase variables:

$$\Phi(\mathbf{r}, t) = \sqrt{n_{\rm c}(\mathbf{r}, t)}e^{i\theta(\mathbf{r},t)}. \tag{2.15}$$

From this we find that, on the one hand,

$$\frac{i\hbar}{\Phi}\frac{\partial\Phi}{\partial t} = -\hbar\frac{\partial\theta}{\partial t} + \frac{i\hbar}{2n_{\rm c}}\frac{\partial n_{\rm c}}{\partial t}. \tag{2.16}$$

On the other hand, the right hand side of (2.5) gives

$$\frac{1}{\Phi}\left(-\frac{\hbar^2\nabla^2}{2m}+V_{\rm trap}+gn_{\rm c}\right)\Phi=-\frac{\hbar^2}{2m}\frac{\nabla^2\sqrt{n_{\rm c}}}{\sqrt{n_{\rm c}}}+\frac{\hbar^2}{2m}|\nabla\theta|^2+V_{\rm trap}+gn_{\rm c}$$
$$-\frac{i\hbar^2}{2mn_{\rm c}}\left(\nabla n_{\rm c}\cdot\nabla\theta+n_{\rm c}\nabla^2\theta\right). \qquad (2.17)$$

Equating the real parts of (2.16) and (2.17), we obtain the equation

$$\hbar\frac{\partial\theta}{\partial t}=-\left(\mu_{\rm c}+\tfrac{1}{2}m\mathbf{v}_{\rm c}^2\right), \qquad (2.18)$$

where we have defined the time-and space-dependent "chemical potential"

$$\mu_{\rm c}(\mathbf{r},t)\equiv-\frac{\hbar^2\nabla^2\sqrt{n_{\rm c}(\mathbf{r},t)}}{2m\sqrt{n_{\rm c}(\mathbf{r},t)}}+V_{\rm trap}(\mathbf{r})+gn_{\rm c}(\mathbf{r},t) \qquad (2.19)$$

and the condensate velocity field

$$\mathbf{v}_{\rm c}(\mathbf{r},t)\equiv\frac{\hbar}{m}\nabla\theta(\mathbf{r},t). \qquad (2.20)$$

Since the curl of a gradient vanishes identically, (2.20) implies that the superfluid motion is irrotational, i.e. $\nabla\times\mathbf{v}_{\rm c}=0$.

Equating the imaginary parts of (2.16) and (2.17), and using the definition of the superfluid velocity in (2.20), we obtain

$$\frac{\partial n_{\rm c}}{\partial t}=-\left(\nabla n_{\rm c}\cdot\mathbf{v}_{\rm c}+n_{\rm c}\nabla\cdot\mathbf{v}_{\rm c}\right)=-\nabla\cdot(n_{\rm c}\mathbf{v}_{\rm c}). \qquad (2.21)$$

In summary, the time-dependent GP equation in (2.5) for the order parameter $\Phi(\mathbf{r},t)$ is completely equivalent to the following two coupled equations for the condensate density $n_{\rm c}(\mathbf{r},t)$ and the velocity field $\mathbf{v}_{\rm c}(\mathbf{r},t)$:

$$\frac{\partial n_{\rm c}}{\partial t}+\nabla\cdot(n_{\rm c}\mathbf{v}_{\rm c})=0, \qquad (2.22)$$

$$m\frac{\partial\mathbf{v}_{\rm c}}{\partial t}=-\nabla\varepsilon_{\rm c}\,, \qquad (2.23)$$

where we have introduced

$$\varepsilon_{\rm c}\equiv\mu_{\rm c}+\tfrac{1}{2}mv_{\rm c}^2\,. \qquad (2.24)$$

This quantity plays the role of the local energy of a condensate atom having potential energy $\mu_{\rm c}$ and kinetic energy $\frac{1}{2}mv_{\rm c}^2$. In Chapter 3, we show that a generalized GP equation, taking into account collisions with the noncondensate atoms, still leads to a set of equations very similar to (2.22) and (2.23). The key equation (2.20) defining the superfluid velocity field of the condensate holds for all Bose superfluids at all temperatures. All aspects

related to the "superfluidity" of a Bose superfluid (as compared with Bose condensation) are tied to the fact that the condensate only exhibits motion that can be associated with the gradient of the phase. This means that the condensate motion is irrotational, as noted above. However, the velocity may have localized singularities corresponding to vortices (see Chapter 9) at which the fluid is not irrotational.

In the formulation given above, a complete description of the GP condensate dynamics is given in terms of two variables $n_c(\mathbf{r}, t)$ and $\mathbf{v}_c(\mathbf{r}, t)$ and leads to two equations of motion reminiscent of the hydrodynamic equations for a classical fluid (see Chapter 14). For this reason, in the BEC literature on Bose gases since 1995 the equations (2.22) and (2.23) are often referred to simply as the "hydrodynamic" theory. In this book, we refer to them as the "quantum hydrodynamic" theory, valid at $T = 0$. This terminology will eliminate confusion when we discuss (in Chapters 15 and 17) the dynamics of the noncondensate cloud of atoms in the collisional hydrodynamic region, where strong collisions (i.e. short collision times) give rise to local equilibrium. In this region, the dynamics of the thermal cloud can also be completely described in terms of a few local variables.

Stringari (1996b) first pointed out that within the *dynamic* TF approximation, one could combine the two equations (2.22) and (2.23) into a single condensate wave equation. Neglecting the kinetic energy term proportional to $-\nabla^2\sqrt{n_c(\mathbf{r}, t)}$ in (2.19) as small compared with the condensate Hartree interaction energy gn_c, we linearize the resulting equations around the equilibrium values:

$$\begin{aligned} n_c &= n_{c0} + \delta n_c, \\ \mathbf{v}_c &= \mathbf{v}_{c0} + \delta \mathbf{v}_c \end{aligned} \tag{2.25}$$

We thus obtain

$$\begin{aligned} \frac{\partial \delta n_c}{\partial t} &= -\nabla \cdot (n_{c0} \delta \mathbf{v}_c) - \nabla \cdot (\mathbf{v}_{c0} \delta n_c), \\ m \frac{\partial \delta \mathbf{v}_c}{\partial t} &= -\nabla \left(\varepsilon_{c0} + g \delta n_c + m \mathbf{v}_{c0} \cdot \delta \mathbf{v}_c \right) , \end{aligned} \tag{2.26}$$

where $\varepsilon_{c0} \equiv \mu_{c0} + \frac{1}{2} m \mathbf{v}_{c0}^2$. We now assume that the solutions under consideration do not involve vortices, and thus we can set $\mathbf{v}_{c0} = 0$. Then (2.26) reduces to the following coupled linearized equations for δn_c and $\delta \mathbf{v}_c$:

$$\begin{aligned} \frac{\partial \delta n_c}{\partial t} &= -\nabla \cdot (n_{c0}(\mathbf{r}) \delta \mathbf{v}_c), \\ \frac{\partial \delta \mathbf{v}_c}{\partial t} &= -\frac{g}{m} \nabla \delta n_c. \end{aligned} \tag{2.27}$$

These equations can be combined to give the well-known $T = 0$ Stringari wave equation (Stringari, 1996b; Dalfovo *et al.*, 1999)

$$\frac{\partial^2 \delta n_{\rm c}(\mathbf{r},t)}{\partial t^2} = \frac{g}{m}\nabla\cdot\left[n_{\rm c0}(\mathbf{r})\nabla\delta n_{\rm c}(\mathbf{r},t)\right]. \tag{2.28}$$

Since in this derivation we have used the TF approximation, in which $n_{\rm c0}(\mathbf{r})$ is given by (2.11) for $r \le R_{\rm TF}$, one can rewrite (2.28) in the equivalent form (assuming an isotropic trap potential for simplicity)

$$\frac{\partial^2 \delta n_{\rm c}}{\partial t^2} = \frac{\mu_{\rm c0}}{m}\nabla\cdot\left[\left(1-\frac{r^2}{R_{TF}^2}\right)\nabla\delta n_{\rm c}\right], \quad r \le R_{\rm TF}, \tag{2.29}$$

where $\mu_{\rm c0}$ is given by (2.13). As we show below, the normal mode solutions $\delta n_{\rm c}(\mathbf{r},t) = \delta n_\omega(\mathbf{r})e^{-i\omega t}$ of (2.29) have frequencies which are independent of both the interaction strength g and the value of $N_{\rm c}$. This is a feature of the underlying TF approximation, which typically starts to break down, as noted earlier, when $N_{\rm c} \lesssim 10^4$ atoms. This is shown by explicit numerical solutions (Hutchinson *et al.*, 1997; Edwards *et al.*, 1996) of the coupled Bogoliubov equations of motion that describe the normal mode solutions of the linearized GP equation, taking the kinetic energy associated with the condensate amplitude $\sqrt{n_{\rm c}}$ fully into account. In the initial experiments on collective modes using ^{87}Rb atoms, one finds (as predicted) that, for $N_{\rm c} \lesssim 10^4$, the normal mode frequencies depend significantly on the magnitude of $N_{\rm c}$. This dependence on $N_{\rm c}$ is shown in Fig. 12.1 of Pitaevskii and Stringari (2003).

Recalling (2.27), we also note that (2.28) can equally well be rewritten in terms of the superfluid velocity $\delta\mathbf{v}_{\rm c}$ or, equivalently, the phase fluctuation $\delta\theta$. This emphasizes that the condensate density oscillations measured in experiments are directly related to the existence of phase fluctuations in the Bose order parameter. The existence of collective modes of a pure condensate may thus be viewed already as "evidence" of superfluidity, the latter being always a consequence of the phase coherence of the macroscopic wavefunction given in (2.15), which gives rise to the irrotational velocity in (2.20).

We conclude this section by reviewing some simple examples of condensate normal modes obtained by solving the $T = 0$ Stringari equation (2.28); more detailed discussion is given in Pitaevskii and Stringari (2003) as well as by Pethick and Smith (2008). This will set the stage for the discussion of analogous solutions of similar equations valid at finite temperatures. A wonderful aspect of the collective oscillations of a condensate in a trapped

gas is that you can "see" them. In Ketterle's words, these "wisps" of Bose-condensed matter are robust – one can kick them or shake them and yet they keep their integrity. The uniform Bose-condensed gas is especially simple, since the solutions are plane waves, $\delta n_{\rm c}(\mathbf{q},t) = \delta n_{q\omega} e^{i(\mathbf{q}\cdot\mathbf{r}-\omega t)}$. In this case, (2.28) reduces to

$$-\omega^2 \delta n_{\mathbf{q}\omega} = \frac{g n_{\rm c0}}{m}(-q^2)\delta n_{\mathbf{q}\omega}, \tag{2.30}$$

with solution $\omega^2 = c_0^2 q^2$, where $c_0 = \sqrt{g n_{\rm c0}/m}$. This solution is the famous Bogoliubov phonon mode (Bogoliubov, 1947) of a uniform Bose condensate. The neglect of the kinetic energy in our TF approximation, of course, prevents us from obtaining the expected particle-like behaviour at large values of the wavevector q.

The Kohn (or sloshing) mode corresponds to oscillation at the trap frequency ω_0 of the centre of mass of the static condensate profile. This mode is described by

$$n_{\rm c}(\mathbf{r},t) = n_{\rm c0}(\mathbf{r} - \boldsymbol{\eta}(t)), \tag{2.31}$$

where $d\boldsymbol{\eta}(t)/dt \equiv \mathbf{v}_{\rm c}(t)$ and

$$\frac{d^2\boldsymbol{\eta}(t)}{dt^2} = -\omega_0^2 \boldsymbol{\eta}(t). \tag{2.32}$$

The proof is simple. Linearizing (2.31), we find

$$n_{\rm c}(\mathbf{r},t) = n_{\rm c0}(\mathbf{r}) - \boldsymbol{\eta}(t)\cdot\nabla n_{\rm c0}(\mathbf{r}), \tag{2.33}$$

which gives an explicit form for the condensate fluctuation (using (2.11)),

$$\delta n_{\rm c}(\mathbf{r},t) = \frac{1}{g} m\omega_0^2 \mathbf{r}\cdot\boldsymbol{\eta}(t). \tag{2.34}$$

Inserting this result into the Stringari equation (2.28), we obtain

$$\begin{aligned} -\nabla n_{\rm c0}\cdot\frac{d^2\boldsymbol{\eta}(t)}{dt^2} &= \omega_0^2 \nabla\cdot[n_{\rm c0}\nabla(\boldsymbol{\eta}(t)\cdot\mathbf{r})] \\ &= \omega_0^2 \nabla\cdot[n_{\rm c0}\boldsymbol{\eta}(t)] \\ &= \omega_0^2 \boldsymbol{\eta}(t)\cdot\nabla n_{\rm c0}. \end{aligned} \tag{2.35}$$

This equation confirms that the centre-of-mass position $\boldsymbol{\eta}(t)$ of the static condensate density profile satisfies the simple harmonic oscillator equation (2.32), with the trap frequency ω_0.

The *breathing* (or monopole) condensate normal mode corresponds to a velocity fluctuation of the form

$$\delta\mathbf{v}_{\rm c}(\mathbf{r},t) = A\mathbf{r}e^{-i\omega t}, \quad r \le R_{\rm TF}. \tag{2.36}$$

Using $\nabla \cdot \mathbf{r} = 3$ and $\nabla \cdot (r^2 \mathbf{r}) = 5r^2$, it is easy to verify from the continuity equation in (2.22) that the associated density fluctuation is

$$\begin{aligned} -i\omega \delta n_{\rm c} &= -\nabla \cdot (n_{\rm c0}(\mathbf{r}) \delta \mathbf{v}_{\rm c}) \\ &= -\frac{\mu_{\rm c0}}{g} A \left(3 - 5\frac{r^2}{R_{\rm TF}^2}\right) e^{-i\omega t}, \end{aligned} \tag{2.37}$$

or

$$\delta n_\omega(\mathbf{r}) = B\left(1 - \frac{5}{3}\frac{r^2}{R_{\rm TF}^2}\right), \tag{2.38}$$

where $\mu_{\rm c0}$ is given by (2.12). Inserting this result into the Stringari wave equation (2.29) gives

$$\begin{aligned} -\omega^2 \delta n_\omega(\mathbf{r}) &= -\frac{\mu_{\rm c0}}{m} \nabla \cdot \left[\left(1 - \frac{r^2}{R_{\rm TF}^2}\right) B \frac{5}{3} \frac{2\mathbf{r}}{R_{\rm TF}^2} \right] \\ &= -5\omega_0^2 \delta n_\omega(\mathbf{r}). \end{aligned} \tag{2.39}$$

Thus the breathing mode for an isotropic parabolic trap has frequency $\omega = \sqrt{5}\,\omega_0$. As noted earlier, this mode frequency is not explicitly dependent on the interaction strength g. However, the origin of this mode is entirely due to mean-field effects. We recall that a noninteracting trapped Bose gas has a breathing mode with frequency $\omega = 2\omega_0$ at all temperatures.

As a final example, we consider the so-called "surface" modes of the condensate, as described in Stringari (1996b):

$$\delta \mathbf{v}_{\rm c}(\mathbf{r}, t) = A \nabla \left[r^l Y_{lm}(\theta, \phi) \right] e^{-i\omega t}. \tag{2.40}$$

These modes correspond to phase fluctuations $\delta\theta_\omega(\mathbf{r}) = mA\, r^l Y_{lm}(\theta, \phi)$. One may easily verify that $\nabla \cdot \delta \mathbf{v}_{\rm c} = 0$ and hence from (2.27) we obtain

$$\frac{\partial^2 \delta \mathbf{v}_{\rm c}}{\partial t^2} = -\omega_0^2 \nabla (\delta \mathbf{v}_{\rm c} \cdot \mathbf{r}). \tag{2.41}$$

This is equivalent to

$$\frac{\partial^2 \delta\theta}{\partial t^2} = -\omega_0^2 (\nabla \delta\theta) \cdot \mathbf{r} \tag{2.42}$$

or

$$-\omega^2 \delta\theta_\omega(\mathbf{r}) = -\omega_0^2 l \delta\theta_\omega(\mathbf{r}). \tag{2.43}$$

Thus the surface oscillations of the condensate phase have a frequency given by $\omega = \sqrt{l}\omega_0 (l = 1, 2, \ldots)$ for an isotropic parabolic trap of frequency ω_0.

One great advantage of the quantum hydrodynamic formalism discussed

in this subsection is that it can be used to treat the normal modes of a Bose gas in an anisotropic trap described by (2.2).

The discussion in this section has been based on the $T = 0$ GP wave equation in (2.28) and has ignored all effects arising from the thermal cloud. However, these results will turn out to be relevant in later chapters, where we deal with finite temperatures. As noted earlier, one obtains a generalized set of GP equations that are formally identical to (2.22) and (2.23), apart from an additional source term on the r.h.s. of the continuity equation. This new term arises from collisions between the condensate and noncondensate atoms. In the simplest approximation for this source term, one can treat the thermal atoms as being in static thermal equilibrium (this approximation is discussed in detail in Chapter 8). In this case it turns out that the condensate oscillations at finite T are again described by a wave equation similar to (2.28), except that now there is a collisional damping term and the equilibrium condensate density $n_{\rm c0}(\mathbf{r})$ is temperature dependent.

2.2 Bogoliubov equations for condensate fluctuations

In this section, we discuss the collective mode solutions of the $T = 0$ GP equation (2.5) in terms of the Bogoliubov equations for quasiparticle excitations. This alternative analysis introduces a general feature of Bose superfluids, namely that the collective mode and elementary excitation spectra are identical (for further discussion of this feature, see Chapter 5). If we linearize around the static equilibrium value of the condensate wavefunction, setting

$$\Phi(\mathbf{r}, t) = e^{-i\mu t/\hbar} \left[\Phi_0(\mathbf{r}) + \delta\Phi(\mathbf{r}, t)\right] \tag{2.44}$$

where $\delta\Phi \ll \Phi_0$, we see that

$$i\hbar \frac{\partial \Phi}{\partial t} = \left[-\frac{\hbar^2\nabla^2}{2m} + V_{\rm trap}(\mathbf{r}) + g\left(|\Phi_0|^2 + \Phi_0^*\delta\Phi + \Phi_0\delta\Phi^*\right)\right](\Phi_0 + \delta\Phi)e^{-i\mu t/\hbar}, \tag{2.45}$$

which gives

$$i\hbar \frac{\partial \delta\Phi}{\partial t} = \left[-\frac{\hbar^2\nabla^2}{2m} + V_{\rm trap}(\mathbf{r}) + 2g|\Phi_0|^2 - \mu\right]\delta\Phi + g\Phi_0^2\delta\Phi^*. \tag{2.46}$$

We obtain a similar equation of motion for $\delta\Phi^*(\mathbf{r},t)$. Solving these two coupled equations using the ansatz[1]

$$\delta\Phi(\mathbf{r},t) = u(\mathbf{r})e^{-i\omega t} - v(\mathbf{r})e^{i\omega t}, \tag{2.47}$$

we find two coupled "Bogoliubov equations" for the amplitudes u_i and v_i

$$\begin{aligned}\left[\hat{T}(\mathbf{r}) + 2gn_{\rm c0}(\mathbf{r})\right] u_i(\mathbf{r}) - g[\Phi_0(\mathbf{r})]^2 v_i(\mathbf{r}) &= E_i u_i(\mathbf{r}) \\ \left[\hat{T}(\mathbf{r}) + 2gn_{\rm c0}(\mathbf{r})\right] v_i(\mathbf{r}) - g[\Phi_0^*(\mathbf{r})]^2 u_i(\mathbf{r}) &= -E_i v_i(\mathbf{r}),\end{aligned} \tag{2.48}$$

where we have defined the operator

$$\hat{T}(\mathbf{r}) \equiv -\frac{\hbar^2\nabla^2}{2m} + V_{\rm trap} - \mu. \tag{2.49}$$

In (2.48), $E_i \equiv \hbar\omega$ are the excitation energies of the condensate. The first careful discussion of the solutions of the Bogoliubov equations (2.48) in the case of a non-uniform Bose gas was given in a classic paper by Fetter (1974).

The equations (2.48) have been solved numerically for various trap potentials, and the observed oscillation frequencies are in good agreement with the predictions for the energy eigenvalues E_i (Edwards *et al.* 1996; Dalfovo *et al.*, 1999). As an illustration of the physics, we consider the case of a uniform Bose gas, where we have the plane-wave solutions

$$\begin{aligned}u(\mathbf{r}) &= ue^{i\mathbf{q}\cdot\mathbf{r}}, \\ v(\mathbf{r}) &= ve^{i\mathbf{q}\cdot\mathbf{r}}.\end{aligned} \tag{2.50}$$

Substituting these into (2.48), one obtains

$$\begin{aligned}(\hbar\omega)^2 &= \left(\frac{\hbar^2q^2}{2m} - \mu_{\rm c0} + 2gn_{\rm c0}\right)^2 - (gn_{\rm c0})^2 \\ &= \varepsilon_{\mathbf{q}}^2 + 2gn_{\rm c0}\varepsilon_q.\end{aligned} \tag{2.51}$$

This is the famous excitation spectrum at $T = 0$ derived by Bogoliubov (1947). Here we have used the result $\mu_{\rm c0} = gn_{\rm c0}$, which follows from (2.9). At long wavelengths, (2.51) reduces to the phonon solution given below (2.30). The crossover from the particle-like region to the collective phonon region occurs at a wavevector $q_{\rm c}$ defined by

$$\frac{\hbar^2q_{\rm c}^2}{2m} = 2gn_{\rm c0} \;\Rightarrow\; q_{\rm c} = \sqrt{\frac{4mn_{\rm c0}g}{\hbar^2}}. \tag{2.52}$$

[1] In this book, we follow the sign convention for the Bogoliubov amplitude $v(\mathbf{r})$ used by Fetter and Walecka (1971) and Pethick and Smith (2008). Pitaevskii and Stringari (2003) use the opposite sign convention, with a plus sign in (2.47).

This shows how the interactions change the qualitative nature of low-energy excitations in a Bose-condensed gas. The fact that the energy spectrum at long wavelengths is phonon-like can be shown to stabilize superfluid motion against the dissipation caused by such excitations (the Landau criterion). See Pitaevskii and Stringari (2003) for a more detailed discussion.

These oscillations of the condensate can be understood as excitations involving the noncondensate. Using (1.4) in the Hamiltonian and keeping terms up to second order in $\tilde{\psi}$ and $\tilde{\psi}^\dagger$, one obtains (see, for example, Griffin, 1996)

$$\begin{aligned}\hat{H} - \mu\hat{N} = &\int d\mathbf{r}\tilde{\psi}^\dagger(\mathbf{r})\left[-\frac{\hbar^2\nabla^2}{2m} + V_{\text{trap}}(\mathbf{r}) - \mu\right]\tilde{\psi}(\mathbf{r}) \\ &+ \int d\mathbf{r}\Phi_0^*(\mathbf{r})\left[-\frac{\hbar^2\nabla^2}{2m} + V_{\text{trap}}(\mathbf{r}) - \mu\right]\Phi_0(\mathbf{r}) \\ &+ 2g\int d\mathbf{r}|\Phi_0(\mathbf{r})|^2\tilde{\psi}^\dagger(\mathbf{r})\tilde{\psi}(\mathbf{r}) \\ &+ \tfrac{1}{2}g\int d\mathbf{r}\Phi_0^2(\mathbf{r})\tilde{\psi}^\dagger(\mathbf{r})\tilde{\psi}^\dagger(\mathbf{r}) \\ &+ \tfrac{1}{2}g\int d\mathbf{r}\Phi_0^{*2}(\mathbf{r})\tilde{\psi}(\mathbf{r})\tilde{\psi}(\mathbf{r}). \end{aligned} \tag{2.53}$$

We can diagonalize this quadratic Hamiltonian using the transformation

$$\tilde{\psi}(\mathbf{r}) = \sum_i \left[u_i(\mathbf{r})\hat{\alpha}_i - v_i^*(\mathbf{r})\hat{\alpha}_i^\dagger\right], \tag{2.54}$$

where $\left[\hat{\alpha}_i, \hat{\alpha}_j^\dagger\right] = \delta_{ij}$. This result corresponds to a noninteracting gas of Bogoliubov quasiparticles with energy $\hbar\omega_i$ (Fetter and Walecka, 1971):

$$\hat{H} - \mu\hat{N} = \text{const.} + \sum_i \hbar\omega_i\hat{\alpha}_i^\dagger\hat{\alpha}_i. \tag{2.55}$$

This transformation shows how the noncondensate part of Hamiltonian can be reduced to a system of noninteracting quasiparticles with a spectrum identical to that of the condensate fluctuations. This equivalence is easy to understand. The condensate fluctuations

$$\delta\Phi \equiv \langle\hat{\psi}(\mathbf{r})\rangle - \Phi_0 \tag{2.56}$$

can be calculated to first order in the symmetry-breaking perturbation in (1.7). Standard linear response theory gives (schematically)

$$\begin{aligned}\delta\Phi &\sim \langle[\hat{\psi}, H_{\text{SB}}]\rangle \\ &\sim \int d\mathbf{r}\left[\langle[\tilde{\psi}, \tilde{\psi}^\dagger]\rangle\eta + \langle[\tilde{\psi}, \tilde{\psi}]\rangle\eta^*\right]. \end{aligned} \tag{2.57}$$

This shows that the single-particle Green's functions of the noncondensate fields have the same spectrum as $\delta\Phi$. This identity of the spectrum of density fluctuations and single-particle excitations is a characteristic signature of Bose-condensed fluids and persists at finite temperatures. For further discussion of this equivalence, see Chapter 5 of this book as well as Chapter 5 of Griffin (1993).

3

Coupled equations for the condensate and thermal cloud

In contrast with Chapter 2, in this chapter we include the dynamics of the thermal cloud. As noted in Chapter 1, we treat the noncondensate atoms using the simplest microscopic model approximation that captures the important physics. In particular, we consider only temperatures high enough ($T \geq 0.4T_{\rm BEC}$) that the noncondensate atoms can be described by a particle-like Hartree–Fock (HF) spectrum. To extend the analysis to very low temperatures is in principle straightforward (see Chapter 7). However, the details are more complicated since the excitations of the thermal cloud take on a collective aspect (i.e. a Bogoliubov-type quasiparticle spectrum must be used). In trapped Bose gases, the HF single-particle spectrum gives a good approximation down to much lower temperatures than in the case of uniform Bose gases, as first emphasized by Giorgini *et al.* (1997).

In Section 3.1, we derive a generalized form of the Gross–Pitaevskii equation for the Bose order parameter $\Phi(\mathbf{r}, t)$ that is valid at finite temperatures. It involves terms that are coupled to the noncondensate component (the thermal cloud) and thus its solution in general requires one to know the equations of motion for the dynamics of the noncondensate atoms. In Section 3.2 we restrict ourselves to finite temperatures high enough that the noncondensate atoms can be described by a quantum kinetic equation for the single-particle distribution function $f(\mathbf{p}, \mathbf{r}, t)$. A detailed microscopic derivation of this kind of kinetic equation is given in Chapters 6 and 7 using the Kadanoff–Baym Green's function formalism.

A characteristic feature of a Bose-condensed gas is that the kinetic equation governing $f(\mathbf{p}, \mathbf{r}, t)$ involves a collision integral $C_{12}[f, \Phi]$ describing collisions between condensate and noncondensate atoms. The generalized GP equation for $\Phi(\mathbf{r}, t)$ also has a term related to these C_{12} collisions, which

leads to the damping (or growth) of condensate fluctuations. The C_{12} collisions will play a crucial role in the rest of this book.

The discussion we give in Sections 3.1 and 3.2 follows the original approach given by Zaremba *et al.* (1999). Throughout this chapter (and the whole book), this paper will be referred to as ZNG. The derivation of the collision integrals given in Appendix A of ZNG follows the approach of Kirkpatrick and Dorfman (1985a). We will not repeat this derivation here since the same results are derived in Chapter 6 using the more transparent and systematic Kadanoff–Baym formalism.

3.1 Generalized GP equation for the condensate

We start with the usual Heisenberg equation of motion for the quantum field operator,

$$i\hbar\frac{\partial\hat{\psi}(\mathbf{r},t)}{\partial t} = \left[-\frac{\hbar^2\nabla^2}{2m} + V_{\rm trap}(\mathbf{r})\right]\hat{\psi}(\mathbf{r},t) + \eta(\mathbf{r}) + g\hat{\psi}^\dagger(\mathbf{r},t)\hat{\psi}(\mathbf{r},t)\hat{\psi}(\mathbf{r},t), \tag{3.1}$$

where $V_{\rm trap}(\mathbf{r})$ is the confining potential. We have here assumed that the interaction potential can be represented as a zero-range pseudopotential of strength $g = 4\pi a\hbar^2/m$, where a is the s-wave scattering length (for further discussion see Pethick and Smith, 2008). The equation for the condensate wavefunction is obtained by taking an average (denoted by angular brackets) of (3.1) with respect to a broken-symmetry nonequilibrium ensemble in which the quantum field operator takes a nonzero expectation value:

$$\Phi(\mathbf{r},t) = \langle\hat{\psi}(\mathbf{r},t)\rangle\,. \tag{3.2}$$

This gives an exact equation of motion for $\Phi(\mathbf{r},t)$

$$i\hbar\frac{\partial\Phi(\mathbf{r},t)}{\partial t} = \left[-\frac{\hbar^2\nabla^2}{2m} + V_{\rm trap}(\mathbf{r})\right]\Phi(\mathbf{r},t) + \eta(\mathbf{r}) + g\langle\hat{\psi}^\dagger(\mathbf{r},t)\hat{\psi}(\mathbf{r},t)\hat{\psi}(\mathbf{r},t)\rangle. \tag{3.3}$$

Introducing the usual definition of the noncondensate field operator $\tilde{\psi}(\mathbf{r},t)$ according to (1.4),

$$\hat{\psi}(\mathbf{r},t) = \Phi(\mathbf{r},t) + \tilde{\psi}(\mathbf{r},t) \tag{3.4}$$

with $\langle\tilde{\psi}(\mathbf{r},t)\rangle = 0$, we have

$$\hat{\psi}^\dagger\hat{\psi}\hat{\psi} = |\Phi|^2\Phi + 2|\Phi|^2\tilde{\psi} + \Phi^2\tilde{\psi}^\dagger + \Phi^*\tilde{\psi}\tilde{\psi} + 2\Phi\tilde{\psi}^\dagger\tilde{\psi} + \tilde{\psi}^\dagger\tilde{\psi}\tilde{\psi}. \tag{3.5}$$

Taking the symmetry-breaking average of (3.5), one has

$$\langle\hat{\psi}^\dagger\hat{\psi}\hat{\psi}\rangle = n_c\Phi + \tilde{m}\Phi^* + 2\tilde{n}\Phi + \langle\tilde{\psi}^\dagger\tilde{\psi}\tilde{\psi}\rangle, \tag{3.6}$$

where we have defined the following local quantities:

$$\begin{aligned} n_c(\mathbf{r},t) &\equiv |\Phi(\mathbf{r},t)|^2, \qquad \text{the local condensate density;} \\ \tilde{n}(\mathbf{r},t) &\equiv \langle\tilde{\psi}^\dagger(\mathbf{r},t)\tilde{\psi}(\mathbf{r},t)\rangle, \qquad \text{the noncondensate density;} \\ \tilde{m}(\mathbf{r},t) &\equiv \langle\tilde{\psi}(\mathbf{r},t)\tilde{\psi}(\mathbf{r},t)\rangle, \qquad \text{the off-diagonal (anomalous) density.} \end{aligned} \tag{3.7}$$

Using (3.6) in (3.3), the "exact" equation of motion for $\Phi(\mathbf{r},t)$ is (Zaremba *et al.*, 1999)

$$\begin{aligned} i\hbar\frac{\partial\Phi(\mathbf{r},t)}{\partial t} = \Big[&-\frac{\hbar^2\nabla^2}{2m} + V_{\rm trap}(\mathbf{r}) + gn_c(\mathbf{r},t) + 2g\tilde{n}(\mathbf{r},t)\Big]\Phi(\mathbf{r},t) \\ &+ g\tilde{m}(\mathbf{r},t)\Phi^* + g\left\langle\tilde{\psi}^\dagger(\mathbf{r},t)\tilde{\psi}(\mathbf{r},t)\tilde{\psi}(\mathbf{r},t)\right\rangle. \end{aligned} \tag{3.8}$$

Besides $n_c(\mathbf{r},t)$ and $\tilde{n}(\mathbf{r},t)$, this equation also involves the off-diagonal noncondensate density $\tilde{m}(\mathbf{r},t) = \langle\tilde{\psi}(\mathbf{r},t)\tilde{\psi}(\mathbf{r},t)\rangle$ and the three-field correlation function $\langle\tilde{\psi}^\dagger\tilde{\psi}\tilde{\psi}\rangle$, both of which have nonzero expectation values because of the assumed Bose broken symmetry. We note that (3.8) is formally exact, within a pseudopotential approximation for the two-body interaction potential.

It is useful to introduce some standard approximations into the equation of motion (3.8). It clearly reduces to the Gross–Pitaevskii (GP) equation discussed in Chapter 2 if all the atoms are in the condensate (i.e. if $\tilde{n} = 0$ and the anomalous correlations $\tilde{m}$ and $\langle\tilde{\psi}^\dagger\tilde{\psi}\tilde{\psi}\rangle$ are absent). This is a very good approximation for $T \ll T_{\rm BEC}$. At $T = 0$ the noncondensate fraction in trapped atomic gases is estimated to be of order 1% (see, for example, Hutchinson *et al.*, 1997) in most BEC experiments done before 2000.

Approximations to the generalized GP equation (3.8) that are used in the current literature on ultracold atoms include the following:

(a) The Hartree–Fock–Bogoliubov (HFB) approximation for Φ corresponds to neglecting the three-field correlation function $\langle\tilde{\psi}^\dagger\tilde{\psi}\psi\rangle$ but keeping the $n_c, (\mathbf{r},t), \tilde{n}(\mathbf{r},t)$ and $\tilde{m}(\mathbf{r},t)$ fluctuations, which in principle must be calculated self-consistently. Formally, the HFB only keeps terms that are first order in the interaction g. This HFB has been widely used, in conjunction with additional equations of motion for $\tilde{n}(\mathbf{r},t)$ and $\tilde{m}(\mathbf{r},t)$ (see for example Griffin, 1996; Giorgini, 1998; Imamović-Tomasović and Griffin, 1999). The time-dependent HFB approximation can be used to generate a density response function

with an excitation spectrum identical to that of the Beliaev second-order self-energy approximation at finite T.

(b) The dynamic Popov approximation corresponds to ignoring both $\langle\tilde{\psi}^\dagger\tilde{\psi}\tilde{\psi}\rangle$ and $\tilde{m} = \langle\tilde{\psi}\tilde{\psi}\rangle$ in (3.8). Theories of this kind (Minguzzi and Tosi, 1997) involve coupled equations for $\Phi(\mathbf{r},t)$ and $\tilde{n}(\mathbf{r},t)$.

(c) The static Popov approximation (Griffin, 1996; Hutchinson *et al.*, 1997; Dalfovo *et al.*, 1999) involves a further simplification, namely, it ignores fluctuations in the density $\tilde{n}(\mathbf{r},t)$ of the thermal cloud so that the noncondensate is always in static thermal equilibrium,

$$\tilde{n}(\mathbf{r},t) \simeq \tilde{n}_0(\mathbf{r}). \tag{3.9}$$

A more complete discussion of these approximations and how they are related to each other is given in Chapter 5.

Using (3.9) in (3.8) corresponds to treating the dynamics of the condensate as if it were moving in a *static* mean field of the noncondensate thermal cloud. One then obtains

$$i\hbar\frac{\partial\Phi}{\partial t} = \left[-\frac{\hbar^2\nabla^2}{2m} + V_{\rm trap}(\mathbf{r}) + 2g\tilde{n}_0(\mathbf{r}) + gn_{\rm c}(\mathbf{r},t)\right]\Phi(\mathbf{r},t). \tag{3.10}$$

There is a considerable literature based on the static thermal cloud in the Popov approximation ($\tilde{m}(\mathbf{r},t) = 0$). The thermal occupation of quasiparticle excitations provides an expression for the equilibrium noncondensate density $\tilde{n}_0(\mathbf{r})$, while the condensate density $n_{\rm c0}(\mathbf{r}) \equiv |\Phi_0(\mathbf{r})|^2$ is obtained from the solution of the static version of the GP equation in (3.10). The self-consistent calculation of these two densities determines the equilibrium properties of the trapped gas within the static Popov approximation. The excitation energies are the quantized fluctuations of the condensate. In physical terms, in the static HF Popov approximation (HFP), only the dynamics of the Hartree mean field of the condensate atoms is treated self-consistently, while the collective dynamics of the noncondensate atoms is completely ignored. This approximation is discussed in detail in Chapters 8 and 9 for several different problems.

Returning to the exact equation (3.8) for the macroscopic wavefunction, it is useful to introduce phase and amplitude variables by setting $\Phi(\mathbf{r},t) = \sqrt{n_{\rm c}(\mathbf{r},t)}e^{i\theta(\mathbf{r},t)}$. After some algebra, one finds that (3.8) is equivalent to

$$\frac{\partial n_{\rm c}}{\partial t} + \nabla\cdot(n_{\rm c}\mathbf{v}_{\rm c}) = \frac{2g}{\hbar}\,{\rm Im}[(\Phi^*)^2\tilde{m} + \Phi^*\langle\tilde{\psi}^\dagger\tilde{\psi}\tilde{\psi}\rangle], \tag{3.11}$$

$$\hbar\frac{\partial\theta}{\partial t} = -\mu_{\rm c} - \frac{\hbar^2}{2m}(\nabla\theta)^2 = -\left(\mu_{\rm c} + \tfrac{1}{2}m\mathbf{v}_{\rm c}^2\right) \equiv -\varepsilon_{\rm c}. \tag{3.12}$$

The calculations are analogous to those given in (2.15)–(2.23) for $T = 0$. Here the time-dependent local variables are the condensate density in (3.7) and the condensate velocity defined by

$$\mathbf{v}_{\rm c}(\mathbf{r},t) \equiv \frac{\hbar}{m}\nabla\theta(\mathbf{r},t). \tag{3.13}$$

The local time-dependent condensate chemical potential in (3.12) is defined by

$$\begin{aligned}\mu_{\rm c}(\mathbf{r},t) \equiv &-\frac{\hbar^2\nabla^2\sqrt{n_{\rm c}}}{2m\sqrt{n_{\rm c}}} + V_{\rm trap}(\mathbf{r}) + gn_{\rm c}(\mathbf{r},t) + 2g\tilde{n}(\mathbf{r},t)\\ &+ \frac{g}{\hbar n_{\rm c}}{\rm Re}[(\Phi^*)^2\tilde{m} + \Phi^*\langle\tilde{\psi}^\dagger\tilde{\psi}\tilde{\psi}\rangle].\end{aligned} \tag{3.14}$$

The condensate velocity as defined in (3.13) is clearly identifiable with the superfluid velocity in the usual discussions of superfluid hydrodynamics.

In Chapters 15 and 17, we will show that the ZNG equations can be used to derive the Landau two-fluid equations when the collisions are strong enough to produce local hydrodynamic equilibrium. In this derivation, it will emerge that in a dilute gas the condensate density $n_{\rm c}$ can be identified at all temperatures with the superfluid density $\rho_{\rm s}/m$, which occurs in Landau two-fluid hydrodynamics. This equivalence is, of course, consistent with our use of the semiclassical Hartree–Fock excitation spectrum of the thermal cloud atoms. It is well known that in a noninteracting gas, the normal fluid density given by Landau's formula reduces to the density of noncondensate atoms ($\rho_{\rm n} = m\tilde{n}$).

One may give a physical interpretation of the generalized condensate equation (3.12). In static thermal equilibrium with $\mathbf{v}_{\rm c} = 0$, all variables are time independent. In this case, the r.h.s. of (3.11) must vanish and (3.12) has the solution $\theta(t) = -\mu_{\rm c0}t/\hbar$. Thus it is clear that in thermal equilibrium a condensate atom has an energy $\varepsilon_{\rm c0}$ equal to $\mu_{\rm c0}$, the equilibrium condensate chemical potential. We next consider a regime where the variables $\mu_{\rm c}(\mathbf{r},t)$ and $\mathbf{v}_{\rm c}(\mathbf{r},t)$ are slowly varying in both space and time. In this situation, we can expand the phase $\theta(\mathbf{r},t)$ around $(\mathbf{r}_0,t_0)$ as follows:

$$\begin{aligned}\theta(\mathbf{r},t) &\simeq \theta(\mathbf{r}_0,t_0) + \frac{\partial\theta}{\partial t}(t-t_0) + \nabla\theta\cdot(\mathbf{r}-\mathbf{r}_0) + \cdots\\ &\equiv \theta(\mathbf{r}_0,t_0) - \frac{1}{\hbar}\varepsilon_{\rm c}(\mathbf{r}_0,t_0)(t-t_0)\\ &\quad + \frac{1}{\hbar}m\mathbf{v}_{\rm c}(\mathbf{r}_0,t_0)\cdot(\mathbf{r}-\mathbf{r}_0) + \cdots,\end{aligned} \tag{3.15}$$

where $\varepsilon_{\rm c}(\mathbf{r}_0,t_0) \equiv \mu_{\rm c}(\mathbf{r}_0,t_0) + \frac{1}{2}mv_{\rm c}^2(\mathbf{r}_0,t_0)$. With this phase variation of the

condensate wavefunction, it is natural to interpret ε_c (defined in (3.12)) as the local energy of a condensate atom and $\mathbf{v}_c$ as its local velocity. This interpretation is consistent with the energy and momentum conservation laws we derive later when we consider two-body collisions between condensate and noncondensate atoms (see Section 3.2).

Formally, (3.8) and hence (3.11), (3.12) are exact results. In this book we will limit ourselves mainly to finite temperatures where the dominant thermal excitations can be approximated as high-energy noncondensate atoms moving in a self-consistent dynamic HF mean field, with local energy

$$\begin{aligned}\tilde{\varepsilon}_p(\mathbf{r},t) &= \frac{p^2}{2m} + V_{\text{trap}}(\mathbf{r}) + 2g\left[n_c(\mathbf{r},t) + \tilde{n}(\mathbf{r},t)\right] \\ &\equiv \frac{p^2}{2m} + U(\mathbf{r},t). \end{aligned} \tag{3.16}$$

We will neglect the effect of the mean field associated with the anomalous (off-diagonal) density $\tilde{m}$ defined in (3.7). One can show that both $\tilde{m}$ and $\langle\tilde{\psi}^\dagger\tilde{\psi}\tilde{\psi}\rangle$ are of order g and thus vanish in a noninteracting Bose gas. Thus one sees the correction terms in (3.11), (3.12) involving these functions are of order at least g^2. In fact, explicit calculations show that, to order g, $\langle\tilde{\psi}^\dagger\tilde{\psi}\tilde{\psi}\rangle$ is imaginary. Our procedure is to keep the imaginary parts of the terms of order g^2 in (3.11) and (3.12). These terms describe collisional damping of the condensate motion.

The final result of the preceding discussion is that the condensate wavefunction $\Phi(\mathbf{r},t)$ is described by

$$\frac{\partial n_c}{\partial t} + \nabla\cdot(n_c\mathbf{v}_c) = -\Gamma_{12}[f,\Phi], \tag{3.17}$$

$$m\left(\frac{\partial \mathbf{v}_c}{\partial t} + \tfrac{1}{2}\nabla v_c^2\right) = -\nabla\mu_c, \tag{3.18}$$

with

$$\mu_c(\mathbf{r},t) = -\frac{\hbar^2\nabla^2\sqrt{n_c(\mathbf{r},t)}}{2m\sqrt{n_c(\mathbf{r},t)}} + V_{\text{trap}}(\mathbf{r}) + gn_c(\mathbf{r},t) + 2g\tilde{n}(\mathbf{r},t). \tag{3.19}$$

Here we have introduced a new function

$$\Gamma_{12}[f,\Phi] \equiv -\frac{2g}{\hbar}\ \text{Im}\left[\Phi^*\langle\tilde{\psi}^\dagger\tilde{\psi}\tilde{\psi}\rangle\right] \equiv \Gamma_{12}(\mathbf{r},t). \tag{3.20}$$

We see that Γ_{12} plays the role of a local "source term" in the condensate continuity equation in (3.17). It is clearly a functional of both the condensate wavefunction $\Phi(\mathbf{r},t)$ and the single-particle distribution function $f(\mathbf{p},\mathbf{r},t)$. It is useful to note that (3.17)–(3.19) are equivalent to a generalized GP

or GGP equation of the kind (see also Proukakis *et al.*, 1998; Stoof, 1999; Walser *et al.*, 1999)

$$i\hbar\frac{\partial\Phi}{\partial t} = \left[-\frac{\hbar^2\nabla^2}{2m} + V_{\rm trap}(\mathbf{r}) + gn_{\rm c}(\mathbf{r},t) + 2g\tilde{n}(\mathbf{r},t) - iR(\mathbf{r},t)\right]\Phi\ , \quad (3.21)$$

where the effect of C_{12} collisions is described by the function

$$R(\mathbf{r},t) \equiv \frac{\hbar\Gamma_{12}[f,\Phi]}{2n_{\rm c}(\mathbf{r},t)} \sim O(g^2). \quad (3.22)$$

As we shall see in later chapters, the function Γ_{12} plays a crucial role in the behaviour of the condensate and can be positive (leading to damping) or negative (leading to growth).

To proceed, we need to calculate the field correlation $\langle\tilde{\psi}^\dagger\tilde{\psi}\tilde{\psi}\rangle$ that determines the source term $\Gamma_{12}(\mathbf{r},t)$ in (3.20). This is discussed in Section 3.2, working to lowest order in g at finite T, where the single-particle HF spectrum (3.16) is a reasonable approximation. The final result is

$$\begin{aligned}\langle\tilde{\psi}^\dagger\tilde{\psi}\tilde{\psi}\rangle = &- ig\frac{\Phi(\mathbf{r},t)}{(2\pi)^5\hbar^6}\int d\mathbf{p}_1\int d\mathbf{p}_2\int d\mathbf{p}_3 \\ &\times \delta(\mathbf{p}_{\rm c}+\mathbf{p}_1-\mathbf{p}_2-\mathbf{p}_3)\delta(\varepsilon_{\rm c}+\tilde{\varepsilon}_1-\tilde{\varepsilon}_2-\tilde{\varepsilon}_3) \\ &\times [f_1(1+f_2)(1+f_3)-(1+f_1)f_2f_3]\,, \end{aligned} \quad (3.23)$$

where $f_i \equiv f(\mathbf{p}_i,\mathbf{r},t)$, $\varepsilon_{\rm c}$ is the local condensate atom energy, and $\mathbf{p}_{\rm c} \equiv m\mathbf{v}_{\rm c}$ is the condensate atom momentum. In Section 3.2, we shall see that (3.23) is closely related to the collision integral $C_{12}[f,\Phi]$ that enters the Boltzmann equation for $f(\mathbf{p},\mathbf{r},t)$. This describes collisions with the noncondensate atoms in which one atom is added to or removed from the condensate. We also note that the expression (3.23) is indeed imaginary, as earlier claimed. Using this result in (3.20), we find an explicit expression for the source term

$$\begin{aligned}\Gamma_{12}(\mathbf{r},t) = \ 2\ g^2\frac{n_{\rm c}(\mathbf{r},t)}{(2\pi)^5\hbar^7}\int d\mathbf{p}_1\int d\mathbf{p}_2\int d\mathbf{p}_3\ \delta(\mathbf{p}_{\rm c}+\mathbf{p}_1-\mathbf{p}_2-\mathbf{p}_3) \\ \times\ \delta\left(\varepsilon_{\rm c}+\tilde{\varepsilon}_1-\tilde{\varepsilon}_2-\tilde{\varepsilon}_3\right) \\ \times\ [f_1(1+f_2)(1+f_3)-(1+f_1)f_2f_3]\,. \end{aligned} \quad (3.24)$$

This expression for Γ_{12} will play a key role in subsequent chapters. Of special interest is the case when the single-particle distribution functions f_i in (3.24) have the functional form of the Bose–Einstein distribution. This includes both static thermal equilibrium (see Chapters 8 and 9) and local hydrodynamic equilibrium (see Chapter 15).

3.2 Boltzmann equation for the noncondensate atoms

In this section, we turn to the dynamics of the noncondensate described by the field operator $\tilde{\psi}(\mathbf{r},t)$ in (3.4). The equation of motion obtained from (3.1) and (3.4) is

$$\begin{aligned} i\hbar\frac{\partial\tilde{\psi}}{\partial t} = \Big(-\frac{\hbar^2\nabla^2}{2m} + V_{\rm trap} + 2gn\Big)\tilde{\psi} - 2g\tilde{n}\tilde{\psi} + g\Phi^2\tilde{\psi}^\dagger \\ + g\Phi^*(\tilde{\psi}\tilde{\psi} - \tilde{m}) + 2g\Phi(\tilde{\psi}^\dagger\tilde{\psi} - \tilde{n}) + g\left(\tilde{\psi}^\dagger\tilde{\psi}\tilde{\psi} - \langle\tilde{\psi}^\dagger\tilde{\psi}\tilde{\psi}\rangle\right), \end{aligned} \tag{3.25}$$

where $n = n_{\rm c} + \tilde{n}$ is the total density. This equation preserves the condition $\langle\tilde{\psi}\rangle = 0$ as a function of time. It allows one to derive a kinetic equation for the noncondensate atoms. Following the pioneering work of Kirkpatrick and Dorfman (1985a) on uniform Bose gases, we define the time evolution of $\tilde{\psi}$ by

$$\tilde{\psi}(\mathbf{r},t) = \hat{S}^\dagger(t,t_0)\tilde{\psi}(\mathbf{r},t_0)\hat{S}(t,t_0)\,, \tag{3.26}$$

where the unitary operator $\hat{S}(t,t_0)$ evolves according to the equation of motion

$$i\hbar\frac{d\hat{S}(t,t_0)}{dt} = \hat{H}_{\rm eff}(t)\hat{S}(t,t_0)\,, \tag{3.27}$$

with $\hat{S}(t_0,t_0) = 1$. Here t_0 is the time at which the initial nonequilibrium density matrix $\hat{\rho}(t_0)$ is specified. The effective Hamiltonian in (3.27) is given by

$$\begin{aligned} \hat{H}_{\rm eff}(t) &= \hat{H}_0(t) + \hat{H}'(t), \\ \hat{H}'(t) &= \hat{H}_1(t) + \hat{H}_2(t) + \hat{H}_3(t) + \hat{H}_4(t)\,, \end{aligned} \tag{3.28}$$

where the leading HF term is

$$\hat{H}_0(t) = \int d\mathbf{r}\,\tilde{\psi}^\dagger\Big[-\frac{\hbar^2\nabla^2}{2m} + U(\mathbf{r},t)\Big]\tilde{\psi}, \tag{3.29}$$

with $U(\mathbf{r},t)$ as defined in (3.16). The perturbation contributions in $\hat{H}'(t)$ are

$$\begin{aligned} \hat{H}_1(t) &= \int d\mathbf{r}\Big[L_1(\mathbf{r},t)\tilde{\psi}^\dagger + L_1^*(\mathbf{r},t)\tilde{\psi}\Big], \\ \hat{H}_2(t) &= \frac{g}{2}\int d\mathbf{r}\Big[\Phi^2(\mathbf{r},t)\tilde{\psi}^\dagger\tilde{\psi}^\dagger + \Phi^{*2}(\mathbf{r},t)\tilde{\psi}\tilde{\psi}\Big], \\ \hat{H}_3(t) &= g\int d\mathbf{r}\Big[\Phi^*(\mathbf{r},t)\tilde{\psi}^\dagger\tilde{\psi}\tilde{\psi} + \Phi(\mathbf{r},t)\tilde{\psi}^\dagger\tilde{\psi}^\dagger\tilde{\psi}\Big], \\ \hat{H}_4(t) &= \frac{g}{2}\int d\mathbf{r}\,\tilde{\psi}^\dagger\tilde{\psi}^\dagger\tilde{\psi}\tilde{\psi} - 2g\int d\mathbf{r}\,\tilde{n}(\mathbf{r},t)\tilde{\psi}^\dagger\tilde{\psi}, \end{aligned} \tag{3.30}$$

with

$$L_1(\mathbf{r},t) \equiv -g\left[2\tilde{n}(\mathbf{r},t)\Phi(\mathbf{r},t) + \tilde{m}(\mathbf{r},t)\Phi^*(\mathbf{r},t) + \langle\tilde{\psi}^\dagger\tilde{\psi}\tilde{\psi}\rangle\right]. \tag{3.31}$$

It is understood that in all these expressions, the arguments of the quantum field operators $\tilde{\psi}, \tilde{\psi}^\dagger$ are $\mathbf{r}$ and t_0. The time dependences of the various terms in the Hamiltonian arise through the mean-field expectation values $n_c, \tilde{n}, L_1$, etc..

It can be shown that (3.26)–(3.31), together with the equal time commutator $[\tilde{\psi}(\mathbf{r},t_0), \tilde{\psi}^\dagger(\mathbf{r}',t_0)] = \delta(\mathbf{r}-\mathbf{r}')$, reproduce the equation of motion (3.25) for $\tilde{\psi}$ as well as that for $\tilde{\psi}^\dagger$. In writing (3.28), $\hat{H}'(t)$ is viewed as a perturbation to the zeroth-order Hamiltonian $\hat{H}_0(t)$. Noting that $U(\mathbf{r},t)$ is the total self-consistent Hartree–Fock (HF) mean field, $\hat{H}_0(t)$ defines excitations of the system at the level of the time-dependent HF approximation. Other choices of the zeroth-order Hamiltonian are also possible. For example, $\hat{H}_0(t)$ in (3.29) could be combined with $\hat{H}_2(t)$ to define a dynamic Bogoliubov Hamiltonian, which would then lead to local Bogoliubov quasiparticle excitations. This extension would be appropriate when considering very low temperatures (see Chapter 7).

We now consider an arbitrary operator $\hat{O}(t)$ made up of some combination of the noncondensate field operators $\tilde{\psi}(\mathbf{r},t)$ and $\tilde{\psi}^\dagger(\mathbf{r},t)$, for example, the local density operator $\tilde{n}(\mathbf{r},t) \equiv \tilde{\psi}^\dagger(\mathbf{r},t)\tilde{\psi}(\mathbf{r},t)$. By making use of (3.26), the expectation value of $\hat{O}(t)$ with respect to the initial density matrix $\hat{\rho}(t_0)$ can be expressed as

$$\begin{aligned}\langle\hat{O}(t)\rangle \equiv \langle\hat{O}\rangle_t &= \mathrm{Tr}\ \hat{\rho}(t_0)\hat{O}(t) \\ &= \mathrm{Tr}\ \tilde{\rho}(t,t_0)\hat{O}(t_0)\,,\end{aligned} \tag{3.32}$$

where $\tilde{\rho}(t,t_0) \equiv \hat{S}(t,t_0)\hat{\rho}(t_0)\hat{S}^\dagger(t,t_0)$ satisfies the following equation:

$$i\hbar\frac{d\tilde{\rho}(t,t_0)}{dt} = [\hat{H}_{\mathrm{eff}}(t), \tilde{\rho}(t,t_0)]\,. \tag{3.33}$$

We refer to Appendix A of ZNG for further discussion of this equation of motion.

Our ultimate objective is to obtain a quantum kinetic equation for the noncondensate atoms. For this purpose, we define the Wigner operator (see for example Kadanoff and Baym, 1962)

$$\hat{f}(\mathbf{p},\mathbf{r},t_0) \equiv \int d\mathbf{r}'\, e^{i\mathbf{p}\cdot\mathbf{r}'/\hbar}\tilde{\psi}^\dagger(\mathbf{r}+\tfrac{1}{2}\mathbf{r}',t_0)\tilde{\psi}(\mathbf{r}-\tfrac{1}{2}\mathbf{r}',t_0)\,. \tag{3.34}$$

Its expectation value (3.32) then yields the Wigner distribution function

$$f(\mathbf{p},\mathbf{r},t) = \mathrm{Tr}\ \tilde{\rho}(t,t_0)\hat{f}(\mathbf{p},\mathbf{r},t_0)\,. \tag{3.35}$$

Knowledge of this function allows one to calculate various nonequilibrium expectation values such as the noncondensate density

$$\tilde{n}(\mathbf{r},t) = \int \frac{d\mathbf{p}}{(2\pi\hbar)^3} f(\mathbf{p},\mathbf{r},t)\,. \tag{3.36}$$

Using (3.33), the equation of motion for f is

$$\begin{aligned}\frac{\partial f(\mathbf{p},\mathbf{r},t)}{\partial t} &= \frac{1}{i\hbar}\mathrm{Tr}\;\tilde{\rho}(t,t_0)[\hat{f}(\mathbf{p},\mathbf{r},t_0),\hat{H}_{\mathrm{eff}}(t)] \\ &= \frac{1}{i\hbar}\mathrm{Tr}\;\tilde{\rho}(t,t_0)[\hat{f}(\mathbf{p},\mathbf{r},t_0),\hat{H}_0(t)] \\ &\quad + \frac{1}{i\hbar}\mathrm{Tr}\;\tilde{\rho}(t,t_0)[\hat{f}(\mathbf{p},\mathbf{r},t_0),\hat{H}'(t)]\,. \end{aligned}\tag{3.37}$$

The first term on the right hand side of (3.37) defines the free-streaming operator in the kinetic equation. With the assumption that the self-consistent mean-field term $U(\mathbf{r},t)$ defined in (3.29) and (3.16) varies slowly in space, we then have

$$\begin{aligned}\frac{\partial f(\mathbf{p},\mathbf{r},t)}{\partial t} &+ \frac{\mathbf{p}}{m}\cdot\nabla_{\mathbf{r}} f(\mathbf{p},\mathbf{r},t) - \nabla U\cdot\nabla_{\mathbf{p}} f(\mathbf{p},\mathbf{r},t) \\ &= \frac{1}{i\hbar}\mathrm{Tr}\;\tilde{\rho}(t,t_0)[\hat{f}(\mathbf{p},\mathbf{r},t_0),\hat{H}'(t)]\,. \end{aligned}\tag{3.38}$$

The right-hand side of this equation clearly represents the effect of interatomic collisions on the single-particle distribution function $f(\mathbf{p},\mathbf{r},t)$. The reduction of this term to the form of a binary collision integral is a lengthy exercise (see Appendix A of Zaremba *et al.* (1999), as well as Chapters 6 and 7). However, the final result has a physically transparent form, which we now discuss. The collision integral can be written as

$$\left.\frac{\partial f}{\partial t}\right|_{\mathrm{coll}} = C_{12}[f,\Phi] + C_{22}[f]\,, \tag{3.39}$$

where

$$\begin{aligned}C_{22}[f] = \frac{2g^2}{(2\pi)^5\hbar^7}&\int d\mathbf{p}_2\int d\mathbf{p}_3\int d\mathbf{p}_4\;\delta(\mathbf{p}+\mathbf{p}_2-\mathbf{p}_3-\mathbf{p}_4) \\ &\times\delta(\tilde{\varepsilon}_p+\tilde{\varepsilon}_{p_2}-\tilde{\varepsilon}_{p_3}-\tilde{\varepsilon}_{p_4}) \\ &\times[(1+f)(1+f_2)f_3f_4 - ff_2(1+f_3)(1+f_4)]\,, \end{aligned}\tag{3.40}$$

$$\begin{aligned}C_{12}[f,\Phi] = \frac{2g^2 n_{\mathrm{c}}}{(2\pi)^2\hbar^4}&\int d\mathbf{p}_1\int d\mathbf{p}_2\int d\mathbf{p}_3\delta(m\mathbf{v}_{\mathrm{c}}+\mathbf{p}_1-\mathbf{p}_2-\mathbf{p}_3) \\ &\times\delta(\varepsilon_{\mathrm{c}}+\tilde{\varepsilon}_{p_1}-\tilde{\varepsilon}_{p_2}-\tilde{\varepsilon}_{p_3})[\delta(\mathbf{p}-\mathbf{p}_1)-\delta(\mathbf{p}-\mathbf{p}_2)-\delta(\mathbf{p}-\mathbf{p}_3)] \\ &\times[(1+f_1)f_2f_3 - f_1(1+f_2)(1+f_3)], \end{aligned}\tag{3.41}$$

with $f \equiv f(\mathbf{p},\mathbf{r},t)$, $f_i \equiv f(\mathbf{p}_i,\mathbf{r},t)$. We note that the variables n_c, $\mathbf{v}_c$, ε_c and $\tilde{\varepsilon}_p$ are all implicit functions of $\mathbf{r}$ and t. For atoms obeying Bose statistics, the creation of an atom in a state i is always associated with the statistical factor $1+f_i$, while the destruction of an atom in state i is always associated with the statistical factor f_i. We call attention to a key difference between C_{12} and C_{22} collisions. On the one hand, C_{22} and C_{12} both conserve energy and momentum in collisions. On the other hand, C_{12} does *not* (but C_{22} does) conserve the number of condensate atoms. It describes how atoms are scattered in and out of the condensate.

The final result is that the distribution function for the noncondensate thermal cloud atoms of momentum $\mathbf{p}$ is described by the kinetic equation

$$\frac{\partial f(\mathbf{p},\mathbf{r},t)}{\partial t}+\frac{\mathbf{p}}{m}\cdot\nabla_{\mathbf{r}} f(\mathbf{p},\mathbf{r},t)-\nabla U\cdot\nabla_{\mathbf{p}} f(\mathbf{p},\mathbf{r},t)=C_{12}[f,\Phi]+C_{22}[f]\,, \quad (3.42)$$

where $U(\mathbf{r},t)$ is the time-dependent HF potential (3.16). From the structure of C_{22} in (3.40), this term clearly describes two-body collisions between excited atoms (2 atoms $\rightleftharpoons$ 2 atoms). In contrast, C_{12} describes collisions between noncondensate atoms which involve one condensate atom (1 atom $\rightleftharpoons$ 2 atoms). This is the origin of the subscripts 22 and 12 notation, which was originally introduced by Kirkpatrick and Dorfman (1985a). The momentum and energy delta functions in the C_{12} collision term take into account the fact that a condensate atom has energy $\varepsilon_c(\mathbf{r},t)$ defined in (3.15) and momentum $m\mathbf{v}_c$ given by (3.13). The thermal atoms have the HF energy defined in (3.16). We note that the HF mean field involves the condensate density $n_c(\mathbf{r},t) \equiv |\Phi(\mathbf{r},t)|^2$. This must be determined self consistently by solving the condensate equations of motion given by (3.17)–(3.20) in conjunction with (3.42).

To complete this introductory discussion of a closed set of equations for both the condensate and noncondensate, we must deal with the off-diagonal self-energy $\tilde{m}$ and the term $\langle\tilde{\psi}^\dagger\tilde{\psi}\tilde{\psi}\rangle$ which appear in (3.11) and (3.14). The ZNG paper shows that $\tilde{m}$ can be neglected to the order in g to which we are working, while the anomalous three-field correlation function is

$$\begin{aligned}\langle\tilde{\psi}^\dagger\tilde{\psi}\tilde{\psi}\rangle = -\frac{ig\Phi}{(2\pi)^5\hbar^6}\int d\mathbf{p}_1\int d\mathbf{p}_2\int d\mathbf{p}_3\;\delta(m\mathbf{v}_c+\mathbf{p}_1-\mathbf{p}_2-\mathbf{p}_3)\\ \times\delta(\varepsilon_c+\tilde{\varepsilon}_{p_1}-\tilde{\varepsilon}_{p_2}-\tilde{\varepsilon}_{p_3})\\ \times[f_1(1+f_2)(1+f_3)-(1+f_1)f_2f_3]. \quad (3.43)\end{aligned}$$

We do not display the real part of $\langle\tilde{\psi}^\dagger\tilde{\psi}\tilde{\psi}\rangle$, which contributes to the local condensate chemical potential defined in (3.14). This contribution is of order g^2 and will be neglected, since our approximations do not adequately

treat all terms of this order (see Appendix A of ZNG for further discussion). However, using (3.20), the imaginary part of $\langle\tilde{\psi}^\dagger\tilde{\psi}\tilde{\psi}\rangle$ in (3.43) contributes an important source term, $-\Gamma_{12}[f,\Phi]$, to the r.h.s of the continuity equation in (3.11) and (3.17). Comparing (3.43) with the expression in (3.41) for $C_{12}[f,\Phi]$, this source term is given by

$$\Gamma_{12}[f,\Phi] \equiv -\frac{2g}{\hbar}\,\mathrm{Im}[\Phi^*\langle\tilde{\psi}^\dagger\tilde{\psi}\tilde{\psi}\rangle] = \int \frac{d\mathbf{p}}{(2\pi\hbar)^3} C_{12}[f(\mathbf{p},\mathbf{r},t),\Phi(\mathbf{r},t)]\,. \quad (3.44)$$

The result for the quantity $\Gamma_{12}[f,\Phi]$ introduced in (3.17) emphasizes that the C_{12} collisions locally change the relative number of condensate (and hence noncondensate) atoms and, as a result, number conservation laws no longer hold separately for $n_{\rm c}$ and $\tilde{n}$. As we shall see, the close connection between $\langle\tilde{\psi}^\dagger\tilde{\psi}\tilde{\psi}\rangle$ in (3.43) and the C_{12} collision integral in (3.41) is essential for the conservation of the total number of particles. This also indicates that the approximations leading to our final equations of motion for the condensate and noncondensate atoms are internally consistent.

3.3 Solutions in thermal equilibrium

It will be useful for later reference to discuss the static equilibrium solutions of the coupled ZNG equations introduced in Section 3.2. For simplicity, here we consider only the case where there are no stationary currents in equilibrium ($\mathbf{v}_{\rm n0} = 0, \mathbf{v}_{\rm c0} = 0$). The more general case (which is relevant when vortices are present and hence $\mathbf{v}_{\rm c0} \neq 0$) is discussed in Chapter 9.

We first discuss the equilibrium solution for the condensate wavefunction, which is obtained from

$$\left[-\frac{\hbar^2\nabla^2}{2m} + V_{\rm trap}(\mathbf{r}) + g n_{\rm c0}(\mathbf{r}) + 2g\tilde{n}_0(\mathbf{r})\right]\Phi_0(\mathbf{r}) = \varepsilon_{\rm c0}\Phi_0(\mathbf{r}). \quad (3.45)$$

Here $\varepsilon_{\rm c0}$ is the energy of the condensate atoms defined by the stationary solution of (3.21)

$$\Phi_0(\mathbf{r},t) = \sqrt{n_{\rm c0}(\mathbf{r})}e^{-i\varepsilon_{\rm c0}t/\hbar}. \quad (3.46)$$

We have used the fact (to be discussed below) that in static thermal equilibrium between the thermal cloud and the condensate, one has

$$\Gamma_{12}[f^0,\Phi_0] = 0, \quad R[f^0,\Phi_0] = 0. \quad (3.47)$$

Substituting (3.46) into (3.45), one can write the condensate atom energy

as

$$\varepsilon_{\rm c0} = -\frac{\hbar^2\nabla^2\sqrt{n_{\rm c0}(\mathbf{r})}}{2m\sqrt{n_{\rm c0}(\mathbf{r})}} + V_{\rm trap}(\mathbf{r}) + gn_{\rm c0}(\mathbf{r}) + 2g\tilde{n}_0(\mathbf{r}), \tag{3.48}$$

which still depends on $n_{\rm c0}(\mathbf{r}) = |\Phi_0(\mathbf{r})|^2$ and $\tilde{n}_{\rm c0}(\mathbf{r})$. However, from (3.18) it follows that $\nabla\varepsilon_{\rm c0} = 0$ and hence that $\varepsilon_{\rm c0}$ is position independent. In the Thomas–Fermi (TF) approximation defined in (2.10), the first term in (3.48) (which is often called the "quantum pressure" associated with the spatial dependence of $\Phi_0(\mathbf{r})$) is neglected.

The collisions between atoms in the thermal cloud described by the C_{22} collision integral in (3.40) force a nonequilibrium distribution $f(\mathbf{p},\mathbf{r},t)$ to evolve to the static equilibrium Bose distribution $f^0(\mathbf{p},\mathbf{r})$. Using the single-particle HF spectrum, this is given by

$$f^0(\mathbf{p},\mathbf{r}) = \frac{1}{e^{\beta_0[p^2/2m+U_0(\mathbf{r})-\tilde{\mu}_0]}-1}, \tag{3.49}$$

where $\beta_0 = 1/k_{\rm B}T_0$ is the inverse uniform temperature. The trapping potential is augmented by the Hartree–Fock mean field,

$$U_0(\mathbf{r}) = V_{\rm trap}(\mathbf{r}) + 2g[n_{\rm c0}(\mathbf{r}) + \tilde{n}_0(\mathbf{r})]. \tag{3.50}$$

Using (3.49), one may verify that the statistical factors in C_{22}, in (3.40), cancel each other:

$$\left[(1+f^0)(1+f_2^0)f_3^0 f_4^0 - f^0 f_2^0(1+f_3^0)(1+f_4^0)\right] = 0, \tag{3.51}$$

where we have used the energy and momentum conservation factors in (3.40). We thus conclude that

$$C_{22}[f^0,\Phi] = 0. \tag{3.52}$$

We note that this result holds as an identity for arbitrary values of the thermal atom chemical potential $\tilde{\mu}_0$ in (3.49). Also it does not depend on any assumption about the Bose condensate wavefunction $\Phi(\mathbf{r},t)$.

Inserting $f^0(\mathbf{p},\mathbf{r})$ into the C_{12} collision integral (3.41), one finds that the R function in (3.22) reduces to

$$\begin{aligned} R[f^0,\Phi] = \frac{g^2}{(2\pi)^5\hbar^5}\int d\mathbf{p}_1\int d\mathbf{p}_2\int d\mathbf{p}_3\delta(\mathbf{p}_{\rm c}+\mathbf{p}_1-\mathbf{p}_2-\mathbf{p}_3) \\ \times\delta(\varepsilon_{\rm c}+\tilde{\varepsilon}_1-\tilde{\varepsilon}_2-\tilde{\varepsilon}_3)(1+f_1^0)f_2^0 f_3^0 \\ \times\left[e^{\beta(\varepsilon_{\rm c0}-\tilde{\mu})}-1\right]. \end{aligned} \tag{3.53}$$

The factor in the square bracket in (3.53) is easily derived. Using the Bose

distribution identity,

$$f^0(x) \equiv \frac{1}{e^x - 1} = e^{-x}[1 + f^0(x)], \tag{3.54}$$

the product of Bose distributions in (3.24) reduces to

$$f_1^0(1 + f_2^0)(1 + f_3^0) = (1 + f_1^0) f_2^0 f_3^0 e^{\beta(\varepsilon_{c0} - \tilde{\mu}_0)}. \tag{3.55}$$

In deriving (3.55), we have made use of the energy and momentum conservation factors in (3.53).

The fact that in general the collision integral $C_{12}[f^0, \Phi]$ does not vanish means that even if the atoms in the thermal cloud are in equilibrium among themselves, as assumed in obtaining (3.49), the condensate and thermal cloud components may not be in diffusive local equilibrium with each other. The latter condition requires that the expression in square brackets in (3.53) vanishes, namely

$$\varepsilon_{c0} = \tilde{\mu}_0. \tag{3.56}$$

This condition shows that ε_{c0} plays the role of the condensate chemical potential μ_{c0}: for a diffusive static equilibrium between the condensate and the noncondensate atoms, we must have $\mu_{c0} = \varepsilon_{c0} = \tilde{\mu}_0$. This condition must be *imposed* in solving for $n_{c0}(\mathbf{r})$ and $\tilde{n}_0(\mathbf{r})$. The equilibrium noncondensate density $\tilde{n}_0(\mathbf{r})$ is obtained by integrating the equilibrium Bose distribution in (3.49) over momentum, to give the textbook result

$$\tilde{n}_0(\mathbf{r}) = \int \frac{d\mathbf{p}}{(2\pi\hbar)^3} f^0(\mathbf{p}, \mathbf{r}) = \frac{1}{\Lambda^3} g_{3/2}(z_0(\mathbf{r})). \tag{3.57}$$

Here

$$\Lambda = (2\pi\hbar^2 \beta_0 / m)^{1/2} \tag{3.58}$$

is the thermal de Broglie wavelength, $g_{3/2}(z)$ is a Bose–Einstein function (see (B.7) in Appendix B) and the local fugacity is $z_0(\mathbf{r}) = e^{\beta_0[\tilde{\mu}_0 - U_0(\mathbf{r})]}$.

We recall that, in the TF approximation, (3.48) reduces to $\varepsilon_{c0} = V_{\rm trap}(\mathbf{r}) + g n_{c0}(\mathbf{r}) + 2g\tilde{n}_0(\mathbf{r})$. Thus the fugacity in (3.57) is

$$z_0(\mathbf{r}) = e^{-\beta_0 g n_{c0}(\mathbf{r})}, \tag{3.59}$$

the TF condensate density being given by

$$n_{c0}(\mathbf{r}) = \frac{1}{g}\left[\varepsilon_{c0} - V_{\rm trap}(\mathbf{r}) - 2g\tilde{n}_0(\mathbf{r})\right] \geq 0. \tag{3.60}$$

The condition that $n_{c0}(\mathbf{r})$ in (3.60) must be positive determines the TF boundary or size of the Bose condensate. Thus (3.57) and (3.60) are two

closed equations for $\tilde{n}_0(\mathbf{r})$ and $n_{c0}(\mathbf{r})$, which must be solved self-consistently (see Minguzzi and Tosi, 1997). This self-consistent solution is discussed in more detail in Section 11.4, both with and without the TF approximation. The density profiles that one finds are illustrated in Fig. 11.5.

The thermodynamic functions of a trapped interacting Bose gas are reviewed in Chapter 13 of Pitaevskii and Stringari (2003) and in Section 13.1 of Pethick and Smith (2008). A high precision experimental study of the temperature dependence of the equilibrium thermodynamic properties was carried out by Gerbier *et al.* (2004b). As discussed by these authors, such measurements require an assessment of many experimental complications, including how the number of atoms in the trap changes with temperature. In Fig. 3.1, we show time-of-flight data from Gerbier *et al.* (2004b) for the temperature dependence of the condensate function for ^{87}Rb atoms in a cigar-shaped trap with $\omega_\perp/2\pi = 413$ Hz and $\omega_z/2\pi = 8.7$ Hz. Figure. 3.1 also gives their calculated values of $N_c(T)/N$, obtained by solving the self-consistent TF equations (3.57) and (3.60) in three different approximations. The curves give the results for (a) a noninteracting Bose gas (broken line), (b) the TF approximation keeping only the condensate HF mean field (dotted line) and (c) the TF approximation keeping the full HF mean field (solid line). One sees that the full HF approximation is in good agreement with the measurements. Earlier quantum Monte Carlo calculations confirmed that the HF approximation described the thermodynamics of trapped Bose-condensed gas with good accuracy (see for example Holzmann *et al.*, 1999).

Gerbier *et al.* (2004a) showed that, in agreement with the results in Fig. 3.1, the decrease in the superfluid transition temperature in a trapped Bose gas due to interactions is well described by the HF mean-field corrections. In the specific experiments of Gerbier *et al.* (2004a), these mean-field corrections to $T_{\rm BEC}$ were as large as 10%, much larger than the finite-size and critical fluctuation effects. For further discussion of such corrections, see Chapter 13 of Pitaevskii and Stringari (2003).

Because the thermal cloud density $\tilde{n}_0(\mathbf{r})$ is always very small relative to the condensate density $n_{c0}(\mathbf{r})$ in trapped Bose gases, we usually ignore the effect of the noncondensate HF field $2g\tilde{n}_0(\mathbf{r})$ in (3.60) in determining the condensate density profile $n_{c0}(\mathbf{r})$.

3.4 Region of validity of the ZNG equations

The ZNG coupled equations introduced in this chapter clearly form an approximation to a more complete description of the coupled dynamics of the condensate and noncondensate atoms. In this concluding section, we make

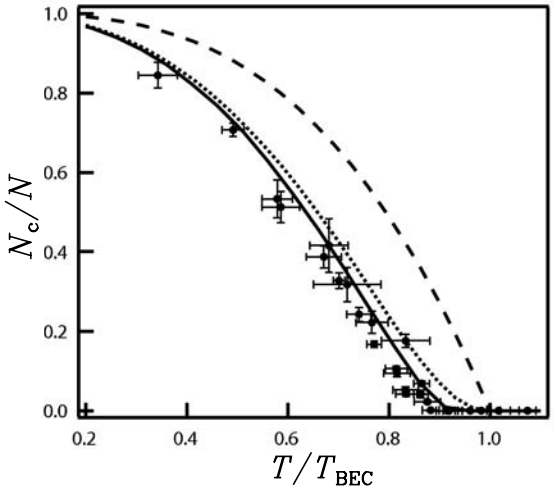


Fig. 3.1. Fraction of condensate atoms vs. the temperature (normalized to the transition temperature of a noninteracting trapped Bose gas). The results of calculations are shown for an ideal Bose gas (broken line) and the Hartree–Fock TF approximations in which the thermal cloud mean field is omitted (dotted line) and retained (solid line) (from Gerbier *et al.*, 2004b).

some further remarks about the underlying basis of the ZNG approximation and its region of validity.

The crucial feature of the ZNG approximation is that it treats the condensate and noncondensate components in qualitatively different ways. As in the $T = 0$ case, the GGP equation (3.21) at finite T must be treated as a quantum mechanical equation, since the quantum nature of the condensate wavefunction can never be ignored. In contrast, the thermal cloud atoms are treated within a semiclassical approximation in which the atoms move in a self-consistent Hartree–Fock potential. In the ZNG treatment, one never deals with the HF eigenfunctions of the thermal atoms, the analogue of the eigenfunctions of the Bogoliubov equations. The ZNG theory is built on the difference between a large number of condensate atoms, all occupying the same single-particle quantum state, and the thermal atoms occupying many different states with low occupation probabilities, justifying a semiclassical treatment.

Treating the condensate and noncondensate at such a different level of approximation might initially seem inconsistent, as well as a serious deficiency of the ZNG theory. However, the ZNG semiclassical treatment of the thermal cloud atoms is based on the fact that at finite temperatures, the

important thermal atoms have relatively high energies. In contrast, if one is interested in very low temperatures, then the ZNG treatment of the thermal excitations is no longer adequate (see Chapter 7).

Within our microscopic model, all the condensate atoms involved in the C_{12} collision term (3.41) have the same energy $\varepsilon_{\rm c}(\mathbf{r},t) = \mu_{\rm c}(\mathbf{r},t) + \frac{1}{2}mv^2(\mathbf{r},t)$ and the same momentum $\mathbf{p}_{\rm c}(\mathbf{r},t) = m\mathbf{v}_{\rm c}(\mathbf{r},t)$. We now discuss in more detail what these approximations involve. We recall that the momentum distribution of atoms in any system is given by the formula (see Pitaevskii and Stringari, 2003)

$$n(\mathbf{p}) = \frac{1}{V}\int d\mathbf{r}_1 d\mathbf{r}_2 \ \langle\hat{\psi}^\dagger(\mathbf{r}_1)\hat{\psi}(\mathbf{r}_2)\rangle e^{i\mathbf{p}\cdot(\mathbf{r}_1-\mathbf{r}_2)/\hbar}, \tag{3.61}$$

where V is the volume of the system. One can separate this into condensate and noncondensate parts, where the condensate momentum distribution is given by

$$n_{\rm c}(\mathbf{p}) = \frac{1}{V}\int d\mathbf{r}_1 \int d\mathbf{r}_2 \ \Phi^*(\mathbf{r}_1)\Phi(\mathbf{r}_2)e^{i\mathbf{p}\cdot(\mathbf{r}_1-\mathbf{r}_2)/\hbar} \ . \tag{3.62}$$

For a nonuniform trapped gas, it is convenient to introduce the relative and centre-of-mass coordinates

$$\mathbf{r}' \equiv \mathbf{r}_1 - \mathbf{r}_2, \ \ \mathbf{r} \equiv \tfrac{1}{2}(\mathbf{r}_1 + \mathbf{r}_2). \tag{3.63}$$

One can then write the condensate momentum distribution (3.62) as

$$n_{\rm c}(\mathbf{p}) = \frac{1}{V}\int d\mathbf{r} \int d\mathbf{r}' \ \Phi^*(\mathbf{r}+\tfrac{1}{2}\mathbf{r}')\Phi(\mathbf{r}-\tfrac{1}{2}\mathbf{r}')e^{i\mathbf{p}\cdot\mathbf{r}'/\hbar} \equiv \frac{1}{V}\int d\mathbf{r}\, n_{\rm c}(\mathbf{p},\mathbf{r}), \tag{3.64}$$

where the local momentum distribution at position $\mathbf{r}$ in the trap is defined as

$$n_{\rm c}(\mathbf{p},\mathbf{r}) \equiv \int d\mathbf{r}' \ \Phi^*(\mathbf{r}+\tfrac{1}{2}\mathbf{r}')\Phi(\mathbf{r}-\tfrac{1}{2}\mathbf{r}')e^{i\mathbf{p}\cdot\mathbf{r}'/\hbar}. \tag{3.65}$$

In terms of the amplitude and the phase of the condensate wavefunction $\Phi(\mathbf{r}) = \sqrt{n_{\rm c}(\mathbf{r})}e^{i\theta(\mathbf{r})}$, one can rewrite (3.65) as

$$n_{\rm c}(\mathbf{p},\mathbf{r}) = \int d\mathbf{r}' \sqrt{n_{\rm c}(\mathbf{r}+\tfrac{1}{2}\mathbf{r}')n_{\rm c}(\mathbf{r}-\tfrac{1}{2}\mathbf{r}')}e^{i[\theta(\mathbf{r}+\frac{1}{2}\mathbf{r}')-\theta(\mathbf{r}-\frac{1}{2}\mathbf{r}')]}e^{i\mathbf{p}\cdot\mathbf{r}'/\hbar} \ . \tag{3.66}$$

As discussed earlier, we limit ourselves to the case where the amplitude and the phase of the condensate vary slowly in space so that we can expand them around $\mathbf{r}$. For the phase difference in (3.66), we write

$$\theta(\mathbf{r}+\tfrac{1}{2}\mathbf{r}') - \theta(\mathbf{r}-\tfrac{1}{2}\mathbf{r}') \simeq \nabla_{\mathbf{r}}\theta(\mathbf{r})\cdot\mathbf{r}' \equiv m\mathbf{v}_{\rm c}(\mathbf{r})\cdot\mathbf{r}'/\hbar, \tag{3.67}$$

where the condensate velocity is defined in (3.13). For the amplitude condensate densities in (3.66), we expand in power of $\mathbf{r}$,

$$n_{\rm c}(\mathbf{r} \pm \tfrac{1}{2}\mathbf{r}') \simeq n_{\rm c}(\mathbf{r}) \pm \tfrac{1}{2}\mathbf{r}' \cdot \nabla_{\mathbf{r}} n_{\rm c}(\mathbf{r}) + \tfrac{1}{8}\sum_{i,j} \frac{\partial^2 n_{\rm c}(\mathbf{r})}{\partial x_i \partial x_j} x_i' x_j' + \cdots, \tag{3.68}$$

where $\mathbf{r}' \equiv (x_1', x_2', x_3')$ and $\mathbf{r} \equiv (x_1, x_2, x_3)$. In the Thomas–Fermi approximation, one neglects the spatial gradients of the condensate density (i.e. the quantum pressure). Thus if we are to be consistent, we should keep only the first term in (3.68). Within this approximation and using (3.67), the expression in (3.66) finally reduces to

$$\begin{aligned} n_{\rm c}(\mathbf{p}, \mathbf{r}) &\simeq \int d\mathbf{r}' n_{\rm c}(\mathbf{r}) e^{i[\mathbf{p} - m\mathbf{v}_{\rm c}(\mathbf{r})]\cdot \mathbf{r}'/\hbar} \\ &= n_{\rm c}(\mathbf{r})(2\pi\hbar)^3 \delta(\mathbf{p} - m\mathbf{v}_{\rm c}(\mathbf{r})). \end{aligned} \tag{3.69}$$

This is the local momentum distribution of the condensate atoms that was implicitly assumed in our earlier derivation of the collision integrals C_{12} and C_{22}. In that derivation, we kept the spatial variation of the phase $\theta(\mathbf{r}, t)$ of the condensate order parameter but ignored the spatial gradients of the amplitude $\sqrt{n_{\rm c}(\mathbf{r}, t)}$. We see from (3.69) that our approximation for the C_{12} collision integral in (3.41) effectively treats the condensate as locally homogeneous, so that all condensate atoms at position $\mathbf{r}$ with density $n_{\rm c}(\mathbf{r})$ have the same momentum, given by $\mathbf{p}_{\rm c}(\mathbf{r}, t) = m\mathbf{v}_{\rm c}(\mathbf{r}, t)$.

One can easily verify that the approximations leading to (3.69) are completely consistent with our treatment of the thermal gas in a high-temperature Hartree–Fock semiclassical approximation. We will show this by calculating the corrections to (3.69) for a static momentum distribution. Using the Thomas–Fermi approximation for the static density profile,

$$n_{\rm c}(\mathbf{r}) = n_{\rm c}(0)\left(1 - \frac{\mathbf{r}^2}{R_{\rm TF}^2}\right), \tag{3.70}$$

where $n_{\rm c}(0)$ is the value at $\mathbf{r} = 0$. To estimate the correction terms in (3.68), the amplitude in (3.66) can be approximated as follows:

$$\sqrt{n_{\rm c}(\mathbf{r} + \tfrac{1}{2}\mathbf{r}') n_{\rm c}(\mathbf{r} - \tfrac{1}{2}\mathbf{r}')} \simeq n_{\rm c}(\mathbf{r}) - \frac{n_{\rm c}(0) r'^2}{4R_{\rm TF}^2} + \frac{n_{\rm c}^2(0)}{2n_{\rm c}(\mathbf{r})} \frac{(\mathbf{r}' \cdot \mathbf{r})^2}{R_{\rm TF}^4} + \cdots \tag{3.71}$$

For the purpose of illustration, we limit ourselves to calculating the local momentum condensate distribution $n_{\rm c}(\mathbf{p}, 0)$ at the centre of the trap. If we approximate the first two terms in (3.71) as a Gaussian, so that

$$n_{\rm c}(0)\left[1 - \frac{r'^2}{4R_{\rm TF}^2}\right] \simeq n_{\rm c}(0) e^{-r'^2/4R_{\rm TF}^2}, \tag{3.72}$$

the integral over $\mathbf{r}'$ in (3.66) is easily done. The final result gives a Gaussian condensate momentum distribution at the trap centre,

$$n_{\rm c}(\mathbf{p},0) \simeq n_{\rm c}(0)\int d\mathbf{r}' e^{-i[\mathbf{p}-m\mathbf{v}_{\rm c}(0)]\cdot\mathbf{r}'/\hbar} e^{-r'^2/4R_{\rm TF}^2}$$
$$= n_{\rm c}(0) R_{\rm TF}^2 \pi^{3/2} e^{-[\mathbf{p}-m\mathbf{v}_{\rm c}(0)]^2/(\Delta p)^2}, \tag{3.73}$$

where the width is $\Delta p \equiv \hbar/R_{\rm TF}$. This estimate shows how keeping the spatial gradient of the nonuniform condensate density profile in (3.68) gives rise to a Gaussian momentum spread of the condensate atoms of width Δp around the peak value $\mathbf{p}_{\rm c} \equiv m\mathbf{v}_{\rm c}(0)$. The latter is determined by the gradient of the condensate phase at the trap centre $\mathbf{r} = 0$.

To be able to ignore the finite width given in (3.73) of the momentum spread of condensate atoms around the value $m\mathbf{v}_{\rm c}(0)$, we require that the width Δp be much smaller than the average momentum $\bar{p}_{\rm th}$ of the thermal atoms. This requires that

$$\bar{p}_{\rm th} \gg \Delta p \text{ or } R_{\rm TF} \gg \lambda_{\rm dB}, \tag{3.74}$$

where $\lambda_{\rm dB} \sim \hbar/\sqrt{mk_{\rm B}T}$ is the thermal de Broglie wavelength. The second inequality given in (3.74) is correct at higher temperatures, precisely where our semiclassical approximation should be valid.

The condensate atoms are described by the coupled equations (3.17)–(3.20). As discussed, these equations are approximations to the exact equations given in (3.11), (3.12) and (3.14). The thermal cloud atoms are described by the kinetic equation in (3.42). The two collision integrals C_{22} and C_{12} are given explicitly by the expressions in (3.40) and (3.41). These equations are coupled through the HF mean fields and must therefore be solved self-consistently. The conditions of validity for our semiclassical kinetic equation for $f(\mathbf{p},\mathbf{r},t)$ are

$$k_{\rm B}T \gg gn_{\rm c}, \quad k_{\rm B}T \gg \hbar\omega_0, \tag{3.75}$$

where $\hbar\omega_0$ is the spacing of the energy levels of the harmonic trap. Similar kinetic equations for trapped thermally excited atoms have also been discussed within different theoretical formulations by Stoof (1999); Walser *et al.* (1999) and Gardiner and Zoller (2000).

For temperatures above the Bose–Einstein transition ($T > T_{\rm BEC}$), where C_{12} vanishes, the kinetic equation (3.42) reduces to the well-known equation discussed in the classic paper by Uehling and Uhlenbeck (1933). Below $T_{\rm BEC}$, (3.42) is analogous to the kinetic equation derived by Kirkpatrick and Dorfman (1985a) for a homogeneous weakly interacting Bose-condensed gas at finite temperatures. One difference is that the KD equation works with

excitations that are defined within the local rest frame of the superfluid (condensate), and thus the excitation energies are measured relative to that of the condensate atoms. In contrast, our kinetic equation (3.42) describes atoms moving in a self-consistent HF field.

It is useful to make this comparison with the KD equation more precise, since it will arise in later chapters. The $C_{12}[f,\Phi]$ collision integral (3.41) involves the energy and momentum conservation factors

$$\delta(\varepsilon_{\rm c}+\tilde{\varepsilon}_1-\tilde{\varepsilon}_2-\tilde{\varepsilon}_3)\delta(m\mathbf{v}_{\rm c}+\mathbf{p}_1-\mathbf{p}_2-\mathbf{p}_3), \tag{3.76}$$

where (in our HF approximation) the thermal atom energy is $\tilde{\varepsilon}_i = p_i^2/2m + V_{\rm trap} + 2g(\tilde{n}+n_{\rm c})$ and $\varepsilon_{\rm c} = \mu_{\rm c} + \frac{1}{2}mv_{\rm c}^2$ is the local energy of an atom in the condensate. However, in the work of Kirkpatrick and Dorfman, $C_{12}[f,\Phi]$ contains the energy–momentum conservation factors

$$\delta(E_1-E_2-E_3)\delta(\mathbf{p}_1'-\mathbf{p}_2'-\mathbf{p}_3')\,. \tag{3.77}$$

Here E_i is the Bogoliubov quasiparticle excitation energy and $\mathbf{p}_i' = \mathbf{p}_i - m\mathbf{v}_{\rm c}$ is the quasiparticle momentum in the "local" rest frame (i.e. the frame in which the superfluid is not moving). In the HF high-temperature limit (large momenta), these excitation energies reduce to $E_i \simeq {p_i'}^2/2m + gn_{\rm c} = \tilde{\varepsilon}_i - \mu_{\rm c} - \mathbf{v}\cdot\mathbf{p}_i + \frac{1}{2}mv_{\rm c}^2$, where $\mu_{\rm c} = V_{\rm trap} + g(n_{\rm c}+2\tilde{n})$ is the local condensate chemical potential (within the TF approximation) as given by (3.19). One can easily verify that the energy and momentum conservation factors in (3.76) and (3.77) are completely equivalent in this high-temperature limit. Thus the two seemingly different formulations, in terms of atoms moving in a HF mean field or in terms of excitations, describe the same physics.

It is apparent that equations (3.17)–(3.19) always have the form of "hydrodynamic" equations for the condensate variables $n_{\rm c}$ and $\mathbf{v}_{\rm c}$. Although these equations have been derived for a trapped gas at finite temperatures, we note that they go over smoothly to the correct $T=0$ theory in that, as $f(\mathbf{p},\mathbf{r},t)$ and hence $\tilde{n}$ become negligible, our condensate equations reduce to the Gross–Pitaevskii dynamics of a pure condensate at $T=0$ (reviewed in Chapter 2). We therefore expect our equations to provide a reasonable description of the dynamics over a wide range of temperatures, from the strongly degenerate limit well below $T_{\rm BEC}$ to the classical high-temperature limit $T \gg T_{\rm BEC}$.

In the case of a trapped Bose-condensed gas, the high-temperature HF approximation that we use for the thermal atoms is not as restrictive as it might seem. In an important early paper, Giorgini *et al.* (1997) showed that the differences arising from using the HF instead of the Bogoliubov excitation spectrum in calculating the thermodynamic properties of a trapped

Bose-condensed gas are very small down to quite low temperatures. This is in contrast with the case of a uniform Bose-condensed gas: in a trapped Bose gas, the dominant thermal excitations are particle-like down to very low temperatures. As a result, our HF spectrum is expected to be valid in a trapped Bose gas over a much larger temperature region than one might have expected initially.

We conclude this chapter with a brief review. We have introduced the coupled equations of motion for the condensate and noncondensate components that will be used throughout the rest of this book. One has a generalized GP equation for the condensate, which has an extra "source" term arising from C_{12} collisions between atoms in the condensate and in the thermal cloud. As we shall discuss at length in Chapters 6 and 7, this simple model for a superfluid Bose gas at finite temperatures can be derived as the first step in a general formalism for nonequilibrium properties developed by Kadanoff and Baym (1962). This is a generalized procedure for dealing with nonequilibrium behaviour in many body systems. In our microscopic model, worked out in detail in a series of papers by the present authors and their coworkers, the thermal gas is treated using a semiclassical approximation, with atoms moving in a self-consistent Hartree–Fock field produced by the condensate and noncondensate atoms. Because of its high density in the centre of the trap, the condensate mean field always dominates over the noncondensate mean field, even at temperatures very close to $T_{\rm BEC}$.

Our simple treatment of the thermal cloud is clearly only valid at higher temperatures, when the collective (Bogoliubov) nature of the thermal excitations can be neglected. While extensions can be made, and have been discussed in the literature, in the present book we will show that our microscopic model for the coupled dynamics in a trapped Bose gas at finite temperatures includes all the essential physics needed to describe nonequilibrium behaviour. In particular, it is able to deal quantitatively with the collisionless region as well as making precise contact with the two-fluid hydrodynamics region. While relatively simple, these coupled equations give considerable insight into many subtle aspects of the dynamics of a Bose superfluid at finite temperatures. Thus, it seems well worth the effort to understand and fully explore the consequences of the simple microscopic theory of a trapped Bose-condensed gas at finite temperatures, which has been summarized in this chapter. This is the main topic of the succeeding chapters.

Chapters 4–7 are devoted to a review of the microscopic theory of an interacting gas of Bose atoms; this is based on the approach developed by Beliaev in 1958. Beliaev's formalism makes use of quantum field theory and

also makes crucial use of the idea of a Bose-broken symmetry, as briefly explained earlier in this chapter. It is found that the characteristic correlations introduced by the existence of a Bose-condensate order parameter are naturally discussed in terms of single-particle Green's functions and linear response functions. These will be reviewed in the next four chapters. In Chapters 6 and 7, we will use nonequilibrium Green's function techniques (developed by Kadanoff and Baym, 1962) to derive the equations of motion for the condensate and noncondensate atoms. We show how the ZNG equations introduced in this chapter emerge as the simplest nontrivial approximation within the Kadanoff–Baym formalism.

4

Green's functions and self-energy approximations

In Chapter 3, we introduced a simple but reasonable approximation for the nonequilibrium dynamics of a Bose-condensed gas based on a generalized GP equation coupled to a kinetic equation. In Chapters 4–7, we turn to the question of how to derive such coupled equations for the condensate and noncondensate components in a way that gives a deeper understanding of the ZNG theory. Chapters 4–7 involve an introduction to Green's functions and field theoretic techniques for nonequilibrium problems. These provide the natural language and formalism to deal with the many subtle aspects of a Bose-condensed gas at finite temperatures. These four chapters are fairly technical. This chapter is mainly based on Kadanoff and Baym (1962) and Imamović-Tomasović (2001). Readers who are not interested in these questions can go straight to Chapter 8, which begins the discussion of applications of the ZNG coupled equations given in Chapter 3.

4.1 Overview of Green's function approach

To derive a microscopic theory of the nonequilibrium behaviour of a dilute weakly interacting Bose-condensed gas at finite temperatures, there are several different approaches available in the literature. We will use the well-known Kadanoff–Baym (KB) nonequilibrium Green's function method. The generalization of this formalism to a Bose-condensed system was first considered by Kane and Kadanoff (1965), whose goal was to derive the Landau two-fluid hydrodynamic equations for a system with a Bose broken symmetry.

The general problem consists of how to calculate the nonequilibrium response of a system induced by an external (space- and time-dependent) perturbation. In response to such an external perturbation, many interest-

ing physical phenomena appear, including the excitation of collective modes and various transport processes.

Kadanoff and Baym formulated a systematic method of calculating the nonequilibrium response of a system evolving under external time-dependent perturbations. They showed that, to lowest order in an external field, one can formulate the method so that it is closely tied to the structure of the usual equilibrium Green's function method (discussed in standard many body theory texts). Indeed, in the limit in which the external time-dependent fields are switched off, the KB nonequilibrium theory reduces to the usual equilibrium theory. Building on the basis of the well-understood equilibrium Green's functions, one first constructs time-ordered single-particle nonequilibrium Green's functions defined for imaginary times. These fictitious nonequilibrium Green's functions satisfy the same boundary conditions as the equilibrium Green's functions and, formally, the same equations of motion for imaginary times. The physical response functions measured experimentally are related to the Green's functions for real times. However, the latter can be obtained from nonequilibrium imaginary-time Green's functions in a well-defined manner and this is the basis of the KB formalism.

In the KB approach, one first calculates the equations of motion for nonequilibrium real-time single-particle Green's functions involving quantum field operators in the presence of the external perturbing fields. The single-particle Green's function is a two-field correlation function and describes the propagation of disturbances in which a single atom is either added to or removed from the many particle system, namely

$$G(1,1') = -i\langle T\hat{\psi}^{\dagger}(1)\hat{\psi}(1')\rangle, \tag{4.1}$$

where we use the shorthand notation $1 \equiv \mathbf{r}_1, t_1$, $1' \equiv \mathbf{r}_{1'}, t_{1'}$ and T is the time ordering operator (to be defined later). Similarly, two-particle Green's functions, which describe disturbances produced by the addition of *two* particles, can be introduced.

An equation of motion for an interacting single-particle Green's function G is given schematically by Dyson's equation

$$G = G_0 + G_0 \Sigma G. \tag{4.2}$$

Here, G_0 is a noninteracting single-particle Green's function and Σ is the single-particle self-energy function that contains all effects associated with the two-particle interaction. The specific approximation we choose for these single-particle self-energies determines all aspects of the resulting microscopic theory. As first shown in great generality by Beliaev (1958a), one can

separate out the condensate degree of freedom $\Phi(\mathbf{r}, t)$ from the noncondensate part of the quantum field operator. In the presence of $\Phi(\mathbf{r}, t)$, it turns out that the single-particle correlations are now most conveniently discussed in terms of two kinds of single-particle Green's functions. The off-diagonal Green's function $\tilde{G}_{12}$ involves new correlations (such as $\langle\tilde{\psi}\tilde{\psi}\rangle$) which arise only in the presence of the condensate order parameter $\Phi(\mathbf{r}, t)$. One can develop a systematic diagrammatic expansion for $\tilde{G}_{\alpha\beta}(1, 1')$ which is formally the same as in a normal system.

Bose broken symmetry plays a crucial role in Bose-condensed fluids because it leads to a *coupling* of the single-particle and density fluctuation excitations; in fact, the poles of single-particle Green's function can be shown to be the the same as the poles of the density correlation function. This equivalence is unique for the Bose-condensed superfluid phase. It explains why one can measure the elementary excitation spectrum given by the poles of single-particle Green's function by measuring the density fluctuation modes that are the poles of the density response function. In normal fluids (non-Bose-condensed, above the superfluid transition temperature) the density response spectrum has no simple relation to the elementary excitation spectrum exhibited by the single-particle Green's function (4.1).

The calculation of the elementary excitation spectrum and/or the density-fluctuation spectrum is, in general, a very complex problem in many body systems. However, in a Bose-condensed fluid one can take advantage of the above-mentioned equivalence of the density fluctuation excitations and the elementary excitations. One can proceed directly by calculating the poles of equilibrium single-particle Green's functions within some given approximation for the self-energies. Alternatively, one can proceed indirectly by calculating the poles of the density response function. Using the above equivalence, the poles of the density response function must correspond to the poles of the single-particle Green's function in some high-order approximation.

As summarized in the classic paper by Hohenberg and Martin (1965), two types of approximation are used in Bose superfluids: gapless approximations and conserving approximations. The advantage of gapless approximations is that the noncondensate single-particle Green's function exhibits the correct spectrum (in a uniform gas, the quasiparticles are phonon-like in the long-wavelength limit, as required in a Bose-condensed phase). In spatially homogeneous systems, a gapless spectrum for long-wavelength excitations is guaranteed if the self-energies satisfy the so-called Hugenholtz–Pines theorem (Hohenberg and Martin, 1965; Griffin, 1993, 1996), namely, if the chemical potential μ is equal to the difference between the diagonal and

off-diagonal elements of the Beliaev self-energy function $\Sigma_{\alpha\beta}$ in the limit of small $\mathbf{q}$ and ω,

$$\mu = \hbar\Sigma_{11}(\mathbf{q}=0,\omega=0) - \hbar\Sigma_{12}(\mathbf{q}=0,\omega=0). \tag{4.3}$$

In contrast, a "conserving approximation" is based on the introduction of a functional Φ from which the self-energy function $\Sigma_{\alpha\beta}$ can be derived by functional differentiation (often also called a Φ-derivable approximation). Using the self-energy, the resulting single-particle Green's function $G_{\alpha\beta}$ can then be used to generate a density response function whose spectrum is guaranteed to satisfy all the usual conservation laws. Using the equivalence of the poles of the single-particle and density fluctuation spectra, the spectrum of the density response function so generated must correspond to a single-particle spectrum given by some improved approximation for the Beliaev self-energies.

In this chapter, we review the Green's functions for Bose-condensed systems in both equilibrium and nonequilibrium situations. We discuss the self-energy functions in the conserving and gapless approximations. These approximations are the building blocks of the KB formalism, setting the stage for Chapters 5–7.

In Chapter 5, we discuss the time-dependent Hartree–Fock–Bogoliubov (HFB) theory as the basis for generating the density response function, which has poles describing excitations with a spectrum given by the Beliaev approximation.

In Chapter 6, we will use the general KB formalism for nonequilibrium Green's functions set up in the present chapter to derive generalized Boltzmann equations. We will derive a kinetic equation for the noncondensate atoms at finite temperatures, including the effect of collisions between the atoms by using simplified versions of the second-order Beliaev self-energy terms. In Chapter 6, we will concentrate on finite temperatures, where the kinetic equation describes noncondensate atoms that can be treated in a self-consistent Hartree–Fock (HF) approximation. As expected, this leads to the kinetic equation with collision integrals C_{12} and C_{22} which was introduced in Chapter 3. We will also derive an equation of motion for the condensate wavefunction, involving a finite-temperature generalization of the well-known Gross–Pitaevskii equation. This generalized GP (GGP) equation includes a dissipative (or growth) term from collisions with the thermal cloud atoms, as well as the Hartree–Fock mean field produced by the noncondensate.

In Chapter 7, we present a generalization of the ZNG results of Chapter 3 that is needed at very low temperatures. Here the single-particle spectrum is

described by the Bogoliubov–Popov approximation instead of the Hartree–Fock particle spectrum valid at higher temperatures. We derive a kinetic equation for the quasiparticle distribution function, with a generalized collision integral C_{12} that now describes scattering between Bogoliubov–Popov quasiparticles and the condensate atoms.

4.2 Nonequilibrium Green's functions in normal systems

The Green's functions that will play a key role in this and the next three chapters are thermodynamic averages of the quantum field operators at two different space–time points, $\hat{\psi}(\mathbf{r},t)$ and $\hat{\psi}^\dagger(\mathbf{r}',t')$. We first review equilibrium Green's functions as an introduction to nonequilibrium Green's functions. In terms of Bose quantum field operators, the many body Hamiltonian $\hat{K}_0 = \hat{H} - \mu\hat{N}$ describing bosons interacting through a two-body potential $v(\mathbf{r}-\mathbf{r}')$ is given by

$$\begin{aligned}\hat{K}_0 = &\int d\mathbf{r}\, \hat{\psi}^\dagger(\mathbf{r}) \left(-\frac{1}{2m}\nabla^2 - \mu\right) \hat{\psi}(\mathbf{r}) \\ &+\tfrac{1}{2}\int d\mathbf{r}d\mathbf{r}'\, \hat{\psi}^\dagger(\mathbf{r})\hat{\psi}^\dagger(\mathbf{r}')v(\mathbf{r}-\mathbf{r}')\hat{\psi}(\mathbf{r})\hat{\psi}(\mathbf{r}'). \end{aligned} \tag{4.4}$$

The equation of motion for the quantum field operator $\hat{\psi}(\mathbf{r},t)$ is given in the Heisenberg representation by (Fetter and Walecka, 1971; Mahan, 1990)

$$i\frac{\partial\hat{\psi}(\mathbf{r},t)}{\partial t} = \left[\hat{\psi}(\mathbf{r},t), \hat{K}_0\right]. \tag{4.5}$$

The solution of (4.5) can be written in the following form

$$\hat{\psi}(\mathbf{r},t) = e^{i\hat{K}_0 t}\hat{\psi}(\mathbf{r})e^{-i\hat{K}_0 t}. \tag{4.6}$$

The simple time dependence of the quantum field operators given by (4.6) is only valid when the grand-canonical Hamiltonian $\hat{K}_0$ is time independent. For simplicity, we set $\hbar = 1$ in this section.

The single-particle equilibrium Green's function is defined as the time-ordered expectation value of the quantum field operators at two different points

$$G(1,1') = -i\langle T(\hat{\psi}(1)\hat{\psi}^\dagger(1'))\rangle, \tag{4.7}$$

while the two-particle Green's function is defined by

$$G_2(12,1'2') = \frac{1}{i^2}\langle T(\hat{\psi}(1)\hat{\psi}(2)\hat{\psi}^\dagger(2')\hat{\psi}^\dagger(1'))\rangle. \tag{4.8}$$

Here, T represents the usual Dyson time-ordering operation, and 1 stands

for $\mathbf{r}_1, t_1$ and $1'$ for $\mathbf{r}_{1'}, t_{1'}$. The time-ordering operator T, when applied to a product of operators, arranges them in chronological order, so that the operator with the earliest time argument acts first (i.e. to the right), namely

$$T\left[\hat{\psi}(1)\hat{\psi}^\dagger(1')\right] \equiv \begin{cases} \hat{\psi}^\dagger(1')\hat{\psi}(1), & \text{if } t_1 < t_{1'} \\ \hat{\psi}(1)\hat{\psi}^\dagger(1'), & \text{if } t_1 > t_{1'}. \end{cases} \tag{4.9}$$

The thermal expectation values in (4.7) and (4.8) are computed using the grand-canonical ensemble

$$\langle \hat{X} \rangle = \frac{\mathrm{Tr}\left(e^{-\beta \hat{K}_0}\hat{X}\right)}{\mathrm{Tr}\left(e^{-\beta \hat{K}_0}\right)}. \tag{4.10}$$

The single-particle Green's function $G(1,1')$ describes the propagation of an atom that is either added to or removed from the many particle system in equilibrium. For example, when $t_1 > t_{1'}$ the creation operator creates an atom at the point $(\mathbf{r}_{1'}, t_{1'})$. This atom propagates to the later time t_1, at which an atom is removed at point $\mathbf{r}_1$ and the system returns to its equilibrium state. Similarly, the two-particle Green's function $G_2(12, 1'2')$ in (4.8) describes, for various time orderings, disturbances produced by the addition of two atoms. For example, when t_1 and $t_{1'}$ are later than t_2 and $t_{2'}$, the two-particle Green's function describes the disturbance produced by the addition of one atom and the removal of another atom, and the subsequent return to equilibrium by the removal of one atom and the addition of another atom. In addition to the one-particle Green's function in (4.7), we define the correlation functions

$$\begin{aligned} G^>(1,1') &\equiv -i\langle \hat{\psi}(1)\hat{\psi}^\dagger(1')\rangle \\ G^<(1,1') &\equiv -i\langle \hat{\psi}^\dagger(1')\hat{\psi}(1)\rangle. \end{aligned} \tag{4.11}$$

We note that $G = G^>$ for $t_1 > t_{1'}$ while $G = G^<$ for $t_1 < t_{1'}$.

The time-evolution operator $e^{-i\hat{K}_0 t}$ in the Heisenberg representation and the weighting factor $e^{-\beta \hat{K}_0}$ that appears in the expression for the grand-canonical average are the same for the imaginary time $t = -i\beta$. This equivalence can be used to derive the Martin–Schwinger boundary condition obeyed by the equilibrium Green's functions,

$$G^<(1,1')|_{t_1=0} = e^{\beta\mu} G^>(1,1')|_{t_1=-i\beta}. \tag{4.12}$$

This boundary condition is crucial in the KB formalism, because it allows one to express a Green's function $G(t-t')$ for imaginary times in the domain $0 < t - t' < i\beta$ as a Fourier series over imaginary frequencies and thus to solve for the Green's function.

One can interpret (4.5) as the equation of motion for the field operators (and hence for the Green's functions as well) for imaginary times τ in the region $0 \leq \tau \equiv it \leq \beta$. We can also generalize the definition of the time-ordering operator T to imaginary times. Later times are defined to be further down the imaginary axis. Therefore in this imaginary-time representation, $G(1,1')$ in (4.7) becomes (see (4.11))

$$G(1,1') = \begin{cases} G^{>}(1,1'), & \text{if } it_1 > it_{1'} \\ G^{<}(1,1'), & \text{if } it_1 < it_{1'}. \end{cases} \tag{4.13}$$

The equation of motion for the quantum field operator $\hat{\psi}(1)$ can be found using (4.5) with $\hat{K}_0$ given by (4.4), namely

$$\left(i\frac{\partial}{\partial t} + \frac{\nabla^2}{2m} + \mu\right)\hat{\psi}(\mathbf{r},t) = \int d\mathbf{r}_1\, v(\mathbf{r}-\mathbf{r}_1)\hat{\psi}^{\dagger}(\mathbf{r}_1,t)\hat{\psi}(\mathbf{r}_1,t)\hat{\psi}(\mathbf{r},t). \tag{4.14}$$

Multiplying (4.14) by $\hat{\psi}^{\dagger}(1')$, applying the T-operator and taking the expectation value of the resulting equation, one obtains

$$\begin{aligned} &\left\langle T\left(i\frac{\partial}{\partial t} + \frac{\nabla^2}{2m} + \mu\right)\hat{\psi}(\mathbf{r},t)\hat{\psi}^{\dagger}(1')\right\rangle \\ &= \int d\mathbf{r}_2\; v(\mathbf{r}_1-\mathbf{r}_2)\langle T[\hat{\psi}(1)\hat{\psi}(2)\hat{\psi}^{\dagger}(2^+)\hat{\psi}^{\dagger}(1')]\rangle|_{t_2=t_1}. \end{aligned} \tag{4.15}$$

The notation 2^+ means that the time argument of $\hat{\psi}^{\dagger}(2)$ is infinitesimally larger than the time arguments of the other $\hat{\psi}$'s. To take the time derivative outside the T-operator, we need to consider the terms containing it carefully. One can show that

$$i\frac{\partial}{\partial t_1}\left\langle T\left(\hat{\psi}(1)\hat{\psi}^{\dagger}(1')\right)\right\rangle - \left\langle T\left(i\frac{\partial}{\partial t_1}\hat{\psi}(1)\hat{\psi}^{\dagger}(1')\right)\right\rangle = \delta(1-1'). \tag{4.16}$$

Using this, we finally obtain the desired equation of motion for $G(1,1')$, namely

$$\begin{aligned} &\left(i\frac{\partial}{\partial t_1} + \frac{\nabla_1^2}{2m} + \mu\right)G(1,1') \\ &\qquad = \delta(1-1') + i\int d\mathbf{r}_2\, v(\mathbf{r}_1-\mathbf{r}_2)G_2(12,1'2^+)|_{t_2=t_1}. \end{aligned} \tag{4.17}$$

Using the same method, one can obtain the adjoint equation of motion containing derivatives with respect to the $1'$ variables:

$$\begin{aligned} &\left(-i\frac{\partial}{\partial t_{1'}} + \frac{\nabla_{1'}^2}{2m} + \mu\right)G(1,1') \\ &\qquad = \delta(1-1') + i\int d\mathbf{r}_2\, v(\mathbf{r}_2-\mathbf{r}_1')G_2(12^-,1'2)|_{t_2=t_{1'}}. \end{aligned} \tag{4.18}$$

We emphasize that (4.17) and (4.18) are for imaginary times and satisfy the boundary condition (4.12).

The single-particle Green's function contains very useful dynamical and statistical mechanical information of the system. For example, it is easily seen that $G^<(\mathbf{r}t, \mathbf{r}t)$ is proportional to the equilibrium expectation value of the density of particles $n(\mathbf{r}, t)$. The two-particle Green's function G_2 defined in (4.8) and appearing in (4.17) and (4.18) describes propagation of two atoms added to the gas. Higher-order Green's functions, defined similarly to G and G_2, describe processes involving more than two atoms.

If we are interested in the effect of disturbances generated by external time-dependent fields, we must introduce the nonequilibrium Green's functions. The system in the presence of an external perturbing field can be described by adding an additional term $\hat{H}'(t)$ to the Hamiltonian (4.4):

$$\hat{K}(t) = \hat{K}_0 + \hat{H}'(t). \tag{4.19}$$

For the external perturbing field, we consider an external time-dependent disturbance $U(\mathbf{r}, t)$ that couples to the local density of atoms $\hat{n}(\mathbf{r}, t)$, so that

$$\hat{H}'(t) \equiv \int d\mathbf{r}\, U(\mathbf{r}, t)\hat{n}(\mathbf{r}, t), \tag{4.20}$$

where the density of atoms is given by

$$\hat{n}(\mathbf{r}, t) = \hat{\psi}^\dagger(\mathbf{r}, t)\hat{\psi}(\mathbf{r}, t). \tag{4.21}$$

The scalar field $U(\mathbf{r}, t)$ has a given time dependence. We assume that this nonequilibrium perturbation vanishes for times $t < t_0$, i.e. that the system is in thermal equilibrium before the time t_0. In the equilibrium case, the time dependence of the operators in the Heisenberg representation is given by (4.6); in the nonequilibrium case, we have to include the time-dependent external potential $\hat{H}'(t)$. Fortunately, one can generalize the well-established equilibrium formalism in such a way that it acts as a basis for the nonequilibrium theory. Using imaginary-time nonequilibrium Green's functions, one can obtain results that can be related to the real-time physical response functions needed to describe nonequilibrium experimental probes.

Physical response functions are defined for real times. However, these are difficult to obtain directly since they do not satisfy a simple boundary condition such as (4.12). Instead of working with real-time Green's functions, the Kadanoff–Baym formalism works with nonequilibrium Green's functions defined on the imaginary time domain ($0 \le \tau = it \le \beta$) that satisfy the same boundary condition, (4.12), at $\tau = 0$ and $\tau = \beta$ as equilibrium Green's functions. Once one obtains the equations of motion for these imaginary-time

Green's functions, one can, by a unique analytic continuation, obtain the equations of motion for the real-time response functions of physical interest. This is the famous Martin–Schwinger approach (Martin and Schwinger, 1959), which is developed in detail in the classic book by Kadanoff and Baym (1962). In this chapter as well as in Chapter 5, we shall deal with these imaginary time Green's functions. In Chapters 6 and 7, we will use these results to calculate nonequilibrium Green's functions for real times; these will be the microscopic basis for the derivation of the KB quantum kinetic equations.

In the nonequilibrium case, the Schrödinger equation for the evolution of a system under the perturbation $\hat{H}'(t)$ (for $t > t_0$) is given by

$$i\frac{\partial|\phi_t\rangle}{\partial t} = \left[\hat{K}_0 + \hat{H}'(t)\right]|\phi_t\rangle. \tag{4.22}$$

In general, it is impossible to solve (4.22). If an external perturbing field $\hat{H}'(t)$ is small, however, one can find its first order effects on the system. We first define the usual interaction representation by writing the time dependence of a state vector as

$$|\phi_t\rangle = e^{-i\hat{K}_0(t-t_0)}|\phi(t)\rangle_I, \tag{4.23}$$

where the subscript I labels the interaction representation. Substituting (4.23) back into (4.22) one obtains

$$i\frac{\partial}{\partial t}|\phi(t)\rangle_I = \hat{H}'_I(t)|\phi(t)\rangle_I. \tag{4.24}$$

Here, the time evolution of the external perturbation is given by

$$\hat{H}'_I(t) = e^{i\hat{K}_0(t-t_0)}\hat{H}'(t)e^{-i\hat{K}_0(t-t_0)}. \tag{4.25}$$

If we integrate both sides of (4.24) from t_0 to t, we find that

$$|\phi(t)\rangle_I = |\phi(t_0)\rangle_I - i\int_{t_0}^{t} dt'\,\hat{H}'(t)|\phi(t')\rangle_I \tag{4.26}$$

To first order in $\hat{H}'$, this gives

$$|\phi(t)\rangle_I = |\phi(t_0)\rangle_I - i\int_{t_0}^{t} dt'\,\hat{H}'_I(t')|\phi(t_0)\rangle_I. \tag{4.27}$$

The second-order correction can be found by substituting the right-hand side of (4.27) into (4.26), and so on by iteration. The general solution of (4.26) can be written as

$$|\phi(t)\rangle_I = T\left(e^{-i\int_{t_0}^{t} dt'\hat{H}'_{\rm I}(t')}\right)|\phi(t_0)\rangle_I \tag{4.28}$$

or

$$|\phi(t)\rangle_I = \hat{v}(t,t_0)|\phi(t_0)\rangle_I, \tag{4.29}$$

where we define the evolution operator

$$\hat{v}(t,t_0) \equiv T\Big[e^{-i\int_{t_0}^{t} dt' \hat{H}'_I(t')}\Big]. \tag{4.30}$$

The time evolution of the expectation value of the operator $\hat{A}$ evolving under the external perturbation $\hat{H}'(t)$ is given by

$$\langle\phi_t|\hat{A}|\phi_t\rangle = \langle\phi_{t_0}|\hat{A}_U(t)|\phi_{t_0}\rangle, \tag{4.31}$$

in the interaction representation, where

$$\hat{A}_U(t) = \hat{\mathcal{U}}^+(t,t_0)\hat{A}\hat{\mathcal{U}}(t,t_0). \tag{4.32}$$

The total evolution operator $\hat{\mathcal{U}}(t,t_0)$ is defined by (see (4.19))

$$\hat{\mathcal{U}}(t,t_0) = T\Big[e^{-i\int_{t_0}^{t} \hat{K}(t')dt'}\Big] = e^{-i\hat{K}_0(t-t_0)}\hat{v}(t,t_0). \tag{4.33}$$

The subscript U on the operator $\hat{A}$ denotes the external scalar potential (see (4.20)). One easily verifies that $\hat{A}_U(t)$ satisfies

$$i\frac{\partial \hat{A}_U(t)}{\partial t} = \Big[\hat{A}_U(t), \hat{K}(t)\Big]. \tag{4.34}$$

Since we are interested in external disturbances of the type given in (4.20), we will label all the quantities developing under $\hat{H}'$ by U. The time evolution of the operator $\hat{A}_U(t)$ can be expressed in terms of the operator $\hat{A}_I$ in the interaction representation

$$\hat{A}_U(t) = \hat{v}^\dagger(t,t_0)\hat{A}_I(t)\hat{v}(t,t_0), \tag{4.35}$$

where, as before

$$\hat{A}_I(t) = e^{i\hat{K}_0(t-t_0)}\hat{A}e^{-i\hat{K}_0(t-t_0)}. \tag{4.36}$$

The evolution operator $\hat{v}(t_0,t_1)$ satisfies the following conditions:

$$\begin{aligned} i\frac{\partial}{\partial t_1}\hat{v}(t_0,t_1) &= \hat{H}'(t_1)\hat{v}(t_0,t_1), \\ \hat{v}(t_0,t_0) &= 1, \\ \lim_{t_0\to\infty}\hat{v}(t_0,t_1) &= \hat{v}(t_1), \\ \lim_{t_0\to-\infty}\hat{v}(t_0,t_0-i\beta) &= 1. \end{aligned} \tag{4.37}$$

The thermal expectation value of an operator $\hat{A}$ evolving in the presence of

external time-dependent perturbation $\hat{H}'(t)$ is given by (using (4.32), (4.33) and (4.35))

$$\begin{aligned}\langle \hat{A}_U(t)\rangle &\equiv Z^{-1}\sum_i e^{-\beta(E_i-\mu N_i)}\langle E_i|\hat{A}_U(t)|E_i\rangle \\ &= Z^{-1}\sum_i e^{-\beta(E_i-\mu N_i)}\langle E_i|\hat{v}^\dagger(t,t_0)\hat{A}_I(t)\hat{v}(t,t_0)|E_i\rangle \\ &= \langle \hat{A}(t)\rangle_U, \end{aligned} \tag{4.38}$$

where an expectation value written without the subscript U denotes the equilibrium expectation value and $|E_i\rangle$ is the initial state at t_0 when the perturbation is turned on. The grand canonical partition function is

$$Z = \sum_i e^{-\beta(E_i-\mu N_i)}. \tag{4.39}$$

A state with energy E_i evolves in time under the effect of external perturbation $\hat{H}'(t)$ as $|\phi_i(t)\rangle = \hat{\mathcal{U}}(t,t_0)|E_i\rangle$. Therefore, all operators develop in time as they would in an equilibrium ensemble.

We introduce a new, fictitious, Green's function in the imaginary-time interval $[t_0, t_0 - i\beta]$ in the following way:

$$G(1,1';U;t_0) \equiv -i\frac{\langle T[\hat{S}\hat{\psi}(1)\hat{\psi}^\dagger(1')]\rangle}{\langle T[\hat{S}]\rangle}, \tag{4.40}$$

where

$$\hat{S} \equiv \exp\left[-i\int_{t_0}^{t_0-i\beta} dt_2\, \hat{H}'(t_2)\right]. \tag{4.41}$$

We note that $T[\hat{S}] = \hat{v}(t_0, t_0 - i\beta)$, where the evolution operator $\hat{v}$ is defined in (4.30). In the interaction representation, the U-dependence is explicit in the $\hat{S}$-operator in (4.41), and the time dependence of the field operators is the same as in the absence of $\hat{H}'(t)$ (see (4.36)). A generalized response function $G(1,1';U;t_0)$ satisfies the boundary condition

$$G(1,1';U;t_0)|_{t_1=t_0} = e^{\beta\mu}G(1,1';U;t_0)|_{t_1=t_0-i\beta}, \tag{4.42}$$

which may be compared with that for an equilibrium Green's function,

$$G(1,1')|_{t_1=0} = e^{\beta\mu}G(1,1')|_{t_1=-i\beta}. \tag{4.43}$$

Physical response functions are defined for real times and are described by the real-time Green's functions

$$g(1,1';U) = -i\langle T[\hat{\psi}_U(1)\hat{\psi}_U^\dagger(1')]\rangle \tag{4.44}$$

where

$$\hat{\mathcal{U}}(t_1) \equiv T \exp\left[-i \int_{-\infty}^{t_1} dt_2 \hat{H}'(t_2)\right]. \tag{4.45}$$

These will be used in Chapters 6 and 7, generalized to deal with a Bose-condensed system.

Having outlined the basic assumptions underlying the Kadanoff–Baym nonequilibrium theory, we now derive the equations of motion for the imaginary-time nonequilibrium Green's functions. As before, we start from the Heisenberg equation of motion (4.5). We first consider the term $T\left[\hat{S}\hat{\psi}(1)\right]$ in (4.40). It can be written in the following form:

$$\begin{aligned} T\left[\hat{S}\hat{\psi}(1)\right] &= T\left[\exp\left(-i\int_{t_0}^{t_0-i\beta} d2\, U(2)\hat{n}(2)\right)\hat{\psi}(1)\right] \\ &= T\left[\exp\left(-i\int_{t_1}^{t_0-i\beta} d2\, U(2)\hat{n}(2)\right)\right]\hat{\psi}(1) \\ &\quad \times T\left[\exp\left(-i\int_{t_0}^{t_1} d2\, U(2)\hat{n}(2)\right)\hat{\psi}(1)\right]. \end{aligned} \tag{4.46}$$

Therefore, the time derivative of (4.46) can be shown to reduce to

$$i\frac{\partial}{\partial t_1} T\left[\hat{S}\hat{\psi}(1)\right] = iT\left[\hat{S}\frac{\partial\hat{\psi}(1)}{\partial t_1}\right] + T\left[\hat{S}\hat{\psi}(1)\right]U(1). \tag{4.47}$$

Hence, we obtain for the equation of motion

$$\begin{aligned} &\left(i\frac{\partial}{\partial t_1} + \frac{\nabla_1^2}{2m} - U(1) + \mu_0\right)G(1,1';U;t_0) \\ &= \delta(1-1') + i\int d\mathbf{r}_2\, v(\mathbf{r}_1-\mathbf{r}_2)G_2(12;1'2^+;U;t_0)|_{t_2=t_1}, \end{aligned} \tag{4.48}$$

where G_2 is defined in (4.8).

Since we want to derive the equation of motion for the nonequilibrium Green's functions for a system disturbed by an external time-dependent potential, we need to find the change in G resulting from an infinitesimal change in the external potential $U(2) \to U(2) + \delta U(2)$. First, the evolution operator $\hat{S}$ changes as follows:

$$\delta\hat{S} = \delta\left[\exp\left(-i\int_{t_0}^{t_0-i\beta} d2\, U(2)\hat{n}(2)\right)\right] = \hat{S}\frac{1}{i}\int_{t_0}^{t_0-i\beta} d2\, \delta U(2)\hat{n}(2). \tag{4.49}$$

Using (4.49), one can show that the change in G resulting from an infinitesimal change in the external potential U is given by

$$\delta G(1,1';U;t_0) = \int_{t_0}^{t_0-i\beta} d2 \Big[G_2(12,1'2^+;U;t_0) - G(1,1';U;t_0)G(2,2^+;U;t_0) \Big] \delta U(2). \quad (4.50)$$

Thus the functional derivative of $G(1,1';U;t_0)$ with respect to $U(2)$ is given by

$$\frac{\delta G(1,1';U;t_0)}{\delta U(2)} = \left[G_2(12,1'2^+;U;t_0) - G(1,1';U;t_0)G(2,2^+;U;t_0) \right]. \quad (4.51)$$

This relation forms the basis for generating the density response function starting from a single-particle Green's function. The generalization of this to a Bose-condensed system is given by (4.87)–(4.90) later in this chapter. Using (4.51), we can rewrite the equation of motion (4.48) in the closed form

$$\left\{ i\frac{\partial}{\partial t_1} + \frac{\nabla_1^2}{2m} - U(1) + \mu_0 - i\int d\mathbf{r}_2\, v(\mathbf{r}_1 - \mathbf{r}_2) \Big[G(\mathbf{r}_2 t_1; \mathbf{r}_2 t_1^+; U; t_0) + \frac{\delta}{\delta U(\mathbf{r}_2, t_1)} \Big] \right\} G(1,1';U;t_0) = \delta(1-1'). \quad (4.52)$$

We define the single-particle self-energy Σ in the presence of the external potential U in the following way:

$$\int d\bar{1} \left[G_0^{-1}(1,\bar{1};U;t_0) - \Sigma(1,\bar{1};U;t_0) \right] G(\bar{1},1';U;t_0) = \delta(1-1'), \quad (4.53)$$

where the inverse matrix $G^{-1}(1,1';U;t_0)$ is defined via

$$\int_{t_0}^{t_0-i\beta} d\bar{1}\, G^{-1}(1,\bar{1};U;t_0) G(\bar{1},1';U;t_0) = \delta(1-1'). \quad (4.54)$$

In the above equations and elsewhere, integration over $d\bar{1}$ means integration over the coordinates $\mathbf{r}_1, t_1$ and, on the r.h.s., $\delta(1-1') \equiv \delta(\mathbf{r}-\mathbf{r}_{1'})\delta(t_1 - t_{1'})$. Using (4.52), (4.53) and (4.54), one can show that the self-energy function in (4.53) can be written in the following way:

$$\Sigma(1,1';U;t_0) = i\int d\bar{1}\, v(1-\bar{1}) G(\bar{1},\bar{1}^+;U;t_0)\delta(1-1') + \int d\bar{1} d\bar{2}\, v(1-\bar{2}) \frac{\delta G(1,\bar{1};U;t_0)}{\delta U(\bar{2})} G^{-1}(\bar{1},1';U;t_0). \quad (4.55)$$

Using $\delta G\, G^{-1} + G\, \delta G^{-1} = 0$, which follows directly from (4.54), one can

rewrite the self-energy function as

$$\Sigma(1,1';U;t_0) = \delta(1-1')i\int d\bar{1}v(1-\bar{1})G(\bar{1},\bar{1}^+;U;t_0) + iv(1-1')G(1,1';U;t_0) + \int d\bar{1}d\bar{2}\,v(1-\bar{2})G(1,\bar{1};U;t_0)\frac{\delta\Sigma(\bar{1},1';U;t_0)}{\delta U(\bar{2})}. \tag{4.56}$$

The Hartree–Fock approximation is obtained by neglecting the $\delta\Sigma/\delta U$ term in (4.56):

$$\Sigma_{\rm HF}(1,1';U;t_0) = \delta(1-1')i\int d\bar{1}\,v(1-\bar{1})G(\bar{1},\bar{1}^+;U;t_0) + iv(1-1')G(1,1';U;t_0). \tag{4.57}$$

The next approximation is obtained by replacing $\delta\Sigma/\delta U$ by $\delta\Sigma_{\rm HF}/\delta U$ and contains the product of three G's:

$$\Sigma(1,1';U;t_0) - \Sigma_{\rm HF}(1,1';U;t_0) = i^2\int d\bar{1}d\bar{2}\,v(1-\bar{1})v(\bar{2}-1')\left[G(1,1';U;t_0)G(\bar{2},\bar{1};U;t_0)G(\bar{1},\bar{2};U;t_0) + G(1,\bar{2};U;t_0)G(\bar{2},\bar{1};U;t_0)G(\bar{1},1';U;t_0)\right]. \tag{4.58}$$

As discussed by Kadanoff and Baym (1962) in great detail, the terms on the r.h.s. of (4.58) give rise to the collision integrals in a Boltzmann equation. Equation (4.56) thus allows one to generate higher-order self-consistent approximations by the simple procedure of iteration of the first-order HF self-energies.

Using the same method, one can obtain the equation of motion for the single-particle Green's function containing derivatives with respect to the other variable, $1'$. Multiplying (4.53) by G_0, one easily finds the equations of motion for the imaginary-time Green's function $G(1,1';U;t_0)$ in the presence of the slowly varying external field U:

$$\left[\,i\frac{\partial}{\partial t_1} + \frac{\nabla_1^2}{2m} - U(1) + \mu_0\right]G(1,1';U;t_0) - \int_{t_0}^{t_0-i\beta} d\bar{1}\,\Sigma(1,\bar{1};U;t_0)G(\bar{1},1';U;t_0) = \delta(1,1') \tag{4.59}$$

and

$$\left[\,-i\frac{\partial}{\partial t_{1'}} + \frac{\nabla_{1'}^2}{2m} - U(1') + \mu_0\right]G(1,1';U;t_0) - \int_{t_0}^{t_0-i\beta} d\bar{1}\,G(1,\bar{1};U;t_0)\Sigma(\bar{1},1';U;t_0) = \delta(1,1'). \tag{4.60}$$

It is convenient to split the self-energy function $\Sigma(1,1';U;t_0)$ in these Dyson equations into two parts:

$$\Sigma(1,1';U;t_0) = \Sigma_{\rm HF}(1,1';U;t_0) + \Sigma_{\rm coll}(1,1';U;t_0), \tag{4.61}$$

where the first-order Hartree–Fock contribution $\Sigma_{\rm HF}$ in (4.57), describes the mean-field effects of an interaction and $\Sigma_{\rm coll}$ denotes the collisional contributions on the r.h.s. of (4.58), which are second and higher order in the interatomic interaction v.

All the results discussed in this section are easily generalized to deal with a Bose-condensed system. The only formal change is that the Green's functions and self-energies will be 2×2 matrices, as explained in Section 4.3 below.

4.3 Green's functions in a Bose-condensed gas

We will separate out the condensate part of the field operator in the usual fashion (Fetter and Walecka, 1971) by setting

$$\hat{\psi}(\mathbf{r}) = \langle\hat{\psi}(\mathbf{r})\rangle + \tilde{\psi}(\mathbf{r}), \tag{4.62}$$

where $\langle\tilde{\psi}(\mathbf{r})\rangle = 0$ and $\langle\hat{\psi}(\mathbf{r})\rangle = \Phi(\mathbf{r},t)$ is the Bose macroscopic wavefunction. The noncondensate (or thermal atom component) field operators $\tilde{\psi}(\mathbf{r})$ and $\tilde{\psi}^\dagger(\mathbf{r})$ satisfy the usual Bose commutation relations.

In a Bose-condensed fluid, the finite value of $\Phi(\mathbf{r},t)$ leads to the appearance of the off-diagonal (or anomalous) propagators $\langle\tilde{\psi}(1)\tilde{\psi}(1')\rangle$ and $\langle\tilde{\psi}^\dagger(1)\tilde{\psi}^\dagger(1')\rangle$. These must be dealt with on an equal basis with the diagonal (or normal) propagators $\langle\tilde{\psi}(1)\tilde{\psi}^\dagger(1')\rangle$. Many body theory is most elegantly formulated in terms of a 2×2 matrix Beliaev single-particle Green's function defined by

$$\hat{G}(1,1';U) = -i\langle T\hat{\Psi}(1)\hat{\Psi}^\dagger(1')\rangle = -i\begin{pmatrix} \langle T\hat{\psi}(1)\hat{\psi}^\dagger(1')\rangle & \langle T\hat{\psi}(1)\hat{\psi}(1')\rangle \\ \langle T\hat{\psi}^\dagger(1)\hat{\psi}^\dagger(1')\rangle & \langle T\hat{\psi}^\dagger(1)\hat{\psi}(1')\rangle \end{pmatrix}, \tag{4.63}$$

where the spinor quantum field operators are defined as

$$\hat{\Psi}(1) = \begin{pmatrix} \hat{\psi}(1) \\ \hat{\psi}^\dagger(1) \end{pmatrix}, \qquad \hat{\Psi}^\dagger(1) = \begin{pmatrix} \hat{\psi}^\dagger(1) & \hat{\psi}(1) \end{pmatrix}. \tag{4.64}$$

Here, as before, T represents the time-ordering operation and we use the usual KB abbreviated notation, $1 \equiv (\mathbf{r},t)$ and $1' \equiv (\mathbf{r}',t')$. Using (4.62), the Beliaev matrix propagator in (4.63) splits into two parts

$$\hat{G}(1,1';U) = \hat{\tilde{G}}(1,1';U) + \hat{h}(1,1';U). \tag{4.65}$$

Here $\hat{\tilde{G}}$ is identical to (4.63) except that it involves the noncondensate part of the field operators,

$$\hat{\tilde{G}}(1,1';U) \equiv -i \begin{pmatrix} \langle T\tilde{\psi}(1)\tilde{\psi}^\dagger(1')\rangle & \langle T\tilde{\psi}(1)\tilde{\psi}(1')\rangle \\ \langle T\tilde{\psi}^\dagger(1)\tilde{\psi}^\dagger(1')\rangle & \langle T\tilde{\psi}^\dagger(1)\tilde{\psi}(1')\rangle \end{pmatrix}. \tag{4.66}$$

The condensate part is given by

$$\hat{h}(1,1';U) \equiv -i \begin{pmatrix} \Phi(1)\Phi^*(1') & \Phi(1)\Phi(1') \\ \Phi^*(1)\Phi^*(1') & \Phi^*(1)\Phi(1') \end{pmatrix}, \tag{4.67}$$

with $\langle\hat{\psi}^\dagger(\mathbf{r})\rangle_t \equiv \Phi^*(\mathbf{r},t)$. One can write (4.65) in the form

$$G_{\alpha\beta}(1,1') = \tilde{G}_{\alpha\beta}(1,1') + G_{1/2\alpha}(1)G^\dagger_{1/2\beta}(1') \ , \tag{4.68}$$

where the condensate Green's function is described by

$$\begin{aligned} \hat{G}_{1/2}(1) &\equiv \sqrt{-i}\langle\hat{\Psi}(1)\rangle = \sqrt{-i}\begin{pmatrix} \Phi(1) \\ \Phi^*(1) \end{pmatrix} \\ \hat{G}^\dagger_{1/2}(1) &\equiv \sqrt{-i}\langle\hat{\Psi}^\dagger(1)\rangle = \sqrt{-i}\begin{pmatrix} \Phi^*(1) & \Phi(1) \end{pmatrix} \ . \end{aligned} \tag{4.69}$$

We see that the dynamics of a Bose-condensed system can formally be separated into two parts. The dynamics of atoms in the condensate are described by a macroscopic wavefunction Φ and the dynamics of atoms that are not Bose-condensed are described by the 2×2 matrix propagator $\hat{\tilde{G}}$. However, the equations of motion describing these two components are coupled and one has to solve them self-consistently. The description of the dynamics of noncondensate atoms is formally identical to the way in which one describes normal Bose gases. The main difference is that now one also has nonzero anomalous averages and these have to be included.

A very useful and elegant way of generating the equations of motion for both $\hat{\tilde{G}}$ and Φ is to use functional derivatives with respect to weak external fields (Cheung and Griffin, 1971),

$$\hat{H}'(t_1) = \frac{1}{2}\int d\mathbf{r}_1 d2\, \hat{\psi}^\dagger(1)U(1,2)\hat{\psi}(2) + \int d\mathbf{r}_1 \left[\hat{\psi}^\dagger(1)\eta_{\rm ext}(1) + \hat{\psi}(1)\eta^*_{\rm ext}(1)\right]. \tag{4.70}$$

Here, we have generalized the external perturbation given in (4.20) to allow the thermal expectation values of the quantum field operators to be nonzero; $U(1,2)$ is an external generating scalar field that is nonlocal in space and time. It represents a perturbation in which a particle is removed from the system at point 1 and added at point 2. The symmetry-breaking fields in

(4.70), $\eta_{\rm ext}$ and $\eta^*_{\rm ext}$, describe particle creation and destruction. All higher-order Green's functions can also be neatly expressed as functional derivatives of single-particle Green's functions with respect to such generating fields.

The equation for the condensate can be written in terms of the two-component Green's function $\hat{G}_{1/2}(1)$ defined in (4.69). One finds (Cheung and Griffin, 1971; Imamović-Tomasović and Griffin, 2000)

$$\int d\bar{1}\, \hat{G}_0^{-1}(1,\bar{1})\hat{G}_{1/2}(\bar{1}) = \sqrt{-i}\hat{\eta}(1) + \sqrt{-i}\hat{\eta}_{\rm ext}(1), \tag{4.71}$$

where the condensate source function $\hat{\eta}$ is defined in terms of the three-field correlation function

$$\sqrt{-i}\hat{\eta}(1) \equiv \frac{1}{2}\int d\bar{2}\sqrt{-i}v(1-\bar{2})\langle T\hat{\Psi}(1)\hat{\Psi}^\dagger(\bar{2})\hat{\Psi}(\bar{2})\rangle. \tag{4.72}$$

The inverse of the noninteracting 2×2 matrix Bose gas propagator in (4.71) is defined by

$$\hat{G}_0^{-1}(1,1') = \left[i\hbar\tau_3\frac{\partial}{\partial t_1} + \frac{\hbar^2\nabla_1^2}{2m} - V_{\rm trap}(\mathbf{r}_1) + \mu_0\right]\hat{\rm I}\ \ \delta(1,1'), \tag{4.73}$$

where τ_3 is the Pauli spin matrix

$$\tau_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

and $\hat{\rm I}$ is the 2×2 identity matrix. The external particle-source fields defined in (4.70) are now described in (4.71) by the spinor

$$\hat{\eta}_{\rm ext}(1) \equiv \begin{pmatrix} \eta_{\rm ext}(1) \\ \eta^*_{\rm ext}(1) \end{pmatrix}. \tag{4.74}$$

Approximations for $\hat{G}_{1/2}(1)$, which give the macroscopic wavefunction $\Phi(1)$, are determined by the choice made for the source function $\hat{\eta}$ in (4.72).

Using (4.62), one can also decompose the three-field correlation function $\hat{\eta}$ defined in (4.72). For example, one has

$$\begin{aligned}\langle T\psi(1)\psi^\dagger(2)\psi(2)\rangle = {}& \Phi(1)n_c(2) + \Phi(2)\langle T\tilde{\psi}(1)\tilde{\psi}^\dagger(2)\rangle \\ & + \Phi^*(2)\langle T\tilde{\psi}(1)\tilde{\psi}(2)\rangle + \langle T\tilde{\psi}(1)\tilde{\psi}^\dagger(2)\tilde{\psi}(2)\rangle.\end{aligned} \tag{4.75}$$

In the first-order Hartree–Fock–Bogoliubov (HFB) approximation, one neglects the three-field correlation function $\langle T\tilde{\psi}\tilde{\psi}^\dagger\tilde{\psi}\rangle$ for the noncondensate atoms in (4.75). In this approximation, three-field correlation functions

such as in (4.75) involve only the condensate density $n_c = |\Phi|^2$ and the noncondensate correlation functions $\langle\tilde{\psi}^\dagger\tilde{\psi}\rangle$ and $\langle\tilde{\psi}\tilde{\psi}\rangle$.

The anomalous Bose source function in (4.71) and (4.72) is usefully rewritten in terms of a 2×2 condensate self-energy function $\hat{S}$ defined by the relation

$$\int d\bar{1}\hat{S}(1,\bar{1})\hat{h}(\bar{1},1') \equiv \sqrt{-i}\hat{\eta}(1)\hat{G}^\dagger_{1/2}(1'), \tag{4.76}$$

where the 2×2 matrix condensate propagator $\hat{h}$ is given in (4.67). In terms of this $\hat{S}$ function, the equation of motion (4.71) is transformed to

$$\begin{aligned}&\int d\bar{1}\,\left[\hat{G}_0^{-1}(1,\bar{1}) - \hat{S}_{\rm HF}(1,\bar{1})\right]\hat{h}(\bar{1},1')\\ &= \int_{-\infty}^{t_1} d\bar{1}(\hat{S}^{>}_{\rm coll}(1,\bar{1}) - \hat{S}^{<}_{\rm coll}(1,\bar{1}))\hat{h}(\bar{1},1').\end{aligned} \tag{4.77}$$

As with the noncondensate self-energies in (4.61), we have here separated the Hartree–Fock contributions $\hat{S}_{\rm HF}$ from the "collisional" or second-order condensate self-energy $\hat{S}_{\rm coll}$.

The general equations of motion for $\hat{\tilde{G}}(1,1')$ and $\hat{G}_{1/2}(1)$ were discussed by Hohenberg and Martin (1965) at $T = 0$ and by Cheung and Griffin (1971) at finite temperatures. If we include only a *static* external potential $V_{\rm trap}(\mathbf{r})$, then the equation of motion for $\hat{\tilde{G}}(1,1')$ is

$$\begin{aligned}&\left[i\hbar\tau_{3,\alpha\beta}\frac{\partial}{\partial t} - \hat{T}(\mathbf{r})\delta_{\alpha\beta}\right]\tilde{G}_{\beta\alpha'}(1,1')\\ &\quad= \delta(1-1')\delta_{\alpha\alpha'} + \int d\bar{2}\,\Sigma_{\alpha\beta}(1,\bar{2})\tilde{G}_{\beta\alpha'}(\bar{2},1'),\end{aligned} \tag{4.78}$$

where the Pauli spin matrix τ_3 is given below (4.73) and $\hat{T}(\mathbf{r})$ is defined in (2.49). Here, repeated Greek indices in a product are summed and a bar represents the usual integration over the relevant space and time variables. The Dyson–Beliaev equation (4.78) for the single-particle Green's functions in a Bose-condensed system depend on the 2×2 matrix self-energy $\Sigma_{\alpha\beta}(1,1')$. The analogous equation of motion for the two components of $\hat{G}_{1/2}$ is given by (for a static external field)

$$-\hat{T}(\mathbf{r})G_{1/2\,\alpha}(\mathbf{r}) = \sqrt{-i}\eta_\alpha(\mathbf{r}), \tag{4.79}$$

where the condensate source function $\eta_\alpha(\mathbf{r})$ in (4.72) is itself is a functional of $\hat{G}_{1/2}$ and $\hat{\tilde{G}}$. Writing (4.79) more explicitly, we can see that this equation corresponds to

$$\left(-\frac{\hbar^2\nabla^2}{2m} + V_{\rm trap}(\mathbf{r}) - \mu\right)\Phi(\mathbf{r}) = -\eta_1(\mathbf{r})\ . \tag{4.80}$$

One may view (4.80) as an "exact" GP equation. Various approximations correspond to different choices for the three-field correlation function in (4.72).

The static HFB approximation is time-independent and is described by the first-order results

$$\hat{\Sigma}_{\rm HFB}(1,1') = g \begin{pmatrix} 2n(\mathbf{r}) & m(\mathbf{r}) \\ m^*(\mathbf{r}) & 2n(\mathbf{r}) \end{pmatrix} \delta(1-1') \tag{4.81}$$

and

$$\hat{\eta}_{\rm HFB}(1) = g \begin{pmatrix} n(\mathbf{r})+\tilde{n}(\mathbf{r}) & \tilde{m}(\mathbf{r}) \\ \tilde{m}^*(\mathbf{r}) & n(\mathbf{r})+\tilde{n}(\mathbf{r}) \end{pmatrix} \begin{pmatrix} \Phi(\mathbf{r}) \\ \Phi^*(\mathbf{r}) \end{pmatrix} . \tag{4.82}$$

In the above equations, we have introduced various time-independent local densities (see also (3.7)),

$$\begin{aligned} n(\mathbf{r}) &\equiv n_{\rm c}(\mathbf{r}) + \tilde{n}(\mathbf{r}) = |\Phi(\mathbf{r})|^2 + \langle \tilde{\psi}^\dagger(\mathbf{r})\tilde{\psi}(\mathbf{r})\rangle, \\ m(\mathbf{r}) &\equiv [\Phi(\mathbf{r})]^2 + \tilde{m}(\mathbf{r}) = [\Phi(\mathbf{r})]^2 + \langle \tilde{\psi}(\mathbf{r})\tilde{\psi}(\mathbf{r})\rangle, \\ m^*(\mathbf{r}) &\equiv [\Phi^*(\mathbf{r})]^2 + \tilde{m}^*(\mathbf{r}) = [\Phi^*(\mathbf{r})]^2 + \langle \tilde{\psi}^\dagger(\mathbf{r})\tilde{\psi}^\dagger(\mathbf{r})\rangle. \end{aligned} \tag{4.83}$$

The essential physics behind (4.81) and (4.82) has already been discussed in connection with (3.5) and (3.6). The point here is that the static HFB approximation (4.81) and (4.82) emerges as the simplest within a Green's function formalism for $G_{1/2\,\alpha}$ and $\tilde{G}_{\alpha\beta}$. This approximation is often called the Girardeau–Arnowitt approximation in the older Bose gas literature (Hohenberg and Martin, 1965).

Solving (4.78) using (4.81) and (4.82), we arrive at the coupled Bogoliubov equations for $\tilde{G}_{11}$ and $\tilde{G}_{21}$,

$$\begin{aligned} \left[i\hbar\frac{\partial}{\partial t} - \hat{T}(\mathbf{r}) - 2gn(\mathbf{r})\right]\tilde{G}_{11}(1,1') - gm(\mathbf{r})\tilde{G}_{21}(1,1') &= \delta(1-1') \\ \left[-i\hbar\frac{\partial}{\partial t} - \hat{T}(\mathbf{r}) - 2gn(\mathbf{r})\right]\tilde{G}_{21}(1,1') - gm^*(\mathbf{r})\tilde{G}_{11}(1,1') &= 0 \ . \end{aligned} \tag{4.84}$$

These equations define the static HFB approximation for the single-particle 2×2 matrix Green's function $\tilde{G}_{\alpha\beta}$. The values of $n(\mathbf{r})$ and $m(\mathbf{r})$ are determined self-consistently by $\tilde{G}_{11}$, $\tilde{G}_{21}$ and $\Phi(\mathbf{r})$, i.e. $\tilde{n} = i\tilde{G}_{11}(1^+,1) = i\tilde{G}^<_{11}(1,1)$, etc. The excitation spectrum within the approximation (4.84) will be discussed further in Section 5.1. A similar set of equations based on the simpler Bogoliubov approximation is given in equations (55.22) and (55.24) of Fetter and Walecka (1971). This approximation corresponds to setting $\tilde{n}(\mathbf{r}) = 0$ and $\tilde{m}(\mathbf{r}) = 0$ in (4.81) and (4.82). The dynamic HFB approximation will be discussed in Chapters 5 and 6.

As discussed in Chapter 7 of Pitaevskii and Stringari (2003) and in Chapter 2 of Griffin (1993), the dynamic structure factor $S(\mathbf{q},\omega)$ is a fundamental correlation function in all many body systems. It is important as a measure of the density fluctuations but also because scattering experiments can measure $S(\mathbf{q},\omega)$ directly. In a uniform system, we normalize $S(\mathbf{q},\omega)$ so that

$$S(\mathbf{q},\omega) = -\frac{1}{\pi n}[f^0(\omega)+1]\,\mathrm{Im}\,L_{11}(\mathbf{q},\omega), \tag{4.85}$$

where $L_{11}(\mathbf{q},\omega)$ is the space–time Fourier transform with respect to the difference between 1 and 2 of the density response function

$$L_{11}(1,2) = -i[\langle T\hat{n}(1)\hat{n}(2)\rangle - \langle\hat{n}(1)\rangle\langle\hat{n}(2)\rangle]. \tag{4.86}$$

Response functions can be expressed in terms of a functional derivative of the single-particle Green's function with respect to an external field such as that given by (4.20), which couples into the local density (4.21). Recalling the Beliaev 2×2 matrix Green's functions as defined in (4.66) and (4.67), the density response function for a Bose superfluid naturally splits into two contributions:

$$L_{11} = \tilde{L}_{11} + L_{\mathrm{S}}. \tag{4.87}$$

Here the noncondensate part is given by

$$\tilde{L}_{11}(1,2) = i\frac{\delta\tilde{G}_{11}(1,1)}{\delta U(2)}\bigg|_{U=0} \tag{4.88}$$

and the condensate (superfluid) part is

$$L_{\mathrm{S}}(1,2) \equiv \sqrt{n_{\mathrm{c0}}}\,\frac{\delta\langle\hat{\psi}(1;U)\rangle}{\delta U(2)}\bigg|_{U=0} + \sqrt{n_{\mathrm{c0}}}\,\frac{\delta\langle\hat{\psi}^\dagger(1;U)\rangle}{\delta U(2)}\bigg|_{U=0}. \tag{4.89}$$

The approach of Hohenberg and Martin (1965) was to calculate correlation functions such as $\tilde{L}_{11}(1,2)$ in (4.88) by taking the functional derivative of $\tilde{G}_{\alpha\beta}$ with respect to the external driving field (4.20) (see Eq. (2.14) of Cheung and Griffin, 1971):

$$\begin{aligned}\tilde{L}_{\alpha\beta}(1,2) &\equiv \frac{i\delta\tilde{G}_{\alpha\beta}(1,1)}{\delta U(2)}\bigg|_{U=0} \\ &= i\tilde{G}_{\alpha\sigma}(1,2)\tilde{G}_{\sigma\beta}(2,1) + i\tilde{G}_{\alpha\sigma}(1,\bar{3})\frac{\delta\Sigma_{\sigma\gamma}(\bar{3},\bar{4})}{\delta U(2)}\bigg|_{U=0}\tilde{G}_{\gamma\beta}(\bar{4},2),\end{aligned} \tag{4.90}$$

where we recall that repeated Greek symbols are summed over. The barred space–time coordinates involve a spatial and an imaginary-time integration,

as in (4.54), but this is left implicit. Equation (4.90) shows how correlation functions $\tilde{L}_{\alpha\beta}(1,2)$ involving the noncondensate atoms are completely determined once we choose some specific approximation for the single-particle matrix self-energy associated with $\tilde{G}_{\alpha\beta}$. The functional derivatives in (4.89) and (4.90) will play a crucial role in our discussion of response functions in Chapter 5.

4.4 Classification of self-energy approximations

In this section, we review two types of approximation that are used to derive the Beliaev matrix self-energies. In the so-called "Φ-derivable" (or conserving) approximation, one assumes that it is possible to find a functional Φ such that one can generate the self-energy function Σ and the condensate source function η by functional differentiation of Φ with respect to the single-particle propagators $\tilde{G}$ and $G_{1/2}$. If such a functional Φ exists, one can prove that the density response function quantities generated by $G_{\alpha\beta}$ will satisfy all conservation laws. The problem with a conserving approximation is that there is no guarantee that the excitation spectrum will be gapless in the low-momentum limit (in a uniform system). In contrast, a gapless approximation for $\Sigma_{\alpha\beta}$ is constructed in such a way that one is guaranteed to have phonons in the long-wavelength (low-momentum) limit even if $G_{\alpha\beta}$ does not lead to a conserving approximation for the density response function generated using (4.89) and (4.90). We note that if an approximation has problems in the case of uniform gases, then a trapped gas will suffer an analogous deficiency.

From now on we will drop the carets on all quantities and leave implicit that G and $\tilde{G}$ are 2×2 matrices and $G_{1/2}$ and η are two-component spinors.

4.4.1 Conserving or "Φ-derivable" approximations

In this section, for clarity, we will use a general interatomic potential v, rather than the standard s-wave pseudopotential approximation. To derive the self-energy and the source function in the so-called "Φ-derivable" approximation, we use the following expressions (Baym and Kadanoff, 1961; Baym, 1962; Kadanoff and Martin, 1963)

$$\begin{aligned} \frac{\delta}{\delta\tilde{G}(11')}\Phi\left[\tilde{G}, G_{1/2}\right]\Big|_{G_{1/2}} &\equiv \Sigma(11'), \\ \frac{1}{2\sqrt{-i}}\frac{\delta}{\delta G_{1/2}(1)}\Phi\left[\tilde{G}, G_{1/2}\right]\Big|_{\tilde{G}} &\equiv \eta(1), \end{aligned} \tag{4.91}$$

where Φ is a functional of the Green's functions $\tilde{G}$ and $G_{1/2}$. As explained above, if such a functional Φ exists, then the resulting $G_{\alpha\beta}$ can be used to generate a density response function that will satisfy conservation laws. Moreover, in general the long-wavelength excitation spectrum of $\tilde{G}_{\alpha\beta}$ generated by (4.91) will have an energy gap at low $\mathbf{q}$.

In the "Φ-derivable" approximation, $\tilde{G}$ and $G_{1/2}$ are independent functions and the relation between the spinors $G_{1/2}$ and $G^\dagger_{1/2}$ is given by the following identities:

$$\begin{aligned} G^\dagger_{1/2\,\alpha}(1) &= \tau_{1,\alpha\beta} G_{1/2\,\beta}(1) \\ \tau_{1,\alpha\beta} G^\dagger_{1/2\,\beta}(1) &= G_{1/2\,\alpha}(1), \end{aligned} \tag{4.92}$$

where τ_1 is the usual Pauli spin matrix,

$$\tau_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Therefore, one obtains

$$\frac{\delta G_{1/2\,\alpha}(1)}{\delta G^\dagger_{1/2\,\sigma}(1)} = \tau_{1,\alpha\beta} \frac{\delta G^\dagger_{1/2\,\beta}(1)}{\delta G^\dagger_{1/2\,\sigma}(1)} = \delta_{\beta\sigma}\tau_{1,\alpha\beta} = \tau_{1,\alpha\sigma}. \tag{4.93}$$

The Hartree–Fock–Bogoliubov (HFB) approximation discussed earlier is generated by[1]

$$\begin{aligned} \Phi_{\rm HFB} = {} & \tfrac{1}{4} i\tilde{G}(11) v(1\bar{1}) \tilde{G}(\bar{2}\bar{2}) + \tfrac{1}{2} i G_{1/2}(1) {G_{1/2}}^\dagger(1) v(1\bar{1}) \tilde{G}(2\bar{2}) \\ & + \tfrac{1}{4} i G_{1/2}(1) {G_{1/2}}^\dagger(1) v(1\bar{1}) G_{1/2}(\bar{2}) G^\dagger_{1/2}(\bar{2}) + \tfrac{1}{2} i\tilde{G}(1\bar{1}) v(1\bar{1}) \tilde{G}(\bar{2}1) \\ & + i G_{1/2}(1) {G_{1/2}}^\dagger(\bar{2}) v(1\bar{1}) \tilde{G}(\bar{2}1), \end{aligned} \tag{4.94}$$

where we recall that repeated 2×2 matrix component labels are summed over. Using (4.91), the HFB approximation (4.94) generates the following results

$$\begin{aligned} \Sigma_{\rm HFB}(11') = {} & \tfrac{1}{2} i v(1\bar{1}) \left[G_{1/2}(\bar{2}) G^\dagger_{1/2}(\bar{2}) + \tilde{G}(\bar{2}\bar{2}) \right] \delta(11') \\ & + i v(11') \left[G_{1/2}(1) G^\dagger_{1/2}(1') + \tilde{G}(11') \right] \end{aligned} \tag{4.95}$$

and

$$\begin{aligned} \sqrt{-i}\eta_{\rm HFB}(1) = {} & \tfrac{1}{2} i v(1\bar{1}) \left[G_{1/2}(1) G_{1/2}(\bar{2}) G^\dagger_{1/2}(\bar{2}) G_{1/2}(1) + \tilde{G}(\bar{2}\bar{2}) \right] \\ & + i v(1\bar{2}) G_{1/2}(\bar{2}) \tilde{G}(1\bar{2}). \end{aligned} \tag{4.96}$$

[1] In this section, to save space we write $\tilde{G}(1,2)$ as $\tilde{G}(12)$ and $v(1-2)$ as $v(12)$ and integrate over all barred variables.

Feynman diagrams representing the first-order HFB approximation appear in Fig. 4.1. The smooth solid lines represent the noncondensate propagator $\tilde{G}$, the wiggly lines represent the condensate propagator h and the broken lines represent the interatomic interaction v. We can rewrite (4.96) in terms of the HFB condensate self-energy S defined in (4.77), obtaining

$$S_{\rm HFB}(11') = \tfrac{1}{2} iv(1\bar{2}) \left[G_{1/2}(\bar{2}) G^{\dagger}_{1/2}(\bar{2}) + \tilde{G}(\bar{2}\bar{2}) \right] \delta(11') + iv(11')\tilde{G}(11'). \tag{4.97}$$

These condensate self-energy diagrams are shown in Fig. 4.2.

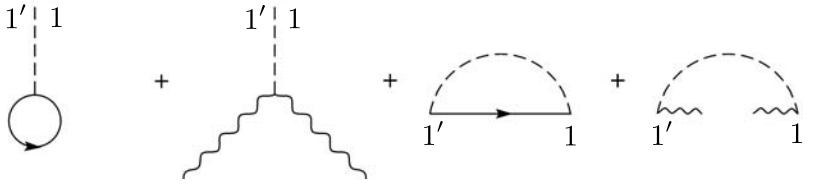


Fig. 4.1. Self-energy diagrams $\Sigma_{\rm HFB}$ in the first-order Hartree–Fock–Bogoliubov approximation (4.95).

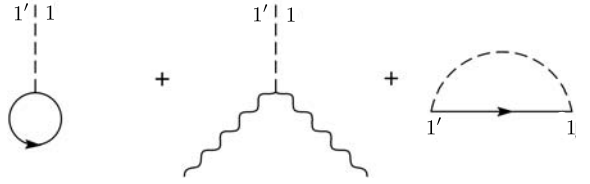


Fig. 4.2. Condensate self-energy $S_{\rm HFB}$ diagrams in the first-order expression (4.97).

To obtain the second-order terms, we generalize the first-order expression for the functional in (4.94) and introduce the second-order terms into Φ:

$$\begin{aligned}
\Phi = {} & \tfrac{1}{4} i\tilde{G}(11)v(1\bar{2})\tilde{G}(\bar{2}\bar{2}) + \tfrac{1}{2} iG_{1/2}(1)G^{\dagger}_{1/2}(1)v(1\bar{2})\tilde{G}(\bar{2}\bar{2}) \\
& + \tfrac{1}{4} iG_{1/2}(1)G^{\dagger}_{1/2}(1)v(1\bar{2})G_{1/2}(\bar{2})G^{\dagger}_{1/2}(\bar{2}) + \tfrac{1}{2} i\tilde{G}(1\bar{2})v(1\bar{2})\tilde{G}(\bar{2}1) \\
& + iG_{1/2}(1){G_{1/2}}^{\dagger}(\bar{2})v(1\bar{2})\tilde{G}(\bar{2}1) \\
& - \tfrac{1}{4}\tilde{G}(\bar{2}1)\tilde{G}(1\bar{2})v(\bar{2}\bar{4})v(\bar{3}1)\tilde{G}(\bar{4}\bar{3})\tilde{G}(\bar{3}\bar{4}) \\
& - \tfrac{1}{4}\tilde{G}(\bar{2}1)\tilde{G}(1\bar{2})v(\bar{2}\bar{4})v(\bar{3}1)\tilde{G}(\bar{4}\bar{3})G_{1/2}(\bar{3})G^{\dagger}_{1/2}(\bar{4}).
\end{aligned} \tag{4.98}$$

Using the defining relations in (4.91), the first-order contributions to the self-energy and the source function are the same as in the HFB approximation given in (4.96) and (4.97). To second order in v, (4.98) gives the *additional*

contributions

$$\Sigma_{\rm coll}(11') = -\tfrac{1}{2}v(\bar{3}1)v(21')\tilde{G}(11')[\tilde{G}(\bar{2}\bar{3})\tilde{G}(\bar{3}\bar{2}) + \tilde{G}(\bar{2}\bar{3})h(\bar{3}\bar{2}) + h(\bar{2}\bar{3})\tilde{G}(\bar{3}\bar{2})] - \tfrac{1}{2}v(\bar{3}1)v(\bar{2}1')h(11')\tilde{G}(\bar{2}\bar{3})\tilde{G}(\bar{3}\bar{2}), \quad (4.99)$$

and

$$\sqrt{-i}\eta_{\rm coll}(1) = -\tfrac{1}{2}v(\bar{3}1)v(\bar{2}\bar{4})\tilde{G}(\bar{3}\bar{2})\tilde{G}(\bar{2}\bar{3})\tilde{G}(1\bar{4})G_{1/2}(\bar{4}). \quad (4.100)$$

In terms of the condensate self-energy function S, (4.100) is equivalent to

$$S_{\rm coll}(1,1') = -\tfrac{1}{2}v(1\bar{3})v(\bar{2}1')\tilde{G}(11')\tilde{G}(\bar{2}\bar{3})\tilde{G}(\bar{3}\bar{2}). \quad (4.101)$$

Kane and Kadanoff (1965) used the conserving approximation based on the self-energies in (4.97), (4.99) and (4.101). However, as they emphasized, this approximation does not satisfy the Hugenholtz–Pines theorem, and this means that the single-particle Green's functions $\tilde{G}_{\alpha\beta}$ do not have the required phonon-like spectrum at long wavelengths.

4.4.2 Gapless approximations

The self-energy in a "gapless" approximation is *defined* by (Hohenberg and Martin, 1965)

$$\sqrt{-i}\frac{\delta\eta(1)}{\delta G_{1/2}(1')} \equiv \Sigma(11'). \quad (4.102)$$

Thus, to derive the second-order Beliaev approximation for the 2×2 matrix self-energy $\Sigma_{\alpha\beta}$, we first need to find a second-order expression for the source function η to be used in (4.102). The source function defined in (4.72) can be written in terms of functional derivatives (Martin, 1963):

$$\sqrt{-i}\eta(1) = iv(1\bar{2})\left\{\tfrac{1}{2}\left[\tilde{G}(\bar{2}\bar{2}) + G_{1/2}(\bar{2})G^{\dagger}_{1/2}(\bar{2})\right]G_{1/2}(1) + i\frac{\delta G_{1/2}(1)}{\delta U(\bar{2}\bar{2})}\right\}. \quad (4.103)$$

We remind the reader that the matrix nature of $\tilde{G}$ and $G_{1/2}$ is left implicit. Equation (4.103) can also be written as:

$$\sqrt{-i}\eta(1) = iv(1\bar{2})\left\{\tfrac{1}{2}\left[\tilde{G}(\bar{2}\bar{2}) + G_{1/2}(\bar{2})G^{\dagger}_{1/2}(\bar{2})\right]G_{1/2}(1) + \tilde{G}(1\bar{2})G_{1/2}(\bar{2})\right\} + \tfrac{1}{2}iv(12)\tilde{G}(1\bar{4})\tilde{G}(\bar{2}\bar{3})\frac{\delta\Sigma(\bar{3}\bar{5})}{\delta G^{\dagger}_{1/2}(4)}\tilde{G}(\bar{5}\bar{2}). \quad (4.104)$$

We will approximate the self-energy Σ in (4.104) by the full Hartree–Fock–Bogoliubov (HFB) first-order self-energy (4.95). Using this self-energy in the

last term in (4.104), we obtain the desired second-order expression for the anomalous source function:

$$\begin{aligned}\sqrt{-i}\eta_{\rm coll}(1) = {}& \tfrac{1}{2}iv(12)\left[G_{1/2}(1)G_{1/2}(\bar{2})G^{\dagger}_{1/2}(\bar{2}) + G_{1/2}(1)\tilde{G}(\bar{2}\bar{2})\right] \\ & + iv(12)G_{1/2}(\bar{2})\tilde{G}(1\bar{2}) \\ & - \tfrac{1}{2}v(1\bar{3})v(2\bar{4})\tilde{G}(\bar{3}\bar{2})\tilde{G}(\bar{2}\bar{3})\tilde{G}(1\bar{4})G_{1/2}(\bar{4}) \\ & - v(13)v(\bar{2}\bar{4})\tilde{G}(\bar{1}\bar{4})\tilde{G}(\bar{4}\bar{3})\tilde{G}(\bar{3}\bar{2})G_{1/2}(\bar{2}). \end{aligned} \tag{4.105}$$

This is equivalent to the second-order Beliaev condensate self-energy

$$\begin{aligned} S_{\rm coll}(1,1') = {}& -\tfrac{1}{2}v(1\bar{3})v(\bar{2}1')\tilde{G}(11')\left[\tilde{G}(\bar{2}\bar{3})\tilde{G}(\bar{3}\bar{2})\right] \\ & - v(1\bar{3})v(\bar{2}1')\tilde{G}(1\bar{2})\left[\tilde{G}(\bar{2}\bar{3})\tilde{G}(\bar{3}1')\right]. \end{aligned} \tag{4.106}$$

Using (4.105) in (4.102), one can generate the Beliaev self-energy terms to second order in v.

The second-order Beliaev collisional part of this self-energy is given explicitly by

$$\begin{aligned}\Sigma_{\rm coll}(1,1') = {}& -\tfrac{1}{2}v(1\bar{3})v(\bar{2}1')\tilde{G}(11')\left[\tilde{G}(\bar{2}\bar{3})\tilde{G}(\bar{3}\bar{2})\right. \\ & \left. + \tilde{G}(\bar{2}\bar{3})h(\bar{3}\bar{2}) + h(\bar{2}\bar{3})\tilde{G}(\bar{3}\bar{2})\right] \\ & - v(1\bar{3})v(\bar{2}1')\tilde{G}(1\bar{2})\left[\tilde{G}(\bar{2}\bar{3})\tilde{G}(\bar{3}1')\right. \\ & \left. + \tilde{G}(\bar{2}\bar{3})h(\bar{3}1') + h(\bar{2}\bar{3})\tilde{G}(\bar{3}1')\right] \\ & - \tfrac{1}{2}v(1\bar{3})v(\bar{2}1')\left[h(11')\tilde{G}(\bar{2}\bar{3})\tilde{G}(\bar{3}\bar{2})\right. \\ & \left. + 2h(1\bar{2})\tilde{G}(\bar{2}\bar{3})\tilde{G}(\bar{3}1')\right]. \end{aligned} \tag{4.107}$$

The diagrams representing the second-order Beliaev self-energies $\Sigma_{\rm coll}$ are shown in Fig. 4.3. The first four diagrams correspond to the second-order contributions in the conserving approximation (4.99) used by Kane and Kadanoff (1965). In this gapless approximation, we note that $\tilde{G}$ and $G_{1/2}$ are not independent, since

$$\frac{\delta\tilde{G}}{\delta G_{1/2}} = \tilde{G}\frac{\delta\Sigma}{\delta G_{1/2}}\tilde{G}. \tag{4.108}$$

We note that the expression for the self-energy in (4.107) goes beyond what Beliaev (1958b) included in his original paper. Above $T_{\rm BEC}$, the original Beliaev approximation reduces to the Hartree–Fock approximation, i.e. it does *not* include the normal second-order self-energy contributions $\tilde{G}\tilde{G}\tilde{G}$

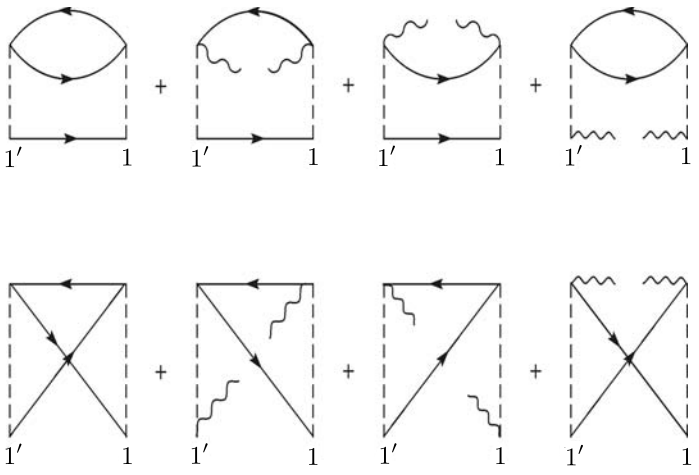


Fig. 4.3. Second-order self-energy $\Sigma_{\rm coll}$ diagrams in the Beliaev approximation.

associated with the C_{22} collisions. Here, we include such terms and, therefore, above $T_{\rm BEC}$ the kinetic equations that we will derive in Chapters 6 and 7 reduce to the usual Boltzmann equation, with collisions described by C_{22}, as discussed in Chapter 6 of Kadanoff and Baym (1962).

In this section, we have shown how one can derive the self-energy functions in both the conserving and gapless approximations. As we shall discuss in Chapters 6 and 7, the specific choice of the single-particle self-energies plays a crucial role in determining the collision terms that appear in the generalized Boltzmann equations as generated by the Kadanoff–Baym formalism.

4.5 Dielectric formalism

More generally, one can prove that the density response function can be written (at all T) as:

$$\chi_{nn}(1,1') = \sum_{\alpha\beta} \int d\bar{2} d\bar{3}\, \Lambda_\alpha(1,\bar{2}) \tilde{G}_{\alpha\beta}(\bar{2},\bar{3}) \Lambda_\beta(\bar{3},1') + \tilde{\chi}_{nn}(1,1'). \quad (4.109)$$

Here $\Lambda_\alpha(1,1')$ is the Bose broken-symmetry vertex function (which vanishes if the condensate is absent, $n_{\rm c0} = 0$). Using the "dielectric formalism" developed in the early 1970s (for a detailed review, see Chapter 5 of Griffin, 1993), one has a systematic diagrammatic method of expressing the self-energies $\Sigma_{\alpha\beta}$, the vertex function Λ_α and the noncondensate density response $\tilde{\chi}_{nn}$ in (4.109) in terms of irreducible proper contributions. These correspond to

diagrams which cannot be split into two parts by cutting either an interaction line or a propagator line. Using these "regular" diagrammatic building blocks, one can prove with great generality that χ_{nn} and $\tilde{G}_{\alpha\beta}$ have the same poles. Any poles specifically associated with $\tilde{\chi}_{nn}$, the last term in (4.109), can be shown to be cancelled by terms arising from the first term in (4.109).

Key papers on the dielectric formalism at $T = 0$ for Bose superfluids are Ma and Woo (1967), Kondor and Szépfalusy (1968) and Wong and Gould (1974). This work was extended to finite temperatures by Griffin and Cheung (1973) and Szépfalusy and Kondor (1974). The formalism was applied to uniform Bose gases by Fliesser *et al.* (2001) to calculate both $\tilde{G}_{\alpha\beta}$ and χ_{nn}. They used a diagrammatic approximation that corresponds to the time-dependent "Hartree plus exchange" for the linear response function χ_{nn}, as calculated by Minguzzi and Tosi (1997). The latter paper is discussed in detail in Section 5.4. The formal extension of the dielectric formalism to a trapped spatially inhomogeneous Bose-condensed gas was given by Reidl *et al.* (2000).

The advantage of this diagrammatic formalism is that one manifestly sees that the poles of $\tilde{G}_{\alpha\beta}$ and χ_{nn} are identical, within a given diagrammatic approximation for the regular "building blocks" defined above. The formalism shows in a clear fashion how a broken-symmetry order parameter $\Phi(\mathbf{r}, t) \neq 0$ guarantees that the single-particle excitation spectrum is the same as the density fluctuation spectrum. As a result the dielectric formalism is a diagrammatic scheme, for systematic calculations of both $\tilde{G}_{\alpha\beta}$ and χ_{nn}, that cuts across the usual division between the gapless and conserving approximations in a Bose-condensed fluid, described in Section 4.4. This formalism also guarantees that various exact Ward identities (see for example Chapter 5 of Griffin, 1993) related to particle-number conservation are satisfied in the presence of a Bose broken symmetry. These and other advantages make this diagrammatic approach the method of choice for future studies of trapped Bose-condensed gases.

5

The Beliaev and the time-dependent HFB approximations

Single-particle Green's functions, density response functions and other correlation functions are calculated in many different ways in the literature on Bose-condensed gases. An in-depth comparison and classification of different approaches was first given in the classic paper by Hohenberg and Martin (1965), with an emphasis on the various exact identities (conservation laws) that are satisfied. A key feature to be included in any theory is that a Bose broken symmetry leads to a hybridization of the single-particle excitations and the collective density fluctuations in such a way that the two excitation spectra become identical. This key feature is demonstrated in Chapter 5 of Griffin (1993).

How to relate and assess various approximations for correlation functions in a Bose superfluid has been a topic of continual interest (and some controversy) since the late 1950s. These questions were largely resolved by the early 1960s at a conceptual level but the detailed applications of the theory were limited to dilute Bose-condensed gases at $T = 0$. Since it was difficult to relate the theory to the properties of superfluid ^{4}He at a quantitative level, this formalism based on a Bose broken symmetry was of little interest to experimentalists. The creation of superfluid Bose-condensed gases in 1995 changed all this and has given new life to the many body theory of dilute weakly interacting Bose-condensed gases.

Various approximations for the Beliaev single-particle self-energies $\Sigma_{\alpha\beta}$ were derived in Sections 4.3 and 4.4. The discussion in Chapter 4 was somewhat abstract. A major goal in the present chapter is to elucidate the subtle relation between the direct diagrammatic evaluation of the single-particle self-energies $\Sigma_{\alpha\beta}(\mathbf{q}, \omega)$ (as first carried out by Beliaev (1958a,b) for a uniform gas at $T = 0$) and the results obtained by solving the time-dependent coupled equations for the condensate and noncondensate components. In Section 5.1, we review the coupled static HFB equations given in (4.84) and

the problems that arise when one uses this first-order self-energy approximation. In Section 5.2, we go on to review the famous second-order Beliaev approximation for the single-particle self-energies, and the resulting excitation frequency and damping. The second-order Beliaev approximation was derived in Section 4.4, to which the reader should refer for background.

In Section 5.3, we discuss how the excitation spectrum given by the Beliaev self-energies to order g^2 can be obtained within a time-dependent linear response solution of the coupled equations, following the approach of Giorgini (2000). Cheung and Griffin (1971) first calculated the density response function at finite T by treating the HFB single-particle self-energies as a time-dependent mean field and showed explicitly that it exhibited the Beliaev single-particle spectrum to order g^2. In Section 5.4, we evaluate the density response function within a "Popov approximation" to the Cheung–Griffin calculation. We prove that this Beliaev–Popov approximation leads to a density response function identical to that given by the self-consistent mean-field approximation for the condensate and noncondensate components (Minguzzi and Tosi, 1997).

In Sections 5.3 and 5.4, the Beliaev excitation spectrum is derived using a time-dependent HFB theory. It is hoped that this will give the reader a deeper insight into the direct diagrammatic evaluation of the equilibrium self-energies given in Section 4.4.

In order to bring out the essential physics, we will limit ourselves in most of this chapter to a *uniform* Bose-condensed gas.

5.1 Hartree–Fock–Bogoliubov self-energies

In this section and in Section 5.2, we describe various approximations to the Dyson–Beliaev equations (4.78) as defined in Section 4.4. For more detail, see the classic treatment in the book by Fetter and Walecka (1971) for $T = 0$ and also Shi and Griffin (1998) for $T \neq 0$. We will describe the various approximations in terms of Feynman diagrams for the Beliaev Green's functions $G_{\alpha\beta}$, shown in Fig. 5.1 (see also Figs. 4.1–4.3). First we review the results of the static first-order HFB approximation at finite T described by (4.81)–(4.84). At finite T, atoms are thermally excited out of the condensate in a dilute Bose gas. Thus "normal" self-energies must be added to the usual $T = 0$ Bogoliubov self-energies given in Fig. 5.2. In the Beliaev self-energy formalism, this was first done by Popov (1965,1987). We have to add the Hartree–Fock self-energies, as shown in Fig. 5.3; this gives

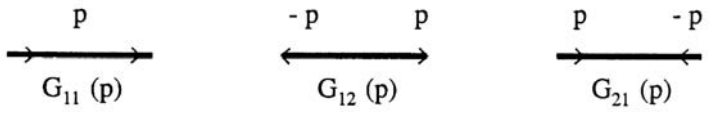


Fig. 5.1. Beliaev Green's functions for a Bose-condensed system.

the first-order "Popov approximation" (g is defined in (2.4)),

$$\begin{aligned}\hbar\Sigma_{11}^{\text{Popov}}(\mathbf{q},\omega) &= 2gn_{\text{c}} + 2g\tilde{n},\\ \hbar\Sigma_{12}^{\text{Popov}}(\mathbf{q},\omega) &= gn_{\text{c}}.\end{aligned} \tag{5.1}$$

In this section, all densities are for static thermal equilibrium. These self-energies lead to the standard results (see Fetter and Walecka, 1971)

$$\tilde{G}_{11}(\mathbf{q},\omega) = \frac{\hbar\omega + \varepsilon_{\mathbf{q}} + \Delta}{(\hbar\omega)^2 - E_{\mathbf{q}}^2}, \qquad \tilde{G}_{12}(\mathbf{q},\omega) = \frac{-\Delta}{(\hbar\omega)^2 - E_{\mathbf{q}}^2} \tag{5.2}$$

with

$$\Delta \equiv \mu_{\text{HP}} - 2g\tilde{n}, \qquad E_{\mathbf{q}}^2 \equiv \varepsilon_{\mathbf{q}}^2 + 2\Delta\varepsilon_{\mathbf{q}}, \tag{5.3}$$

where $\varepsilon_{\mathbf{q}} = q^2/2m$ is the free atom energy. In the Bose gas literature, it is customary to define a chemical potential as follows (see (4.3)):

$$\mu_{\text{HP}} \equiv \hbar\Sigma_{11}(\mathbf{q}=0,\omega=0) - \hbar\Sigma_{12}(\mathbf{q}=0,\omega=0). \tag{5.4}$$

This was introduced by Hugenholtz and Pines (1959) at $T = 0$ (and generalized to $T \neq 0$ by Hohenberg and Martin, 1965).

Hugenholtz and Pines showed that if $\tilde{G}_{\alpha\beta}(\mathbf{q},\omega)$ has a *gapless* excitation spectrum in the $\mathbf{q} \to 0$, $\omega \to 0$ limit then the *true* chemical potential μ must equal μ_{HP} as defined in (5.4). In the Popov approximation described by (5.1), we have (Griffin, 1999a)

$$\mu_{\text{HP}} = 2gn_{\text{c}} + 2g\tilde{n} - gn_{\text{c}} = gn_{\text{c}} + 2g\tilde{n}, \tag{5.5}$$

and thus $\Delta = gn_{\text{c}}$. The generalized GP equation introduced in Chapter 3 determines the chemical potential μ_{c}. In the Popov approximation, this finite-temperature GP equation is given by

$$\left(-\frac{\hbar^2\nabla^2}{2m} + V_{\text{trap}} + gn_{\text{c}} + 2g\tilde{n}\right)\Phi = \mu_{\text{c}}\Phi. \tag{5.6}$$

One sees that in a uniform gas we obtain $\mu_{\text{c}} = \mu_{\text{HP}}$, as required for a gapless

spectrum. In the Popov approximation, the single-particle Green's functions $\tilde{G}_{\alpha\beta}$ in (5.2) have poles given by

$$E_{\mathbf{q}} = [\varepsilon_{\mathbf{q}}^2 + 2gn_{\rm c}(T)\varepsilon_{\mathbf{q}}]^{1/2}. \tag{5.7}$$

This is formally the same as the $T = 0$ Bogoliubov excitation frequency, except that now the condensate density $n_{\rm c}(T)$ is temperature dependent and decreases to zero as $T \to T_{\rm BEC}$. A more detailed discussion of the Popov approximation is given in Chapter 3 of the review article by Shi and Griffin (1998).

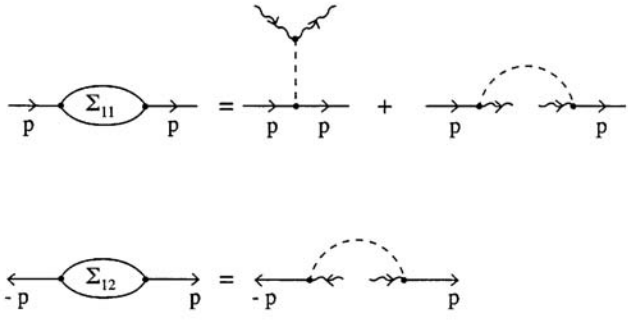


Fig. 5.2. Self-energy diagrams in the $T = 0$ Bogoliubov approximation.

The full Hartree–Fock–Bogoliubov (HFB) approximation involves calculating $\Sigma_{\alpha\beta}$ self-consistently using the complete 2×2 matrix Beliaev propagator; see Fig. 5.4. We note that above $T_{\rm BEC}$ the HFB approximation reduces to the standard self-consistent Hartree–Fock approximation. It is useful to note that, in a variational sense, the HFB is the best *single-particle* approximation for a Bose-condensed system. It will give the best results for thermodynamic functions when these are calculated within a single-particle excitation picture. The HFB is a natural generalization of the standard Hartree–Fock approximation used for normal quantum gases.

However, there is one undesirable aspect of the full HFB, which was noticed in the 1960s and has been the subject of many theoretical papers on Bose-condensed gases. Namely, the HFB does *not* obey the Hugenholtz–Pines (HP) theorem and, as a result, the single-particle excitations have an energy gap in the long-wavelength limit $\mathbf{q} \to 0$. It is easy to check this by calculating the two chemical potentials introduced earlier. Using the definition in (5.4), one finds

$$\mu_{\rm HP} = 2g(n_{\rm c} + \tilde{n}) - g(n_{\rm c} + \tilde{m}) = g(n + \tilde{n} - \tilde{m}), \tag{5.8}$$

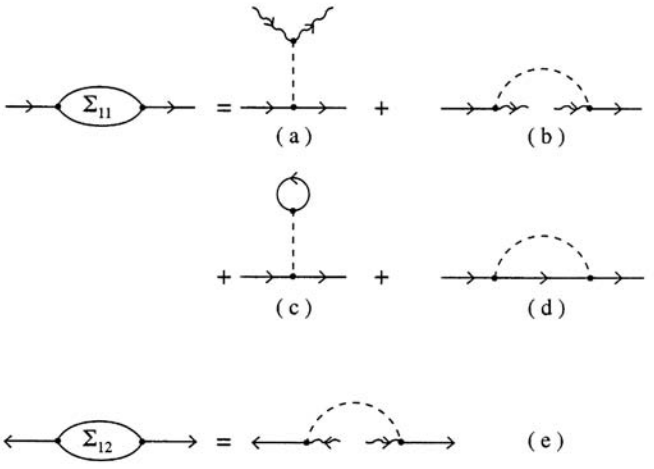


Fig. 5.3. Self-energy diagrams in the Popov approximation.

where $n = n_c + \tilde{n}$. The noncondensate densities $\tilde{n} = i\tilde{G}_{11}(1^+, 1)$ and $\tilde{m} = i\tilde{G}_{12}(1^+, 1)$ were defined in (4.83). Using the Bogoliubov–Popov Green's functions to evaluate them, one finds (see for example Talbot and Griffin, 1983; Shi and Griffin, 1998) the explicit expressions

$$\begin{aligned}\tilde{n}^{(1)} &= \int \frac{d\mathbf{k}}{(2\pi)^3}\left[v_{\mathbf{k}}^2 + (u_{\mathbf{k}}^2 + v_{\mathbf{k}}^2) f^0(E_{\mathbf{k}})\right] \\ &= \int \frac{d\mathbf{k}}{(2\pi)^3}\left\{\frac{\varepsilon_{\mathbf{k}} + \Delta}{2E_{\mathbf{k}}}\left[1 + 2f^0(E_{\mathbf{k}})\right] - \frac{1}{2}\right\} \end{aligned} \tag{5.9}$$

and

$$\begin{aligned}\tilde{m}^{(1)} &= -\int{}' \frac{d\mathbf{k}}{(2\pi)^3} u_{\mathbf{k}} v_{\mathbf{k}}\left[1 + 2f^0(E_{\mathbf{k}})\right] \\ &= -gn_c \int \frac{d\mathbf{k}}{(2\pi)^3}\left[\frac{1 + 2f^0(E_{\mathbf{k}})}{2E_{\mathbf{k}}} - \frac{1}{2\varepsilon_{\mathbf{k}}}\right]. \end{aligned} \tag{5.10}$$

The prime on the momentum integral in the first line of (5.10) denotes the fact that the anomalous density $\tilde{m}^{(1)}$ must be renormalized to remove the ultraviolet divergence coming from high-momentum contributions. The second line in (5.10) gives the explicit form of the renormalized value of $\tilde{m}^{(1)}$. In a uniform gas, the static equation of motion for $\Phi = \sqrt{n_c}$ in the

HFB approximation,

$$\left(-\frac{\hbar^2\nabla^2}{2m} + V_{\rm trap} + gn_{\rm c} + 2g\tilde{n} + g\tilde{m} \right)\sqrt{n_{\rm c}} = \mu_{\rm c}\sqrt{n_{\rm c}}, \tag{5.11}$$

gives a condensate chemical potential

$$\mu_{\rm c} = g(n + \tilde{n} + \tilde{m}). \tag{5.12}$$

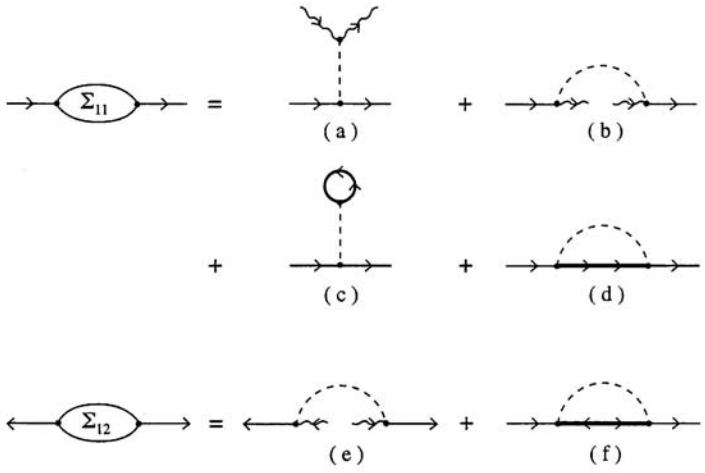


Fig. 5.4. The HFB Beliaev self-energy diagrams.

Clearly (5.12) is not the same as the result in (5.8). One "solution" of this problem is to omit the anomalous density $\tilde{m}$ (which is the source of the problem) but keep $\tilde{n}$. This corresponds to the self-consistent Hartree–Fock–Popov approximation (Griffin, 1996; Hutchinson *et al.*, 1997; Dalfovo *et al.*, 1999). While this simple approximation is somewhat *ad hoc*, it has several nice features. It gives the correct excitation spectrum for both $T \to 0$ and for $T > T_{\rm BEC}$; moreover, this spectrum is gapless at all temperatures.

The origin of the above-mentioned problem with the HFB is clear. The static HFB formally keeps all self-energies that are first order in the interaction g. However, by computing these self-energy diagrams using self-consistent $\tilde{G}_{\alpha\beta}$ propagators, one is clearly bringing in terms to all orders in g. Specifically, one can easily check that $\tilde{m}$ must be at least of order g. Thus the $g\tilde{m}$ contribution to Σ_{12} is at least $O(g^2)$. This shows that to improve on the static HFB as a theory of excitations, we have to include self-energy contributions to at least second order in g. This is precisely what is accomplished in the Beliaev second-order calculation, as we now describe in Section 5.2.

5.2 Beliaev self-energy approximation

We now review the results obtained by Beliaev (1958a), who evaluated the $\Sigma_{\alpha\beta}$ self-energies keeping contributions up to order g^2 (see Section 4.4). Beliaev's work has been extended to finite temperatures and to trapped gases (Shi and Griffin, 1998; Fedichev *et al.*, 1998). When one includes the second-order diagrams for $\Sigma_{\alpha\beta}$, one must also be careful to treat the HFB first-order diagrams correctly to order g^2 since there are many contributions of order g^2 that cancel each other (Shi and Griffin, 1998).

The Beliaev-type second-order calculation cures all the problems of the first-order static HFB approximation reviewed in Section 5.1. Explicit results for the second-order corrections to the self-energies $\Sigma_{11}(\mathbf{q},\omega)$ and $\Sigma_{12}(\mathbf{q},\omega)$ at finite temperatures are discussed in Shi and Griffin (1998) as well as in Talbot and Griffin (1983). Both terms contain infrared-divergent contributions that cancel out in physical quantities (at finite T as well as at $T=0$). Writing

$$\begin{aligned}\hbar\Sigma_{11}(\mathbf{q}=0,\omega=0) &= 2gn_{\rm c} + 2g\tilde{n}^{(1)} + g^2A,\\ \hbar\Sigma_{12}(\mathbf{q}=0,\omega=0) &= gn_{\rm c} + g\tilde{m}^{(1)} + g^2B,\end{aligned} \tag{5.13}$$

one finds

$$\mu_{\rm HP} = gn_{\rm c} + 2g\tilde{n}^{(1)} - g\tilde{m}^{(1)} + g^2(A-B). \tag{5.14}$$

Using the results of Shi and Griffin (1998), we have, to order g^2,

$$g^2(A-B) = 2g\tilde{m}^{(1)}. \tag{5.15}$$

Substituting this into (5.14), one sees that $\mu_{\rm HP}$ reduces precisely to the HFB expression for $\mu_{\rm c}$ in (5.12). This result shows that satisfying the Hugenholtz–Pines relation (and hence achieving a gapless phonon spectrum at low q) requires a careful treatment of all second-order contributions to the self-energies.

It is useful to give some explicit results at $T=0$ that Beliaev (1958a) first worked out (see also Fetter and Walecka, 1971, p. 218). One can prove using (5.9) and (5.10) that, at $T=0$,

$$\begin{aligned}\tilde{m}^{(1)} = 3\tilde{n}^{(1)} &= 3\left[n_{\rm c}\frac{m^{3/2}}{3\pi^2}\left(n_{\rm c}g^3\right)^{1/2}\right]\\ &= 3\left[n_{\rm c}\frac{8}{3}\left(n_{\rm c}a^3/\pi\right)^{1/2}\right].\end{aligned} \tag{5.16}$$

Here we have used $g = 4\pi a/m$, where a is the s-wave scattering length and

also Bogoliubov's famous result for the depletion $\tilde{n}$ at $T = 0$. The Beliaev self-energies in (5.13) reduce to (see for example Griffin and Cheung, 1973)

$$\begin{aligned}\hbar\Sigma_{11}(\mathbf{q}=0,\omega=0) &= 2gn_{\rm c} + \frac{14}{3\pi^2}m^{3/2}gn_{\rm c}\left(n_{\rm c}g^3\right)^{1/2} + \cdots,\\ \hbar\Sigma_{12}(\mathbf{q}=0,\omega=0) &= gn_{\rm c} + \frac{3}{\pi^2}m^{3/2}gn_{\rm c}\left(n_{\rm c}g^3\right)^{1/2} + \cdots.\end{aligned} \tag{5.17}$$

In fact, both Σ_{11} and Σ_{12} contain the same additional infrared-divergent contributions, which are not included in (5.17) (see Shi and Griffin, 1998). Using these results, one obtains[1]

$$\begin{aligned}\mu_{\rm HP} &= gn_{\rm c} + \frac{40}{3}n_{\rm c}a\left(n_{\rm c}a^3/\pi\right)^{1/2}\\ &= \frac{4\pi an}{m}\left[1 + \frac{32}{3}\left(n_{\rm c}a^3/\pi\right)^{1/2}\right].\end{aligned} \tag{5.18}$$

In the second line, we have given $\mu_{\rm HP}(n)$ in terms of the total density n,

$$n = n_{\rm c} + n_{\rm c}\frac{8}{3}\left(n_{\rm c}a^3/\pi\right)^{1/2} \equiv n_{\rm c} + \tilde{n}. \tag{5.19}$$

Using (5.16) in (5.12), the HFB chemical potential reduces to

$$\mu_{\rm HFB} = gn_{\rm c} + 5g\tilde{n}^{(1)} = gn_{\rm c} + \frac{40}{3}n_{\rm c}a\left(n_{\rm c}a^3/\pi\right)^{1/2}, \tag{5.20}$$

which is identical to $\mu_{\rm HP}$ in (5.18).

One also finds that the phonon excitation given by the pole of $\tilde{G}^{\rm Bel}_{\alpha\beta}$ has a velocity given by

$$c^2_{\rm Bel} = \frac{4\pi na}{m^2}\left[1 + 16\left(n_{\rm c}a^3/\pi\right)^{1/2}\right], \tag{5.21}$$

again expressed in terms of the total density n. One may easily check by calculating $d\mu_{\rm HP}(n)/dn$ using (5.18) that

$$c^2_{\rm Bel} = \frac{n}{m}\frac{d\mu_{\rm HP}(n)}{dn} \equiv c^2_{\rm comp}; \tag{5.22}$$

that is, the Beliaev phonon velocity at $T = 0$ given by (5.21) is precisely equal to the *compressional* sound velocity defined in (5.22).

As first realized in the 1960s, the fact that the Beliaev phonon velocity in (5.21) equals the compressional velocity is a fundamental signature of Bose superfluids. It raises a basic question: why does the pole of the single-particle Green's function $\tilde{G}_{\alpha\beta}$ have same frequency as the pole of the density response function χ_{nn}? This "equivalent spectra" theorem was proved to be

[1] This is equivalent to the result obtained by Lee and Yang (1958) using a different approach.

correct at $T = 0$ to *all* orders of perturbation theory by Gavoret and Nozières (1964). To be precise, at $T = 0$, in any Bose-condensed fluid (liquid or gas), Gavoret and Nozières showed that (for $\mathbf{q}, \omega \to 0$)

$$\tilde{G}_{\alpha\beta}(\mathbf{q},\omega) \simeq \frac{a}{(\hbar\omega)^2 - (cq)^2}, \qquad \chi_{nn}(\mathbf{q},\omega) \simeq \frac{b}{(\hbar\omega)^2 - (cq)^2}, \tag{5.23}$$

where c^2 is the compressional sound velocity in (5.22). One can prove this equivalence to be a direct consequence of Bose broken symmetry, i.e. that $\Phi(\mathbf{r}) \equiv \langle\hat{\psi}\rangle$ is nonzero. This was only proved and its significance understood in the early 1960s (Gavoret and Nozières, 1964; Hohenberg and Martin, 1965; Bogoliubov, 1970).

The above equivalence was implicitly assumed in the classic papers by Landau (1941) and by Feynman (1953, 1954), before the distinction between elementary excitations and density fluctuations in many body systems was clarified. In these papers, it was simply assumed without proof that, in superfluid ^{4}He, the important single-particle excitations determining the thermodynamics at low temperatures were density fluctuations, corresponding to compressional sound waves. Since the 1960s, we understand that this key assumption in the pioneering work of Landau and of Feynman has its microscopic basis in the existence of a finite Bose condensate, i.e. $\Phi(\mathbf{r}) \neq 0$.

The equivalence of density and single-particle excitations holds at all temperatures in Bose superfluids and not just at $T = 0$. The equivalence of the poles of the single-particle Green's functions $\tilde{G}_{\alpha\beta}$ and the density response χ_{nn} is exhibited most clearly using the dielectric formalism (see Section 4.5 of this book and Chapter 5 of Griffin, 1993). This diagrammatic formalism shows how a Bose broken symmetry couples and alters these two kinds of correlation function. This feature will play a crucial role in Sections 5.3 and 5.4.

It is easy to understand the origin of this equivalence of the field fluctuation and density fluctuation spectra. The density operator in a Bose-condensed system can be decomposed as follows:

$$\begin{aligned}\hat{n}(\mathbf{r}) &= \hat{\psi}^\dagger(\mathbf{r})\hat{\psi}(\mathbf{r}) \\ &= |\Phi_0(\mathbf{r})|^2 + \Phi_0(\mathbf{r})\tilde{\psi}^\dagger(\mathbf{r}) + \Phi_0^*(\mathbf{r})\tilde{\psi}(\mathbf{r}) + \tilde{\psi}^\dagger(\mathbf{r})\tilde{\psi}(\mathbf{r}),\end{aligned} \tag{5.24}$$

where the first three terms correspond to the condensate density $\hat{n}_{\rm c}(\mathbf{r}) \equiv n_{\rm c}(\mathbf{r}) + \delta n_{\rm c}(\mathbf{r})$ and the last term is the noncondensate density operator $\tilde{n}(\mathbf{r})$. Clearly density fluctuations in a Bose-condensed system (i.e. one with $\Phi_0 \neq 0$) have a direct coupling to the single-particle field fluctuations, owing to the possibility of atoms coming in or out of the condensate reservoir. In

momentum space, (5.24) is equivalent to

$$\hat{n}_{\mathbf{q}} = \sum_{\mathbf{k}} \hat{a}^{\dagger}_{\mathbf{k}} \hat{a}_{\mathbf{k}-\mathbf{q}} = |\hat{a}_0|^2 + (\hat{a}^{\dagger}_0 \hat{a}_{-\mathbf{q}} + \hat{a}_0 \hat{a}^{\dagger}_{\mathbf{q}}) + \tilde{n}_{\mathbf{q}}. \tag{5.25}$$

Making the same decomposition as in (5.24) for the density response function gives

$$\begin{aligned} \chi_{nn}(1,1') &\simeq \langle \hat{n}(1)\hat{n}(1') \rangle \\ &= \langle \hat{n}(1) \rangle \langle \hat{n}(1') \rangle + \langle \delta n_{\mathrm{c}}(1) \delta n_{\mathrm{c}}(1') \rangle \\ &\quad + \langle \delta n_{\mathrm{c}}(1) \delta \tilde{n}(1') \rangle + \langle \delta \tilde{n}(1) \delta n_{\mathrm{c}}(1') \rangle + \langle \delta \tilde{n}(1) \delta \tilde{n}(1') \rangle. \end{aligned} \tag{5.26}$$

We note that in the Bogoliubov approximation we set $\delta \tilde{n} = 0$. Using the definition of δn_{c} given in (5.24), the equivalence of the χ_{nn} and $\tilde{G}_{\alpha\beta}$ spectra is then trivial:

$$\chi_{nn}(\mathbf{q},\omega) \;\Rightarrow\; \chi^{\mathrm{B}}_{nn}(\mathbf{q},\omega) \equiv n_{\mathrm{c}} \sum_{\alpha,\beta} \tilde{G}^{\mathrm{B}}_{\alpha\beta}(\mathbf{q},\omega). \tag{5.27}$$

However, this kind of relation between χ_{nn} and $\tilde{G}_{\alpha\beta}$ is always valid (see Section 4.5) in the presence of a Bose broken symmetry.

The equivalence of the single-particle excitations and the density fluctuations due to a Bose condensate lies at the heart of the phenomenon of superfluidity. It essentially *restricts* the possible single-particle excitations to density fluctuations; in turn, this ensures the *stability* of irrotational superfluid flow by eliminating decay processes (Nozières and Pines, 1990). However, as Nozières (1966) emphasized, the fact that "the $T = 0$ superfluid equations merge with those of ordinary hydrodynamics but this does not alter the fact that real understanding must be based on a microscopic description based on long-range order induced by a condensate".

The full time-dependent HFB equation of motion for the condensate wavefunction $\Phi(\mathbf{r},t) \equiv \langle \hat{\psi}(\mathbf{r}) \rangle_t$ (when restricted to the case $\Phi = \Phi^*$ for simplicity) is given by

$$i\hbar \frac{\partial \Phi(\mathbf{r},t)}{\partial t} = \Big[-\frac{\hbar^2 \nabla^2}{2m} + V_{\mathrm{trap}}(\mathbf{r}) + g n_{\mathrm{c}}(\mathbf{r},t) + 2g\tilde{n}(\mathbf{r},t) + g\tilde{m}(\mathbf{r},t) \Big] \Phi(\mathbf{r},t). \tag{5.28}$$

This time-dependent equation is derived in Sections 4.3 and 6.2. Various approximations to this key equation that can be used to classify different theories of the condensate dynamics (the collective modes) of trapped Bose gases include the following:

- *GP* (*Gross–Pitaevskii*): Ignore $\tilde{n}$ and $\tilde{m}$ completely. This is only valid at $T = 0$ in a weakly interacting dilute Bose gas.

- *Static HFP* (*Hartree–Fock–Popov*): Keep $n_c(\mathbf{r},t)$ but set $\tilde{n}(\mathbf{r},t) = \tilde{n}_0(\mathbf{r})$ and $\tilde{m}(\mathbf{r},t) = 0$ (Hutchinson *et al.*, 1997; Dodd *et al.*, 1998).
- *Static HFB* (*Hartree–Fock–Bogoliubov*): Keep $n_c(\mathbf{r},t)$ but set $\tilde{n}(\mathbf{r},t) = \tilde{n}_0(\mathbf{r})$ and $\tilde{m}(\mathbf{r},t) = \tilde{m}_0(\mathbf{r})$ (Hohenberg and Martin, 1965; Griffin, 1996). As noted in Section 5.1, this produces an energy gap in the single-particle excitation spectrum (since $\mu_{\rm HP} \neq \mu_c$).
- *Dynamic HFB*: Treat all dynamic mean fields due to $n_c(\mathbf{r},t)$, $\tilde{n}(\mathbf{r},t)$ and $\tilde{m}(\mathbf{r},t)$ on an equal basis in a generalized mean-field calculation of the density response function $\chi_{nn}(1,1')$. This will be discussed in Sections 5.3 and 5.4. One can prove that the poles of χ_{nn} given by this kind of calculation ($\chi_{nn} \sim ((\hbar\omega)^2 - E^2)^{-1}$) are identical to the poles of the single-particle Green's function $\tilde{G}_{\alpha\beta}$ given by the second-order Beliaev approximation ($\tilde{G}_{\alpha\beta} \sim ((\hbar\omega)^2 - E^2)^{-1}$). This proof was first given by Hohenberg and Martin (1964) at $T = 0$ and generalized to finite temperatures by Cheung and Griffin (1971).

Various theories of excitations at finite temperatures in trapped gases are easily understood in terms of this type of generalized linear response calculation of χ_{nn} based on (5.28). For example, in a paper to be discussed in Section 5.4, Minguzzi and Tosi (1997) allowed fluctuations in δn_c and $\delta\tilde{n}$ but completely ignored the fluctuations $\delta\tilde{m}$ (we will refer to this as the dynamic HFP approximation). In contrast, as we review in Section 5.3, the papers by Giorgini (1998, 2000) keep fluctuations involving $\delta\tilde{n}$ and $\delta\tilde{m}$ that are induced by the condensate mean-field fluctuations δn_c.

In Section 5.4, we will discuss in detail how the hybridization of single-particle and density fluctuations arise. As a simple introduction to the basic physics involved, we now sketch the argument using the time-dependent HFP approximation (Minguzzi and Tosi, 1997). In a simple mean-field approximation (MFA) the thermal cloud density response $\delta\tilde{n}$ induced by a small time-dependent field $U_{\rm ex}$ is given by

$$\delta\tilde{n} = \chi^0_{\tilde{n}}[U_{\rm ex} + 2g(\delta n_c + \delta\tilde{n})]. \tag{5.29}$$

Here $\chi^0_{\tilde{n}}$ is the thermal cloud density response function for a gas of noninteracting excitations. Taking the limit $U_{\rm ex} \to 0$, (5.29) gives

$$\delta\tilde{n} = \frac{\chi^0_{\tilde{n}} 2g\delta n_c}{1 - 2g\chi^0_{\tilde{n}}}. \tag{5.30}$$

Using this result in the HFP equation of motion (5.28), in which we set

$\tilde{m} = 0$, we find

$$2g\delta\tilde{n} + g\delta n_{\rm c} = g\left(\frac{1+2g\chi_{\tilde{n}}^0}{1-2g\chi_{\tilde{n}}^0}\right)\delta n_{\rm c} \equiv g'(\mathbf{q},\omega)\delta n_{\rm c}. \tag{5.31}$$

In this approximation, in which one retains only the $\delta\tilde{n}$ fluctuations induced by the condensate, the equation of motion for $\delta\Phi(\mathbf{r},t)$ reduces to the GP equation but with g now given by the quantity g' defined in (5.31). Thus the generalized Bogoliubov condensate excitations (i.e. with g replaced by $g'(\mathbf{q},\omega)$) are now given by the zeros of

$$\begin{aligned}(\hbar\omega)^2 - (\varepsilon_{\mathbf{q}}^2 + 2g' n_{\rm c}\varepsilon_{\mathbf{q}}) &\propto ((\hbar\omega)^2 - \varepsilon_{\mathbf{q}}^2)(1-2g\chi_{\tilde{n}}^0) - 2gn_{\rm c}\varepsilon_{\mathbf{q}}(1+2g\chi_{\tilde{n}}^0) \\ &= [(\hbar\omega)^2 - (\varepsilon_{\mathbf{q}}^2 + 2gn_{\rm c}\varepsilon_{\mathbf{q}})](1-2g\chi_{\tilde{n}}^0) - 8g^2 n_{\rm c}\varepsilon_{\mathbf{q}}\chi_{\tilde{n}}^0;\end{aligned} \tag{5.32}$$

(see also (5.65)). These hybridized excitations are precisely identical to the poles of the mean-field response functions obtained by Minguzzi and Tosi (1997). This simple calculation illustrates how the Bogoliubov condensate fluctuations and the noncondensate density fluctuations are coupled and hybridized by the Bose condensate. This brief discussion is intended as an introduction to the detailed treatment given in Section 5.4.

5.3 Beliaev as time-dependent HFB

Giorgini (2000) presents a derivation of the Beliaev second-order approximation for the frequency and damping of condensate excitations based on a self-consistent solution of a time-dependent HFB theory. From our point of view, Giorgini's formulation (see Section II of his paper) has the advantage that it is based on two coupled time-dependent equations for the condensate and noncondensate atoms. This approach can be used to understand the generalized GP and kinetic equations which form the basis of the ZNG formalism introduced in Chapter 3. In this section, we will summarize the key steps of Giorgini's analysis, with an emphasis on the structure of the theory. For further details, we refer the reader to Giorgini (2000).

The linearized generalized GP equation of Giorgini (see Section III of his paper) is

$$\begin{aligned}\hbar\omega\delta\Phi(\mathbf{r},\omega) = &\left\{-\frac{\hbar^2\nabla^2}{2m} + V_{\rm trap}(\mathbf{r}) - \mu + 2g[n_{\rm c}(\mathbf{r}) + \tilde{n}(\mathbf{r})]\right\}\delta\Phi(\mathbf{r},\omega) \\ &+ [gn_{\rm c}(\mathbf{r}) + g\tilde{m}_0(\mathbf{r})]\,\delta\Phi^*(\mathbf{r},\omega) \\ &+ 2g\Phi_0(\mathbf{r})\delta\tilde{n}(\mathbf{r},\omega) + g\Phi_0(\mathbf{r})\delta\tilde{m}(\mathbf{r},\omega).\end{aligned} \tag{5.33}$$

The order parameter $\delta\Phi(\mathbf{r},\omega)$ is defined by $\Phi(\mathbf{r},t) \equiv \Phi_0(\mathbf{r}) + \delta\Phi(\mathbf{r},t)$. As noted in Section 5.2, this equation is based on the time-dependent HFB approximation, which keeps all the interaction terms that are first order in g. A similar equation of motion describes $\delta\Phi^*(\mathbf{r},\omega)$. This shows how the condensate fluctuations $\delta\Phi$ and $\delta\Phi^*$ are coupled to the noncondensate fluctuations $\delta\tilde{n}$ and $\delta\tilde{m}$. To determine the latter two quantities, we need the generalized kinetic equations (16) and (17) of Giorgini (2000). These give the normal and anomalous Bogoliubov quasiparticle distribution functions, defined by $f_{ij}(\mathbf{r},t) \equiv \langle\alpha_i^\dagger(t)\alpha_j(t)\rangle - f_j^0\delta_{ij}$ and $g_{ij}(\mathbf{r},t) \equiv \langle\alpha_i(t)\alpha_j(t)\rangle$. Here $\alpha_i^\dagger$ and α_j are the creation and destruction operators for the Bogoliubov quasiparticles; the labels i and j represent the eigenstates of the coupled Bogoliubov equations. We emphasize that these generalized kinetic equations do *not* include the analogue of the collision integrals C_{12} and C_{22}.

As one can see from Giorgini's results, the linearized coupled equations of motion for $f_{ij}(\mathbf{r},\omega)$ and $g_{ij}(\mathbf{r},\omega)$ involve terms that are proportional to $g\delta\Phi(\mathbf{r},\omega)$ and $g\delta\Phi^*(\mathbf{r},\omega)$. Contributions from the terms $g\delta\tilde{n}(\mathbf{r},\omega)$ and $g\delta\tilde{m}(\mathbf{r},\omega)$ turn out to be higher order and thus can be omitted (For a uniform gas these kinetic equations are given by (5.37) and (5.38)). Once one has solved the equations for $f_{ij}(\mathbf{r},\omega)$ and $g_{ij}(\mathbf{r},\omega)$, the results can be used to calculate the fluctuations in $\delta\tilde{n}(\mathbf{r},\omega)$ and $\delta\tilde{m}(\mathbf{r},\omega)$ from the expressions

$$\begin{aligned}
\delta\tilde{n}(\mathbf{r},\omega) = \sum_{i,j} \Big\{ & \left[u_i^*(\mathbf{r})u_j(\mathbf{r}) + v_i^*(\mathbf{r})v_j(\mathbf{r})\right] f_{ij}(\mathbf{r},\omega) \\
& - u_i(\mathbf{r})v_j(\mathbf{r})g_{ij}(\mathbf{r},\omega) - u_i^*(\mathbf{r})v_j^*(\mathbf{r})g_{ij}^*(\mathbf{r},\omega)\Big\} \\
\delta\tilde{m}(\mathbf{r},\omega) = \sum_{i,j} \Big\{ & -2v_i^*(\mathbf{r})u_j(\mathbf{r})f_{ij}(\omega) + u_i(\mathbf{r})u_j(\mathbf{r})g_{ij}(\mathbf{r},\omega) \\
& + v_i^*(\mathbf{r})v_j^*(\mathbf{r})g_{ij}^*(\mathbf{r},\omega)\Big\}.
\end{aligned} \tag{5.34}$$

Here u_i, v_i are the usual Bogoliubov quasiparticle amplitudes. amplitudes. This linearized time-dependent mean-field approximation gives a closed set of self-consistent equations for the oscillations around the static Hartree–Fock–Bogoliubov (HFB) solution specified by $\Phi_0(\mathbf{r}), \tilde{n}(\mathbf{r})$ and $\tilde{m}(\mathbf{r})$.

This theory based on the coupled equations (5.33) and (5.34), in addition to the equations of motion for f_{ij} and g_{ij}, corresponds to the time-dependent HFB theory (TDHFB). It is a natural extension to a Bose superfluid of the well-known time-dependent Hartree and Hartree–Fock (TDHF) theories of collective modes in normal quantum gases. A clear discussion of the time-dependent Hartree theory is given in Chapter 7 of Kadanoff and Baym (1962). This type of theory of collective modes in Bose superfluids was cat-

egorized in the classic paper by Hohenberg and Martin (1965). Hohenberg and Martin show how one can start from a single-particle Green's function $G^{\rm HFB}_{\alpha\beta}$ including only self-energy terms of order in g and generate, using (4.89) and (4.90), a density response function that has poles corresponding to collective modes including terms of second order in g. This response function can be shown to satisfy various conservation laws, including the Hugenholtz–Pines theorem discussed in Section 5.1. In the paragraphs following equation (17) of Giorgini (2000), a useful discussion is given of various approximations to the TDHFB theory outlined above, making contact with simpler theories for the excitations in an interacting Bose-condensed gas.

In order to obtain collective modes corresponding to the second-order Beliaev approximation, one needs only the coupling to the condensate fluctuations $\delta\Phi$ and $\delta\Phi^*$ in solving the equations of motion for f_{ij} and g_{ij}. The reasoning is that to $O(g^2)$ we need expressions for $\delta\tilde{n}(\mathbf{r},\omega)$ and $\delta\tilde{m}(\mathbf{r},\omega)$ in (5.33) only to first order in g. This means that, in the equations of motion for f_{ij} and g_{ij}, keeping terms that are proportional to $g\delta\tilde{n}(\mathbf{r},\omega)$ and $g\delta\tilde{m}(\mathbf{r},\omega)$ would lead to results for $\delta\tilde{n}$ and $\delta\tilde{m}$ from (5.34) that are of second order in g^2. Thus to $O(g^2)$ we may omit the terms proportional to $g\delta\tilde{n}(\mathbf{r},\omega)$ and $g\delta\tilde{m}(\mathbf{r},\omega)$ in the generalized collisionless kinetic equations for f_{ij} and g_{ij}. Combining (5.33) and (5.34), this procedure gives a closed set of equations for the fluctuations $\delta\Phi, \delta\Phi^*, \delta\tilde{n}$ and $\delta\tilde{m}$. In this approximation, the single-particle Bogoliubov excitations are treated within a self-consistent linear response associated *only* with the time-dependent condensate mean field. This is a key simplification. As a result, only the condensate mean fields proportional to $g\delta\Phi(\mathbf{r},\omega)$ and $g\delta\Phi^*(\mathbf{r},\omega)$ are retained in Giorgini's derivation of the Beliaev approximation.

To illustrate the preceding discussion, we consider Giorgini's formulation for a uniform gas (see Section III of his paper). In this case, we take

$$\begin{aligned}\delta\Phi(\mathbf{r},t) &= \delta\Phi(\mathbf{q},\omega)e^{i(\mathbf{q}\cdot\mathbf{r}/\hbar-\omega t)},\\ \delta\Phi^*(\mathbf{r},t) &= \delta\Phi^*(\mathbf{q},\omega)e^{i(\mathbf{q}\cdot\mathbf{r}/\hbar-\omega t)}\end{aligned} \tag{5.35}$$

and

$$v_{\mathbf{q}}(\mathbf{r}) = u_{\mathbf{q}}e^{i\mathbf{q}\cdot\mathbf{r}/\hbar}, \quad v_{\mathbf{q}}(\mathbf{r}) = v_{\mathbf{q}}e^{i\mathbf{q}\cdot\mathbf{r}/\hbar}, \tag{5.36}$$

where the Bogoliubov quasiparticle amplitudes $u_{\mathbf{q}}$ and $v_{\mathbf{q}}$ are assumed to be real. One finds that the two equations in (5.34) reduce to (note that we

use a sign convention for $v_{\mathbf{q}}$ different from that of Giorgini)

$$\begin{aligned}\delta\tilde{n}(\mathbf{q},\omega) &= \sum_{\mathbf{p}} \Big[(u_{\mathbf{p}}u_{\mathbf{p+q}} + v_{\mathbf{p}}v_{\mathbf{p+q}})f_{\mathbf{p},\mathbf{p+q}}(\omega) \\ &\qquad - u_{\mathbf{p}}v_{\mathbf{q-p}}g_{\mathbf{p},\mathbf{q-p}}(\omega) - u_{\mathbf{p}}v_{-\mathbf{p}-\mathbf{q}}g^*_{\mathbf{p},-\mathbf{q}-\mathbf{p}}(\omega)\Big], \\ \delta\tilde{m}(\mathbf{q},\omega) &= \sum_{\mathbf{p}} \Big[-2v_{\mathbf{p}}u_{\mathbf{p+q}}f_{\mathbf{p},\mathbf{p+q}}(\omega) + u_{\mathbf{p}}u_{\mathbf{q-p}}g_{\mathbf{p},\mathbf{q-p}}(\omega) \\ &\qquad + v_{\mathbf{p}}v_{\mathbf{p+q}}g^*_{\mathbf{p},-\mathbf{q}-\mathbf{p}}(\omega)\Big].\end{aligned} \tag{5.37}$$

In a uniform gas, the distribution functions f_{ij} and g_{ij} satisfy the equations of motion

$$\begin{aligned}&f_{\mathbf{p},\mathbf{p+q}}(\omega)\left[\hbar\omega - (E_{\mathbf{p+q}} - E_{\mathbf{p}}) + i0^+\right] \\ &= 2g\sqrt{n_{\mathrm{c}}}(f^0_{\mathbf{p}} - f^0_{\mathbf{q-p}})\Big[\delta\Phi(\mathbf{q},\omega)\left(u_{\mathbf{p}}u_{\mathbf{q+p}} + v_{\mathbf{p}}v_{\mathbf{q+p}} - v_{\mathbf{p}}u_{\mathbf{q+p}}\right) \\ &\qquad + \delta\Phi^*(\mathbf{q},\omega)\left(u_{\mathbf{p}}u_{\mathbf{q+p}} + v_{\mathbf{p}}v_{\mathbf{q+p}} - u_{\mathbf{p}}v_{\mathbf{q+p}}\right)\Big], \\ &g_{\mathbf{p},\mathbf{q-p}}(\omega)[\hbar\omega - (E_{\mathbf{p+q}} + E_{\mathbf{p}}) + i0^+] \\ &= 2g\sqrt{n_{\mathrm{c}}}(1 + f^0_{\mathbf{p}} + f^0_{\mathbf{q-p}})\Big[\delta\Phi(\mathbf{q},\omega)\left(u_{\mathbf{p}}u_{\mathbf{q-p}} + u_{\mathbf{p}}v_{\mathbf{q-p}} + v_{\mathbf{p}}u_{\mathbf{q-p}}\right) \\ &\qquad + \delta\Phi^*(\mathbf{q},\omega)\left(u_{\mathbf{p}}u_{\mathbf{q-p}} - u_{\mathbf{p}}v_{\mathbf{q-p}} - v_{\mathbf{p}}u_{\mathbf{q-p}}\right)\Big],\end{aligned} \tag{5.38}$$

along with a similar equation for $g^*_{\mathbf{p},\mathbf{q-p}}(\omega)$. In these generalized "kinetic equations", the terms on the r.h.s. can be viewed as the effect of the time-dependent condensate mean field on the dynamics of the Bogoliubov HFB quasiparticles of energy $E_{\mathbf{p}}$. The terms $E_{\mathbf{p+q}} \pm E_{\mathbf{p}}$ on the l.h.s. of the two equations in (5.38) describe the free time evolution of the Bogoliubov excitations.

Giorgini shows that solving the linearized coupled equations (5.33), (5.37) and (5.38) gives rise to collective modes (in a uniform gas) with an energy and damping precisely equivalent to the second-order Beliaev approximation (as reviewed in Section 5.2). More specifically, one finds

$$E^{\mathrm{Bel}}_{\mathbf{q}} = E^{\mathrm{B}}_{\mathbf{q}} + \delta E_{\mathbf{q}} - i\hbar\gamma_{\mathbf{q}}, \tag{5.39}$$

where the corrections of order g^2 are given by

$$\begin{aligned}\delta E_{\mathbf{q}} - i\hbar\gamma_{\mathbf{q}} = u^2_{\mathbf{q}}\hbar\Sigma_{11}(\mathbf{q},\omega = E^{\mathrm{B}}_{\mathbf{q}}/\hbar) - 2u_{\mathbf{q}}v_{\mathbf{q}}\hbar\Sigma_{12}(\mathbf{q},\omega = E^{\mathrm{B}}_{\mathbf{q}}/\hbar) \\ + v^2_{\mathbf{q}}\hbar\Sigma_{22}(\mathbf{q},\omega = -E^{\mathrm{B}}_{\mathbf{q}}/\hbar).\end{aligned} \tag{5.40}$$

Here $\Sigma_{ij}(\mathbf{q},\omega)$ are the equilibrium Beliaev second-order self-energies, which

can be derived directly using diagrammatic techniques at $T = 0$ (Beliaev, 1958b) and at finite T (Shi and Griffin, 1998; Fedichev and Shlyapnikov, 1998). These were discussed in Section 4.4.

In Chapter 3, in our semiclassical kinetic equation for $f(\mathbf{p}, \mathbf{r}, t)$ for atoms moving in a self-consistent Hartree–Fock mean field, we ignored the anomalous correlation function (this is the Popov approximation), setting $\delta\tilde{m} = 0$. This meant that the function $g_{\mathbf{p},\mathbf{q}-\mathbf{p}}$ vanished, like the Bogoliubov amplitude $v_{\mathbf{p}}$, leaving us with the HF excitation spectrum $E_{\mathbf{p}} = \varepsilon_{\mathbf{p}} + gn_{\mathrm{c}}$. In this approximation, we see that (5.37) and (5.38) can be combined to give

$$\begin{aligned}\delta\tilde{n}(\mathbf{q},\omega) &= \sum_{\mathbf{p}} f_{\mathbf{p},\mathbf{p}+\mathbf{q}}(\omega) \\ &= \sum_{\mathbf{p}} 2g\frac{\sqrt{n_{\mathrm{c}}}}{V}\frac{(f^0_{\mathbf{p}} - f^0_{\mathbf{p}+\mathbf{q}})[\delta\Phi(\mathbf{q},\omega) + \delta\Phi^*(\mathbf{q},\omega)]}{\hbar\omega - (\varepsilon_{\mathbf{p}+\mathbf{q}} - \varepsilon_{\mathbf{p}}) + i0^+} \\ &\equiv \chi^0_{\tilde{n}}(\mathbf{q},\omega)2g\delta n_{\mathrm{c}}(\mathbf{q},\omega). \end{aligned} \tag{5.41}$$

In the last line, $\chi^0_{\tilde{n}}(\mathbf{q},\omega)$ is identified as the well-known density response function for the HF atoms forming the thermal cloud,

$$\chi^0_{\tilde{n}}(\mathbf{q},\omega) = \sum_{\mathbf{p}} \frac{f^0_{\mathbf{p}} - f^0_{\mathbf{p}+\mathbf{q}}}{\hbar\omega - (\varepsilon_{\mathbf{p}+\mathbf{q}} - \varepsilon_{\mathbf{p}}) + i0^+}. \tag{5.42}$$

In (5.41), we have also taken advantage of the fact that, in a linearized theory, the fluctuations in the condensate $n_{\mathrm{c}} = |\Phi|^2$ are given by

$$\begin{aligned} n_{\mathrm{c}} &\simeq |\Phi_0|^2 + \Phi_0\,(\delta\Phi + \delta\Phi^*) \\ &= n_{\mathrm{c}} + \delta n_{\mathrm{c}}. \end{aligned} \tag{5.43}$$

We now summarize the structure of the preceding calculations, which have led to an alternative derivation of the equilibrium Beliaev second-order excitation spectrum given by (5.40). As noted above, Giorgini's coupled equations for the fluctuations in the condensate and noncondensate components can be simplified so as to reduce to the generalized GP and kinetic equations given in Chapter 3. A crucial approximation in our semiclassical microscopic theory was the neglect of the off-diagonal density $\tilde{m}(\mathbf{r}, t)$, which is usually referred to as the "Popov approximation" (with apologies to Popov, whose work generally emphasized rigorous results). In contrast, Giorgini's equations of motion start with a description of the condensate and noncondensate based on keeping all the HFB interaction terms, which are formally of first order in g. Consistently with this, we note that there are no collision integrals C_{12} and C_{22}, which involve terms of order g^2 in the generalized kinetic equations given by (5.37) and (5.38).

The key point is that, while the coupled equations (5.33), (5.37) and (5.38) are based on the first-order time-dependent HFB, treating the time-dependent mean fields for the fluctuations $\delta\Phi_0, \delta\Phi^*, \delta\tilde{n}$ and $\delta\tilde{m}$ within linear response theory enables us to generate an improved theory of the collective modes, namely, one with an excitation spectrum which is correct to order g^2. This collective mode spectrum turns out to be in precise agreement, as can be seen from (5.40), with the spectrum of the single-particle Green's functions $G_{\alpha\beta}$ computed in the Beliaev second-order approximation. Thus starting with the single-particle Beliaev Green's functions $G^{\rm HFB}_{\alpha\beta}$ based on HFB self-energies, one can generate a density response function whose excitation spectrum is identical to that of an improved approximation for $G_{\alpha\beta}$, namely $G^{\rm Bel}_{\alpha\beta}$.

This feature, of generating an improved (gapless) collective-mode spectrum using a time-dependent theory based on a lower-order approximation (which may have an unphysical single-particle spectrum) for mean fields, is well known in many body physics. What can cause confusion in Bose-condensed fluids is that one can prove (see, for example, Chapter 5 of Griffin, 1993) that the density response function shares precisely the same poles as the single-particle Beliaev Green's function $G_{\alpha\beta}$, owing to the hybridization effects of the Bose order parameter. In summary, the calculations discussed by Giorgini (2000) are a way of generating $G^{\rm Bel}_{\alpha\beta}$ starting from $G^{\rm HFB}_{\alpha\beta}$. This same physics is discussed by Hohenberg and Martin (1965) at $T = 0$ and by Cheung and Griffin (1971) at finite temperatures, using functional differentiation Green's function techniques based on (4.89) and (4.90).

In connection with the Beliaev approximation, we note again that the single-particle self-energies $\Sigma^{\rm Bel}_{\alpha\beta}(\mathbf{q}, \omega)$ do not include the diagrams of order g^2 that one deals with in a non-Bose-condensed gas (see equation (5.28) in Kadanoff and Baym, 1962). That is, above the superfluid transition, the "Beliaev approximation" reduces to the standard first-order HF self-energies, without any contributions of order g^2.

As we have emphasized, Giorgini (2000) starts with collisionless quantum kinetic equations for the distribution functions f_{ij} and g_{ij}. In the kinetic equation derived by Imamović-Tomasović and Griffin (2001) using the HFB single-particle spectrum (see Chapter 7), and by ZNG using the simpler HF spectrum (see Chapter 6), one finds a C_{12} collision integral of order g^2. As discussed by Kadanoff and Baym (1962, p. 119), kinetic equations traditionally treat the mean-field dynamics of the excitations on the l.h.s. to first order in the interactions, while the effects of collisions on the r.h.s. (which describes the transition rate from one state to another) involve terms of order g^2. If one solves kinetic equations that explicitly include the effects of

collisions (of order g^2) using a self-consistent time-dependent approach (of the kind Giorgini used), one generates a density response function that is better than the Beliaev approximation, since it includes the effect of collisions in a highly nonperturbative manner.

Of course, to obtain such a "post-Beliaev" spectrum one must first derive a kinetic equation for Beliaev excitations (i.e. based on $G^{\rm Bel}_{\alpha\beta}$ rather than $G^{\rm HFB}_{\alpha\beta}$). This gives a well-defined way of deriving an excitation spectrum which is the next improvement over the Beliaev approximation. The pioneering work of Kane and Kadanoff (1965) discussed a conserving approximation for the self-energies that included terms of order g^2. Thus their work in fact leads to a kinetic equation that could be used to derive *corrections* to the Beliaev excitation spectrum. Equally, one could calculate $\chi_{nn}(\mathbf{q},\omega)$ from (4.89) and (4.90) using $G^{\rm Bel}_{\alpha\beta}$ instead of $G^{\rm HFB}_{\alpha\beta}$ (such a calculation has not been carried out).

In Section 7.3, we will show that if we linearize the collision term C_{12} in the kinetic equation, we find that it leads to damping (of order g^2) of the HFB excitations that is precisely equal to the damping of the poles of $G^{\rm Bel}_{\alpha\beta}$. This result is to be expected since, as we show in Chapter 6 (see (6.42)), the C_{12} collision integral in a generalized kinetic equation can be expressed in terms of the *nonequilibrium* second-order Beliaev self-energies. When we linearize C_{12}, we obtain results related to the equilibrium Beliaev self-energies given in (5.40). These features are all consistent with the fact that, in the presence of a Bose condensate, the collective mode spectrum at any given level of approximaton must correspond to the spectrum of some improved approximation for the single-particle Green's functions (see Hohenberg and Martin, 1965; Chapter 5 of Griffin, 1993; Griffin, 1996).

5.4 Density response in the Beliaev–Popov approximation

In this section, we show that the time-dependent collisionless (which means setting C_{22}, C_{12} and hence Γ_{12} to zero) ZNG equations lead to a theory of collective modes that is the "Popov version" of the Beliaev excitation spectrum. More specifically, the density response function $\chi_{nn}(\mathbf{q},\omega)$ of a uniform Bose superfluid gas will be calculated within the Popov version of the Beliaev approximation. Cheung and Griffin (1971) calculated the density response function from (4.89) and (4.90) using $G^{\rm HFB}_{\alpha\beta}$. We now show that the Popov version of this paper, involving the omission of off-diagonal correlations, gives a density response identical to the mean-field or random phase approximation (RPA) calculation of Minguzzi and Tosi (1997). In the

following discussion, we assume the reader has some knowledge of the paper Cheung and Griffin (1971), which will be referred to as CG.[2]

We first summarize the mean-field linear response calculation of the density response function first given by Minguzzi and Tosi (1997). This discussion is based on the Popov approximation in that it neglects off-diagonal mean fields associated with $\tilde{m}(\mathbf{r},t)$. It is convenient to introduce two distinct external time-dependent potentials that couple into the condensate and noncondensate local density operators $n_{\rm c}(\mathbf{r})$ and $\tilde{n}(\mathbf{r})$, to give

$$U_{\rm ex} = \int d\mathbf{r}[\tilde{n}(\mathbf{r})\delta\tilde{U}(\mathbf{r},t) + n_{\rm c}(\mathbf{r})\delta U_{\rm c}(\mathbf{r},t)]. \tag{5.44}$$

Within a standard mean-field linear response calculation (for a uniform gas), we have

$$\begin{aligned}\delta n_{\rm c}(\mathbf{q},\omega) &= \chi_{\rm c}^0(\mathbf{q},\omega)[\delta U_{\rm c}(\mathbf{q},\omega) + g\delta n_{\rm c}(\mathbf{q},\omega) + 2g\delta\tilde{n}(\mathbf{q},\omega)],\\ \delta\tilde{n}(\mathbf{q},\omega) &= \chi_{\tilde{n}}^0(\mathbf{q},\omega)[\delta\tilde{U}(\mathbf{q},\omega) + 2g\delta n_{\rm c}(\mathbf{q},\omega) + 2g\delta\tilde{n}(\mathbf{q},\omega)].\end{aligned} \tag{5.45}$$

Here $\chi_{\rm c}^0$ and $\chi_{\tilde{n}}^0$ are the response functions of the condensate and noncondensate, defined later. They are calculated in the absence of Hartree and exchange interactions since these have been incorporated into the mean fields produced by the condensate and noncondensate, as can be seen in (5.45). The factors 2 in (5.45) are due to the equal effects of the Hartree and exchange fields in the case of a contact interaction of strength g. In contrast, the condensate feels only the condensate Hartree field because all its atoms are in the same single-particle quantum state. These different numerical factors (2 and 1) play an important role in a Bose-condensed gas at finite temperatures. The pair of equations (5.45) gives two coupled algebraic equations for the induced fluctuations in $\delta n_{\rm c}$ and $\delta\tilde{n}$. These are easily solved, with the result

$$\begin{aligned}\delta n_{\rm c}(\mathbf{q},\omega) &= \frac{\chi_{\rm c}^0(\mathbf{q},\omega)[1-2g\chi_{\tilde{n}}^0(\mathbf{q},\omega)]\delta U_{\rm c} + 2g\chi_{\rm c}^0(\mathbf{q},\omega)\chi_{\tilde{n}}^0(\mathbf{q},\omega)\delta\tilde{U}}{D(\mathbf{q},\omega)},\\ \delta\tilde{n}(\mathbf{q},\omega) &= \frac{\chi_{\tilde{n}}^0(\mathbf{q},\omega)[1-g\chi_{\rm c}^0(\mathbf{q},\omega)]\delta\tilde{U} + 2g\chi_{\rm c}^0(\mathbf{q},\omega)\chi_{\tilde{n}}^0(\mathbf{q},\omega)\delta U_{\rm c}}{D(\mathbf{q},\omega)},\end{aligned} \tag{5.46}$$

where the denominator is given by

$$\begin{aligned}D(\mathbf{q},\omega) &= (1-g\chi_{\rm c}^0)(1-2g\chi_{\tilde{n}}^0) - 4g^2\chi_{\tilde{n}}^0\chi_{\rm c}^0\\ &= 1 - g[2\chi_{\tilde{n}}^0 + \chi_{\rm c}^0(1+2g\chi_{\tilde{n}}^0)].\end{aligned} \tag{5.47}$$

The zeros of $D(\mathbf{q},\omega)$ determine the frequency of the collective oscillations.

[2] To avoid confusion, we note that CG set the mass of the atoms involved to unity ($m = 1$). This is common practice in the older theoretical literature on Bose gases.

The structure of (5.47) shows clearly how the condensate excitations hybridize with the noncondensate excitations. We will discuss this hybridized excitation spectrum later in the present section.

At this point, we can compare the mean-field results (5.46) with the exact linear response functions associated with a perturbation of the kind given in (5.44),

$$\begin{aligned} \delta n_{\rm c}(\mathbf{q},\omega) &= \chi_{cc}(\mathbf{q},\omega)\delta U_{\rm c}(\mathbf{q},\omega) + \chi_{c\tilde{n}}(\mathbf{q},\omega)\delta\tilde{U}(\mathbf{q},\omega), \\ \delta\tilde{n}(\mathbf{q},\omega) &= \chi_{\tilde{n}\tilde{n}}(\mathbf{q},\omega)\delta\tilde{U}(\mathbf{q},\omega) + \chi_{\tilde{n}c}(\mathbf{q},\omega)\delta U_{\rm c}(\mathbf{q},\omega). \end{aligned} \tag{5.48}$$

Here, for example, the response function $\chi_{c\tilde{n}}$ stands for a correlation function involving the local condensate and noncondensate density operators $n_{\rm c}(\mathbf{r})$ and $\tilde{n}(\mathbf{r})$. Comparing (5.48) with (5.46), one finds

$$\begin{aligned} \chi_{cc}(\mathbf{q},\omega) &= \frac{\chi_{\rm c}^0(1-2g\chi_{\tilde{n}}^0)}{D}, \\ \chi_{c\tilde{n}}(\mathbf{q},\omega) &= \chi_{\tilde{n}c}(\mathbf{q},\omega) = \frac{2g\chi_{\tilde{n}}^0\chi_{\rm c}^0}{D}, \\ \chi_{\tilde{n}\tilde{n}}(\mathbf{q},\omega) &= \frac{\chi_{\tilde{n}}^0(1-g\chi_{\rm c}^0)}{D}, \end{aligned} \tag{5.49}$$

where D is given by (5.47). Setting $n \equiv n_{\rm c} + \tilde{n}$, the total density response function is given by

$$\begin{aligned} \chi_{nn}(\mathbf{q},\omega) &\equiv \chi_{cc} + 2\chi_{\tilde{n}c} + \chi_{\tilde{n}\tilde{n}} \\ &= \frac{\chi_{\rm c}^0 + \chi_{\tilde{n}}^0 + g\chi_{\tilde{n}}^0\chi_{\rm c}^0}{D(\mathbf{q},\omega)}. \end{aligned} \tag{5.50}$$

This density response function $\chi_{nn}(\mathbf{q},\omega)$ is the same as $L_{11}(\mathbf{q},\omega)$ in (4.86) and in the Green's function analysis of Cheung and Griffin (1971).

The analogous results for a trapped gas found by Minguzzi and Tosi (1997) reduce precisely to (5.49) and (5.50) in the case of a uniform gas. In the analysis of Minguzzi and Tosi, however, the "noninteracting" response functions $\chi_{\rm c}^0$ and $\chi_{\tilde{n}}^0$ appearing in (5.45) are not uniquely specified. Two limiting cases of (5.50) are useful to work out. At $T = 0$, where there are no thermal atoms ($\tilde{n} = 0$) and hence $\chi_{\tilde{n}}^0(\mathbf{q},\omega)$ vanishes, (5.50) reduces to the expected Bogoliubov result for a pure condensate:

$$\chi_{nn}(\mathbf{q},\omega) \to \chi_{cc}(\mathbf{q},\omega) = \frac{\chi_{\rm c}^0}{1-g\chi_{\rm c}^0}. \tag{5.51}$$

Above the superfluid transition $T > T_{\rm BEC}$, where there is no condensate

($n_c = 0$) and hence $\chi_c^0(\mathbf{q},\omega) = 0$ vanishes, we have

$$\chi_{nn}(\mathbf{q},\omega) \to \chi_{\tilde{n}\tilde{n}}(\mathbf{q},\omega) = \frac{\chi_{\tilde{n}}^0}{1 - 2g\chi_{\tilde{n}}^0}. \tag{5.52}$$

We now turn to an alternative way, based on Green's functions, of deriving the above mean-field linear response results. This gives additional insight into their structure. Cheung and Griffin (1971) first extended the $T = 0$ calculations of Hohenberg and Martin (1965) (HM) to the case of finite temperatures. The density response function $\chi_{nn}(\mathbf{q},\omega)$ is calculated in both papers by taking the functional derivative of the single-particle HFB Beliaev Green's function with respect to a weak auxiliary scalar field $U(\mathbf{r},t)$, as in (4.89) and (4.90). This approach gives rise to a conserving (gapless) collective spectrum for $\chi_{nn}(\mathbf{q},\omega)$, even though the excitation spectrum of $G_{\alpha\beta}^{\rm HFB}(\mathbf{q},\omega)$ has an energy gap (in the $\mathbf{q} \to 0$ limit) as discussed in Section 5.1. The HM–CG procedure is clearly a more formal Green's function version of the analysis given by Giorgini (2000) reviewed in Section 5.3. In CG, the resulting density response function is evaluated to second order in g^2 and is shown to have an excitation spectrum identical to $G_{\alpha\beta}^{\rm Bel}(\mathbf{q},\omega)$, as given by the Beliaev second-order self-energies $\Sigma_{\alpha\beta}$ (see Section 5.2).

Using Green's functions, we will now review the formalism developed in CG. In the time-dependent HFB approximation, the first-order self-energies used in CG are given by (see Chapter 4 for the notation)

$$\hbar\Sigma_{\alpha\beta}^{\rm HFB}(1,1') = igG_{11}^{\rm HFB}(1,1')\delta(1-1')\delta_{\alpha\beta} + igG_{\alpha\beta}^{\rm HFB}(1,1')\delta(1-1'). \tag{5.53}$$

The first term is the Hartree self-energy and the second term is the exchange self-energy; $G_{\alpha\beta}^{\rm HFB}(1,1')$ as defined in (4.65) involves both the condensate and noncondensate contributions. Using the HFB self-energies (5.53) in (4.90), one generates coupled equations for the correlation functions $L_{\alpha\beta}$ and $\tilde{L}_{\alpha\beta}$ defined at the end of Section 4.3. These are given by equations (2.20)–(2.22) in CG, and it is shown that this set of equations can be used to find the density response function L_{11} in (4.86). In particular, in CG it is proved that the poles of this response function are identical to the poles of the single-particle Green's function $\tilde{G}_{\alpha\beta}(\mathbf{q},\omega)$ calculated within the Beliaev second-order approximation.

In the following analysis, we will simplify the CG calculation described above by ignoring the mean-field contributions coming from the off-diagonal density $\tilde{m}(\mathbf{r})$. We christen this theory the *Beliaev–Popov approximation* for the density response function. As might be expected, this approximation will lead precisely to the Minguzzi–Tosi (MT) results given above. The reason for

giving the present derivation is that it exhibits the MT mean-field theory as an approximation of microscopic many body theory. For more details about the Green's function techniques used, we refer the reader to CG as well as Chapter 4 of this book.

The CG analysis starts from the self-energies in (5.53). The Beliaev–Popov approximation corresponds to keeping the diagonal terms $\delta\Sigma^{\rm HFB}_{\sigma\sigma}/\delta U$ in the second term on the r.h.s. of (4.90); that is, the off-diagonal ($\sigma \neq \gamma$) terms are neglected, $\delta\Sigma^{\rm HFB}_{\sigma\gamma}/\delta U = 0$. One can show that the Popov approximation corresponds to setting $\tilde{L}_{12}$ and $\tilde{L}_{21}$ to zero. The final result is that equation (2.20) in CG reduces to (setting the generating time-dependent potential U to zero at the end)

$$\begin{aligned}\tilde{L}_{11}(\mathbf{q},\omega) = {} & \tilde{\chi}^0_{11}(\mathbf{q},\omega) + \tilde{\chi}^0_{12}(\mathbf{q},\omega) \\ & + g\tilde{\chi}^0_{11}(\mathbf{q},\omega)2L_{11}(\mathbf{q},\omega) \\ & + g\tilde{\chi}^0_{12}(\mathbf{q},\omega)[L_{22}(\mathbf{q},\omega) + L_{11}(\mathbf{q},\omega)]. \end{aligned} \tag{5.54}$$

The response functions in (5.54) are defined as follows:

$$\begin{aligned}\tilde{\chi}^0_{11}(1-2) &\equiv i\tilde{G}_{11}(1,2)\tilde{G}_{11}(2,1), \\ \tilde{\chi}^0_{12}(1-2) &\equiv i\tilde{G}_{12}(1,2)\tilde{G}_{21}(2,1),\end{aligned} \tag{5.55}$$

the equilibrium Green's functions $\tilde{G}_{11}$ and $\tilde{G}_{12}$ being given by the static HFB approximation. One can simplify (5.54) by using the exact identities for the correlation functions:

$$\begin{aligned}\tilde{G}_{12}(1,2) &= \tilde{G}_{21}(1,2), \\ L_{22}(1,2) &= L_{11}(1,2).\end{aligned} \tag{5.56}$$

Within the HFB self-energy approximation (5.53), the components of $L_{\rm S}$ in (4.89) are determined by the coupled equations (2.28) and (2.29) in CG. In our Beliaev–Popov approximation (in which, we recall, $L_{12} = L_{21} = 0$), these equations simplify and can be combined to give

$$L_{\rm S}(\mathbf{q},\omega) = \chi^0_{\rm c}(\mathbf{q},\omega)\left\{1 + g[L_{11}(\mathbf{q},\omega) + \tilde{L}_{11}(\mathbf{q},\omega)]\right\}, \tag{5.57}$$

where we have introduced the condensate propagator

$$\chi^0_{\rm c}(\mathbf{q},\omega) \equiv \frac{2\varepsilon_{\mathbf{q}} n_{\rm c}}{(\hbar\omega)^2 - \varepsilon^2_{\mathbf{q}}}. \tag{5.58}$$

Before proceeding, several comments should be made. The correlation functions defined in (5.55) involve the single-particle Green's functions based on the static HFB approximation, which exhibits an energy spectrum with a gap in the $\mathbf{q} \to 0$ limit (see Section 5.1). However, in the HFP approximation

these single-particle Green's functions exhibit a gapless spectrum, namely the Bogoliubov–Popov spectrum, with chemical potential

$$\mu = gn_{\rm c} + 2g\tilde{n}. \tag{5.59}$$

A second comment is that while the expression (5.58) appears to be the condensate response for a noninteracting Bose gas, in fact it arises in our HFP approximation. In particular, the nontrivial result (5.59) is required to arrive at the expression for $L_{\rm S}$ given in (5.57).

The coupled algebraic equations in (5.54) and (5.57) can be easily solved. One can write the solutions in several alternative ways. From (4.87) and (5.54), one obtains

$$\begin{aligned} L_{11}(\mathbf{q},\omega) &= \tilde{L}_{11} + L_{\rm S} \\ &= \frac{\bar{\chi}^0 + L_{\rm S}}{1 - 2g\bar{\chi}^0}, \end{aligned} \tag{5.60}$$

where we define

$$\bar{\chi}^0(\mathbf{q},\omega) \equiv \tilde{\chi}^0_{11}(\mathbf{q},\omega) + \tilde{\chi}^0_{12}(\mathbf{q},\omega). \tag{5.61}$$

Using (5.57), one can show that the condensate part of the total density response function is given by

$$\begin{aligned} L_{\rm S}(\mathbf{q},\omega) &= \frac{\chi^0_{\rm c}}{1 - g\chi^0_{\rm c}} \left(1 + 2g\tilde{L}_{11}\right) \\ &= \frac{2n_{\rm c}\varepsilon_{\mathbf{q}}}{(\hbar\omega)^2 - (E^{\rm B}_{\mathbf{q}})^2} \left(1 + 2g\tilde{L}_{11}\right), \end{aligned} \tag{5.62}$$

where

$$E^{\rm B}_{\mathbf{q}} = (\varepsilon^2_{\mathbf{q}} + 2gn_{\rm c}\varepsilon_{\mathbf{q}})^{1/2} \tag{5.63}$$

is the usual Bogoliubov–Popov excitation energy. The expression (5.62) makes clear how the two parts of the density response function ($L_{\rm S}$ and $\tilde{L}_{11}$) are coupled together. The results (5.60) and (5.62) reduce to the expected results for $T > T_{\rm BEC}$ (where $L_{\rm S} = 0$ and $\tilde{G}_{12} = 0$). At $T = 0$ (where $\tilde{L}_{11} = 0$) we see that $L_{\rm S}$ reduces to χ_{cc} in (5.51), as expected.

Combining (5.54) and (5.57) as two equations for $\tilde{L}_{11}$ and $L_{\rm S}$, we can eliminate $\tilde{L}_{11}$ and obtain an explicit expression for $L_{\rm S}$,

$$L_{\rm S}(\mathbf{q},\omega) = \frac{2n_{\rm c}\varepsilon_{\mathbf{q}}}{D'(\mathbf{q},\omega)}, \tag{5.64}$$

where the denominator D' is defined as follows:

$$D'(\mathbf{q},\omega) \equiv (1 - 2g\bar{\chi}^0)[(\hbar\omega)^2 - (E^{\rm B}_q)^2] - 8g^2 n_{\rm c}\varepsilon_{\mathbf{q}}\bar{\chi}^0. \tag{5.65}$$

Using (5.64) in (5.60), one can show that the total density response function can be reduced to

$$L_{11}(\mathbf{q},\omega) = \frac{2n_{\rm c}\varepsilon_{\mathbf{q}} + \bar{\chi}^0[(\hbar\omega)^2 - (\varepsilon_{\mathbf{q}}^2 - 2gn_{\rm c}\varepsilon_{\mathbf{q}})]}{D'(\mathbf{q},\omega)}. \tag{5.66}$$

If one leaves out the contribution χ_{12}^0 in the Beliaev–Popov approximation for $\bar{\chi}^0$ in (5.61), setting $\bar{\chi}^0 = \tilde{\chi}_{11}^0 \equiv \chi_{\tilde{n}}^0$, one can show that the expression in (5.66) is identical to the Minguzzi–Tosi linear response result given in (5.50). Within this approximation for $\bar{\chi}^0$, one can show that the Minguzzi–Tosi denominator $D(\mathbf{q},\omega)$ in (5.47) is related to $D'(\mathbf{q},\omega)$ in (5.65) as follows:

$$\begin{aligned} D' &= \left(1 - 2g\bar{\chi}^0\right)\left[(\hbar\omega)^2 - \varepsilon_{\mathbf{q}}^2\right] - 2gn_{\rm c}\varepsilon_{\mathbf{q}}\left(1 + 2g\bar{\chi}^0\right) \\ &= \left[(\hbar\omega)^2 - \varepsilon_{\mathbf{q}}^2\right] D. \end{aligned} \tag{5.67}$$

Using this relation, one can show that the Beliaev–Popov expression for L_{11} in (5.66) can be written in the form:

$$L_{11}(\mathbf{q},\omega) = \frac{\bar{\chi}^0 + \chi_{\rm c}^0(1 + g\bar{\chi}^0)}{1 - g[2\bar{\chi}^0 + \chi_{\rm c}^0(1 + 2g\bar{\chi}^0)]}. \tag{5.68}$$

This is precisely the result (5.50) obtained from the linear response theory, as developed by Minguzzi and Tosi (1997).

The zeros of the denominator $D'(\mathbf{q},\omega)$ in (5.67) give the hybridized collective mode spectrum (at finite temperatures) of the Beliaev–Popov density response function. Working it out, one finds that the final result can be written as

$$(\hbar\omega)^2 = \varepsilon_{\mathbf{q}}^2 + 2gn_{\rm c}\varepsilon_{\mathbf{q}}\left[\frac{1 + 2g\bar{\chi}^0(\mathbf{q},\omega)}{1 - 2g\bar{\chi}^0(\mathbf{q},\omega)}\right]. \tag{5.69}$$

This is the usual Bogoliubov spectrum, but with the interaction g replaced by g' as defined in (5.31). In the limit $\mathbf{q} \to 0$ and weak interactions, this expression can be approximated by

$$(\hbar\omega)^2 = (c_0 q)^2[1 + 4g\bar{\chi}^0(\mathbf{q}, \omega = c_0 q/\hbar)], \tag{5.70}$$

where $c_0^2 \equiv n_{\rm c}(T)g/m$ is the phonon velocity in the Bogoliubov–Popov approximation. We recall that $\bar{\chi}^0(\mathbf{q},\omega)$ has real and imaginary parts. The expression in (5.70) is used in Section 13.1 in our discussion of Landau damping.

The Hartree and exchange contributions can be easily distinguished in our explicit results for $L_{11}(\mathbf{q},\omega)$ given above. In particular, if we consider only the *Hartree* self-energies, then we find the expected RPA result (compare

with the expression in (5.68))

$$L_{11}(\mathbf{q},\omega) = \frac{\bar{\chi}^0 + \chi_{\rm c}^0}{1 - g(\bar{\chi}^0 + \chi_{\rm c}^0)}. \tag{5.71}$$

In this limit, the phonon spectrum analogous to the Hartree–Fock result (5.70) can be shown to reduce to

$$(\hbar\omega)^2 = (c_0 q)^2[1 + g\bar{\chi}^0(\mathbf{q}, \omega = c_0 q/\hbar)]. \tag{5.72}$$

Thus, the factor 4 in (5.70) must arise from the inclusion of exchange mean fields coming from the noncondensate atoms.

We can relate the MT linear response mean-field results given in (5.49) to the condensate and noncondensate components $L_{\rm S}$ and $\tilde{L}_{11}$ calculated from the CG theory. By direct comparison with our results based on a Popov approximation of the more complete Beliaev theory discussed in CG, we obtain

$$\begin{aligned} L_{\rm S} &= \frac{\chi_{\rm c}^0}{D} = \chi_{cc} + \chi_{c\tilde{n}}, \\ \tilde{L}_{11} &= \frac{\chi_{\tilde{n}}^0(1 + g\chi_{\rm c}^0)}{D} = \chi_{\tilde{n}c} + \chi_{\tilde{n}\tilde{n}}. \end{aligned} \tag{5.73}$$

As in our derivation of (5.68), we have approximated $\bar{\chi}_0$ in (5.61) by $\chi_{\tilde{n}}^0$ as given in (5.45). The MT response functions $\chi_{\tilde{n}c}, \chi_{cc}$ and $\chi_{\tilde{n}\tilde{n}}$ in (5.73) are given in (5.49).

Fliesser *et al.* (2001) used the dielectric formalism (see Section 4.5) to evaluate both the density response function $L_{11}(\mathbf{q},\omega)$ and the single-particle Green's function $\tilde{G}_{\alpha\beta}(\mathbf{q},\omega)$. As noted in Section 4.5, this powerful approach ensures that both $L_{11}(\mathbf{q},\omega)$ and $\tilde{G}_{\alpha\beta}(\mathbf{q},\omega)$ have the same poles, thus giving an approximation which is both conserving and gapless. Reidl *et al.* (2000) and Fliesser *et al.* (2001) obtained results for $L_{11}(\mathbf{q},\omega)$ that agree precisely with those of MT as given by (5.68). Thus their analysis may be viewed as a microscopic basis for the time-dependent Hartree–Fock theory used in MT.

The results given in this section are only valid in the Beliaev–Popov approximation for the density response function as given in (5.54). However, they have the advantage of exhibiting, in an explicit manner, how the condensate modes at $T = 0$ hybridize with the dynamics of the thermal cloud at finite temperatures. The dynamical role of the symmetry-breaking field associated with a Bose condensate is also made clear in the detailed structure of the response functions given in this section.

Liu *et al.* (2004) extended the linear response dynamic mean-field approach of MT to include off-diagonal mean fields associated with the anomalous correlation functions $\tilde{m}(\mathbf{r}, t)$ in a trapped gas at finite temperatures. As expected, the spectrum of the resulting collective modes agrees with the results of Giorgini (2000) discussed in Section 5.3 and with the finite-temperature version of the diagrammatic approach of Beliaev (1958b) for the case of a trapped Bose gas (Fedichev and Shlyapnikov, 1998; Morgan, 2005).

6

Kadanoff–Baym derivation of the ZNG equations

In Chapter 4, we introduced the Kadanoff–Baym equations of motion for the imaginary-time nonequilibrium Green's functions for a Bose gas, as given by (4.59) and (4.60). In this chapter, we will use the generalization of these equations of motion to find the equivalent equations of motion for the real-time Green's functions. These can be written in a natural way in the form of a kinetic equation. Using a simple Hartree–Fock approximation, we show how the coupled equations for the condensate and thermal cloud given in Chapter 3 emerge naturally from the Kadanoff–Baym (KB) formalism. This chapter is based on Imamović-Tomasović and Griffin (2001) and Imamović-Tomasović (2001), building on the pioneering work of Kane and Kadanoff (1965).

In this chapter and Chapter 7 we review the KB formalism. However, we also encourage the reader to read the original account given by Kadanoff and Baym (1962). The goals and accomplishments of their seminal book are beautifully captured by the following quote from p. 138:

> Our rather elaborate Green's function arguments provide a means of describing transport phenomena in a self-contained way, starting from a dynamical approximation, i.e. an approximation for $G_2(U)$ in terms of $G_1(U)$. These calculations require no extra assumptions. The theory provides at the same time a description of the transport processes that occur and a determination of the quantities which appear in the transport equations.

A closely related way of treating the nonequilibrium dynamics of a Bose-condensed gas is based on the two-particle irreducible (2PI) effective action together with the Schwinger–Keldysh closed-time path formalism. Berges (2004) gives a detailed review of this approach, which allows one to derive the nonequilibrium action on the basis of controllable approximations. Minimizing the 2PI effective action gives equations of motion for the condensate wavefunction and noncondensate Green's functions. Baier and Stock-

amp (2004) used this 2PI formalism to derive a kinetic equation that is equivalent to the ZNG kinetic equations based on the Hartree–Fock excitation spectrum, to be discussed in this chapter.

6.1 Kadanoff–Baym formalism for Bose superfluids

We first discuss the key question, how the imaginary-time Green's functions G defined in (4.40) and the *real-time* Green's functions g defined in (4.44) are related. It is useful to write $G(1,1';U;t_0)$ in the following form, assuming that $i(t_1 - t_0) < i(t_{1'} - t_0)$:

$$\begin{aligned} &G(1,1';U;t_0) \equiv G^<(1,1';U;t_0) \\ &= -i\frac{\left\langle \mathcal{U}(t_0,t_0-i\beta)[\mathcal{U}^\dagger(t_0,t_{1'})\psi^\dagger(1')\mathcal{U}(t_0,t_{1'})]\,\mathcal{U}^\dagger(t_0,t_1)\psi(1)\,\mathcal{U}(t_0,t_1)\right\rangle}{\langle \mathcal{U}(t_0,t_0-i\beta)\rangle}. \end{aligned} \tag{6.1}$$

We may compare this with the expression for $g^<(1,1';U)$, namely

$$g^<(1,1';U) = -i\left\langle v^\dagger(t_1)\psi(1)v(t_1)v^\dagger(t_1')\psi^\dagger(1')v(t_1')\right\rangle. \tag{6.2}$$

The evolution operators $\mathcal{U}(t,t_0)$ and $v(t)$ are defined in (4.33) and (4.37), respectively.[1]

One sees directly that the real-time Green's function $g(1,1';U)$ is identical to $G(1,1';U;t_0)$ in the limit $t_0 \to -\infty$, i.e.

$$\begin{aligned} \lim_{t_0\to-\infty} G^<(1,1';U;t_0) &= g^<(1,1';U) \\ \lim_{t_0\to-\infty} G^>(1,1';U;t_0) &= g^>(1,1';U). \end{aligned} \tag{6.3}$$

These are the key formulas of the KB formalism, relating an imaginary-time response function to a real-time response function. In the KB approach, to find equations of motion for response functions defined by the real-time Green's functions g, one starts with equations of motion for the related imaginary-time Green's functions G. The real-time Green's functions are then obtained by analytic continuation. One writes an imaginary-time Green's function as a Fourier series over the discrete frequencies. Starting from the Fourier coefficients defined on the discrete set of imaginary frequencies, an analytical continuation to all frequencies gives us the required functions on the real frequency axis. From these, we can obtain the real-time Green's functions $g(1,1',U)$.

[1] In this chapter, the operators $\mathcal{U}(t,t_0)$ and $v(t)$ are not denoted by a caret, as they are in Chapter 4. Later in this chapter, a caret over a function denotes a 2×2 matrix which is needed to describe Bose superfluids (see Chapter 4).

We denote the collisional self-energy contributions (those of second and higher order in g) as $\Sigma_{\rm coll}$, and define

$$\Sigma_{\rm coll}(1,1';U;t_0) = \begin{cases} \Sigma^{>}_{\rm coll}(1,1';U;t_0), & \text{if } i(t_1 - t_{1'}) > 0 \\ \\ \Sigma^{<}_{\rm coll}(1,1';U;t_0), & \text{if } i(t_1 - t_{1'}) < 0. \end{cases} \tag{6.4}$$

Physical response functions are related to the correlation functions defined in (4.11). Therefore, we write (4.59) for the case $i(t_1 - t_0) < i(t_{1'} - t_0)$, obtaining

$$\begin{aligned} \left[i\hbar\frac{\partial}{\partial t_1} + \frac{\hbar^2\nabla_1^2}{2m} - U_{\rm eff}(1;t_0) \right] G^{<}(1,1';U;t_0) \\ = \int_{t_0}^{t_1} d\bar{1}\, \Sigma^{>}_{\rm coll}(1,\bar{1};U;t_0) G^{<}(\bar{1},1';U;t_0) \\ + \int_{t_1}^{t_{1'}} d\bar{1}\, \Sigma^{<}_{\rm coll}(1,\bar{1};U;t_0) G^{<}(\bar{1},1';U;t_0) \\ + \int_{t_{1'}}^{t_0 - i\beta} d\bar{1}\, \Sigma^{<}_{\rm coll}(1,\bar{1};U;t_0) G^{>}(\bar{1},1';U;t_0). \end{aligned} \tag{6.5}$$

Here $U_{\rm eff}(1)$ is the effective mean field which contains an external potential U, the Hartree–Fock part of the self-energy $\Sigma_{\rm HF}$ and the chemical potential μ_0. To obtain the real-time Green's function (4.44), we take the limit $t_0 \to -\infty$ of (6.5) and use a key identity from (6.3). The result is the equation of motion

$$\begin{aligned} \left[i\hbar\frac{\partial}{\partial t_1} + \frac{\hbar^2\nabla_1^2}{2m} - U_{\rm eff}(1) \right] g^{<}(1,1';U) \\ = \int_{-\infty}^{t_1} d\bar{1}\, \left[\Sigma^{>}_{\rm coll}(1,\bar{1};U) - \Sigma^{<}_{\rm coll}(1,\bar{1};U) \right] g^{<}(\bar{1},1';U) \\ - \int_{-\infty}^{t_{1'}} d\bar{1} \Sigma^{<}_{\rm coll}(1,\bar{1};U) \left[g^{>}(\bar{1},1';U) - g^{<}(\bar{1},1';U) \right], \end{aligned} \tag{6.6}$$

where we define

$$\begin{aligned} U_{\rm eff}(1) &\equiv U_{\rm eff}(1;-\infty), \\ \Sigma^{<}(1,\bar{1};U) &\equiv \Sigma^{<}(1,\bar{1};U;-\infty). \end{aligned} \tag{6.7}$$

Now, considering (4.60) for $i(t_1 - t_0) < i(t_{1'} - t_0)$, the analogous equation

of motion for $g^<(1,1')$ with respect to the other variable $1'$ is

$$\begin{aligned}
&\left[-i\hbar\frac{\partial}{\partial t_{1'}} + \frac{\hbar^2\nabla_{1'}^2}{2m} - U_{\rm eff}(1')\right] g^<(1,1';U) \\
&= \int_{-\infty}^{t_1} d\bar{1}\,\left[g^>(1,\bar{1};U) - g^<(1,\bar{1};U)\right]\Sigma^<_{\rm coll}(\bar{1},1;U) \\
&\quad - \int_{-\infty}^{t_{1'}} d\bar{1}\, g^<(1,\bar{1};U)\left[\Sigma^>_{\rm coll}(\bar{1},1';U) - \Sigma^<_{\rm coll}(\bar{1},1';U)\right]. \qquad (6.8)
\end{aligned}$$

Applying the same arguments as above, we can find similar equations of motion for $g^>$ in the case $i(t_1 - t_0) > i(t_{1'} - t_0)$, namely

$$\begin{aligned}
&\left[i\hbar\frac{\partial}{\partial t_1} + \frac{\hbar^2\nabla_1^2}{2m} - U_{\rm eff}(1)\right] g^>(1,1';U) \\
&= \int_{-\infty}^{t_1} d\bar{1}\,\left[\Sigma^>_{\rm coll}(1,\bar{1};U) - \Sigma^<_{\rm coll}(1,\bar{1};U)\right] g^>(\bar{1},1';U) \\
&\quad - \int_{-\infty}^{t_{1'}} d\bar{1}\,\Sigma^>_{\rm coll}(1,\bar{1};U)\left[g^>(\bar{1},1';U) - g^<(\bar{1},1';U)\right] \qquad (6.9)
\end{aligned}$$

and

$$\begin{aligned}
&\left[-i\hbar\frac{\partial}{\partial t_{1'}} + \frac{\hbar^2\nabla_{1'}^2}{2m} - U_{\rm eff}(1')\right] g^>(1,1';U) \\
&= \int_{-\infty}^{t_1} d\bar{1}\,\left[g^>(1,\bar{1};U) - g^<(1,\bar{1};U)\right]\Sigma^>_c(\bar{1},1;U) \\
&\quad - \int_{-\infty}^{t_{1'}} d\bar{1}\, g^>(1,\bar{1};U)\left[\Sigma^>_{\rm coll}(\bar{1},1';U) - \Sigma^<_{\rm coll}(\bar{1},1';U)\right]. \qquad (6.10)
\end{aligned}$$

The equations of motion derived above can be generalized to describe the 2×2 real-time Green's function matrix $\hat{\tilde{g}}$ for the noncondensate atoms (see Section 4.3). These equations are:

$$\begin{aligned}
&\int d\bar{1}\left[\hat{g}_0^{-1}(1,\bar{1}) - \hat{\Sigma}_{\rm HFB}(1,\bar{1})\right]\hat{\tilde{g}}^<(\bar{1},1') \\
&= \int_{-\infty}^{t_1} d\bar{1}\,\hat{\Gamma}(1,\bar{1})\hat{\tilde{g}}^<(\bar{1},1') - \int_{-\infty}^{t_{1'}} d\bar{1}\,\hat{\Sigma}^<_{\rm coll}(1,\bar{1})\hat{a}(\bar{1},1') \qquad (6.11)
\end{aligned}$$

and

$$\begin{aligned}
&\int d\bar{1}\,\hat{\tilde{g}}^<(1,\bar{1})\left[\hat{g}_0^{-1}(\bar{1},1') - \hat{\Sigma}_{\rm HFB}(\bar{1},1')\right] \\
&= \int_{-\infty}^{t_1} d\bar{1}\,\hat{a}(1,\bar{1})\hat{\Sigma}^<_{\rm coll}(\bar{1},1') - \int_{-\infty}^{t_{1'}} d\bar{1}\,\hat{\tilde{g}}^<(1,\bar{1})\,\hat{\Gamma}(\bar{1},1'). \qquad (6.12)
\end{aligned}$$

Here the 2×2 matrices $\hat{a}(1,1')$ and $\hat{\Gamma}(1,1')$ are defined by the matrix elements

$$\begin{aligned} a_{\alpha\beta}(1,1') &\equiv \tilde{g}^{>}_{\alpha\beta}(1,1') - \tilde{g}^{<}_{\alpha\beta}(1,1'), \\ \Gamma_{\alpha\beta}(1,1') &\equiv \Sigma^{>}_{\alpha\beta}(1,1') - \Sigma^{<}_{\alpha\beta}(1,1'). \end{aligned} \tag{6.13}$$

The single-particle self-energy in (6.11) and (6.12) can be split into two parts (Kane and Kadanoff, 1965)

$$\hat{\Sigma}(1,1') = \hat{\Sigma}_{\rm HF}(1,1') + \hat{\Sigma}_{\rm coll}(1,1'), \tag{6.14}$$

where the Hartree–Fock self energy $\hat{\Sigma}_{\rm HF}$ is given by (4.57). In (6.11) and (6.12), the self-energies $\hat{\Sigma}^{>}_{\rm coll}$ and $\hat{\Sigma}^{<}_{\rm coll}$ refer to the second-order collisional self-energy $\hat{\Sigma}_{\rm coll}$. We have thus obtained the nonequilibrium form of the Dyson–Beliaev equations of motion for the noncondensate atoms. These depend on the nonequilibrium condensate wavefunction, which is governed by its own equation of motion. In thermal equilibrium, these equations reduce to the usual Beliaev equations discussed in many body theory texts (Abrikosov *et al.*, 1963; Fetter and Walecka, 1971). They are the starting point for our derivation of Boltzmann-like transport equations in this chapter, as well as in Chapter 7.

6.2 Hartree–Fock–Bogoliubov equations

In Section 5.1, we derived the nonequilibrium Dyson–Beliaev equations of motion for the noncondensate atoms, given by (6.11) and (6.12), and in Section 4.3 the equation of motion (4.71) for the condensate. In this section, we will omit the second-order self-energy contribution $\hat{\Sigma}_{\rm coll}$. The resulting HFB equations of motion can be conveniently written in 2×2 matrix form for the noncondensate atoms

$$\begin{aligned} \int d\bar{1} \left[\hat{g}_0^{-1}(1,\bar{1}) - \hat{\Sigma}_{\rm HFB}(1,\bar{1})\right] \hat{\tilde{g}}^{<}(\bar{1},1') &= 0, \\ \int d\bar{1}\, \hat{\tilde{g}}^{<}(1,\bar{1}) \left[\hat{g}_0^{-1}(\bar{1},1') - \hat{\Sigma}_{\rm HFB}(\bar{1},1')\right] &= 0, \end{aligned} \tag{6.15}$$

and for the condensate atoms

$$\int d\bar{1} \left[\hat{g}_0^{-1}(1,\bar{1}) - \hat{S}_{\rm HFB}(1,\bar{1})\right] \hat{h}(\bar{1},1') = -i\hat{\eta}_{\rm ext}(1)\langle\hat{\Psi}^{\dagger}(1')\rangle. \tag{6.16}$$

In the HFB approximation, the 2×2 self-energies in (6.15) and (6.16) are

$$\hat{\Sigma}_{\rm HFB}(1,1') = g \begin{pmatrix} 2n(1) & m(1) \\ m^*(1) & 2n(1) \end{pmatrix} \delta(1-1'),$$
$$\hat{S}_{\rm HFB}(1,1') = g \begin{pmatrix} 2\tilde{n}(1)+n_{\rm c}(1) & \tilde{m}(1) \\ \tilde{m}^*(1) & 2\tilde{n}(1)+n_{\rm c}(1) \end{pmatrix} \delta(1-1'). \tag{6.17}$$

In the above equations, $n(1)$ is the *total* local density given by

$$\begin{aligned} n(1) &\equiv i\left[\tilde{g}^<_{11}(1,1^+) + h_{11}(1,1)\right] \\ &= \langle \tilde{\psi}^\dagger(1)\tilde{\psi}(1)\rangle + |\Phi(1)|^2 \equiv \tilde{n}(1) + n_{\rm c}(1), \end{aligned} \tag{6.18}$$

where $\tilde{n}(1)$ and $n_{\rm c}(1)$ are the noncondensate and condensate densities, respectively. Similarly, $m(1)$ is the "anomalous" local density, defined by

$$\begin{aligned} m(1) &\equiv i\left[\tilde{g}^<_{12}(1,1) + h_{12}(1,1)\right] \\ &= \langle \tilde{\psi}(1)\tilde{\psi}(1)\rangle + [\Phi(1)]^2 \\ &= \tilde{m}(1) + [\Phi(1)]^2 . \end{aligned} \tag{6.19}$$

In (6.18) and (6.19), we have used the definitions of $G^>$ and $G^<$ in (4.11).

The results (6.15)–(6.19) correspond to the nonequilibrium time-dependent version of the *static* HFB approximation given by (4.81)–(4.84). These HFB results are identical to the equations obtained by Kane and Kadanoff (1965). The associated equations of motion for the order parameter $\Phi(1)$ are

$$\begin{aligned} \left\{ i\hbar\frac{\partial}{\partial t_1} + \frac{\hbar^2\nabla_1^2}{2m} + \mu_0 - V_{\rm trap}(\mathbf{r}_1) - U(1) - g\left[2\tilde{n}(1) + n_{\rm c}(1)\right] \right\} \Phi(1) \\ = g\tilde{m}(1)\Phi^*(1) + \eta(1), \end{aligned} \tag{6.20}$$

and its complex conjugate. Equations (6.15)–(6.20) form a closed set of equations and give the *dynamic* HFB approximation, the densities $\tilde{n}, n_{\rm c}$ and $\tilde{m}$ all being time-dependent. The *static* HFB approximation is discussed in Section 4.3, and is summarized by (4.81)–(4.84).

We now turn to solving these HFB equations of motion for atoms in the presence of a trapping potential. If the external generating fields induce a disturbance with a wavelength much longer than thermal wavelengths and frequencies much smaller than characteristic particle energies then the propagator $g(1,1') = g(\mathbf{r},t;\mathbf{R},T)$ can be expected to vary *slowly* as a function of the *centre-of-mass* coordinates

$$\mathbf{R} = \tfrac{1}{2}(\mathbf{r}_1 + \mathbf{r}_{1'}), \quad T = \tfrac{1}{2}(t_1 + t_{1'}) \tag{6.21}$$

and to be dominated by small values of the *relative* coordinates

$$\mathbf{r} = \mathbf{r}_1 - \mathbf{r}_{1'}, \quad t = t_1 - t_{1'}. \tag{6.22}$$

More precisely, Fourier transforming with respect to $\mathbf{r}$ and t, the function $g(\mathbf{p}, \omega; \mathbf{R}, T)$ describes the density of elementary excitations of momentum $\mathbf{p}$ and energy ω at the point $(\mathbf{R}, T)$. These quasiparticles are assumed to have high momentum and energy (relative to the collective modes that we will discuss using the kinetic equations), which means that the single-particle Green's function $g(\mathbf{r}, t; \mathbf{R}, T)$ is weighted mainly at small values of the relative coordinates $\mathbf{r}$ and t. This is a crucial assumption in the ensuing discussion.

If the Bose-condensate order parameter $\langle\psi(1)\rangle$ is written in terms of amplitude and phase variables

$$\langle\psi(1)\rangle = \sqrt{n_{\rm c}(1)}e^{i\theta(1)}, \tag{6.23}$$

we can generalize the usual definitions for the superfluid velocity and local chemical potential to nonequilibrium systems by writing down the following identities (Kane and Kadanoff, 1965):

$$\begin{aligned} \nabla_1\theta(1) &\equiv m\mathbf{v}_{\rm s}(1) \\ \frac{\partial\theta(1)}{\partial t_1} &\equiv -\left[\mu(1) - \mu_0 + \tfrac{1}{2}m{\mathbf{v}_{\rm s}}^2(1)\right]. \end{aligned} \tag{6.24}$$

The superfluid velocity $\mathbf{v}_{\rm s}$ enters as the gradient of the phase of the condensate wavefunction, and the local chemical potential $\mu(1)$ is determined by the time derivative of the phase.

However, a problem arises in that the phase is a rapidly varying function of the centre-of-mass coordinates $\mathbf{R}, T$ that induces strong variations in the off-diagonal elements of the condensate propagator $\hat{h}$ in (4.67), and these are then coupled to the components of the noncondensate propagator $\tilde{g}$. To remove this strong $(\mathbf{R}, T)$-dependence from the phase θ in (6.23), we apply well-known (Kane and Kadanoff, 1965) gauge transformations to $\hat{h}(1, 1')$ and $\hat{\tilde{g}}^{<}_{>}(1, 1')$:

$$\begin{aligned} \hat{h}'(1, 1') &= e^{-i\theta(1)\tau_3}\hat{h}(1, 1')e^{i\theta(1)\tau_3}, \\ \hat{\tilde{g}'}^{<}_{>}(1, 1') &= e^{-i\theta(1)\tau_3}\hat{\tilde{g}}^{<}_{>}(1, 1')e^{i\theta(1)\tau_3}, \end{aligned} \tag{6.25}$$

where τ_3 is the Pauli spin matrix given below (4.73). The primes refer to the transformed quantities. The physical interpretation of the equations (6.25) is that they involve a transformation to a coordinate system in which noncondensate atoms are moving with average velocity $\mathbf{v}_{\rm s}$ with respect to a

stationary condensate. The gauge transformation (6.25) removes the strong $(\mathbf{R}, T)$-dependence associated with the order parameter phase θ and leaves the equations of motion (6.15)–(6.20) invariant if we replace g_0^{-1} (the real-time version of G_0^{-1} in (4.73)) by

$$g_0'^{-1}(1,1') = \Big\{ i\hbar\tau_3 \frac{\partial}{\partial t_1} - \frac{\partial\theta(1)}{\partial t_1} + \frac{\hbar^2}{2m}[\nabla_1 + i\tau_3\nabla_1\theta(1)]^2 - V_{\rm trap}(\mathbf{r}_1) - U(1) + \mu_0 \Big\}\delta(1,1'). \tag{6.26}$$

The gauge transformation (6.25) changes the momentum $\mathbf{p}$ to $\mathbf{p} - m\mathbf{v}_{\rm s}$, as expected for the momentum in the local rest frame. After carrying out this gauge transformation, the HFB equations for $\tilde{g}^{<}_{11}(1,1';U)$ and $\tilde{g}^{<}_{12}(1,1';U)$ are (again showing the transformed quantities by primes)

$$\begin{aligned}
&\Big\{ i\hbar\frac{\partial}{\partial t_1} - \frac{\partial\theta(1)}{\partial t_1} + \frac{\hbar^2}{2m}[\nabla_1 + i\,\nabla_1\theta(1)]^2 + \mu_0 - U_{\rm eff}(1) \Big\}\tilde{g}'^{<}_{11}(1,1';U) \\
&\qquad = gm'(1)\tilde{g}'^{<}_{21}(1,1';U) \\
&\Big\{ i\hbar\frac{\partial}{\partial t_1} - \frac{\partial\theta(1)}{\partial t_1} + \frac{\hbar^2}{2m}[\nabla_1 + i\,\nabla_1\theta(1)]^2 + \mu_0 - U_{\rm eff}(1) \Big\}\tilde{g}'^{<}_{12}(1,1';U) \\
&\qquad = gm'(1)\tilde{g}'^{<}_{22}(1,1';U) \\
&\Big\{ -i\hbar\frac{\partial}{\partial t_{1'}} - \frac{\partial\theta(1')}{\partial t_{1'}} + \frac{\hbar^2}{2m}[\nabla_{1'} - i\,\nabla_{1'}\theta(1')]^2 + \mu_0 - U_{\rm eff}(1') \Big\}\tilde{g}'^{<}_{11}(1,1';U) \\
&\qquad = gm'^{*}(1')\tilde{g}'^{<}_{12}(1,1';U) \\
&\Big\{ i\hbar\frac{\partial}{\partial t_{1'}} - \frac{\partial\theta(1')}{\partial t_{1'}} + \frac{\hbar^2}{2m}[\nabla_{1'} + i\,\nabla_{1'}\theta(1')]^2 + \mu_0 - U_{\rm eff}(1') \Big\}\tilde{g}'^{<}_{12}(1,1';U) \\
&\qquad = gm'(1')\tilde{g}'^{<}_{11}(1,1';U).
\end{aligned} \tag{6.27}$$

In the above expressions,

$$U_{\rm eff}(1) \equiv V_{\rm trap}(\mathbf{r}_1) + U(1) + 2gn'(1) \tag{6.28}$$

is the effective self-consistent Hartree–Fock dynamic mean field. The condensate part $2gn_{\rm c}(1)$ in (6.28) can be viewed as an additional "external field" acting on the noncondensate. Since we will always work with these gauge-transformed correlation functions in this chapter, we drop the primes on $\tilde{g}_{\alpha\beta}$, n and m to simplify the notation.

The corresponding equation of motion for the condensate amplitude $\sqrt{n_{\rm c}}$

in the moving frame of reference can be written in the form

$$\left[i\hbar\frac{\partial}{\partial t_1}-\frac{\partial\theta(1)}{\partial t_1}+\frac{\hbar^2\nabla_1^2}{2m}-\tfrac{1}{2}m\mathbf{v}_{\rm s}^2+\mu_0-V_{\rm trap}(\mathbf{r}_1)-U(1)-g\left[2\tilde{n}(1)+n_{\rm c}(1)\right]\right.$$
$$\left.+i\mathbf{v}_{\rm s}(1)\cdot\nabla_1+\tfrac{i}{2}\nabla_1\cdot\mathbf{v}_{\rm s}(1)\right]\sqrt{n_{\rm c}(1)}=g\tilde{m}(1)\sqrt{n_{\rm c}(1)}+\eta'(1), \qquad (6.29)$$

where $\eta'(1)\equiv\eta(1)e^{-i\theta(1)}$ is the symmetry-breaking source function in the moving frame of reference. We note that the condensate and noncondensate mean fields enter (6.27) and (6.29) with different numerical factors (either 1 or 2). The reason is that the atoms in the condensate are each in the same quantum state and thus there is no exchange term. In the case of the noncondensate atoms, both Hartree and Fock terms arise since we are dealing with atoms in different quantum states.

6.3 Derivation of a kinetic equation with collisions

In this section, starting from the general results in Section 6.1 we will extend the analysis based on the HFB self-energy in Section 6.2 to include collisions between the atoms. This will give kinetic equations that include second-order collision processes. We will use the Beliaev self-energy approximation as defined in (4.99). We limit our discussion to temperatures high enough where the important thermal excitations can be treated as free atoms moving in a time-dependent self-consistent Hartree–Fock field. As in Section 5.4, we will refer to this as the *Beliaev–Popov* approximation for the second-order self-energies. This discussion will set the stage for Chapter 7, where we treat the more complete Beliaev second-order approximation.

We start our analysis with the Dyson–Beliaev equations of motion for the real-time noncondensate propagators $\hat{\tilde{g}}(1,1')$ written in the 2×2 matrix form given in Section 6.1. We note that the equations (6.11) and (6.12) already have the structure of a kinetic equation. The Hartree–Fock self-energy contribution is included in the l.h.s., giving the mean-field contribution to the so-called "streaming" term. The second-order self-energy describing binary collisions is included on the r.h.s. and will eventually be shown to give rise to the collision integrals C_{11} and C_{12} defined in Chapter 3.

We will write equation (4.71) for the order parameter in terms of the condensate self-energy function S defined in (4.76). Using the HFB approximation (6.17) for the condensate self-energy, we have

$$\hat{S}_{\rm HFB}(1,1')=g\begin{pmatrix}n(1)+\tilde{n}(1) & \tilde{m}(1)\\ \tilde{m}^*(1) & n(1)+\tilde{n}(1)\end{pmatrix}\delta(1-1'). \qquad (6.30)$$

Similarly, one can find an expression for the second-order contribution to the condensate self-energies $S^{<}_{>}$. If we recall the definition for the condensate propagator $\hat{h}$ in (4.67) and use the explicit form for the matrix condensate Hartree–Fock self energy $\hat{S}_{\rm HFB}$ in (6.30), then we obtain a generalized GP equation for the macroscopic wavefunction $\Phi(\mathbf{r}, t)$, namely

$$\begin{aligned}\Big[-i\hbar\frac{\partial}{\partial t_1} + \frac{\hbar^2\nabla_1^2}{2m} - V_{\rm trap}(\mathbf{r}_1) + \mu_0 - g[n_{\rm c}(1) + 2\tilde{n}(1)] \Big]\Phi(1) \\ = g\tilde{m}(1)\Phi^*(1) + \int_{-\infty}^{t_1} d\bar{1}(S^{>}_{11} - S^{<}_{11})(1,\bar{1})\Phi(\bar{1}) \\ + \int_{-\infty}^{t_1} d\bar{1}(S^{>}_{12} - S^{<}_{12})(1,\bar{1})\Phi^*(\bar{1}). \qquad (6.31)\end{aligned}$$

The specific forms of $\Sigma_{\rm coll}$ and $\hat{S}^{<}_{>}$ will depend on the second-order approximation that we use. In this chapter, we work with the second-order self-energy given by the Beliaev (gapless) approximation rather than with a conserving (Φ-derivable) approximation (see Section 4.4). The Beliaev approximation has the important feature that the noncondensate Green's function exhibits the correct excitation spectrum (phonon-like in the long-wavelength limit in a uniform gas).

To derive a generalized kinetic equation including coupling to a condensate, we first gauge-transform (6.11) and (6.12) to the local rest frame to remove the phase of the macroscopic wavefunction (Kane and Kadanoff, 1965), as explained in Section 6.2. To illustrate how this method works, in this section we will neglect the off-diagonal (anomalous) Green's functions $\tilde{g}_{12}$ and $\tilde{g}_{21}$. In this approximation, the 2×2 matrix equations (6.11) and (6.12) can be reduced to scalar equations for the $\tilde{g}^{>}_{<11}$ component:

$$\begin{aligned}\left[i\hbar\frac{\partial}{\partial t_1} - \frac{\partial\theta(1)}{\partial t_1} + \frac{\hbar^2}{2m}[\nabla_1 + im\mathbf{v}_{\rm s}(1)]^2 - U(1) + \mu_0 \right] \tilde{g}^{>}_{<11}(\bar{1}, 1') \\ = \int_{-\infty}^{t_1} d\bar{1}\ \Gamma_{11}(1,\bar{1})\tilde{g}^{>}_{<11}(\bar{1}, 1') - \int_{-\infty}^{t_{1'}} d\bar{1}\ \Sigma^{>}_{<11}(1,\bar{1})a_{11}(\bar{1}, 1') \qquad (6.32)\end{aligned}$$

and

$$\begin{aligned}\left[-i\hbar\frac{\partial}{\partial t_{1'}} - \frac{\partial\theta(1')}{\partial t_{1'}} + \frac{\hbar^2}{2m}[\nabla_{1'} - im\mathbf{v}_{\rm s}(1')]^2 - U(1') + \mu_0 \right] \tilde{g}^{>}_{<11}(\bar{1}, 1') \\ = \int_{-\infty}^{t_1} d\bar{1}\ a_{11}(1,\bar{1})\Sigma^{>}_{<11}(\bar{1}, 1') - \int_{-\infty}^{t_{1'}} d\bar{1}\ \tilde{g}^{>}_{<11}(1,\bar{1})\Gamma_{11}(\bar{1}, 1'). \qquad (6.33)\end{aligned}$$

The effective self-consistent HF dynamic mean field $U(1)$ is defined as in (6.28).

In the KB approach, we express correlation functions in terms of relative and centre-of-mass coordinates, defined in (6.21) and (6.22). Correlation functions ($\tilde{g}, \Sigma$, etc.) are assumed to be dominated by small values of the relative coordinates $\mathbf{r}, t$ (or high momenta and frequencies in the Fourier transforms) but to vary *slowly* as functions of the centre-of-mass coordinates $\mathbf{R}, T$. Using these key properties of correlation functions to simplify the equations, we can write (6.32) and (6.33) in terms of the centre-of-mass and relative coordinates to obtain

$$\begin{aligned}
&\left\{ i\hbar\frac{\partial}{\partial T} + \left(\mathbf{r}\cdot\nabla_{\mathbf{R}} + t\frac{\partial}{\partial T}\right)\left[\mu(\mathbf{R},T) - U(\mathbf{R},T)\right] + i\nabla_{\mathbf{R}}\cdot\mathbf{v}_{\mathrm{s}}(\mathbf{R},T) \right. \\
&\left. + \frac{1}{m}\nabla_{\mathbf{R}}\cdot\nabla_{\mathbf{r}} + i\left[\left(\mathbf{r}\cdot\nabla_{\mathbf{R}} + t\frac{\partial}{\partial T}\right)\mathbf{v}_{\mathrm{s}}(\mathbf{R},T)\right]\cdot\nabla_{\mathbf{r}} + i\mathbf{v}_{\mathrm{s}}(\mathbf{R},T)\cdot\nabla_{\mathbf{R}}\right\}\tilde{g}^{<}_{11}(\mathbf{r},t) \\
&\qquad = \int d\mathbf{r}_1 dt_1 \Big[\tilde{g}^{>}_{11}(\mathbf{r},t)\Sigma^{<}_{11}(\mathbf{r}-\mathbf{r}_1, t-t_1; \mathbf{R}, T) \\
&\qquad\qquad - \tilde{g}^{<}_{11}(\mathbf{r}_1,t_1)\Sigma^{>}_{11}(\mathbf{r}-\mathbf{r}_1, t-t_1; \mathbf{R}, T)\Big]. \qquad (6.34)
\end{aligned}$$

The $(\mathbf{R},T)$-dependence of $\tilde{g}_{11}(\mathbf{r},t)$ has been left implicit in the above equations. The double Fourier transform of (6.34) gives an equation for the quantity $\tilde{g}_{11}(\mathbf{p},\omega;\mathbf{R},T)$, namely

$$\begin{aligned}
&\left[\frac{\partial}{\partial T} + \nabla_{\mathbf{p}}\left(\tilde{\varepsilon}_p + \mathbf{p}\cdot\mathbf{v}_{\mathrm{s}}\right)\cdot\nabla_{\mathbf{R}} - \nabla_{\mathbf{R}}\left(\tilde{\varepsilon}_p + \mathbf{p}\cdot\mathbf{v}_{\mathrm{s}}\right)\cdot\nabla_{\mathbf{p}} \right. \\
&\qquad \left. + \frac{\partial}{\partial T}\left(\tilde{\varepsilon}_p + \mathbf{p}\cdot\mathbf{v}_{\mathrm{s}}\right)\frac{\partial}{\partial\omega}\right]\tilde{g}^{<}_{11}(\mathbf{p},\omega;\mathbf{R},T) \qquad (6.35) \\
&\qquad = \tilde{g}^{>}_{11}(\mathbf{p},\omega;\mathbf{R},T)\Sigma^{<}_{11}(\mathbf{p},\omega;\mathbf{R},T) - \tilde{g}^{<}_{11}(\mathbf{p},\omega;\mathbf{R},T)\Sigma^{>}_{11}(\mathbf{p},\omega;\mathbf{R},T).
\end{aligned}$$

Here we define the "normal" HF single-particle energy by

$$\tilde{\varepsilon}_p(\mathbf{R},T) \equiv \frac{p^2}{2m} + U(\mathbf{R},T) - \mu(\mathbf{R},T). \qquad (6.36)$$

In the Thomas–Fermi approximation, this single-particle HF energy reduces to

$$\tilde{\varepsilon}_p(\mathbf{R},T) \equiv \frac{p^2}{2m} + gn_{\mathrm{c}}(\mathbf{R},T). \qquad (6.37)$$

Using (6.35) to calculate the single-particle spectral density a_{11} from (6.13), we find that the self-energies $\Sigma^{<}$ cancel out, leaving

$$\begin{aligned}
&\left[\frac{\partial}{\partial T} + \nabla_{\mathbf{p}}\left(\tilde{\varepsilon}_p + \mathbf{p}\cdot\mathbf{v}_{\mathrm{s}}\right)\cdot\nabla_{\mathbf{R}} - \nabla_{\mathbf{R}}\left(\tilde{\varepsilon}_p + \mathbf{p}\cdot\mathbf{v}_{\mathrm{s}}\right)\cdot\nabla_{\mathbf{p}} \right. \\
&\qquad \left. + \frac{\partial}{\partial T}\left(\tilde{\varepsilon}_p + \mathbf{p}\cdot\mathbf{v}_{\mathrm{s}}\right)\frac{\partial}{\partial\omega}\right]a_{11}(\mathbf{p},\omega;\mathbf{R},T) = 0. \qquad (6.38)
\end{aligned}$$

This equation is a key step in deriving a kinetic equation for the distribution function for the noncondensate atoms. One may explicitly verify that the HF quasiparticle approximation for the spectral density,

$$a_{11}(\mathbf{p},\omega;\mathbf{R},T) = 2\pi\delta\left(\omega - \mathbf{p}\cdot\mathbf{v}_{\rm s} - \tilde{\varepsilon}_p(\mathbf{R},T)\right), \tag{6.39}$$

does indeed satisfy (6.38). We also recall that the assumption that the correlation functions $\tilde{g}^{<}_{>}(\mathbf{r},t;\mathbf{R},T)$ are peaked at small values of $\mathbf{r}$ and t implies that large values of $\mathbf{p}$ and ω determine $\tilde{g}^{<}_{>}(\mathbf{p},\omega;\mathbf{R},T)$, where the particle-like spectrum (6.36) and hence (6.39) are reasonable approximations.

Following Kadanoff and Baym (1962), we define a *quasiparticle* distribution function f in the following way:

$$\begin{aligned}
\tilde{g}^{<}_{11}(\mathbf{p},\omega;\mathbf{R},T) &\equiv a_{11}(\mathbf{p},\omega;\mathbf{R},T) f(\mathbf{p},\mathbf{R},T) \\
\tilde{g}^{>}_{11}(\mathbf{p},\omega;\mathbf{R},T) &\equiv a_{11}(\mathbf{p},\omega;\mathbf{R},T)[1+f(\mathbf{p},\mathbf{R},T)],
\end{aligned} \tag{6.40}$$

where the spectral density a_{11} is given by the HF approximation (6.39). We recall that the usual Wigner distribution function is related to the KB diagonal single-particle Green's function by

$$\begin{aligned}
f(\mathbf{p},\mathbf{R},T) &\equiv \int d\mathbf{r}\; e^{-i\mathbf{p}\cdot\mathbf{r}} i\tilde{g}^{<}_{11}(\mathbf{r},t=0;\mathbf{R},T) \\
&= \int_{-\infty}^{\infty} d\omega\; \tilde{g}^{<}_{11}(\mathbf{p},\omega;\mathbf{R},T).
\end{aligned} \tag{6.41}$$

Using (6.40) in conjunction with the explicit result for the spectral density a_{11} given in (6.39), we see that the quasiparticle distribution function (6.40) does indeed reduce to the Wigner distribution function $f(\mathbf{p},\mathbf{R},T)$. This is correct in the high-temperature limit only, where the HF quasiparticle energy spectrum reduces to the HF energy spectrum for atoms.

Using (6.38), (6.39) and (6.40) and integrating over ω, (6.35) reduces to a kinetic equation for the distribution function f, namely

$$\begin{aligned}
&\left[\frac{\partial}{\partial T} + \nabla_{\mathbf{p}}\left(\tilde{\varepsilon}_p + \mathbf{p}\cdot\mathbf{v}_{\rm s}\right)\cdot\nabla_{\mathbf{R}} - \nabla_{\mathbf{R}}\left(\tilde{\varepsilon}_p + \mathbf{p}\cdot\mathbf{v}_{\rm s}\right)\cdot\nabla_{\mathbf{p}}\right] f(\mathbf{p},\mathbf{R},T) \\
&\quad = [1+f(\mathbf{p},\mathbf{R},T)]\Sigma^{<}_{11}(\mathbf{p},\omega=\tilde{\varepsilon}_p+\mathbf{p}\cdot\mathbf{v}_{\rm s};\mathbf{R},T) \\
&\qquad - f(\mathbf{p},\mathbf{R},T)\Sigma^{>}_{11}(\mathbf{p},\omega=\tilde{\varepsilon}_p+\mathbf{p}\cdot\mathbf{v}_{\rm s};\mathbf{R},T).
\end{aligned} \tag{6.42}$$

This kinetic equation is the major result of this chapter. The r.h.s. is completely determined by the second-order collisional self-energies $\Sigma^{<}_{\rm coll>}$. In Section 6.4 we consider the simplest approximation for these, that is, one consistent with the HF approximation used in the l.h.s. of (6.42).

A similar generalized quantum kinetic equation was also derived by Stoof

(1999, 2001) for Bose superfluids, using the related Keldysh formalism based on a path integral formalism. Stoof works with a probability distribution over different Bose order parameters within a (globally) number conserving theory. This approach leads to an additional Gaussian noise term in a generalized GP equation (see Section 6.5) associated with these order parameter fluctuations (see Duine and Stoof, 2001). For further discussion, see the end of Section 8.1.

6.4 Collision integrals in the Hartree–Fock approximation

Neglecting the off-diagonal Green's functions, the Fourier transform of the collisional part of the self-energies $\Sigma^{<}_{>}$ in (4.107) is given by (using (6.4))

$$\begin{aligned}
&\Sigma^{<}_{>11}(\mathbf{p},\omega)\\
&= 2g^2\int \frac{d\mathbf{p}_i d\omega_i}{(2\pi)^{12}}(2\pi)^4\delta(\omega+\omega_1-\omega_2-\omega_3)\delta(\mathbf{p}+\mathbf{p}_1-\mathbf{p}_2-\mathbf{p}_3)\\
&\times\Big[\tilde{g}^{>}_{<11}(\mathbf{p}_1,\omega_1)\tilde{g}^{<}_{>11}(\mathbf{p}_2,\omega_2)h_{11}(\mathbf{p}_3,\omega_3)+\tilde{g}^{>}_{<11}(\mathbf{p}_1,\omega_1)h_{11}(\mathbf{p}_2,\omega_2)\tilde{g}^{<}_{>11}(\mathbf{p}_3,\omega_3)\\
&+h_{11}(\mathbf{p}_1,\omega_1)\tilde{g}^{<}_{>11}(\mathbf{p}_2,\omega_2)\tilde{g}^{<}_{>11}(\mathbf{p}_3,\omega_3)+\tilde{g}^{>}_{<11}(\mathbf{p}_1,\omega_1)\tilde{g}^{<}_{>11}(\mathbf{p}_2,\omega_2)\tilde{g}^{<}_{>11}(\mathbf{p}_3,\omega_3)\Big].
\end{aligned}\tag{6.43}$$

Here and elsewhere, the $(\mathbf{R},T)$-dependence of the Green's functions and self-energies is left implicit. Integration has been performed over all $\mathbf{p}_i$ and ω_i variables in (6.43). The simplest approximation is to calculate $\Sigma^{>}$ and $\Sigma^{<}$ appearing on the r.h.s. of (6.42) using the HF propagators (6.39) and (6.40).

The last term in the self-energy in (6.43) which involves three propagators for noncondensate atoms is the origin of the C_{22} collision integral. The associated self-energies in (6.42) are given by

$$\begin{aligned}
\Sigma^{<}_{11}(\mathbf{p},\omega=\tilde{\varepsilon}_p+\mathbf{p}\cdot\mathbf{v}_{\rm s}) = \frac{2g^2}{(2\pi)^5\hbar^7}&\int d\mathbf{p}_1\int d\mathbf{p}_2\int d\mathbf{p}_3\\
&\times\delta(\tilde{\varepsilon}_p+\tilde{\varepsilon}_{p_1}-\tilde{\varepsilon}_{p_2}-\tilde{\varepsilon}_{p_3})\delta(\mathbf{p}+\mathbf{p}_1-\mathbf{p}_2-\mathbf{p}_3)\\
&\times(1+f_1)f_2f_3,
\end{aligned}\tag{6.44}$$

$$\begin{aligned}
\Sigma^{>}_{11}(\mathbf{p},\omega=\tilde{\varepsilon}_p+\mathbf{p}\cdot\mathbf{v}_{\rm s}) = \frac{2g^2}{(2\pi)^5\hbar^7}&\int d\mathbf{p}_1\int d\mathbf{p}_2\int d\mathbf{p}_3\\
&\times\delta(\tilde{\varepsilon}_p+\tilde{\varepsilon}_{p_1}-\tilde{\varepsilon}_{p_2}-\tilde{\varepsilon}_{p_3})\delta(\mathbf{p}+\mathbf{p}_1-\mathbf{p}_2-\mathbf{p}_3)\\
&\times f_1(1+f_2)(1+f_3).
\end{aligned}\tag{6.45}$$

The terms involving $\mathbf{p}_i \cdot \mathbf{v}_\mathrm{s}$ cancel out of the energy delta functions as a result of the momentum delta functions. For example, $\mathbf{v}_\mathrm{s} \cdot (\mathbf{p} + \mathbf{p}_1 - \mathbf{p}_2 - \mathbf{p}_3)$ vanishes because of the momentum conservation condition $\mathbf{p} + \mathbf{p}_1 = \mathbf{p}_2 + \mathbf{p}_3$. Using (6.44) and (6.45), the $C_{22}[f, \Phi]$ collisional contribution on the r.h.s. of (6.42) is given by

$$\begin{aligned} C_{22}[f,\Phi] &= (1+f)\Sigma_{11}^{<} - f\Sigma_{11}^{>} \\ &= \frac{2g^2}{(2\pi)^5\hbar^7}\int d\mathbf{p}_1 \int d\mathbf{p}_2 \int d\mathbf{p}_3 \\ &\quad \times \delta(\tilde{\varepsilon}_p + \tilde{\varepsilon}_{p_1} - \tilde{\varepsilon}_{p_2} - \tilde{\varepsilon}_{p_3})\delta(\mathbf{p} + \mathbf{p}_1 - \mathbf{p}_2 - \mathbf{p}_3) \\ &\quad \times \Big[(1+f)(1+f_1)f_2 f_3 - f f_1 (1+f_2)(1+f_3)\Big], \end{aligned} \tag{6.46}$$

where $f \equiv f(\mathbf{p}, \mathbf{R}, T)$ and $f_i \equiv f(\mathbf{p}_i, \mathbf{R}, T)$. We recall that C_{22} describes collisions where 2 excited (i.e. noncondensate) atoms are scattered into 2 excited atoms, hence the subscript '22'. The collision integral $C_{12}[f, \Phi]$, involving collisions between noncondensate atoms and condensate atoms, comes from the terms involving h_{11} in (6.43). One can show that for slowly varying external disturbances (Kane and Kadanoff, 1965), one has

$$h_{11}(\mathbf{p}, \omega; \mathbf{R}, T) = (2\pi)^4 n_\mathrm{c}(\mathbf{R}, T)\delta(\mathbf{p})\delta(\omega). \tag{6.47}$$

This approximation for the condensate propagator is consistent with the discussion in Chapter 3 that led to the result (3.69). Using the HF quasiparticle approximation in (6.39) and (6.40) in conjunction with (6.47), the C_{12} collisional contribution to (6.43) is

$$\begin{aligned} &\Sigma_{11}^{<}(\mathbf{p}, \omega = \tilde{\varepsilon}_p + p \cdot \mathbf{v}_\mathrm{s}) \\ &\quad = \frac{2g^2 n_\mathrm{c}}{(2\pi)^2\hbar^4}\int d\mathbf{p}_2 \int d\mathbf{p}_3 \Big[\delta(\tilde{\varepsilon}_p - \tilde{\varepsilon}_{p_2} - \tilde{\varepsilon}_{p_3})\delta(\mathbf{p} - \mathbf{p}_2 - \mathbf{p}_3) f_2 f_3 \\ &\qquad\qquad + 2\delta(\tilde{\varepsilon}_p - \tilde{\varepsilon}_{p_2} + \tilde{\varepsilon}_{p_3})\delta(\mathbf{p} - \mathbf{p}_2 + \mathbf{p}_3) f_2 (1+f_3)\Big], \end{aligned} \tag{6.48}$$

$$\begin{aligned} &\Sigma_{11}^{>}(\mathbf{p}, \omega = \tilde{\varepsilon}_p + \mathbf{p} \cdot \mathbf{v}_\mathrm{s}) \\ &\quad = \frac{2g^2 n_\mathrm{c}}{(2\pi)^2\hbar^4}\int d\mathbf{p}_2 \int d\mathbf{p}_3 \Big[\delta(\tilde{\varepsilon}_p - \tilde{\varepsilon}_{p_2} - \tilde{\varepsilon}_{p_3})\delta(\mathbf{p} - \mathbf{p}_2 - \mathbf{p}_3)(1+f_2)(1+f_3) \\ &\qquad\qquad + 2\delta(\tilde{\varepsilon}_p - \tilde{\varepsilon}_{p_2} + \tilde{\varepsilon}_{p_3})\delta(\mathbf{p} - \mathbf{p}_2 + \mathbf{p}_3) f_3 (1+f_2)\Big]. \end{aligned} \tag{6.49}$$

As with the C_{22} collisional contributions above, the terms involving $\mathbf{p}_i \cdot \mathbf{v}_\mathrm{s}$ in the energy conservation factor cancel out. Using these results, one finds

that the C_{12} collisional contribution to the r.h.s. of (6.42) is

$$\begin{aligned} C_{12}[f,\Phi] &= (1+f)\Sigma_{11}^{<} - f\Sigma_{11}^{>} \\ &= \frac{2g^2 n_{\rm c}(\mathbf{R},T)}{(2\pi)^2\hbar^4}\int d\mathbf{p}_1 \int d\mathbf{p}_2 \Big\{ [(1+f)f_1 f_2 - f(1+f_1)(1+f_2)] \\ &\quad \times \delta(\tilde{\varepsilon}_p - \tilde{\varepsilon}_1 - \tilde{\varepsilon}_2)\delta(\mathbf{p}-\mathbf{p}_1-\mathbf{p}_2) \\ &\quad + 2\,[(1+f)(1+f_1)f_2 - f f_1(1+f_2)] \\ &\quad \times \delta(\tilde{\varepsilon}_p + \tilde{\varepsilon}_1 - \tilde{\varepsilon}_2)\delta(\mathbf{p}+\mathbf{p}_1-\mathbf{p}_2)\Big\}. \end{aligned} \tag{6.50}$$

A compact version of this collision term is often used in the literature, namely (Kirkpatrick and Dorfman, 1985a)

$$\begin{aligned} &C_{12}[f,\Phi] \\ &= \frac{2g^2 n_{\rm c}(\mathbf{R},T)}{(2\pi)^2\hbar^4}\int d\mathbf{p}_1 \int d\mathbf{p}_2 \int d\mathbf{p}_3\,[\delta(\mathbf{p}-\mathbf{p}_1) - \delta(\mathbf{p}-\mathbf{p}_2) - \delta(\mathbf{p}-\mathbf{p}_3)] \\ &\quad \times \delta(\tilde{\varepsilon}_{p_1} - \tilde{\varepsilon}_{p_2} - \tilde{\varepsilon}_{p_3})\delta(\mathbf{p}_1-\mathbf{p}_2-\mathbf{p}_3)\Big[(1+f_1)f_3 f_2 - f_1(1+f_3)(1+f_2)\Big]. \end{aligned} \tag{6.51}$$

We note that C_{12} describes collisions where we go from 1 excited and 1 condensate atom to 2 excited atoms, hence the subscript "12". In Fig. 6.1, we show the various collision processes which are described by the different terms in (6.50).

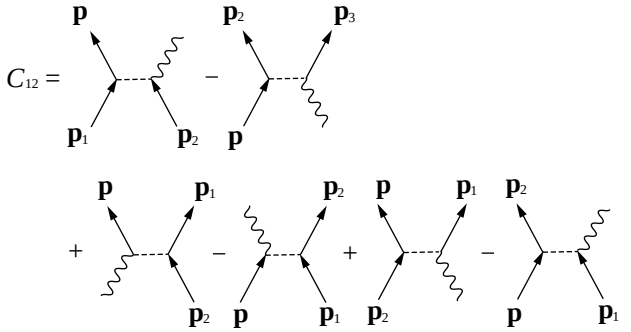


Fig. 6.1. Various collision processes described by the C_{12} collision integral (6.50). The solid, wiggly and broken lines are the same as in Figs. 4.1–4.3 and are defined below (4.96).

Inserting (6.46) and (6.51) into (6.42), we arrive at a kinetic equation

equivalent to that discussed in Chapter 3,

$$\left[\frac{\partial}{\partial T}+\nabla_{\mathbf{p}}\left(\tilde{\varepsilon}_p+\mathbf{p}\cdot\mathbf{v}_{\rm s}\right)\cdot\nabla_{\mathbf{R}}-\nabla_{\mathbf{R}}\left(\tilde{\varepsilon}_p+\mathbf{p}\cdot\mathbf{v}_{\rm s}\right)\cdot\nabla_{\mathbf{p}}\right]f(\mathbf{p},\mathbf{R},T)$$
$$=C_{22}[f(\mathbf{p},\mathbf{R},T),\Phi(\mathbf{r},t)]+C_{12}[f(\mathbf{p},\mathbf{R},T),\Phi(\mathbf{r},t)]. \qquad (6.52)$$

The l.h.s of (6.52) is the expected collisionless "streaming term" in the frame of reference in which the superfluid velocity is zero (Zaremba *et al.* (1999) derived the equivalent of (6.52) in the lab frame). For a *uniform* gas at finite temperatures, (6.52) was first obtained by Kirkpatrick and Dorfman (1985a) using a completely different approach (See Appendix A of Zaremba *et al.*, 1999).

6.5 Generalized GP equation

We finally turn to the derivation of an equation of motion for the condensate propagator or, equivalently, the order parameter. We first write (6.31) for $\Phi(\mathbf{r},t)$ using coordinates defined in the local rest frame:

$$\left[i\frac{\partial}{\partial t}-\frac{\partial\theta(1)}{\partial t}+\frac{\hbar^2}{2m}[\nabla+im\mathbf{v}_{\rm s}(1)]^2+\mu_0-V_{\rm trap}(\mathbf{r})-g\left[2\tilde{n}(1)+n_{\rm c}(1)\right]\right]\Phi(1)$$
$$=\int_{-\infty}^{t}d\bar{1}\left[S_{11}^{>}-S_{11}^{<}\right]\left(\mathbf{r}-\mathbf{r}_1,t-t_1;\tfrac{1}{2}(\mathbf{r}+\mathbf{r}_1),\tfrac{1}{2}(t+t_1)\right)\Phi(\mathbf{r}_1,t_1). \qquad (6.53)$$

In this section, we drop the subscript "coll" denoting the collisional or second-order contribution to the diagonal element of the matrix condensate self-energy, S_{11}. As usual in the KB approach, we assume that the condensate S_{11} correlation functions introduced in Sections 4.3 and 4.4 (see (4.106)) are dominated by *small* values of the relative coordinates $(\mathbf{r}-\mathbf{r}_1,t-t_1)$. This means we can approximate the centre-of-mass coordinates $\frac{1}{2}(\mathbf{r}+\mathbf{r}_1)$ and $\frac{1}{2}(t+t_1)$ by $\mathbf{r}$ and t, respectively, in the S_{11} functions in (6.53), to give $S^{\gtrless}(\mathbf{r}-\mathbf{r}_1,t-t_1;\mathbf{r},t)$. For the same reason, we can approximate the macroscopic wavefunction $\Phi(\mathbf{r}_1,t_1)$ on the r.h.s. of (6.53) by $\Phi(\mathbf{r},t)\equiv\sqrt{n_{\rm c}(\mathbf{r},t)}$. Introducing Fourier transforms with respect to the relative coordinates $\mathbf{r}-\mathbf{r}_1$ and $t-t_1$, (6.53) can be written as

$$\left[i\frac{\partial}{\partial t}-\frac{\partial\theta(\mathbf{r},t)}{\partial t}+\frac{\hbar^2}{2m}\left[\nabla+im\mathbf{v}_{\rm s}(\mathbf{r},t)\right]^2+\mu_0-V_{\rm trap}(\mathbf{r})\right.$$
$$\left.-g\left[2\tilde{n}(\mathbf{r},t)+n_{\rm c}(\mathbf{r},t)\right]\right]\Phi(\mathbf{r},t)$$
$$=\int_{-\infty}^{t}d\mathbf{r}_1dt_1e^{i[\mathbf{p}\cdot(\mathbf{r}-\mathbf{r}_1)-i\omega(t-t_1)]}\Phi(\mathbf{r},t)\int\frac{d\mathbf{p}d\omega}{(2\pi)^4}\left[S_{11}^{>}-S_{11}^{<}\right](\mathbf{p},\omega;\mathbf{r},t). \qquad (6.54)$$

In the Beliaev–Popov approximation used to obtain (6.43), $S_{11}^{><}(\mathbf{p},\omega;\mathbf{r},t)$ in (4.106) is given by

$$S_{11}^{><}(\mathbf{p},\omega;\mathbf{r},t) = 2g^2 \int \frac{d\mathbf{p}_i d\omega_i}{(2\pi)^{12}} (2\pi)^4 \delta(\omega+\omega_1-\omega_2-\omega_3)\delta(\mathbf{p}+\mathbf{p}_1-\mathbf{p}_2-\mathbf{p}_3) \times \tilde{g}_{11}^{><}(\mathbf{p}_1,\omega_1;\mathbf{r},t)\tilde{g}_{11}^{<>}(\mathbf{p}_2,\omega_2;\mathbf{r},t)\tilde{g}_{11}^{><}(\mathbf{p}_3,\omega_3;\mathbf{r},t). \quad (6.55)$$

In evaluating the r.h.s. of (6.54), we use the Hartree–Fock approximation for the single-particle Green's functions given by (6.39) and (6.40) as well as the following identity:

$$\lim_{\delta\to 0^+} \int_{-\infty}^{T} dt\, e^{i(\omega-i\delta)(T-t)} = \pi\delta(\omega) + iP\left(\frac{1}{\omega}\right). \quad (6.56)$$

After some calculation, we finally obtain the generalized Gross–Pitaevskii equation

$$i\hbar \frac{\partial\sqrt{n_{\rm c}(\mathbf{r},t)}}{\partial t} = \left[\frac{\partial\theta(\mathbf{r},t)}{\partial t} - \frac{\hbar^2}{2m}\left[\nabla + im\mathbf{v}_{\rm s}(\mathbf{r},t)\right]^2 - \mu_0 + V_{\rm trap}(\mathbf{r}) + g\left[2\tilde{n}(\mathbf{r},t) + n_{\rm c}(\mathbf{r},t)\right] - iR(\mathbf{r},t)\right]\sqrt{n_{\rm c}(\mathbf{r},t)}. \quad (6.57)$$

The new source term R in this GGP equation is directly related to the C_{12} collision term (6.50) in the kinetic equation (6.52), namely

$$R(\mathbf{r},t) \equiv \int \frac{d\mathbf{p}}{(2\pi)^3} \frac{C_{12}[f(\mathbf{p},\mathbf{r},t)]}{2n_{\rm c}(\mathbf{r},t)}. \quad (6.58)$$

In Chapters 8 and 9, we shall see that this R term describes the damping (or growth if R is negative) of condensate amplitude fluctuations due to collisions with the thermal cloud atoms. The appearance of this term in (6.57) is expected, since the C_{12} collisions can change the local density of the atoms in the condensate and hence modify the magnitude of the macroscopic wavefunction $\sqrt{n_{\rm c}(\mathbf{r},t)}$.

The real part of the right-hand side of (6.54) has been omitted since we are working only to first order in the interaction, as far as renormalized energies are concerned. If we transform back into the lab frame (recall that in the lab frame, one has $\Phi = \sqrt{n_{\rm c}}e^{i\theta}$), (6.57) reduces to the time-dependent generalized Gross–Pitaevskii equation for $\Phi(\mathbf{r},t)$, as discussed in Chapter 3.

The coupled ZNG equations for the condensate and the thermal cloud atoms were also derived using a two-particle irreducible (2PI) formalism by Baier and Stockamp (2004). The analogous equations for a Bose gas in an optical lattice have been discussed using the 2PI approach by Konabe and Nikuni (2008).

6.6 Linearized collision integrals in collisionless theories

In contrast with the collisionless kinetic equations discussed in Section 5.3, the Kadanoff–Baym kinetic equation (6.52) derived in Section 6.4 includes the collision integrals C_{12} and C_{22}. These are formally of order g^2 and lead to damping. As shown in Section 6.3, within the KB formalism the collision terms we derived are directly related to the nonequilibrium collisional Beliaev self-energies. This is shown explicitly by the r.h.s. of the generalized kinetic equations (6.42). Within the approximations used by ZNG, these second-order Beliaev self-energies $\Sigma_{11}^{<}$ and $\Sigma_{11}^{>}$ are given by (6.48) and (6.49), leading to the expression for $C_{12}[f, \Phi]$ given in (6.51). Similarly, $C_{22}[f, \Phi]$ in (6.46) is related to the self-energy contributions $\Sigma_{11}^{<}$ and $\Sigma_{11}^{>}$ given in (6.44) and (6.45), respectively.

The ZNG kinetic equations and generalized GP equation derived in Sections 6.4 and 6.5 are approximate. However, collision integrals such as (6.51) are highly nonlinear functions of the single-particle distribution functions. These coupled equations are used to discuss the collisionless region (where mean fields dominate) in Chapters 8–13. The other extreme of collisional hydrodynamics, where the collisions produce local equilibrium, is discussed in Chapters 15–19. The ZNG coupled equations are not limited to dealing with *small* deviations from equilibrium, in which the response functions can be expressed in terms of correlation functions averaged over a thermal equilibrium ensemble. Simulations of the ZNG equations to be discussed in Chapters 11 and 12 can deal with the region outside linear response theory, a region often probed in experiments on collective modes in trapped Bose gases.

In this section, we evaluate the collision integrals C_{12} and C_{22} for the case when $f(\mathbf{p}, \mathbf{r}, t)$ is only slightly different from the value in thermal equilibrium $f^0(\mathbf{p}, \mathbf{r})$. Since C_{12} and C_{22} vanish when the thermal cloud and condensate are in static equilibrium (see Chapter 3), this means that we can *linearize* the collision integrals so that they are proportional to the deviation $\delta f(\mathbf{p}, \mathbf{r}, t)$. In this linearized theory, it is clear that the r.h.s. of the kinetic equation (6.42) reduces to

$$-\delta f(\mathbf{p}, \mathbf{r}, t)[\Sigma_{11,0}^{>}(\mathbf{p}, \mathbf{r}) - \Sigma_{11,0}^{<}(\mathbf{p}, \mathbf{r})], \tag{6.59}$$

where $\Sigma_{11,0}^{<}$ are the equilibrium second-order self-energies. We note that these self-energies can be shown to satisfy the condition

$$(1 + f^0)\Sigma_{11,0}^{<} - f^0\Sigma_{11,0}^{>} = 0, \tag{6.60}$$

which is equivalent to the statement that the collision integrals vanish in thermal equilibrium.

Before working out the linearized form for C_{12} and C_{22} in the way described above, we stress that this discussion is for the case when collisions can be viewed as a *perturbative* second-order correction to a collisionless kinetic equation for $f(\mathbf{p}, \mathbf{r}, t)$. More precisely, we assume that in dealing with thermal cloud atoms in a momentum state $\mathbf{p}$, we can approximate the collision integrals assuming that all the other atoms (condensate and noncondensate) are in complete thermodynamic equilibrium. This situation is to be contrasted with the two-fluid collisional hydrodynamic domain to be studied in Chapters 15 and 17. This region corresponds to local equilibrium brought about by strong collisions, and is intrinsically *nonperturbative*. When discussing this collisional hydrodynamic region in Chapter 15, we shall "linearize" the C_{12} collision integral but, as will be seen, this involves the difference between *partial* and *complete* local equilibrium. As we shall show in Chapter 15, this difference depends on the small difference between the local chemical potentials of the condensate and the thermal cloud.

We begin with the linearized approximation for the collision integral $C_{22}[f(\mathbf{p}, \mathbf{r}, t), \Phi(\mathbf{r}, t)]$ given by (6.46). As we have discussed in Chapter 3, in complete (thermal and diffusive) equilibrium, C_{22} vanishes:

$$\begin{aligned}
&[(1+f^0)(1+f_2^0)f_3^0 f_4^0 - f^0 f_2^0 (1+f_3^0)(1+f_4^0)]\delta(\tilde{\varepsilon}_p + \tilde{\varepsilon}_2 - \tilde{\varepsilon}_3 - \tilde{\varepsilon}_4) \\
&= f^0 f_2^0 f_3^0 f_4^0 \left[e^{\beta(\tilde{\varepsilon}_p + \tilde{\varepsilon}_2 - 2\tilde{\mu}_0)} - e^{\beta(\tilde{\varepsilon}_3 + \tilde{\varepsilon}_4 - 2\tilde{\mu}_0)}\right] \delta(\tilde{\varepsilon}_p + \tilde{\varepsilon}_2 - \tilde{\varepsilon}_3 - \tilde{\varepsilon}_4) \\
&= 0. \qquad (6.61)
\end{aligned}$$

However, if the l.h.s. of (6.61) involves the nonequilibrium distribution $f = f^0 + \delta f$, to first order in δf it reduces to

$$-\delta f(\mathbf{p}, \mathbf{r}, t) \left[f_2^0(1+f_3^0)(1+f_4^0) - (1+f_2^0) f_3^0 f_4^0\right] \delta(\tilde{\varepsilon}_p + \tilde{\varepsilon}_2 - \tilde{\varepsilon}_3 - \tilde{\varepsilon}_4). \quad (6.62)$$

Using this linearized approximation, we find that (6.46) reduces to

$$C_{22}[f, \Phi] = -\frac{2}{\tau_{22}^0(\mathbf{p}, \mathbf{r})} \delta f(\mathbf{p}, \mathbf{r}, t), \quad (6.63)$$

where

$$\begin{aligned}
\frac{1}{\tau_{22}^0(\mathbf{p}, \mathbf{r})} \equiv \frac{2g^2}{(2\pi)^5 \hbar^7} &\int d\mathbf{p}_2 \int d\mathbf{p}_3 \int d\mathbf{p}_4 \, \delta(\mathbf{p} + \mathbf{p}_1 - \mathbf{p}_2 - \mathbf{p}_3) \\
&\times \delta(\tilde{\varepsilon}_p + \tilde{\varepsilon}_{p2} - \tilde{\varepsilon}_{p3} - \tilde{\varepsilon}_{p4}) \\
&\times \left[f_2^0(1+f_3^0)(1+f_4^0) - (1+f_2^0) f_3^0 f_4^0\right]. \quad (6.64)
\end{aligned}$$

The first term in the square brackets in (6.64) describes the scattering-out

(destruction) of a noncondensate atom in state $\mathbf{p}$ while the second term describes the scattering-in (creation) of a noncondensate atom in state $\mathbf{p}$. It is clear that (6.64) describes the lifetime τ_{22}^0 of a state $\mathbf{p}$ in a gas in thermal equilibrium given by the transition rate obtained from the usual Golden Rule second-order perturbation result. This kind of linearized approximation for the C_{22} collision integral was discussed by Kadanoff and Baym (1962, p. 30) in the case of a normal quantum gas.

We next turn to a similar treatment of the $C_{12}[f,\Phi]$ collision term in (6.51). This term plays a crucial role in a Bose-condensed gas since it describes collisions that either *add* or *remove* atoms from the condensate. An analogous expression for C_{12} written in the lab frame is

$$\begin{aligned} C_{12}[f,\Phi] = \frac{2g^2 n_{\rm c}(\mathbf{r},t)}{(2\pi)^2\hbar^4}\int d\mathbf{p}_2 \int d\mathbf{p}_3 \\ \times\Big\{ \left[(1+f)f_2 f_3 - f(1+f_2)(1+f_3)\right]\delta(\varepsilon_{\rm c}+\tilde{\varepsilon}_p-\tilde{\varepsilon}_2-\tilde{\varepsilon}_3) \\ \times\delta(\mathbf{p}_{\rm c}+\mathbf{p}-\mathbf{p}_2-\mathbf{p}_3) \\ -2\left[(1+f_2)f f_3 - f_2(1+f)(1+f_3)\right]\delta(\varepsilon_{\rm c}+\tilde{\varepsilon}_2-\tilde{\varepsilon}_p-\tilde{\varepsilon}_4) \\ \times\delta(\mathbf{p}_{\rm c}+\mathbf{p}_2-\mathbf{p}-\mathbf{p}_3)\Big\}. \end{aligned} \tag{6.65}$$

As in the preceding discussion of $C_{22}[f,\Phi]$, we now linearize (6.65), assuming that *only* $f(\mathbf{p},\mathbf{r},t)$ in (6.65) deviates from its thermal equilibrium value. We also assume that the condensate atoms are all in their equilibrium state, with energy

$$\varepsilon_{\rm c0} = \mu_{\rm c0} + \tfrac{1}{2}m\mathbf{v}_{\rm c0}^2(\mathbf{r}). \tag{6.66}$$

Recalling that $C_{12}[f^0,\Phi^0]=0$ in complete thermal equilibrium, to first order in the deviation $\delta f(\mathbf{p},\mathbf{r},t)\equiv f-f^0$ one finds that (6.65) reduces to

$$\begin{aligned} C_{12}^{\rm lin}[f,\Phi^0] = -\delta f(\mathbf{p},\mathbf{r},t)\frac{2g^2 n_{\rm c0}(\mathbf{r})}{(2\pi)^2\hbar^4}\int d\mathbf{p}_2\int d\mathbf{p}_3 \\ \times\Big\{\left[(1+f_2^0)(1+f_3^0)-f_2^0 f_3^0)\right]\delta(\varepsilon_{\rm c0}+\tilde{\varepsilon}_p^0-\tilde{\varepsilon}_2^0-\tilde{\varepsilon}_3^0) \\ \times\delta(\mathbf{p}_{\rm c0}+\mathbf{p}-\mathbf{p}_2-\mathbf{p}_3) \\ +2\left[(1+f_2^0)f_3^0 - f_2^0(1+f_3^0)\right]\delta(\varepsilon_{\rm c0}+\tilde{\varepsilon}_2^0-\tilde{\varepsilon}_p^0-\tilde{\varepsilon}_3^0) \\ \times\delta(\mathbf{p}_{\rm c0}+\mathbf{p}_2-\mathbf{p}-\mathbf{p}_3)\Big\}. \end{aligned} \tag{6.67}$$

The superscript "lin" on C_{12} is a reminder that (6.67) is a *linearized* approximation, as discussed above. Note that the second term has a factor $+2$, in contrast with the factor -2 in (6.65).

The first term in (6.67) describes collisions in which an atom in a momentum state $\mathbf{p}$ decays into two states $\mathbf{p}_2$ and $\mathbf{p}_3$, kicking one atom out of the condensate. This clearly involves scattering processes analogous to the Beliaev damping discussed in Section 5.3. We note that in the first term the factor in square brackets reduces to $1 + f_2^0 + f_3^0$.

In contrast, the second term in (6.67) involves an atom with momentum $\mathbf{p}$ absorbing an atom with momentum $\mathbf{p}_3$ and creating one in state $\mathbf{p}_2$, thus adding an atom to the condensate. The scattering processes involved in this case are analogous to those in "Landau damping". The factor in square brackets reduces to $f_3^0 - f_2^0$. Writing (6.67) as $C_{12} = -2\delta f/\tau_{12}$, one sees that $\tau_{12}(\mathbf{p}, \mathbf{r})$ is the lifetime of the state $\mathbf{p}$ that one would obtain from a Born approximation transition-rate calculation, assuming that all the other atoms in the thermal cloud are in thermal equilibrium. Landau damping calculated in various approximations is discussed in Chapter 13.

If we are dealing with a stationary condensate, where $\mathbf{v}_{c0}(\mathbf{r}) = 0$, then all the condensate atoms involved in the C_{12} collision term have the same energy μ_{c0} and have momentum $\mathbf{p}_{c0} = 0$. To be explicit in this case, in (6.67) we have

$$\begin{aligned} \varepsilon_{c0} &\equiv \mu_{c0} = V_{\rm trap}(\mathbf{r}) + g n_{c0}(\mathbf{r}) + 2g\tilde{n}_0(\mathbf{r}), \\ \tilde{\varepsilon}^0_{\mathbf{p}} &\equiv \frac{p^2}{2m} + V_{\rm trap}(\mathbf{r}) + 2g[n_{c0}(\mathbf{r}) + \tilde{n}_0(\mathbf{r})]. \end{aligned} \tag{6.68}$$

To summarize, we have found that, working to first order in the deviation $\delta f_{\mathbf{p}}(\mathbf{r}, t)$, the sum of the collision integrals $C_{22} + C_{12}$ in the kinetic equation (3.42) can be approximated by

$$-2\left(\frac{1}{\tau^0_{22}} + \frac{1}{\tau^0_{12}}\right)\delta f_{\mathbf{p}}(\mathbf{r}, t), \tag{6.69}$$

where the reciprocal collision times (of order g^2) are given by (6.64) and (6.67), evaluated for all atoms in thermal equilibrium. If we consider solutions of the kind $\delta f_{\mathbf{p}}(\mathbf{r}, t) = \delta f_{\mathbf{p}}(\mathbf{r}) e^{-i\omega t}$, we see that (6.69) combined with the $\delta f/\partial t$ term on the l.h.s. of the kinetic equation (3.42) leads to

$$-i\omega\delta f \to -i\left[\omega + 2i\left(\frac{1}{\tau^0_{22}} + \frac{1}{\tau^0_{12}}\right)\right]\delta f. \tag{6.70}$$

Using the linearized collision integrals given in (6.69) corresponds to including the damping of the atoms described by the kinetic equation.

We recall that the C_{12} integral also enters into the GGP equation (3.21) through the dissipative term $-iR$, where $R(\mathbf{r}, t)$ is defined in (3.22). Using

our linearized approximation (6.67) for C_{12} to calculate R, we find

$$-iR(\mathbf{r},t) = \frac{i}{n_{c0}(\mathbf{r})}\frac{\hbar}{\bar{\tau}_{12}(\mathbf{r})}\delta\tilde{n}(\mathbf{r},t). \tag{6.71}$$

Here $1/\bar{\tau}_{12}$ is the collision time in (6.67) momentum averaged over the deviation $\delta f_{\mathbf{p}}(\mathbf{r},t)$. The term (6.71) extends the HF term $2g\delta\tilde{n}$ in the GGP equation (3.21) to include collisional damping. In Chapter 8, we will evaluate Γ_{12} to first order in the deviation δn_c of the condensate atom density from equilibrium, assuming that all the thermal atoms are in thermal equilibrium (this is the so-called static thermal cloud approximation). The full linearized expression for $C_{12}[f,\Phi]$ thus contains terms proportional to the deviations δf and δn_c, the coefficients in each case being evaluated in thermal equilibrium.

We conclude with some comments about the linearized C_{12} integral. In Sections 5.3 and 5.4, we showed that one can start from a collisionless kinetic equation with time-dependent HFB mean fields and generate a density response function that includes terms of second order in g. One may view the KB formalism as similarly generating an improved kinetic equation with analogous second-order corrections, although now not limited to thermal equilibrium. It is no surprise that if we linearize the second-order collision term C_{12} we reproduce (as shown by (6.59) and (6.67)) the Beliaev second-order damping terms, evaluated for thermal equilibrium and within the Popov approximation. In Section 7.3, we will linearize the analogous expression for C_{12} for the case when the thermal cloud is described in terms of Bogoliubov–Popov excitations. This will be shown to give rise to Beliaev and Landau damping terms in complete agreement with the full Beliaev approximation (5.40).

7

Kinetic equation for Bogoliubov thermal excitations

In Chapter 6, we derived a generalized Gross–Pitaevskii condensate equation which is coupled to a kinetic equation for the distribution function for the thermal atoms. However, the kinetic equation in Chapter 6 is only valid in the semiclassical limit. It involves the assumption that the thermal energy $k_{\rm B}T$ is much greater than the spacing between the harmonic trap energy levels ($k_{\rm B}T \gg \hbar\omega_0$ where ω_0 is the trap frequency) and also much greater than the average interaction energy ($k_{\rm B}T \gg gn$). The ZNG model, based on HF excitations, is still expected to be adequate down to quite low temperatures in trapped Bose gases, as will be shown by the results in Chapter 12. However, the ZNG model will break down at very low temperatures, where the Hartree–Fock excitations must be replaced by the Bogoliubov spectrum. To deal with this, one has to derive a kinetic equation for the Bogoliubov quasiparticle excitations. This is the goal of the present chapter.

In this chapter, we use the second-order Beliaev approximation to discuss the *nonequilibrium dynamics* of a trapped Bose-condensed gas at finite temperatures. In doing to, we combine the second-order Beliaev self-energies with the lower-order Bogoliubov excitation spectrum, including off-diagonal single-particle propagators but still omitting the anomalous correlation functions $\tilde{m}$. This last condition defines the Bogoliubov–Popov approximation. In this chapter, we consider only the damping which arises from collisions. We will not explicitly calculate corrections that are second-order in g to the quasiparticle energy spectrum or to the condensate chemical potential, both of which are associated with the real parts of the second-order Beliaev self-energies.

The present chapter is a natural generalization of work presented in Chapter 6 for the simpler HF excitation spectrum assumed in the ZNG analysis. In Section 7.1, following the Kadanoff–Baym (KB) approach (Kane and Kadanoff, 1965; Kane, 1966), we derive a generalized quantum Boltzmann

equation for the quasiparticle distribution function $f(\mathbf{p},\omega;\mathbf{R},T)$, which is now frequency dependent. In Section 7.2, we use this generalized KB quantum Boltzmann equation to derive the kinetic equation for quasiparticles at low temperatures, in which the C_{12} collision integral describes collisions between the condensate atoms and the normal fluid quasiparticles in the Bogoliubov–Popov approximation. In Section 7.3 we *linearize* the collision integrals, assuming that the distribution function is very close to thermal equilibrium, and show that in this limit, they describe both Landau and Beliaev damping.

A related derivation of these generalized kinetic equations based on the 2PI formalism (see the introduction to Chapter 6) was given by Rey *et al.* (2005). Their paper treats the case of a condensate and a thermal cloud in an optical lattice.

7.1 Generalized kinetic equation

In Section 6.2, we wrote down the general equations of motion (6.11) and (6.12) for the nonequilibrium real-time Green's functions. We will now use these to derive a kinetic equation for an appropriately defined quasiparticle distribution function. We recall that (6.11) and (6.12) are 2×2 matrix equations. In addition, (6.11) involves differential operators with the respect to the coordinates $\mathbf{r},t$ and (6.12) involves derivatives with respect to the coordinates $\mathbf{r}',t'$. Since our single-particle Green's functions are functions of both the coordinates 1 and $1'$, one has to find a way to combine the two equations so as to derive a *single* kinetic equation for the quasiparticle distribution function. We will discuss later how these Green's functions can be related to the quasiparticle distribution function.

As in Chapter 6, we write (6.11) and (6.12) in terms of the centre-of-mass and relative coordinates, $\mathbf{R},T$ and $\mathbf{r},t$, defined in (6.21) and (6.22). Taking the trace of the resulting matrix equation, we obtain

$$\begin{aligned}
&\hat{\mathcal{L}}_{11}\tilde{g}^{<}_{11}(\mathbf{r},t)+\hat{\mathcal{L}}_{22}\tilde{g}^{<}_{22}(\mathbf{r},t)\\
&=g\left[\left(\mathbf{r}\cdot\nabla_{\mathbf{R}}+t\frac{\partial}{\partial T}\right)m\,\tilde{g}^{<}_{21}(\mathbf{r},t)+\left(\mathbf{r}\cdot\nabla_{\mathbf{R}}+t\frac{\partial}{\partial T}\right)m^{*}\,\tilde{g}^{<}_{12}(\mathbf{r},t)\right]\\
&\quad+\int_{-\infty}^{\infty}d\bar{\mathbf{r}}d\bar{t}\,\mathrm{Tr}\left[\Sigma^{>}_{\mathrm{coll}}(\mathbf{r}-\bar{\mathbf{r}},t-\bar{t}\,)\tilde{g}^{<}(\bar{\mathbf{r}},\bar{t}\,)-\Sigma^{<}_{\mathrm{coll}}(\mathbf{r}-\bar{\mathbf{r}},t-\bar{t}\,)\tilde{g}^{>}(\bar{\mathbf{r}},\bar{t}\,)\right].
\end{aligned}\tag{7.1}$$

Here g is the interaction strength, while the off-diagonal density $m(\mathbf{R},T)$ was defined in (6.19). The $(\mathbf{R},T)$-dependence of these off-diagonal densities, the real-time Green's functions $\tilde{g}_{ij}$ and the associated collisional self-energies

$\Sigma_{\rm coll}$ is left implicit. The new operators $\hat{\mathcal{L}}_{11}$ and $\hat{\mathcal{L}}_{22}$ on the l.h.s. of (7.1) are defined as follows:

$$\begin{aligned}\hat{\mathcal{L}}_{11} \equiv & \, i\frac{\partial}{\partial T} + \left(\mathbf{r}\cdot\nabla_{\mathbf{R}} + t\frac{\partial}{\partial T}\right)\Big[\mu(\mathbf{R},T) - U(\mathbf{R},T)\Big] + \frac{1}{m}\,\nabla_{\mathbf{R}}\cdot\nabla\mathbf{r} \\ & + i\left(\mathbf{r}\cdot\nabla_{\mathbf{R}} + t\frac{\partial}{\partial T}\right)\mathbf{v}_{\rm s}(\mathbf{R},T)\cdot\nabla_{\mathbf{r}} + i\mathbf{v}_{\rm s}(\mathbf{R},T)\cdot\nabla_{\mathbf{R}} + i\,\nabla_{\mathbf{R}}\cdot\mathbf{v}_{\rm s}(\mathbf{R},T), \\ \hat{\mathcal{L}}_{22} \equiv & -i\frac{\partial}{\partial T} + \left(\mathbf{r}\cdot\nabla_{\mathbf{R}} + t\frac{\partial}{\partial T}\right)\Big[\mu(\mathbf{R},T) - U(\mathbf{R},T)\Big] + \frac{1}{m}\,\nabla_{\mathbf{R}}\cdot\nabla\mathbf{r} \\ & - i\left(\mathbf{r}\cdot\nabla_{\mathbf{R}} + t\frac{\partial}{\partial T}\right)\mathbf{v}_{\rm s}(\mathbf{R},T)\cdot\nabla_{\mathbf{r}} - i\mathbf{v}_{\rm s}(\mathbf{R},T)\cdot\nabla_{\mathbf{R}} - i\,\nabla_{\mathbf{R}}\cdot\mathbf{v}_{\rm s}(\mathbf{R},T),\end{aligned} \tag{7.2}$$

where the self-consistent time-dependent HF field $U(\mathbf{R},T)$ is given by

$$U(\mathbf{R},T) = V_{\rm trap}(\mathbf{R}) + 2g[n_{\rm c}(\mathbf{R},T) + \tilde{n}(\mathbf{R},T)]. \tag{7.3}$$

We emphasize that in the expansion for small values of the relative coordinates $\mathbf{r}, t$, we have not kept all terms of order $\partial/\partial T$ and $\nabla_{\mathbf{R}}$ in (7.1). These additional terms, which we have neglected, contribute to the many-body renormalization effects, i.e. the changes in the dispersion relation of the quasiparticles due to two-body interactions. Such corrections involve the *real part* of the second-order Beliaev self-energies. The Bogoliubov–Popov quasiparticle approximation that we shall use for the Beliaev spectral densities $a_{\alpha\beta}$ defined in (6.13) neglects such second-order effects. In the present chapter, we will concentrate on the damping associated with the collisional self-energies $\Sigma^{>}_{\rm coll}$ on the right-hand side of (7.1). We note that Kane and Kadanoff (1965) and Kane (1966), albeit working within a truncated but "conserving" second-order self-energy approximation, derived equations of motion that include many body renormalization effects to second order in g. Their work (see the remarks after (4.101)) was thus a more sophisticated implementation of the Kadanoff–Baym formalism than that used in this chapter.

A double Fourier transform of (7.1) with respect to the relative coordinates $\mathbf{r}$ and t gives

$$\begin{aligned}& \hat{\mathcal{L}}_{11}\tilde{g}^{<}_{11} + \hat{\mathcal{L}}_{22}\tilde{g}^{<}_{22} - g\nabla_{\mathbf{R}}m^{*}\cdot\nabla_{\mathbf{p}}\tilde{g}^{<}_{12} - g\nabla_{\mathbf{R}}m\cdot\nabla_{\mathbf{p}}\tilde{g}^{<}_{21} \\ & \qquad + g\frac{\partial m^{*}}{\partial T}\frac{\partial\tilde{g}^{<}_{12}}{\partial\omega} + g\frac{\partial m}{\partial T}\frac{\partial\tilde{g}^{<}_{21}}{\partial\omega} \\ & = {\rm Tr}\left[\Sigma^{>}_{\rm coll}(\mathbf{p},\omega;\mathbf{R},T)\tilde{g}^{<}(\mathbf{p},\omega;\mathbf{R},T) - \Sigma^{<}_{\rm coll}(\mathbf{p},\omega;\mathbf{R},T)\tilde{g}^{>}(\mathbf{p},\omega;\mathbf{R},T)\right],\end{aligned} \tag{7.4}$$

with $\tilde{g}^{<}_{\alpha\beta} \equiv \tilde{g}^{<}_{\alpha\beta}(\mathbf{p},\omega;\mathbf{R},T)$. We have defined the operators

$$\begin{aligned}\hat{\mathcal{L}}_{11} &\equiv \frac{\partial}{\partial T} + \nabla_{\mathbf{p}}\tilde{\varepsilon}_{+}\cdot\nabla_{\mathbf{R}} - \nabla_{\mathbf{R}}\tilde{\varepsilon}_{+}\cdot\nabla_{\mathbf{p}} + \frac{\partial\tilde{\varepsilon}_{+}}{\partial T}\frac{\partial}{\partial\omega},\\ \hat{\mathcal{L}}_{22} &\equiv -\frac{\partial}{\partial T} + \nabla_{\mathbf{p}}\tilde{\varepsilon}_{-}\cdot\nabla_{\mathbf{R}} - \nabla_{\mathbf{R}}\tilde{\varepsilon}_{-}\cdot\nabla_{\mathbf{p}} + \frac{\partial\tilde{\varepsilon}_{-}}{\partial T}\frac{\partial}{\partial\omega},\end{aligned} \tag{7.5}$$

where

$$\tilde{\varepsilon}_{\pm} \equiv \tilde{\varepsilon}_{\mathbf{p}} \pm \mathbf{p}\cdot\mathbf{v}_{\mathrm{s}} \tag{7.6}$$

and $\tilde{\varepsilon}_{\mathbf{p}}$ is the Hartree–Fock single particle energy,

$$\tilde{\varepsilon}_{\mathbf{p}}(\mathbf{R},T) = \frac{p^2}{2m} + V_{\mathrm{trap}}(\mathbf{R}) + 2gn(\mathbf{R},T) - \mu_{\mathrm{c}}(\mathbf{R},T). \tag{7.7}$$

The condensate chemical potential $\mu_{\mathrm{c}}(\mathbf{R},T)$ in (7.7) is given by

$$\mu_{\mathrm{c}} \equiv -\frac{\hbar^2\nabla^2_{\mathbf{R}}\sqrt{n_{\mathrm{c}}(\mathbf{R},T)}}{2m\sqrt{n_{\mathrm{c}}(\mathbf{R},T)}} + V_{\mathrm{trap}}(\mathbf{R}) + g\left[2\tilde{n}(\mathbf{R},T) + n_{\mathrm{c}}(\mathbf{R},T)\right]. \tag{7.8}$$

We note that the result (7.4) already gives an equation whose structure is characteristic of a kinetic equation. The l.h.s. is the collisionless or "streaming" part, while the r.h.s. will be shown to describe collisions.

A kinetic equation for thermally excited atoms in a trapped Bose gas can be written in terms of distribution functions for the atoms directly or for quasiparticle excitations. In Chapter 6, we transformed the equations of motion for nonequilibrium Green's functions at high temperatures into a kinetic equation for a distribution function $f(\mathbf{p},\mathbf{R},T)$ that describe noncondensate atoms with a Hartree–Fock energy spectrum. If one is dealing with the Bogoliubov spectrum appropriate to a thermal cloud at very low temperatures, it is more convenient to work within a Bogoliubov quasiparticle picture. In the theory of Bose-condensed trapped gases, one introduces quasiparticles by expressing the quantum field operators for the noncondensate atoms as a coherent superposition of creation and annihilation operators for Bose quasiparticles, with the weights given by the usual Bogoliubov coherence factors u and v (see also (2.47)):

$$\tilde{\psi}(\mathbf{R},T) \equiv \sum_i \left[u_i(\mathbf{R})\hat{\alpha}_i e^{-iE_iT/\hbar} - v_i^*(\mathbf{R})\hat{\alpha}_i^\dagger e^{iE_iT/\hbar}\right]. \tag{7.9}$$

In our semiclassical approximation, (7.9) becomes

$$\tilde{\psi}(\mathbf{R},T) \equiv \int \frac{d\mathbf{p}}{(2\pi)^3}\left[u_{\mathbf{p}}(\mathbf{R})\hat{\alpha}_{\mathbf{p}} e^{-iE_pT/\hbar} - v_{\mathbf{p}}^*(\mathbf{R})\hat{\alpha}_{\mathbf{p}}^\dagger e^{iE_pT/\hbar}\right]. \tag{7.10}$$

Here $\hat{\alpha}^\dagger_{\mathbf{p}}$ and $\hat{\alpha}_{\mathbf{p}}$ are the Bogoliubov quasiparticle creation and annihilation operators, respectively, which obey the usual Bose commutation relations. One can see that creating an atom with momentum $\mathbf{p}$ is equivalent to creating a quasiparticle with momentum $\mathbf{p}$ with amplitude $u_{\mathbf{p}}$ and, at the same time, destroying a quasiparticle with momentum $-\mathbf{p}$ with amplitude $v_{\mathbf{p}}$. The quasiparticle distribution function is given by the statistical average of the quasiparticle operators, i.e. $f(\mathbf{p}) \equiv \langle \hat{\alpha}^\dagger_{\mathbf{p}} \hat{\alpha}_{\mathbf{p}} \rangle$. We recall (Imamović-Tomasović and Griffin, 2000) that the distribution function for atoms $f_{\rm at}(\mathbf{p}, \omega; \mathbf{R}, T)$ is directly related to the diagonal Green's function, namely

$$f_{\rm at}(\mathbf{p}, \omega; \mathbf{R}, T) = -i\tilde{g}^<_{11}(\mathbf{p}, \omega; \mathbf{R}, T). \tag{7.11}$$

In terms of quantum field operators, $\tilde{g}^<_{11}$ is given by (see Chapter 9 of Kadanoff and Baym, 1962)

$$-i\tilde{g}^<_{11}(\mathbf{p}, \omega; \mathbf{R}, T) = \int d\mathbf{r} dt\, e^{-i\mathbf{p}\cdot\mathbf{r} + i\omega t} \Big\langle \tilde{\psi}^\dagger(\mathbf{R} - \tfrac{1}{2}\mathbf{r}, T - \tfrac{1}{2}t) \tilde{\psi}(\mathbf{R} + \tfrac{1}{2}\mathbf{r}, T + \tfrac{1}{2}t) \Big\rangle. \tag{7.12}$$

The usual Boltzmann equation is expressed in terms of the Wigner distribution function $f_{\rm W}(\mathbf{p}, \mathbf{R}, T)$ (Imamović-Tomasović and Griffin, 2000). This limits the description to the semiclassical approximation, since it is assumed that the position and momentum of the atoms can be defined simultaneously. Thus in order to use this kind of distribution function for quantum systems, it is necessary to perform some type of averaging in order to remove effects due to the uncertainty principle. In this chapter, we want to derive a kinetic equation for the quasiparticles that is valid at low temperatures, where the semiclassical approximation is no longer valid. To include the quantum effects, we must introduce a *quasiparticle* distribution function $f(\mathbf{p}, \omega; \mathbf{R}, T)$ with an additional variable ω in the following way (Kane and Kadanoff, 1965; Mahan, 1990; Haug and Jauko, 1996; Zubarev *et al.*, 1997):

$$\begin{aligned} -i\tilde{g}^<_{\alpha\beta}(\mathbf{p}, \omega; \mathbf{R}, T) &\equiv a_{\alpha\beta}(\mathbf{p}, \omega; \mathbf{R}, T) f(\mathbf{p}, \omega; \mathbf{R}, T), \\ -i\tilde{g}^>_{\alpha\beta}(\mathbf{p}, \omega; \mathbf{R}, T) &\equiv a_{\alpha\beta}(\mathbf{p}, \omega; \mathbf{R}, T)[1 + f(\mathbf{p}, \omega; \mathbf{R}, T)], \end{aligned} \tag{7.13}$$

where the spectral densities $a_{\alpha\beta}$ were defined in (6.13). By means of the Bogoliubov–Popov approximation for the spectral density (see (7.19) below Section 7.2), one can use equations (7.13) to obtain the well-known relation between the quasiparticle distribution function $f(\mathbf{p}, \omega; \mathbf{R}, T)$ and the atom distribution function $f_{\rm at}$ (7.11), namely

$$f_{\rm at}(\mathbf{p}, \omega; \mathbf{R}, T) = \left[u^2_{\mathbf{p}}(\mathbf{R}, T) + v^2_{\mathbf{p}}(\mathbf{R}, T) \right] f(\mathbf{p}, \omega; \mathbf{R}, T) + v^2_{\mathbf{p}}(\mathbf{R}, T). \tag{7.14}$$

The semiclassical Wigner distribution function is obtained by integrating over the frequency ω.

One can show, using (6.11) and (6.12), that the spectral function $a_{\alpha\beta}$ defined in (6.13) satisfies

$$\hat{\mathcal{L}}_{11}a_{11} + \hat{\mathcal{L}}_{22}a_{22} - g\nabla_{\mathbf{R}}m^* \cdot \nabla_{\mathbf{p}}a_{12} - g\nabla_{\mathbf{R}}m \cdot \nabla_{\mathbf{p}}a_{21} - g\frac{\partial m^*}{\partial T}\frac{\partial a_{12}}{\partial \omega} - g\frac{\partial m}{\partial T}\frac{\partial a_{21}}{\partial \omega} = 0. \tag{7.15}$$

Using (7.13) and (7.15), we can rewrite our original kinetic equation (7.4) for $\tilde{g}^{<}_{\alpha\beta}$ to finally obtain a new kinetic equation specifically for the quasiparticle distribution function $f(\mathbf{p}, \omega; \mathbf{R}, T)$, namely

$$a_{11}\hat{\mathcal{L}}_{11}f + a_{22}\hat{\mathcal{L}}_{22}f - a_{12}g\nabla_{\mathbf{R}}m^* \cdot \nabla_{\mathbf{p}}f - a_{21}g\nabla_{\mathbf{R}}m \cdot \nabla_{\mathbf{p}}f + a_{12}\frac{\partial m^*}{\partial T}\frac{\partial f}{\partial \omega} + a_{21}\frac{\partial m}{\partial T}\frac{\partial f}{\partial \omega} = f\,\mathrm{Tr}\,(\hat{\Sigma}^{>}\hat{a}) - (1+f)\,\mathrm{Tr}\,(\hat{\Sigma}^{<}\hat{a}). \tag{7.16}$$

We note that the additional variable ω in $f(\mathbf{p}, \omega; \mathbf{R}, T)$ results in new streaming terms involving $\partial/\partial\omega$ on the l.h.s. of (7.16). These terms are not present in the semiclassical kinetic equation (which can be obtained from (7.16) by integrating over ω). This reminds us that the terms involving $\partial/\partial\omega$ are of quantum origin (Kadanoff and Baym, 1962).

Equation (7.16) is the most general form for a kinetic equation for the quasiparticle distribution function f within the KB formalism. To derive (7.16), we have assumed only that the external perturbation varies slowly in space and time, and therefore that all relevant physical quantities vary slowly as functions of the centre-of-mass coordinates $\mathbf{R}, T$ defined in (6.21). Another necessary assumption is that one can introduce the quasiparticle distribution function f through the definition in (7.13). The generalized kinetic equation (7.16) for a Bose-condensed system was first discussed by Kane (1966) in his thesis.

From the r.h.s. of (7.16) we can deduce that the general structure of the collision integral I has the following form (Kane and Kadanoff, 1965; Kane, 1966; Stoof, 2001):

$$I[f(\mathbf{p}, \mathbf{R}, T)] \equiv \int \frac{d\omega}{2\pi}\left[f\,\mathrm{Tr}(\hat{\Sigma}^{>}\hat{a}) - (1+f)\,\mathrm{Tr}(\hat{\Sigma}^{<}\hat{a})\right]. \tag{7.17}$$

Using (7.13) in the general expression for the nonequilibrium Beliaev self-energies $\Sigma^{>}_{<}$ given below in Section 7.2, one can prove that the collision integral given by (7.17) conserves momentum (see the appendix of Imamović-

Tomasović, 2001),

$$\int d\mathbf{p}\,\mathbf{p} I[f(\mathbf{p},\mathbf{R},T)] = 0. \tag{7.18}$$

One can prove that the collision integral in (7.17) conserves energy as well.

7.2 Kinetic equation in the Bogoliubov–Popov approximation

In this section, we use the generalized kinetic equation (7.16) to derive a kinetic equation for quasiparticles described by the Bogoliubov–Popov approximation. More precisely, we use the Bogoliubov–Popov quasiparticle spectral densities

$$\begin{aligned}
a_{11}(\mathbf{p},\omega;\mathbf{R},T) &= 2\pi[u_{\mathbf{p}}^2\delta(\omega-\mathbf{v}_{\mathrm{s}}\cdot\mathbf{p}-E_p) - v_{\mathbf{p}}^2\delta(\omega-\mathbf{v}_{\mathrm{s}}\cdot\mathbf{p}+E_p)],\\
a_{12}(\mathbf{p},\omega;\mathbf{R},T) &= -2\pi u_{\mathbf{p}}v_{\mathbf{p}}[\delta(\omega-\mathbf{v}_{\mathrm{s}}\cdot\mathbf{p}-E_p) - \delta(\omega-\mathbf{v}_{\mathrm{s}}\cdot\mathbf{p}+E_p)],\\
a_{21}(\mathbf{p},\omega;\mathbf{R},T) &= a_{12}(\mathbf{p},\omega;\mathbf{R},T),\\
a_{22}(\mathbf{p},\omega;\mathbf{R},T) &= -a_{11}(-\mathbf{p},-\omega;\mathbf{R},T).
\end{aligned} \tag{7.19}$$

The Bogoliubov coherence factors $u_{\mathbf{p}}(\mathbf{R},T)$ and $v_{\mathbf{p}}(\mathbf{R},T)$ are given by (see for example, Fetter and Walecka, 1971)

$$\begin{aligned}
u_{\mathbf{p}}^2(\mathbf{R},T) &= \frac{\tilde{\varepsilon}_p(\mathbf{R},T)+E_p(\mathbf{R},T)}{2E_p(\mathbf{R},T)},\\
u_{\mathbf{p}}^2 - v_{\mathbf{p}}^2 = 1, &\quad u_{\mathbf{p}}v_{\mathbf{p}} = \frac{gn_{\mathrm{c}}(\mathbf{R},T)}{2E_p(\mathbf{R},T)},
\end{aligned} \tag{7.20}$$

and the quasiparticle energy E_p is given by

$$E_p(\mathbf{R},T) = \sqrt{\tilde{\varepsilon}_p^2(\mathbf{R},T) - (gn_{\mathrm{c}}(\mathbf{R},T))^2}. \tag{7.21}$$

The HF energy $\tilde{\varepsilon}_p$ was defined in (7.8). We note that in the Bose gas literature, $v_{\mathbf{p}}$ is sometimes defined with the opposite sign, so that $u_{\mathbf{p}}v_{\mathbf{p}}$ in (7.20) is negative. We also emphasize that the spectral densities (7.19) can be derived in the quasiparticle approximation from the general equations of motion for the Green's functions, as shown explicitly by Kane (1966). In this chapter, we simply take these results as inputs for our general KB equation of motion formalism.

In the Thomas–Fermi approximation, where one neglects the "quantum pressure" term in (7.8), (7.7) reduces to $\tilde{\varepsilon}_p = p^2/2m + gn_{\rm c}$. In this case, the quasiparticle energy E_p in (7.21) is the Bogoliubov–Popov excitation energy:

$$E_p(\mathbf{R},T) = \sqrt{\varepsilon_p^2 + 2gn_{\rm c}(\mathbf{R},T)\varepsilon_p}, \tag{7.22}$$

where $\varepsilon_p = p^2/2m$. We note that the single-particle spectral densities $a_{\alpha\beta}(\mathbf{p},\omega;\mathbf{R},T)$ given in (7.19) exhibit both positive and negative energy poles, a characteristic feature of a Bose-condensed system. The Hartree–Fock approximation used in Chapter 6 is reproduced by (7.19) on setting $u_{\mathbf{p}}^2 = 1$ and $v_{\mathbf{p}}^2 = 0$. Physically, (7.19) corresponds to the assumption that the thermal cloud can be considered as a gas of weakly interacting single-particle excitations with excitation energy (7.22). One can check explicitly that the spectral densities (7.19) do satisfy the general equation of motion (7.15).

Substituting the spectral densities (7.19) into (7.16), we derive a kinetic equation for the quasiparticles. Defining the *quasiparticle* distribution function by

$$f_{\rm qp}(\mathbf{p},\mathbf{R},T) \equiv f(\mathbf{p},\, \omega = E_p + \mathbf{v}_{\rm s}\cdot\mathbf{p}\,;\, \mathbf{R},T), \tag{7.23}$$

the final result is

$$\frac{\partial f_{\rm qp}}{\partial T} + \nabla_{\mathbf{p}}\,(E_p + \mathbf{v}_{\rm s}\cdot\mathbf{p})\cdot\nabla_{\mathbf{R}} f_{\rm qp} - \nabla_{\mathbf{R}}\,(E_p + \mathbf{v}_{\rm s}\cdot\mathbf{p})\cdot\nabla_{\mathbf{p}} f_{\rm qp} = I[f_{\rm qp}]. \tag{7.24}$$

The collision integral $I[f_{\rm qp}(\mathbf{p},\mathbf{R},T)]$ defined in (7.17) is given explicitly by

$$\begin{aligned} I[f_{\rm qp}(\mathbf{p},\mathbf{R},T)] \equiv \int_{-\infty}^{\infty}\frac{d\omega}{2\pi}\Big\{ & u_{\mathbf{p}}^2\Big[(1+f_{\rm qp})\Sigma_{11}^{<} - f_{\rm qp}\Sigma_{11}^{>}\Big] \\ & + v_{\mathbf{p}}^2\Big[(1+f_{\rm qp})\Sigma_{22}^{<} - f_{\rm qp}\Sigma_{22}^{>}\Big] \\ & + u_{\mathbf{p}}v_{\mathbf{p}}\Big[(1+f_{\rm qp})\Big(\Sigma_{12}^{<}+\Sigma_{21}^{<}\Big) - f_{\rm qp}(\Sigma_{12}^{>}+\Sigma_{21}^{>})\Big]\Big\}. \end{aligned} \tag{7.25}$$

The kinetic equation (7.24) was first obtained by Kane (1966). A similar quasiparticle kinetic equation has been derived by Stoof (1999, 2001) using the Schwinger–Keldysh formalism for nonequilibrium processes.

What remains to be done is to choose a specific approximation for the second-order self-energies $\Sigma_{\alpha\beta}$, so that we can evaluate the collision integral (7.25). We use the second-order Beliaev approximation (4.107), the Fourier

transform of which is

$$
\begin{aligned}
&\hat{\Sigma}^{<}(\mathbf{p},\omega;\mathbf{R},T)\\
&= -\frac{g^2}{2\hbar^4}\int \frac{d\mathbf{p}_i d\omega_i}{(2\pi)^8}\delta(\omega+\omega_1-\omega_2-\omega_3)\delta(\mathbf{p}+\mathbf{p}_1-\mathbf{p}_2-\mathbf{p}_3)\\
&\quad\times\Big\{\hat{\tilde{g}}^{>}(\mathbf{p}_2,\omega_2)\,\mathrm{Tr}\Big[\hat{\tilde{g}}^{>}(\mathbf{p}_1,\omega_1)\hat{\tilde{g}}^{<}(\mathbf{p}_3,\omega_3)\Big]\\
&\qquad+2\hat{\tilde{g}}^{>}(\mathbf{p}_2,\omega_2)\hat{\tilde{g}}^{>}(\mathbf{p}_1,\omega_1)\hat{\tilde{g}}^{<}(\mathbf{p}_3,\omega_3)\\
&\qquad+\hat{\tilde{g}}^{>}(\mathbf{p}_2,\omega_2)\,\mathrm{Tr}\Big[\hat{h}(\mathbf{p}_1,\omega_1)\hat{\tilde{g}}^{<}(\mathbf{p}_3,\omega_3)+\hat{\tilde{g}}^{>}(\mathbf{p}_1,\omega_1)\hat{h}(\mathbf{p}_3,\omega_3)\Big]\\
&\qquad+\hat{h}(\mathbf{p}_2,\omega_2)\,\mathrm{Tr}[\hat{\tilde{g}}^{>}(\mathbf{p}_1,\omega_1)\hat{\tilde{g}}^{<}(\mathbf{p}_3,\omega_3)]\\
&\qquad+2\hat{h}(\mathbf{p}_2,\omega_2)\hat{\tilde{g}}^{>}(\mathbf{p}_1,\omega_1)\hat{\tilde{g}}^{<}(\mathbf{p}_3,\omega_3)\\
&\qquad+2\hat{\tilde{g}}^{>}(\mathbf{p}_2,\omega_2)\Big[\hat{h}(\mathbf{p}_1,\omega_1)\hat{\tilde{g}}^{<}(\mathbf{p}_3,\omega_3)+\hat{\tilde{g}}^{>}(\mathbf{p}_1,\omega_1)\hat{h}(\mathbf{p}_3,\omega_3)\Big]\Big\}.
\end{aligned}
\tag{7.26}
$$

Note that in (7.26) integration is carried out over all the variables $\mathbf{p}_i$ and ω_i, $i=1,2,3$. As usual, the $(\mathbf{R},T)$-dependence of the functions Σ, $\tilde{g}$ and h on the r.h.s. has been suppressed for simplicity of notation. The quasiparticle energy $E_p(\mathbf{R},T)$ in (7.24) is the energy of the quasiparticles in the local rest frame ($\mathbf{v}_\mathrm{s}=0$). The Beliaev second-order expression (7.26) consists of two kinds of contribution: (a) terms that include both the condensate propagator h *and* the noncondensate propagators $\tilde{g}$; (b) terms that include the noncondensate propagators $\tilde{g}$ only. The first type of contribution describes collisions where one condensate atom interacts with the thermally excited quasiparticles. These terms form part of the collision integral C_{12}, which comprises processes that go from 1 thermally excited quasiparticle (and one condensate atom) to 2 thermally excited quasiparticles. The first two terms in (7.26) involve three noncondensate propagators $\tilde{g}$. This part of the collision integral is called C_{22} and describes collisions where 2 thermally excited quasiparticles are scattered into 2 excited quasiparticles. At low temperatures, when the number of thermally-excited quasiparticles is very small, we can neglect the C_{22} collision integral as being much smaller than C_{12}.

From (7.18), it follows that both C_{12} and C_{22} conserve momentum, i.e.

$$
\int d\mathbf{p}\,\mathbf{p}\,C_{12}=0,\quad \int d\mathbf{p}\,\mathbf{p}\,C_{22}=0. \tag{7.27}
$$

In addition, using (7.19) in (7.26) one can show that the collision integral

in (7.17) conserves quasiparticle energy E_p and therefore both C_{12} and C_{22} will satisfy the conditions

$$\begin{aligned} \int d\mathbf{p}\, E_p C_{12} &= 0, \\ \int d\mathbf{p}\, E_p C_{22} &= 0. \end{aligned} \tag{7.28}$$

For slowly varying external disturbances, the condensate propagator in (4.67) can be approximated by

$$\hat{h}(\mathbf{p},\omega;\mathbf{R},T) = n_{\rm c}(\mathbf{R},T)(2\pi)^4\delta(\mathbf{p})\delta(\omega)\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}. \tag{7.29}$$

To evaluate the collision integral C_{12} in terms of the Bose coherence factors u and v in (7.20), we need to substitute (7.19) and (7.29) into (7.26). One can simplify (7.26) greatly using the following exact symmetry relations:

$$\begin{aligned} \tilde{g}^{>}_{22}(\mathbf{p},\omega;\mathbf{R},T) &= \tilde{g}^{<}_{11}(-\mathbf{p},-\omega;\mathbf{R},T), \\ \tilde{g}^{>}_{12}(\mathbf{p},\omega;\mathbf{R},T) &= \tilde{g}^{<}_{12}(-\mathbf{p},-\omega;\mathbf{R},T), \\ \tilde{g}^{>}_{21}(\mathbf{p},\omega;\mathbf{R},T) &= \tilde{g}^{<}_{21}(-\mathbf{p},-\omega;\mathbf{R},T). \end{aligned} \tag{7.30}$$

After considerable algebra, one finds the following expressions for the non-equilibrium self-energies (limiting ourselves to the C_{12} terms in (7.26) that include one condensate propagator $\hat{h}$):

$$\begin{aligned} &\Sigma^{<}_{11}(\mathbf{p},\omega;\mathbf{R},T) \\ &= -\frac{g^2}{\hbar^4}\int \frac{d\mathbf{p}_2 d\omega_2}{(2\pi)^4} n_{\rm c}(\mathbf{R},T) \\ &\times \Big[2\tilde{g}^{>}_{11}(\mathbf{p}_2,\omega_2)\tilde{g}^{<}_{11}(\mathbf{p}-\mathbf{p}_2,\omega-\omega_2) + 4\tilde{g}^{>}_{11}(\mathbf{p}_2,\omega_2)\tilde{g}^{<}_{11}(\mathbf{p}_2-\mathbf{p},\omega_2-\omega) \\ &\quad + 8\tilde{g}^{>}_{12}(\mathbf{p}_2,\omega_2)\tilde{g}^{>}_{11}(\mathbf{p}-\mathbf{p}_2,\omega-\omega_2) + 4\tilde{g}^{>}_{12}(\mathbf{p}_2,\omega_2)\tilde{g}^{<}_{12}(\mathbf{p}-\mathbf{p}_2,\omega-\omega_2) \Big], \\ &\Sigma^{>}_{22}(\mathbf{p},\omega;\mathbf{R},T) \\ &= -\frac{g^2}{\hbar^4}\int \frac{d\mathbf{p}_2 d\omega_2}{(2\pi)^4} n_{\rm c}(\mathbf{R},T) \\ &\times \Big[4\tilde{g}^{>}_{12}(\mathbf{p}_2,\omega_2)\tilde{g}^{>}_{12}(\mathbf{p}-\mathbf{p}_2,\omega-\omega_2) + 4\tilde{g}^{>}_{11}(\mathbf{p}_2,\omega_2)\tilde{g}^{<}_{11}(\mathbf{p}_2-\mathbf{p},\omega_2-\omega) \\ &\quad + 8\tilde{g}^{>}_{12}(\mathbf{p}_2,\omega_2)\tilde{g}^{<}_{11}(\mathbf{p}_2-\mathbf{p},\omega_2-\omega) + 2\tilde{g}^{<}_{11}(-\mathbf{p}_2,-\omega_2)\tilde{g}^{>}_{11}(\mathbf{p}_2-\mathbf{p},\omega_2-\omega) \Big], \end{aligned}$$

$$\Sigma^{\gtrless}_{12}(\mathbf{p},\omega;\mathbf{R},T) = -\frac{g^2}{\hbar^4}\int \frac{d\mathbf{p}_2 d\omega_2}{(2\pi)^4} n_{\rm c}(\mathbf{R},T)$$
$$\times\Big[6\tilde{g}^{\gtrless}_{12}(\mathbf{p}_2,\omega_2)\tilde{g}^{\gtrless}_{12}(\mathbf{p}-\mathbf{p}_2,\omega-\omega_2) + 4\tilde{g}^{\gtrless}_{12}(\mathbf{p}_2,\omega_2)\tilde{g}^{\gtrless}_{11}(\mathbf{p}-\mathbf{p}_2,\omega-\omega_2)$$
$$+ 4\tilde{g}^{\gtrless}_{11}(\mathbf{p}_2,\omega_2)\tilde{g}^{\lessgtr}_{11}(\mathbf{p}_2-\mathbf{p},\omega_2-\omega) + 4\tilde{g}^{\gtrless}_{12}(\mathbf{p}_2,\omega_2)\tilde{g}^{\lessgtr}_{11}(\mathbf{p}_2-\mathbf{p},\omega_2-\omega)\Big]. \tag{7.31}$$

We note that $\Sigma^{>}_{12} = \Sigma^{>}_{21}$. As usual, the $(\mathbf{R},T)$-dependence of the $\tilde{g}$'s is suppressed on the r.h.s. of these equations. These expressions have the same structure as the equilibrium self-energies obtained by Shi and Griffin (1998), Fedichev and Shlyapnikov (1998) and Giorgini (2000). This is expected since the entire KB nonequilibrium formalism reduces to the usual equilibrium self-energies in this limit, which is one of its strengths.

One can show using the general properties of the nonequilibrium Green's functions in (7.30) in conjunction with (7.13) that the quasiparticle distribution function $f(\mathbf{p},\omega;\mathbf{R},T)$ satisfies the exact relation

$$f(-\mathbf{p},-\omega;\mathbf{R},T) = -[1+f(\mathbf{p},\omega;\mathbf{R},T)]. \tag{7.32}$$

If we introduce the following abbreviations for the Bogoliubov coherence factors

$$A_{\mathbf{p}} \equiv u^2_{\mathbf{p}}, \quad B_{\mathbf{p}} \equiv v^2_{\mathbf{p}}, \quad C_{\mathbf{p}} \equiv -u_{\mathbf{p}}v_{\mathbf{p}}, \tag{7.33}$$

then, using the key identity in (7.32), the self-energies in (7.31) can be finally written as

$$\Sigma^{\gtrless}_{11}(\mathbf{p},\omega;\mathbf{R},T)$$
$$= \frac{g^2}{\hbar^4}\int \frac{d\mathbf{p}_2 d\omega_2}{(2\pi)^2} n_{\rm c}(\mathbf{R},T) \begin{pmatrix} (1+f_1)(1+f_2) \\ f_1 f_2 \end{pmatrix}$$
$$\times\Big[\,(2A_1A_2 + 8A_1C_2 + 4C_1C_2 + 4B_1A_2)\,\delta(\omega_2-E_2)\delta(\omega_1-E_1)$$
$$-\,(2B_1A_2 + 8B_1C_2 + 4C_1C_2 + 4A_1A_2)\,\delta(\omega_2-E_2)\delta(\omega_1+E_1)$$
$$-\,(2A_1B_2 + 8A_1C_2 + 4C_1C_2 + 4B_1B_2)\,\delta(\omega_2+E_2)\delta(\omega_1-E_1)$$
$$+\,(2B_1B_2 + 8B_1C_2 + 4C_1C_2 + 4A_1B_2)\,\delta(\omega_2+E_2)\delta(\omega_1+E_1)\Big], \tag{7.34}$$

$$
\begin{aligned}
&\Sigma^{>}_{12<}(\mathbf{p},\omega;\mathbf{R},T)\\
&=\frac{g^2}{\hbar^4}\int\frac{d\mathbf{p}_2 d\omega_2}{(2\pi)^2}n_{\rm c}(\mathbf{R},T)\begin{pmatrix}(1+f_1)(1+f_2)\\ f_1 f_2\end{pmatrix}\\
&\quad\times\Big[\left(6C_1C_2+4A_1C_2+4B_1A_2+4B_1C_2\right)\delta(\omega_2-E_2)\delta(\omega_1-E_1)\\
&\qquad-\left(4B_1C_2+4A_1C_2+6C_1C_2+4A_1A_2\right)\delta(\omega_2-E_2)\delta(\omega_1+E_1)\\
&\qquad-\left(4A_1C_2+4B_1C_2+6C_1C_2+4B_1B_2\right)\delta(\omega_2+E_2)\delta(\omega_1-E_1)\\
&\qquad+\left(4A_1B_2+4B_1C_2+6C_1C_2+4A_1C_2\right)\delta(\omega_2+E_2)\delta(\omega_1+E_1)\Big],
\end{aligned}
\tag{7.35}
$$

and

$$
\begin{aligned}
&\Sigma^{>}_{22<}(\mathbf{p},\omega;\mathbf{R},T)\\
&=\frac{g^2}{\hbar^4}\int\frac{d\mathbf{p}_2 d\omega_2}{(2\pi)^2}n_{\rm c}(\mathbf{R},T)\begin{pmatrix}(1+f_1)(1+f_2)\\ f_1 f_2\end{pmatrix}\\
&\quad\times\Big[\left(2B_1B_2+8B_1C_2+4C_1C_2+4B_1A_2\right)\delta(\omega_2-E_2)\delta(\omega_1-E_1)\\
&\qquad-\left(2A_1B_2+8A_1C_2+4C_1C_2+4A_1A_2\right)\delta(\omega_2-E_2)\delta(\omega_1+E_1)\\
&\qquad-\left(2B_1A_2+8B_1C_2+4C_1C_2+4B_1B_2\right)\delta(\omega_2+E_2)\delta(\omega_1-E_1)\\
&\qquad+\left(2A_1A_2+8A_1C_2+4C_1C_2+4A_1B_2\right)\delta(\omega_2+E_2)\delta(\omega_1+E_1)\Big],
\end{aligned}
\tag{7.36}
$$

where we have defined the new variables $\mathbf{p}_1 \equiv \mathbf{p}-\mathbf{p}_2$ and $\omega_1 \equiv \omega-\omega_2$. In these expressions, f_i denotes the quasiparticle distribution function $f(\mathbf{p}_i,\omega_i;\mathbf{R},T)$.

Using these results, the C_{12} collision integral (7.25) can finally be reduced to

$$
\begin{aligned}
C_{12}[f,\Phi]=\frac{g^2 n_{\rm c}(\mathbf{R},T)}{2\hbar^4}\int\frac{d\mathbf{p}_1 d\mathbf{p}_2}{(2\pi)^2}\Big\{&\Big[(u_1-v_1)(u_{\mathbf{p}}u_2+v_{\mathbf{p}}v_2)\\
&+(u_2-v_2)(u_{\mathbf{p}}u_1+v_{\mathbf{p}}v_1)-(u_{\mathbf{p}}-v_{\mathbf{p}})(u_1v_2+v_1u_2)\Big]^2\\
&\times\delta(\mathbf{p}-\mathbf{p}_1-\mathbf{p}_2)\delta(E_p-E_1-E_2)\Big[(1+f)f_1f_2-f(1+f_1)(1+f_2)\Big]\\
&+2\Big[(u_1-v_1)(u_{\mathbf{p}}u_2+v_{\mathbf{p}}v_2)+(u_{\mathbf{p}}-v_{\mathbf{p}})(u_1u_2+v_1v_2)\\
&-(u_2-v_2)(u_{\mathbf{p}}v_1+u_1v_{\mathbf{p}})\Big]^2\delta(\mathbf{p}+\mathbf{p}_1-\mathbf{p}_2)\delta(E_p+E_1-E_2)\\
&\times\Big[(1+f)(1+f_1)f_2-ff_1(1+f_2)\Big]\Big\}.
\end{aligned}
\tag{7.37}
$$

The first term in (7.37) describes the collision of a condensate atom with an excitation of momentum $\mathbf{p}$, producing two excitations with momenta $\mathbf{p}_1$ and $\mathbf{p}_2$. At $T = 0$, this is the only scattering process possible since there are no thermally excited excitations. The second term describes an excitation of momentum $\mathbf{p}$ absorbing a thermal excitation of momentum $\mathbf{p}_1$, producing an excitation with momentum $\mathbf{p}_2 = \mathbf{p} + \mathbf{p}_1$ plus a condensate atom. The various collision processes (7.37) are shown in Fig. 6.1. We recall that all u's, v's and the quasiparticle energy E_p have an implicit $(\mathbf{R}, T)$ dependence.

The expression (7.37) for the C_{12} collision integral was first written down by Eckern (1984) and soon afterwords by Kirkpatrick and Dorfman (1985a), who gave a more detailed derivation.[1] The Kadanoff–Baym approach given above allows a cleaner derivation of C_{12}, in a form which is also valid for a trapped Bose-condensed gas. In addition, the KB formalism gives a systematic way of including further improvements.

One can rewrite (7.37) in the following more compact (but less transparent) form, originally used by KD:

$$\begin{aligned} C_{12}[f] = \frac{g^2 n_{\rm c}(\mathbf{R},T)}{2\hbar^4} \int \frac{d\mathbf{p}_1 d\mathbf{p}_2 d\mathbf{p}_3}{(2\pi)^2} |A(2,3;1)|^2 \delta(\mathbf{p}_1 - \mathbf{p}_2 - \mathbf{p}_3) \\ \times \delta(E_1 - E_2 - E_3)\Big[\delta(\mathbf{p} - \mathbf{p}_1) - \delta(\mathbf{p} - \mathbf{p}_2) - \delta(\mathbf{p} - \mathbf{p}_3)\Big] \\ \times \Big[(1+f_1)f_2 f_3 - f_1(1+f_2)(1+f_3)\Big]. \end{aligned} \tag{7.38}$$

Here the scattering amplitude in $|A|^2$ is given in terms of the Bogoliubov coherence factors u and v:

$$\begin{aligned} A(2,3;1) \equiv (u_3 - v_3)(u_1 u_2 + v_1 v_2) + (u_2 - v_2)(u_1 u_3 + v_1 v_3) \\ - (u_1 - v_1)(u_2 v_3 + v_2 u_3). \end{aligned} \tag{7.39}$$

The first term in (7.37) corresponds to the first term in (7.38), while the second term in (7.37) is the sum of the second and third terms in (7.38).

To summarize, the thermally excited quasiparticle distribution function defined by (7.23) satisfies the kinetic equation

$$\begin{aligned} \left[\frac{\partial}{\partial T} + \nabla_{\mathbf{p}} (E_p + \mathbf{v}_{\rm s}\cdot\mathbf{p})\cdot\nabla_{\mathbf{R}} - \nabla_{\mathbf{R}} (E_p + \mathbf{v}_{\rm s}\cdot\mathbf{p})\cdot\nabla_{\mathbf{p}}\right] f_{\rm qp}(\mathbf{p},\mathbf{R},T) \\ = C_{12}[f_{\rm qp}], \end{aligned} \tag{7.40}$$

where C_{12} is given by (7.38). To remove the rapidly varying phase of the order parameter, we have gauge-transformed to the local rest frame, where the

[1] The expression given in (7.37) corrects some typographical errors in equations (76)–(78) of Imamović-Tomasović and Griffin (2001).

condensate is at rest. Hence, the energy of the thermally excited quasiparticles in (7.40) is measured relative to this local frame. Since the thermal excitations are moving with superfluid velocity $\mathbf{v}_{\rm s}$ relative to the condensate, the energy of quasiparticles measured relative to the condensate is $E_p + \mathbf{v}_{\rm s} \cdot \mathbf{p}$. As a result, the expression $E_p + \mathbf{v}_{\rm s} \cdot \mathbf{p}$ in the streaming term on the left-hand side of (7.40) is as expected. Similarly, if we denote the quasiparticle distribution in the coordinate system where the quasiparticles are at rest by $f(\mathbf{p}, \omega; \mathbf{R}, T)$, the quasiparticles moving with velocity $\mathbf{v}_{\rm s}$ relative to the condensate are described by the distribution function $f(\mathbf{p}, \omega = E_p + \mathbf{v}_{\rm s} \cdot \mathbf{p}; \mathbf{R}, T) \equiv f_{\rm qp}(\mathbf{p}, \mathbf{R}, T)$, which in fact appears in (7.40). If we used a reference frame where the quasiparticles are at rest, the streaming term would include the energy of the quasiparticles only, i.e. the $\mathbf{v}_{\rm s} \cdot \mathbf{p}$ term in (7.40) would not be present. This lab frame of reference is the one originally used by Zaremba *et al.* (1999), as reviewed in Chapter 3.

To understand the C_{12} collision integral in (7.37) and (7.38) better, it is useful to consider two limiting cases. We define $p_{\rm c}^2 \equiv 2mgn_{\rm c}$ as the characteristic momentum for the crossover between the linear and the quadratic part of the quasiparticle spectrum (see (2.52)):

(1) If all momenta $p_i \gg p_{\rm c}$ then the quasiparticle spectrum E_p defined by (7.22) reduces to the single-particle spectrum $\tilde{\varepsilon}_p$ in (7.7). Moreover, in this limit, it follows from (7.20) that $u \to 1$ and $v \to 0$, where u and v are the Bogoliubov amplitudes, and the scattering amplitude A in (7.39) reduces to the value 2. The collision integral (7.38) then reduces to the ZNG result in Chapter 6. This approximation is only valid at temperatures where the dominant thermal excitations can be described by the Hartree–Fock single-particle spectrum (7.7).

(2) In the opposite limit, when all three momenta p_i are smaller than $p_{\rm c}$, one can expand the Bogoliubov coherence factors u and v as follows:

$$u_{\mathbf{p}} \simeq \left(\frac{gn_{\rm c}}{2E_p}\right)^{1/2} + \frac{1}{2}\left(\frac{E_p}{2gn_{\rm c}}\right)^{1/2},$$
$$v_{\mathbf{p}} \simeq \left(\frac{gn_{\rm c}}{2E_p}\right)^{1/2} - \frac{1}{2}\left(\frac{E_p}{2gn_{\rm c}}\right)^{1/2}, \tag{7.41}$$

where $E_p \simeq cp$ and $c = \sqrt{gn_{\rm c}/m}$ is the usual Bogoliubov sound speed. In this limit, the scattering amplitude (7.39) reduces to (Eckern, 1984)

$$A(1;2,3) \simeq \frac{3}{2^{7/4}}\sqrt{\frac{p_1 p_2 p_3}{p_{\rm c}^3}}. \tag{7.42}$$

This approximation is valid at low temperatures, where only low-momentum

excitations are relevant. (The sign of $v_{\mathbf{p}}$ in (7.41) is opposite from that used in Pitaevskii and Stringari, 1997).

7.3 Comments on improved theory

In Section 6.5, we derived the generalized GP equation by making use of the Popov approximation for the Beliaev source self-energies $S_{11}^{<}$, as given by (6.55). As discussed by Imamović-Tomasović and Griffin (2001), this calculation can be easily extended to deal with the second-order source self-energies based on the full Beliaev approximation. We refer the reader to this paper for calculational details. The final result is that we again obtain a GGP equation for the condensate wavefunction, with the form (6.57). However, the source function $R(\mathbf{r}, t)$ given in (6.58) now involves the C_{12} collision integral (7.38), based on Bogoliubov–Popov excitations.

It is important to emphasize that the GGP equation and the kinetic equation discussed in this chapter involve a *nonequilibrium* generalization of the Beliaev second-order self-energies. This is quite different from treatments that calculate the poles of *equilibrium* Green's functions within the second-order Beliaev approximation (Shi and Griffin, 1998; Giorgini, 2000; Fedichev and Shlyapnikov, 1998). As discussed in Section 5.3, the results can also be generated by starting from a time-dependent collisionless kinetic equation which does not include the collision terms C_{22} and C_{12}. In contrast with the ZNG equations, these treatments are restricted to the linear response region described by small deviations from thermal equilibrium.

In Section 6.6 we gave a detailed discussion of the linearized collision integral C_{12} in (6.50) on the basis of the simple ZNG HF approximation for the second-order Beliaev self-energies. By the same linearization procedure used in deriving (6.67), in which the atoms are treated in the Hartree–Fock approximation, one can reduce the expression for C_{12} in (7.38) to

$$\begin{aligned} &C_{12}^{\rm lin}[f(\mathbf{p},\mathbf{r},t),\Phi^0(\mathbf{r})] \\ &= -\delta f_{\mathbf{p}}(\mathbf{r},t)\frac{g^2 n_{\rm c0}(\mathbf{r})}{2(2\pi)^2}\frac{1}{\hbar^4}\int d\mathbf{p}_2\int d\mathbf{p}_3 \\ &\quad\times\Big[|A(2,3;\mathbf{p})|^2\,\delta(\mathbf{p}-\mathbf{p}_2-\mathbf{p}_3)\delta(E^0_{\mathbf{p}}-E^0_2-E^0_3)\Big(1+f^0_2+f^0_3\Big) \\ &\qquad+2\,|A(\mathbf{p},3;2)|^2\,\delta(\mathbf{p}+\mathbf{p}_3-\mathbf{p}_2)\delta(E^0_{\mathbf{p}}+E^0_3-E^0_2)\Big(f^0_3-f^0_2\Big)\Big]. \end{aligned} \tag{7.43}$$

We can use this *linearized* collision integral to define the lifetime of an

excitation of momentum $\mathbf{p}$ arising from C_{12} collisions, namely

$$C_{12}[f,\Phi^0] \equiv -\frac{2}{\tau_{12}(\mathbf{p},\mathbf{r})}\delta f_{\mathbf{p}}(\mathbf{r},t). \tag{7.44}$$

The second term in the square brackets in (7.43) gives the lifetime $\tau_{12}^{\rm L}(\mathbf{p},\mathbf{r})$ due to Landau damping

$$\begin{aligned}\frac{1}{\tau_{12}^{\rm L}(\mathbf{p},\mathbf{r})} = \frac{g^2 n_{\rm c0}(\mathbf{r})}{2(2\pi)^2}\frac{1}{\hbar^4}\int d\mathbf{p}_2 \int d\mathbf{p}_3 \\ \times \delta(\mathbf{p}+\mathbf{p}_3-\mathbf{p}_2)\delta(E^0_{\mathbf{p}}+E^0_3-E^0_2)\left(f^0_3-f^0_2\right) \\ \times 4\Big[u_{\mathbf{p}}(u_2u_3+v_2v_3-u_2v_3)-v_{\mathbf{p}}(u_2u_3+v_2v_3-v_2u_3)\Big]^2. \end{aligned}\tag{7.45}$$

If we take a spatial average over the trap, this expression gives the Landau damping of a Bogoliubov-Popov excitation $E^0_{\mathbf{p}}$ which is analogous to the treatment of Landau damping of condensate collective modes given by Pethick and Smith (2008, p. 282), Jackson and Zaremba (2003b), Pitaevskii and Stringari (1997) and Giorgini (2000). We note that the latter two papers define the Bogoliubov amplitude $v_{\mathbf{p}}(\mathbf{r})$ with a different sign convention from that used here (see footnote 1 in Chapter 2).

Treating the excitations labelled by 2 and 3 in (7.45) in the HF approximation appropriate to a thermal gas at high temperatures, this corresponds to setting the Bogoliubov amplitudes $v_2(\mathbf{r})$ and $v_3(\mathbf{r})$ to zero, and $u_2(\mathbf{r}) = u_3(\mathbf{r}) = 1$. In this case, the matrix element in (7.45) reduces to $[u_{\mathbf{p}}(\mathbf{r})-v_{\mathbf{p}}(\mathbf{r})]u_2(\mathbf{r})u_3(\mathbf{r})$ with $u_2(\mathbf{r}) = u_3(\mathbf{r}) = 1$. This is the matrix element used in the standard calculation of the Landau damping of a Bogoliubov excitation due to the absorption and emission of atoms in the thermal cloud, the latter being treated in the HF approximation (see Chapter 13).

Chapter 13 is devoted to a detailed discussion of the Landau damping of condensate collective modes given by the GP equation. As noted above, the results are formally the same as those found for Bogoliubov excitations as obtained from the linearized C_{12} collision integral. We also show in Chapter 13 that the results for Landau damping of collective modes are in excellent agreement with the full numerical solutions described in Chapter 12.

The first term in the square brackets in (7.43), in contrast, corresponds to Beliaev damping, in which an excitation $\mathbf{p}$ decays into two thermal excitations, $\mathbf{p}_2$ and $\mathbf{p}_3$. The matrix element $A(2,3;\mathbf{p})$ in (7.43) describing this Beliaev decay process is in precise agreement with the results given in the literature (Shi and Griffin, 1998; Fedichev and Shlyapnikov, 1998). As reviewed in Section 5.3, Giorgini (2000) obtained the same result (taking into account his different sign convention for $v_{\mathbf{p}}(\mathbf{r})$, noted above).

In this chapter we have derived an improved kinetic equation for the quasiparticle distribution function based on the Beliaev second-order approximation, which should be valid at very low temperatures. This improved kinetic equation involves the Bogoliubov–Popov quasiparticle spectrum and the usual coherence factor (7.39) involving the u and v functions, as shown above.

The kinetic equation for the quasiparticle distribution and the generalized Gross–Pitaevskii equation are coupled and have to be solved self-consistently. They should provide a sound basis for future studies of the nonequilibrium response of a trapped Bose gas at very low temperatures. One could use these results to derive the Landau–Khalatnikov two-fluid hydrodynamic equations in the collision-dominated region. This would extend the derivation discussed in Chapters 15 and 17, which is based on the simpler ZNG kinetic equation. The latter assumes a Hartree–Fock thermal excitation spectrum within a semiclassical approximation. The approach developed in the well known book by Khalatnikov (1965) for superfluid helium is based on a quasiparticle kinetic equation (for phonons and rotons) that is analogous to that given in (7.40). We note that once we have chosen a specific approximation for the equilibrium single-particle self-energies, we can use the KB formalism to construct the analogous nonequilibrium self-energies and so derive a Boltzmann-like equation for the dynamics of the noncondensate atoms. Thus, while we have used the Beliaev self-energy approximation in this chapter, our KB analysis can be generalized to deal with improved approximations in a well-defined manner.

8

Static thermal cloud approximation

The collective oscillations of a condensate at zero temperature are well described by the solutions of the linearized time-dependent Gross–Pitaevskii (GP) equation of motion for the condensate wavefunction $\Phi(\mathbf{r}, t)$. At finite temperatures, however, the condensate dynamics is modified by interactions with the noncondensate atoms that comprise the thermal cloud. To account for these interactions in detail involves a sophisticated numerical analysis, which will be described in Chapter 11. However, some qualitative understanding of the effect of collisions between the condensate and noncondensate components can be gained by treating the thermal cloud within an approximation that ignores its dynamics. This approximation, referred to as the *static thermal cloud approximation*, is the topic of the present chapter. As explained in more detail below, it is defined by the assumption that the condensate moves in the presence of a thermal cloud that remains in a state of thermal equilibrium. Thus, if the condensate is induced to oscillate, it initially departs from equilibrium with the thermal cloud, but collisions lead to a damping of the condensate oscillation and ultimately equilibrate the two components. This collisional damping is in addition to the usual Landau and Beliaev damping, which is present even in the "collisionless" regime.

The approximate version of the fully coupled ZNG equations to be discussed here provides the simplest finite-temperature extension of the theory of condensate dynamics based on the usual GP equation. The extent to which the treatment gives a reasonable first approximation will be examined in Chapter 11. It will be shown that the static thermal cloud approximation does provide a qualitative understanding of the damping of modes in which the condensate is the main participant. Of course, the approximation can be used only for modes in which the motion of the thermal cloud can be ignored. For example, the most basic dipole (or "sloshing") mode, in which

the centre of mass oscillates at the harmonic trap frequency, exists at all temperatures. Above $T_{\rm BEC}$ this mode is associated with the dynamics of the thermal cloud, while below the transition both components participate; clearly it cannot be accounted for properly in the static thermal cloud approximation. The same applies to other modes that are a continuation of the dynamics of the thermal cloud from above the superfluid transition to below.

In Section 8.1 we develop a generalized GP equation that describes the dynamics of the condensate within the static thermal cloud approximation (based on Williams and Griffin, 2000). As an illustration of the approach, we then use this equation to investigate the damping of a particular condensate oscillation, the radial breathing mode. We defer until Chapter 9 a discussion of another application, the problem of vortex nucleation induced by a rotating thermal cloud. In Section 8.2, we discuss various phenomenological equations which can be shown to follow naturally from the static thermal cloud approximation. In Section 8.3, we review the pioneering work of Pitaevskii (1959) on superfluid relaxation and show that his phenomenological equation has a basis, at least for a dilute Bose-condensed gas, in the collisional exchange of atoms between the condensate and thermal cloud. The results in Sections 8.2 and 8.3 have not been published before.

8.1 Condensate collective modes at finite temperatures

Our starting point is the generalized GP equation (3.21),

$$i\hbar\frac{\partial\Phi(\mathbf{r},t)}{\partial t} = \Big[-\frac{\hbar^2\nabla^2}{2m} + V_{\rm trap}(\mathbf{r}) + gn_{\rm c}(\mathbf{r},t) + 2g\tilde{n}(\mathbf{r},t) - iR(\mathbf{r},t)\Big]\Phi(\mathbf{r},t). \tag{8.1}$$

Here $n_{\rm c}(\mathbf{r},t) = |\Phi(\mathbf{r},t)|^2$ is the condensate density and $\tilde{n}(\mathbf{r},t)$ is the density of the thermal cloud, given by (3.36). The intercomponent collisional term

$$R(\mathbf{r},t) \equiv \frac{\hbar}{2n_{\rm c}(\mathbf{r},t)}\Gamma_{12}[f,\Phi] = \frac{\hbar}{2n_{\rm c}(\mathbf{r},t)}\int\frac{d\mathbf{p}}{(2\pi\hbar)^3}C_{12}[f(\mathbf{p},\mathbf{r},t),\Phi(\mathbf{r},t)] \tag{8.2}$$

involves the $C_{12}[f,\Phi]$ collision integral appearing in the kinetic equation (3.42) for the single-particle distribution function. As discussed in Chapter 3, $C_{12}[f,\Phi]$ accounts for the collisional transfer of atoms between the condensate and noncondensate and, through $R(\mathbf{r},t)$, is responsible for a change in normalization of the condensate wavefunction. In contrast, the $C_{22}[f,\Phi]$ collision integral in (3.40), which describes binary collisions between noncondensate atoms, has no direct effect on the number of condensate atoms.

Hence there is no term analogous to $R(\mathbf{r},t)$ in (8.1) coming from $C_{22}[f,\Phi]$ collisions.

It is clear from (8.2) that some knowledge of the distribution function $f(\mathbf{p},\mathbf{r},t)$ is required in order to evaluate $R(\mathbf{r},t)$. Within the static thermal cloud approximation, f is approximated by an equilibrium distribution. To motivate this approximation, it is convenient to consider first a somewhat more general situation corresponding to what we will refer to as "partial local equilibrium". This is defined by the Bose distribution

$$\tilde{f}(\mathbf{p},\mathbf{r},t) = \frac{1}{e^{\beta[(\mathbf{p}-m\mathbf{v}_{\rm n})^2/2m+U-\tilde{\mu}]}-1}\,, \tag{8.3}$$

in which the temperature parameter $\beta = (k_{\rm B}T)^{-1}$, the local fluid velocity $\mathbf{v}_{\rm n}$, the chemical potential $\tilde{\mu}$ and the thermal cloud effective potential U, as given by (3.16), are all functions of $\mathbf{r}$ and t. As will be discussed in Chapter 17, C_{22} collisions drive the distribution function $f(\mathbf{p},\mathbf{r},t)$ towards this partial local equilibrium form, which satisfies the condition

$$C_{22}[\tilde{f},\Phi] = 0\,. \tag{8.4}$$

This result is easily proved using the identity

$$f(x) \equiv \frac{1}{e^x-1} = e^{-x}\,[1+f(x)]\,; \tag{8.5}$$

in particular, we find that

$$\begin{aligned}(1+\tilde{f})(1+\tilde{f}_2)\tilde{f}_3\tilde{f}_4 &= e^{\beta[\tilde{\varepsilon}_p+\tilde{\varepsilon}_{p_2}-\tilde{\varepsilon}_{p_3}-\tilde{\varepsilon}_{p_4}+(\mathbf{p}+\mathbf{p}_2-\mathbf{p}_3-\mathbf{p}_4)\cdot m\mathbf{v}_{\rm n}]} \\ &\qquad \times \tilde{f}\tilde{f}_2(1+\tilde{f}_3)(1+\tilde{f}_4) \\ &= \tilde{f}\tilde{f}_2(1+\tilde{f}_3)(1+\tilde{f}_4)\,,\end{aligned} \tag{8.6}$$

where the last line follows on taking into account the energy- and momentum-conserving delta functions in (3.40). This equality implies (8.4), which can be viewed as the defining equation of partial local equilibrium.

We next examine the implications of the local equilibrium distribution for the C_{12} collision integral. Inserting $\tilde{f}(\mathbf{p},\mathbf{r},t)$ into (3.41), one finds that (8.2) reduces to

$$\begin{aligned}R(\mathbf{r},t) = \frac{g^2}{(2\pi)^5\hbar^6}\left[e^{\beta(\mu_{\rm c}+\frac{1}{2}m(\mathbf{v}_{\rm c}-\mathbf{v}_{\rm n})^2-\tilde{\mu})}-1\right]\int d\mathbf{p}_1\int d\mathbf{p}_2\int d\mathbf{p}_3 \\ \times\,\delta(\mathbf{p}_{\rm c}+\mathbf{p}_1-\mathbf{p}_2-\mathbf{p}_3)\delta(\varepsilon_{\rm c}+\tilde{\varepsilon}_1-\tilde{\varepsilon}_2-\tilde{\varepsilon}_3)(1+\tilde{f}_1)\tilde{f}_2\tilde{f}_3.\end{aligned} \tag{8.7}$$

The condensate chemical potential $\mu_{\rm c}(\mathbf{r},t)$ and the local condensate velocity $\mathbf{v}_{\rm c}(\mathbf{r},t)$ are given by (3.19) and (3.13), respectively. The factor in square brackets in (8.7) is again derived using energy and momentum conservation

and the identity in (8.5). It is clear that $R(\mathbf{r},t)$ can vanish only if this factor is zero, which requires that

$$\tilde{\mu}(\mathbf{r},t) = \mu_{\rm c}(\mathbf{r},t) + \tfrac{1}{2}m[\mathbf{v}_{\rm c}(r,t) - \mathbf{v}_{\rm n}(\mathbf{r},t)]^2 \equiv \varepsilon'_{\rm c}(\mathbf{r},t)\,. \tag{8.8}$$

The quantity $\varepsilon'_{\rm c}$ defined in (8.8) is the energy of a condensate atom in the local rest frame of the thermal cloud. We thus see that the condition of complete local equilibrium between the condensate and the thermal cloud requires this condensate energy to be equal to the local thermal cloud chemical potential $\tilde{\mu}$. When (8.8) is satisfied, there is no net exchange of particles between the two components. This is not the case for the condition of *partial* local equilibrium as defined by (8.4), which does not imply any relation between the condensate and thermal cloud chemical potentials.

Unless (8.8) is satisfied, (8.3) will in general lead to $R \neq 0$, implying particle exchange between the two components. Inserting (8.7) into (8.1) results in an equation for the condensate $\Phi(\mathbf{r},t)$, but one which still depends on variables describing the thermal cloud. As such, the equation must be supplemented by dynamical equations for these additional variables. In Chapter 15 we show that the assumption of partial local equilibrium results in a set of hydrodynamic equations for the thermal cloud, which together with (8.1) provide a closed set of equations for all quantities of interest. These coupled equations will be studied in detail in Chapter 15. In the remainder of this chapter, we consider a simplified theory in which the dynamics of the thermal cloud is omitted.

We will suppose that the condensate and thermal cloud are initially in absolute thermal equilibrium, as discussed in Section 3.3. The system is then perturbed by imposing an external potential that couples only to the condensate. This of course drives the system out of equilibrium and the dynamics should in principle be described by the full set of coupled ZNG equations. However, if we assume that the collision rate between thermal atoms is sufficiently high, then the state of the thermal cloud will not deviate significantly from absolute thermal equilibrium. Under these conditions, the distribution function of the thermal cloud can be taken to have the static absolute equilibrium form

$$f^0(\mathbf{p},\mathbf{r}) = \frac{1}{e^{\beta_0[p^2/2m + U_0(\mathbf{r}) - \tilde{\mu}_0]} - 1}\,, \tag{8.9}$$

where $\beta_0 = 1/k_{\rm B}T_0$ is the uniform temperature and $U_0(\mathbf{r}) = V_{\rm trap}(\mathbf{r}) + 2g[n_{\rm c0}(\mathbf{r}) + \tilde{n}_0(\mathbf{r})]$. This, of course, is a special case of (8.3) with $\mathbf{v}_{\rm n} = 0$ and all other variables time-independent. In effect, we are taking the thermal cloud to be a fixed reservoir of thermal atoms to which the condensate

is coupled. Equation (8.9) defines in precise mathematical terms what is meant by the *static thermal cloud approximation*. In Chapters 11–19, we will consider the full dynamics of both the condensate and the thermal cloud.

Within the static thermal cloud approximation, the dynamic HF-field term in (8.1) produced by the thermal cloud is replaced by the static mean field $2g\tilde{n}_0(\mathbf{r})$. The condensate dynamics is then governed by the GGP equation

$$i\hbar\frac{\partial\Phi(\mathbf{r},t)}{\partial t} = \Big[-\frac{\hbar^2\nabla^2}{2m} + V_{\rm trap}(\mathbf{r}) + gn_{\rm c}(\mathbf{r},t) + 2g\tilde{n}_0(\mathbf{r}) - iR_0(\mathbf{r},t)\Big]\Phi(\mathbf{r},t)\,. \tag{8.10}$$

The collisional exchange term R_0 is obtained from (8.7) by replacing $\tilde{f}$ by f^0 and the energy $\tilde{\varepsilon}_p$ of an atom in the thermal cloud by its equilibrium value $\tilde{\varepsilon}^0_p = p^2/2m + U_0(\mathbf{r})$. This yields

$$R_0(\mathbf{r},t) = \frac{\hbar}{2\tau^c_{12}(\mathbf{r},t)}\left(e^{\beta[\varepsilon_{\rm c}(\mathbf{r},t)-\tilde{\mu}_0]} - 1\right)\,, \tag{8.11}$$

where we have defined a C_{12} collision time via

$$\begin{aligned}\frac{1}{\tau^c_{12}(\mathbf{r},t)} \equiv \frac{2g^2}{(2\pi)^5\hbar^7}\int d\mathbf{p}_1\int d\mathbf{p}_2\int d\mathbf{p}_3\,\delta(\mathbf{p}_{\rm c}+\mathbf{p}_1-\mathbf{p}_2-\mathbf{p}_3)\\ \times\,\delta(\varepsilon_{\rm c}+\tilde{\varepsilon}^0_{p_1}-\tilde{\varepsilon}^0_{p_2}-\tilde{\varepsilon}^0_{p_3})(1+f^0_1)f^0_2 f^0_3\,.\end{aligned} \tag{8.12}$$

This expression represents the rate of collisions of a condensate atom with the thermal cloud.[1] The condensate atom energy was defined in (3.12) and the condensate momentum $\mathbf{p}_{\rm c}$ is $m\mathbf{v}_{\rm c}$. The thermal cloud density $\tilde{n}_0(\mathbf{r})$ that appears in $\tilde{\varepsilon}^0_p$ and in (8.10) is given by (3.57). We note that $R_0(\mathbf{r},t)$ now depends on time only through the condensate variables $n_{\rm c}(\mathbf{r},t)$, $\mathbf{v}_{\rm c}(\mathbf{r},t)$ and $\varepsilon_{\rm c}(\mathbf{r},t)$, and thus (8.10) is a closed, albeit highly nonlinear, equation for $\Phi(\mathbf{r},t)$. From now on the equilibrium temperature parameter β_0^{-1} of the thermal cloud will be denoted by β^{-1}.

The approximate theory outlined above was first introduced and used by Williams and Griffin (2000). By making use of the amplitude–phase representation in (2.15), the same analysis that led to (2.22) and (2.23) gives the following coupled equations for the condensate density and velocity:

$$\frac{\partial n_{\rm c}}{\partial t} + \nabla\cdot(n_{\rm c}\mathbf{v}_{\rm c}) = -\Gamma^0_{12}[\Phi]\,, \tag{8.13}$$

[1] The superscript c distinguishes this collision time from the relaxation time τ_{12} to be introduced in Chapter 11. The latter defines the rate at which a *thermal cloud atom* suffers a collision with the condensate. These collision times differ since the local densities of the two components are different.

$$m\frac{\partial \mathbf{v}_{\rm c}}{\partial t} = -\nabla\left(\mu_{\rm c} + \tfrac{1}{2}mv_{\rm c}^2\right). \tag{8.14}$$

Here $\Gamma_{12}^0[\Phi] = 2n_{\rm c}(\mathbf{r},t)R_0(\mathbf{r},t)/\hbar$ and

$$\mu_{\rm c}(\mathbf{r},t) = -\frac{\hbar^2\nabla^2\sqrt{n_{\rm c}}}{2m\sqrt{n_{\rm c}}} + V_{\rm trap}(\mathbf{r}) + 2g\tilde{n}_0(\mathbf{r}) + gn_{\rm c}(\mathbf{r},t). \tag{8.15}$$

These coupled equations are completely equivalent to (8.10).

For small-amplitude oscillations, equations (8.13) and (8.14) can be linearized in terms of the condensate fluctuations[2] $\delta n_{\rm c}$ and $\delta\mathbf{v}_{\rm c}$. Since $\mu_{\rm c0} = \tilde{\mu}_0$ in equilibrium, the terms in the square bracket in (8.7) reduce to $\beta\delta\mu_{\rm c}$ and we have

$$\delta\Gamma_{12}^0 = \frac{\beta n_{\rm c0}(\mathbf{r})}{\tau_{12}^{c0}(\mathbf{r})}\delta\mu_{\rm c}(\mathbf{r},t)\,. \tag{8.16}$$

The equilibrium collision time $\tau_{12}^{c0}(\mathbf{r})$ is given by (8.12), but now with all condensate variables taking their equilibrium values ($\mathbf{p}_{\rm c} = 0$, $\varepsilon_{\rm c} = \mu_{\rm c0}$),

$$\frac{1}{\tau_{12}^{c0}(\mathbf{r})} \equiv \frac{2g^2}{(2\pi)^5\hbar^7}\int d\mathbf{p}_1\int d\mathbf{p}_2\int d\mathbf{p}_3\delta(\mathbf{p}_1-\mathbf{p}_2-\mathbf{p}_3) \times\delta\left(\mu_{\rm c0} - U_0(\mathbf{r}) + \frac{p_1^2}{2m} - \frac{p_2^2}{2m} - \frac{p_3^2}{2m}\right)(1+f_1^0)f_2^0f_3^0\,. \tag{8.17}$$

To proceed, we make use of the TF approximation (see Section 2.1), in which the first term (quantum pressure) in (8.15) is neglected. Since $n_{\rm c0}(\mathbf{r})$ is always much larger than $\tilde{n}_0(\mathbf{r})$ in the region where the condensate density is appreciable, the treatment can be simplified further by neglecting the mean field of the thermal cloud. In this case, the equilibrium condensate density is given by

$$n_{\rm c0}(\mathbf{r}) \simeq \frac{1}{g}\left[\mu_{\rm c0} - V_{\rm trap}(\mathbf{r})\right], \tag{8.18}$$

which is the TF density at $T = 0$ given in (2.11), but normalized to the temperature-dependent value of $N_{\rm c0}(T)$. With $U_0(\mathbf{r}) \simeq V_{\rm trap}(\mathbf{r}) + 2gn_{\rm c0}(\mathbf{r})$ and $\mu_{\rm c0} = \tilde{\mu}_0$, the thermal cloud distribution function in the condensate region is given by

$$f_i^0 = f^0(\mathbf{p}_i,\mathbf{r}) = \frac{1}{e^{\beta[p_i^2/2m + gn_{\rm c0}(\mathbf{r})]} - 1}. \tag{8.19}$$

In the evaluation of τ_{12}^{c0} from (8.17), $\mu_{\rm c0} - U_0(\mathbf{r})$ can be replaced by $-gn_{\rm c0}(\mathbf{r})$ at positions where the condensate density is nonzero. Thus, $\tau_{12}^{c0}(\mathbf{r})$ depends

[2] We recall that we have assumed that $\mathbf{v}_{\rm n0} = 0$. At this point, we also assume that there is no superfluid flow in the equilibrium state and thus set $\mathbf{v}_{\rm c0}(\mathbf{r})$ to zero. A vortex state (see Chapter 9) involves a situation in which $\mathbf{v}_{\rm c0}(\mathbf{r})$ will not be zero.

on $\mathbf{r}$ only through its implicit dependence on $n_{c0}(\mathbf{r})$. Outside the condensate (where $\Phi = 0$), the distribution is, of course, still given by (8.9). The equilibrium densities of the two components are then determined by adjusting the common chemical potential in order to ensure that the total number of atoms is N (see Sections 3.3 and 11.4 for further discussion).

With these simplifications, the linearized equations for the condensate variables δn_c and $\delta \mathbf{v}_c$ obtained from (8.13) and (8.14) are

$$\frac{\partial \delta n_c}{\partial t} + \nabla \cdot (n_{c0} \delta \mathbf{v}_c) = -\frac{1}{\tau'} \delta n_c \,, \tag{8.20}$$

$$m \frac{\partial \delta \mathbf{v}_c}{\partial t} = -g \nabla \delta n_c \,, \tag{8.21}$$

where the relaxation time $\tau'(\mathbf{r})$ is defined via

$$\frac{1}{\tau'(\mathbf{r})} \equiv \frac{\beta g n_{c0}(\mathbf{r})}{\tau_{12}^{c0}(\mathbf{r})} \,. \tag{8.22}$$

Like $\tau_{12}^{c0}(\mathbf{r})$, $\tau'(\mathbf{r})$ depends on $\mathbf{r}$ only through its dependence on the static equilibrium condensate density $n_{c0}(\mathbf{r})$. Note also that $1/\tau'$ is of order at least g^3, in contrast with the Landau and Beliaev damping rates. Since the right hand side of (8.20) is a source term, it is clear that τ' is associated with the absence of diffusive equilibrium between the condensate and the thermal cloud. As we shall see, this leads to the damping of the condensate collective modes.

Combining (8.20) and (8.21), we obtain the wave equation (Williams and Griffin, 2000)

$$\frac{\partial^2 \delta n_c}{\partial t^2} - \frac{g}{m} \nabla \cdot (n_{c0} \nabla \delta n_c) = -\frac{1}{\tau'} \frac{\partial \delta n_c}{\partial t} \,. \tag{8.23}$$

This reduces to the wave equation derived by Stringari (1996b) at $T = 0$, if the right-hand side is neglected. The latter equation has oscillatory normal mode solutions $\delta n_c(\mathbf{r}, t) = \delta n_i(\mathbf{r}) e^{-i\omega_i t}$, where the mode amplitudes are determined (Stringari, 1996b) by the solutions of

$$-\frac{g}{m} \nabla \cdot [n_{c0}(\mathbf{r}) \nabla \delta n_i(\mathbf{r})] = \omega_i^2 \delta n_i(\mathbf{r}). \tag{8.24}$$

A review of these zero-temperature condensate modes is given in Chapter 2.

The effect of finite temperatures is two-fold. One effect is the appearance of the collisional relaxation time τ' in (8.23), which leads to the damping of the modes. This will be discussed shortly. The second effect is the temperature dependence of the condensate number $N_{c0}(T)$ due to thermal depletion. This effect, however, is of secondary importance. Since the $T = 0$ mode frequencies obtained from the Stringari equation do not depend on N_c,

these same normal mode frequencies are valid at finite temperatures, apart from the above-mentioned mode damping. As the temperature approaches $T_{\rm BEC}$, the total number of condensate atoms becomes small and eventually the criterion $N_{\rm c}\, a \gg a_{\rm ho}$ (see (2.14)) for the validity of the TF approximation breaks down. At this point, the mode frequencies deviate from the TF values ω_i given by (8.24) and become dependent on $N_{\rm c0}(T)$.

The undamped Stringari modes given by (8.24) form an orthonormal basis satisfying the condition $\int d\mathbf{r}\, \delta n_i(\mathbf{r}) \delta n_j(\mathbf{r}) = \delta_{ij}$. The solution of (8.23) in the presence of collisional relaxation can be obtained by expanding the density fluctuation as $\delta n_{\rm c}(\mathbf{r}) = \sum_i c_i \delta n_i(\mathbf{r})$. Inserting this expansion into (8.23), one obtains a set of linear equations

$$\omega^2 c_i = \omega_i^2 c_i - i\omega\lambda \sum_j \gamma_{ij} c_j \tag{8.25}$$

for the expansion coefficients, where we have defined

$$\gamma_{ij} \equiv \int d\mathbf{r} \frac{\delta n_i(\mathbf{r}) \delta n_j(\mathbf{r})}{\tau'(\mathbf{r})}\,, \tag{8.26}$$

the matrix elements of the relaxation rate. In (8.25) we have introduced a perturbation parameter λ (to be set to unity at the end of the calculation) to keep track of the effects of the damping term. Expanding ω and c_i in powers of λ, we find that the mode frequency to first order in λ is given by $\Omega_i = \omega_i - i\Gamma_i$, where

$$\Gamma_i \equiv \frac{\gamma_{ii}}{2} = \frac{1}{2} \int d\mathbf{r} \frac{\delta n_i^2(\mathbf{r})}{\tau'(\mathbf{r})}. \tag{8.27}$$

This shows that the damping rate Γ_i involves a spatial average of the relaxation rate $1/\tau'(\mathbf{r})$ weighted by the square of the undamped density fluctuation.

For illustration, we consider the $n = 1$, $l = 0$ radial breathing mode solution of (8.24) in an isotropic trap with trap frequency $\omega_0/2\pi = 10$ Hz and containing $N = 2 \times 10^6$ ^{87}Rb atoms. The s-wave scattering length is $a = 57$ Å. The mode has a frequency $\omega_{10} = \sqrt{5}\omega_0$, and its density fluctuation, shown in Fig. 8.1(b), is $\delta n_{10}(\mathbf{r}) \propto 1 - \frac{5}{3}(r/R_{\rm TF})^2$. The inset of Fig. 8.1(b) shows the equilibrium condensate and noncondensate densities. In Fig. 8.1(a) we plot $1/\tau'(\mathbf{r})$ vs. position for $T = 0.5 T_{\rm BEC}$ and $T = 0.9 T_{\rm BEC}$. We see that the relaxation rate increases monotonically up to the condensate boundary, but less rapidly at higher T. This behaviour of $1/\tau'(\mathbf{r})$ mimics the behaviour of the noncondensate density $\tilde{n}_0(\mathbf{r})$. The sharp cusp of $\tilde{n}_0(\mathbf{r})$ and the sudden drop of $\delta n_{10}(\mathbf{r})$ and $1/\tau'(\mathbf{r})$ at the condensate boundary are all artifacts of the TF approximation; inclusion of the kinetic energy related to the quantum

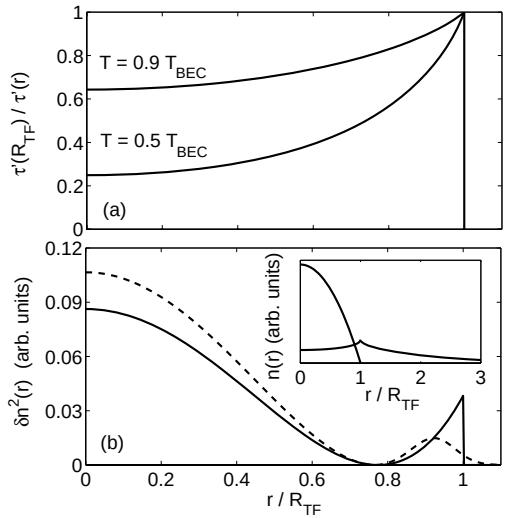


Fig. 8.1. (a) Graph of $1/\tau'(\mathbf{r})$, normalized by its value at the Thomas–Fermi radius $R_{\rm TF}$, vs. position. (b) The square of the density fluctuation $\delta n_{10}(\mathbf{r})$ of the Stringari breathing mode (solid line) and the $T = 0$ Bogoliubov mode (broken line) for $N_{\rm c0}(T = 0.9T_{\rm BEC}) = 2.3 \times 10^5$. Both solutions are normalized by $\int \delta n_{10}^2(\mathbf{r}) d\mathbf{r} = 1$. The densities of the inverted parabolic-shaped condensate and the thermal cloud (with a cusp) are shown in the inset for $T = 0.9T_{\rm BEC}$ (from Williams and Griffin, 2000).

pressure in a more accurate calculation has the effect of smoothing out this behaviour at the boundary. This is illustrated in Fig. 8.1(b) where the breathing mode density fluctuation, as obtained by solving the Bogoliubov equations at $T = 0$, is compared with the analogous TF result.

In Fig. 8.2(a) we plot the damping rate Γ_{10} calculated from (8.27) using the TF breathing mode density fluctuation shown in Fig. 8.1(b). We only show results up to $T = 0.95T_{\rm BEC}$ (where $N_{\rm c}/N \simeq 0.04$) since beyond this point the TF approximation starts to break down, and the mode frequencies exhibit a strong temperature dependence (Hutchinson *et al.*, 1997; Dodd *et al.*, 1998). We show in Fig. 8.2(b) some results for quadrupole modes in an anisotropic trap. The overall behaviour is qualitatively similar to that of the radial breathing mode in an isotropic trap shown in Fig. 8.2(a).

We emphasize that the damping shown by the solid line in Fig. 8.2(a), and by the solid and broken-and-dotted lines in Fig. 8.2(b), is due to the collisional exchange contributions calculated in the static thermal cloud approximation. Other contributions, such as Landau damping, arise when the dynamics of the thermal cloud is taken into account. This more complete

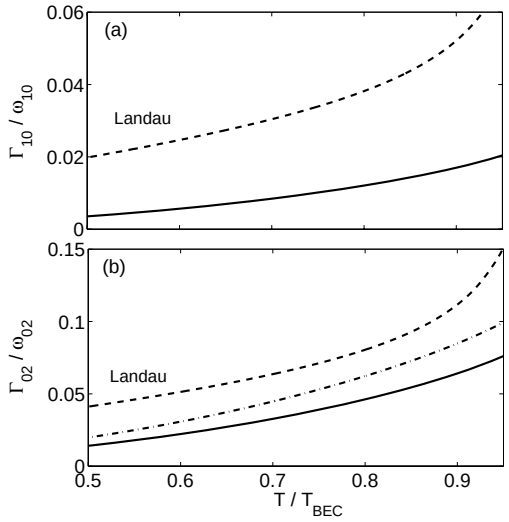


Fig. 8.2. Normal mode damping rates vs. temperature. The damping rates are normalized by their corresponding mode frequencies. (a) The results for the $n = 1$, $l = 0$ breathing mode for an isotropic trap ($\omega_0/2\pi = 10$ Hz, $N = 2 \times 10^6$); the solid line gives the collisional exchange rate (8.27) and the broken line gives an estimate of the Landau damping rate (as explained in the main text). (b) The damping rates for the quadrupole mode $n = 0$, $l = 2$ in an anisotropic trap ($\omega_\perp/2\pi = 23$ Hz, $\lambda = \sqrt{8}$, $N = 10^6$). The solid line corresponds to $m = 0$ and the broken-and-dotted line to $m = 2$ (from Williams and Griffin, 2000).

analysis will be given in later chapters, both in the weakly collisional region ($\omega\tau \gg 1$, Chapters 11–13) and in the collision-dominated hydrodynamic region ($\omega\tau \ll 1$, Chapters 15 and 17). Here we present a rough estimate of the Landau damping contribution that makes use of the damping rate as calculated for a uniform gas. One finds that, at finite temperatures, the Landau damping of a Bogoliubov phonon of wavevector q is given by (Pitaevskii and Stringari, 1997; Shi and Griffin, 1998; Fedichev *et al.*, 1998)

$$\Gamma_{\rm L} = \left(\frac{3\pi}{8}\right) \frac{a k_{\rm B} T\, q}{\hbar}. \tag{8.28}$$

In applying this expression to the trapped gas, we approximate the wavevector as ω_{10}/c, where $c = \sqrt{g n_{\rm c0}(0)/m}$ is the Bogoliubov phonon velocity corresponding to the density at the centre of the trap (Pitaevskii and Stringari, 1997). This gives the results shown by the broken lines in Fig. 8.2. The Landau damping is larger than, but comparable with, the collisional exchange damping that we have been considering. A fully satisfactory theory of the

finite-T damping of the normal modes of the condensate must include *both* the collisional exchange and Landau contributions (see Chapters 12 and 13). The expression (8.28) for the Landau damping is discussed in more detail in Section 13.1.

The results presented so far are based on the TF approximation, in which the quantum pressure term in (8.15) is neglected. This approximation leads to the damped wave equation given in (8.23). If the quantum pressure term is included, the linearized equations of motion for the condensate take the form of coupled Bogoliubov equations, generalized to include the effects of damping from the R_0 term in (8.10). This extension of the theory was given by Williams and Griffin (2001). They found only minor differences from the TF results except near $T_{\rm BEC}$ where, as expected, the TF approximation starts to break down owing to the small number of condensate atoms.

We mention here two other applications of the static thermal cloud approximation. The first concerns the formation of vortices induced by a rigidly rotating thermal cloud (see Haljan *et al.*, 2001). The theory explaining this (Williams *et al.*, 2002) will be discussed in Chapter 9. A second application is found in the work of Konabe and Nikuni (2006), who considered the effect of a static thermal cloud on the stability of a current-carrying condensate in an optical lattice. Their calculations show how the interaction with the thermal cloud provides a microscopic mechanism for the breakdown of superfluidity (the so-called Landau instability).

The virtue of the static thermal cloud approximation is that it provides a simple first estimate of the damping effects arising from the thermal cloud at finite temperatures. It has the further advantage of allowing one to study the effect of C_{12} collisions in isolation from other contributions. However, as already emphasized, there are other contributions, such as Landau damping, that are associated with the dynamics of the thermal cloud. To account for these, a full numerical simulation of the ZNG equations is required. In Chapter 12, we present examples of such simulations and compare the results with those based on the static thermal cloud approximation. Although we find significant differences, the collisional exchange of atoms between the condensate and thermal cloud remains an important contribution to the total damping rate.

We noted at the end of Section 6.3 that Stoof (1999, 2001) developed a general theory for the nonequilibrium behaviour of a condensate at finite temperatures using a path-integral formulation. However, this general theory has so far only been studied assuming that the thermal cloud can be treated in a self-consistent HF approximation and, moreover, is always in static thermal equilibrium (Duine and Stoof, 2001; Proukakis and Jackson,

2008). This is clearly the equivalent of the static thermal cloud approximation used in this chapter. The stochastic GP equation discussed by Stoof and coworkers is indeed very similar to our generalized GP equation (8.10), apart from an additional Gaussian noise term. We emphasize that the dissipative term R in our GGP equation is sufficient to ensure that the condensate will relax to thermal equilibrium.

8.2 Phenomenological GP equations with dissipation

The static thermal cloud approximation eliminates the thermal cloud as a dynamical component and leads to a closed generalized GP equation (8.10) for the condensate itself. This equation includes the effect of atom exchange between the condensate and thermal cloud, which, as discussed in Section 8.1, gives rise to the damping of condensate modes. From another point of view, particle exchange also allows the condensate to either grow or decrease in size. This can be seen most clearly in terms of the hydrodynamic equations for $n_{\rm c}(\mathbf{r},t)$ and $\mathbf{v}_{\rm c}(\mathbf{r},t)$ in (8.13) and (8.14). The continuity equation (8.13), in particular, contains the extra term $\Gamma_{12}^0[\Phi]$ which acts as a source or sink for condensate atoms. Thus (8.10) can be viewed as an equation for the growth of a Bose condensate that, while approximate, has a well-defined microscopic basis. A more realistic calculation of condensate growth including the full dynamics of the thermal cloud in the ZNG theory is discussed in Chapter 12.

Equation (8.10) gives a microscopic basis for various phenomenological GP equations that have been proposed to deal with the relaxation and growth of a condensate. We can make contact with these different formulations through some minor approximations to (8.10). Using the phase–density representation for $\Phi(\mathbf{r},t)$, one can easily verify that

$$\frac{i\hbar}{\Phi}\frac{\partial\Phi}{\partial t} = -\hbar\frac{\partial\theta}{\partial t} + \frac{i\hbar}{2}\frac{\partial \ln n_{\rm c}}{\partial t} = \varepsilon_{\rm c} + \frac{i\hbar}{2}\frac{\partial \ln n_{\rm c}}{\partial t}\,. \tag{8.29}$$

The last equality follows from (3.12). The term involving the time derivative of the density is linear in the condensate velocity $\mathbf{v}_{\rm c}$. If we assume this velocity to be small, we arrive at the approximate relation

$$\varepsilon_{\rm c}(\mathbf{r},t) \simeq \frac{i\hbar}{\Phi(\mathbf{r},t)}\frac{\partial\Phi(\mathbf{r},t)}{\partial t}\,. \tag{8.30}$$

We now use (8.30) to obtain an alternate form of the particle exchange term R_0 given in (8.11). Expanding the exponential under the assumption

that $\beta|\varepsilon_{\rm c}(\mathbf{r},t)-\tilde{\mu}_0| \ll 1$, we obtain

$$R_0(\mathbf{r},t) \simeq \frac{\hbar\beta}{2\tau_{12}^{c0}(\mathbf{r})}[\varepsilon_{\rm c}(\mathbf{r},t)-\tilde{\mu}_0] \simeq \frac{\hbar\beta}{2\tau_{12}^{c0}(\mathbf{r})}\left(\frac{i\hbar}{\Phi}\frac{\partial\Phi}{\partial t}-\tilde{\mu}_0\right). \tag{8.31}$$

Substituting this expression into (8.10), we arrive at the following approximate equation for the evolution of the condensate wavefunction in the presence of atom exchange with the thermal cloud:

$$i\hbar\frac{\partial\Phi(\mathbf{r},t)}{\partial t} = \mathcal{H}_{\rm c}(t)\Phi(\mathbf{r},t) + i\gamma(\mathbf{r})\left(\tilde{\mu}_0 - i\hbar\frac{\partial}{\partial t}\right)\Phi(\mathbf{r},t). \tag{8.32}$$

Here, we have defined

$$\mathcal{H}_{\rm c}(t) \equiv -\frac{\hbar^2\nabla^2}{2m} + V_{\rm trap}(\mathbf{r}) + gn_c(\mathbf{r},t) + 2g\tilde{n}_0(\mathbf{r}) \tag{8.33}$$

and the relaxation factor

$$\gamma(\mathbf{r}) \equiv \frac{\beta\hbar}{2\tau_{12}^{c0}(\mathbf{r})}\,. \tag{8.34}$$

Equation (8.32) is equivalent to an equation derived by Gardiner *et al.* (2002), who refer to it as the "phenomenological condensate growth equation". Their derivation is based on a completely different theoretical formalism, which is reviewed, with extensive references, by Bradley *et al.* (2004) and in the thesis of Penckwitt (2004). The extension of this equation to include superfluid flow in the equilibrium state is given in Chapter 9 in connection with vortex lattice formation.

It should be noted that the approximations that we have used to obtain (8.32) from (8.10) are not entirely consistent. Whereas previously we linearized the generalized GP equation systematically to obtain (8.20) and (8.21), only the particle exchange term R_0 is linearized in obtaining (8.32). Thus the range of validity of this nonlinear equation is not entirely clear. Nevertheless, it can be used in the spirit of obtaining a qualitative understanding of condensate relaxation and growth phenomena.

With the transformation $\Phi = e^{-i\tilde{\mu}_0 t/\hbar}\overline{\Phi}$, (8.32) becomes

$$i\hbar\frac{\partial\overline{\Phi}(\mathbf{r},t)}{\partial t} = [\mathcal{H}_{\rm c}(t)-\tilde{\mu}_0]\,\overline{\Phi}(\mathbf{r},t) + \hbar\gamma(\mathbf{r})\frac{\partial\overline{\Phi}(\mathbf{r},t)}{\partial t}, \tag{8.35}$$

which can be rearranged trivially as

$$i\hbar[1+i\gamma(\mathbf{r})]\frac{\partial\overline{\Phi}(\mathbf{r},t)}{\partial t} = [\mathcal{H}_{\rm c}(t)-\tilde{\mu}_0]\overline{\Phi}(\mathbf{r},t). \tag{8.36}$$

An equation of this form (generalized to a rotating frame of reference and with γ equal to a constant) was used by Tsubota *et al.* (2002) and by

Kasamatsu *et al.*(2003, 2005) in the study of vortex-lattice formation. This topic will be discussed at length in Chapter 9.

Equation (8.36) has a number of useful properties. First, one can easily check that it admits a time-independent solution $\overline{\Phi}(\mathbf{r},t) = \Phi_0(\mathbf{r})$, where $\mathcal{H}_{\rm c0}\Phi_0 = \varepsilon_{\rm c0}\Phi_0$ with $\varepsilon_{\rm c0} = \tilde{\mu}_0$. Owing to the nonlinearity of the equation, the eigenvalue $\varepsilon_{\rm c0}$ depends on the normalization of the condensate wavefunction. Thus specifying the eigenvalue imposes a constraint on the normalization. Less obvious are the relaxation properties of (8.36). To determine these, it is convenient to introduce the eigenstates of the instantaneous Hamiltonian: $\mathcal{H}_{\rm c}(t)\phi_n = \varepsilon_n\phi_n$. Expanding the wavefunction as $\overline{\Phi}(\mathbf{r},t) = \sum_n a_n(t)\phi_n(\mathbf{r})$ and assuming the relaxation factor γ to be independent of position, the expansion coefficients evolve over a short time interval Δt as

$$a_n(t+\Delta t) = e^{-i(\varepsilon_n-\tilde{\mu}_0)\Delta t/\hbar(1+\gamma^2)}e^{-\gamma(\varepsilon_n-\tilde{\mu}_0)\Delta t/\hbar(1+\gamma^2)}a_n(t)\,. \tag{8.37}$$

Components with $\varepsilon_n > \tilde{\mu}_0$ are seen to decay in time, while those with $\varepsilon_n < \tilde{\mu}_0$ grow. If on the one hand the spectrum is such that $\varepsilon_n > \tilde{\mu}_0$ for all n, the amplitude of the state whose energy is closest to $\tilde{\mu}_0$ decays least rapidly and will become the dominant term in the expansion. On the other hand, if some eigenvalues lie below $\tilde{\mu}_0$, then the one furthest away grows the most rapidly. In either case, the state with the lowest eigenvalue will become the dominant term in the expansion. In the following discussion, we will refer to this particular eigenvalue ε_n as the dominant eigenvalue.

We next consider the behaviour of the eigenvalue spectrum associated with (8.36) as a function of time. To this end, we note that the norm of the condensate wavefunction $N_{\rm c}(t) = \int d\mathbf{r}\,|\overline{\Phi}(\mathbf{r},t)|^2$ satisfies the equation

$$\frac{dN_{\rm c}}{dt} = \frac{2\gamma[\tilde{\mu}_0 - \bar{\varepsilon}_{\rm c}(t)]}{\hbar(1+\gamma^2)}N_{\rm c}\,, \tag{8.38}$$

where we have defined

$$\bar{\varepsilon}_{\rm c}(t) = \frac{\langle\overline{\Phi}(t)|\mathcal{H}_{\rm c}(t)|\overline{\Phi}(t)\rangle}{\langle\overline{\Phi}(t)|\overline{\Phi}(t)\rangle}\,. \tag{8.39}$$

From (8.38), we see that if $\bar{\varepsilon}_{\rm c}(t) < \tilde{\mu}_0$, then $dN_{\rm c}/dt$ will be positive and $N_{\rm c}(t)$ will *increase* with time. Conversely, if $\bar{\varepsilon}_{\rm c}(t) > \tilde{\mu}_0$, then $dN_{\rm c}/dt$ will be negative and $N_{\rm c}(t)$ will *decrease* in time. Introducing $\overline{\Phi}(\mathbf{r},t) \equiv \sqrt{N_{\rm c}(t)}\phi(\mathbf{r},t)$, where by definition $\langle\phi|\phi\rangle = 1$, (8.39) can be written as

$$\bar{\varepsilon}_{\rm c}(t) = \langle\phi| - \hbar^2\nabla^2/2m + V_{\rm trap}(\mathbf{r}) + 2g\tilde{n}_0(\mathbf{r}) + g|\phi|^2N_{\rm c}(t)|\phi\rangle. \tag{8.40}$$

For a repulsive interaction, it is clear that $\bar{\varepsilon}_{\rm c}(t)$ is an increasing function of $N_{\rm c}(t)$. The same is true for each eigenvalue ε_n.

If the spectrum is such that $\varepsilon_n > \tilde{\mu}_0$ for all n, then we have $\overline{\varepsilon}_{\rm c}(t) > \tilde{\mu}_0$, which implies that $N_{\rm c}(t)$ will decrease in time and the eigenvalue spectrum shifts down in energy. In particular, the dominant eigenvalue will shift towards $\tilde{\mu}_0$. However, if $\overline{\varepsilon}_{\rm c}(t) < \tilde{\mu}_0$, then the spectrum necessarily extends below $\tilde{\mu}_0$ and in this case $N_{\rm c}(t)$ will increase in time and the dominant eigenvalue will shift up towards $\tilde{\mu}_0$. We thus conclude that the dynamical evolution drives the dominant eigenvalue towards $\tilde{\mu}_0$ and that the solution $\overline{\Phi}(\mathbf{r},t)$ of (8.36) will necessarily relax to the stationary state $\Phi_0(\mathbf{r})$. This same behaviour also applies to the solutions of (8.32) and (8.36). Although the above analysis is based on a constant value of γ, we would expect the solution to behave in a similar way even when γ is spatially-dependent.

Another form of the relaxation equation for the condensate can be obtained by considering the last term in (8.35) as a small perturbation. A single iteration yields a simplified version of (8.38),

$$i\hbar\frac{\partial\overline{\Phi}(\mathbf{r},t)}{\partial t} = (1-i\gamma)[\mathcal{H}_{\rm c}(t)-\tilde{\mu}_0]\overline{\Phi}(\mathbf{r},t). \tag{8.41}$$

This is the phenomenological equation used by Choi *et al.* (1998) to study the damping of collective modes. As discussed above, the solution of this equation relaxes to the equilibrium state Φ_0.[3] A deviation from this state that corresponds to a collective mode excitation will relax to zero. Thus the parameter γ provides a qualitative description of the damping of the condensate mode. Choi *et al.* (1998) introduced the parameter γ to represent dissipation in a generic sense, without any microscopic estimate of its value. However, as shown in the present chapter, one possible source of dissipation that would lead to an equation of the form (8.41) is the collisional exchange of atoms between the condensate and the thermal cloud, the latter being described within the static thermal cloud approximation.

8.3 Relation to Pitaevskii's theory of superfluid relaxation

In obtaining (8.41), Choi *et al.* (1998) followed the pioneering work of Pitaevskii (1959) (this approach is also summarized in Chapter 17 of the book by Khalatnikov, 1965). Pitaevskii's original goal was to extend the usual two-fluid equations of Landau and Khalatnikov (which we will derive in Chapter 17) to account for the damping of hydrodynamic modes near the

[3] Choi *et al.* (1998) pointed out that the time evolution of (8.41) provides an efficient method of obtaining the stationary state Φ_0. In fact, the use of (8.41) is a variation on the method of imaginary-time propagation often used to obtain the ground state of the GP equation. The imaginary-time propagation scheme is obtained from (8.41) by replacing $1-i\gamma$ with $-i$ (see for example Minguzzi *et al.*, 2004).

superfluid transition temperature in liquid helium, which is associated with the relaxation of the superfluid density. In his approach the superfluid is described by a complex "wavefunction" (see the remarks below)

$$\Psi_{\rm s}(\mathbf{r},t) \equiv \sqrt{n_{\rm s}(\mathbf{r},t)}e^{i\theta(\mathbf{r},t)}, \tag{8.42}$$

with superfluid velocity $\mathbf{v}_{\rm s}(\mathbf{r},t) = \hbar\nabla\theta(\mathbf{r},t)/m$. Using a combination of physical reasoning and general conservation laws, Pitaevskii arrived at a time-dependent equation for $\Psi_{\rm s}(\mathbf{r},t)$ of the form[4]

$$\begin{aligned} i\hbar\frac{\partial\Psi_{\rm s}}{\partial t} = &\left[-\frac{\hbar^2\nabla^2}{2m} + m\left(\frac{\partial\epsilon}{\partial\rho}\right)_{\rho_{\rm s},s} + m\left(\frac{\partial\epsilon}{\partial\rho_{\rm s}}\right)_{\rho,s}\right]\Psi_{\rm s}(\mathbf{r},t) \\ &- i\Lambda\left[-\frac{\hbar^2\nabla^2}{2m} + m\left(\frac{\partial\epsilon}{\partial\rho_{\rm s}}\right)_{\rho,s}\right]\Psi_{\rm s}(\mathbf{r},t)\,. \end{aligned} \tag{8.43}$$

Here $\epsilon(\rho,\rho_{\rm s},s)$ is the energy density of the fluid expressed in terms of the total mass density ρ, the superfluid mass density is $\rho_{\rm s} = m|\Psi_{\rm s}|^2$ and s is the entropy density. The imaginary term on the right-hand side is introduced to account for the relaxation of the superfluid density to its equilibrium value. No microscopic expression is given for the dimensionless phenomenological parameter Λ, but for a dilute Bose gas we shall see that it can be identified with γ in (8.34). Pitaevskii's method of including superfluid relaxation in (8.43) stimulated a considerable literature, which was reviewed by Geurst (1980).

It should be emphasized that the "superfluid wavefunction" $\Psi_{\rm s}(\mathbf{r},t)$ defined in (8.42) is itself a phenomenological variable, introduced in the context of formulating a two-fluid collisional hydrodynamic theory of superfluid helium. It has no fundamental relation to the microscopic Bose order parameter $\Phi(\mathbf{r},t) = \sqrt{n_{\rm c}(\mathbf{r},t)}e^{i\theta(\mathbf{r},t)}$ introduced by Beliaev (1958a,b). In particular, the amplitude of the Bose order parameter involves the local condensate density $n_{\rm c}(\mathbf{r},t)$, not the superfluid density $n_{\rm s}(\mathbf{r},t)$.

In order to identify Λ with γ in the ZNG theory, we will establish a correspondence between various thermodynamic quantities. Pitaevskii (1959) gave the thermodynamic relation (see equation (2.6) of this paper)

$$P + \epsilon - Ts = \rho\left(\frac{\partial\epsilon}{\partial\rho}\right)_{\rho_{\rm s},s} + \rho_{\rm s}\left(\frac{\partial\epsilon}{\partial\rho_{\rm s}}\right)_{\rho,s}. \tag{8.44}$$

[4] This is equation (1.10) in Pitaevskii (1959) for $\mathbf{v}_{\rm n} = 0$, which is the case being considered here. The extra term denoted in this paper by "g" does not appear in (8.43). This term can be neglected in the region where superfluid relaxation is dominant.

Introducing the normal density $\rho_{\rm n} \equiv \rho - \rho_{\rm s}$, we have the relation

$$\left(\frac{\partial \epsilon}{\partial \rho_{\rm s}}\right)_{\rho_{\rm n},s} = \left(\frac{\partial \epsilon}{\partial \rho}\right)_{\rho_{\rm s},s} + \left(\frac{\partial \epsilon}{\partial \rho_{\rm s}}\right)_{\rho,s} . \tag{8.45}$$

We now define the chemical potentials

$$\mu_{\rm n} \equiv m\left(\frac{\partial \epsilon}{\partial \rho_{\rm n}}\right)_{\rho_{\rm s},s} = m\left(\frac{\partial \epsilon}{\partial \rho}\right)_{\rho_{\rm s},s} \text{ and } \quad \mu_{\rm s} \equiv m\left(\frac{\partial \epsilon}{\partial \rho_{\rm s}}\right)_{\rho_{\rm n},s} . \tag{8.46}$$

Equation (8.44) can then be written as

$$P + \epsilon - Ts = n\mu_{\rm n} + n_{\rm s}(\mu_{\rm s} - \mu_{\rm n}) = n_{\rm n}\mu_{\rm n} + n_{\rm s}\mu_{\rm s} . \tag{8.47}$$

The analogous relation within the ZNG model (to be discussed in detail in Section 15.2) is

$$P + \epsilon - Ts = \tilde{n}\tilde{\mu} + n_{\rm c}\mu_{\rm c} . \tag{8.48}$$

A term by term comparison of (8.47) and (8.48) makes clear the correspondence between the superfluid variables used by Pitaevskii and those appearing in the ZNG theory describing two-fluid hydrodynamics. As a further check on the identification of these variables, we note that the internal energy in the ZNG theory (see equation (68) of Zaremba *et al.* (1999) and (15.51) in Chapter 15) is given by

$$\epsilon(n, n_{\rm c}) = \tfrac{1}{2}g\left(n^2 + 2n\tilde{n} - \tilde{n}^2\right) = gn^2 - \tfrac{1}{2}gn_{\rm c}^2. \tag{8.49}$$

From this expression, we obtain

$$\left(\frac{\partial \epsilon}{\partial n}\right)_{n_{\rm c}} + \left(\frac{\partial \epsilon}{\partial n_{\rm c}}\right)_{n} = 2gn - gn_{\rm c} = gn_{\rm c} + 2g\tilde{n}. \tag{8.50}$$

This expression is precisely the condensate chemical potential $\mu_{\rm c}$ in the ZNG theory. This result is consistent with (8.45) and (8.46), confirming that the superfluid and condensate densities are equal in the case of a dilute Bose-condensed gas.

Making use of (8.45) and the definitions in (8.46), Pitaevskii's equation (8.43) can be rewritten as

$$i\hbar\frac{\partial \Psi_{\rm s}}{\partial t} = \left(-\frac{\hbar^2\nabla^2}{2m} + \mu_{\rm s}\right)\Psi_{\rm s}(\mathbf{r},t) - i\Lambda\left(-\frac{\hbar^2\nabla^2}{2m} + \mu_{\rm s} - \mu_{\rm n}\right)\Psi_{\rm s}(\mathbf{r},t) . \tag{8.51}$$

In two-fluid hydrodynamic theory, $\mu_{\rm n}$ is a dynamical variable and corresponds to the variable $\tilde{\mu}$ in the ZNG theory. Thus, the analogue of the static thermal cloud approximation would be the replacement of $\mu_{\rm n}$ in (8.51) by a

constant $\mu_{\rm n0}$. Making this replacement, and introducing the transformation $\Psi_{\rm s} = e^{-i\mu_{\rm n0}t/\hbar}\overline{\Psi}_{\rm s}$, we obtain

$$i\hbar\frac{\partial\overline{\Psi}_{\rm s}}{\partial t} = (1-i\Lambda)\Big(-\frac{\hbar^2\nabla^2}{2m}+\mu_{\rm s}-\mu_{\rm n0}\Big)\overline{\Psi}_{\rm s}(\mathbf{r},t)\,. \tag{8.52}$$

This equation is precisely of the form (8.41), with $(-\hbar^2\nabla^2/2m+\mu_{\rm s})$ playing the role of the Hamiltonian $\mathcal{H}_{\rm c}$. Equation (8.41) can therefore be viewed as a derivation of Pitaevskii's phenomenological equation for the relaxation of the superfluid component, within the context of the ZNG theory based on a static normal fluid. In this case, the phenomenological relaxation parameter Λ is identified with γ in (8.34), and the mechanism for superfluid relaxation is C_{12} collisions.

In Chapter 15, we will generalize the present discussion and show that Pitaevskii's extension of the two-fluid hydrodynamic equations naturally emerges from the ZNG coupled equations even when the dynamics of the normal fluid is retained. The "extra equation" that Pitaevskii obtains (see his equation (3.5)) to describe the relaxation of the superfluid density close to the transition temperature is essentially the same as the generalized continuity equation for the condensate that we give in (15.33). This equation involves the quantity $\delta\Gamma_{12}$ given by (15.36), where $\mu_{\rm diff} \equiv \tilde{\mu} - \mu_{\rm c}$ is the difference between the chemical potentials of the thermal cloud and the condensate in partial local equilibrium. The relaxation time τ_μ in (15.36) governs the rate at which $\tilde{\mu}$ and $\mu_{\rm c}$ equilibrate. In the context of Pitaevskii's equation, one finds $\mu_{\rm n} - \mu_{\rm s} = \mu_{\rm diff}$. This shows how the damping term proportional to Λ in (8.51) has its origin in the Γ_{12} source term even in the case when the dynamics of both the condensate (superfluid) and thermal cloud (normal fluid) are included in the collisional hydrodynamic region.

9

Vortices and vortex lattices at finite temperatures

The goal of creating and observing quantized vortices in trapped Bose gases arose almost immediately following the first achievements of Bose–Einstein condensation. The motivation for doing so was the obvious analogy with vortices in liquid helium and in type-II superconductors, and the fact that the quantization of circulation is directly associated with superfluid flow. It was recognized that the observation of quantized vortices could be taken as indisputable evidence for the existence of superfluidity in these systems.

The review by Fetter and Svidzinsky (2001) contains a summary of the early experiments and the theoretical background for understanding the vortex state in a weakly interacting Bose gas based on the GP equation. This material will not be repeated here apart from those aspects that have a direct bearing on the focus of the present chapter, namely the properties of vortices at *finite* temperature. Although there have been some theoretical contributions to this subject, much remains to be done. The discussion in this chapter provides a framework for addressing the finite-temperature properties of vortex formation and vortex lattices in the context of the ZNG theory. The results in this chapter have not been published before, apart from those in subsection 9.8.1, which are based on Williams *et al.* (2002).

There are several issues that relate to finite temperatures. First, there is the nucleation and formation of vortices from an initial highly nonequilibrium state. Second, there is the interaction of vortices with thermal excitations, which is responsible for the dissipative dynamics of a nonequilibrium vortex state. Third, there is the question of the final equilibrium state, with respect to the condensate and noncondensate densities in the vicinity of a vortex and to the geometrical arrangement of vortices in space. The experiments performed to date have been at such low temperatures that the effects of the thermal component have been relatively unimportant. However, in

principle the thermal component must always be taken into account, especially when considering the nonequilibrium situations referred to above.

The creation of vortices and vortex lattices in trapped gases is typically achieved experimentally by means of a rotating anisotropic potential. The rotating potential acts as a "stirrer" which imparts angular momentum to the system. After thermalization, a (near) steady state is reached in a frame of reference rotating with the external potential. In the following sections we outline the key ideas required to describe the rotating trap situation. We begin with a classical description appropriate to the thermal component and then reconsider the same problem from a quantum point of view. In Section 9.4, we derive a general form of the ZNG equations valid in the presence of rotations. We then apply these equations by making use of the static thermal cloud approximation discussed in Chapter 8.

9.1 Rotating frames of reference: classical treatment

In this and in the following sections, we will discuss in detail the subject of rotating reference frames. Much of the material can be found scattered throughout various basic texts, however a comprehensive discussion which treats both the classical and quantum situations is lacking. Since this material is fundamental to understanding vortices in trapped Bose-condensed gases at finite temperatures, it is appropriate to review it here. These results are then applied to trapped gases starting in Section 9.3. The confident reader can begin there.

We consider an inertial frame of reference S that we refer to as the *laboratory* (or *lab*) frame. It is defined by Cartesian coordinate axes and a corresponding set of orthonormal unit vectors $\mathbf{u}_i$ $(i = x, y, z)$. A second frame of reference S', initially coincident with S, is rotated with respect to S about an axis passing through the origin. For purposes of illustration, we show in Fig. 9.1(a) the special case where S' undergoes a rotation about the z-axis through an angle θ. To describe a continuous uniform rotation, we set $\theta = \Omega t$, where Ω is the angular velocity of S'. Our objective in the following discussion is to describe the system dynamics in this noninertial frame of reference S'.

It is convenient to introduce the notions of *passive* and *active* rotations. These are distinguished by the way in which one views the transformation of vectors under rotations. Figure 9.1(a) depicts a passive rotation, in which the frame of reference S' is rotated with respect to the laboratory frame S but the position vector $\mathbf{r} = x\mathbf{u}_x + y\mathbf{u}_y + z\mathbf{u}_z$ remains *fixed* with respect to the frame S. This *same* vector can be expressed in terms of the S' basis

vectors as $\mathbf{r} = x'\mathbf{u}'_x + y'\mathbf{u}'_y + z'\mathbf{u}'_z$. The new coordinates are related to the old by the transformation (for this special case)

$$\begin{aligned} x' &= x\cos\theta + y\sin\theta, \\ y' &= -x\sin\theta + y\cos\theta, \\ z' &= z. \end{aligned} \tag{9.1}$$

If we imagine the vector $\mathbf{r}$ as specifying a point on a rigid body, the act of rotating S' has no effect on the state of the rigid body, hence the term *passive*.

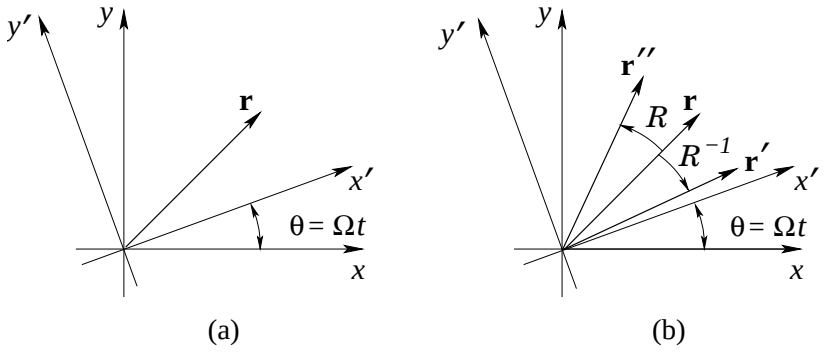


Fig. 9.1. Transformation of a vector $\mathbf{r}$ under (a) a passive rotation and (b) an active rotation. See the main text for a detailed explanation.

If, however, the rigid body is fixed in S' and rotates with it then the physical state of the object of interest does change (with respect to the laboratory frame) and the rotation is by definition *active*. In this case, the vector $\mathbf{r}$ fixed to the rigid body rotates into the vector $\mathbf{r}''$ shown in Fig. 9.1(b). It might seem more natural to denote this vector as $\mathbf{r}'$, but instead we choose to call $\mathbf{r}'$ the vector into which $\mathbf{r}$ rotates under the inverse rotation (labelled by R^{-1} in the figure).[1] This notational choice turns out to be more convenient when we consider the rotation of quantum states in Section 9.2. We observe that the components of $\mathbf{r}'$ with respect to the S coordinate system are the same as the components of $\mathbf{r}$ with respect to the S' coordinate system. Thus the relations between the components of $\mathbf{r}$ and $\mathbf{r}'$ *with respect to the same frame of reference* S are also given by the transformation (9.1). Finally, when dealing with the *components* of $\mathbf{r}$ with respect to S' in the remainder of this section, we will refer to them

[1] Conversely, $\mathbf{r}'$ rotates into $\mathbf{r}$ under the rotation R.

collectively as $\mathbf{r}'$ (for lack of a better notation), with the rationale that these components are the same as those of $\mathbf{r}'$ with respect to S in Fig. 9.1(b).[2]

We now consider an anisotropic potential which rotates at angular velocity Ω and is thus time-independent in the rotating frame. In terms of the coordinates in S', this potential takes the form

$$V'_{\rm trap}(\mathbf{r}') = \tfrac{1}{2}m\omega_\perp^2\left[(1+\epsilon)x'^2 + (1-\epsilon)y'^2 + \lambda^2 z'^2\right], \tag{9.2}$$

where ϵ is a parameter specifying the anisotropy in the xy-plane and $\lambda = \omega_z/\omega_\perp$. For the rotation depicted in Fig. 9.1(a), the components of $\mathbf{r}$ in the two frames of reference are related by (9.1) with $\theta = \Omega t$. Substituting these relations into (9.2) gives the time-dependent trapping potential in the laboratory frame

$$\begin{aligned} V_{\rm trap}(\mathbf{r},t) = \tfrac{1}{2}m\omega_\perp^2\big[&(1+\epsilon)(x\cos\Omega t + y\sin\Omega t)^2 \\ &+ (1-\epsilon)(-x\sin\Omega t + y\cos\Omega t)^2 + \lambda^2 z^2\big]. \end{aligned} \tag{9.3}$$

One can easily check that this potential is time independent if $\epsilon = 0$.

The dynamics of a particle in the laboratory frame S is described by the time-dependent Lagrangian

$$\mathcal{L}(t) = \tfrac{1}{2}mv^2 - V_{\rm trap}(\mathbf{r},t), \tag{9.4}$$

where the vector $\mathbf{r} = \mathbf{r}(t)$ specifies the time-dependent position of the particle, and $\mathbf{v} = d\mathbf{r}/dt$ is the particle velocity. To obtain the Lagrangian in the rotating frame, we note that the velocity of the particle as measured by an observer in the rotating frame is $\mathbf{v}' \equiv \dot{x}'\mathbf{u}'_x + \dot{y}'\mathbf{u}'_y + \dot{z}'\mathbf{u}'_z$. This velocity is related to $\mathbf{v}$ by

$$\mathbf{v} = \mathbf{v}' + \mathbf{\Omega}\times\mathbf{r}', \tag{9.5}$$

where we note that the unit vectors in the rotating frame satisfy $\dot{\mathbf{u}}'_i = \mathbf{\Omega}\times\mathbf{u}'_i$. Substituting this expression for $\mathbf{v}$ into (9.4), the Lagrangian describing the dynamics in the frame S' is given by

$$\mathcal{L}' = \tfrac{1}{2}mv'^2 + m\mathbf{v}'\cdot(\mathbf{\Omega}\times\mathbf{r}') + \tfrac{1}{2}m(\mathbf{\Omega}\times\mathbf{r}')^2 - V'_{\rm trap}(\mathbf{r}'). \tag{9.6}$$

This Lagrangian should be thought of as a function of the dynamical variables r'_i and $\dot{r}'_i$. When expressed in terms of these variables, it is time independent. The Lagrangian equations of motion (see, for example, Landau and Lifshitz, 1969b),

$$\frac{d}{dt}\left(\frac{\partial\mathcal{L}'}{\partial\dot{r}'_i}\right) = \frac{\partial\mathcal{L}'}{\partial r'_i}, \tag{9.7}$$

[2] Although we are risking some confusion, $\mathbf{r}' \equiv \mathbf{r}$ in this section. The notation $\mathbf{r}'$ simply means that we are thinking of the components of $\mathbf{r}$ with respect to S'.

then lead to

$$m\frac{d\mathbf{v}'}{dt} = -\nabla' V'_{\rm trap}(\mathbf{r}') + 2m\mathbf{v}' \times \mathbf{\Omega} + m\mathbf{\Omega} \times (\mathbf{r}' \times \mathbf{\Omega}). \tag{9.8}$$

This is the familiar equation of motion containing the noninertial Coriolis force $2m\mathbf{v}' \times \mathbf{\Omega}$ and the centrifugal force $m\mathbf{\Omega} \times (\mathbf{r}' \times \mathbf{\Omega})$.[3]

The Hamiltonian in the rotating frame is defined in the usual way as (Landau and Lifshitz, 1969b)

$$\mathcal{H}' \equiv \mathbf{v}' \cdot \mathbf{p}' - \mathcal{L}', \tag{9.9}$$

where the canonical momentum is defined by

$$\mathbf{p}' \equiv \frac{\partial \mathcal{L}'}{\partial \mathbf{v}'} = m\mathbf{v}' + m\mathbf{\Omega} \times \mathbf{r}' = m\mathbf{v} = \mathbf{p}. \tag{9.10}$$

Thus the momentum vector is the same in both frames of reference.[4] Making use of the Lagrangian in (9.6), we have

$$\mathcal{H}' = \frac{p'^2}{2m} + V'_{\rm trap}(\mathbf{r}') - \mathbf{\Omega} \cdot \mathbf{L}', \tag{9.11}$$

where $\mathbf{L}' = \mathbf{r}' \times \mathbf{p}'$ is the angular momentum in the rotating frame. By virtue of the fact that $\mathbf{r}' = \mathbf{r}$ and $\mathbf{p}' = \mathbf{p}$, the angular momentum is also the same in the two frames of reference. Noting that $V'_{\rm trap}(\mathbf{r}') = V_{\rm trap}(\mathbf{r}, t)$, (9.11) can also be written as

$$\begin{aligned} \mathcal{H}' &= \frac{p^2}{2m} + V_{\rm trap}(\mathbf{r}, t) - \mathbf{\Omega} \cdot \mathbf{L} \\ &= \mathcal{H}(t) - \mathbf{\Omega} \cdot \mathbf{L}\,, \end{aligned} \tag{9.12}$$

where $\mathcal{H}(t) = \mathbf{v} \cdot \mathbf{p} - \mathcal{L}(t)$ is the (generally *time-dependent*) Hamiltonian in the lab frame. One will often see $\mathcal{H}(t)$ written simply as "$\mathcal{H}$" but it should be understood that it is only time independent when expressed in terms of the primed variables, as in (9.11).

Equation (9.8) encapsulates the classical dynamics of a single particle in the rotating frame of reference. We now turn to the behaviour of a gas of thermal atoms in the semiclassical approximation. The dynamics is conveniently described in terms of a distribution function $f'(\mathbf{p}', \mathbf{r}', t)$ in the rotating frame. As indicated, it is considered a function of the components

[3] It should be noted that the time derivative in (9.8) is understood to be $d\mathbf{v}'/dt = \ddot{x}'\mathbf{u}'_x + \ddot{y}'\mathbf{u}'_y + \ddot{z}'\mathbf{u}'_z$, that is, the basis vectors are *not* differentiated. This quantity is, in fact, the acceleration of the particle as measured in the rotating frame.

[4] One should resist the temptation of associating preconceived notions with the word "momentum"; $\mathbf{p}'$ is a momentum because it is *defined* by (9.10). Even if the particle is stationary in S', it still has a nonzero (canonical) momentum. This is one of the many peculiarities that arise when dealing with noninertial frames of reference.

p_i' and r_i' in the rotating frame. The Boltzmann kinetic equation for f' is obtained by noting that the particle dynamics is governed by the Hamiltonian $\mathcal{H}'$. We therefore have (Huang, 1987)

$$\frac{\partial f'}{\partial t} + \{f', \mathcal{H}'\} = C_{22}'[f'], \tag{9.13}$$

where the Poisson bracket is defined as (Sakurai, 1994)

$$\{A, B\} = \sum_i \left(\frac{\partial A}{\partial r_i'} \frac{\partial B}{\partial p_i'} - \frac{\partial A}{\partial p_i'} \frac{\partial B}{\partial r_i'} \right). \tag{9.14}$$

As we will prove in Section 9.4, the collision integral $C_{22}'[f']$ has the same form as in (3.40), but is expressed in terms of the rotating frame variables. Evaluating the Poisson bracket in (9.13) leads to

$$\frac{\partial f'}{\partial t} + \left(\frac{\mathbf{p}'}{m} - \mathbf{\Omega} \times \mathbf{r}' \right) \cdot \nabla_{\mathbf{r}'} f' - \left(\nabla_{\mathbf{r}'} V_{\rm trap}' + \mathbf{\Omega} \times \mathbf{p}' \right) \cdot \nabla_{\mathbf{p}'} f' = C_{22}'[f']. \tag{9.15}$$

The terms involving $\mathbf{\Omega}$ are the new terms associated with the *rotating* frame of reference.

The steady-state equilibrium solution of (9.15) for the rotating frame is of particular interest. For such a state, $\partial f'/\partial t = 0$. Referring to (9.13), we see that this is true if $f' = f'(\mathcal{H}')$ and $C_{22}'[f'] = 0$. Both these conditions are satisfied by the equilibrium Bose distribution[5]

$$\begin{aligned} f_0'(\mathbf{p}', \mathbf{r}') &= \frac{1}{\exp \beta(\mathcal{H}' - \tilde{\mu}_0') - 1} \\ &= \frac{1}{\exp \beta[p'^2/2m + V_{\rm trap}'(\mathbf{r}') - \mathbf{\Omega} \cdot \mathbf{r}' \times \mathbf{p}' - \tilde{\mu}_0'] - 1} \\ &= \frac{1}{\exp \beta[(\mathbf{p}' - m\mathbf{v}_{\rm rb})^2/2m + V_{\rm trap}'(\mathbf{r}') + V_{\rm cent}(\mathbf{r}') - \tilde{\mu}_0'] - 1} \end{aligned} \tag{9.16}$$

where in the last line we have introduced the *rigid-body* rotation velocity

$$\mathbf{v}_{\rm rb}(\mathbf{r}') \equiv \mathbf{\Omega} \times \mathbf{r}' \tag{9.17}$$

and the centrifugal potential

$$V_{\rm cent}(\mathbf{r}') \equiv -\tfrac{1}{2} m (\mathbf{\Omega} \times \mathbf{r}')^2 = -\tfrac{1}{2} m v_{\rm rb}^2 . \tag{9.18}$$

We refer to the quantity $\tilde{\mu}_0'$ as the *rotating frame* chemical potential of the thermal cloud.

Since $\mathbf{p}' - m\mathbf{v}_{\rm rb} = m\mathbf{v}'$, the velocity distribution given by (9.16) depends

[5] Because of the primes, we denote the equilibrium Bose distribution by f_0 in this chapter, rather than f^0 as used in the rest of the book.

only on the magnitude of $\mathbf{v}'$ and hence is isotropic in the rotating frame. The density distribution, however, is anisotropic (although stationary) and is given by

$$\begin{aligned}\tilde{n}'_0(\mathbf{r}') &= \int \frac{d\mathbf{p}'}{(2\pi\hbar)^3} f'_0(\mathbf{p}', \mathbf{r}') \\ &= \frac{1}{\Lambda^3} g_{3/2}(z'_0(\mathbf{r}'))\,, \end{aligned} \tag{9.19}$$

where Λ_0 is the thermal de Broglie wavelength (3.58), and the local equilibrium fugacity is

$$z'_0(\mathbf{r}') = e^{\beta[\tilde{\mu}'_0 - V'_{\rm trap}(\mathbf{r}') - V_{\rm cent}(\mathbf{r}')]}\,. \tag{9.20}$$

Owing to the centrifugal potential, the density bulges out in the radial direction. Since the fugacity must lie in the range $0 < z'_0 < 1$, we must have $V'_{\rm trap} + V_{\rm cent} > \tilde{\mu}'_0$ for (9.19) to describe a physically meaningful density profile. If Ω is greater than $\omega_\perp\sqrt{1-\epsilon}$, then this condition will be violated and a stable stationary state is no longer possible.

The lab frame distribution function corresponding to (9.16) is obtained by using the transformation of coordinates given in (9.1) and $\mathbf{p}' = \mathbf{p}$. Since the Jacobian of the transformation is unity, we have $d\mathbf{p}'d\mathbf{r}' = d\mathbf{p}d\mathbf{r}$ and thus

$$f_0(\mathbf{p}, \mathbf{r}, t) = \frac{1}{\exp\beta[(\mathbf{p} - m\mathbf{v}_{\rm rb})^2/2m + V_{\rm trap}(\mathbf{r}, t) + V_{\rm cent}(\mathbf{r}) - \tilde{\mu}'_0] - 1}\,. \tag{9.21}$$

This shows that the velocity field in the lab frame is $\mathbf{v}(\mathbf{r}) = \mathbf{v}_{\rm rb} = \boldsymbol{\Omega} \times \mathbf{r}$; that is, the density distribution $\tilde{n}_0(\mathbf{r}, t)$ rotates *rigidly* with angular velocity Ω. We note that the chemical potential in the lab frame is still $\tilde{\mu}'_0$.

9.2 Rotating frames of reference: quantum treatment

In the classical description of Section 9.1, the trajectory of a particle can be followed in space and, as such, the particle's position vector is a well-defined dynamical variable. It is represented as the *same* vector in the two frames of reference S and S', and it is thus natural to adopt the passive-rotation point of view. However, quantum states are not described in terms of trajectories, and so a different approach is needed in this case. As we shall see, the active-rotation point of view is the most convenient when describing the transformation of quantum states in Hilbert space under rotations.

Figure 9.1(b) illustrates the *active* rotation of the vector $\mathbf{r}$ into the vector $\mathbf{r}''$. (As mentioned in Section 9.1, we reserve the notation $\mathbf{r}'$ for the vector obtained by a rotation in the opposite direction.) For purposes of illustration

we consider a rotation about the z axis, but more general rotations can be handled in the same way. The same rotation of the basis vectors $\mathbf{u}_i$, which define the S-coordinate system, generates the basis vectors $\mathbf{u}'_i$ of the S'-coordinate system. The components of the vector $\mathbf{r}'' = r''_i \hat{\mathbf{u}}_i$ in the S-basis are in general given by $r''_i = \mathcal{R}_{ij} r_j$ where $\mathcal{R}_{ij}$ is a real orthogonal matrix. For the case being considered, we have

$$\begin{aligned} x'' &= x\cos\theta - y\sin\theta, \\ y'' &= x\sin\theta + y\cos\theta, \\ z'' &= z. \end{aligned} \tag{9.22}$$

Note that the transformation (9.22) differs from (9.1), where r'_i represents a component of $\mathbf{r}$ (the unrotated vector) with respect to the rotated basis. In the following, we will denote the transformation of the vector components as $\mathbf{r}'' = \mathcal{R}[\mathbf{r}]$, with the understanding that all components are with respect to the S-basis.

The connection of the above with quantum mechanics is revealed most directly by considering the rotation of the position eigenket $|\mathbf{r}\rangle$. The rotated state is denoted as $\hat{R}(\theta)|\mathbf{r}\rangle$, where $\hat{R}(\theta)$ is, by definition, the rotation operator. It is physically clear that when we measure the position of the particle in the rotated state we must obtain the result $\mathbf{r}''$. Thus, $\hat{R}(\theta)|\mathbf{r}\rangle = |\mathbf{r}''\rangle = |\mathcal{R}[\mathbf{r}]\rangle$. The rotation of an arbitrary physical state $|\psi\rangle$ is then represented as

$$|\psi'\rangle = \hat{R}(\theta)|\psi\rangle\,. \tag{9.23}$$

As shown in standard texts (for example, Sakurai, 1994), the angular momentum operator is the generator of infinitesimal rotations. For the special case being considered, we have

$$\hat{R}(\theta) = e^{-i\theta \hat{L}_z/\hbar}\,, \tag{9.24}$$

where $\hat{L}_z$ is the z-component of the angular momentum operator.

The transformation of quantum operators under rotation is defined by the requirement that the expectation value of some physical quantity $\hat{O}$ is unchanged if both the quantum state and measuring apparatus (with which the quantum operator is associated) undergo the same rotation. The mathemetical statement of this condition is

$$\langle\psi'|\hat{O}'|\psi'\rangle = \langle\psi|\hat{O}|\psi\rangle\,. \tag{9.25}$$

The left hand side of this equation is $\langle\psi|\hat{R}^\dagger(\theta)\hat{O}'\hat{R}(\theta)|\psi\rangle$ and, since the state $|\psi\rangle$ is arbitrary, we have $\hat{R}^\dagger(\theta)\hat{O}'\hat{R}(\theta) = \hat{O}$. Noting that $\hat{R}(\theta)$ is unitary, we

see that the rotated operator is given by

$$\hat{O}' = \hat{R}(\theta)\hat{O}\hat{R}^{\dagger}(\theta)\,, \tag{9.26}$$

which depends parametrically on the rotation angle θ. As such, it satisfies the "equation of motion"

$$i\hbar\frac{d\hat{O}'}{d\theta} = [\hat{L}_z, \hat{O}']\,. \tag{9.27}$$

Applying (9.27) to the components of the position operator $\hat{\mathbf{r}}$ yields the equations

$$\begin{aligned}\frac{d\hat{x}'}{d\theta} &= \hat{y}',\\ \frac{d\hat{y}'}{d\theta} &= -\hat{x}',\end{aligned} \tag{9.28}$$

which have the solution

$$\begin{aligned}\hat{x}' &= \hat{x}\cos\theta + \hat{y}\sin\theta\,,\\ \hat{y}' &= -\hat{x}\sin\theta + \hat{y}\cos\theta\,.\end{aligned} \tag{9.29}$$

In contrast with the vector components r_i'' in (9.22), we see that the operator components transform according to the transposed rotation matrix, $\mathcal{R}^{\mathrm{T}} \equiv \mathcal{R}^{-1}$; that is, a rotation in the *opposite* sense. Thus in general, $\hat{r}_i' = \mathcal{R}_{ij}^{-1}\hat{r}_j$.

To clarify the meaning of (9.29), we note that

$$\hat{x}'|\mathbf{r}\rangle = (x\cos\theta + y\sin\theta)|\mathbf{r}\rangle\,. \tag{9.30}$$

Referring to (9.1), we see that the $\hat{x}'$ eigenvalue is just x', the component of $\mathbf{r}$ in the direction $\mathbf{u}_x'$. In other words, a measurement of the particle's position in S' yields the result $\mathbf{r}$ expressed in terms of the S' basis. We also observe that

$$\hat{x}'|\mathbf{r}''\rangle = (\hat{x}\cos\theta + \hat{y}\sin\theta)|\mathbf{r}''\rangle = x|\mathbf{r}''\rangle\,, \tag{9.31}$$

where have used (9.22) for the components of $\mathbf{r}''$. This simply confirms that a measurement of $\hat{x}'$ in the rotated state yields the same result as a measurement of $\hat{x}$ in the original state.

The transformation in (9.29) is actually valid for *any* vector operator (Sakurai, 1994), including the momentum operator $\hat{\mathbf{p}}$. In all cases, the components of the rotated vector operator are the components of the original operator with respect to the rotated axes, that is, $\hat{p}_i' = \mathcal{R}_{ij}^{-1}\hat{p}_j$. Recall that in the classical formulation we had the relation $\mathbf{p}' = \mathbf{p}$. The analogous quantum result is $\hat{p}_i'\mathbf{u}_i' = \mathbf{u}_i'\mathcal{R}_{ij}^{-1}\hat{p}_j = \hat{p}_j\mathbf{u}_j$, where we have noted that the basis

vectors transform as $\mathbf{u}'_i = \mathbf{u}_j \mathcal{R}_{ji}$. It is in this sense that we have $\hat{\mathbf{p}}' = \hat{\mathbf{p}}$ quantum mechanically.

We can now define a trap potential operator corresponding to the trap potential in (9.2). This operator is given by

$$V'_{\rm trap}(\hat{\mathbf{r}}') \equiv \tfrac{1}{2} m\omega_\perp^2 \left[(1+\epsilon)\hat{x}'^2 + (1-\epsilon)\hat{y}'^2 + \lambda^2 \hat{z}'^2\right], \tag{9.32}$$

where the primed operators are the rotated components given in (9.29). In view of (9.26), this can be written as

$$V'_{\rm trap}(\hat{\mathbf{r}}') = \hat{R}(\theta) V'_{\rm trap}(\hat{\mathbf{r}}) \hat{R}^\dagger(\theta)\,. \tag{9.33}$$

Using the transformation in (9.29) and setting $\theta = \Omega t$, the coordinate representation of this potential in the lab frame is precisely the expression given in (9.3). In other words, the lab frame trap potential operator is $V_{\rm trap}(\hat{\mathbf{r}}, t) = \hat{R}(\Omega t) V'_{\rm trap}(\hat{\mathbf{r}}) \hat{R}^\dagger(\Omega t)$.

We next consider the dynamics of a single atom in a rotating frame. The time development of its quantum state $|\psi\rangle$ in the lab frame in the presence of the rotating trap potential in (9.32) is given by the equation

$$\begin{aligned} i\hbar \frac{\partial |\psi(t)\rangle}{\partial t} &= \left[\frac{\hat{p}^2}{2m} + \hat{R}(\Omega t) V'_{\rm trap}(\hat{\mathbf{r}}) \hat{R}^\dagger(\Omega t)\right] |\psi(t)\rangle \\ &= \hat{R}(\Omega t) \left[\frac{\hat{p}^2}{2m} + V'_{\rm trap}(\hat{\mathbf{r}})\right] \hat{R}^\dagger(\Omega t) |\psi(t)\rangle\,, \end{aligned} \tag{9.34}$$

where we note that $\hat{p}^2$ is a scalar and therefore invariant under rotations. The state vector

$$|\psi'(t)\rangle \equiv \hat{R}^\dagger(\Omega t) |\psi(t)\rangle \tag{9.35}$$

evolves according to the equation

$$i\hbar \frac{\partial |\psi'(t)\rangle}{\partial t} = \left[\frac{\hat{p}^2}{2m} + V'_{\rm trap}(\hat{\mathbf{r}}) - \Omega \hat{L}_z\right] |\psi'(t)\rangle\,, \tag{9.36}$$

where the term $\Omega \hat{L}_z$ comes from taking the time derivative of $\hat{R}(\Omega t)$. We see that the time evolution of the state $|\psi'(t)\rangle$ is governed by a *time-independent* trapping potential.[6] We can therefore consider $|\psi'(t)\rangle$ as the state of the system as viewed from the rotating frame S'. The state in the lab frame is given by $|\psi(t)\rangle = \hat{R}(\Omega t)|\psi'(t)\rangle$; it is obtained by applying an *active* rotation to the state $|\psi'(t)\rangle$.

[6] When suitably generalized, (9.36) is also valid for a many body system in which the particles interact via central forces.

Taking the inner product of (9.36) with the ket vector $|\mathbf{r}'\rangle$, we have

$$i\hbar\frac{\partial\psi'(\mathbf{r}',t)}{\partial t} = \left[-\frac{\hbar^2\nabla'^2}{2m} + V'_{\rm trap}(\mathbf{r}') - \Omega L'_z\right]\psi'(\mathbf{r}',t)\,. \tag{9.37}$$

The quantity in brackets is just the Hamiltonian $\mathcal{H}'$ defined in (9.11). The prime on the position vector $\mathbf{r}'$ in (9.37) may seem superfluous since $\mathbf{r}'$ is an independent variable. However, the notation becomes more meaningful when we consider the wavefunction in the lab frame. This is given by

$$\psi(\mathbf{r},t) = \langle\mathbf{r}|\psi(t)\rangle = \langle\mathbf{r}|\hat{R}(\Omega t)|\psi'(t)\rangle = \langle\mathcal{R}^{-1}[\mathbf{r}]|\psi'(t)\rangle = \psi'(\mathbf{r}',t)\,, \tag{9.38}$$

where $\mathbf{r}' = \mathcal{R}^{-1}[\mathbf{r}]$ is the vector into which $\mathbf{r}$ rotates under the inverse rotation $\mathcal{R}^{-1}$ (see Fig. 9.1(b)). The prescription for obtaining the lab frame wavefunctions is the following. First, (9.37) is solved as a function of the independent variable $\mathbf{r}'$. Once the function $\psi'(\mathbf{r}',t)$ is known, the variable $\mathbf{r}'$ is replaced by $\mathcal{R}^{-1}[\mathbf{r}]$, where $\mathcal{R}$ is the time-dependent rotation matrix. The result is the lab frame wavefunction $\psi(\mathbf{r},t)$. In the case of a stationary solution of (9.37), $|\psi'(\mathbf{r}')|^2$ will be time independent, but $|\psi'(\mathcal{R}^{-1}[\mathbf{r}])|^2 = |\psi(\mathbf{r},t)|^2$ will in general exhibit a time dependence that is associated with the rotation of the system with respect to the lab frame.

9.3 Transformation of the kinetic equation

Having defined rotation operators, we are now in a position to examine the state of the thermal component in a trapped gas. A kinetic description involves the Wigner operator defined as (see (3.34))

$$\hat{f}(\mathbf{p},\mathbf{r}) = \int d\mathbf{r}_1\, e^{i\mathbf{p}\cdot\mathbf{r}_1/\hbar}\hat{\psi}^\dagger\left(\mathbf{r}+\tfrac{1}{2}\mathbf{r}_1\right)\hat{\psi}\left(\mathbf{r}-\tfrac{1}{2}\mathbf{r}_1\right)\,. \tag{9.39}$$

According to (9.26), the rotated Wigner operator is given by

$$\hat{f}'(\mathbf{p},\mathbf{r}) = \hat{R}(\theta)\hat{f}(\mathbf{p},\mathbf{r})\hat{R}^\dagger(\theta)\,. \tag{9.40}$$

Notice that the phase-space coordinates $\mathbf{p}$ and $\mathbf{r}$ are fixed independent variables labelling the operator and are unaffected by the rotation. As such, these variables also appear as labels for the rotated Wigner operator. Since the quantum field operator $\hat{\psi}^\dagger(\mathbf{r})$ creates a particle at position $\mathbf{r}$, the rotated operator must have the effect of creating a particle at position $\mathcal{R}[\mathbf{r}]$, namely

$$\hat{R}(\theta)\hat{\psi}^\dagger(\mathbf{r})\hat{R}^\dagger(\theta) = \hat{\psi}^\dagger(\mathcal{R}[\mathbf{r}])\,. \tag{9.41}$$

The transformation in (9.40) is then given by

$$\begin{aligned}\hat{f}'(\mathbf{p},\mathbf{r}) &= \int d\mathbf{r}_1\, e^{i\mathbf{p}\cdot\mathbf{r}_1/\hbar}\hat{\psi}^\dagger\left(\mathcal{R}[\mathbf{r}]+\tfrac{1}{2}\mathcal{R}[\mathbf{r}_1]\right)\hat{\psi}\left(\mathcal{R}[\mathbf{r}]-\tfrac{1}{2}\mathcal{R}[\mathbf{r}_1]\right)\\ &= \int d\mathbf{r}_1\, e^{i\mathbf{p}\cdot\mathcal{R}^{-1}[\mathbf{r}_1]/\hbar}\hat{\psi}^\dagger\left(\mathcal{R}[\mathbf{r}]+\tfrac{1}{2}\mathbf{r}_1\right)\hat{\psi}\left(\mathcal{R}[\mathbf{r}]-\tfrac{1}{2}\mathbf{r}_1\right)\\ &= \int d\mathbf{r}_1\, e^{i\mathcal{R}[\mathbf{p}]\cdot\mathbf{r}_1/\hbar}\hat{\psi}^\dagger\left(\mathcal{R}[\mathbf{r}]+\tfrac{1}{2}\mathbf{r}_1\right)\hat{\psi}\left(\mathcal{R}[\mathbf{r}]-\tfrac{1}{2}\mathbf{r}_1\right)\\ &= \hat{f}(\mathcal{R}[\mathbf{p}],\mathcal{R}[\mathbf{r}])\,. \end{aligned} \tag{9.42}$$

This is equivalent to the relation

$$\hat{f}'(\mathbf{p}',\mathbf{r}') = \hat{f}(\mathbf{p},\mathbf{r})\,, \tag{9.43}$$

where $\mathbf{r}' = \mathcal{R}^{-1}[\mathbf{r}]$ and $\mathbf{p}' = \mathcal{R}^{-1}[\mathbf{p}]$. The primed variables in (9.43) are time dependent because of the time dependence of the rotation matrix.

According to (3.35), the semiclassical distribution function $f(\mathbf{p},\mathbf{r},t)$ is obtained by performing an average of $\hat{f}(\mathbf{p},\mathbf{r})$ with respect to the density operator $\hat{\rho}(t)$. The same ensemble average of $\hat{f}'(\mathbf{p}',\mathbf{r}')$ generates the rotating-frame distribution function $f'(\mathbf{p}',\mathbf{r}',t)$ and from (9.43), we have $f'(\mathbf{p}',\mathbf{r}',t) = f(\mathbf{p},\mathbf{r},t)$. In the semiclassical limit, $f(\mathbf{p},\mathbf{r},t)$ satisfies the kinetic equation in (3.42). The corresponding kinetic equation for $f'(\mathbf{p}',\mathbf{r}',t)$ in the rotating frame can be obtained by a straightforward transformation of the various terms in (3.42). For example, we have

$$\frac{\partial f}{\partial t} = \frac{\partial f'}{\partial t} + \frac{\partial f'}{\partial p_i'}\frac{dp_i'}{dt} + \frac{\partial f'}{\partial r_i'}\frac{dr_i'}{dt}\,. \tag{9.44}$$

Noting that the time derivatives are classical counterparts of (9.27), we have (see (9.13) and (9.14))

$$\frac{dp_i'}{dt} = \{\boldsymbol{\Omega}\cdot\mathbf{L}, p_i'\} = \frac{\partial(\boldsymbol{\Omega}\cdot\mathbf{L})}{\partial r_i'}\,, \tag{9.45}$$

$$\frac{dr_i'}{dt} = \{\boldsymbol{\Omega}\cdot\mathbf{L}, r_i'\} = -\frac{\partial(\boldsymbol{\Omega}\cdot\mathbf{L})}{\partial p_i'}\,. \tag{9.46}$$

We thus find that

$$\frac{\partial f}{\partial t} = \frac{\partial f'}{\partial t} + \{\boldsymbol{\Omega}\cdot\mathbf{L}, f'\}\,. \tag{9.47}$$

Similarly, the other terms on the r.h.s. of (3.42) are given by

$$p_i\frac{\partial f}{\partial r_i} = p_i\frac{\partial f'}{\partial r_j'}\frac{\partial r_j'}{\partial r_i} = p_i\frac{\partial f'}{\partial r_j'}\mathcal{R}^{-1}_{ji} = p_j'\frac{\partial f'}{\partial r_j'}\,, \tag{9.48}$$

and

$$\frac{\partial V_{\rm trap}}{\partial r_i}\frac{\partial f}{\partial p_i} = \frac{\partial V'_{\rm trap}}{\partial r'_j}\frac{\partial r'_j}{\partial r_i}\frac{\partial f'}{\partial p'_k}\frac{\partial p'_k}{\partial p_i} = \frac{\partial V'_{\rm trap}}{\partial r'_j}\mathcal{R}^{-1}_{ji}\frac{\partial f'}{\partial p'_k}\mathcal{R}^{-1}_{ki} = \frac{\partial V'_{\rm trap}}{\partial r'_j}\frac{\partial f'}{\partial p'_j}\,. \tag{9.49}$$

Putting all these results together, the streaming terms on the l.h.s. of the Boltzmann equation in (3.42) are transformed in the rotating frame,

$$\begin{aligned}&\frac{\partial f}{\partial t} + \frac{\mathbf{p}}{m}\cdot\nabla_{\mathbf{r}} f - \nabla_{\mathbf{r}} V_{\rm trap}\cdot\nabla_{\mathbf{p}} f \\ &\quad = \frac{\partial f'}{\partial t} + \left(\frac{\mathbf{p}'}{m} - \boldsymbol{\Omega}\times\mathbf{r}'\right)\cdot\nabla_{\mathbf{r}'} f' - \left(\nabla_{\mathbf{r}'} V'_{\rm trap} + \boldsymbol{\Omega}\times\mathbf{p}'\right)\cdot\nabla_{\mathbf{p}'} f'\,.\end{aligned} \tag{9.50}$$

We note that this result is identical to the left hand side of (9.15) found in the classical treatment.

9.4 Zaremba–Nikuni–Griffin equations in a rotating frame

With the discussion in the previous sections in mind, we now consider the full set of ZNG equations in the rotating frame. The generalized GP equation is similar to the Schrödinger equation in (9.37) but with the addition of the mean-field interaction terms and the collisional particle-exchange term. To illustrate how these terms are transformed to the rotating frame, we consider the example of the mean-field coupling to the thermal cloud, which in operator form is $2g\tilde{n}(\hat{\mathbf{r}},\mathbf{t})$. This term leads to the additional contribution

$$\hat{R}^\dagger(\Omega t) 2g\tilde{n}(\hat{\mathbf{r}},t)\hat{R}(\Omega t) = 2g\tilde{n}(\mathcal{R}[\hat{\mathbf{r}}],t) \tag{9.51}$$

inside the square brackets in the second line of (9.34). Notice that the position operator is here transformed according to the *inverse* of the rotation operator defined in (9.24). In the coordinate representation, we have

$$2g\tilde{n}(\mathcal{R}[\mathbf{r}'],t) \equiv 2g\tilde{n}'(\mathbf{r}',t)\,. \tag{9.52}$$

In other words, $\tilde{n}'(\mathbf{r}',t) = \tilde{n}(\mathbf{r},t)$ with $\mathbf{r}' = \mathcal{R}^{-1}[\mathbf{r}]$. This is precisely the relation between the wavefunctions in the two frames of reference that we derived in (9.38). With this observation, the generalized GP equation for the condensate in the rotating frame is clearly given by

$$\begin{aligned}i\hbar\frac{\partial\Phi'(\mathbf{r}',t)}{\partial t} = \Bigg[&-\frac{\hbar^2\nabla'^2}{2m} + V'_{\rm trap}(\mathbf{r}') + gn'_{\rm c}(\mathbf{r}',t) + 2g\tilde{n}'(\mathbf{r}',t) \\ &- \boldsymbol{\Omega}\cdot\mathbf{L}' - iR'(\mathbf{r}',t)\Bigg]\Phi'(\mathbf{r}',t)\,,\end{aligned} \tag{9.53}$$

where $n'_{\rm c}(\mathbf{r}',t) = n_{\rm c}(\mathbf{r},t)$ and $R'(\mathbf{r}',t) = R(\mathbf{r},t)$. The trapping potential $V'_{\rm trap}$ in (9.53) is time independent in the rotating frame, but the mean fields are still dynamic, i.e. time dependent. As in (9.38), the condensate wavefunction in the lab frame is obtained from $\Phi'(\mathbf{r}',t)$ using the prescription $\Phi(\mathbf{r},t) = \Phi'(\mathcal{R}^{-1}[\mathbf{r}],t)$.

Making use of the phase–amplitude representation in the rotating frame,

$$\Phi'(\mathbf{r}',t) = \sqrt{n'_{\rm c}(\mathbf{r}',t)}e^{i\theta'(\mathbf{r}',t)}\,, \tag{9.54}$$

the condensate velocity is defined in the usual way, with components

$$v'_{ci}(\mathbf{r}',t) \equiv \frac{\hbar}{m}\frac{\partial\theta'}{\partial r'_i} = \frac{\hbar}{m}\frac{\partial\theta}{\partial r_j}\frac{\partial r_j}{\partial r'_i} = v_{cj}(\mathbf{r},t)\mathcal{R}_{ji} = \mathcal{R}^{-1}_{ij}v_{cj}(\mathbf{r},t)\,. \tag{9.55}$$

Thus the local fluid velocity is given by $\mathbf{v}'_{\rm c} = \mathcal{R}^{-1}[\mathbf{v}_{\rm c}]$,[7] involving the same transformation as that used to obtain $\mathbf{r}'$ and $\mathbf{p}'$. Using (2.16), (2.17) and the relation

$$\frac{1}{\Phi'}(\mathbf{\Omega}\cdot\mathbf{L}')\Phi' = m\mathbf{v}'_{\rm c}\cdot(\mathbf{\Omega}\times\mathbf{r}') - \frac{i\hbar}{2n'_{\rm c}}\nabla'\cdot[n'_{\rm c}(\mathbf{\Omega}\times\mathbf{r}')] \tag{9.56}$$

in (9.53), we obtain the rotating frame condensate "hydrodynamic equations" (see also Recati *et al.*, 2001)

$$\frac{\partial n'_{\rm c}}{\partial t} = -\nabla'\cdot[n'_{\rm c}(\mathbf{v}'_{\rm c} - \mathbf{\Omega}\times\mathbf{r}')] - \Gamma'_{12}[f',\Phi'] \tag{9.57}$$

$$m\frac{\partial\mathbf{v}'_{\rm c}}{\partial t} = -\nabla'\left[\varepsilon'_{\rm c} - m\mathbf{v}'_{\rm c}\cdot(\mathbf{\Omega}\times\mathbf{r}')\right], \tag{9.58}$$

where $\Gamma'_{12} \equiv 2n'_{\rm c}R'/\hbar$ and

$$\varepsilon'_{\rm c}(\mathbf{r}',t) = \mu'_{\rm c}(\mathbf{r}',t) + \tfrac{1}{2}mv'^2_{\rm c}(\mathbf{r}',t) = \mu_{\rm c}(\mathbf{r},t) + \tfrac{1}{2}mv^2_{\rm c}(\mathbf{r},t)\,. \tag{9.59}$$

The condensate chemical potential $\mu_{\rm c}(\mathbf{r},t)$ was defined in (3.19).

We next consider the transformation of the full Boltzmann kinetic equation in the ZNG theory to the rotating frame. Using the results in Sec. 9.3 for the transformation of the streaming terms, we find

$$\frac{\partial f'}{\partial t} + \left(\frac{\mathbf{p}'}{m} - \mathbf{\Omega}\times\mathbf{r}'\right)\cdot\nabla_{\mathbf{r}'}f' - \left(\nabla_{\mathbf{r}'}U'(\mathbf{r}',t) + \mathbf{\Omega}\times\mathbf{p}'\right)\cdot\nabla_{\mathbf{p}'}f' = C'_{12}[f',\Phi'] + C'_{22}[f']. \tag{9.60}$$

[7] We emphasize that $\mathbf{v}'_{\rm c}$ is *not* the velocity of the fluid as measured by an observer in the rotating frame. Rather, v'_{ci} are the components of the lab frame velocity $\mathbf{v}_{\rm c}$ with respect to the rotating basis. The fluid velocity as measured by an observer in S' is $\mathbf{v}'_{\rm c} - \mathbf{\Omega}\times\mathbf{r}'$, which, not surprisingly, is the velocity appearing in the rotating frame continuity equation (9.57).

The left-hand side of this equation differs from the streaming terms in (9.15) in that $V'_{\rm trap}$ is replaced by the total mean-field potential

$$U'(\mathbf{r}',t) = V'_{\rm trap}(\mathbf{r}') + 2g[n'_{\rm c}(\mathbf{r}',t) + \tilde{n}'(\mathbf{r}',t)]\,. \tag{9.61}$$

The factor in parentheses in the second term of (9.60) is the velocity relative to the rotating frame S'.[7] This is clearly the correct velocity to be used when considering the streaming of particles in phase space.

The transformation of the collision integrals is also straightforward and can be illustrated for the case of $C_{12}[f,\Phi]$, as defined in (3.41). First we make the direct substitutions $f(\mathbf{p},\mathbf{r},t) = f'(\mathbf{p}',\mathbf{r}',t)$ and $n_{\rm c}(\mathbf{r},t) = n'_{\rm c}(\mathbf{r}',t)$. Next we note that the Jacobian of the transformation $\mathbf{p}' = \mathcal{R}^{-1}[\mathbf{p}]$ is unity, so that $d\mathbf{p} = d\mathbf{p}'$. We then observe that the transformation of the momentum delta functions is given by

$$\delta(\mathbf{p}-\mathbf{p}_i) = \delta(\mathcal{R}[\mathbf{p}'-\mathbf{p}'_i]) = \delta(\mathbf{p}'-\mathbf{p}'_i)\,. \tag{9.62}$$

The last step follows from the rotational invariance of the delta function, and can be proved by considering the delta function as the limit of a Gaussian distribution. Finally, the variables in the energy conserving delta function transform as follows:

$$\tilde{\varepsilon}_p(\mathbf{r},t) = \frac{p^2}{2m} + U(\mathbf{r},t) = \frac{p'^2}{2m} + U'(\mathbf{r}',t) \equiv \tilde{\varepsilon}'_{p'}(\mathbf{r}',t); \tag{9.63}$$

$\varepsilon_{\rm c}$ undergoes a similar transformation, according to (9.59). With these changes, the transformed collision integral $C'_{12}[f',\Phi']$ is found to have *exactly* the same form as $C_{12}[f,\Phi]$, with all the unprimed variables merely replaced by primed variables. The same prescription clearly applies to $C'_{22}[f']$, as noted after (9.13). For completeness, in the rotating frame, R' in (9.53) is given by

$$R'(\mathbf{r}',t) = \frac{\hbar}{2n'_{\rm c}(\mathbf{r}',t)} \int \frac{d\mathbf{p}'}{(2\pi\hbar)^3} C'_{12}[f']. \tag{9.64}$$

In summary, (9.53) and (9.60) constitute the generalization of the ZNG equations introduced in Chapter 3 to a rotating frame. These equations provide the basis of future calculations of vortices and vortex lattices which fully include the dynamical effects of the thermal cloud.[8] Our detailed discussion has shown how all variables are transformed from the lab to the rotating frame. The transformation of the C_{12} and C_{22} collision integrals to the rotating frame have not been discussed previously in the literature. In

[8] From a operational point of view, it should be clear that all the primes in equations (9.53) and (9.60) can be dropped when solving these equations. Then, to obtain the lab-frame quantities, the position and momentum variables in all functions must be replaced by $\mathcal{R}^{-1}[\mathbf{r}]$ and $\mathcal{R}^{-1}[\mathbf{p}]$, respectively.

the rest of this chapter, we will make use of the ZNG equations to discuss the formation of vortices and vortex lattices, emphasizing the role of the thermal cloud.

9.5 Stationary states

In this section we examine the possibility of realizing a *stationary* condensate state in the rotating frame. For the condensate, such states will have the form $\Phi'(\mathbf{r}',t) = \Phi'_0(\mathbf{r}')e^{-i\tilde{\varepsilon}'_{c0}t/\hbar}$, where $\Phi'_0(\mathbf{r}')$ satisfies (see (9.53))

$$\left(-\frac{\hbar^2\nabla'^2}{2m} + V'_{\rm trap}(\mathbf{r}') + gn'_{c0}(\mathbf{r}') + 2g\tilde{n}'_0(\mathbf{r}') - \mathbf{\Omega}\cdot\mathbf{L}'\right)\Phi'_0(\mathbf{r}') = \tilde{\varepsilon}'_{c0}\Phi'_0(\mathbf{r}'), \tag{9.65}$$

with $n'_{c0}(\mathbf{r}') = |\Phi'_0(\mathbf{r}')|^2$ and

$$\tilde{n}'_0(\mathbf{r}') = \int \frac{d\mathbf{p}'}{(2\pi\hbar)^3} f'_0(\mathbf{p}',\mathbf{r}'). \tag{9.66}$$

In writing (9.65), we have here anticipated the fact (to be proved shortly) that $C'_{12}[f']$ must vanish in equilibrium and hence $R'_0(\mathbf{r}',t) = 0$.

It is clear that a stationary solution of the GP equation also requires a time-independent distribution function $f'_0(\mathbf{p}',\mathbf{r}')$ describing the thermal cloud. It is important to establish the conditions under which such time-independent solutions arise. For this purpose, it is convenient to rewrite the l.h.s. of (9.60) in terms of Poisson brackets, defined in (9.14):

$$\frac{\partial f'}{\partial t} + \{f', \mathcal{H}'_{\rm eff}\} = C'_{12}[f', \Phi'] + C'_{22}[f'], \tag{9.67}$$

where

$$\mathcal{H}'_{\rm eff} \equiv \frac{p'^2}{2m} + V'_{\rm trap}(\mathbf{r}') + 2g[n'_{\rm c}(\mathbf{r}',t) + \tilde{n}'(\mathbf{r}',t)] - \mathbf{\Omega}\cdot\mathbf{L}'. \tag{9.68}$$

Equation (9.67) is an obvious generalization of the classical kinetic equation (9.13) by the addition of the mean-field interaction terms and the collision integral $C'_{12}[f', \Phi']$.

A stationary solution of (9.67) is obtained if each term in the equation is separately zero. The Poisson bracket will vanish if $f' = f'(\mathcal{H}'^0_{\rm eff})$ with $\mathcal{H}'^0_{\rm eff} \equiv p'^2/2m + U'_0(\mathbf{r}') - \mathbf{\Omega}\cdot\mathbf{L}'$. For $C'_{22}[f']$ to vanish, f' must have the form of the Bose distribution

$$f'_0(\mathbf{p}',\mathbf{r}') = \frac{1}{\exp\beta(\mathcal{H}'^0_{\rm eff} - \tilde{\mu}'_0) - 1}. \tag{9.69}$$

The thermal cloud chemical potential $\tilde{\mu}'_0$ is a parameter determined by the

additional requirement that the condensate is in equilibrium with the thermal cloud. Substituting (9.69) into $C'_{12}[f', \Phi']$ and making use of energy and momentum conservation, we find (compare with (3.53))

$$\begin{aligned} C'_{12}[f'_0, \Phi'_0] = \frac{2g^2 n'_{\rm c0}}{(2\pi)^2\hbar^4}\left[1 - e^{\beta(\varepsilon'_{\rm c0} - m\mathbf{v}'_{\rm c0}\cdot\mathbf{\Omega}\times\mathbf{r}' - \tilde{\mu}'_0)}\right] \int d\mathbf{p}'_1 \int d\mathbf{p}'_2 \int d\mathbf{p}'_3 \\ \times \delta(m\mathbf{v}'_{\rm c0} + \mathbf{p}'_1 - \mathbf{p}'_2 - \mathbf{p}'_3)\delta(\varepsilon'_{\rm c0} + \tilde{\varepsilon}'^0_{p'_1} - \tilde{\varepsilon}'^0_{p'_2} - \tilde{\varepsilon}'^0_{p'_3}) \\ \times [\delta(\mathbf{p}' - \mathbf{p}'_1) - \delta(\mathbf{p}' - \mathbf{p}'_2) - \delta(\mathbf{p}' - \mathbf{p}'_3)] \\ \times [1 + f'_0(\mathbf{p}'_1, \mathbf{r}')]f'_0(\mathbf{p}'_2, \mathbf{r}')f'_0(\mathbf{p}'_3, \mathbf{r}')\,, \end{aligned} \tag{9.70}$$

with $\tilde{\varepsilon}'^0_{p'} = p'^2/2m + U'_0(\mathbf{r}')$. As in the discussion of Section 3.3, $C'_{12}[f'_0, \Phi'_0]$ can be made to vanish by requiring the leading square bracket in (9.70) to be zero. Using the phase–amplitude representation of the condensate wavefunction, and noting that $\bar{\varepsilon}'_{\rm c0}$ is real, we find from (9.65) that

$$\begin{aligned} \bar{\varepsilon}'_{\rm c0} &= -\frac{\hbar^2}{2m}\frac{\nabla'^2\sqrt{n'_{\rm c0}}}{\sqrt{n'_{\rm c0}}} + V'_{\rm trap} + gn'_{\rm c0} + 2g\tilde{n}'_0 + \tfrac{1}{2}mv'^2_{\rm c0} - m\mathbf{v}'_{\rm c0}\cdot\mathbf{\Omega}\times\mathbf{r}' \\ &= \varepsilon'_{\rm c0} - m\mathbf{v}'_{\rm c0}\cdot\mathbf{\Omega}\times\mathbf{r}'\,, \end{aligned} \tag{9.71}$$

where $\varepsilon'_{\rm c0}$ is the stationary state version of the energy $\varepsilon'_{\rm c}$ defined in (9.59). We thus see that $C'_{12}[f'_0, \Phi'_0]$ will vanish if we simply choose

$$\bar{\varepsilon}'_{\rm c0} = \tilde{\mu}'_0\,. \tag{9.72}$$

This is the defining condition for absolute equilibrium in the rotating frame. Of course, if $C'_{12}[f'_0, \Phi'_0] = 0$ then $R'_0 = 0$ and our neglect of this term in the generalized GP equation in (9.65) is internally consistent.

The general stationary state in the rotating frame thus consists of a stationary solution to the generalized GP equation and an equilibrium Bose distribution, given by (9.69), the parameters being determined self-consistently by the requirement $\bar{\varepsilon}'_{\rm c0} = \tilde{\mu}'_0$. Since it is possible for the condensate described by (9.65) to have solutions corresponding to vortices, we arrive at the important conclusion that dissipationless vortex states can exist even in the presence of a thermal cloud.

To make contact with our discussion of local equilibrium in Chapter 8, it is convenient to write the distribution function $f'_0(\mathbf{p}, \mathbf{r}, t)$ for the rotating thermal cloud in the *lab* frame. This is given by

$$f_0(\mathbf{p}, \mathbf{r}, t) = \frac{1}{\exp\beta\left[(\mathbf{p} - m\mathbf{v}_{\rm rb})^2/2m + U_0(\mathbf{r}, t) - \tilde{\mu}_{\rm loc}(\mathbf{r})\right] - 1}\,, \tag{9.73}$$

where we have defined the *local* chemical potential

$$\tilde{\mu}_{\rm loc}(\mathbf{r}) \equiv \tilde{\mu}_0' + \tfrac{1}{2}mv_{\rm rb}^2(\mathbf{r}) = \tilde{\mu}_0' + \tfrac{1}{2}m\Omega^2\rho^2\,, \tag{9.74}$$

which increases quadratically with the distance ρ from the rotation axis.[9] The lab frame distribution function in (9.73) clearly corresponds to a thermal cloud undergoing rigid-body rotation ($\mathbf{v}_{\rm n} = \mathbf{v}_{\rm rb}$) with a time-dependent density $\tilde{n}_0(\mathbf{r},t)$ determined by $U_0(\mathbf{r},t) = V_{\rm trap}(\mathbf{r},t) + 2g[n_{\rm c0}(\mathbf{r},t) + \tilde{n}_0(\mathbf{r},t)]$.

The condition of absolute equilibrium in the rotating frame given in (9.72) can also be written, using the relations in (9.59), as

$$\tilde{\varepsilon}_{\rm c0}' = \mu_{\rm c0}(\mathbf{r},t) + \tfrac{1}{2}mv_{\rm c0}^2(\mathbf{r},t) - m\mathbf{v}_{\rm c0}(\mathbf{r},t)\cdot\boldsymbol{\Omega}\times\mathbf{r} = \tilde{\mu}_0'\,, \tag{9.75}$$

or alternatively,

$$\mu_{\rm c0}(\mathbf{r},t) + \tfrac{1}{2}m[\mathbf{v}_{\rm c0}(\mathbf{r},t) - \mathbf{v}_{\rm rb}(\mathbf{r})]^2 = \tilde{\mu}_{\rm loc}(\mathbf{r})\,. \tag{9.76}$$

This is the analogue of the condition for complete local equilibrium given in (8.8). In the present case, the lab frame thermal cloud variables $\tilde{\mu}_{\rm loc}$ and $\mathbf{v}_{\rm rb}$ are time independent, as they correspond to a steady rigid-body rotation of the thermal cloud. The condensate variables, however, are in general space *and* time dependent as a result of the transformation from the rotating frame. The centrifugal effect of rotation leads to a local chemical potential $\tilde{\mu}_{\rm loc}(\mathbf{r})$ that is spatially dependent, but the thermal cloud chemical potential $\tilde{\mu}_0'$ in (9.75) characterizing the equilibrium state in the rotating frame is a true constant, independent of position and time.

9.6 Stationary vortex states at zero temperature

As background for our discussion of vortices at finite temperatures, we will briefly review here some properties of vortex states in the $T = 0$ limit, where the system is described by the usual GP equation without the $2g\tilde{n}_0'$ term in (9.65). A more complete discussion can be found in the review by Fetter and Svidzinsky (2001) and the monograph by Aftalion (2006).

Whether vortices are stable is a matter of energetics. In the $T = 0$ limit, the energy of the system in the rotating frame is given by

$$E' = E - \Omega\langle L_z\rangle\,, \tag{9.77}$$

[9] Alternatively, the term $-\frac{1}{2}m\Omega^2\rho^2$ can be combined with $U_0(\mathbf{r},t)$, where it would be called a centrifugal potential.

where[10]

$$E = \int d\mathbf{r}\, \Phi_0'^*(\mathbf{r}) \left[-\frac{\hbar^2 \nabla^2}{2m} + V'_{\rm trap}(\mathbf{r}) + \frac{g}{2} |\Phi_0'(\mathbf{r})|^2 \right] \Phi_0'(\mathbf{r}) \tag{9.78}$$

and

$$\langle L_z \rangle = \int d\mathbf{r}\, \Phi_0'^*(\mathbf{r}) L_z \Phi_0'(\mathbf{r})\,. \tag{9.79}$$

These equations are valid for an arbitrary anisotropic potential $V'_{\rm trap}$. Because of the L_z operator, the solutions of (9.65) in general have a phase (i.e. are complex), implying that the state carries a superfluid current. Such currents arise whenever the system is confined by a rotating anisotropic potential, even if no vortices are present. Fetter (1974) gives some examples of such superfluid states in confined liquid helium; similar states in a trapped Bose gas are considered by Recati *et al.* (2001).

The situation is simpler in the axisymmetric trapping potential limit ($\epsilon \to 0$ in (9.32)). In this case, the GP equation admits solutions that are simultaneous eigenfunctions of L_z and the GP Hamiltonian. These are the simplest eigenfunctions but by no means the most general. Owing to the nonlinearity of the GP equation, there are stationary states (in the rotating frame) that break the rotational symmetry of the confining potential, both with (Lundh, 2002) and without (Recati *et al.*, 2001) vortices.

For a trap with cylindrical symmetry, an axial vortex state will have the form

$$\Phi'_q(\mathbf{r}) = f(\rho, z) e^{iq\phi}\,, \tag{9.80}$$

where $\rho = \sqrt{x^2 + y^2}$, ϕ is the azimuthal angle and q is the "charge" of the vortex. The function $f(\rho, z)$ vanishes at $\rho = 0$ and thus the density goes to zero on the symmetry axis, defining the core of the vortex. The core has a radius given approximately by the healing length

$$\xi = \frac{1}{\sqrt{8\pi n a}}\,, \tag{9.81}$$

where n is the local density in the absence of the vortex and a is the s-wave scattering length. Since the density decreases as one approaches the edge of the condensate along the symmetry axis, the size of the vortex core increases.

A singly charged vortex ($q = 1$) is the most stable and we will confine our

[10] We drop the prime on the spatial variable for convenience, but retain the prime on the wavefunction to remind the reader that Φ'_0 is the solution of the GP equation in the rotating frame.

attention to this case. This axisymmetric state has a condensate velocity

$$\mathbf{v}_{\rm c0} = \frac{\hbar}{m\rho}\hat{\boldsymbol{\phi}}\,, \tag{9.82}$$

and a quantized circulation

$$\kappa \equiv \oint \mathbf{v}_{\rm c0}\cdot d\mathbf{l} = \frac{h}{m}\,, \tag{9.83}$$

where the closed contour encloses the axial vortex. For this state, $\langle L_z\rangle = N\hbar$ and the energy is $E_1' = E_1 - N\hbar\Omega$. (We are assumimg that $f(\rho,z)$ in (9.80) is the solution that yields the *lowest* possible energy E_1'.) In the absence of a vortex, we have $\langle L_z\rangle = 0$ and hence $E_0' = E_0$. The change in energy is thus

$$\Delta E_{10}' = \Delta E_{10} - N\hbar\Omega\,, \tag{9.84}$$

where $\Delta E_{10} = E_1 - E_0$; $\Delta E_{10}'$ becomes negative for Ω greater than some critical angular velocity $\Omega_{\rm c} \equiv \Delta E_{10}/N\hbar$. The single vortex state is *thermodynamically stable* in the rotating frame above this value. In the TF limit, the critical angular velocity is approximately given by (Lundh *et al.*, 1997)

$$\Omega_{\rm c} \simeq \frac{5}{2}\frac{\hbar^2}{mR_\perp^2}\ln\left(\frac{0.67R_\perp}{\xi}\right)\,, \tag{9.85}$$

where $R_\perp$ is the TF radius of the condensate at $z = 0$.

At this point, one might wonder why a discussion in terms of rotating frames is even relevant when the trapping potential is axisymmetric ($\epsilon = 0$). It is always possible to observe the system from a noninertial frame of reference, but why would one want to do so if the trapping potential is already time independent in the lab frame? And if this is the case, what does it mean to say that the vortex state is thermodynamically stable?

The answers to these questions require a reinterpretation of (9.77). One can pose the following question: what is the minimum energy of the system if it is constrained to have a certain amount of angular momentum $\langle L_z\rangle$? The mathematical formulation of this problem would be to minimize the energy functional as defined in (9.77), with Ω interpreted as a Lagrange parameter. If $\langle L_z\rangle$ is allowed to take on arbitrary values, then the problem is more difficult, and certainly more general, than that considered in the derivation of (9.84). To conform to this latter case, we can restrict the minimization by using only vortex-free states with $\langle L_z\rangle = 0$ and single-vortex states with $\langle L_z\rangle = N\hbar$. We then find that the vortex-free state is favoured up to a certain critical value $\Omega_{\rm c}$ and the single-vortex state above that value. From this perspective, the interpretation of Ω as an "angular

velocity" is not necessary. However, since the mathematical (and physical) conclusions of the constrained minimization are the same in the end, we can consider Ω_c as a critical angular velocity even when the trap potential is strictly axisymmetric. If one prefers, one can always revert to the rotating reference frame picture by simply imagining a rotating potential with an arbitrarily small, but finite, anisotropy parameter ϵ.

With increasing Ω, more complex vortex configurations arise. The number of vortices increases with Ω and the vortices eventually arrange themselves in a triangular vortex lattice (Aftalion, 2006). Even if $\epsilon = 0$, the rotational symmetry of the condensate is lost and the rotating frame picture is needed. These configurations are stationary solutions in the rotating frame and as a result they rotate at angular velocity Ω in the lab frame.

9.7 Equilibrium vortex state at finite temperatures

The properties of vortices at finite temperatures have been studied to a much lesser extent. Some calculations have been done (Isoshima and Machida, 1999; Mizushima *et al.*, 2001; Virtanen *et al.*, 2001) to determine the equilibrium structure of the condensate and the thermal cloud. In these studies, the thermal cloud is described by the Hartree–Fock–Bogoliubov–Popov (HFBP) excitation spectrum. These excitations are also collective modes of the condensate but, as discussed in Chapters 5 and 6, the full dynamics of the condensate and thermal cloud are not taken into account in a consistent manner.[11] Nevertheless, the theory should provide a reasonably accurate description of the equilibrium properties.

We consider a single axial vortex, for which $\mathbf{v}_{c0}$ is given by (9.82) in an axisymmetric trap. In the ZNG theory, the thermal gas is described by the lab frame distribution (see (9.73))

$$f_0(\mathbf{p},\mathbf{r}) = \frac{1}{\exp\beta[(\mathbf{p}-m\mathbf{v}_{\rm rb})^2/2m + U_0(\mathbf{r}) + V_{\rm cent}(\mathbf{r}) - \tilde{\mu}_0'] - 1}\,. \qquad (9.86)$$

Note that we are still thinking of the thermal gas as being in equilibrium in

[11] Although the core density is small, it has a profound effect on the lowest, $m = -1$, mode. In the Bogoliubov (or linearized GP) theory, the frequency of this mode is *negative* (Dodd *et al.*, 1997) corresponding to a *counterclockwise* motion of the associated density fluctuation ($e^{i(m\phi-\omega_m t)} = e^{-i(\phi-|\omega_{-1}|t)}$). As explained by Fetter and Svidzinsky (2001), the mode corresponds physically to the precessional motion of an off-centre vortex. However, in the HFBP approximation, this mode has a *positive* frequency (Isoshima and Machida, 1999; Mizushima *et al.*, 2001; Virtanen *et al.*, 2001), corresponding to a *clockwise* precession of the vortex. This is an artifact of the HFBP approximation and is analogous to the violation of the Kohn theorem for the centre-of-mass dipole mode. This behaviour is a result of the *static* noncondensate density in the core region, which acts as a pinning potential (Isoshima and Machida, 1999). Including the dynamics of the noncondensate fully would restore the counterclockwise precessional motion of an off-centre vortex, as observed experimentally (Anderson *et al.*, 2000).

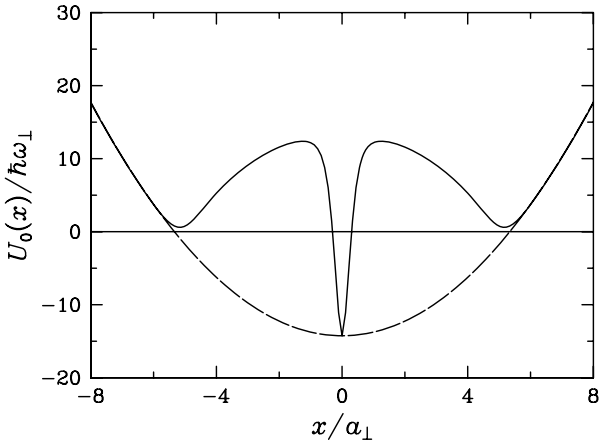


Fig. 9.2. The thermal cloud effective potential, $U_0 = V_{\rm trap} + 2gn_{\rm c0}$, along the x-axis for a condensate with an axial vortex. The bare harmonic trapping potential is the broken line. The potentials are plotted relative to the chemical potential $\mu_{\rm c0}$. The unit of energy is $\hbar\omega_\perp$ and the unit of length is $a_\perp = \sqrt{\hbar/m\omega_\perp}$. The trap parameters are $\omega_\perp/2\pi = 200$ Hz, $\lambda = 0.16$, $N = 10^6$ and $T = 200$ nK. The condensate fraction is $N_{\rm c}/N = 0.87$.

the rotating frame, so that the thermal cloud in the lab frame is undergoing *rigid-body rotation*. For the single-particle distribution function in (9.86) to be physically meaningful, we require $U_0 + V_{\rm cent} > \tilde{\mu}_0'$. As will be discussed in Section 11.4, this condition can be violated because of the semiclassical approximation used for the thermal component. The situation with a vortex present is even more problematic.

In Fig. 9.2 we show the potential $U_0 = V_{\rm trap} + 2gn_{\rm c0}$ obtained from a self-consistent calculation of the condensate and thermal cloud density profiles in the $\Omega \to 0$ limit. This potential acts as the external potential for the thermal cloud. The dip below the condensate chemical potential is due to the fact that the density of the condensate with a vortex goes to zero on the z-axis. It is clear that the semiclassical Bose distribution in (9.86) makes no sense in this region, where the local fugacity becomes larger than unity. Therefore, to perform a self-consistent calculation one must restrict the occupation of thermal states to energies above the chemical potential. With this prescription, the fugacity is given by $z_0 = 1$ in the core region and the thermal cloud density has a limiting value $\Lambda_0^{-3} g_{3/2}(z_0 = 1)$. The resulting self-consistent density profiles are plotted in Fig. 9.3. These results are in close agreement with HFBP calculations (Virtanen *et al.*, 2001), except for the magnitude of the thermal cloud density in the core region, where the

density is underestimated by a factor of $2 \sim 3$. However, this is not a serious error since the fraction of thermal atoms in the core region is very small[11].

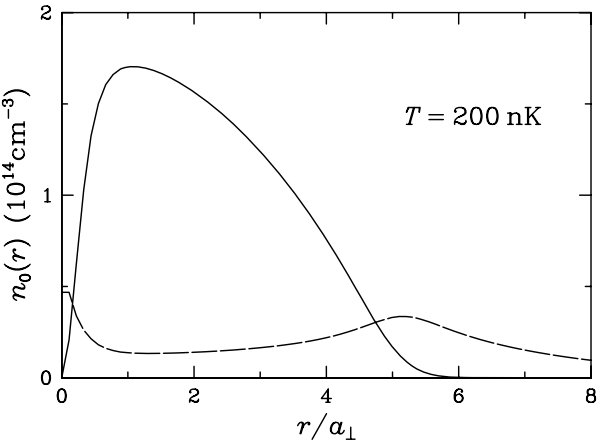


Fig. 9.3. The condensate (solid line) and thermal cloud (broken line) densities as functions of the radial distance from the axis of the vortex at $z = 0$. For clarity, the thermal cloud density has been multiplied by 10. The conditions are the same as in Fig. 9.2.

Determining the stability of the vortex state at finite temperatures requires a calculation of the *total* energy of the system. In addition to the condensate energy E' given by (9.77)–(9.79), we have to calculate the thermal cloud energy

$$\tilde{E}' = \tilde{E} - \Omega \langle \tilde{L}_z \rangle \,, \tag{9.87}$$

where one has

$$\tilde{E} = \int \frac{d\mathbf{r}d\mathbf{p}}{(2\pi\hbar)^3} \left\{ \frac{p^2}{2m} + V_{\mathrm{trap}}(\mathbf{r}) + g\left[\tilde{n}_0(\mathbf{r}) + 2n_{\mathrm{c0}}(\mathbf{r})\right] \right\} f_0(\mathbf{p}, \mathbf{r}) \tag{9.88}$$

and

$$\langle \tilde{L}_z \rangle \equiv \int \frac{d\mathbf{r}d\mathbf{p}}{(2\pi\hbar)^3} (\mathbf{r} \times \mathbf{p})_z f_0(\mathbf{p}, \mathbf{r}) \,. \tag{9.89}$$

The full mean-field interaction energy of the thermal cloud with the condensate is included in $\tilde{E}$. The interaction of the thermal cloud with itself is reduced by a factor 2 relative to the condensate term, in order to avoid double counting this self-interaction energy.

Since the thermal cloud is undergoing rigid-body rotation, its angular

momentum can be expressed as

$$\langle \tilde{L}_z \rangle = \int d\mathbf{r}\, (\mathbf{r} \times m\mathbf{v}_{\rm rb})_z \tilde{n}_0(\mathbf{r}) = \tilde{I}_z \Omega\,, \tag{9.90}$$

where the thermal cloud moment of inertia is

$$\tilde{I}_z = m \int d\mathbf{r}\, \rho^2 \tilde{n}_0(\mathbf{r})\,. \tag{9.91}$$

We note that the thermal atom distribution (9.86) is valid with or without a vortex present in the condensate. What distinguishes the two situations is the form of U_0, which includes the condensate mean field $2gn_{\rm c0}$, and the value of the thermal cloud chemical potential $\tilde{\mu}_0'$. This implies that it is possible for the thermal cloud to have a nonzero angular momentum even when the condensate does not have a vortex ($q = 0$ in (9.80)). Conversely, it is also possible for the condensate to contain a vortex even in the $\Omega \to 0$ limit. In this case, the condensate has angular momentum but, since $\mathbf{v}_{\rm rb} \equiv \boldsymbol{\Omega} \times \mathbf{r} = 0$, the thermal cloud is nonrotating. One can also imagine the thermal cloud rigidly rotating at a rate Ω' different from Ω. In this case, the exponent in the square bracket in (9.70) would be

$$\varepsilon_{\rm c0}' - m\mathbf{v}_{\rm c0}' \cdot \boldsymbol{\Omega'} \times \mathbf{r}' - \tilde{\mu}_0' = \overline{\varepsilon}_{\rm c0}' + m\mathbf{v}_{\rm c0}' \cdot (\boldsymbol{\Omega} - \boldsymbol{\Omega'}) \times \mathbf{r}' - \tilde{\mu}_0'\,, \tag{9.92}$$

where $\overline{\varepsilon}_{\rm c0}'$ is defined in (9.71). As discussed earlier, the requirement for absolute equilibrium between the condensate and thermal cloud is that $C_{12}'[f_0', \Phi_0']$ vanishes, which in turn requires that the expression in (9.92) vanishes. This condition can be satisfied only for the special case of an axial vortex, whose velocity is given by (9.82). Assuming that $\boldsymbol{\Omega}$ and $\boldsymbol{\Omega'}$ both point in the z-direction, we have $(\boldsymbol{\Omega} - \boldsymbol{\Omega'}) \times \mathbf{r}' = (\Omega - \Omega')\rho' \hat{\boldsymbol{\phi}}$, and thus the expression in (9.92) vanishes if $\tilde{\mu}_0'$ is equal to $\overline{\varepsilon}_{\rm c0}' + \hbar(\Omega - \Omega')$.

9.8 Nonequilibrium vortex states

The creation of vortices in trapped Bose gases by means of stirring is a complex and highly nonequilibrium process. In fact, there initially was some scepticism about whether the method would work; it was feared that stirring the condensate would destroy BEC before vortices could actually appear. These fears were put to rest with the experimental realization of vortex states by stirring (Matthews *et al.*, 1999; Madison *et al.*, 2000). It was found that the condensate was quite robust even when stirred vigorously enough to impart a large amount of angular momentum to it (Abo-Shaeer *et al.*, 2001; Coddington *et al.*, 2004).

The "stirring" is usually provided by a rotating optical potential, although

a purely magnetic potential can also be used (Hodby, 2002). Two routes to the vortex state have been followed successfully. In the first, a stirrer is applied only after the system has been prepared in a low-temperature condensed phase. The act of stirring excites the system to a final state which, after a period of equilibration, contains a condensate with vortices in equilibrium with a thermal component. Despite the delicate nature of the experiments, this is a "brute-force" method: the stirrer simply "torques" the system and nature fortunately complies by allowing the system to evolve to the final state of interest.

In the other approach (Haljan *et al.*, 2001), a nondegenerate Bose gas is first spun up to a state with a large amount of angular momentum. After the stirrer is turned off, this angular momentum is preserved and the gas eventually equilibrates to a distribution given by (9.69) (or (9.86) in the lab frame). Since the mean-field interaction in the thermal cloud is very weak, the cloud essentially finds itself in the *bare* trapping potential augmented by the centrifugal potential $-m\Omega^2\rho^2/2$. Evaporative cooling is then used to cool the gas below $T_{\rm BEC}$. This is done in a clever way that selectively removes atoms that are close to the rotation axis and thus has little effect on the total angular momentum of the cloud. Therefore, even though energy (carried away by the atoms) is removed from the thermal cloud, a significant amount of angular momentum is still retained by the system.

As the cooling progresses, the gas eventually begins to form a Bose condensate. Within the ZNG theory, the generalized GP equation can be used to show that the rate of condensate growth is given by

$$\frac{dN_{\rm c}(t)}{dt} = -\int d\mathbf{r}\Gamma_{12}(\mathbf{r},t) = -\frac{2}{\hbar}\int d\mathbf{r}\, n_{\rm c}(\mathbf{r},t)R(\mathbf{r},t)\,, \qquad (9.93)$$

where $N_{\rm c}(t)$ is the number of condensate atoms and R is the particle-exchange term in the generalized GP equation associated with C_{12} collisions. Since the Bose condensate forms in the presence of a rotating thermal cloud, there is the possibility that the thermal cloud will impart a certain amount of angular momentum to the condensate, allowing vortices to form. In fact, it is found experimentally (Haljan *et al.*, 2001) that at the end of the cooling process the condensate equilibrates into a state with a well-defined vortex lattice.

The route to this final equilibrium vortex state is very complex. In principle it could be described by simulated solutions of the coupled ZNG equations, of the kind to be discussed in Chapter 12. However, this would be a very demanding calculation and has yet to be performed. A less ambitious approach is to make use of the kind of phenomenological equation

for the condensate wavefunction discussed in Section 8.2, based on treating the thermal gas as being always in static thermal equilibrium. Calculations along these lines have been made by several groups and provide considerable insight into the nucleation of vortices and the formation of a vortex lattice as the final equilibrium state. Before reviewing these calculations, however, it is useful to consider first a simplified semi-analytic model that provides some qualitative understanding of the vortex-nucleation process. The present discussion is based on Williams *et al.* (2002).

9.8.1 Dynamic instability due to a rotating thermal cloud

We assume that the system arrives at a "quasi-equilibrium" state after forming a Bose condensate in which the condensate is nonrotating but finds itself in the presence of a rigidly rotating thermal cloud.[12] In this situation, the lab frame distribution function of the thermal cloud is given by (9.86), with the effective trapping potential

$$\begin{aligned} U_{\rm eff}(\mathbf{r}) &\equiv U_0(\mathbf{r}) + V_{\rm cent}(\mathbf{r}) \\ &= \tfrac{1}{2}m(\omega_\perp^2 - \Omega^2)\rho^2 + \tfrac{1}{2}m\omega_z^2 z^2 + 2g[n_{\rm c0}(\mathbf{r}) + \tilde{n}_0(\mathbf{r})]\,. \end{aligned} \tag{9.94}$$

The condensate density profile is $n_{\rm c0}(\mathbf{r}) = |\Phi_0(\mathbf{r})|^2$, where the condensate wavefunction Φ_0 is the solution of (3.45) or equivalently of (9.65) with $\Omega = 0$. The density profiles of the two components are determined self-consistently with the equilibrium constraint $\mu_{\rm c0} = \tilde{\mu}_0'$ (see (9.75) with $\mathbf{v}_{\rm c0} = 0$).

Although the condensate and thermal cloud are in mechanical equilibrium, this "quasi-equilibrium" state may not be dynamically stable. To establish whether it is, we must consider small-amplitude dynamical perturbations away from the stationary state. If a collective excitation exists whose amplitude grows with time, then the state is *dynamically unstable*. Such instabilities are an indication that the condensate is able to make a transition to another, more stable, structure.

To study the properties of small-amplitude perturbations from the "quasi-equilibrium" state, we make use of the *static thermal cloud approximation* described at length in Chapter 8. In this approximation, the distribution function of the rigidly rotating thermal cloud is given by (9.86). The condensate dynamics is described by the quantum hydrodynamic equations (8.13) and (8.14). The source term $\Gamma_{12}^0[\Phi] = 2n_{\rm c}R_0/\hbar$ (see the r.h.s. of (8.13)) is

[12] This is only one possible scenario. Vortices may also appear in the condensate even as it is being formed.

given in the present situation by

$$\Gamma_{12}[f^0,\Phi] = \frac{n_{\rm c}(\mathbf{r},t)}{\tau_{12}^c(\mathbf{r},t)}\left(e^{\beta[\varepsilon_{\rm c}(\mathbf{r},t)-m\mathbf{v}_{\rm c}(\mathbf{r},t)\cdot\mathbf{v}_{\rm rb}(\mathbf{r})-\tilde{\mu}_0']}-1\right). \tag{9.95}$$

where $\varepsilon_{\rm c}$ is found from (3.12) and τ_{12}^c from (8.12). The result (9.95) is obtained from the particle-exchange source term in (8.7) by noting that (9.86) has the same form as the local equilibrium distribution (8.3) with the replacements $\tilde{\mu}\to\tilde{\mu}_0'-V_{\rm cent}(\mathbf{r})$, $\mathbf{v}_{\rm n}\to\mathbf{v}_{\rm rb}$ and $U\to U_0$. Making these substitutions in (8.7) and using $V_{\rm cent}=-mv_{\rm rb}^2/2$, one obtains (9.95).

Linearizing (8.13) and (8.14) about the stationary solution, one obtains the equations (Williams *et al.*, 2002)

$$\frac{\partial\delta n_{\rm c}}{\partial t} = -\nabla\cdot(n_{\rm c0}\delta\mathbf{v}_{\rm c}) - \delta\Gamma_{12}^0\,, \tag{9.96}$$

$$\frac{\partial\delta\mathbf{v}_{\rm c}}{\partial t} = -\frac{g}{m}\nabla\delta n_{\rm c}\,, \tag{9.97}$$

$$\delta\Gamma_{12}^0 = \frac{1}{\tau'}\left(\delta n_{\rm c}-\frac{m}{g}\mathbf{v}_{\rm rb}\cdot\delta\mathbf{v}_{\rm c}\right), \tag{9.98}$$

with the C_{12} relaxation time τ' defined by (8.17) and (8.22).

Inserting (9.98) into (9.96), we find two coupled equations for the fluctuations $\delta n_{\rm c}$ and $\delta\mathbf{v}_{\rm c}$. Eliminating the variable $\delta\mathbf{v}_{\rm c}$ from these equations gives a single equation

$$\frac{\partial^2\delta n_{\rm c}}{\partial t^2}-\frac{g}{m}\nabla\cdot(n_{\rm c0}\nabla\delta n_{\rm c}) = -\frac{1}{\tau'}\left(\frac{\partial\delta n_{\rm c}}{\partial t}+\Omega\frac{\partial\delta n_{\rm c}}{\partial\phi}\right). \tag{9.99}$$

This differs from (8.23) by the second term on the r.h.s., associated with the rigid rotation of the thermal cloud. Following the method of solution discussed in Section 8.1, the r.h.s. of (9.99) can be treated as a perturbation and so $\delta n_{\rm c}$ is expanded in terms of the Stringari mode eigenfunctions defined by (8.24). For the axisymmetric trap under consideration, these eigenfunctions are proportional to $e^{im\phi}$.[13] To lowest order in the perturbation, the mode frequency is given by $\Omega_m=\omega_m-i\Gamma_m$, where ω_m is the $T=0$ mode frequency in the absence of coupling to a thermal cloud, with damping

$$\Gamma_m = \frac{\gamma_{mm}}{2}\left(1-\frac{m\Omega}{\omega_m}\right). \tag{9.100}$$

Here γ_{mm} refers to the matrix element of $1/\tau'$ defined in (8.26).

[13] The azimuthal index is denoted by the same symbol, m, as the atomic mass, but the distinction should be clear from the context.

The damping of the condensate excitation given by (9.100) has the interesting feature that it depends on the mode's polarity. The density fluctuation of the condensate mode is proportional to $\cos(m\phi - \omega_m t)$. Thus, on the one hand, a mode with $m < 0$ has a density fluctuation that rotates in a clockwise sense, which is opposite to the sense of rotation of the thermal cloud. For these modes, the damping *increases* with Ω. On the other hand, for modes rotating in the same direction as the thermal cloud ($m > 0$), the damping *decreases*. In fact, the damping changes sign for $\Omega > \omega_m/m$, indicating a dynamical instability. The earliest onset of an instability corresponds to the critical angular velocity given by

$$\Omega_{\rm cr} \equiv \left(\frac{\omega_m}{m}\right)_{\rm min} . \tag{9.101}$$

Depending on the trap anisotropy λ, the critical mode index $m_{\rm cr}$ is found to lie approximately in the range 5–12 (Dalfovo and Stringari, 2001). For $\Omega > \Omega_{\rm cr}$, the collective mode with $m = m_{\rm cr}$ will grow the most rapidly and will dominate the subsequent time evolution. Although the preceding linear stability analysis cannot predict what the final state might be, Williams *et al.* (2002) argued that such a surface mode instability is a precursor to vortex nucleation. Numerical calculations described in the following section, which take into account the nonlinearities in the condensate dynamics, confirm this idea.

The criterion for a dynamical instability given by (9.101) coincides with the Landau criterion for the excitation of surface collective oscillations, as discussed in detail by Dalfovo and Stringari (2001). The energy of an excitation in the rotating frame is found to be

$$\varepsilon'_m = \hbar\omega_m - m\hbar\Omega \,, \tag{9.102}$$

and this excitation energy becomes *negative* when $\Omega > \omega_m/m$. This is clearly a generalization, to the case of rotations, of the usual Landau criterion (Landau, 1941) for the stability of a superfluid undergoing uniform flow. The energy of a phonon excitation in the frame of reference of the superfluid is given by (Pitaevskii and Stringari, 2003)

$$\varepsilon'_q = \varepsilon_q - \hbar\mathbf{q}\cdot\mathbf{v}_{\rm c} \,, \tag{9.103}$$

where $\varepsilon_q = \hbar c q$ and $\mathbf{v}_{\rm c}$ is the superfluid velocity in the lab frame. When $\varepsilon'_q < 0$, phonon excitations can be created spontaneously, resulting in the breakdown of superfluidity. A negative excitation energy is sometimes referred to as an *energetic instability* and is in general distinct from a dynamical instability. However, in the present example, the two instabilities

are seen to coincide. A similar result was found by Konabe and Nikuni (2006, 2008) for the stability of superfluid flow in an optical lattice at finite temperatures.

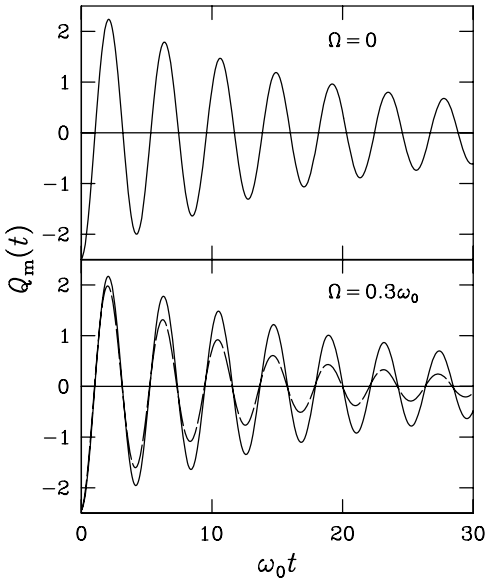


Fig. 9.4. Numerical simulation of quadrupole modes wthin the static thermal cloud approximation for a condensate in the presence of a rotating thermal cloud. The upper panel shows the quadrupole moment in the absence of rotation; the lower panel shows the behaviour for a rotating cloud, for the $m = 2$ (solid line) and $m = -2$ (broken line) modes. The system paramters are $N = 10^5$ ($N_c \simeq 5 \times 10^4$), $T = 250$ nK, $\omega_0/2\pi = 187$ Hz.

As an illustration of the damping result in (9.100) we consider the case of the quadrupole mode ($m = 2$) which exhibits a dynamical instability at $\Omega_{m=2} = \sqrt{2}\omega_\perp/2 = \omega_\perp/\sqrt{2}$. Simulations of the kind to be described in Chapter 12 were performed within the static thermal cloud approximation. Figure 9.4 shows the time dependence of the quadrupole moment initially excited in a gas containing $N \simeq 1.0 \times 10^5$ ^{87}Rb atoms at a temperature of 250 nK in an isotropic trap. The upper panel is for $\Omega = 0$, while the lower panel shows the results for $\Omega = 0.3\omega_\perp$. For the latter case, one can see that the counter-rotating mode has a larger damping than the co-rotating mode, as (9.100) predicts.

9.8.2 Phenomenological relaxation in a rotating frame

We will now develop a phenomenological equation for the rotating frame at finite temperatures, extending the discussion in Section 8.2. Using (9.95) in the rotating frame and the relation $R'_0 = \hbar\Gamma'_{12}/2n'_{\rm c}$, we have

$$R'_0(\mathbf{r}',t) = \frac{\hbar}{2\tau'^{c}_{12}(\mathbf{r}',t)}\left(e^{\beta[\varepsilon'_{\rm c}(\mathbf{r}',t)-m\mathbf{v}'_{\rm c}(\mathbf{r}',t)\cdot\mathbf{\Omega}\times\mathbf{r}'-\tilde{\mu}'_0]}-1\right). \tag{9.104}$$

To lowest order in deviations from equilibrium, (9.104) reduces to

$$R'_0(\mathbf{r}',t) \simeq \frac{\hbar\beta}{2\tau^{c0}_{12}(\mathbf{r}')}(\varepsilon'_{\rm c}-\mathbf{\Omega}\cdot\mathbf{r}'\times m\mathbf{v}'_{\rm c}-\tilde{\mu}'_0), \tag{9.105}$$

where the intercomponent collision time $\tau^{c0}_{12}(\mathbf{r}')$ is defined by (8.17). Following the analysis of Section 8.2, and using the approximation $i\hbar\partial\Phi'/\partial t \simeq (\varepsilon'_{\rm c}-\mathbf{\Omega}\cdot\mathbf{r}'\times m\mathbf{v}'_{\rm c})\Phi'$, we arrive at the phenomenological GP equation

$$i\hbar\frac{\partial\Phi'(\mathbf{r}',t)}{\partial t} = \left[\mathcal{H}'_{\rm c}(t)-\Omega L'_z\right]\Phi'(\mathbf{r}',t)+i\gamma'(\mathbf{r}')\left(\tilde{\mu}'_0-i\hbar\frac{\partial}{\partial t}\right)\Phi'(\mathbf{r}',t), \tag{9.106}$$

where we define

$$\mathcal{H}'_{\rm c}(t) \equiv -\frac{\hbar^2\nabla'^2}{2m}+V'_{\rm trap}(\mathbf{r}')+gn'_c(\mathbf{r}',t)+2g\tilde{n}'_0(\mathbf{r}'), \tag{9.107}$$

and (see (8.34))

$$\gamma'(\mathbf{r}') \equiv \frac{\hbar\beta}{2\tau^{c0}_{12}(\mathbf{r}')}. \tag{9.108}$$

The result in (9.106) is an obvious generalization of (8.32) to the rotating frame and agrees with the equation obtained by Gardiner *et al.* (2002).

One can easily check that (9.106) admits a stationary solution given by $\Phi'(\mathbf{r}',t) = \Phi'_0(\mathbf{r}')e^{-i\bar{\varepsilon}'_{\rm c0}t/\hbar}$, where $(\mathcal{H}'_{\rm c0}-\Omega L'_z)\Phi'_0 = \bar{\varepsilon}'_{\rm c0}\Phi'_0$ and $\bar{\varepsilon}'_{\rm c0} = \tilde{\mu}'_0$. The state of the condensate described by Φ'_0 need not be an eigenstate of the angular momentum operator L'_z and in general will exhibit an array of vortices whose number will depend on Ω. As discussed in Section 8.2, one can show that an initial time-dependent solution of (9.106) will relax towards the stationary state solution $\Phi'_0(\mathbf{r}')$.

With the transformation $\Phi'(\mathbf{r}',t) = e^{-i\tilde{\mu}'_0t/\hbar}\,\overline{\Phi}'(\mathbf{r}',t)$, (9.106) reduces to

$$i\hbar(1+i\gamma')\frac{\partial\overline{\Phi}'(\mathbf{r}',t)}{\partial t} = \left[\mathcal{H}'_{\rm c}(t)-\tilde{\mu}'_0-\Omega L'_z\right]\overline{\Phi}'(\mathbf{r}',t), \tag{9.109}$$

which is the generalization of (8.36) to a rotating frame. Taking γ' as a phenomenological constant, (9.109) is the equation solved by Tsubota *et al.*

(2002) and Kasamatsu *et al.* (2003, 2005) in their simulations of vortex nucleation and vortex-lattice formation. These authors considered a trapping potential with a weak anisotropy and began with the ground state obtained with $\Omega = 0$. At $t = 0$ the rotation is switched on, which amounts to introducing a finite value Ω into (9.109). The initial state is no longer an eigenstate of the Hamiltonian when the term $-\Omega L'_z$ is included, and the state begins to evolve in time. As discussed in Section 8.2, the wavefunction $\overline{\Phi}'$ will relax towards an eigenstate of $\mathcal{H}'_{c0} - \Omega L'_z$ with eigenvalue $\bar{\varepsilon}'_{c0} = \tilde{\mu}'_0$, which fixes the normalization of the condensate wavefunction. In the simulations carried out by Tsubota *et al.* (2002), $\tilde{\mu}'_0$ was adjusted during the course of the evolution in order to keep the atom number constant.

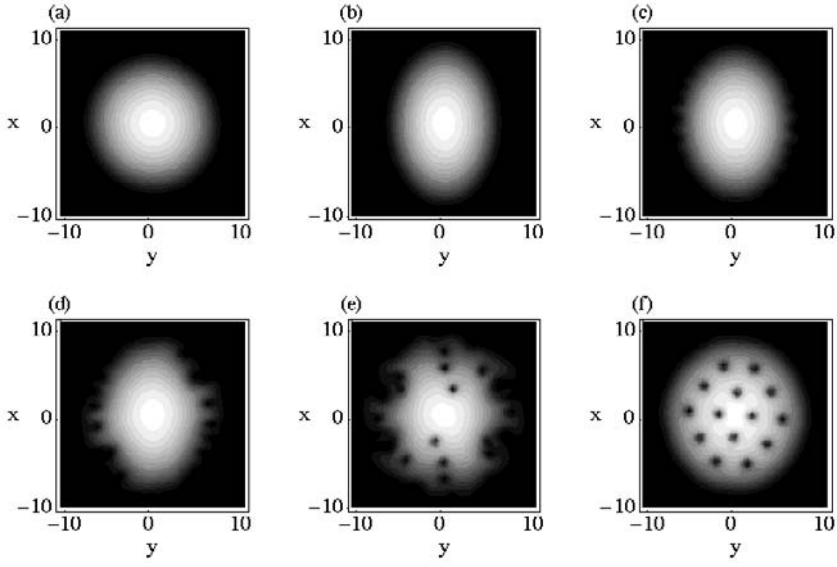


Fig. 9.5. Time development of the condensate density $|\Phi'(\mathbf{r}', t)|^2$ after the rotation of an anisotropic trap is switched on at $t = 0$. The rotation rate is $\Omega = 0.7\omega_\perp$, where $\omega_\perp/2\pi = 219$ Hz. The times are (a) $t = 0$ ms, (b) 21 ms, (c) 107 ms, (d) 114 ms, (e) 123 ms and (f) 262 ms. Lengths are measured in units of the harmonic oscillator length $a_{ho} = \sqrt{\hbar/m\omega_\perp} = 0.724$ μm (from Tsubota *et al.*, 2002).

Figure 9.5 shows the results of two-dimensional (2D) simulations of (9.109) performed by Tsubota *et al.* (2002). Frame (a) is the original equilibrium density distribution, which shows a small elongation in the x direction due to the trap anisotropy. After about five trap periods, the cloud exhibits a pronounced elliptical elongation, as shown in frame (b). The aspect ratio

R_x/R_y is actually an oscillatory function of time, oscillating about a steady-state value R_x^0/R_y^0 that corresponds to a stationary state in which the condensate is strongly distorted, but is undergoing irrotational flow (Recati *et al.*, 2001). The state in (b) is not dynamically stable, and in frame (c) we begin to see the growth of surface excitations. This dynamical instability is distinct from that considered in subsection 9.8.1; as shown by Recati *et al.* (2001) and Sinha and Castin (2001), it is a property of the GP equation in the rotating frame (without dissipation, i.e. $\gamma' = 0$) that arises for sufficiently large Ω. As the amplitude of the condensate excitations increases in time, vortices begin to penetrate the surface of the condensate (d), become incorporated in the condensate (e) and then equilibrate into a vortex lattice (f). These simulations provide a reasonably good account of experiments in which a vortex lattice is produced by means of a rotating anisotropic potential (Madison *et al.*, 2000).

Penckwitt *et al.* (2002) also performed 2D simulations and found very similar behaviour to Tsubota *et al.* (2002). They also performed simulations that correspond more closely to the experiments of Haljan *et al.* (2001), where one starts with a rotating thermal cloud above $T_{\rm BEC}$ and then evaporatively cools to form a condensate with a vortex lattice. This is the situation already considered in subsection 9.8.1, but the simulations include the full *nonlinear* dynamics of the condensate. Penckwitt *et al.* (2002) also considered a slightly more general situation, in which the thermal cloud rotates rigidly about the same axis as the condensate but at a different rate Ω'. In this case, (9.105) becomes

$$R_0'(\mathbf{r}',t) \simeq -\frac{\hbar\beta}{2\tau_{12}^{c0}(\mathbf{r}')}\left(\varepsilon_{\rm c}' - \mathbf{\Omega}'\cdot\mathbf{r}'\times m\mathbf{v}_{\rm c}' - \tilde{\mu}_0'\right) . \tag{9.110}$$

With the further approximation $(\mathbf{r}'\times m\mathbf{v}_{\rm c}')_z\Phi' \simeq L_z'\Phi'$, we then obtain

$$\begin{aligned} i\hbar\frac{\partial\Phi'(\mathbf{r}',t)}{\partial t} &= \left[\mathcal{H}_{\rm c}'(t) - \Omega L_z'\right]\Phi'(\mathbf{r}',t) \\ &\quad + i\gamma'(\mathbf{r}')\left[\tilde{\mu}_0' + (\Omega'-\Omega)L_z' - i\hbar\frac{\partial}{\partial t}\right]\Phi'(\mathbf{r}',t) . \end{aligned} \tag{9.111}$$

For a stationary state, the term multiplying γ' must vanish, which requires Φ' to be an eigenstate of L_z'. Bearing in mind the approximation made below (9.110), this condition is equivalent to that established earlier in (9.92). Thus the possible stationary states allowed by (9.111), with Ω' different from Ω, are quite limited.

Penckwitt *et al.* (2002) discussed a simulation where the thermal cloud is rigidly rotating, but the condensate is nonrotating ($\Omega = 0$). This corresponds

to the experiment performed by Haljan *et al.* (2001). The trapping potential is axisymmetric and the simulation starts with the ground state solution of the GP equation. Equation (9.111) with $\Omega = 0$ is then solved numerically[14] for a spatially independent value of γ' and some chosen value of the thermal cloud rotation frequency Ω'. The initial state is also perturbed by a superposition of angular modes localized in the surface region of the condensate. This perturbation "seeds" the surface modes, some of which are susceptible to the dynamic instability discussed in subsection 9.8.1.

As the simulation progresses, vortices appear in a ring outside the condensate and these eventually penetrate into the condensate and equilibrate into a vortex lattice. However, as implied by the discussion below (9.111), this cannot be a true stationary state solution since the vortex state is not an eigenstate of L'_z. In spite of this the vortex lattice is finally found to rotate with angular velocity Ω'. This can be understood by noting that a vortex lattice effectively gives rise to rigid-body rotation of the condensate. That is, the average flow velocity at a distance ρ from the rotation axis is $\langle v_\phi \rangle = \Omega' \rho$ and the average angular momentum of the condensate is $\langle L_z \rangle = \langle \Phi_0 | L_z | \Phi_0 \rangle = I_z \Omega'$, where I_z is the moment of inertia of the condensate about the z axis. Thus, once the vortex lattice has formed, the solution of (9.111) is stationary in an *average* sense, which is consistent with the simulations carried out by Penckwitt *et al.* (2002).

The simulations just described are closely related to those of Tsubota *et al.* (2002). One can understand this similarity in the following way. With the transformation $\Phi' = e^{-i(\tilde{\mu}'_0 + \Omega' L'_z)t/\hbar} \overline{\Phi}'$, (9.111), with $\Omega = 0$, becomes

$$i\hbar(1 + i\gamma') \frac{\partial \overline{\Phi}'(\mathbf{r}', t)}{\partial t} = \left(e^{i\Omega' L'_z t/\hbar} \mathcal{H}'_{\rm c}(t) e^{-i\Omega' L'_z t/\hbar} - \tilde{\mu}'_0 - \Omega' L'_z \right) \overline{\Phi}'(\mathbf{r}', t) \,. \tag{9.112}$$

Penckwitt *et al.* (2002) considered this equation in the case of an axisymmetric trap. In this case, the only term of $\mathcal{H}'_{\rm c}(t)$ in (9.107) that does not commute with L'_z is the condensate mean field $g n'_{\rm c}(\mathbf{r}', t)$. However, the condensate density is axisymmetric in an average sense, even in the presence of vortices and, as a first approximation, one can ignore this noncommutativity. In this case, (9.112) takes the same form as (9.109), the angular velocity Ω' of the rotating thermal cloud playing the role of the angular velocity Ω of a rotating potential. In view of this, it is not surprising that the results of simulations based on (9.109) and on (9.112) are quite similar. However, the physical origin of the dynamic instabilities arising in

[14] With $\Omega = 0$, (9.111) can be taken as the equation in the lab frame and so all primes on the variables can be dropped.

the two sets of calculations is different. For the axisymmetric trap considered by Penckwitt *et al.* (2002), the imposition of an initial perturbation on the starting wavefunction acts as a seed for the dynamical instability discussed in subsection 9.8.1. In contrast, Tsubota *et al.* (2002) worked with a weakly anisotropic trapping potential whose rotation induces a quasi-stationary state (Recati *et al.*, 2001) that itself is dynamically unstable (Recati *et al.*, 2001; Sinha and Castin, 2001). In both simulations, however, a surface mode instability is the precursor to the appearance of vortices and the dissipation associated with the parameter γ is essential for allowing the vortices to equilibrate into a vortex lattice.

Although the above simulations are suggestive of a vortex-nucleation and vortex-lattice formation mechanism, the full dynamics of the thermal cloud was not included. What effect this might have on the results requires further study. As explained in Section 8.2, a dissipative GP equation such as (9.112) has the property that time-dependent solutions relax to a final stationary equilibrium state. However, this dynamical behaviour is a consequence of the way in which the phenomenological relaxation parameter γ enters the equation. We have shown in this chapter that the dissipation can have a well-defined physical basis in the C_{12} collisions between the condensate and thermal atoms. More experimental work is needed to see whether it accounts for the thermal cloud dynamics on the growth and stabilization of vortex lattices, as a function of temperature.

10

Dynamics at finite temperatures using the moment method

The moment method has often been used to obtain approximate solutions of a kinetic equation. The basic strategy is to use the kinetic equation to obtain equations of motion for the moments of various dynamical quantities, denoted by $\chi(\mathbf{p}, \mathbf{r})$. These moments involve an average with respect to the single-particle distribution function $f(\mathbf{p}, \mathbf{r}, t)$. For example, to derive hydrodynamic equations one takes moments with respect to the momentum variable only. However, it is also possible to perform an average over both position and momentum to obtain the time-dependent variable

$$\langle \chi(t) \rangle \equiv \frac{1}{N} \int \frac{d\mathbf{r} d\mathbf{p}}{(2\pi\hbar)^3} \chi(\mathbf{p}, \mathbf{r}) f(\mathbf{p}, \mathbf{r}, t). \tag{10.1}$$

The main advantage of this method is that the equations of motion one obtains are relatively simple and can be analyzed semi-analytically. Moreover, it provides a convenient, qualitative description of the coupled dynamics of the condensate and the thermal cloud in both the collisionless and hydrodynamic regimes. However, the approach cannot easily be used to include Landau damping, which is usually the dominant source of damping in the collisionless regime.

As a simple illustration of the moment method, we discuss in Section 10.1 the $m = 0$ monopole–quadrupole mode above $T_{\rm BEC}$, following the treatment of Guéry-Odelin *et al.* (1999). We then consider in Section 10.2 the scissors modes in a trapped superfluid Bose gas. These modes were first discussed by Guéry-Odelin and Stringari (1999) using the moment method for a pure superfluid and for a classical gas. Their analysis was extended by Nikuni (2002) to the whole superfluid region using the ZNG coupled equations for the condensate and thermal cloud. This latter work complements the numerical simulation of the ZNG equations performed by Jackson and Zaremba (2001), which is decribed in Chapter 12. The scissors modes in the

superfluid phase have been studied experimentally in considerable detail by the Oxford group (Maragò, 2001; Maragò *et al.*, 2000, 2001, 2002).

As discussed by Guéry-Odelin and Stringari (1999) and Maragò *et al.* (2002), the scissors modes in a trapped superfluid gas are of fundamental interest since they reflect the different dynamical properties of the condensate and the normal component. A pure condensate at $T = 0$ exhibits a single scissors mode because of the irrotational nature of the superfluid velocity. In contrast, a normal gas in an anisotropic trap exhibits two distinct modes since the flow has both rotational and irrotational components. In the Bose-condensed gas at finite temperatures, these modes are coupled mainly through mean-field interactions. A calculation of this coupled dynamics using the moment method is presented in Section 10.2.

In Section 10.3, we discuss the general relation between the moment of inertia of a superfluid at finite temperature and the quadrupole response function (Zambelli and Stringari, 2001). This exact relation gives a direct connection between the scissors modes (the poles of the quadrupole response function) and the moment of inertia of the condensate and noncondensate components of the trapped gas. The results of Section 10.2 are used to provide an explicit expression for the quadrupole response function in Section 10.3.

10.1 Bose gas above $T_{\rm BEC}$

In this section, we illustrate the moment method to study the $m = 0$ monopole–quadrupole mode in a normal Bose gas confined in an axisymmetric trap. The moment of interest in this case is $\langle x^2 + y^2 \rangle = \langle r^2 - z^2 \rangle$, which is a measure of the extent of the cloud in the radial direction. To see how the moment method works, we consider

$$\frac{d\langle r^2 \rangle}{dt} = \frac{1}{N} \int \frac{d\mathbf{p} d\mathbf{r}}{(2\pi\hbar)^3} r^2 \frac{\partial f}{\partial t} \,. \tag{10.2}$$

Substituting the other terms into the kinetic equation for $\partial f/\partial t$, the only nonvanishing contribution is

$$\frac{d\langle r^2 \rangle}{dt} = -\frac{1}{N} \int \frac{d\mathbf{p} d\mathbf{r}}{(2\pi\hbar)^3} r^2 \frac{\mathbf{p}}{m} \cdot \nabla f = 2 \left\langle \frac{\mathbf{r} \cdot \mathbf{p}}{m} \right\rangle \,. \tag{10.3}$$

Thus we see that the $\langle r^2 \rangle$ moment is coupled to the $\langle \mathbf{r} \cdot \mathbf{p}/m \rangle$ moment, and so it is clear that we should seek an equation of motion for the latter.

Continuing in this way, we arrive at the following six coupled equations

$$\begin{aligned}
&\frac{d\langle\chi_1\rangle}{dt} - 2\langle\chi_3\rangle = 0, \qquad \frac{d\langle\chi_2\rangle}{dt} - 2\langle\chi_4\rangle = 0\\
&\frac{d\langle\chi_3\rangle}{dt} - \langle\chi_5\rangle + \tfrac{1}{3}(2\omega_\perp^2 + \omega_z^2)\langle\chi_1\rangle + \tfrac{1}{3}(\omega_z^2 - \omega_\perp^2)\langle\chi_2\rangle = 0,\\
&\frac{d\langle\chi_4\rangle}{dt} - \langle\chi_6\rangle + \tfrac{2}{3}(\omega_z^2 - \omega_\perp^2)\langle\chi_1\rangle + \tfrac{1}{3}(\omega_\perp^2 + 2\omega_z^2)\langle\chi_2\rangle = 0,\\
&\frac{d\langle\chi_5\rangle}{dt} + \tfrac{2}{3}(\omega_z^2 + 2\omega_\perp^2)\langle\chi_3\rangle + \tfrac{1}{3}(2\omega_z^2 - \omega_\perp^2)\langle\chi_4\rangle = 0,\\
&\frac{d\langle\chi_6\rangle}{dt} + \tfrac{4}{3}(\omega_z^2 - \omega_\perp^2)\langle\chi_3\rangle + \tfrac{2}{3}(2\omega_z^2 + \omega_\perp^2)\langle\chi_4\rangle = \langle\chi_6\rangle_{\rm coll}.
\end{aligned} \tag{10.4}$$

where we define the quantities $\chi_i(\mathbf{p}, \mathbf{r})$

$$\begin{aligned}
&\chi_1 = r^2, \qquad &&\chi_2 = 3z^2 - r^2,\\
&\chi_3 = \frac{\mathbf{r}\cdot\mathbf{p}}{m}, \qquad &&\chi_4 = \frac{2zp_z}{m} - \frac{\mathbf{r}_\perp\cdot\mathbf{p}_\perp}{m},\\
&\chi_5 = \frac{p^2}{m^2}, \qquad &&\chi_6 = \frac{2p_z^2}{m^2} - \frac{p_\perp^2}{m^2}.
\end{aligned} \tag{10.5}$$

Apart from χ_6, all these are collisional invariants. This means that for $i = 1, \ldots, 5$, we have

$$\langle\chi_i\rangle_{\rm coll} \equiv \frac{1}{N}\int \frac{d\mathbf{p}d\mathbf{r}}{(2\pi\hbar)^3}\chi_i C_{22}[f] = 0. \tag{10.6}$$

If it were not the dependence on $\langle\chi_6\rangle_{\rm coll}$, the six moments in (10.4) would satisfy a closed set of linear algebraic equations.

In order to calculate the collisional contribution $\langle\chi_6\rangle$, (10.6), it is convenient to write the distribution function as (compare (17.25))

$$f(\mathbf{p}, \mathbf{r}, t) = f^0(\mathbf{p}, \mathbf{r}) + f^0(\mathbf{p}, \mathbf{r})[1 + f^0(\mathbf{p}, \mathbf{r})]\psi(\mathbf{p}, \mathbf{r}, t), \tag{10.7}$$

where f^0 is the thermal equilibrium Bose distribution. Treating ψ as a small perturbation and noting that $C_{22}[f^0] = 0$, the C_{22} collision integral can be linearized in ψ to obtain

$$\begin{aligned}
C_{22}[f] \simeq\ &\frac{2g^2}{(2\pi)^5\hbar^7}\int d\mathbf{p}_2\int d\mathbf{p}_3\int d\mathbf{p}_4\\
&\times \delta(\mathbf{p} + \mathbf{p}_2 - \mathbf{p}_3 - \mathbf{p}_4)\delta(\tilde{\varepsilon}_p + \tilde{\varepsilon}_{p_2} - \tilde{\varepsilon}_{p_3} + \tilde{\varepsilon}_{p_4})\\
&\times f^0 f_2^0\Big(1 + f_3^0\Big)\Big(1 + f_4^0\Big)(\psi_3 + \psi_4 - \psi_2 - \psi)\\
\equiv\ &L_{22}[\psi]\,,
\end{aligned} \tag{10.8}$$

where $f_i^0 = f^0(\mathbf{p}_i, \mathbf{r})$. We observe that any terms in ψ that are proportional to the collisional invariants $\chi_i (i = 1, \ldots, 5)$ make no contribution to $L_{22}[\psi]$. Although such dependences exist (see Chapters 17 and 18), we need consider

here only terms that are *not* collisional invariants. The simplest nontrivial ansatz for $\psi(\mathbf{p},\mathbf{r},t)$ is a term proportional to χ_6:

$$\psi(\mathbf{p},\mathbf{r},t) \simeq \alpha(2p_z^2 - p_\perp^2)\,. \tag{10.9}$$

Here the time-dependent parameter $\alpha(t)$ characterizes the anisotropy in the nonequilibrium momentum distribution. With this ansatz, we find that

$$\langle\chi_6\rangle = \frac{\alpha}{N}\int \frac{d\mathbf{p}d\mathbf{r}}{(2\pi\hbar)^3}\frac{1}{m^2}(2p_z^2 - p_\perp^2)^2 f^0(\mathbf{p},\mathbf{r})[1+f^0(\mathbf{p},\mathbf{r})]\,, \tag{10.10}$$

which provides a relation between the parameter α and the $\langle\chi_6\rangle$ moment.

However, the ansatz (10.9) also provides an explicit expression for $\langle\chi_6\rangle_{\rm coll}$. Using (10.8) and (10.9) in (10.6) gives

$$\langle\chi_6\rangle_{\rm coll} = \frac{\alpha}{N}\int \frac{d\mathbf{p}d\mathbf{r}}{(2\pi\hbar)^3}\frac{1}{m^2}(2p_z^2 - p_\perp^2)L_{22}[2p_z^2 - p_\perp^2]\,. \tag{10.11}$$

Eliminating α by means of (10.10), we finally obtain

$$\langle\chi_6\rangle_{\rm coll} = -\frac{\langle\chi_6\rangle}{\tau}, \tag{10.12}$$

which closes the set of equations (10.4). The inverse of the (quadrupole) relaxation time τ introduced in (10.12) is defined by

$$\frac{1}{\tau} \equiv -\frac{\displaystyle\int \frac{d\mathbf{p}d\mathbf{r}}{(2\pi\hbar)^3}(2p_z^2 - p_\perp^2)L_{22}[2p_z^2 - p_\perp^2]}{\displaystyle\int \frac{d\mathbf{p}d\mathbf{r}}{(2\pi\hbar)^3}(2p_z^2 - p_\perp^2)^2 f^0(\mathbf{p},\mathbf{r})[1+f^0(\mathbf{p},\mathbf{r})]}. \tag{10.13}$$

The only difference between the expression for τ in a classical gas (Guéry-Odelin *et al.*, 1999) and expression (10.13) for a degenerate (but normal) Bose gas is in the form of the equilibrium distribution function f^0 entering (10.13).

Nikuni (2002) showed that the inverse of this quadrupole relaxation time τ can be expressed as a weighted spatial average of the inverse of the viscous relaxation time $\tau_\eta(\mathbf{r})$ to be discussed in Chapter 18 (see (18.28)), which is associated with the shear viscosity in the collisional hydrodynamic regime of a normal Bose gas. Using this connection, one can write (10.13) in the more transparent form

$$\frac{1}{\tau} = \frac{\int d\mathbf{r}\tilde{P}_0(\mathbf{r})/\tau_\eta(\mathbf{r})}{\int d\mathbf{r}\tilde{P}_0(\mathbf{r})}\,, \tag{10.14}$$

where $\tilde{P}_0(\mathbf{r}) = g_{5/2}(z_0)/\beta_0\Lambda_0^3$ is the local equilibrium pressure in the gas (the equilibrium version of (15.21)). For a (nondegenerate) Maxwell–Boltzmann

gas, one finds that (Smith and Jensen, 1989; Kavoulakis *et al.*, 1998b, 2000; Nikuni and Griffin, 2001a)

$$\tau_\eta(\mathbf{r}) = \frac{5}{4}\tau_{\rm MB}(\mathbf{r}) \tag{10.15}$$

where $\tau_{\rm MB}$ is the usual mean collision time for a classical gas and is given by

$$\tau_{\rm MB}^{-1}(\mathbf{r}) \equiv \sqrt{2}\sigma\tilde{n}_0(\mathbf{r})\bar{v} = \sqrt{2}(8\pi a^2)\tilde{n}_0(\mathbf{r})\left(\frac{8k_{\rm B}T}{\pi m}\right)^{1/2}. \tag{10.16}$$

Inserting (10.15) into (10.14), the quadrupole relaxation time reduces to

$$\frac{1}{\tau} = \frac{4}{5}\Gamma_{\rm coll}\,, \tag{10.17}$$

where we have defined the collision rate $\Gamma_{\rm coll} \equiv \sqrt{2}\sigma\bar{n}\bar{v}$ in terms of the spatially averaged density in the trap,

$$\bar{n} \equiv \frac{\int d\mathbf{r}\,\tilde{n}_0^2(\mathbf{r})}{\int d\mathbf{r}\,\tilde{n}_0(\mathbf{r})} = \frac{\tilde{n}_0(0)}{2\sqrt{2}}. \tag{10.18}$$

The result (10.17) for $1/\tau$ in terms of $\Gamma_{\rm coll}$ was obtained by Guéry-Odelin *et al.* (1999) for a classical gas.

Assuming the time dependence $e^{-i\omega t}$, the six coupled equations (10.4) together with (10.12) can be solved. The frequencies are then given by the solutions of

$$(\omega^2 - 4\omega_z^2)(\omega^2 - 4\omega_\perp^2) + \frac{i}{\omega\tau}\left[\omega^4 - \tfrac{2}{3}\omega^2(5\omega_\perp^2 + 4\omega_z^2) + 8\omega_\perp^2\omega_z^2\right] = 0. \tag{10.19}$$

Solving (10.19), we obtain the solution $\omega = \Omega - i\Gamma$, describing damped modes. In Fig. 10.1, we plot Ω and Γ for the low-frequency mode as a function of $\omega_z\tau$. One can see that in the *collisionless* or mean-field limit defined by $\omega\tau \gg 1$, the solutions of (10.19) are given by $\omega_{\rm mf} = 2\omega_z,\, 2\omega_\perp$.

In the opposite *collisional* hydrodynamic limit $\omega\tau \ll 1$, the two solutions are given by

$$\omega_{\rm coll}^2 = \tfrac{1}{3}\left(5\omega_\perp^2 + 4\omega_z^2 \pm \sqrt{25\omega_\perp^4 + 16\omega_z^4 - 32\omega_z^2\omega_\perp^2}\right). \tag{10.20}$$

These frequencies are precisely what collisional hydrodynamics predicts, as discussed at the end of Section 15.4. For a spherical trap $\omega_\perp = \omega_z = \omega_0$, one finds the two solutions $\omega_{\rm coll} = \sqrt{2}\omega_0$ and $2\omega_0$, which correspond to the uncoupled quadrupole and monopole modes. For a cigar-shaped trap

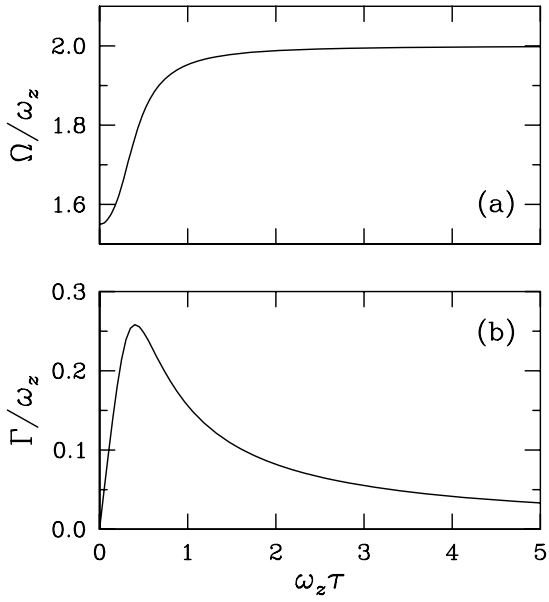


Fig. 10.1. Moment calculations of (a) the frequency Ω and (b) the damping Γ of the $m = 0$ mode as a function of the quadrupole relaxation time τ. These results were found by solving (10.19). The trap frequencies are $\omega_\perp/2\pi = 474$ Hz and $\omega_z/2\pi = 16.8$ Hz, as used by Buggle *et al.* (2005) .

$\omega_\perp \gg \omega_z$, one finds $\omega_{\rm coll} = \sqrt{\frac{12}{5}}\,\omega_z$ and $\sqrt{\frac{10}{3}}\,\omega_\perp$. For a disk-shaped trap $\omega_\perp \ll \omega_z$, one finds $\omega_{\rm coll} = \sqrt{\frac{8}{3}}\omega_z$ and $\sqrt{3}\,\omega_\perp$.

In the intermediate crossover-frequency region, the frequency Ω and damping Γ obtained by solving (10.19) can be expressed in terms of the spatially averaged quadrupole relaxation time τ defined in (10.13), for different trap geometries. The solutions can be summarized by a simple formula, in the case of a spherical, cigar, or disk-shaped trap, namely

$$\omega = \omega_{\rm mf}^2 + \frac{\omega_{\rm coll}^2 - \omega_{\rm mf}^2}{1 + i\omega\bar{\tau}}. \tag{10.21}$$

In this formula, $\bar{\tau}$ is related to the expression for τ in (10.13) simply by a numerical factor. For example, one finds $\bar{\tau} = \tau$ for the quadrupole mode in a spherical trap and $\bar{\tau} = 6\tau/5$ for the lowest mode in a cigar-shaped trap.

Buggle *et al.* (2005) made careful measurements of the frequency Ω and damping Γ of the quadrupole mode in a trapped nondegenerate gas. They mapped out the transition from the collisionless to the hydrodynamic region described by (10.19). A summary of their final results is shown in Fig.

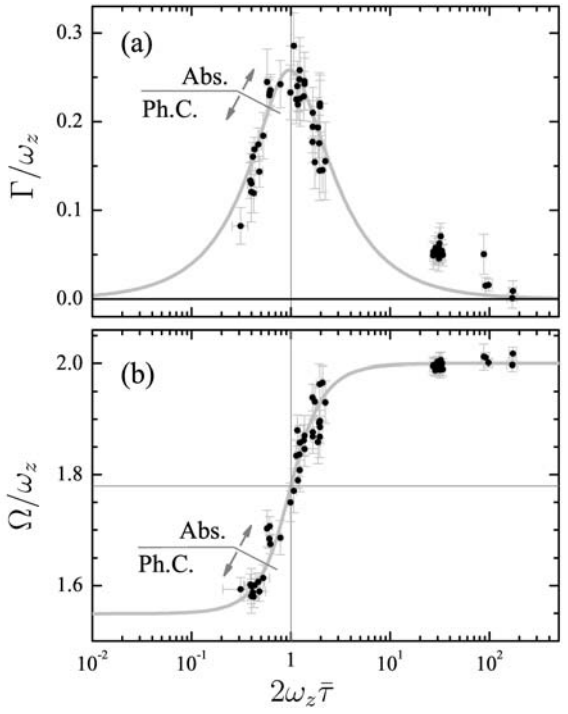


Fig. 10.2. Experimental data for (a) the damping Γ and (b) the frequency Ω of the quadrupole mode in a nondegenerate trapped Bose gas as a function of the relaxation time defined in (10.13) and (10.21). The crossover between the collisionless and hydrodynamic regions is shown. Note that the horizontal axes are in logarithmic units. The solid lines are based on the real and imaginary parts of the solution of (10.21). The label "Ph.C." stands for phase contrast imaging, while the label "Abs." stands for absorptive imaging (from Buggle *et al.*, 2005).

10.2. We refer to this paper for further discussion of how these results were obtained. The trap parameters of Buggle *et al.* (2005) are used in calculating the damping shown in Figs. 19.1 and 19.2.

10.2 Scissors oscillations in a two-component superfluid

The observation of the scissors mode in trapped Bose gases (Maragò *et al.*, 2000) is a dramatic demonstration of the irrotational nature of a trapped superfluid Bose gas. In the scissors mode, the atomic cloud oscillates about an axis of an anisotropic trap potential. This mode is induced by a sudden rotation of the trap about this axis (Guéry-Odelin and Stringari, 1999).

Above the Bose–Einstein transition temperature ($T_{\rm BEC}$), the thermal cloud exhibits two normal mode frequencies, corresponding to rotational and irrotational motions.

In contrast, the pure condensate motion at $T = 0$ exhibits only a single frequency, since only irrotational motion is allowed. This difference between the condensate (superfluid) scissors oscillation at $T \sim 0$ and the thermal gas (normal fluid) scissors oscillation above $T_{\rm BEC}$ has been nicely shown in the experiments of the Oxford group (Maragò *et al.*, 2000). The observed frequencies of oscillations are in good agreement with the theoretical predictions by Guéry-Odelin and Stringari (1999) for both $T = 0$ and $T > T_{\rm BEC}$.

At finite temperatures in the Bose-condensed phase, where an appreciable fraction of the atoms are excited out of the condensate, one expects *coupled* motions of both the superfluid and normal fluid components. In the experiments at Oxford (Maragò *et al.*, 2001) such coupled scissors mode oscillations were observed at *finite* temperatures, and from these Maragò and coworkers determined the temperature dependences of the frequency and damping rate of the oscillations of each component. In this section, we discuss the scissors mode in a trapped Bose-condensed gas at finite temperatures using the ZNG coupled equations.

We will restrict ourselves to the collisionless regime, defined as the region in which the mean collision rate is much smaller than the collective mode frequencies. Generalizing the moment-calculation approach used by Guéry-Odelin and Stringari (1999), we will derive coupled equations describing the oscillations of the quadrupole moments of the condensate and noncondensate components at finite temperatures in the superfluid phase. This section is based on the work of Nikuni (2002).

In Section 12.4, we will solve the same coupled ZNG equations numerically using FFT Monte-Carlo simulations developed by Jackson and Zaremba (2001) to discuss the temperature-dependent oscillations associated with the scissors mode. The analytical calculations presented here complement the numerical results in Section 12.4.

We consider a Bose-condensed gas confined in an anisotropic harmonic trap potential described by

$$V_{\rm trap}(\mathbf{r}) = \tfrac{1}{2}m[\omega_\perp^2(x^2+y^2) + \omega_z^2 z^2], \tag{10.22}$$

with $\omega_z^2 \equiv (1+\epsilon)\omega_0^2$ and $\omega_\perp^2 \equiv (1-\epsilon)\omega_0^2$, where $\omega_0^2 \equiv \frac{1}{2}(\omega_\perp^2 + \omega_z^2)$. The parameter ϵ characterizes the size of the *deformation* of the trap potential

in the zx-plane,

$$\epsilon = \frac{\omega_z^2 - \omega_\perp^2}{\omega_z^2 + \omega_\perp^2}. \tag{10.23}$$

The coupled dynamics of the condensate and noncondensate are, as usual, described by the generalized Gross–Pitaevskii equation (3.21) for the condensate wavefunction $\Phi(\mathbf{r}, t)$ and the kinetic equation (3.42) for the noncondensate distribution function $f(\mathbf{p}, \mathbf{r}, t)$. The experimental data of the Oxford group, especially the dependence on temperature, was described in great detail by Maragò (2001), Maragò *et al.* (2000) and Hodby (2002).

We use the Thomas–Fermi (TF) approximation, which neglects the quantum pressure term in the condensate chemical potential. Within the TF approximation, the equilibrium condensate density profile is given by

$$n_{\rm c0}(\mathbf{r}) = \frac{1}{g}[\mu_{\rm c0} - V_{\rm trap}(\mathbf{r})] - 2\tilde{n}_0(\mathbf{r}). \tag{10.24}$$

The equilibrium Bose distribution $f^0(\mathbf{p}, \mathbf{r})$ describing the noncondensate atoms is given by the Bose–Einstein distribution function (3.49), and $\tilde{n}_0(\mathbf{r})$ is given by (3.57). We note that the equilibrium TF densities for the condensate and noncondensate are isotropic in the renormalized coordinates defined by $(x', y', z') \equiv ((\omega_\perp/\bar{\omega})x, (\omega_\perp/\bar{\omega})y, (\omega_z/\bar{\omega})z)$, where $\bar{\omega} \equiv (\omega_\perp^2 \omega_z)^{1/3}$. Thus both $n_{\rm c0}(\mathbf{r})$ and $\tilde{n}_0(\mathbf{r})$ can be expressed as functions of $r' = (x'^2 + y'^2 + z'^2)^{1/2}$. Of course, the expression for $n_{\rm c0}(\mathbf{r})$ in (10.24) is valid only within the TF condensate radius $r' < R'_{\rm TF}$, where

$$R'_{\rm TF} \equiv \left\{ \frac{2}{m\bar{\omega}^2} \left[\mu_{\rm c0} - 2g \frac{1}{\Lambda^3} g_{3/2}(z_0 = 1) \right] \right\}^{1/2}. \tag{10.25}$$

Outside the TF condensate radius ($r' > R'_{\rm TF}$), the condensate density $n_{\rm c0}(\mathbf{r}')$ is zero.

The scissors mode can be excited by a sudden rotation of the anisotropic trap potential (10.22) in the zx-plane by a small angle $-\theta_0$. We will follow the convention of Maragò *et al.* (2000, 2001) that the rotation is about the y-axis.[1] This perturbation induces time-dependent density fluctuations which enter into the quadrupole moment

$$Q(t) \equiv \int d\mathbf{r}\, zx n(\mathbf{r}, t), \tag{10.26}$$

where the total density $n(\mathbf{r}, t) = n_{\rm c}(\mathbf{r}, t) + \tilde{n}(\mathbf{r}, t)$ includes the contributions from the condensate and the thermal cloud. Following Guèry-Odelin and

[1] We note that in the discussion by Guéry-Odelin and Stringari (1999) the small rotation is about the z-axis, which in our case is the axis of rotational symmetry. In our analysis we take the rotation to be about the y-axis.

Stringari (1999), one can derive coupled moment equations involving the relevant quadrupole variables for the condensate and noncondensate, starting from the ZNG coupled equations of motion given by (3.17), (3.18), and (3.42). In the linearized theory, the six exact moment equations are

$$\frac{d}{dt}\langle zx\rangle_{\rm c} = \langle zv_x + xv_z\rangle_{\rm c} - \langle zx\rangle_{12}, \tag{10.27}$$

$$\frac{d}{dt}\langle zv_x + xv_z\rangle_{\rm c} = -2\omega_0^2\langle zx\rangle_{\rm c} - \frac{2g}{m}\langle\nabla\tilde{n}_0\cdot\nabla(zx)\rangle_{\rm c} + \frac{2g}{m}\langle\nabla n_{\rm c0}\cdot\nabla(zx)\rangle_{\rm n}, \tag{10.28}$$

$$\frac{d}{dt}\langle zx\rangle_{\rm n} = \langle zv_x + xv_z\rangle_{\rm n} + \langle zx\rangle_{12}, \tag{10.29}$$

$$\frac{d}{dt}\langle zv_x + xv_z\rangle_{\rm n} = -2\omega_0^2\langle zx\rangle_{\rm n} - \frac{2g}{m}\langle\nabla n_{\rm c0}\cdot\nabla(zx)\rangle_{\rm n} + \frac{2g}{m}\langle\nabla\tilde{n}_0\cdot\nabla(zx)\rangle_{\rm c} + \frac{2}{m}\langle\bar{P}_{zx}\rangle_{\rm n}, \tag{10.30}$$

$$\frac{d}{dt}\langle zv_x - xv_z\rangle_{\rm n} = -2\epsilon\omega_0^2\langle zx\rangle_{\rm n} - \epsilon\frac{2g}{m}\langle\nabla n_{\rm c0}\cdot\nabla(zx)\rangle_{\rm n} + \epsilon\frac{2g}{m}\langle\nabla\tilde{n}_0\cdot\nabla(zx)\rangle_{\rm c}, \tag{10.31}$$

$$\frac{d}{dt}\langle\bar{P}_{zx}\rangle_{\rm n} = -m\omega_0^2\langle zv_x + xv_z\rangle_{\rm n} - m\epsilon\omega_0^2\langle zv_x - xv_z\rangle_{\rm n} - \frac{2g}{m}\left\langle v_z\frac{\partial n_0}{\partial x} + v_x\frac{\partial n_0}{\partial z}\right\rangle_{\rm n} + \left\langle\frac{p_zp_x}{m}\right\rangle_{\rm coll}. \tag{10.32}$$

In these equations, ω_0^2 takes the value given below (10.22). The various condensate and noncondensate moments of some function $\chi(\mathbf{r})$ are defined by

$$\langle\chi\rangle_{\rm c} \equiv \frac{1}{N}\int d\mathbf{r}\, n_{\rm c}(\mathbf{r},t)\chi(\mathbf{r}), \quad \langle\chi\rangle_{\rm n} \equiv \frac{1}{N}\int d\mathbf{r}\,\tilde{n}(\mathbf{r},t)\chi(\mathbf{r}).$$

$$\langle\chi\mathbf{v}\rangle_{\rm c} \equiv \frac{1}{N}\int d\mathbf{r}\, n_{\rm c}(\mathbf{r},t)\chi(\mathbf{r})\mathbf{v}_{\rm c}(\mathbf{r},t), \tag{10.33}$$

$$\langle\chi\mathbf{v}\rangle_{\rm n} \equiv \frac{1}{N}\int d\mathbf{r}\,\tilde{n}(\mathbf{r},t)\chi(\mathbf{r})\mathbf{v}_{\rm n}(\mathbf{r},t).$$

The local velocity $\mathbf{v}_{\rm n}$ and the local density of the kinetic pressure tensor $\bar{P}_{zx}$ of the thermal cloud are defined in terms of the single-particle distribution function as

$$\tilde{n}(\mathbf{r},t)\mathbf{v}_{\rm n} \equiv \int\frac{d\mathbf{p}}{(2\pi\hbar)^3}\frac{\mathbf{p}}{m}f(\mathbf{p},\mathbf{r},t), \tag{10.34}$$

$$\tilde{n}(\mathbf{r},t)\bar{P}_{zx}(\mathbf{r},t) \equiv \int\frac{d\mathbf{p}}{(2\pi\hbar)^3}\frac{1}{m}(p_z - mv_{{\rm n}z})(p_x - mv_{{\rm n}x})f(\mathbf{p},\mathbf{r},t). \tag{10.35}$$

As an illustration, we will outline the derivation of the second moment equation in (10.28). The *linearized* form of the moment $\langle zv_x + xv_z\rangle_{\rm c}$ is given by

$$\langle zv_x + xv_z\rangle_{\rm c} = \frac{1}{N}\int d\mathbf{r}\, n_{\rm c0}(\mathbf{r})[zv_{{\rm c}x}(\mathbf{r},t) + xv_{{\rm c}z}(\mathbf{r},t)]. \tag{10.36}$$

Using the linearized equation of motion for $\mathbf{v}_{\rm c}$, we obtain

$$\begin{aligned} m\frac{d}{dt}\langle zv_x + xv_z\rangle_{\rm c} &= -\frac{1}{N}\int d\mathbf{r}\, n_{\rm c0}(\mathbf{r})\left(z\frac{\partial}{\partial x} + x\frac{\partial}{\partial z}\right)\delta\mu_{\rm c}(\mathbf{r}) \\ &= \frac{1}{N}\int d\mathbf{r}\, n_{\rm c0}(\mathbf{r})\left(z\frac{\partial}{\partial x} + x\frac{\partial}{\partial z}\right) g(\delta n_{\rm c} + 2\delta\tilde{n}) \\ &= g\langle\nabla n_{\rm c0}\cdot\nabla(zx)\rangle_{\rm c} + 2g\langle\nabla n_{\rm c0}\cdot\nabla(zx)\rangle_{\rm n}, \end{aligned} \tag{10.37}$$

where we have used the TF expression for the nonequilibrium chemical potential $\delta\mu_{\rm c}$. In the first term we now use the TF expression for $n_{\rm c0}$ and obtain $\nabla n_{\rm c0} = -\nabla V_{\rm trap} - 2g\nabla\tilde{n}_0$. We also note that $\nabla V_{\rm trap}\cdot\nabla(zx) = m(\omega_z^2 + \omega_\perp^2)zx \equiv 2m\omega_0^2 zx$. Using these relations in (10.37), we arrive at the result given in (10.28). Similar manipulations are involved in deriving (10.29)–(10.32).

In (10.27)–(10.32), one also has moments involving the collision integrals C_{12} and C_{22}, defined by

$$\langle zx\rangle_{12} \equiv \frac{1}{N}\int d\mathbf{r}\, zx\Gamma_{12}(\mathbf{r},t), \tag{10.38}$$

$$\left\langle\frac{p_z p_x}{m}\right\rangle_{\rm coll} \equiv \frac{1}{N}\int d\mathbf{r}\int\frac{d\mathbf{p}}{(2\pi\hbar)^3}\frac{p_z p_x}{m}\{C_{12}[f(\mathbf{p},\mathbf{r},t),\Phi(\mathbf{r},t)] + C_{22}[f(\mathbf{p},\mathbf{r},t),\Phi(\mathbf{r},t)]\}. \tag{10.39}$$

The condensate moments and noncondensate moments are coupled through the HF mean fields as well as through the collisional exchange term $\langle zx\rangle_{12}$ in (10.38). If we neglect the coupling between the two components by setting both the interaction strength g and $\langle zx\rangle_{12}$ equal to 0 in (10.27)–(10.32), we recover the uncoupled moment equations derived by Guéry-Odelin and Stringari (1999). That is, we obtain the $T = 0$ condensate equations in (10.27) and (10.28) and the thermal cloud gas equations in (10.29)–(10.32) originally derived for $T > T_{\rm BEC}$.

To be explicit, the scissors mode equations of a pure condensate at $T = 0$ are

$$\frac{d}{dt}\langle zx\rangle_{\rm c} = \langle zv_x + xv_z\rangle_{\rm c}, \tag{10.40}$$

$$\frac{d}{dt}\langle zv_x + xv_z\rangle_{\rm c} = -2\omega_0^2\langle zx\rangle_{\rm c}. \tag{10.41}$$

The normal mode solution of these equations with time dependence $e^{-i\omega t}$ is given by $\omega = \sqrt{2}\omega_0 = \sqrt{\omega_\perp^2 + \omega_z^2}$. For $T > T_{\rm BEC}$, the scissors mode equations for the normal cloud reduce to

$$\frac{d}{dt}\langle zx\rangle_{\rm n} = \langle zv_x + xv_z\rangle_{\rm n}, \tag{10.42}$$

$$\frac{d}{dt}\langle zv_x + xv_z\rangle_{\rm n} = -2\omega_0^2\langle zx\rangle_{\rm n}, \tag{10.43}$$

$$\frac{d}{dt}\langle zv_x - xv_z\rangle_{\rm n} = -2\epsilon\omega_0^2\langle zx\rangle_{\rm n}. \tag{10.44}$$

10.2.1 Truncated set of moment equations

The moment equations in (10.27)–(10.32) are exact consequences of the coupled ZNG equations within the TF approximation, but they are not closed because they depend on the HF mean-field terms and the collision terms. To obtain a closed set of equations, we must introduce an ansatz for the forms of the condensate and noncondensate oscillations. For the condensate, we assume that the velocity field has the same irrotational form as the $T = 0$ result, namely

$$\mathbf{v}_{\rm c}(\mathbf{r}, t) = \alpha_{\rm c}(t)\nabla(zx) = \alpha_{\rm c}(t)(z, 0, x). \tag{10.45}$$

The small parameter $\alpha_{\rm c}(t)$ characterizes the amplitude of the condensate velocity, which we must calculate.

The distribution function of the noncondensate atoms is written in the form (10.7), in terms of the function $\psi(\mathbf{p}, \mathbf{r}, t)$ that represents the small fluctuations of the noncondensate around static thermal equilibrium. We make an ansatz in which ψ is written in the form

$$\psi(\mathbf{r}, \mathbf{p}, t) = \frac{\beta g\delta\tilde{n}(\mathbf{r}, t)}{\tilde{\gamma}_0(\mathbf{r})} + \beta\mathbf{v}_{\rm n}(\mathbf{r}, t)\cdot\mathbf{p} + \alpha_{zx}(t)\beta\frac{p_z p_x}{m}, \tag{10.46}$$

with $\tilde{\gamma}_0 \equiv (\beta g/\Lambda^3)g_{1/2}(z_0)$ (see (15.66)) and $\beta = 1/k_{\rm B}T$. The first term in (10.46) is associated with *density* fluctuations, while the second term is associated with *velocity* fluctuations. We also assume that the noncondensate velocity field involved in the scissors mode has the form

$$\mathbf{v}_{\rm n}(\mathbf{r}, t) = (\alpha_x(t)z,\, 0,\, \alpha_z(t)x). \tag{10.47}$$

The third term in (10.46) represents the quadrupole deformation in the momentum distribution, characterized by the small parameter $\alpha_{zx}(t)$. The ansatz given by (10.46) and (10.47) is motivated by the exact solution (Kavoulakis *et al.*, 1998b, 2000) of the *collisionless* kinetic equation above

$T_{\rm BEC}$ (i.e. the equation derived by setting $C_{22} = 0$ and ignoring the HF mean fields). The two normal modes for $T > T_{\rm BEC}$ in this case have frequencies given by $\omega_+ = |\omega_\perp \pm \omega_z|$.

Taking the time derivative of (10.28), (10.30) and (10.31), and using (10.45) and (10.47), we finally obtain the following set of coupled moment equations:

$$\frac{d^2}{dt^2}\langle zv_x + xv_z\rangle_{\rm c} = -\omega_0^2(2-\Delta_1)\langle zv_x + xv_z\rangle_{\rm c} - \omega_0^2\Delta_2\langle zv_x + xv_z\rangle_{\rm n} + \frac{g}{m}\langle\nabla n_{\rm c0}\cdot\nabla(zx)\rangle_{12}, \tag{10.48}$$

$$\frac{d^2}{dt^2}\langle zv_x + xv_z\rangle_{\rm n} = -\omega_0^2(4+2\Delta_3-\Delta_2)\langle zv_x + xv_z\rangle_{\rm n} - \omega_0^2\Delta_1\langle zv_x + xv_z\rangle_{\rm c} - 2\epsilon\omega_0^2(1+\Delta_3)\langle zv_x - xv_z\rangle_{\rm n} - \frac{g}{m}\langle\nabla n_{\rm c0}\cdot\nabla(zx)\rangle_{12} + \frac{2}{m}\left\langle\frac{p_z p_x}{m}\right\rangle_{\rm coll}, \tag{10.49}$$

$$\frac{d^2}{dt^2}\langle zv_x - xv_z\rangle_{\rm n} = \epsilon\omega_0^2(2-\Delta_2)\langle zv_x + xv_z\rangle_{\rm n} + \epsilon\omega_0^2\Delta_1\langle zv_x + xv_z\rangle_{\rm c} - \epsilon\frac{g}{m}\langle\nabla n_{\rm c0}\cdot\nabla(zx)\rangle_{12}. \tag{10.50}$$

The effect of the HF mean field has been parameterized in (10.48)–(10.50) by the following three quantities:

$$\Delta_1 \equiv \frac{1}{I_{\rm c,rb}}\left\{\frac{2g}{\omega_0^2}\int d\mathbf{r}[\nabla\tilde{n}_0\cdot\nabla(zx)][\nabla n_{\rm c0}\cdot\nabla(zx)]\right\} = \Delta_2\frac{I_{\rm n,rb}}{I_{\rm c,rb}}, \tag{10.51}$$

$$\Delta_3 \equiv \frac{1}{I_{\rm n,rb}}\left[\frac{1}{1-\epsilon^2}\frac{2g}{\omega_0^2}\int d\mathbf{r}\,\tilde{n}_0(\mathbf{r})\left(z\frac{\partial n_0}{\partial z} + x\frac{\partial n_0}{\partial x}\right)\right] = \frac{1}{I_{\rm n,rb}}\left[\frac{2}{(1-\epsilon^2)\omega_0^2}\int d\mathbf{r}\,\tilde{P}_0(\mathbf{r})\right] - 1. \tag{10.52}$$

Here $\tilde{P}_0 \equiv (k_{\rm B}T/\Lambda^3)g_{5/2}[z_0(\mathbf{r})]$ is the equilibrium kinetic pressure. In obtaining the last line of (10.52), we have used the equilibrium relation $\nabla\tilde{P}_0 + \tilde{n}_0\nabla U_0 = 0$ (see Chapter 15). $I_{\rm c,rb}$ and $I_{\rm n,rb}$ are the rigid-body values for the moment of inertia of the condensate and noncondensate components (Stringari, 1996a; Zambelli and Stringari, 2001) and are defined by

$$\begin{aligned} I_{\rm c,rb} &\equiv m\int d\mathbf{r}(z^2+x^2)n_{\rm c0}(\mathbf{r}), \\ I_{\rm n,rb} &\equiv m\int d\mathbf{r}(z^2+x^2)\tilde{n}_0(\mathbf{r}), \end{aligned} \tag{10.53}$$

where m is the mass of the atoms.

We will now give the details of the derivation of the moment equation (10.49) to illustrate the kind of manipulation used. First, we need to relate

the time-dependent parameters $\alpha_{\rm c}$, α_x and α_z to the moment variables. Using (10.45) and the definition of the moments in (10.33) and (10.53), we obtain

$$\langle zv_x + xv_z\rangle_{\rm c} = \alpha_{\rm c}\frac{1}{N}\int d\mathbf{r}(z^2+x^2)n_{\rm c0}(\mathbf{r}) = \alpha_{\rm c}\frac{I_{\rm c,rb}}{mN}. \tag{10.54}$$

Similarly, using (10.47) we find

$$\langle zv_x\rangle_{\rm n} = \alpha_x\frac{1}{N}\int d\mathbf{r}\, z^2\tilde{n}_0 = \alpha_x\frac{1-\epsilon}{2mN}I_{\rm n,rb}, \tag{10.55}$$

$$\langle xv_z\rangle_{\rm n} = \alpha_z\frac{1}{N}\int d\mathbf{r}\, x^2\tilde{n}_0 = \alpha_z\frac{1+\epsilon}{2mN}I_{\rm n,rb}. \tag{10.56}$$

Here we have used the fact (noted earlier) that the noncondensate density is isotropic in terms of the renormalized coordinates (x', y', z'). The rigid-body moment of inertia in the above equations is defined in (10.53). It can be written as

$$\begin{aligned} I_{\rm n,rb} &= m\int d\mathbf{r}'\bar{\omega}^2\left(\frac{z'^2}{\omega_z^2}+\frac{x'^2}{\omega_\perp^2}\right)\tilde{n}_0(\mathbf{r}') = \frac{m\bar{\omega}^2}{3}\int d\mathbf{r}'\frac{\omega_z^2+\omega_\perp^2}{\omega_z^2\omega_\perp^2}r'^2\tilde{n}_0(r') \\ &= \frac{2m\bar{\omega}^2}{3(1-\epsilon^2)\omega_0^2}\int d\mathbf{r}'r'^2\tilde{n}_0(\bar{r}'). \end{aligned} \tag{10.57}$$

We have used (10.23) to obtain the identity $\omega_0^4(1-\epsilon^2) = \omega_\perp^2\omega_z^2$, which is used in deriving the last line in (10.57). Therefore, we find

$$\begin{aligned} \int d\mathbf{r}\, z^2\tilde{n}_0(\mathbf{r}) &= \int d\mathbf{r}'\frac{\bar{\omega}^2}{\omega_z^2}z'^2\tilde{n}_0(\mathbf{r}') = \frac{\bar{\omega}^2}{3\omega_z^2}\int d\mathbf{r}'r'^2\tilde{n}_0(r') \\ &= \frac{\bar{\omega}^2}{3(1+\epsilon)\omega_0^2}\int d\mathbf{r}'r'^2\tilde{n}_0 = \frac{1-\epsilon}{2m}I_{\rm n,rb}, \end{aligned} \tag{10.58}$$

$$\int d\mathbf{r}\, x^2\tilde{n}_0(\mathbf{r}) = \frac{\bar{\omega}^2}{3(1-\epsilon)\omega_0^2}\int d\mathbf{r}'r'^2\tilde{n}_0 = \frac{1+\epsilon}{2m}I_{\rm n,rb}. \tag{10.59}$$

Combining (10.55) and (10.56), we obtain

$$\begin{aligned} \langle zv_x + xv_z\rangle_{\rm n} &= \tfrac{1}{2}\Big[(1+\epsilon)\alpha_z(t) + (1-\epsilon)\alpha_x(t)\Big]\frac{I_{\rm n,rb}}{mN}, \\ \langle zv_x - xv_z\rangle_{\rm n} &= \tfrac{1}{2}\Big[(1-\epsilon)\alpha_z(t) - (1+\epsilon)\alpha_x(t)\Big]\frac{I_{\rm n,rb}}{mN}, \end{aligned} \tag{10.60}$$

or, equivalently,

$$\alpha_z(t) = \frac{mN}{(1+\epsilon)I_{\rm n,rb}}(\langle zv_x + xv_z\rangle_{\rm n} - \langle zv_x - xv_z\rangle_{\rm n}),$$

$$\alpha_x(t) = \frac{mN}{(1-\epsilon)I_{\rm n,rb}}(\langle zv_x + xv_z\rangle_{\rm n} + \langle zv_x - xv_z\rangle_{\rm n}). \tag{10.61}$$

The results (10.54) and (10.61) give us what we need, namely the time-dependent parameters $\alpha_{\rm c}, \alpha_x$ and α_y in (10.45)–(10.47) in terms of moments. We also note that (10.58) and (10.59) can also be combined to give a useful expression for ϵ, which was defined in (10.23), in terms of moments:

$$\epsilon = \frac{m\int d\mathbf{r}(z^2-x^2)\tilde{n}_0(\mathbf{r})}{I_{\rm n,rb}} = \frac{\langle z^2-x^2\rangle_{\rm n0}}{\langle z^2+x^2\rangle_{\rm n0}}. \tag{10.62}$$

Using the preceding results, we now return to the derivation of (10.49). The time derivative of (10.30) is given by

$$\begin{aligned}\frac{d^2}{dt^2}\langle zv_x + xv_z\rangle_{\rm n} = &-2\omega_0^2\frac{d}{dt}\langle zx\rangle_{\rm n} - \frac{2g}{m}\frac{d}{dt}\langle\nabla n_{\rm c0}\cdot\nabla(zx)\rangle_{\rm n}\\ &+\frac{2g}{m}\frac{d}{dt}\langle\nabla\tilde{n}_0\cdot\nabla(zx)\rangle_{\rm c} + \frac{2}{m}\frac{d}{dt}\langle\bar{P}_{xy}\rangle_{\rm n},\end{aligned} \tag{10.63}$$

where

$$\begin{aligned}\frac{d}{dt}\langle zx\rangle_{\rm n} &= \frac{1}{N}\int d\mathbf{r}\, zx\frac{\partial\delta\tilde{n}}{\partial t} = \frac{1}{N}\int d\mathbf{r}\, zx[-\nabla\cdot(\tilde{n}_0\mathbf{v}_{\rm n}) + \delta\Gamma_{12}]\\ &= \langle zv_x + xv_z\rangle_{\rm n} + \langle zx\rangle_{12},\end{aligned} \tag{10.64}$$

$$\begin{aligned}\frac{d}{dt}\langle\nabla n_{\rm c0}\cdot\nabla(zx)\rangle_{\rm n} &= \frac{1}{N}\int d\mathbf{r}[\nabla n_{\rm c0}\cdot\nabla(zx)]\frac{\partial\delta\tilde{n}}{\partial t}\\ &= -\frac{1}{N}\int d\mathbf{r}[\nabla n_{\rm c0}\cdot\nabla(zx)]\nabla\cdot(\tilde{n}_0\mathbf{v}_{\rm n})\\ &\quad + \langle\nabla n_{\rm c0}\cdot\nabla(zx)\rangle_{12}.\end{aligned} \tag{10.65}$$

$$\begin{aligned}\frac{d}{dt}\langle\nabla\tilde{n}_0\cdot\nabla(zx)\rangle_{\rm c} &= \frac{1}{N}\int d\mathbf{r}[\nabla\tilde{n}_0\cdot\nabla(zx)]\frac{\partial\delta n_{\rm c}}{\partial t}\\ &= -\frac{1}{N}\int d\mathbf{r}[\nabla\tilde{n}_0\cdot\nabla(zx)]\nabla\cdot(n_{\rm c0}\mathbf{v}_{\rm c})\\ &\quad - \langle\nabla\tilde{n}_0\cdot\nabla(zx)\rangle_{12}.\end{aligned} \tag{10.66}$$

All terms in these equations have to be expressed in terms of moments.

We first consider the first term in the r.h.s of the last line of (10.66). Using the ansatz for $\mathbf{v}_{\rm c}$ given in (10.45), the relation (10.54) and the definition of Δ_1 in (10.52), one can show that

$$\frac{2g}{m}\frac{1}{N}\int d\mathbf{r}[\nabla\tilde{n}_0\cdot\nabla(zx)]\nabla\cdot(n_{\rm c0}\mathbf{v}_{\rm c}) = \Delta_1\omega_0^2\langle zv_x + xv_z\rangle_{\rm c}. \tag{10.67}$$

We next consider the first term on the r.h.s. of the last line of (10.65). One finds that

$$\begin{aligned}&\frac{2g}{m}\frac{1}{N}\int d\mathbf{r}[\nabla n_{\rm c0}\cdot\nabla(zx)]\nabla\cdot(\tilde{n}_0\mathbf{v}_n)\\&=\frac{2g}{m}\frac{1}{N}\int d\mathbf{r}[\nabla n_{\rm c0}\cdot\nabla(zx)]\left(\alpha_z x\frac{\partial\tilde{n}_0}{\partial z}+\alpha_x z\frac{\partial\tilde{n}_0}{\partial x}\right).\end{aligned}\tag{10.68}$$

Using the decompositions

$$\begin{aligned}x\frac{\partial\tilde{n}_0}{\partial z}&=\frac{\omega_z}{\omega_\perp}x'\frac{\partial\tilde{n}_0}{\partial r'}\frac{\partial r'}{\partial z'}=\frac{\omega_z}{\omega_\perp}\frac{z'x'}{r'}\frac{\partial\tilde{n}_0}{\partial r'},\\z\frac{\partial\tilde{n}_0}{\partial x}&=\frac{\omega_\perp}{\omega_z}z'\frac{\partial\tilde{n}_0}{\partial r'}\frac{\partial r'}{\partial x'}=\frac{\omega_\perp}{\omega_z}\frac{z'x'}{r'}\frac{\partial\tilde{n}_0}{\partial r'},\\\nabla\tilde{n}_0\cdot\nabla(zx)&=\left(\frac{\omega_\perp}{\omega_z}+\frac{\omega_z}{\omega_\perp}\right)\frac{z'x'}{r'}\frac{\partial\tilde{n}_0}{\partial r'}=\frac{2\omega_0^2}{\omega_z\omega_\perp}\frac{z'x'}{r'}\frac{\partial\tilde{n}_0}{\partial r'},\end{aligned}\tag{10.69}$$

one can show that the terms within the large parentheses in (10.68) are

$$x\frac{\partial\tilde{n}_0}{\partial z}=\tfrac{1}{2}(1+\epsilon)\nabla\cdot\tilde{n}_0\nabla(zx),\quad z\frac{\partial\tilde{n}_0}{\partial x}=\tfrac{1}{2}(1-\epsilon)\nabla\tilde{n}_0\cdot\nabla(zx).\tag{10.70}$$

Using these relations along with the ansatz for $\mathbf{v}_{\rm n}$ in (10.47), the expression in (10.68) can finally be reduced to

$$\begin{aligned}&\tfrac{1}{2}[(1+\epsilon)\alpha_z+(1-\epsilon)\alpha_x]\frac{1}{N}\int d\mathbf{r}[\nabla n_{\rm c0}\cdot\nabla(zx)][\nabla\tilde{n}_0\cdot\nabla(zx)]\\&\qquad=\Delta_2\omega_0^2\langle zv_x+xv_z\rangle_{\rm n}.\end{aligned}\tag{10.71}$$

The r.h.s. is obtained using (10.60) as well as the definition of Δ_2 given in (10.51). By a similar calculation, one can also show that

$$\frac{2g}{m}\left\langle v_z\frac{\partial n}{\partial x}+v_x\frac{\partial n}{\partial z}\right\rangle_{\rm n}=\Delta_3\omega_0^2\left(\langle zv_x+xv_z\rangle_{\rm n}+\epsilon\langle zv_x-xv_z\rangle_{\rm n}\right).\tag{10.72}$$

Using the above relations and (10.32), we can finally reduce (10.63) to the form given in (10.49). The moment equations (10.48) and (10.50) can be derived in a similar manner.

The last terms in the moment equations (10.48)–(10.50), involving the moments (10.38) and (10.39), are related to the collision integrals C_{12} and C_{22}. These terms turn out to be very small in the collisionless limit (Nikuni, 2002), and so we can neglect them. In this limit, (10.48)–(10.50) provide a closed set of equations for the three moments $\langle zv_x+xv_z\rangle_{\rm c}$, $\langle zv_x+xv_z\rangle_{\rm n}$ and $\langle zv_x-xv_z\rangle_{\rm n}$. These can be solved and used to calculate the undamped scissors mode oscillations involving the condensate and noncondensate components at finite temperatures. The effect of interactions is contained in the

three parameters Δ_i defined in (10.51) and (10.52), which characterize the HF mean-field coupling.

It is convenient to introduce a three-component vector describing the dynamical moment variables:

$$\mathbf{w} \equiv \begin{pmatrix} \langle zv_x + xv_z \rangle_{\rm c} \\ \langle zv_x + xv_z \rangle_{\rm n} \\ \langle zv_x - xv_z \rangle_{\rm n} \end{pmatrix}. \tag{10.73}$$

The coupled moment equations (10.48)–(10.50) can then be written as a 3×3 matrix equation

$$\frac{d^2}{dt^2}\mathbf{w} + \omega_0^2 \hat{\rm K}\mathbf{w} = 0, \tag{10.74}$$

where the 3×3 matrix $\hat{\rm K}$ is defined as

$$\hat{\rm K} = \begin{pmatrix} 2 - \Delta_1 & \Delta_2 & 0 \\ \Delta_1 & 4 + 2\Delta_3 - \Delta_2 & 2\epsilon(1 + \Delta_3) \\ -\epsilon\Delta_1 & -\epsilon(2 - \Delta_2) & 0 \end{pmatrix}. \tag{10.75}$$

Looking for normal mode solutions of (10.74) with time dependence $e^{-i\omega t}$, (10.74) reduces to

$$\hat{\rm K}\mathbf{w} = \tilde{\omega}^2 \mathbf{w}, \tag{10.76}$$

where $\tilde{\omega} \equiv \omega/\omega_0$ is the mode frequency normalized to ω_0, defined below (10.22). The solution of (10.76) will be discussed in the next section.

We note that for an axisymmetric trap potential above $T_{\rm BEC}$, one has $\epsilon = 0$ and $\Delta_1 = \Delta_2 = 0$ in (10.75), and thus the eigenvalue equation (10.76) predicts a mode frequency $\omega^2 = \omega_0^2(4+2\Delta_3)$. In the high-temperature Maxwell–Boltzmann gas limit $T \gg T_{\rm BEC}$, the HF mean-field parameter defined in (10.52) is given by $\Delta_3 \approx -gn(0)/(2\sqrt{2}k_{\rm B}T)$, where $n(0)$ is the density at the centre of the trap. This frequency agrees with that obtained in Pedri *et al.* (2003), where the mean field is treated by a scaling ansatz to first order in the interaction g.

Nikuni (2002) extended this analysis to give a detailed account of the damping of the normal modes (10.76) arising from C_{12} and C_{22} collisions. However, this collisional damping does not explain the damping of the scissors mode observed in the Oxford experiments (Maragò *et al.*, 2001), because a truncated moment calculation such as we have discussed cannot easily deal with Landau damping. The latter is always the dominant contribution to the collective mode damping in the collisionless, i.e. mean-field, regime (Guilleumas and Pitaevskii, 2000; Jackson and Zaremba, 2001; Williams and Griffin, 2001). This Landau damping will be calculated in Chapter 13. In

order to incorporate Landau damping in the moment method, one has to find a solution of the set of moment equations (10.27)–(10.32) that takes into account the *incoherent* motion of the thermal cloud. Our simple ansatz in (10.46) only describes the *coherent* thermal cloud dynamics.

10.2.2 Oscillations of the two components

In this subsection, we will solve (10.76) for the collective oscillations of the condensate and noncondensate components. If we neglect the HF mean fields (i.e. we set $\Delta_i = 0$) then a nontrivial solution of (10.76) is obtained by setting the determinant of the matrix to zero:

$$\begin{vmatrix} 2-\tilde{\omega}^2 & 0 & 0 \\ 0 & 4-\tilde{\omega}^2 & 2\epsilon \\ 0 & -2\epsilon & -\tilde{\omega}^2 \end{vmatrix} = (2-\tilde{\omega}^2)(\tilde{\omega}^4 - 4\tilde{\omega}^2 + 4\epsilon^2) = 0. \tag{10.77}$$

In this simple limit, the mode frequencies are (Guéry-Odelin and Stringari, 1999)

$$\omega_{\rm c} = \sqrt{2}\omega_0, \qquad \omega_{\pm} = \sqrt{2}\omega_0(1 \pm \sqrt{1-\epsilon^2})^{1/2}. \tag{10.78}$$

The thermal gas modes $\omega_{\pm}$ exhibit a high-frequency and a low-frequency solution. Keeping Δ_i finite, we solve the eigenvalue problem (10.76) numerically, using the trap parameters of the Oxford experiments (Maragò *et al.*, 2000).

In Fig. 10.3 we plot the three scissors normal mode frequencies ($\omega_{\rm c}$, ω_+ and ω_-) as a function of the temperature. In the low-temperature limit, $\omega_{\rm c}$ approaches the well-known $T = 0$ condensate scissors mode frequency $\omega_{\rm c} = \sqrt{2}\omega_0$ (Guéry-Odelin and Stringari, 1999). At finite temperatures, the frequency is shifted owing to the HF mean field of the noncondensate atoms. Expanding the solution of (10.76) to first order in the interaction parameters Δ_i, one finds

$$\omega_{\rm c}^2 \simeq \omega_0^2(2 - \Delta_1). \tag{10.79}$$

This simple expression for $\omega_{\rm c}$ is found to be a good approximation to the numerical solution of (10.76). This result can also be obtained directly by neglecting the noncondensate fluctuations in (10.48), i.e. setting $\langle zv_x + xv_z\rangle_{\rm n} = 0$. In this case, the positive frequency shift in the condensate mode ($\omega_{\rm c}$) in Fig. 10.3 is mainly due to the mean field $2g\tilde{n}_0$ of the static thermal cloud. As for the noncondensate modes, the frequencies $\omega_{\pm}$ approach the thermal cloud frequencies found above $T_{\rm BEC}$, namely $\omega_{\pm} = |\omega_z \pm \omega_{\perp}| =$

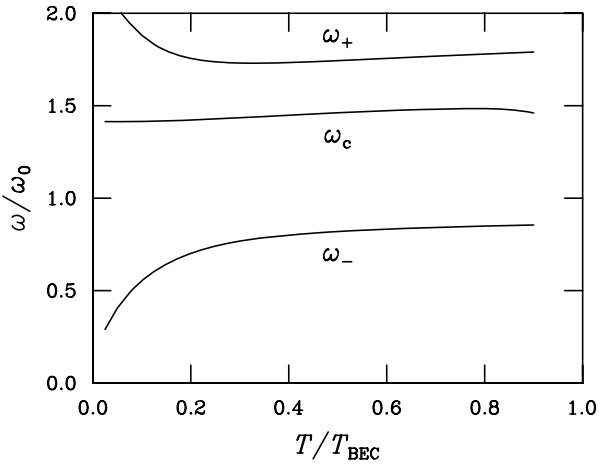


Fig. 10.3. Calculated temperature dependence of three scissors-mode frequencies for $N = 5 \times 10^4$ ^{87}Rb atoms in an anisotropic trap, with $\omega_z/2\pi = 126$ Hz and $\omega_\perp = \sqrt{8}\omega_z$, as used in the experiments of Maragò *et al.* (2000, 2001). The frequency is normalized by $\omega_0 \equiv [(\omega_z^2 + \omega_\perp^2)/2]^{1/2}$ and the temperature is renormalized by the ideal Bose gas transition temperature $T_{\rm BEC}$. The transition temperature measured in the experiments was $0.93T_{\rm BEC}$ (from Nikuni, 2002).

$\sqrt{2}\omega_0(1 \pm \sqrt{1-\epsilon^2})^{1/2}$ as T approaches $T_{\rm BEC}$. At temperatures well below $T_{\rm BEC}$, however, there are large frequency shifts in these noncondensate modes because of the coupling to an increasingly large condensate.

The solutions of (10.76) also give the relative amplitude of the quadrupole oscillations of the condensate and noncondensate components. It is more convenient to use the rotation angles $\theta_{\rm c}$ and $\theta_{\rm n}$ of the condensate and noncondensate clouds, assuming that the two components rotate without distortion. This gives

$$\begin{aligned} n_{\rm c}(\mathbf{r},t) &\simeq n_{\rm c0}(-z\sin\theta_{\rm c} + x\cos\theta_{\rm c},\, y,\, z\cos\theta_{\rm c} + x\sin\theta_{\rm c}) \\ &\simeq n_{\rm c0}(\mathbf{r}) + \theta_{\rm c}\left(x\frac{\partial n_{\rm c0}}{\partial z} - z\frac{\partial n_{\rm c0}}{\partial x}\right), \end{aligned} \tag{10.80}$$

$$\begin{aligned} \tilde{n}(\mathbf{r},t) &\simeq \tilde{n}_0(-z\sin\theta_{\rm n} + x\cos\theta_{\rm n},\, y,\, z\cos\theta_{\rm n} + x\sin\theta_{\rm n}) \\ &\simeq \tilde{n}_0(\mathbf{r}) + \theta_{\rm n}\left(x\frac{\partial \tilde{n}_0}{\partial z} - z\frac{\partial \tilde{n}_0}{\partial x}\right). \end{aligned} \tag{10.81}$$

The quadrupole moments can be expressed in terms of these rotation angles $\theta_{\rm c}$ and $\theta_{\rm n}$ as follows:

$$\langle zx\rangle_{\rm c} = \theta_{\rm c}\frac{1}{N}\int d\mathbf{r}\left(zx^2\frac{\partial n_{\rm c0}}{\partial z} - z^2x\frac{\partial n_{\rm c0}}{\partial x}\right) = \theta_{\rm c}\langle z^2 - x^2\rangle_{\rm c0}, \tag{10.82}$$

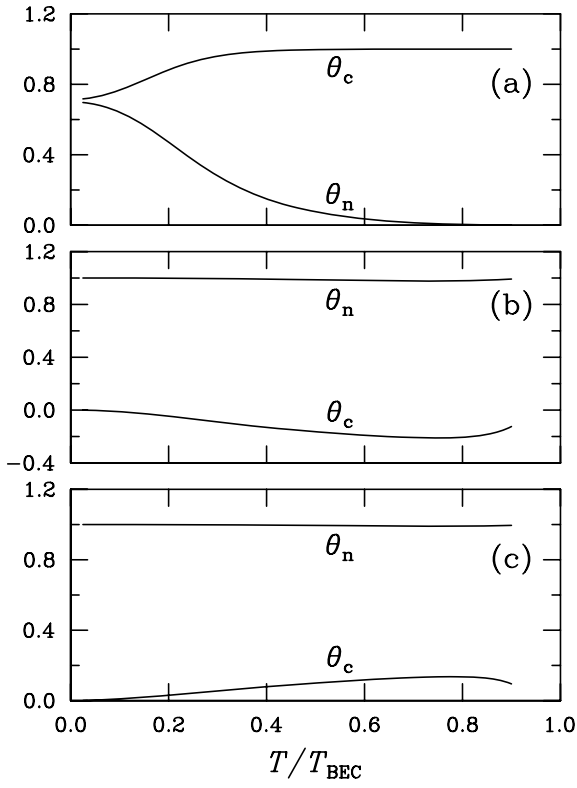


Fig. 10.4. Calculated temperature dependence of the rotation angles $\theta_{\rm c}$ and $\theta_{\rm n}$ (normalized so that $\theta_{\rm c}^2+\theta_{\rm n}^2=1$) of the two components associated with (a) the $\omega_{\rm c}$ mode, (b) the ω_+ mode and (c) the ω_- mode (from Nikuni, 2002).

$$\langle zx\rangle_{\rm n}=\theta_{\rm n}\frac{1}{N}\int d\mathbf{r}\left(zx^2\frac{\partial\tilde{n}_0}{\partial z}-z^2x\frac{\partial\tilde{n}_0}{\partial x}\right)=\theta_{\rm n}\langle z^2-x^2\rangle_{\rm n0}. \tag{10.83}$$

These quadrupole moments are obtained from the solution of (10.76) using the relation $\langle zx\rangle_{\rm c,n}=i\omega\langle zv_x+xv_z\rangle_{\rm c,n}$ (neglecting collisions). One can then express the rotation angles in terms of the quadrupole moments ($i=$ c or n)

$$\theta_i(t)=\frac{\langle zx\rangle_i}{\langle z^2-x^2\rangle_{i0}}=\frac{i\omega\langle zv_x+xv_z\rangle_i}{\langle z^2-x^2\rangle_{i0}}. \tag{10.84}$$

We note that using the TF density profiles for $n_{\rm c0}(\mathbf{r})$ and $\tilde{n}_0(\mathbf{r})$ in the expressions for the rigid-body moments of inertia given in (10.53), one obtains

$$Nm\langle z^2-x^2\rangle_{\rm c0}=-\epsilon I_{\rm c,rb},\quad Nm\langle z^2-x^2\rangle_{\rm n0}=-\epsilon I_{\rm n,rb}. \tag{10.85}$$

In Fig. 10.4(a)–(c), we plot the temperature dependence of the rotation

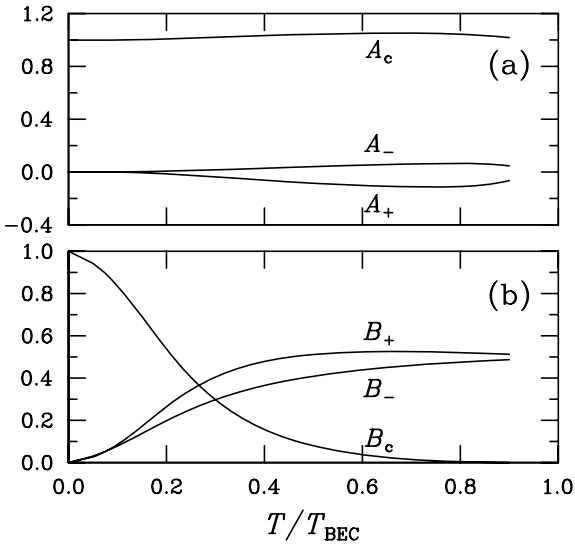


Fig. 10.5. Temperature dependence of coefficients A_i and B_i in (10.90)–(10.92) for the coupled scissors mode oscillations induced by a sudden rotation of the trap potential. (a) Plot of the A_i coefficients in the expression for the condensate rotation angle $\theta_c(t)$ in (10.87). (b) Plot of the B_i coefficients in the expression for the noncondensate angle rotation $\theta_n(t)$ in (10.87) (from Nikuni, 2002).

amplitudes in (10.84) of the condensate and noncondensate oscillations associated with the ω_c and $\omega_\pm$ modes. We find that both the ω_c and ω_- modes involve mainly in-phase oscillations of the two components, while the ω_+ mode involves mainly out-of-phase oscillations. At $T > 0.5T_{BEC}$, the ω_c mode mostly involves condensate oscillations. For decreasing T, the amplitude of the noncondensate component of the ω_c mode increases and, at low temperatures, the thermal cloud moves almost together with the condensate cloud. In contrast, the $\omega_\pm$ modes are seen to involve mainly oscillations of the noncondensate at all temperatures.

We can use these calculations to discuss the experiments by Maragò *et al.* (2000, 2001). In these experiments, the scissors modes were excited by a sudden rotation of the trap potential by an angle $-\theta_0$ at time $t = 0$. We assume that the atoms are initially in thermal equilibrium for $t < 0$. The corresponding initial conditions for the angles $\theta_c(t)$ and $\theta_n(t)$ and their time derivatives are found from the moment equations in (10.76),

$$\theta_c(0) = \theta_0, \quad \theta_c''(0) = -2\omega_0^2\theta_0, \quad \theta_c''''(0) = 4\omega_0^2\theta_0,$$

$$\theta_c'(0) = \theta_c'''(0) = \theta_c'''''(0) = 0,$$

$$\theta_{\rm n}(0)=\theta_0,\quad \theta_{\rm n}''(0)=-2\omega_0^2\theta_0,\quad \theta_{\rm n}''''(0)=4\omega_0^2[2+\Delta_3-(1+\Delta_3)\epsilon^2]\theta_0,$$

$$\theta_{\rm n}'(0)=\theta_{\rm n}'''(0)=\theta_{\rm n}'''''(0)=0. \tag{10.86}$$

The solutions for the angles $\theta_{\rm c}(t)$ and $\theta_{\rm n}(t)$ for $t>0$ can be written as

$$\begin{aligned}\theta_{\rm c}(t)/\theta_0&=A_{\rm c}\cos\omega_{\rm c}t+A_+\cos\omega_+t+A_-\cos\omega_-t,\\ \theta_{\rm n}(t)/\theta_0&=B_{\rm c}\cos\omega_{\rm c}t+B_+\cos\omega_+t+B_-\cos\omega_-t.\end{aligned} \tag{10.87}$$

The amplitudes A_i, B_i for the contribution of the three normal-mode frequencies in (10.87) must satisfy the initial conditions in (10.86). As a result, the amplitudes must satisfy the relations

$$\begin{aligned}A_{\rm c}+A_++A_-&=1,\\ \omega_{\rm c}^2A_{\rm c}+\omega_+^2A_++\omega_-^2A_-&=2\omega_0^2,\\ \omega_{\rm c}^4A_{\rm c}+\omega_+^4A_++\omega_-^4A_-&=4\omega_0^4\end{aligned} \tag{10.88}$$

and

$$\begin{aligned}B_{\rm c}+B_++B_-&=1,\\ \omega_{\rm c}^2B_{\rm c}+\omega_+^2B_++\omega_-^2B_-&=2\omega_0^2,\\ \omega_{\rm c}^4B_{\rm c}+\omega_+^4B_++\omega_-^4B_-&=4\omega_0^4[(2+\Delta_3)-(1+\Delta_3)\epsilon^2].\end{aligned} \tag{10.89}$$

Solving these equations, we obtain

$$\begin{aligned}A_{\rm c}&=\frac{\omega_+^2\omega_-^2-2\omega_0^2(\omega_+^2+\omega_-^2)+4\omega_0^4}{(\omega_{\rm c}^2-\omega_+^2)(\omega_{\rm c}^2-\omega_-^2)},\\ A_+&=\frac{\omega_-^2\omega_{\rm c}^2-2\omega_0^2(\omega_-^2+\omega_{\rm c}^2)+4\omega_0^4}{(\omega_+^2-\omega_-^2)(\omega_+^2-\omega_{\rm c}^2)},\\ A_-&=\frac{\omega_{\rm c}^2\omega_+^2-2\omega_0^2(\omega_{\rm c}^2+\omega_+^2)+4\omega_0^4}{(\omega_-^2-\omega_{\rm c}^2)(\omega_-^2-\omega_+^2)},\end{aligned} \tag{10.90}$$

$$\begin{aligned}B_{\rm c}&=\frac{\omega_+^2\omega_-^2-2\omega_0^2(\omega_+^2+\omega_-^2)+4\omega_0^4[(2+\Delta_3)-(1+\Delta_3)\epsilon^2]}{(\omega_{\rm c}^2-\omega_+^2)(\omega_{\rm c}^2-\omega_-^2)},\\ B_+&=\frac{\omega_-^2\omega_{\rm c}^2-2\omega_0^2(\omega_-^2+\omega_{\rm c}^2)+4\omega_0^4[(2+\Delta_3)-(1+\Delta_3)\epsilon^2]}{(\omega_+^2-\omega_-^2)(\omega_+^2-\omega_{\rm c}^2)},\\ B_-&=\frac{\omega_{\rm c}^2\omega_+^2-2\omega_0^2(\omega_{\rm c}^2+\omega_+^2)+4\omega_0^4[(2+\Delta_3)-(1+\Delta_3)\epsilon^2]}{(\omega_-^2-\omega_{\rm c}^2)(\omega_-^2-\omega_{\rm c}^2)}.\end{aligned} \tag{10.91}$$

The condensate and noncondensate oscillations (10.87) involve the three scissors mode frequencies $\omega_{\rm c}$, ω_+ and ω_-. In Fig. 10.5, we plot the temperature dependences of these amplitudes A_i and B_i ($i={\rm c}, +, -$), which

determine the rotation angles given in (10.87). The oscillations of the condensate θ_c are found to have a large contribution from the frequency ω_c at all temperatures. In contrast, the thermal cloud oscillations have a large contribution from $\omega_\pm$ frequencies for $T > 0.5T_{\rm BEC}$. However, the condensate frequency ω_c makes the dominant contribution to θ_n at low temperatures. This temperature dependence of the relative weights of these two contributions is consistent with the numerical simulations of the ZNG coupled equations reported by Jackson and Zaremba (2001), which are discussed in Section 12.4.

10.3 The moment of inertia and superfluid response

Our results for the scissors mode in Section 10.2 can be used to derive an expression for the moment of inertia I, a quantity that characterizes the rotational properties of a trapped superfluid Bose gas. Zambelli and Stringari (2001) derived an exact relation between the moment of inertia and the quadrupole response function χ_Q. We now briefly review their work. The moment of inertia I is defined as the linear response of the system to an external rotational field $-\Omega\hat{L}_y$, namely

$$I = \frac{\langle \hat{L}_y \rangle}{\Omega}, \tag{10.92}$$

where, in a second-quantized notation,

$$\hat{L}_y \equiv -i\hbar \int d\mathbf{r}\, \hat{\psi}^\dagger(\mathbf{r}) \left(z\frac{\partial}{\partial x} - x\frac{\partial}{\partial z} \right) \hat{\psi}(\mathbf{r}) \tag{10.93}$$

is the y-component of the angular momentum operator. Using the results of standard linear response theory, the expression (10.92) can be reduced to

$$I = \frac{1}{Z} \sum_{n,m} (e^{-\beta E_m} - e^{-\beta E_n}) \frac{|\langle m|\hat{L}_y|n\rangle|^2}{E_n - E_m}, \tag{10.94}$$

where $|n\rangle$ and E_n are the exact eigenstates and eigenvalues of the unperturbed Hamiltonian of the system, and $Z \equiv \sum_n e^{-\beta E_n}$ is the partition function.

In the case of the anisotropic harmonic trap potential (10.22), one can relate the angular momentum operator $\hat{L}_y$ to the quadrupole operator by

$$[\hat{H}, \hat{L}_y] = -2i\hbar m \epsilon \omega_0^2 \hat{Q}, \tag{10.95}$$

where m is the mass of an atom and the quadrupole operator is defined by

(10.26),

$$\hat{Q} \equiv \int d\mathbf{r}\, zx\hat{\psi}^\dagger(\mathbf{r})\hat{\psi}(\mathbf{r}). \tag{10.96}$$

The quadrupole response function is defined by (see also (12.8))

$$\chi_Q(\omega) = \frac{2}{\hbar Z}\sum_{m,n} e^{-\beta E_m}|\langle m|\hat{Q}|n\rangle|^2 \frac{(E_n - E_m)/\hbar}{(\omega + i0^+)^2 - (E_n - E_m)^2/\hbar^2}. \tag{10.97}$$

Using this response function and (10.95), one can express the moment of inertia I in (10.94) as

$$I = -\frac{4m^2\epsilon^2\omega_0^4}{\pi}\int_{-\infty}^{\infty} d\omega\, \frac{\chi_Q''(\omega)}{\omega^3}. \tag{10.98}$$

Here $\chi_Q''(\omega)$ is the imaginary part of the response function $\chi_Q(\omega + i0^+)$ in (10.97), namely

$$\chi_Q''(\omega) = \frac{\pi}{\hbar Z}\sum_{m,n}(e^{-\beta E_n} - e^{-\beta E_m})|\langle m|\hat{Q}|n\rangle|^2\delta(\omega - (E_n - E_m)/\hbar); \tag{10.99}$$

in writing the coefficient in (10.98), we have used the identity $4\epsilon^2\omega_0^4 = (\omega_z^2 - \omega_\perp^2)^2 = \omega_+^2\omega_-^2$.

A sudden rotation of the trap potential at $t = 0$ corresponds to a perturbation of the form

$$\hat{H}_{\rm pert} = -\lambda\hat{Q}\Theta(-t), \quad \lambda \equiv -2\theta_0\epsilon m\omega_0^2, \tag{10.100}$$

where $\Theta(-t) = 1$ for $t < 0$ and 0 for $t > 0$. Standard linear response theory shows that the induced quadrupole moment $Q(t) = \langle\hat{Q}\rangle_t$ is given by

$$Q(t > 0) = -\frac{\lambda}{\pi}\int_{-\infty}^{\infty} d\omega\, \frac{\chi_Q''(\omega)}{\omega}e^{-i\omega t} \equiv \int_{-\infty}^{\infty} d\omega\, Q(\omega)e^{-i\omega t}. \tag{10.101}$$

Thus $\chi_Q''(\omega)$ in (10.99) and the Fourier transform $Q(\omega)$ defined in (10.101) are related, namely

$$\chi_Q''(\omega) = -\frac{\pi}{\lambda}\omega Q(\omega). \tag{10.102}$$

Combining (10.98) and (10.102), one obtains a useful expression for the moment of inertia I in terms of the Fourier transform of the quadrupole moment $Q(\omega)$,

$$I = -\frac{2m\epsilon\omega_0^2}{\theta_0}\int_{-\infty}^{\infty} d\omega\, \frac{Q(\omega)}{\omega^2}. \tag{10.103}$$

We now use the results for the undamped coupled scissors mode oscillations derived in subsection 10.2.2 to derive an explicit formula for the

moment of inertia I. We recall from (10.84) and (10.85) that the induced quadrupole moment is given by

$$\begin{aligned} Q(t) &= N[\langle zx\rangle_{\rm c} + \langle zx\rangle_{\rm n}] \\ &= -\frac{\epsilon}{m}[I_{\rm c,rb}\theta_{\rm c}(t) + I_{\rm n,rb}\theta_{\rm n}(t)]. \end{aligned} \tag{10.104}$$

In terms of the solutions for $\theta_{\rm c}(t)$ and $\theta_{\rm n}(t)$ given in (10.87), we find that the Fourier transform of $Q(t)$ in (10.104) is given by

$$\begin{aligned} Q(\omega) = -\frac{\theta_0\epsilon}{2m}\Big\{ & I_{\rm c,rb}\sum_{i={\rm c},+,-} A_i[\delta(\omega-\omega_i)+\delta(\omega+\omega_i)] \\ & + I_{\rm n,rb}\sum_{i={\rm c},+,-} B_i[\delta(\omega-\omega_i)+\delta(\omega+\omega_i)]\Big\}, \end{aligned} \tag{10.105}$$

where the coefficients A_i and B_i have been calculated using the moment technique in Section 10.2 and are given by (10.91) and (10.92).

The expression for $\chi''_Q(\omega)$ in (10.102) satisfies an exact frequency sum rule relation discussed by Zambelli and Stringari (2001), namely

$$\int d\omega\, \chi''_Q(\omega)\omega = -\frac{\pi}{m^2} I_{\rm rb}. \tag{10.106}$$

Here $I_{\rm rb} \equiv I_{\rm c,rb} + I_{\rm n,rb}$ is the *rigid-body* value of the moment of inertia of the total system (see (10.53)), defined by

$$I_{\rm rb} = m\int d\mathbf{r}(z^2+x^2)n_0(\mathbf{r}), \tag{10.107}$$

where $n_0(\mathbf{r}) = n_{\rm c0}(\mathbf{r}) + \tilde{n}_0(\mathbf{r})$. The frequency sum rule (10.106) can be obtained directly from the expression in (10.99) by making use of the double commutator operator identity

$$\langle[\hat{Q},[\hat{H},\hat{Q}]]\rangle = \frac{I_{\rm rb}}{m^2}. \tag{10.108}$$

One can verify using (10.102) that our moment-technique expression for $Q(\omega)$ in (10.105) satisfies the sum rule (10.106).

Inserting the explicit expression for $Q(\omega)$ given in (10.105) into (10.103), one finds that the moment of inertia at finite temperatures is

$$I = 2\epsilon^2\bigg(I_{\rm c,rb}\sum_{i={\rm c},+,-}\frac{A_i}{\tilde{\omega}_i^2} + I_{\rm n,rb}\sum_{i={\rm c},+,-}\frac{B_i}{\tilde{\omega}_i^2}\bigg). \tag{10.109}$$

Using the formulas for A_i and B_i, the sums in (10.109) can be calculated

explicitly to give

$$\begin{aligned}\sum_{i=c,+,-}\frac{2\epsilon^2 A_i}{\tilde{\omega}_i^2} &= \epsilon^2 - \frac{(1-\epsilon^2)\Delta_1}{(2-\Delta_1-\Delta_2)}\\ \sum_{i=c,+,-}\frac{2\epsilon^2 B_i}{\tilde{\omega}_i^2} &= 1 + \frac{(1-\epsilon^2)\Delta_2}{(2-\Delta_1-\Delta_2)}.\end{aligned} \tag{10.110}$$

We note the important identity $I_{\mathrm{c,rb}}\Delta_1 = I_{\mathrm{n,rb}}\Delta_2$, which follows from the definitions of Δ_1 and Δ_2 in (10.51). Using this and the results in (10.110), the expression in (10.109) finally reduces to a simple formula for the total moment of inertia, namely

$$I = \epsilon^2 I_{\mathrm{c,rb}} + I_{\mathrm{n,rb}}. \tag{10.111}$$

An expression analogous to (10.111) was derived by Stringari (1996a) for a *noninteracting* trapped Bose-condensed gas. The simplicity of the final expression in (10.111) results from a subtle cancellation of terms related to the interaction terms depending on Δ_1 and Δ_2 in (10.110). Thus the physical meaning of the simple expression given in (10.111), which was derived while retaining interactions, is nontrivial. In spite of this, as we discuss below, the two contributions $\epsilon^2 I_{\mathrm{c,rb}}$ and $I_{\mathrm{n,rb}}$ in (10.111) can be associated separately with the irrotational contribution from the condensate and the rotational contribution from the noncondensate, respectively. In Fig. 10.6, we plot the moment of inertia given by (10.111) as a function of temperature, as well as the separate irrotational and rotational contributions, $\epsilon^2 I_{\mathrm{c,rb}}$ and $I_{\mathrm{n,rb}}$. We also remark, that in contrast with the noninteracting gas results of Stringari (1996a), interactions enter into the expressions for $I_{\mathrm{c,rb}}$ and $I_{\mathrm{n,rb}}$ in (10.53) through their effect on the static density profiles.

The result for the total moment of inertia given in (10.111) can be understood by considering the anisotropic trap potential in (10.22) to be rotating with angular frequency Ω in the zx-plane. In this rotating frame, the stationary Bose distribution of the noncondensate atoms is given by (see Section 9.5)

$$f_\Omega(\mathbf{p},\mathbf{r}) = \frac{1}{\exp\beta[p^2/2m + U(\mathbf{r}) - \Omega\mathbf{p}\cdot(\hat{\mathbf{y}}\times\mathbf{r}) - \tilde{\mu}] - 1}. \tag{10.112}$$

In terms of this distribution function, the total angular momentum along the y axis of the noncondensate component is given by (Dalfovo *et al.*, 1996)

$$L_{\mathrm{n},y} \equiv \int \frac{d\mathbf{r}d\mathbf{p}}{(2\pi\hbar)^3}(zp_x - xp_z) f_\Omega(\mathbf{p},\mathbf{r}). \tag{10.113}$$

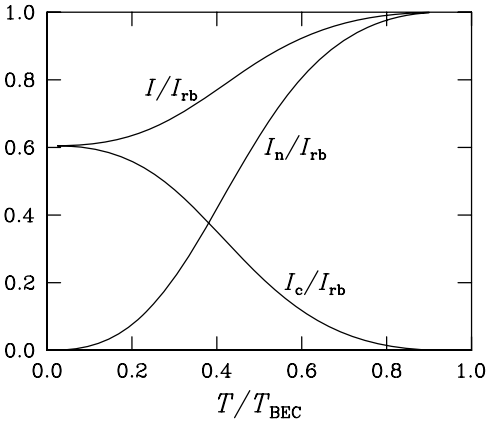


Fig. 10.6. Temperature dependence of the moment of inertia I as given in (10.111), normalized to $I_{\rm rb}$ defined in (10.107). We also plot the irrotational condensate component $I_{\rm c} = \epsilon^2 I_{\rm c,rb}$ and the rotational noncondensate component $I_{\rm n} = I_{\rm n,rb}$, where $I_{\rm c,rb}$ and $I_{\rm n,rb}$ are defined in (10.53) (from Nikuni, 2002).

To first order in Ω, (10.112) reduces to

$$f_\Omega(\mathbf{p},\mathbf{r}) = f^0(\mathbf{p},\mathbf{r}) - \beta\Omega(zp_x - xp_z)f^0(1+f^0). \tag{10.114}$$

Using this in (10.113), we find

$$\begin{aligned} L_{{\rm n},y} &\simeq \Omega \int \frac{d\mathbf{r}d\mathbf{p}}{(2\pi\hbar)^3}(zp_x - xp_z)^2 \beta f^0(1+f^0) \\ &= \Omega \int d\mathbf{r}(z^2+x^2) \int \frac{d\mathbf{p}}{(2\pi\hbar)^3}\frac{p^2}{3}\beta f^0(1+f^0) \\ &= \Omega \int d\mathbf{r}\, m(z^2+x^2)\tilde{n}_0(\mathbf{r}) \\ &\equiv \Omega I_{\rm n,rb}. \end{aligned} \tag{10.115}$$

Here we have used the identity (well-known for an ideal Bose gas)

$$\begin{aligned} \tilde{n}_0(\mathbf{r}) &\equiv \int \frac{d\mathbf{p}}{(2\pi\hbar)^3} f^0(\mathbf{p},\mathbf{r}) = \int \frac{d\mathbf{p}}{(2\pi\hbar)^3}\frac{p^2}{3m}\Big(-\frac{\partial f^0}{\partial \varepsilon_p}\Big) \\ &= \int \frac{d\mathbf{p}}{(2\pi\hbar)^3}\frac{p^2}{3m}\beta f^0(1+f^0), \end{aligned} \tag{10.116}$$

which holds for our HF excitation spectrum. The last line in (10.115) means that the noncondensate part of the moment of inertia is given by the *rigid-body* value defined in (10.53), i.e. $I_{\rm n} \equiv L_{{\rm n},y}/\Omega = I_{\rm n,rb}$.

For the condensate contribution, one has to find the stationary solution

of the GP equation in the rotating frame (Zambelli and Stringari, 2001). In the linear regime, the stationary velocity field is determined from

$$\nabla \cdot [n_{\rm c0}(\mathbf{v}_{\rm c} - \Omega\hat{\mathbf{y}} \times \mathbf{r})] = 0. \tag{10.117}$$

For the equilibrium TF condensate density $n_{\rm c0}$ determined from (10.24), one can show that the stationary condensate velocity field is given by

$$\mathbf{v}_{\rm c} = -\epsilon\Omega\nabla(zx) = \epsilon\Omega(z\hat{\mathbf{x}} + x\hat{\mathbf{z}}). \tag{10.118}$$

Using (10.118), the y-component of the total angular momentum of the condensate component is given by

$$\begin{aligned} L_{{\rm c},y} &= \int d\mathbf{r}\ m(\mathbf{r} \times \mathbf{v}_{\rm c})_y n_{\rm c0}(\mathbf{r}) \\ &= -\epsilon\Omega \int d\mathbf{r}\ m(z^2 - x^2) n_{\rm c0}(\mathbf{r}) \\ &= \epsilon^2 \Omega I_{\rm c,rb}. \end{aligned} \tag{10.119}$$

We have used (10.85) in the last line. Thus the condensate part of the moment of inertia is given by $I_{\rm c} \equiv L_{{\rm c},y}/\Omega = \epsilon^2 I_{\rm c,rb}$. This has the same form as the $T = 0$ expression for the irrotational value of the moment of inertia of the TF condensate derived by Zambelli and Stringari (2001).

From the results (10.115) and (10.119), we conclude that the first term in (10.111) represents the irrotational value of the condensate component in the Thomas–Fermi limit, while the second term represents the rigid-body value of the noncondensate component. As we have noted, the expression (10.111) is analogous to the noninteracting gas expression, $\Delta_i = 0$, as first obtained by Stringari (1996a). However, this equivalence is *nontrivial* when the mean field terms Δ_i are included, because of the cancellation of contributions in going from (10.109) to (10.111). It means that the two terms in (10.109) (based on calculating the expression for I in (10.103) using the moment calculation result in (10.105)) cannot be naively identified with the superfluid and normal fluid contributions. We will return to this point at the end of Section 12.4, where I is computed using numerical simulations of the coupled ZNG equations.

In this section, the coupled equations for the condensate and thermal cloud have been used to study the scissors mode in a trapped Bose-condensed gas at finite temperatures. Our coupled moment equations are the natural extension to superfluids at finite temperatures of similar calculations carried out by Guéry-Odelin and Stringari (1999) for $T = 0$ and $T > T_{\rm BEC}$. We solved these coupled moment equations to give explicit expressions for the three scissors mode frequencies. These solutions were used to calculate the

quadrupole response function and the moment of inertia, using an exact frequency moment sum rule derived by Zambelli and Stringari (2001). It was shown that the moment of inertia is given by the simple expression in (10.111). This result clearly shows that the linear response of a trapped superfluid Bose gas at finite temperatures involves both the irrotational motion of the condensate component as well as the rotational motion of the thermal cloud component.

In spite of its approximate nature, which results from the somewhat *ad hoc* truncation of the equations of motion, the moment method developed in this chapter is useful in understanding the detailed dynamics of a trapped Bose-condensed gas at finite temperatures. In particular, it provides a simple physical picture of the *coupled* motions of the condensate and thermal gas components involved in the scissors modes and the contribution of these to the moment of inertia. The moment method can be extended to study other collective modes at finite temperatures.

11

Numerical simulation of the ZNG equations

In this chapter we describe the numerical methods that can be used to solve the ZNG equations in the context of a dynamical simulation. These equations consist of a generalized GP equation (3.21) for the condensate and a Boltzmann equation (3.42) for the thermal component. The fact that the two equations are coupled makes their numerical solution more complex than when either is considered on its own. Indeed, the distinct quantum and classical aspects of the problem require specifically tailored numerical methods. Although most of these methods are well established and described elsewhere (Taha and Ablowitz, 1984; Sanz-Serna and Calvo, 1994), we provide in this chapter a detailed pedagogical discussion that will serve as a guide to those interested in carrying out such calculations for trapped Bose gases. This chapter is based on the papers of Jackson and Zaremba (2002a,b).

There are two main parts to the numerical problem. The first is developing a method for solving the time-dependent GP equation for an arbitrary three-dimensional geometry. This we take up in Section 11.1. Second, a method is needed for solving the Boltzmann equation that accounts for the dynamics of the thermal component. Here one must deal both with the Hamiltonian dynamics of the thermal atoms, as they move in the self-consistent mean field of the condensate and thermal cloud, and with the collisions that take place between the thermal atoms themselves (the C_{22} collisions) and between the thermal atoms and the condensate (the C_{12} collisions). The methods used to account for these two distinct collisional processes are taken up in Section 11.3. As we shall see, collisions play an important role and cannot be neglected even when the dynamical behaviour is dominated by mean-field interactions. However, when the collision rate is high they are of paramount importance. One then enters a qualitatively new regime in which collisions are responsible for establishing local hydrodynamic equilibrium. The dy-

namical behaviour characteristic of this regime will be discussed in Chapters 14–17.

Although most of this chapter is devoted to numerical methods, some numerical results regarding equilibrium collision rates are presented in Section 11.5. These serve as a test of the numerical methods developed but are of interest in their own right. Since most of these results deal with systems which are only weakly perturbed from equilibrium, it is important to know the equilibrium state of the system. Section 11.4 thus includes some numerical details needed to calculate equilibrium properties.

In Chapter 12, we will use these simulation techniques to discuss various collective modes at finite temperatures.

11.1 The generalized Gross–Pitaevskii equation

The numerical methods needed to solve the generalized GP equation at finite temperatures are essentially the same as those used to solve the simpler GP equation at $T = 0$ (Taha and Ablowitz, 1984). However, some modifications are needed to account for the dynamic mean field of the thermal cloud and the non-Hermitian collisional exchange rate $R(\mathbf{r}, t)$ defined in (3.22). To develop the algorithm it is convenient to rewrite (3.21) in the compact operator form

$$i\hbar \frac{\partial}{\partial t} |\Phi(t)\rangle = \hat{H}(t) |\Phi(t)\rangle \,. \tag{11.1}$$

Here, $\hat{H}(t) = \hat{T} + \hat{V}(t)$ is the time-dependent Hamiltonian operator, where the potential operator $\hat{V}(t)$ includes $\hat{R}(t)$. In most cases the time dependence of the condensate wavefunction is dominated by the nonlinear condensate potential itself. In fact, when the number of condensate atoms is large this interaction can lead to numerical instabilities. Thus, an important requirement is that the numerical algorithm be stable. Of course, the accuracy and efficiency of the algorithm are also important considerations.

A formal solution of (11.1) is given by

$$|\Phi(t + \Delta t)\rangle = \hat{U}(t + \Delta t, t) |\Phi(t)\rangle \,, \tag{11.2}$$

where the evolution operator $\hat{U}$ has the expansion (Fetter and Walecka, 1971)

$$\hat{U}(t+\Delta t, t) = 1 + \frac{1}{i\hbar} \int_t^{t+\Delta t} dt' \hat{H}(t') - \frac{1}{\hbar^2} \int_t^{t+\Delta t} dt' \int_t^{t'} dt'' \hat{H}(t') \hat{H}(t'') + \cdots \,. \tag{11.3}$$

Since we are generally dealing with a nonequilibrium situation, $\hat{U}$ depends

on t as well as on the time difference Δt. Expanding the Hamiltonian as a Taylor series,

$$\begin{aligned}\hat{H}(t') &= \hat{H}(t) + \frac{d\hat{H}}{dt}(t'-t) + \frac{1}{2}\frac{d^2\hat{H}}{dt^2}(t'-t)^2 + \cdots \\ &\equiv \hat{\alpha} + \hat{\beta}(t'-t) + \tfrac{1}{2}\hat{\gamma}(t'-t)^2 \cdots ,\end{aligned} \tag{11.4}$$

we obtain

$$\hat{U}(t+\Delta t, t) = 1 + \frac{\hat{\alpha}}{i\hbar}\Delta t + \frac{\hat{\beta}}{2i\hbar}\Delta t^2 - \frac{\hat{\alpha}^2}{2\hbar^2}\Delta t^2 + \mathcal{O}(\Delta t^3). \tag{11.5}$$

The lowest-order exponential approximant to this expansion is given by

$$\hat{U}(t+\Delta t, t) \simeq e^{-i\hat{H}(t)\Delta t/\hbar} + \mathcal{O}(\Delta t^2)\,. \tag{11.6}$$

For a time-independent Hamiltonian this approximant is in fact exact, but more generally it contains errors of second order in Δt. The effect of these errors can be reduced by decreasing the time step Δt, but at the expense of the computational effort needed to complete a simulation. It is therefore advantageous to have an approximant whose errors are of higher order than second order in Δt.

To achieve this we consider the exponential approximant

$$\hat{U}(t+\Delta t, t) \simeq e^{-i(\hat{\alpha}+\frac{1}{2}\hat{\beta}\Delta t)\Delta t/\hbar}. \tag{11.7}$$

An expansion of the exponential in powers of Δt confirms that (11.5) is reproduced to within errors of order Δt^3. To this same order of accuracy, we can make use of (11.4) to estimate $\hat{\beta}$ in (11.4) by reverse differencing,

$$\hat{\beta} \simeq \frac{\hat{H}(t) - \hat{H}(t-\Delta t)}{\Delta t}\,. \tag{11.8}$$

We thus obtain

$$\hat{U}(t+\Delta t, t) \simeq e^{-i\tilde{H}(t)\Delta t/\hbar} + \mathcal{O}(\Delta t^3)\,, \tag{11.9}$$

where

$$\tilde{H}(t) = \hat{T} + \tilde{V}(t) \tag{11.10}$$

with

$$\tilde{V}(t) \equiv \tfrac{1}{2}[3\hat{V}(t) - \hat{V}(t-\Delta t)]\,. \tag{11.11}$$

This result can be thought of as an approximation to the potential at time $t + \Delta t/2$, the midpoint of the current time step, as obtained by a linear extrapolation from the potential values at times $t - \Delta t$ and t. The only

additional overhead associated with using this scheme is that of storing the potential from the previous time step.

The actual numerical implementation of the evolution operator can be achieved by various methods. One approach is to use Cayley's form (Press *et al.*, 1992)

$$e^{-i\tilde{H}(t)\Delta t/\hbar} \simeq \frac{1 - i\tilde{H}(t)\Delta t/2\hbar}{1 + i\tilde{H}(t)\Delta t/2\hbar}\,, \tag{11.12}$$

which is unitary and accurate to second order in time; it implies that the wavefunction must be found by solving the equation

$$[1 - i\tilde{H}(t)\Delta t/2\hbar]|\Phi(t + \Delta t)\rangle = [1 + i\tilde{H}(t)\Delta t/2\hbar]|\Phi(t)\rangle\,. \tag{11.13}$$

In the coordinate representation, the kinetic energy operator $\hat{T}$ is the Laplacian. Approximating it by finite-differencing leads to a tridiagonal matrix equation for each spatial direction. This is essentially the Crank–Nicholson method (Press *et al.*, 1992) which is known to be stable and accurate to second order in space and time steps.

An alternative method is based on the Baker–Campbell–Hausdorff (BCH) formula. This yields the so-called split-operator form of the evolution operator, given by

$$e^{-i\tilde{H}(t)\Delta t/\hbar} = e^{-i\tilde{V}(t)\Delta t/2\hbar} e^{-i\hat{T}\Delta t/\hbar} e^{-i\tilde{V}(t)\Delta t/2\hbar} + \mathcal{O}(\Delta t^3)\,, \tag{11.14}$$

where the error is of the same order as that in (11.9). In principle, higher-order schemes can be constructed by splitting the evolution operator into more elaborate combinations of the $\tilde{V}(t)$ and $\hat{T}$ operators. However this would necessitate an improved approximation to $\tilde{H}(t)$, (11.10). There is evidently a trade-off between increasing the complexity of the scheme and the improved accuracy that it affords. We will therefore make use of (11.14) in the following and observe that the accuracy of the scheme can be improved, if necessary, by simply decreasing the time step.

The implementation of (11.14) is straightforward. The two potential factors $e^{-i\tilde{V}(t)\Delta t/2\hbar}$ in the split-operator form are local in real space and their action on the real space wavefunction is therefore multiplicative. The kinetic energy factor $e^{-i\hat{T}\Delta t/\hbar}$ is local in momentum space, however, and is most conveniently applied using the momentum representation of the wavefunction. The wavefunction at time $t + \Delta t$ is then obtained using the following sequence of steps. With the wavefunction at time t defined on a spatial grid of points (usually a Cartesian grid), the first potential factor is applied by multiplication. The wavefunction is then Fourier transformed and the kinetic energy factor is applied in momentum space. An inverse Fourier

transform brings the wavefunction back to real space and the final potential factor is applied. In this way the wavefunction is evolved over a sequence of time steps. The limiting steps in the calculation are the forward and inverse Fourier transforms, but efficient fast Fourier transform (FFT) routines are available for this.[1]

The split-operator approach can be used also to obtain the equilibrium ground state of the GP equation, starting from some non-self-consistent approximation. In this case, the evolution takes place in imaginary time $t \to -i\tau$ (Minguzzi *et al.*, 2004). If the wavefunction $\Phi(\mathbf{r}, \tau)$ is expanded in terms of the eigenstates of the instantaneous Hamiltonian $\tilde{H}(\tau)$, the lowest energy component decays the least rapidly on application of $\exp[-\tau\tilde{H}(\tau)/\hbar]$. Thus, with the repeated application of this operator followed by a wavefunction renormalization, one eventually arrives at the stationary ground state solution. Stability is not an issue in this case, but the step size must be chosen sufficiently small to achieve the desired accuracy of the solution.

The relative merits of the higher-order approximant (11.9) as compared to the lower-order scheme (11.6) can be determined by means of test simulations. With (11.6), one finds that the energy of the system typically increases monotonically with time. This is clearly an undesirable effect if one is interested in the damping of collective modes at finite temperatures. More importantly, the rate of increase scales with the mode energy. Thus, higher-frequency excitations, initially generated at a low level by the numerics, build in amplitude over a sufficiently long simulation and eventually result in numerical instabilities. These problems are essentially eliminated with the higher-order scheme (11.9). The improved stability allows larger time steps to be taken without compromising accuracy, leading to a considerable saving in computational effort.

11.2 Collisionless particle evolution

In this section we will discuss the solution of the collisionless Boltzmann equation (3.42), where we set $C_{12} = C_{22} = 0$. The effect of collisions will be dealt with in Sec. 11.3. With the right-hand side of (3.42) set to zero, the phase-space density $f(\mathbf{p}, \mathbf{r}, t)$ evolves according to the free-streaming operator. Known also as the Vlasov equation when mean-field interactions are present, the collisionless Boltzmann equation arises in many disparate fields, such as plasma physics, condensed matter physics and astrophysics. Since the potential term $U(\mathbf{r}, t)$ itself depends on f through the

[1] See Press *et al.* (1992) for a general discussion. The routines used in our calculations can be obtained from http:/www.fftw.org

noncondensate mean-field interaction, this nonlinear six-dimensional partial differential equation is very difficult to solve directly. However, an alternative approach ideally suited to numerical calculations is available. Rather than considering f as a function of the independent phase-space variables $\mathbf{p}$ and $\mathbf{r}$, one adopts a Lagrangian point of view whereby one follows the motion of a phase point as a function of time. The position and momentum simply evolve according to Newton's equation of motion. If $\tilde{N}_{\rm tp}$ phase points are initially distributed in phase space according to the phase-space density $f(\mathbf{p}, \mathbf{r}, t_0)$, the subsequent density of phase points will represent the actual evolution of f. We can think of the $\tilde{N}_{\rm tp}$ phase points as a cloud of discrete *test particles* (Hockney and Eastwood, 1981) whose motion constitutes an N-body simulation.

For the cloud of test particles the phase-space distribution is given by

$$f(\mathbf{p}, \mathbf{r}, t) \simeq \gamma (2\pi\hbar)^3 \sum_{i=1}^{\tilde{N}_{\rm tp}} \delta(\mathbf{r} - \mathbf{r}_i(t)) \delta(\mathbf{p} - \mathbf{p}_i(t)) \,, \tag{11.15}$$

where the weighting factor $\gamma \equiv \tilde{N}/\tilde{N}_{\rm tp}$ is fixed by the requirement that the phase-space distribution is normalized to the number of physical atoms, $\tilde{N} = \int d\mathbf{r} d\mathbf{p} f/(2\pi\hbar)^3$. There is no need for $\tilde{N}_{\rm tp}$ to equal the actual number of particles $\tilde{N}$. The only requirement is that $\tilde{N}_{\rm tp}$ be sufficiently large to minimize the effects of a discrete particle description and thus provide a reasonably accurate representation of the continuous phase-space distribution. In fact, if $\tilde{N}$ is relatively small, then we need $\tilde{N}_{\rm tp} \gg \tilde{N}$ to properly represent f. Conversely, if $\tilde{N}$ is very large, then $\tilde{N}_{\rm tp}$ can be chosen to be significantly smaller. Each test particle then represents a proportionately larger number of real particles and, in effect, becomes a "superparticle".

The time evolution of f is determined by the time-dependent position and momentum variables of each test particle, given by the equations

$$\frac{d\mathbf{r}_i(t)}{dt} = \frac{\mathbf{p}_i(t)}{m}, \tag{11.16}$$

$$\frac{d\mathbf{p}_i(t)}{dt} = -\nabla U(\mathbf{r}, t)|_{\mathbf{r}=\mathbf{r}_i(t)} \,. \tag{11.17}$$

The density in a particular phase-space cell of volume $\Delta\mathbf{p}\Delta\mathbf{r}/(2\pi\hbar)^3$ can be estimated by simply counting the number of phase points that lie in the cell and multiplying the result by γ.

The numerical solution of (11.16) and (11.17) is more subtle than might appear at first glance. Conventional integration schemes for ordinary differential equations, such as classical Runge–Kutta methods, can lead to the non-conservation of energy over long-time simulations when applied to

Hamiltonian systems. This is clearly a problem, for example, when one is trying to understand the damping of collective excitations. Symplectic integrators (Sanz-Serna and Calvo, 1994; Yoshida, 1993), used extensively in molecular dynamics (MD) simulations, avoid this problem, having the desirable property of conserving phase-space volume and energy. They are the classical analogue of the split-operator method discussed earlier. To develop the method, it is convenient to work within the Lie formalism (Sanz-Serna and Calvo, 1994). For simplicity, we consider the dynamics of a single particle with classical Hamiltonian $H = T + V$. The evolution of its phase-space coordinate $\mathbf{z} = (\mathbf{p}, \mathbf{r})$ is then determined by the equation

$$\frac{d\mathbf{z}}{dt} = \{\mathbf{z}, H\} \equiv -i\mathcal{L}\mathbf{z}, \tag{11.18}$$

where $\{F, G\} = \sum_i (\partial_{r_i} F \partial_{p_i} G - \partial_{p_i} F \partial_{r_i} G)$ is a Poisson bracket and $\mathcal{L}$ is the Liouville operator (Prigogine, 1962). The formal solution of (11.18) is

$$\mathbf{z}(t + \Delta t) = e^{-i\mathcal{L}\Delta t}\mathbf{z}(t). \tag{11.19}$$

The BCH formula (11.14) can be used again to show that (Yoshida, 1993)

$$\begin{aligned} e^{-i\mathcal{L}\Delta t} = {} & e^{-i\mathcal{L}_T \Delta t/2} e^{-i\mathcal{L}_V \Delta t} e^{-i\mathcal{L}_T \Delta t/2} \\ & - \frac{\Delta t^3}{12}\Big\{\{T, V\}, \Big(V + \tfrac{1}{2}T\Big)\Big\} + \mathcal{O}(\Delta t^4)\,, \end{aligned} \tag{11.20}$$

where the Liouville operators are defined via $\mathcal{L}_V \mathbf{z} = i\{\mathbf{z}, V\}$ and $\mathcal{L}_T \mathbf{z} = i\{\mathbf{z}, T\}$. One can see the analogy with the quantum operator in (11.14); both are accurate to order Δt^2. The effect of the classical operator (11.20) in the simulations is to update the particle positions and velocities in three steps. In the first step, $\mathcal{L}_T$ advances the positions of the particles to

$$\tilde{\mathbf{r}}_i = \mathbf{r}_i(t) + \tfrac{1}{2}\mathbf{v}_i(t)\Delta t, \tag{11.21}$$

leaving the velocities unchanged. In the next step, $\mathcal{L}_V$ acts to change the velocities to

$$\mathbf{v}_i(t + \Delta t) = \mathbf{v}_i(t) - \frac{1}{m}\nabla V(\tilde{\mathbf{r}}_i)\Delta t\,, \tag{11.22}$$

and lastly, the particle positions are updated to their final values of

$$\mathbf{r}_i(t + \Delta t) = \tilde{\mathbf{r}}_i + \tfrac{1}{2}\mathbf{v}_i(t + \Delta t)\Delta t. \tag{11.23}$$

When the potential has an explicit time dependence, V is evaluated at $t + \Delta t/2$, the midpoint of the interval, producing a value analogous to $\tilde{V}(t)$ in (11.11). Repeating these steps determines the phase-space trajectories of the test particles.

11.2.1 Thermal cloud potential

In the simulations for a trapped Bose gas, the potential acting on the test particles consists of the external trapping potential and the mean-field interaction,

$$U(\mathbf{r},t) = V_{\rm trap}(\mathbf{r},t) + 2gn(\mathbf{r},t)\,, \tag{11.24}$$

with $n(\mathbf{r},t) = n_{\rm c}(\mathbf{r},t) + \tilde{n}(\mathbf{r},t)$. The effective potential U must be determined self-consistently as the system evolves in time. The mean field of the thermal cloud ($2g\tilde{n}$) is in general much weaker than that of the condensate ($2gn_{\rm c}$), owing to the larger spatial extent (and therefore lower density) of the thermal cloud. Nevertheless, it is necessary to include this term to ensure conservation of the total energy of the system. In addition, from the perspective of the condensate, the noncondensate mean field is necessary to account for the temperature-dependent damping and frequency shifts of the condensate collective modes.

Although the calculation of the condensate mean field is straightforward, the use of discrete particles with a contact interatomic potential creates a problem in determining the noncondensate mean field. Taken literally, the mean field consists of a series of delta peaks

$$\tilde{U}_{\rm tp}(\mathbf{r},t) = 2g\gamma \sum_{i=1}^{\tilde{N}_{\rm tp}} \delta[\mathbf{r} - \mathbf{r}_i(t)] \equiv 2g\tilde{n}_{\rm tp}(\mathbf{r},t)\,. \tag{11.25}$$

This expression clearly cannot be used to generate the forces acting on the test particles that are required in the MD simulation since the interaction only comes into play when particles coincide. Since the ensemble averaged thermal cloud density is smooth and differentiable, a smoothing operation must be performed on $\tilde{n}_{\rm tp}(\mathbf{r},t)$.

As a first step, we divide space into cubic cells Δx on a side and determine the mean density within each cell by binning the test particles appropriately. This histogram distribution provides a useful visual representation of the density, but is not differentiable, having spatial discontinuities on the scale of the three-dimensional (3D) grid. In addition, the cell densities exhibit jumps in time as particles migrate from one cell to another. Such jumps impart temporal fluctuations to the thermal cloud mean field that are an artifact of the test-particle description. In fact, the relative amplitude of these fluctuations depends on the number of test particles used in the simulation and will decrease as this number is increased. It is clear, however, that the binned density has some undesirable properties associated with the statistical fluctuations in the number and positions of particles in each cell.

A partial remedy for this problem is to use the cloud-in-cell method (Hockney and Eastwood, 1981), which involves a slightly different binning procedure. Rather than assigning particles to the cell as a whole, they are assigned to the cell corners that coincide with the spatial grid. This is most readily explained in one dimension (1D). A particle at a position x between two grid points at x_k and x_{k+1} is assigned to both points with weightings $1-\alpha$ and α respectively, where $\alpha = (x - x_k)/(x_{k+1} - x_k)$. This can be viewed as a more sophisticated binning procedure in that it takes into account the actual position of a particle within the cell. In the generalization to 3D a particle is assigned to the eight corners of a cell. Thus the density at a grid point is effectively an interpolation of the densities in the eight adjoining cells and, as such, is smoother than the histogram distribution. Furthermore, the density varies continuously in time as particles migrate from one cell to another.

Although smoother, the cloud-in-cell density defined above still exhibits undesirable spatial fluctuations, which makes calculation of the forces on the test particles problematic. What is needed is an additional smoothing procedure that generates a smooth and differentiable density. This can be achieved by performing a convolution with a sampling function $S(\mathbf{r})$ that is normalized to unity. Specifically, we define

$$\tilde{U}_S(\mathbf{r}, t) \equiv \int d\mathbf{r}' S(\mathbf{r} - \mathbf{r}') \tilde{U}_{\rm tp}(\mathbf{r}', t) = 2g\gamma \sum_{i=1}^{\tilde{N}_{\rm tp}} S(\mathbf{r} - \mathbf{r}_i), \tag{11.26}$$

where we choose $S(\mathbf{r}) = (\pi\eta^2)^{-3/2} e^{-r^2/\eta^2}$, i.e. an isotropic Gaussian of width η. Since $\nabla S|_{\mathbf{r}=0} = 0$, no force is exerted by a particle on itself and the sum can extend over all particles in the ensemble. Ideally, the width of $S(\mathbf{r})$ should be small compared with the scale on which the noncondensate density varies. If, at the same time, the number of particles contributing to the sum at a given position $\mathbf{r}$ is large, then it is clear that the sampled potential will be relatively smooth. Note that the smoothing operation is equivalent to replacing the contact interaction $g\delta(\mathbf{r})$ by a finite-ranged interatomic potential. Ideally, this model potential should yield results very similar to the contact interaction. The extent to which this is true can easily be checked by repeating a calculation for a range of η values.

The sampled potential (or its gradient) is needed at the position of each test particle and at the mesh points on which the condensate wavefunction is defined. However a direct summation of (11.26) over all test particles for each spatial point of interest would be prohibitive. We therefore proceed by again making use of an FFT. The advantage of doing so is that the

Fourier transform of a convolution is the product of the Fourier transforms. The Fourier transform of the cloud-in-cell density is first performed using an FFT and the result is multiplied by the analytic FT of the sampling function (also a Gaussian). An inverse FFT then generates the sampled potential on the 3D grid. This potential is used directly in the GP evolution, while the forces on the test particles are obtained by taking a numerical derivative and interpolating to the positions of the particles.

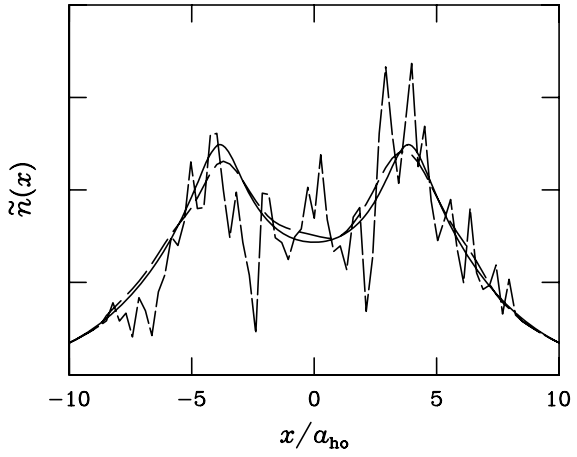


Fig. 11.1. Equilibrium noncondensate density vs. position, along a line through the centre of an isotropic trap (the system parameters are given in the main text). The distribution of $\tilde{N}_{\rm tp} = 4.0 \times 10^5$ test particles is generated by sampling according to the actual equilibrium density (solid line). The rapidly fluctuating broken line is the result of binning the test particles using a cloud-in-cell method, while the smooth broken line shows the effect of convolving the cloud-in-cell density with a Gaussian (from Jackson and Zaremba, 2002b).

To see how this scheme works, it is useful to consider a specific example. We consider an isotropic harmonic trap with trapping frequency $\omega_0/2\pi = 187\,\text{Hz}$, containing in total $N = 5 \times 10^4$ ^{87}Rb atoms, for which the ideal-gas transition temperature is $T_{\rm BEC} = 310.6$ nK. The equilibrium condensate and noncondensate densities are then calculated self-consistently for a temperature $T = 250\,\text{nK}$. For these conditions we have $\tilde{N} \simeq 4.0 \times 10^4$ thermal atoms, approximately 80% of the total number. A distribution of $\tilde{N}_{\rm tp} = 4.0 \times 10^5$ test particles is then generated using the method described at the end of Sec. 11.4, and the cloud-in-cell density is then constructed. The jagged broken line in Fig. 11.1 shows this density along a line through the centre of the trap. When compared with the correct equilibrium profile, shown by the solid line, the extent of the statistical fluctuations in the cloud-

in-cell density is evident. However, after applying the smoothing operation with a width parameter $\eta \simeq 0.76\, a_{\rm ho}$, where $a_{\rm ho} = \sqrt{\hbar/M\omega_0} \simeq 0.79\, \mu$m is the harmonic oscillator length for the trap under consideration, the convolved density shown by the smooth broken line is obtained. (For comparison, the mesh size is $\Delta x \simeq 0.27\, a_{\rm ho}$.)

It should be noted that the dramatic smoothing of the density achieved is partly a consequence of performing a full 3D convolution. A 1D convolution of the cloud-in-cell density with the same width parameter would not reduce the amplitude of the spatial fluctuations to the same degree. A comparison of the convolved density with the actual equilibrium density shows only minor differences, which are partly due to the statistical sampling of the test particles. In addition, one sees the expected broadening of the peaks in the thermal cloud density at the edges of the condensate. However, these small differences would not be expected to affect the dynamics of the system significantly. This can in fact be checked; small variations in η about the value chosen are found to have little effect. Since the thermal atoms also feel the mean field of the condensate, the $n_{\rm c}(\mathbf{r})$ term appearing in the effective potential $U(\mathbf{r})$ should also be convolved, for consistency.

11.3 Collisions

The methods outlined so far allow one to follow the condensate wavefunction and trajectories of thermal atoms subject to time-dependent effective potentials. However, in general, the collision integrals C_{12} and C_{22} in the Boltzmann equation will be nonzero and their effect on the dynamics must be included. In other words, during each time step there is a certain probability that a given test particle will collide with another thermal atom or with an atom in the condensate. If the typical collision time scale τ is such that $\tau \gg \Delta t$, one can treat the free particle evolution and collisions sequentially. Each particle's trajectory is first followed using the methods discussed in the previous section, and the possibility that collisions have occurred is then included at the end of the time step.

11.3.1 The C_{22} collision integral

The C_{22} collision integral in (3.40) or (6.46) corresponds physically to the scattering of pairs of thermal atoms into final thermal atom states. Within the N-body simulation, these collisions are accounted for by assigning a probability that two particles will collide when they are sufficiently close together. The determination of this probability is discussed in this subsec-

tion. As we shall see, Monte Carlo sampling is the essential link between the analytical expression for the C_{22} collision integral and the numerical simulations. Details of the Monte Carlo sampling technique are given in Appendix A.

Before making these connections, however, it is useful first to elaborate the physical significance of the C_{22} collision integral. Referring to (3.40), it is clear that $[d\mathbf{r}d\mathbf{p}/(2\pi\hbar)^3]C_{22}^{\rm out}$ represents the number of atoms leaving the phase-space volume element $d\mathbf{r}d\mathbf{p}/(2\pi\hbar)^3$ per unit time as a result of collisions. An integration over momenta thus gives the mean number of atoms in $d\mathbf{r}$ suffering a collision per unit time. Thus the quantity

$$\Gamma_{22}^{\rm out}(\mathbf{r},t) = \int \frac{d\mathbf{p}}{(2\pi\hbar)^3} C_{22}^{\rm out} \tag{11.27}$$

represents the mean number of atoms suffering a collision per unit volume and per unit time at the point $\mathbf{r}$ in space. In a general nonequilibrium situation, $\Gamma_{22}^{\rm out}$ is a function of both $\mathbf{r}$ and t. For convenience, we will often omit this explicit dependence. We note that $\Gamma_{22}^{\rm out}/2$ gives the *mean number of collisions* per unit volume and per unit time since two atoms are involved in each collision.

We now relate $\Gamma_{22}^{\rm out}(\mathbf{r},t)$ to the mean collision time $\tau_{22}(\mathbf{r},t)$. By definition, dt/τ_{22} is the probability that an atom chosen at random at position $\mathbf{r}$ will experience a collision in the time interval dt. In terms of this quantity, the number of atoms in $d\mathbf{r}$ experiencing a collision in the time interval dt is therefore $\tilde{n}(\mathbf{r},t)d\mathbf{r}(dt/\tau_{22})$, i.e. the number of atoms in the volume element $d\mathbf{r}$ multiplied by the probability that any of these atoms suffers a collision in the time interval dt. Comparing this with the definition of $\Gamma_{22}^{\rm out}$, we obtain the relation

$$\Gamma_{22}^{\rm out}(\mathbf{r},t) = \frac{\tilde{n}(\mathbf{r},t)}{\tau_{22}(\mathbf{r},t)}\,. \tag{11.28}$$

The explicit form of $\Gamma_{22}^{\rm out}$ is given by

$$\begin{aligned}\Gamma_{22}^{\rm out} = {} & \frac{\sigma}{\pi(2\pi\hbar)^6 m^2}\int d\mathbf{p}_1 f_1 \int d\mathbf{p}_2 f_2 \int d\mathbf{p}_3 \int d\mathbf{p}_4 \delta(\mathbf{p}_1+\mathbf{p}_2-\mathbf{p}_3-\mathbf{p}_4) \\ & \times \delta(\varepsilon_1+\varepsilon_2-\varepsilon_3-\varepsilon_4)(1+f_3)(1+f_4).\end{aligned} \tag{11.29}$$

For a given pair of incoming momenta $\mathbf{p}_1$ and $\mathbf{p}_2$, the outgoing momenta $\mathbf{p}_3$ and $\mathbf{p}_4$ are restricted by the momentum- and energy-conserving delta functions. To perform the integrations, it is convenient to introduce the

new momentum variables

$$\mathbf{p}_1 = \tfrac{1}{\sqrt{2}}(\mathbf{p}_0 + \mathbf{p}), \quad \mathbf{p}_2 = \tfrac{1}{\sqrt{2}}(\mathbf{p}_0 - \mathbf{p}),$$
$$\mathbf{p}_3 = \tfrac{1}{\sqrt{2}}(\mathbf{p}'_0 + \mathbf{p}'), \quad \mathbf{p}_4 = \tfrac{1}{\sqrt{2}}(\mathbf{p}'_0 - \mathbf{p}'). \tag{11.30}$$

Thus $\mathbf{p}_0$ is proportional to the centre-of-mass momentum $\mathbf{p}_1+\mathbf{p}_2$ of the incoming 1 and 2 particles, while $\mathbf{p}$ is proportional to the relative momentum $\mathbf{p}_1-\mathbf{p}_2$. The factors $1/\sqrt{2}$ ensure that the Jacobian of the transformation is unity. With these variables, the momentum and energy delta functions reduce to

$$\delta(\mathbf{p}_1+\mathbf{p}_2-\mathbf{p}_3-\mathbf{p}_4)\delta(\varepsilon_1+\varepsilon_2-\varepsilon_3-\varepsilon_4) = \frac{m}{\sqrt{2}}\delta(\mathbf{p}_0-\mathbf{p}'_0)\delta(p^2-p'^2). \tag{11.31}$$

Integrating over $\mathbf{p}'_0$ and p', (11.29) takes the simpler form

$$\Gamma_{22}^{\rm out} = \int \frac{d\mathbf{p}_1}{(2\pi\hbar)^3} f_1 \int \frac{d\mathbf{p}_2}{(2\pi\hbar)^3} f_2 \int \frac{d\Omega}{4\pi}\sigma|\mathbf{v}_1-\mathbf{v}_2|(1+f_3)(1+f_4), \tag{11.32}$$

where $\mathbf{p}_{3,4} = [\mathbf{p}_0 \pm p\hat{\mathbf{u}}(\Omega)]/\sqrt{2}$, with $\hat{\mathbf{u}}(\Omega)$ a unit vector in the direction specified by the solid angle Ω.

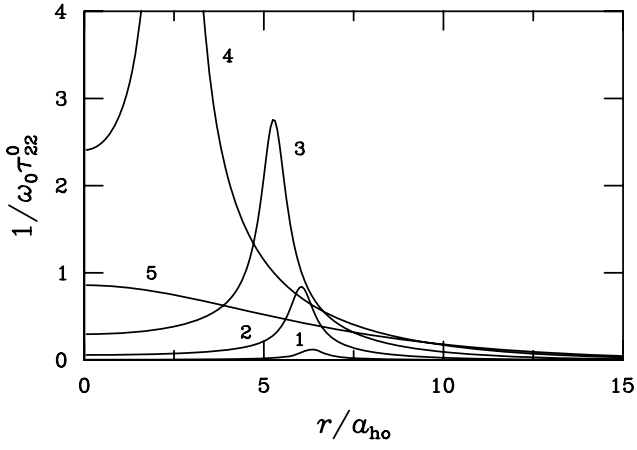


Fig. 11.2. The equilibrium local C_{22} collision rate $1/\tau_{22}^0$ in units of the trap frequency ω_0 as a function of the radial distance (in units of the harmonic oscillator length $a_{\rm ho} = \sqrt{\hbar/m\omega_0}$). The harmonic trap has a frequency $\omega_0/2\pi = 200$ Hz and contains 10^5 ^{87}Rb atoms. The curves labelled 1 through 5 correspond to temperatures from 100 nK to 500 nK, in steps of 100 nK (from Jackson and Zaremba, 2002a).

In equilibrium, $f = f^0$ and all quantities are independent of time. In this

case we have

$$\Gamma_{22}^0 \equiv \int \frac{d\mathbf{p}_1}{(2\pi\hbar)^3} f_1^0 \int \frac{d\mathbf{p}_2}{(2\pi\hbar)^3} f_2^0 \int \frac{d\Omega}{4\pi} \sigma |\mathbf{v}_1 - \mathbf{v}_2| (1 + f_3^0)(1 + f_4^0)$$
$$= \frac{\tilde{n}_0(\mathbf{r})}{\tau_{22}^0(\mathbf{r})}, \tag{11.33}$$

where $\tilde{n}_0(\mathbf{r})$ is the equilibrium thermal cloud density and $\tau_{22}^0(\mathbf{r})$ is the equilibrium version of the mean collision time defined via (11.28). In the classical (i.e. Maxwell–Boltzmann) limit, $1/\tau_{22}^0$ reduces to $\sqrt{2}\sigma v_{\rm th}\tilde{n}_0$, with $\sigma = 8\pi a^2$ and $v_{\rm th} = (8kT/\pi m)^{1/2}$.

It is instructive to plot $1/\tau_{22}^0$ as a function of position in a typical trapped gas. For this purpose we consider an isotropic trap with frequency $\omega_0/2\pi =$ 200 Hz containing 10^5 ^{87}Rb atoms. In Fig. 11.2 we show $1/\tau_{22}^0$ for a range of temperatures. Below $T_{\rm BEC} \simeq 419$ nK, we see that it is a strong function of position and is peaked at the edge of the condensate. With increasing temperature the peak moves to smaller radii as the condensate depletes and shrinks in size. The peak is due to a corresponding peak in the thermal atom distribution at this position and to the Bose enhancement of the collision process. Above $T_{\rm BEC}$ the collision rate drops dramatically.

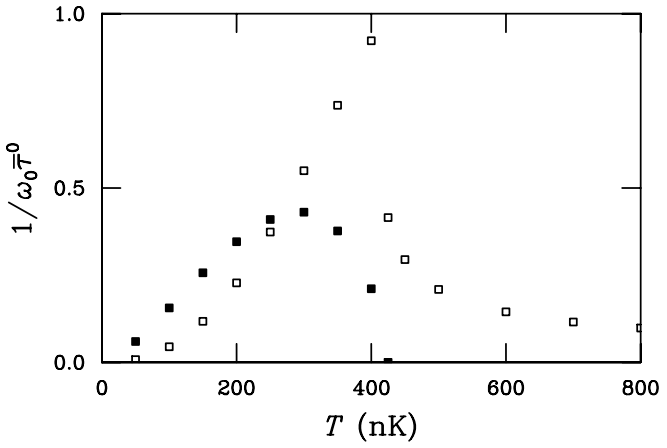


Fig. 11.3. Average collision rates per atom in units of the trap frequency ω_0, as a function of temperature. The solid squares are for C_{12} collisions and the open squares are for C_{22} collisions. The trap parameters are the same as in Fig. 11.2 (from Jackson and Zaremba, 2002a).

The fact that $\omega_0\tau_{22}^0 \ll 1$ in the region of the peak indicates that locally the gas is in the collisional hydrodynamic regime. However, in order to

determine the global conditions in the gas, it is useful to define an average collision rate for the thermal atoms:

$$\frac{1}{\bar{\tau}_{22}^0} \equiv \frac{1}{\tilde{N}} \int d\mathbf{r}\, \frac{\tilde{n}_0(\mathbf{r})}{\tau_{22}^0(\mathbf{r})}\,. \tag{11.34}$$

This average rate is plotted in Fig. 11.3, which shows the rate increasing monotonically with temperature up to $T_{\rm BEC}$ and then dropping suddenly as the gas becomes normal. Near $T_{\rm BEC}$, $\omega_0 \bar{\tau}_{22}^0 \simeq 1$ for these particular conditions. This limiting value increases with increasing N and indicates that the collisional hydrodynamic regime is easily reached near $T_{\rm BEC}$.

At this point it is worth repeating and emphasizing the physical significance of the collision times discussed in this section. The local mean collision time $\tau_{22}(\mathbf{r}, t)$ is related to the probability that a thermal atom at position $\mathbf{r}$ suffers a collision with another thermal atom. In other words, it is the mean time between collisions for an atom in the gas at this position and this instant of time. Thus, it has a concrete physical meaning even though its direct experimental measurement may be difficult. In Chapters 17–19, we discuss true hydrodynamic phenomena and the associated transport coefficients. Each of these has its own characteristic relaxation time, which is defined specifically for the transport property of interest. Although all these times differ from each other and from the collision time $\tau_{22}(\mathbf{r}, t)$ in their magnitudes and in their detailed spatial dependences, they all reflect the rate at which collisions in the thermal cloud are taking place.

Up to this point we have discussed the direct evaluation of the equilibrium collision rates using the analytical expressions provided. However, it is far from obvious how these rates are to be incorporated into the N-body simulations. The key idea in answering this question emerges when we consider the evaluation of (11.32) using a Monte Carlo sampling technique, as discussed in detail in Appendix A.

The evaluation of the $\Gamma_{22}^{\rm out}$ collision rate in (11.32) involves an integration over all possible incoming momenta $\mathbf{p}_1$ and $\mathbf{p}_2$. These integrals can be estimated by means of an appropriate Monte Carlo sampling of the integration variables. Using this procedure, we show in Appendix A that the average number of collisions occurring in $\Delta\mathbf{r}$ in a time interval Δt can be expressed as

$$\tfrac{1}{2}\Gamma_{22}^{\rm out} \Delta\mathbf{r} \Delta t \simeq \sum_{(ij)} P_{ij}^{22}\,, \tag{11.35}$$

where the sum extends over all pairs ij of particles in the volume element $\Delta\mathbf{r}$ taken two at a time, and P_{ij}^{22} is the probability that the pair ij suffers

a collision in the time interval Δt. This probability is given by

$$P_{ij}^{22} = \gamma\tilde{n}(\mathbf{r})\sigma|\mathbf{v}_i - \mathbf{v}_j| \int \frac{d\Omega}{4\pi}(1+f_3)(1+f_4)\Delta t\,, \tag{11.36}$$

where the factor $\gamma = \tilde{N}/\tilde{N}_{\rm tp}$ accounts for the fact that the number of test particles in the simulation is not the same as the actual number of atoms in the thermal cloud. Equation (11.36) involves an average over Ω corresponding to all possible pairs of outgoing velocities at opposite ends of a diameter of a sphere of radius $\frac{1}{2}|\mathbf{v}_1 - \mathbf{v}_2|$ centred on $\frac{1}{2}(\mathbf{v}_1 + \mathbf{v}_2)$. However, in a simulation the final velocities of the colliding pair ij must actually take on specific, albeit random, values. This is achieved by specifying the scattering angle $\Omega_{\rm R}$ in terms of uniformly distributed random values of the scattering variables $\cos\theta$ and ϕ and then defining the collision probability for this particular scattering event as

$$P_{ij}^{22} = \gamma\tilde{n}\sigma|\mathbf{v}_i - \mathbf{v}_j|(1+f_3^{\Omega_{\rm R}})(1+f_4^{\Omega_{\rm R}})\Delta t. \tag{11.37}$$

This probability depends upon the phase-space densities of the final states, $f_3^{\Omega_{\rm R}}$, $f_4^{\Omega_{\rm R}}$, reflecting Bose enhancement of the collision process. If this event-specific scattering probability is averaged over a random distribution of scattering angles $\Omega_{\rm R}$, then we recover the average probability defined in (11.36).

The simulation of C_{22} collisions thus proceeds as follows. In order to estimate the phase-space density $f(\mathbf{p}, \mathbf{r}, t)$ at each time step, the test particles are assigned to cells of volume $\Delta\mathbf{r}$ in position space and, within each cell, are binned in momentum space. The latter is done on a Cartesian grid, where a cloud-in-cell method is used to apportion particles to the momentum grid points. Since collisions are treated one cell at a time, the phase-space density need only be calculated and stored for the particular position-space cell of interest.

A pair of atoms ij within a given cell is then selected at random and a collision probability is assigned according to (11.37) using the phase-space density obtained from the binning procedure. To determine whether a collision actually takes place, a random number X_{ij}^{22}, uniformly distributed between 0 and 1, is chosen. If $X_{ij}^{22} < P_{ij}^{22}$, then the collision event occurs, and the velocities of the test particles are updated accordingly. If $X_{ij}^{22} > P_{ij}^{22}$, then the collision does not occur and the velocities of the colliding pair remain as they were. In either case, the pair ij is eliminated from the set and the next pair is selected at random. The above procedure is then repeated until all $\tilde{N}_{\rm tp}/2$ pairs of particles in the cell have been used. All the other cells are processed in the same way.

11.3.2 The C_{12} collision integral

The C_{12} collisions are treated in a similar manner to the C_{22} collisions. The key difference is that one collision partner is a condensate atom in a definite state, and it is necessary to distinguish the collisional processes that either transfer an atom into or out of the condensate. For example, the "out" collision rate is given by

$$\Gamma_{12}^{\rm out} = \frac{\sigma n_{\rm c}}{\pi m^2 (2\pi\hbar)^3} \int d\mathbf{p}_2 d\mathbf{p}_3 d\mathbf{p}_4 \, \delta(\mathbf{p}_{\rm c} + \mathbf{p}_2 - \mathbf{p}_3 - \mathbf{p}_4)\delta(\varepsilon_{\rm c} + \varepsilon_2 - \varepsilon_3 - \varepsilon_4) \times f_2(1+f_3)(1+f_4). \tag{11.38}$$

This represents the scattering of an incoming thermal atom (2) from the condensate to produce two outgoing thermal atoms (3 and 4). The designation "out" refers to the fact that an atom is leaving the condensate. Thus, $\Gamma_{12}^{\rm out}$ is the rate of increase of the number of thermal atoms per unit volume and per unit time as a result of the collision with a condensate atom. The reverse process gives the "in" collision rate of a thermal atom

$$\Gamma_{12}^{\rm in} = \frac{\sigma n_{\rm c}}{\pi m^2 (2\pi\hbar)^3} \int d\mathbf{p}_2 d\mathbf{p}_3 d\mathbf{p}_4 \, \delta(\mathbf{p}_{\rm c} + \mathbf{p}_3 - \mathbf{p}_2 - \mathbf{p}_4)\delta(\varepsilon_{\rm c} + \varepsilon_3 - \varepsilon_2 - \varepsilon_4) \times f_2(1+f_3)f_4. \tag{11.39}$$

In obtaining (11.39) we have interchanged the labels 2 and 3 in order to define an integral having the same f_2 weighting factor as in (11.38). Here the incoming thermal atoms are labelled 2 and 4 and their collision results in one outgoing thermal atom (3) with the addition of an atom to the condensate.

We consider $\Gamma_{12}^{\rm out}$ first. Using the same transformation as in (11.30), but with $\mathbf{p}_1$ replaced by $\mathbf{p}_{\rm c}$, we have

$$\Gamma_{12}^{\rm out} = \int \frac{d\mathbf{p}_2}{(2\pi\hbar)^3} f_2 n_{\rm c} \sigma v_{\rm r}^{\rm out} \int \frac{d\Omega}{4\pi}(1+f_3)(1+f_4), \tag{11.40}$$

where $v_{\rm r}^{\rm out} = \sqrt{|\mathbf{v}_{\rm c} - \mathbf{v}_2|^2 - 4(U-\mu_{\rm c})/m}$ plays the role of the relative velocity $|\mathbf{v}_1 - \mathbf{v}_2|$ in (11.32). It is understood that the integration over $\mathbf{p}_2$ in (11.40) is restricted by the condition that $|\mathbf{v}_{\rm c} - \mathbf{v}_2|^2 > 4(U-\mu_{\rm c})/m$. In the TF approximation, $U - \mu_{\rm c} = g n_{\rm c}$, which is the difference between the mean-field energy $g(n_{\rm c} + 2\tilde{n})$ of a condensate atom, and that of a thermal atom $2g(n_{\rm c} + \tilde{n})$. Thus, with the transfer of a condensate atom to the thermal cloud there is an increase $g n_{\rm c}$ in the potential energy. This requires the incoming thermal atom to have an excess kinetic energy (assuming that g is positive) and is the reason for the restriction on the range of integration.

The analysis of $\Gamma_{12}^{\rm in}$ is carried out somewhat differently. In this case, incoming thermal atoms 2 and 4 collide to produce an outgoing thermal

atom 3 and a condensate atom. The integral over $\mathbf{p}_3$ in (11.39) is carried out first to give

$$\Gamma_{12}^{\rm in} = \int \frac{d\mathbf{p}_2}{(2\pi\hbar)^3} f_2 \int \frac{d\mathbf{p}_4}{(2\pi\hbar)^3} f_4 \frac{n_{\rm c}\sigma h^3}{\pi m} \delta[(\mathbf{p}_2 - \mathbf{p}_{\rm c})\cdot(\mathbf{p}_4 - \mathbf{p}_{\rm c}) - mgn_{\rm c}](1+f_3)\,, \tag{11.41}$$

where $\mathbf{p}_3 = \mathbf{p}_2 + \mathbf{p}_4 - \mathbf{p}_{\rm c}$. The delta function dictates that $(\mathbf{p}_{\rm c} - \mathbf{p}_2)\cdot(\mathbf{p}_{\rm c} - \mathbf{p}_4) = mgn_{\rm c}$, which places a constraint on the allowed $\mathbf{p}_2$ and $\mathbf{p}_4$ momenta. Thus, unlike the case of C_{22} collisions, one cannot arbitrarily specify the momenta of the incoming atoms 2 and 4. To proceed, we perform the integration involving the delta function to obtain

$$\Gamma_{12}^{\rm in} = \int \frac{d\mathbf{p}_2}{(2\pi\hbar)^3} f_2 \frac{n_{\rm c}\sigma}{\pi v_{\rm r}^{\rm in}} \int d\mathbf{v}_\perp\, f_4(1+f_3), \tag{11.42}$$

where $\mathbf{v}_{\rm r}^{\rm in} \equiv \mathbf{v}_2 - \mathbf{v}_{\rm c}$ is the velocity of thermal atom 2 relative to the local condensate velocity. The second integral is with respect to a two-dimensional velocity vector $\mathbf{v}_\perp$ that is perpendicular to $\mathbf{v}_{\rm r}^{\rm in}$. The velocity of the other incoming thermal atom, particle 4, is given by

$$\mathbf{v}_4 = \mathbf{v}_{\rm c} + \mathbf{v}_\perp + \frac{gn_{\rm c}}{mv_{\rm r}^{\rm in}} \hat{\mathbf{v}}_{\rm r}^{\rm in}\,, \tag{11.43}$$

while the velocity of the outgoing thermal atom is

$$\mathbf{v}_3 = \mathbf{v}_2 + \mathbf{v}_\perp + \frac{gn_{\rm c}}{mv_{\rm r}^{\rm in}} \hat{\mathbf{v}}_{\rm r}^{\rm in}\,. \tag{11.44}$$

Although the expressions for $\Gamma_{12}^{\rm out}$ and $\Gamma_{12}^{\rm in}$ in (11.40) and (11.42) look very different, the two must be equivalent in equilibrium. We can thus use either expression to define an equilibrium C_{12} collision time that is analogous to Γ_{22}^0, (11.33). Using (11.40), we have

$$\frac{\tilde{n}_0(\mathbf{r})}{\tau_{12}^0(\mathbf{r})} = \int \frac{d\mathbf{p}_2}{(2\pi\hbar)^3} f_2^0 n_{\rm c}\sigma v_{\rm r}^{\rm out} \int \frac{d\Omega}{4\pi}(1+f_3^0)(1+f_4^0), \tag{11.45}$$

and the average C_{12} collision rate per thermal atom is (compare (11.34))

$$\frac{1}{\bar{\tau}_{12}^0} = \frac{1}{\tilde{N}} \int d\mathbf{r}\, \frac{\tilde{n}_0(\mathbf{r})}{\tau_{12}^0(\mathbf{r})}\,. \tag{11.46}$$

In Fig. 11.4 we show $1/\omega_0\tau_{12}^0$ as a function of the position in the trap for the same conditions as those of Fig. 11.2. This collision rate again has a maximum at the edge of the condensate but, unlike $1/\tau_{22}^0$ in Fig. 11.2, falls off rapidly beyond this point, being proportional to the condensate density $n_{\rm c}$. It is interesting to note that, for T approaching $T_{\rm BEC}$, we have $\omega_0\tau_{12}^0 \ll 1$ over the whole region of the condensate, indicating that a thermal atom has

a high probability of making a C_{12} collision in the time it takes to traverse the condensate. However, the number of thermal atoms overlapping the condensate is relatively small and the average collision rate defined in (11.46) is reduced considerably from the local value. This can be seen in Fig. 11.3, where $1/\omega_0\bar{\tau}_{12}^0$ is plotted as a function of temperature. This figure shows that $\omega_0\bar{\tau}_{12}^0 \simeq 1$ at elevated temperatures, and thus the thermal cloud is in the transition regime rather than the truly hydrodynamic regime, as Fig. 11.4 might lead one to believe. Of course the importance of these collisions increases with N, and one can expect to reach the collisional hydrodynamic regime for this particular trap configuration for N values that are about an order of magnitude larger (10^6).

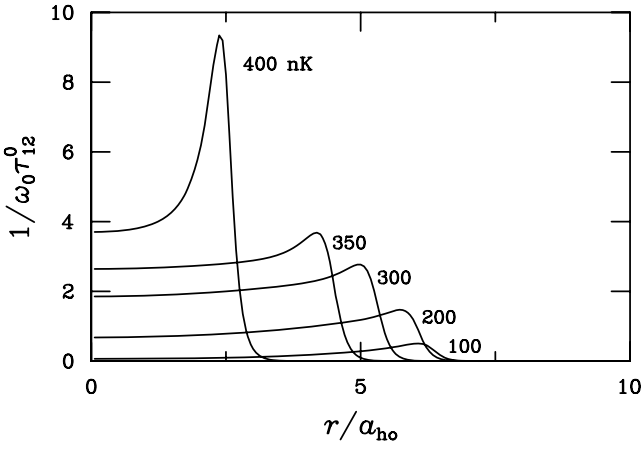


Fig. 11.4. The equilibrium local C_{12} collision rate, $1/\tau_{12}^0$, in a trap (in units of the trap frequency ω_0) as a function of the radial distance (in units of the harmonic oscillator length $a_{\rm ho}$). The physical conditions are the same as in Fig. 11.2 (from Jackson and Zaremba, 2002a).

The implementation of C_{12} collisions in the simulations is again achieved via a Monte Carlo sampling of the collision rates in (11.40) and (11.42). In Appendix A, we show that

$$\Gamma_{12}^{\rm out(in)}\Delta\mathbf{r}\Delta t \simeq \sum_{i=1}^{N_{\rm cell}} P_i^{\rm out(in)}\,, \tag{11.47}$$

where

$$P_i^{\rm out} = \gamma n_{\rm c}\sigma v_{\rm r}^{\rm out}\int\frac{d\Omega}{4\pi}(1+f_3)(1+f_4)\Delta t \tag{11.48}$$

is the probability that the ith particle in the cell is involved in an "out"-type

of collision and

$$P_i^{\rm in} = \gamma \frac{g n_{\rm c}}{\pi v_{\rm r}^{\rm in}} \int d\mathbf{v}_\perp \, f_4 (1 + f_3) \Delta t \tag{11.49}$$

is the probability that the ith particle is involved in an "in"-type of collision.

To make use of these probabilities in the simulations we replace the integrations in (11.48) and (11.49) by a random variable. In the case of "out" collisions, we use

$$P_i^{\rm out} = \gamma n_{\rm c} \sigma v_{\rm r}^{\rm out} (1 + f_3^{\Omega_{\rm R}})(1 + f_4)^{\Omega_{\rm R}} \Delta t \,, \tag{11.50}$$

where $\Omega_{\rm R}$ is a randomly selected scattering angle for the outgoing pair of atoms, whose velocities $\mathbf{v}_3$ and $\mathbf{v}_4$ lie on a sphere of radius $v_{\rm r}^{\rm out}/2$ centred on $(\mathbf{v}_{\rm c} + \mathbf{v}_2)/2$. Similarly, for "in" collisions we use

$$P_i^{\rm in} = \gamma \frac{g n_{\rm c} \mathcal{A}_v}{\pi v_{\rm r}^{\rm in}} f_4^{\mathbf{v}_\perp^{\rm R}} (1 + f_3^{\mathbf{v}_\perp^{\rm R}}) \Delta t \,, \tag{11.51}$$

where $\mathbf{v}_\perp^{\rm R}$ is a random velocity vector uniformly distributed over an area $\mathcal{A}_v$ sufficiently large to sample completely the occupied regions of phase space. If this probability is averaged over N_v random $\mathbf{v}_\perp^{\rm R}$ points then one obtains a Monte Carlo estimate of the $\mathbf{v}_\perp$ integral in (11.49). This average must, of course, be independent of $\mathcal{A}_v$, which implies that the simulation based on (11.51) is also independent of $\mathcal{A}_v$. As a final point, we note that we have defined the "in" collision probability for *each* atom in the cell, despite the fact that *two* incoming thermal atoms are involved in the collision. This procedure is correct since, from the point of view of the simulation, it is only necessary to ensure that the rate at which these collisions are occurring is properly reproduced. It is possible to define collision probabilities for *pairs* of atoms, as in the case of C_{22} collisions, but the present scheme is more convenient since it can then be implemented in the same way as the "out" collisions, which only involve one incoming thermal atom.

Whether an "out" or "in" collision actually takes place in the simulation is determined using a Monte Carlo sampling technique. Having selected a test particle in a given cell, we choose a random number X_i^{12} between 0 and 1. If $X_i^{12} < P_i^{\rm out}$ then an "out" collision occurs: the incoming test particle is removed from the ensemble and two new test particles are created. However, if $P_i^{\rm out} < X_i^{12} < P_i^{\rm out} + P_i^{\rm in}$ then an "in" collision takes place and the test particle is eliminated. Finally, if $X_i^{12} > P_i^{\rm out} + P_i^{\rm in}$, then neither type of collision occurs and the algorithm moves on to the next particle in the cell. Thus, only one of these three mutually exclusive outcomes occurs with each successive particle selection.

To complete the "in" collision event, a second test particle, atom 4, must

also be removed and a new thermal atom, atom 3, must be created. However, to define $P_i^{\rm in}$, f_4 must be known at the specific velocity given by (11.43). In the simulation f is represented by a binned distribution of test particles and in general there will not be a particle in the momentum cell with momentum $m\mathbf{v}_4$. We therefore expand the search for the required test particle to neighbouring phase-space cells, and remove one if it is found. Otherwise, $P_i^{\rm in}$ is set to zero. One can argue that this approach is justified as long as the collisional event is properly described in a statistical sense. The procedure that we have outlined does ensure that the probability for an "in" collision occurring is, on average, proportional to the local phase-space density f_4.

The change in the number of thermal atoms arising from C_{12} collisions must be compensated by a corresponding change in the number of condensate atoms. This occurs as a result of the non-Hermitian R-term in the generalized GP equation which is given by

$$R(\mathbf{r},t) = \frac{\hbar}{2n_{\rm c}(\mathbf{r},t)} \int \frac{d\mathbf{p}}{(2\pi\hbar)^3} C_{12} = \frac{\hbar}{2n_{\rm c}(\mathbf{r},t)} \left(\Gamma_{12}^{\rm out} - \Gamma_{12}^{\rm in}\right) . \qquad (11.52)$$

It is the net particle transfer rate that affects the normalization of the GP wavefunction. In a small time step Δt, the wavefunction change arising from R is

$$\Phi(\mathbf{r}, t+\Delta t) = e^{-R(\mathbf{r},t)\Delta t/\hbar}\Phi(\mathbf{r},t)\,, \qquad (11.53)$$

that is, it decreases in magnitude if $R > 0$ corresponding to $\Gamma_{12}^{\rm out} > \Gamma_{12}^{\rm in}$. In the simulation, R at a particular grid point $\mathbf{r}_{jkl}$ is estimated in terms of the "in" and "out" collision probabilities, i.e.

$$R(\mathbf{r}_{jkl},t)\Delta\mathbf{r}\Delta t = \frac{\hbar}{2n_{\rm c}} \sum_i (P_i^{\rm out} - P_i^{\rm in}). \qquad (11.54)$$

We note that this gives a finite value to R independently of the Monte Carlo sampling used to determine whether C_{12} collisions actually take place. The values of R at grid points can be obtained using a cloud-in-cell method similar to that described earlier in defining the thermal atom density.

According to (11.53) and (11.54), the normalization of the condensate wavefunction varies quasicontinuously in time as a result of the net collisional exchange rate. However, the same collisional events lead to a stochastic variation in the thermal atom number, which can change by only an integral number of atoms in each time step. Thus strict number conservation must necessarily be violated at any given instant of time. However, the cumulative effect of R ensures that the total particle number, $N = N_{\rm c} + \tilde{N}$, is conserved in a statistical sense. This can be confirmed by running a simulation for an

equilibrium state. Both the thermal atom and condensate numbers are found to fluctuate from one time step to the next, but the total particle number remains constant to within $\sqrt{N}$ statistical fluctuations. These fluctuations have no effect on the long-time dynamical evolution of interest.

Before leaving this general description we will mention one final detail in the practical implementation of the numerical simulations. Equations (11.38) and (11.39) give the true "out" and "in" collision rates in terms of the particle distributions. The cubic terms $f_2 f_3 f_4$ in these expressions cancel exactly in the total C_{12} collision integral. However, since the "in" and "out" collisional probabilities are calculated quite differently, this cancellation will not be precise numerically. The only way to ensure cancellation is to drop the cubic terms from the definition of the collision probabilities. Doing so results in new collision probabilities (distinguished by a bar)

$$\overline{P}_i^{\rm out} = n_{\rm c}\sigma v_{\rm r}^{\rm out}(1 + f_3^{\Omega_{\rm R}} + f_4^{\Omega_{\rm R}})\Delta t \tag{11.55}$$

and

$$\overline{P}_i^{\rm in} = \alpha \frac{g n_{\rm c}\mathcal{A}_v}{\pi v_{\rm r}^{\rm in}} f_4^{\mathbf{v}_\perp^{\rm R}} \Delta t\,, \tag{11.56}$$

which are to be compared with (11.50) and (11.51). These new definitions do change slightly the absolute number of "in" or "out" collisions that occur in the course of a simulation, but the *net* rate at which atoms are exchanged between the condensate and thermal cloud is given correctly.

11.4 Self-consistent equilibrium properties

An important application of the methods described above is to the calculation of small-amplitude collective oscillations around the equilibrium state. It is therefore necessary to know the equilibrium thermal cloud distribution and the condensate wavefunction. These quantities were discussed in Section 3.3. In this section, we discuss how these equilibrium densities are determined.

Since the condensate and thermal cloud are coupled by mean fields, the equilibrium densities have to be found self-consistently using an iterative procedure. The fact that the GP equation is itself nonlinear also dictates that an iterative method is needed. The general strategy for obtaining such a self-consistent solution can be summarized as follows. We suppose that, at some stage of the iteration process, an approximate condensate wavefunction and thermal cloud density are known. We denote these by $\Phi_0^{(i)}$ and $\tilde{n}_0^{(i)}$, respectively, where the superscript i is an iteration index. With

these quantities in hand, a new condensate effective potential is defined as

$$U_{\rm c}^{(i+1)} = V_{\rm trap} + g n_{\rm c}^{(i)} + 2g\tilde{n}_0^{(i)} \tag{11.57}$$

and the GP equation is solved as a linear Schrödinger equation to obtain $\Phi_0^{(i+1)}$ and the chemical potential $\mu_{\rm c}^{(i+1)}$. At this point the normalization of the wavefunction is retained as $N_{\rm c}^{(i)}$, and a new noncondensate effective potential is defined as

$$U^{(i+1)} = V_{\rm trap} + 2g\overline{n}_{\rm c}^{(i+1)} + 2g\tilde{n}_0^{(i)}, \tag{11.58}$$

where $\overline{n}_{\rm c}^{(i+1)} = |\Phi_0^{(i+1)}|^2$. The bar distinguishes this intermediate density from the one, yet to be defined, used in the next iteration of the wavefunction calculation. The potential $U^{(i+1)}$ together with the chemical potential $\mu_{\rm c}^{(i+1)}$ are now used to define a new thermal cloud density $\tilde{n}_0^{(i+1)}$, which integrates to $\tilde{N}^{(i+1)}$. The number of condensate atoms is then set to $N_{\rm c}^{(i+1)} = N - \tilde{N}^{(i+1)}$, the wavefunction $\Phi_0^{(i+1)}$ is renormalized to this value and a new condensate density $n_{\rm c}^{(i+1)} = |\Phi_0^{(i+1)}|^2$ is constructed using the renormalized wavefunction. With these ingredients, a new condensate effective potential can be defined and the cycle repeated.

Although this process is conceptually straightforward, there are a few subtle points. The first concerns the way in which iterations are initiated. At a given temperature T, neither the number of condensate atoms $N_{\rm c}$ nor the number of noncondensate atoms $\tilde{N}$ is known a *priori*. These values must emerge from the self-consistent calculation itself, with the constraint that $N_{\rm c} + \tilde{N} = N$, the total number of atoms. In this situation, the following strategy is found to be effective. The atom numbers are initially set as follows: $N_{\rm c}^{(0)} = N[1-(T/T_{\rm BEC})^3]$ and $\tilde{N}^{(0)} = 0$. The initial condensate wavefunction can be taken arbitrarily to be the ground state harmonic oscillator state for the trap of interest, with the chemical potential set to the ground state eigenvalue.[2] With these choices, the initial thermal cloud potential is $U^{(1)}(\mathbf{r}) = V_{\rm trap}(\mathbf{r}) + 2g n_{\rm c}^{(0)}(\mathbf{r})$, and the calculation then proceeds as described.

A second point is that the iterative procedure may fail to converge when the change in potentials from one iteration to the next is large. This problem can usually be remedied by mixing the new and old potentials, thereby reducing the size of the change. Specifically, we construct

$$\alpha U_{\rm c}^{(i+1)} + (1-\alpha) U_{\rm c}^{(i)} \rightarrow U_{\rm c}^{(i+1)}, \tag{11.59}$$

where α is a mixing parameter in the range 0 to 1. On the l.h.s. $U_{\rm c}^{(i+1)}$ is

[2] If a converged condensate wavefunction and thermal cloud density are available from an earlier calculation, these can be used to initiate the calculation at a new temperature.

given by (11.57) but is reduced by a factor α and mixed with the potential of the previous iteration to define the potential to be used in the next iteration. The value of α that yields the fastest rate of convergence can be found by experimentation, although in practice it is often sufficient to choose a value, other than this optimal value, that simply ensures convergence.

One final point has less to do with the numerical methods than with the semiclassical approximation used for the thermal cloud itself. We have assumed implicitly that the chemical potential lies below the minimum of the thermal cloud effective potential, $U_{\rm min}$, but there is no fundamental reason why this should be the case. Since the condensate is determined quantum mechanically by the GP equation, the chemical potential is necessarily above the minimum of $U_{\rm c}$. Since $U - U_{\rm c} = gn_{\rm c}$, $U_{\rm min} > (U_c)_{\rm min}$ if we are considering repulsive interactions. Nevertheless, it is possible that $(U_c)_{\rm min} < U_{\rm min} < \mu_{\rm c}$. For isotropic traps this turns out to be the case for temperatures approaching $T_{\rm BEC}$ when the condensate density is small. The situation for highly anisotropic traps is somewhat more restrictive, since the inequality is violated at lower temperatures. When this happens one must artificially set U equal to the chemical potential in regions where U falls below the chemical potential.

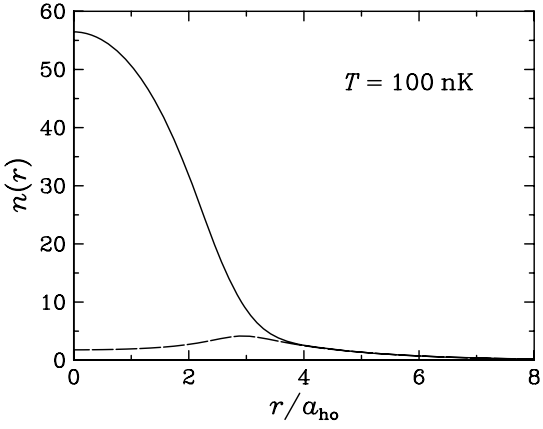


Fig. 11.5. Density of atoms (in units of $a_{\rm ho}^{-3}$) as a function of position (in units of $a_{\rm ho} = \sqrt{\hbar/m\omega_0}$) in an isotropic parabolic trap containing 5000 ^{87}Rb atoms. The trap frequency is $\omega_0/2\pi = 200$ Hz. The solid line is the total density and the broken line is the noncondensate density (from Zaremba *et al.*, 1999).

We show in Fig. 11.5 an example of such a self-consistent calculation. The suppression of the thermal component in the region of the condensate is due to the repulsive condensate mean field. This effect is clearly seen in

the effective potentials plotted in Fig. 11.6. The mean-field potential of the condensate acting on itself leads to the flat-bottomed potential shown. This implies that the condensate wavefunction has a nearly constant curvature in this region, which in turn implies the nearly parabolic profile characteristic of the Thomas–Fermi approximation. The effective potential for the noncondensate, however, shows a local minimum at the edge of the condensate which leads to a peak in the thermal cloud density. Notice that the minimum of the noncondensate potential does lie above the chemical potential in this case.

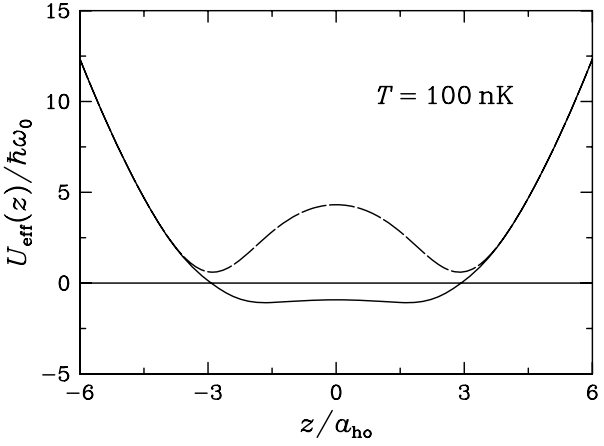


Fig. 11.6. The condensate (solid line) and noncondensate (broken line) effective potentials (in units of $\hbar\omega_0$), relative to the condensate chemical potential $\mu_{\rm c}$, as a function of position (in units of the oscillator length $a_{\rm ho}$). The system parameters are the same as in Fig. 11.5.

To represent the thermal cloud in the simulations, an ensemble of test particles must be defined. In the equilibrium case, the ensemble should be representative of the Bose equilibrium distribution in (3.49). This can be achieved using the following *rejection algorithm* (see Press *et al.*, 1992). First, particles are distributed in position space according to the density $\tilde{n}(\mathbf{r})$. This is done by selecting three random numbers uniformly and independently distributed between $-r_{\max}$ and $r_{\max}$, which define the Cartesian coordinates, $\mathbf{r}_i$, of a particle in the occupied region of position space. Another uniform deviate is then chosen from $R_i^1 \in [0, \tilde{n}_{\max}]$, where $\tilde{n}_{\max} \geq \max\{\tilde{n}(\mathbf{r})\}$, and compared with the density $\tilde{n}(\mathbf{r}_i)$ at that point. If $R_i^1 > \tilde{n}(\mathbf{r}_i)$, then the particle is discarded and another set of position coordinates is selected. If $R_i^1 < \tilde{n}(\mathbf{r}_i)$, however, the particle is included in the test particle ensemble. The particle's momentum is then specified according to the equilibrium

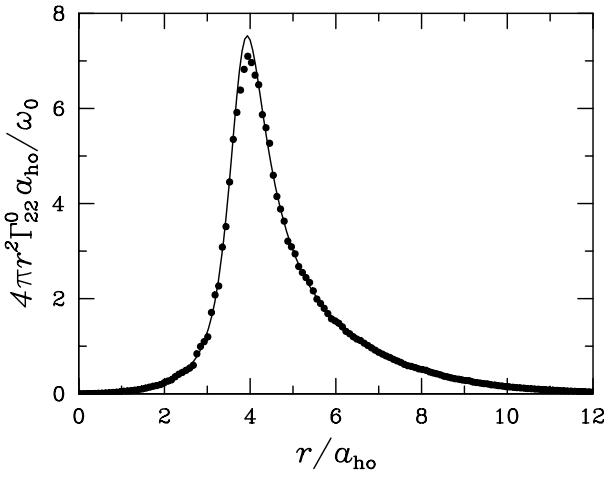


Fig. 11.7. The collision rate $4\pi r^2\Gamma^0_{22}(r)$ ($\times 10^{-3}a_{\rm ho}/\omega_0$) as a function of position, for the system considered in Fig. 11.1. The solid line is obtained using (11.33), while the points are the results of a Monte Carlo calculation using (11.60) (from Jackson and Zaremba, 2002b).

Bose distribution. Since the distribution is isotropic in momentum space, the magnitude of the momentum, p_i, is first selected at random from a uniform distribution on $[0, p_{\rm max}]$. A random number $R_i^2 \in [0, f_{\rm max}]$, where $f_{\rm max} \geq z(\mathbf{r}_i)/[1 - z(\mathbf{r}_i)]$ with $z(\mathbf{r}_i)$ the local fugacity, is compared with $f(p_i, \mathbf{r}_i)$ to decide whether the momentum is to be accepted or rejected. In the case of rejection another p_i is chosen, while if the momentum is accepted, then two random angles are selected, $\phi \in [0, 2\pi]$, $\cos\theta \in [-1, 1]$, which in turn define the momentum vector $\mathbf{p}_i$. This procedure is repeated until a total of $\tilde{N}_{\rm tp}$ particles in the ensemble have been accumulated.

Self-consistent calculations of the kind discussed in this section were carried out by Gerbier *et al.* (2004b) in connection with their measurements, but within the TF approximation for the condensate. We refer to Fig. 3.1 and the discussion at the end of Section 3.3.

11.5 Equilibrium collision rates

In this section we present the results of calculations that check the Monte Carlo sampling technique used to evaluate the C_{12} and C_{22} collision rates in a dynamical simulation. The system we consider is the same as that described in connection with Fig. 11.1. The equilibrium collision rate Γ^0_{22} can be evaluated numerically directly from the expression (11.33). The

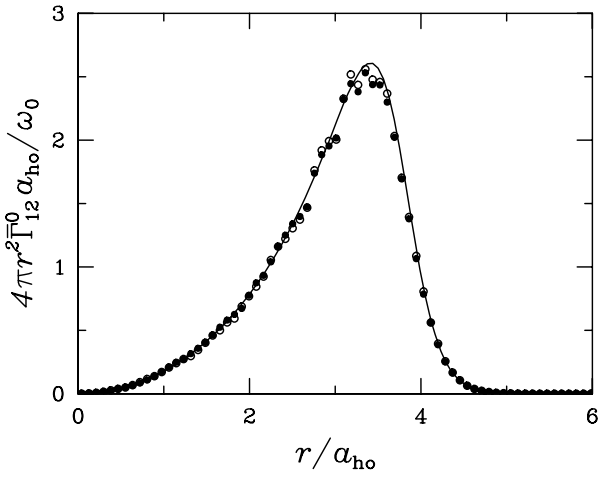


Fig. 11.8. Same parameters as in Fig. 11.7, for $\bar{\Gamma}^0_{12}$ collisions between the condensate and thermal cloud in equilibrium. (The bar signifies that Γ^0_{12} is calculated using the probabilites in (11.55) and (11.56).) The solid line displays the result of a direct evaluation of (11.38) ($\times 10^{-3}$), while the circles show a Monte Carlo calculation for the "out" rate (solid) and "in" rate (open) (from Jackson and Zaremba, 2002b).

result as a function of the radial coordinate r is shown as the solid line in Fig. 11.7. Since $4\pi r^2 \Gamma^0_{22} = 4\pi r^2 \tilde{n}_0/\tau^0_{22}$, the ordinate variable in Fig. 11.7 is the integrand defining the average collision rate in (11.34). Thus the area under the graph divided by the number of thermal atoms gives the mean collision rate $1/\bar{\tau}^0_{22}$.

The equilibrium C_{12} collision rates can also be calculated using the equilibrium thermal cloud distribution f^0 and the equilibrium condensate density $n^0_{\rm c}(r)$ in (11.38) or (11.39). The "in" and "out" rates are in fact equal to each other in equilibrium and will be denoted $\bar{\Gamma}^0_{12}$ (the bar is a reminder that these rates are calculated ignoring the cubic terms in the full expression). The result of the calculation as a function of r is shown as the solid line in Fig. 11.8. Both the C_{12} and C_{22} collision rates exhibit a maximum near the condensate surface (the r^2 weighting exaggerates the effect), where the fugacity z approaches unity and the equilibrium Bose distribution is strongly peaked at $\mathbf{p} = 0$. However, in the case of C_{22} collisions the tail of the distribution falls off more slowly since the thermal cloud extends to larger distances than the condensate.

The Monte Carlo collision rates are obtained by performing a dynamical simulation for a sample of test particles that represents the equilibrium distribution. The particles move in the equilibrium effective potential and

their trajectories are followed using the classical equations of motion. Since collisions should not change the equilibrium distribution, it is sufficient to consider the collisionless dynamics of the particles in this particular simulation. This dynamics provides an ergodic sampling of the equilibrium collision rates along the phase-space trajectories of each test particle. Throughout these calculations the condensate wavefunction $\Phi_0(\mathbf{r})$ remains fixed in time.

At each time step Δt in the evolution, the distribution of particles is binned in cells of volume $\Delta\mathbf{r}$ and the collision probabilities (11.37), (11.50) and (11.51) are calculated and summed to obtain a realization of the collision rates in each cell at a particular instant of time t_n. In the case of C_{22} collisions we have

$$\Gamma_{22}^0(t_n)\Delta\mathbf{r} \simeq 2\sum_{(ij)} \frac{P_{ij}^{22}(t_n)}{\Delta t}\,,$$

where the sum extends over all pairs of test particles in a particular cell. To obtain a histogram of the radial distribution $4\pi r^2\Delta r\Gamma_{22}^0$ as a function of the radial position r, it is clearly sufficient to allocate the individual collision probabilities to bins of width Δr according to the radial position of the colliding pair. By repeating the calculation over M successive time steps we can obtain the ergodic average

$$\langle\Gamma_{22}^0\rangle 4\pi r^2\Delta r = \frac{2}{M}\sum_{n=1}^{M}\sum_{(ij)} \frac{P_{ij}^{22}(t_n)}{\Delta t}\,, \tag{11.60}$$

where it is understood that the summand is accumulated in the radial bin of interest. These values can then be plotted against the radial coordinate of the midpoint of the radial bin. The points shown in Fig. 11.7 were obtained with only $M = 200$ time steps of size $\omega_0\Delta t = 0.002$. The results already show good statistics and agree very well with the direct numerical calculations. The main error arises from estimating $f(\mathbf{p},\mathbf{r},t)$ in real time by binning particles into phase-space cells. This was confirmed by repeating the simulation but calculating the collision probabilities using the actual equilibrium Bose distribution f^0 rather than the binned approximation to it. In this case the distribution is smoother and does not exhibit the clustering of points seen in Fig. 11.7. One can try to improve the binned distribution but there is a trade-off between using smaller phase-space cells which would provide a more accurate representation of the distribution, and larger cells which contain more particles and thus improve the statistics. The cell size used to obtain the results in Fig. 11.7 was chosen to optimize these opposing requirements.

The same procedure for the C_{12} "out" and "in" collision rates leads to the results in Fig. 11.8. In this case there are two sets of points corresponding to the different forms of the "out" and "in" probabilities in (11.50) and (11.51). The similarity of the two sets of points confirms the equivalence of these expressions for the equilibrium probabilities. The calculations also confirm that the "in" rate is independent of the arbitrary area $\mathcal{A}_v$ in (11.51).

It is of course important that the difference between equilibrium "in" and "out" rates be small since any imbalance implies a net, and spurious, transfer of atoms between the condensate and thermal cloud. In the present case, on integrating the data in Fig. 11.8 we find a small discrepancy between the total "in" and "out" rates, of about 1%. This imbalance can be minimized by a judicious choice of the shape of the phase-space cells [3] and by using a larger sample of test particles, but a residual numerical imbalance is unavoidable. Fortunately, collective mode frequencies and damping rates are only weak functions of the number of condensate atoms, and a small extraneous drift in the condensate number will not affect the results significantly.

[3] It is advantageous to use real-space cells that enclose regions of almost constant thermal density, using a spatial grid that reflects the quasi-elliptical geometry of the cloud.

12

Simulation of collective modes at finite temperature

In this chapter we present several dynamical simulations that make use of the numerical methods discussed in the previous chapter. Some of these are model simulations that are not directly linked to experiment but are designed to investigate some aspect of the dynamical behaviour. Others are performed with the express purpose of explaining specific experimental data. Both kinds of simulation serve to illustrate the range of nonequilibrium phenomena that can be studied in ultracold Bose gases using the ZNG equations.

A dynamical simulation is typically initiated in one of two ways. Either an appropriate nonequilibrium initial state is imposed on the system, or the system, initially in equilibrium, is dynamically excited by the application of an external perturbation. The latter parallels the procedure used experimentally to study small-amplitude collective excitations and usually amounts to some parametric modulation of the trapping potential. However, this approach may not always be feasible if the excitation phase requires a prohibitively long simulation time. In this case, the best one can do is to specify some initial nonequilibrium state that represents the experimental situation as closely as possible. This is not an issue in model simulations, where we are at liberty to specify the initial state in whatever way serves our purpose.

In Sections 12.1–12.3 we present three examples of model simulations. All essentially check some aspect of the numerical procedures. By studying the equilibration of an initial nonequilibrium state in Section 12.1, we confirm that the total number of atoms is conserved to a very good approximation during the course of the evolution. This is a nontrivial result since, as explained at the end of subsection 11.3.2, the numbers of condensate and thermal atoms change in quite different ways. In Section 12.2, we study dipolar oscillations of the trapped gas and demonstrate that the simulations

respect a known exact property, namely the separation of the centre-of-mass and internal degrees of freedom in a harmonic trap (the so-called generalized Kohn theorem). Finally, in Section 12.3 we examine in detail the effect of collisions on the radial breathing mode in an isotropic trap. One advantage of numerical simulations over real experiments is that the different collisional mechanisms can be turned off at will and can thus be studied in isolation.

We then proceed in Sections 12.4–12.6 with a discussion of simulations which correspond to real experiments. Three different modes are considered: the scissors mode, the quadrupole mode and the transverse breathing mode. Each has its own points of interest regarding the coupled dynamics of the condensate and thermal cloud. The results of simulations based on the ZNG theory are found to be in excellent quantitative agreement with experiment. These comparisons confirm the accuracy of the theory in dealing with finite-temperature dynamics in the weakly-collisional, as opposed to hydrodynamic, regime.

12.1 Equilibration

Our first example corresponds to the "quench" of an initial distribution of atoms in thermal equilibrium. Such a quench can be realized experimentally using evaporative cooling (Ketterle and van Druten, 1996), which selectively removes high energy atoms from the trap. We consider here, however, a somewhat different kind of quench that cannot be realized experimentally but is easy to impose in a simulation.

We start with the system considered at the end of Section 11.2 (see Fig. 11.1). This consists of $N = 5 \times 10^4$ ^{87}Rb atoms in an isotropic trap with frequency $\omega_0/2\pi = 187$ Hz, at an initial temperature $T_0 = 200$ nK. At this temperature, the number of condensate atoms is 2.58×10^4, roughly half the total number in the trap. The equilibrium density of the thermal cloud is given by (3.57) in terms of the local equilibrium fugacity $z_0(\mathbf{r}) = \exp[\beta_0(\mu_0 - U_0(\mathbf{r})]$. The temperature of the system is now suddenly reduced by a factor 2. This is achieved by assigning momenta according to the distribution $f(\mathbf{p}, \mathbf{r}) = (\beta/\beta_0)^{3/2}[z_0(\mathbf{r})^{-1} \exp(\beta p^2/2m) - 1]^{-1}$, where $\beta = 2\beta_0$, using the rejection algorithm described in Section 11.4. The prefactor ensures that this initial nonequilibrium state has the same density $\tilde{n}_0(\mathbf{r})$ as the original equilibrium state. Since the cloud at the temperature $T = T_0/2$ is too "cold" to be at positions of high potential energy, it begins to collapse in real space. At the same time, there is a net transfer of atoms from the thermal cloud to the condensate as a result of C_{12} collisions.

In Fig. 12.1, we show the evolution of the condensate and noncondensate

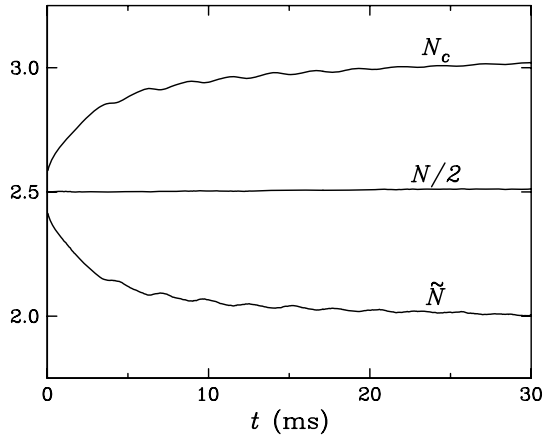


Fig. 12.1. The number ($\times 10^{-4}$) of condensate (N_c) and noncondensate ($\tilde{N}$) atoms and half the total number ($N/2$) plotted as a function of time following a momentum quench. The trap frequency is $\omega_0/2\pi = 187$ Hz, $N = 5 \times 10^4$ ^{87}Rb atoms and the initial temperature is $T_0 = 200$ nK (from Jackson and Zaremba, 2002a).

numbers as a function of time. One can see that the condensate number relaxes to a higher value, corresponding to a final equilibrium temperature $T_f < T_0$. This happens on two time scales. The shorter time scale is about 2–3 ms, which is consistent with the mean equilibrium C_{12} collision rate at 200 nK shown in Fig. 11.3. The longer time scale, of the order of tens of ms, is associated, as we shall see, with the relaxation of the internal dynamics of the cloud. As N_c increases there is a compensating decrease in $\tilde{N}$. The figure also gives half the total number of atoms, with $N = N_c + \tilde{N}$, as a function of time in the simulation. There are small fluctuations in $N/2$ superimposed on a very slight upward drift, both of which can just be discerned. As explained at the end of subsection 11.3.2, the variation in N is due to the different ways in which the the condensate and noncondensate numbers change in the simulation. This variation is too small to have a significant effect on the time dependences of the other variables plotted in Fig. 12.1 and so, for all practical purposes, the total particle number N is conserved.

We see in Fig. 12.1 a periodic ripple superimposed on the exponential-like relaxation of the particle numbers. This modulation is associated with a radial oscillation of the thermal cloud and condensate that is induced by the quench. In Fig. 12.2 we show the root-mean-square radii $\bar{r}_i \equiv \sqrt{\langle r^2(t)\rangle_i}$

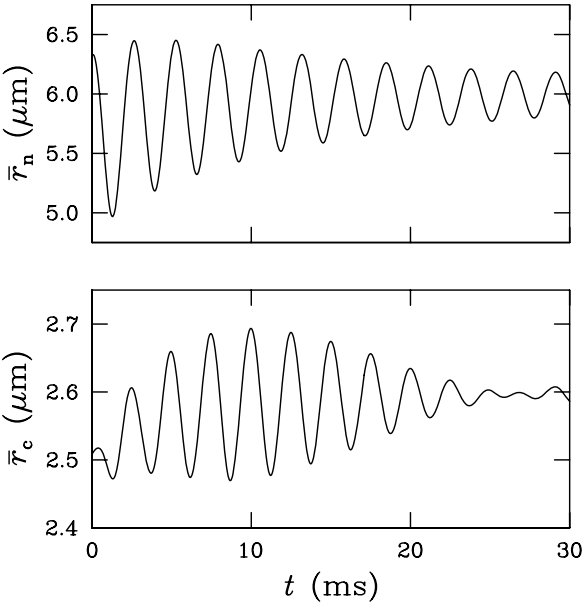


Fig. 12.2. The root-mean-square radius, $\bar{r} = \sqrt{\langle r^2 \rangle}$, as a function of time following a momentum quench. $\bar{r}_n$ and $\bar{r}_c$ are the radii of the thermal cloud and condensate, respectively (from Jackson and Zaremba, 2002a).

of the thermal cloud and condensate, where

$$\langle r^2(t) \rangle_n \equiv \frac{1}{\tilde{N}_{tp}(t)} \sum_{i=1}^{\tilde{N}_{tp}} r_i(t)^2 \quad \text{and} \quad \langle r^2(t) \rangle_c \equiv \frac{1}{N_c(t)} \int d\mathbf{r}\, r^2 n_c(\mathbf{r}, t)\,. \tag{12.1}$$

The thermal cloud (upper panel) at first decreases in size, reaches a maximally compressed state and then rebounds. Subsequently, it oscillates at a frequency very close to the monopole mode frequency $2\omega_0$ of a harmonically trapped noninteracting thermal cloud. The amplitude of the oscillations diminishes on a time scale of 10–20 ms that is determined by the rate of transfer of energy from the collective motion of the thermal cloud to the random thermal motion of individual atoms. In the lower panel of Fig. 12.2 we see similar oscillations in the radius of the condensate; these first build up in time and then damp out as the system approaches its final state of equilibrium. This clearly demonstrates the important effect that the dynamic mean field of the thermal atoms has on the condensate. As a result of the net transfer of particles from the thermal cloud to the condensate, mentioned earlier, the condensate grows in size and its mean radial extent

approaches a value about 0.1 μm (4%) larger than the intial value. The frequency of the condensate oscillations is slightly higher than that of the thermal cloud and about 5% lower than the TF frequency, $\sqrt{5}\omega_0$ (Stringari, 1996b), of the monopole mode in a pure condensate. This downward shift of the mode frequency is due to the fact that the condensate oscillation is being driven by the thermal cloud.

It should be emphasized that simulations of the kind discussed here could be used to analyze the condensate growth behaviour observed experimentally (Miesner *et al.*, 1998). This would require imposing a quench that corresponded more closely to the evaporative cooling procedure used in the experiments. Some work in this direction has been done, but only within the so-called ergodic approximation for the thermal cloud (Bijlsma *et al.*, 2000). In this approximation, it is assumed that the distribution function f depends on the phase-space variables only through the energy variable $E(\mathbf{p}, \mathbf{r}, t)$, i.e. $f(\mathbf{p}, \mathbf{r}, t) \equiv g(E(\mathbf{p}, \mathbf{r}, t), t)$. Even these rather approximate calculations account qualitatively for the condensate growth curves observed in the experiments. However, since the full dynamics of the thermal cloud is not included in the ergodic approximation, such calculations cannot describe the oscillations in the size of the thermal cloud that are seen following the quench (Miesner *et al.*, 1998). It would clearly be of interest to investigate the condensate growth problem in more detail by performing full dynamical simulations of the ZNG equations.

12.2 Dipole oscillations

Our second example serves as a very stringent test of the numerical procedures. For a harmonically confined gas, the centre-of-mass degree of freedom is separable from all other internal degrees of freedom. As a result, the centre of mass exhibits a dipole oscillation at the trap frequency ω_0 that is independent of interactions and the internal dynamical state of the gas. Similar behaviour is observed in other systems, for example, interacting electrons in a harmonic potential, where the term "generalized Kohn theorem" was introduced to describe this behaviour (for a review see Dobson, 1994).

A particularly simple example of dipole oscillations occurs when a trapped gas is displaced rigidly from its equilibrium position. The internal state of the gas is not affected by the displacement and the gas oscillates without change in shape, i.e. $n(\mathbf{r}, t) = n_0(\mathbf{r} - \boldsymbol{\eta}(t))$, with $\boldsymbol{\eta}(t) = \boldsymbol{\eta}_0 \cos \omega_0 t$. As proved in Section VI of Zaremba *et al.* (1999), the ZNG theory is formally consistent with the generalized Kohn theorem. Our purpose here is to demonstrate

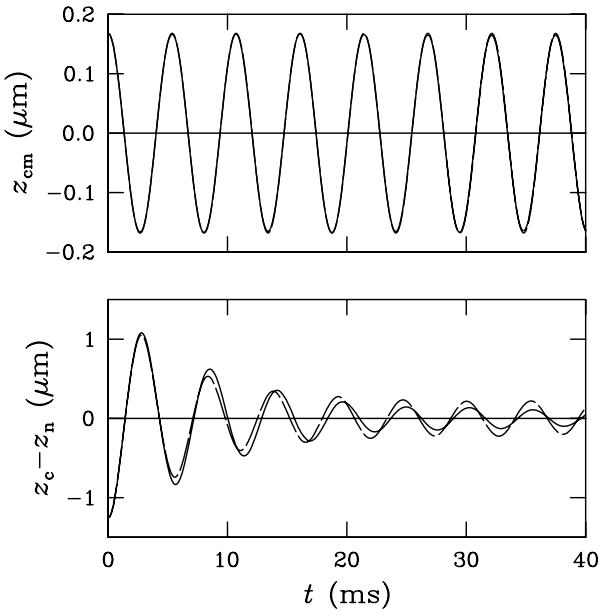


Fig. 12.3. The centre-of-mass position $z_{\rm cm}$, and the separation between the mean condensate and noncondensate positions $z_{\rm c} - z_{\rm n}$, as a function of time. The initial conditions are given in the main text. The broken curves correspond to the absence of collisions, while the solid curves include both C_{12} and C_{22} collisions (from Jackson and Zaremba, 2002a).

that our numerical solution of the coupled ZNG equations also satisfies this requirement.

The simulations were carried out for the system described in the previous section but at a higher temperature ($T_0 = 225$ nK), at which the difference between the condensate and noncondensate numbers is greater.[1] Once the equilibrium condensate wavefunction and thermal atom distribution have been calculated, the dipole oscillation is excited by displacing each component by a small distance Δz relative to the trap, but in opposite directions. This is easily achieved by shifting the condensate wavefunction by a distance $-\Delta z$ corresponding to a certain number of grid points on the spatial mesh and adding a constant displacement Δz to the position of each test particle. The two components are then released and their subsequent motion is followed in time.

In the first simulation to be described, all collisional processes are turned off. The two components then interact only through the mean-field poten-

[1] A larger number difference is desirable since it gives rise to an appreciable value of $z_{\rm cm}(0)$ for the particular excitation being used.

tials. The mean displacement of the condensate, $z_{\rm c}(t) = \int d\mathbf{r} z n_{\rm c}(\mathbf{r},t)/N_{\rm c}$, and that of the thermal cloud, $z_{\rm n}(t) = \sum_{i=1}^{\tilde{N}_{\rm tp}} z_i(t)/\tilde{N}_{\rm tp}$, are then calculated as a function of time. In Fig. 12.3, we plot the centre-of-mass position

$$z_{\rm cm}(t) = \frac{1}{N}\left(N_{\rm c} z_{\rm c}(t) + \tilde{N} z_{\rm n}(t)\right), \tag{12.2}$$

and the relative displacement

$$z(t) = z_{\rm c}(t) - z_{\rm n}(t). \tag{12.3}$$

The $t = 0$ amplitude of the centre-of-mass motion, $z_{\rm cm}(0) = \Delta z(\tilde{N} - N_{\rm c})/N$, is nonzero because of the different values of $\tilde{N}$ and $N_{\rm c}$ in the equilibrium state.

We see that $z_{\rm cm}(t)$ oscillates at the trap frequency with no change in amplitude, as required by the generalized Kohn theorem. But the simplicity of the centre-of-mass oscillation belies the complexity of the internal dynamics of the two components. This is revealed by the decay of the relative displacement $z(t)$ as a function of time. Since the two components are moving through each other, the mean field that each exerts on the other is continually changing in time and as a result, the density distributions of the individual components will show strong temporal variations. This would not be true if both components were initially displaced by $z_{\rm cm}(0)$ in the *same* direction. In this case, there would not be any relative motion of the two components ($z(t)$ would be constant), and their density distributions would retain their shapes. The fact that the centre-of-mass oscillation is indeed independent of the complex internal dynamics demonstrates that the numerics is consistent with the generalized Kohn theorem.

These results can be interpreted in terms of a two-mode model (Zaremba *et al.*, 1999). In the centre-of-mass mode, the condensate and noncondensate oscillate together without distortion at frequency ω_1, so that $z_{\rm c}^{(1)}(t)/z_{\rm n}^{(1)}(t) = 1$. The second mode at frequency ω_2 is an out-of-phase dipole mode in which the centre-of-mass remains fixed in position, so that $z_{\rm c}^{(2)}(t)/z_{\rm n}^{(2)}(t) = -\tilde{N}/N_{\rm c}$. Within this picture, the initial displacements $\pm\Delta z$ of the two components excite a superposition of the two modes. Expressing the individual displacements as a linear combination of the mode eigenvectors, we have

$$\begin{aligned} z_{\rm c}(t) &= \left(\frac{\tilde{N} - N_{\rm c}}{N}\cos\omega_1 t - \frac{2\tilde{N}}{N}\cos\omega_2 t\right)\Delta z\,, \\ z_{\rm n}(t) &= \left(\frac{\tilde{N} - N_{\rm c}}{N}\cos\omega_1 t + \frac{2N_{\rm c}}{N}\cos\omega_2 t\right)\Delta z\,. \end{aligned} \tag{12.4}$$

In this model, $z_{\rm cm}(t) = \Delta z(\tilde{N} - N_{\rm c})/N\cos\omega_1 t$ and $z(t) = -2\Delta z\cos\omega_2 t$.

These expressions describe qualitatively the results of the full simulation, although the relative displacement does not exhibit any damping since in the model there is no coupling to other internal degrees of freedom.

Next we include both the C_{12} and C_{22} collision processes. The qualitative behaviour is very similar to that found previously. In fact, in the top panel of Fig. 12.3, the solid and broken curves, for the centre-of-mass oscillation, with and without collisions, coincide. This demonstrates that collisions do not affect the centre-of-mass motion. However, the noticeable change in the damping of the out-of-phase dipole mode (lower panel) indicates that collisions affect the internal state of the thermal cloud. For further remarks see Chapter 13, where we discuss Landau damping in more detail. As a final observation, we note that the frequency of the out-of-phase dipole mode is slightly smaller than that of the in-phase mode. Although our results are for a different trap geometry, the shift in frequency is in the same direction as that observed experimentally (Stamper-Kurn *et al.*, 1998). This result is only found if the full dynamics of the thermal cloud is taken into account. In an earlier calculation using two-fluid hydrodynamic equations (Zaremba *et al.*, 1999), a shift in the *opposite* direction was found. The difference can be traced to the fact that, in the latter calculation, no allowance was made for a change in shape of the noncondensate density profile as it moves relative to the condensate. Thus, rather subtle adjustments in the internal state of the thermal cloud can have observable consequences.

12.3 Radial breathing mode

In this section, we discuss simulations of the monopole "breathing" mode in isotropic traps. To initiate the simulation, we were guided by the known properties of the monopole mode of the condensate at $T = 0$ in the TF approximation (Stringari, 1996b), and those of a harmonically trapped Boltzmann gas. In both cases, the velocity field of the mode in the linearized (i.e. small-amplitude) regime has the form $\mathbf{v}(\mathbf{r}) = a(t)\mathbf{r}$, where $a(t)$ is an oscillatory function of time. In other words, the gas moves in the radial direction and has a local velocity that increases in magnitude in proportion to its distance from the origin. However, when considering the mean-field coupled dynamics of the condensate and noncondensate, the separate velocity fields of the two components will not be precisely of this form. Thus, if the velocity field $a(t)\mathbf{r}$ is used as an initial condition, then in general the gas will be excited into a superposition of normal modes having the same symmetry. Fortunately, these other modes will usually have distinct frequencies and can be separated from the mode of interest by spectral analysis. In

the present case, we will find that the assumed velocity field predominantly excites the lowest monopole mode.

The above velocity field can be imposed on the condensate by simply multiplying the ground state wavefunction by the phase factor $\exp(imar^2/2\hbar)$. In the case of the thermal cloud, the velocity of the ith particle in the equilibrium ensemble is incremented by $a\mathbf{r}_i$. These initial conditions define the starting point of a simulation. However, an alternative approach is also available. The assumed velocity field implies that the density undergoes a dilation, i.e. its time dependence is given by $n(\mathbf{r},t) = \alpha^{-3}n_0(\mathbf{r}/\alpha)$, where $\alpha(t)$ is a time-dependent scaling parameter and $n_0(\mathbf{r})$ is the initial density profile. To verify this, we choose $\alpha(t) = 1+\epsilon(t)$ and expand $n(\mathbf{r},t)$ to lowest order in ϵ; the resulting density fluctuation can then be shown to satisfy the linearized continuity equation if $\mathbf{v}(\mathbf{r}) = a(t)\mathbf{r}$, with $a(t) = \dot{\epsilon}(t)$. The second option thus amounts to specifying the initial condensate wavefunction as $\Phi(r,0) = \alpha^{-3/2}\Phi_0(r/\alpha)$ and to redefining the position of each thermal atom in the equilibrium ensemble as $\alpha\mathbf{r}_i$. Both approaches for defining the initial conditions have been tried and give similar results but, in the following, only the method in which a density fluctuation is imposed will be used.

The first simulation to be described is performed within the static thermal cloud approximation (STCA) discussed in Chapter 8. In this approximation, one follows the dynamics of the condensate in the presence of a static equilibrium distribution of thermal atoms. Thus the noncondensate mean field $2g\tilde{n}_0(\mathbf{r})$, which appears in the GP equation, acts as a fixed external potential. The other quantity needed to evolve the condensate wavefunction is the collisional exchange term R given by (11.52). If the condensate is perturbed from equilibrium, then C_{12} will no longer be zero and, as explained in Chapter 8, R leads to damping of the condensate motion. This collisional exchange of atoms between the condensate and thermal cloud is the mechanism by which the condensate relaxes towards equilibrium.

Although it is possible to evaluate R in the STCA directly from (8.11) and (8.12) using the equilibrium distribution f^0, instead we evaluate it using a N-body simulation as discussed in subsection 11.3.2. An equilibrium distribution of test particles is set up initially and their motion, in the effective potential defined by the condensate and noncondensate equilibrium densities, is followed in time. In principle, this evolution maintains the equilibrium distribution of thermal atoms even in the absence of C_{22} collisions. The "out" and "in" C_{12} collision probabilities are calculated at each time step from (11.55) and (11.56), respectively, and these results are then used to calculate $R(\mathbf{r},t)$ using (11.54). These simulations can be compared directly with the calculations of Williams and Griffin (2000) and thus provide

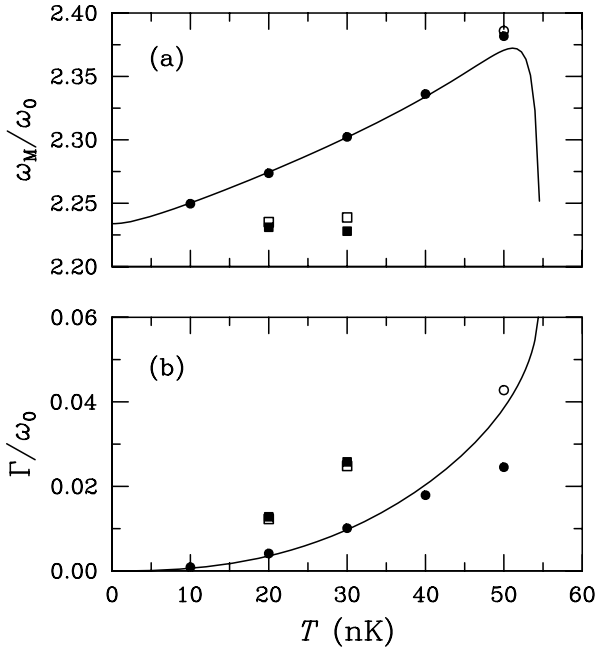


Fig. 12.4. Temperature dependent (a) frequency shifts and (b) damping rates of the condensate monopole mode in a spherical trap ($\omega_0/2\pi = 10$ Hz), in the presence of a static thermal cloud. The total number of atoms is $N = 2 \times 10^6$. The critical temperature for the corresponding ideal gas is $T_{\rm BEC} = 57\,{\rm nK}$. The solid circles give the simulation results, while the solid line shows the prediction of Williams and Griffin (2000). The open squares (at 20 and 30 nK) give results corresponding to the full inclusion of the thermal cloud dynamics, but with only C_{22} collisions; the solid squares include both C_{22} and C_{12} collisions (from Jackson and Zaremba, 2002b).

a further test of our simulation methods, specifically the method of calculating the C_{12} collision probabilities.

The monopole mode is excited using the density-scaling method discussed above. The scale parameter used in the calculations was $\alpha = 0.95$, corresponding to a 5% compression of the condensate in the radial direction. The widths of the condensate wavefunction in the x, y and z directions are defined by the mean-square deviations, e.g. $\sigma_x \equiv \sqrt{\langle x^2\rangle - \langle x\rangle^2}$, where the moments are given by $\langle\chi\rangle = N_{\rm c}^{-1}\int d\mathbf{r}\chi n_{\rm c}(\mathbf{r})$. Plots of these widths show a damped oscillation (an example of this behaviour is shown in Fig. 12.5); a fit of the data to an exponentially decaying sinusoid (see (12.7) in Section 12.4) allows one to extract the frequency ω and damping rate γ. Since each direction gives a slightly different fit, owing to statistical fluctuations, an average

of the three values is plotted in Fig. 12.4. The overall good agreement with Williams and Griffin (2000) of the frequencies and damping rates provides a useful check of the Monte Carlo sampling method we have used to calculate the C_{12} collision probabilities.[2]

The static thermal cloud approximation demonstrates that the collisional exchange of atoms between the condensate and the thermal cloud contributes to the damping of condensate collective modes. However, it is difficult to assess the accuracy of the approximation from the calculations presented so far. The assumption that the thermal cloud remains in equilibrium precludes the possibility of damping due to the transfer of energy between the two components. To allow for this, one must perform a full simulation in which the thermal cloud and condensate evolve self-consistently in the presence of their mutual mean-field interactions.

Full simulations were performed at $T = 20$ nK and 30 nK and the results, along with those of the STCA, are given in Fig. 12.4. The open squares show the results obtained ignoring C_{12} collisions but including the full dynamics of the thermal cloud together with the C_{22} collisions between the thermal atoms. The damping rates are seen to be more than a factor 2 larger than the STCA results. When C_{12} collisions are also included, the results for the frequency and damping rate change only slightly. However, it should be emphasized that collisions are actually playing a relatively minor role in these simulations. They affect the damping only indirectly through their effect on the thermal cloud distribution. As will be discussed in more detail shortly, the damping in the full simulations is predominantly Landau damping, which is associated with the mean-field interaction between the condensate and the thermal cloud.

As regards the frequency of the mode, the full simulation shows a slight downward shift with increasing temperature, in contrast with the much stronger increasing trend within the STCA. The explanation is that the condensate in the STCA is oscillating in the presence of the *static* mean field of the equilibrium thermal cloud (see Fig. 11.5), which enhances the oscillator frequency of the trap. This effect increases with temperature because of the increasing number of thermal atoms. When the thermal cloud is allowed to respond to the dynamic mean field of the condensate, this "stiffening" of the confining potential does not arise. These comparisons again

[2] The discrepancy in the damping rate at $T = 50$ nK arises from errors in the binning procedure used to obtain the phase-space density at positions near the condensate surface. This was checked by simply repeating the simulation using the analytical expression for f^0, instead of the binned phase-space density, in the calculation of the C_{12} collision probabilities. The result of this calculation is indicated by the open circle in Fig. 12.4(b), which is in much better agreement with the result of Williams and Griffin (2000).

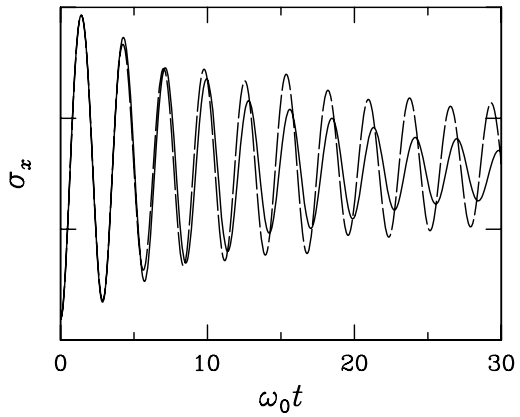


Fig. 12.5. Time-dependent width σ_x of the condensate (arbitrary units) after excitation of the monopole mode at $T = 200\,\mathrm{nK}$. The broken line shows collisionless evolution, while the result of a full simulation (including both C_{12} and C_{22}) is indicated by the solid line (from Jackson and Zaremba, 2002b).

demonstrate that the dynamics of the thermal cloud is crucial in making quantitative predictions for mode frequencies and damping rates. Nevertheless, the static thermal cloud approximation is useful since it allows one to estimate the effects of C_{12} collisions on collective modes without the need for a detailed solution of the kinetic equation (Duine and Stoof, 2001; Penckwitt *et al.*, 2002; Williams *et al.*, 2002).

To investigate the effects of collisions more systematically, we performed additional simulations for the isotropic trap containing 5×10^4 ^{87}Rb atoms, as considered in Fig. 11.1. The monopole mode was excited by scaling the condensate wavefunction by a scale factor $\alpha = 0.9$, while the thermal cloud was left in its equilibrium state. The condensate width as a function of time is shown in Fig. 12.5 for a temperature $T = 200$ nK ($T/T_{\mathrm{BEC}} = 0.644$). The broken curve was obtained by including mean-field effects but ignoring collisions entirely, while the solid curve is the result obtained when C_{12} and C_{22} collision terms are also included. In both cases, one observes regular oscillations with an amplitude that tends to decrease with time. The simulation with collisions, however, follows more closely a simple exponential decay, as can be confirmed by fitting a damped sinusoid to the results.

Damping of the condensate oscillation is, of course, a sign that the energy associated with the collective motion of the condensate is being dissipated. In the absence of collisions, the only mechanism for this is the transfer of energy to the thermal cloud via the mean-field interaction. By analogy with

plasma oscillations, where the energy associated with a collective plasma wave is transferred to single-particle electronic excitations, this mean-field-mediated decay is referred to as Landau damping. A more complete discussion of Landau damping is presented in Chapter 13.

From the point of view of the thermal atoms, the dynamic condensate mean field $2gn_c(\mathbf{r}, t)$ acts as an external perturbation. The rate at which this mean field can do work on the thermal cloud depends on precisely how the atoms are distributed in phase space. In the simulations of Fig. 12.5, the thermal cloud is initially in an equilibrium state but, as time progresses, interactions with the condensate drive it away from equilibrium. The degree to which this occurs depends on whether collisions are taken into account, hence the differences between the two curves in Fig. 12.5. Collisions, and in particular C_{22} collisions, drive the thermal cloud towards a state of *local equilibrium* (see Chapters 15 and 17). This has the effect of maintaining the damping rate at its initial value and accounts for the more or less uniform damping rate exhibited in Fig. 12.5 by the curve that includes collisions. Without collisions the distribution function has larger temporal fluctuations, with the result that the damping of the monopole mode becomes more irregular.

Results for the monopole mode frequencies and damping rates over a range of temperatures are presented in Fig. 12.6. These were obtained by fitting an exponentially decaying sinusoid over the range of times shown in Fig. 12.5. At each temperature, simulations were performed including (i) no collisions, (ii) only C_{22} collisions and (iii) both C_{12} and C_{22} collisions. All three give similar results at low temperatures, where the fraction of thermal atoms is small. The frequency is close to the TF value, $\sqrt{5}\omega_0$, and the damping is seen to go to zero. However, with increasing temperature the differences between the three simulations become significant. Without collisions, the monopole frequency tends to increase with temperature over almost the full range of temperatures shown. The condensate confinement effectively becomes "stiffer" as a result of its mean-field interaction with the thermal cloud. However, when all collisions are included the frequency *decreases*. The effect is as if the thermal cloud imparts extra mass to the condensate. This explanation is consistent with the fact that the relative fraction of thermal atoms increases with temperature.

The damping rate in units of the trap frequency is shown in Fig. 12.6(b). It vanishes as $T \to 0$ since, as explained above, the only dissipation mechanism is the coupling to the thermal cloud. As the thermal population decreases, so does the damping rate. The decrease in the rate observed in the collisionless limit as T approaches T_{BEC} has a different explanation. The damping rate

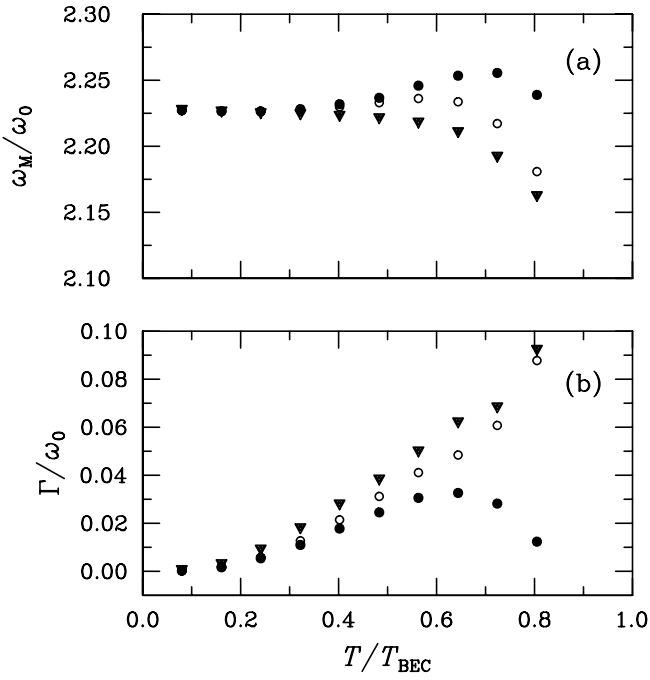


Fig. 12.6. (a) Frequency and (b) damping rate of a monopole mode ($\omega_0/2\pi = 187\,\mathrm{Hz}$, $N = 5 \times 10^4$), including the thermal cloud dynamics. Results are shown for simulations with no collisions, $C_{22} = C_{12} = 0$ (solid circles), with C_{22} collisions only (open circles) and wth both C_{12} and C_{22} collisions (triangles) (from Jackson and Zaremba, 2002b).

in this limit is not well defined since the decay of the oscillations, as shown in Fig. 12.5, is not purely exponential. In fact, the nonequilibrium distribution of thermal atoms can re-excite the condensate mode at certain times, thereby reducing the damping rate. When C_{22} collisions are included, the thermal distribution is forced to be more equilibrium-like, and as a result, a smooth monotonic increase in the damping rate is observed.

In the absence of C_{12} collisions, the damping is due purely to the Landau mechanism. When C_{12} collisions are turned on, atoms begin to be exchanged between the condensate and thermal cloud and we see a noticeable increase in the damping rate. The observed increase, however, cannot be attributed only to the collisional exchange of atoms between the condensate and thermal cloud. C_{12} collisions also modify the thermal atom distribution function, which in turn modifies the Landau damping. Thus, the mean-field

and collisional effects are interrelated and both must be included to account completely for the observed damping rates.

As a final comment, we note that the damping rates from the full simulations in Figs. 12.4 and 12.6 are similar, for a given value of $T/T_{\rm BEC}$, when expressed in units of the trap frequency ω_0, despite the large difference in the number of trapped atoms N. Thus, the damping rate is dependent more on the fraction of thermal atoms $\tilde{N}/N$, which is set by the temperature, than on the absolute number of atoms in the trap. The damping rate will depend on the mode being investigated, of course, and several other collective modes of interest will be considered in the following sections.

12.4 Scissors mode oscillations

The scissors mode was discussed in Chapter 10, when the ZNG equations were solved approximately using the moment method. Here we present a more complete analysis based on numerical simulations of the same equations.

The experiments of Maragò *et al.* (2000, 2001) made use of an axially symmetric trap potential of the form $V_{\rm trap}(\mathbf{r}) = \frac{1}{2}m(\omega_\perp^2\rho^2 + \omega_z^2 z^2)$. To excite the scissors mode, the trap potential was rotated adiabatically through a small angle θ_0 about the y-axis, and then suddenly in the opposite direction about the y-axis through an angle $-2\theta_0$. To simulate this process, we first determine the equilibrium state of the system at some temperature T using the procedure outlined in Section 11.4. We then actively rotate the particle coordinates and condensate wavefunction relative to the trapping potential through an angle $2\theta_0$ about the y axis. This defines the initial nonequilibrium state used in the dynamical simulations.

The relevant dynamical variables that describe the density fluctuation are the xz quadrupole moment of the condensate defined by

$$Q_{\rm c}(t) \equiv \int d\mathbf{r}\, xz n_{\rm c}(\mathbf{r},t), \tag{12.5}$$

and that of the thermal cloud (see Section 11.2 for the notation)

$$\tilde{Q}(t) \equiv \frac{\tilde{N}}{\tilde{N}_{\rm tp}} \sum_{i=1}^{\tilde{N}_{\rm tp}} x_i(t) z_i(t)\,. \tag{12.6}$$

A rigid-body rotation of the two components through a small angle θ about the y axis would result in the quadrupole moments $Q_\alpha = \theta Q_\alpha^0$, where $Q_\alpha^0 = \langle x^2 - z^2\rangle_\alpha^0$ is the xx-component of the equilibrium quadrupole moment tensor of the αth component. We thus define the "rotation angle"

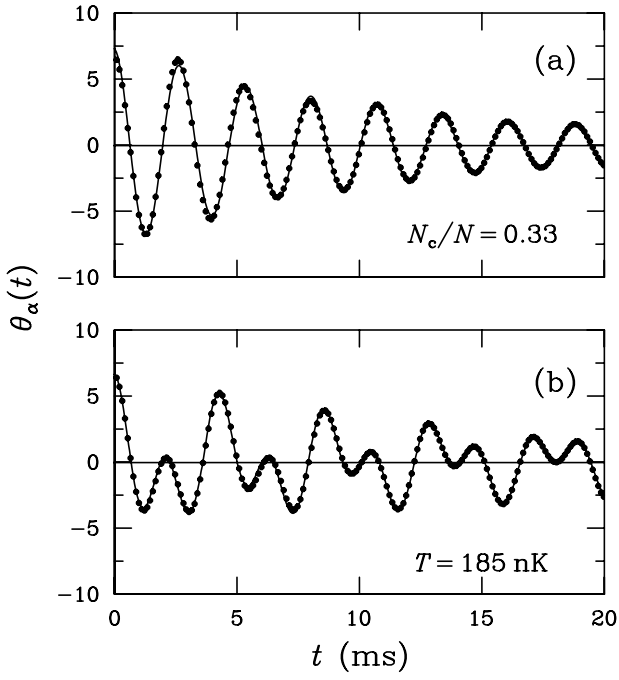


Fig. 12.7. Angular displacements of the condensate (a) and thermal cloud (b) vs time. The points are the results of the simulation and the solid lines are three-mode fits to these results. The trap parameters are $\omega_\perp/2\pi = 126$ Hz, $\omega_z = \sqrt{8}\omega_\perp$ (Maragò *et al.*, 2001) and $N = 5 \times 10^4$ (from Jackson and Zaremba, 2002a).

$\theta_\alpha(t) \equiv Q_\alpha(t)/Q_\alpha^0$ as the variable representing the angular displacement of the two components in our simulations. One advantage of using these angular variables is that the range of angular displacements is similar for both components and independent of temperature, in contrast with the quadrupole moments themselves. In addition, this is the variable that can be extracted directly from the experimental images of the cloud (Maragò *et al.*, 2001).

In Fig. 12.7, we show numerical results for the angular displacements $\theta_\alpha(t)$ of the two components as a function of time at an intermediate temperature of $T = 185$ nK, for a condensate fraction $N_c/N = 0.33$. The condensate exhibits a simple damped oscillation while the noncondensate is seen to oscillate in a regular, but more complex manner. To analyze the data, the curves are fitted to a superposition of damped exponentials:

$$\theta(t) = \sum_i A_i e^{-\gamma_i t} \cos(\omega_i t + \phi_i)\,, \tag{12.7}$$

where the amplitude A_i, frequency ω_i, damping rate γ_i and phase constant ϕ_i are fitting parameters. The solid curves shown in Fig. 12.7 are obtained using a three-mode approximation in which both curves are fitted by a common set of three frequencies and damping rates. At this intermediate temperature, however, a single mode approximation works very well for the condensate whereas two independent modes are sufficient to fit the noncondensate oscillation.

The quality of the three-mode fit in Fig. 12.7 implies that the initial conditions essentially excite three normal modes of the system. One mode is a condensate-like mode in which the quadrupole-moment fluctuation of the condensate is much larger than that of the noncondensate. The other two modes involve mainly noncondensate fluctuations. In view of the moment analysis in Chapter 10, these results would follow if the coupling between the two components were relatively weak. We would then expect only small shifts from the frequencies obtained when the two components are treated independently. The frequency of the condensate-like mode in Fig. 12.7 is, in fact, close to the condensate scissors mode frequency $\omega_{\rm sc} = \sqrt{\omega_\perp^2 + \omega_z^2}$ found in Chapter 10 within the TF approximation.[3] Similarly, the angular displacement of the noncondensate has two frequency components which are close to the two scissors mode frequencies $\omega_\pm = \omega_z \pm \omega_\perp$ of the Boltzmann gas above $T_{\rm BEC}$.

The main signature in Fig. 12.7 of the coupling between the condensate and the thermal cloud is the damping of the condensate mode. This is again an example of Landau damping and would be absent at $T = 0$. This is illustrated in Fig. 12.8, which presents results of simulations done at a lower temperature, $T = 55$ nK, where $N_{\rm c}/N = 0.95$. Since the fraction of thermal atoms is small in this case, the condensate mode is only weakly damped, with a frequency very close to the $T = 0$ limit. However, the behaviour of the thermal cloud is quite different from what is seen at the higher temperature in Fig. 12.7. The dominant frequency component is now that of the condensate oscillation, which indicates that the thermal cloud is being driven by the more massive condensate. As can be seen, the three-mode fit for the noncondensate is reasonable but not as good as in Fig. 12.7. This suggests that the condensate mean field is exciting other internal modes of oscillation of the thermal cloud beyond those accounted for by the three-mode fit. With increasing temperature, the number of

[3] Only part of the frequency shift is due to the mean-field coupling between the two components. A simulation of the GP equation at $T = 0$ for $N = 2 \times 10^4$ yields a frequency 1.5% *larger* than the TF result. This is an indication of the error incurred by the TF approximation when N is not sufficiently large (see Section IV of Dalfovo *et al.*, 1999).

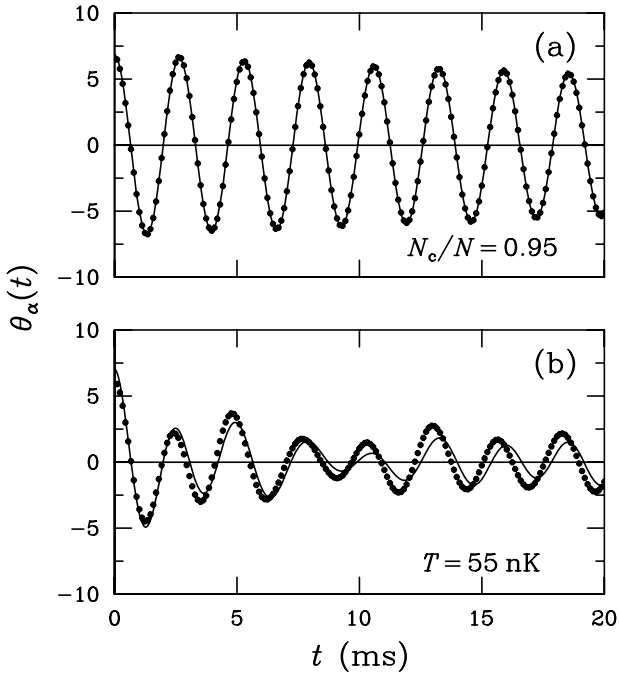


Fig. 12.8. As in Fig. 12.7, but at a lower temperature, $T = 55$ nK (from Jackson and Zaremba, 2002a).

thermal atoms increases and the fraction in the proximity of the condensate diminishes. The relative coupling to the condensate thus weakens, and the thermal cloud then begins to display its own scissors mode oscillations, as seen in Fig. 12.7.

The situation for T very close to $T_{\rm BEC}$ is also of interest. In Fig. 12.9 we show the scissors mode oscillations of the condensate and thermal cloud at temperature 296 nK, where $N_{\rm c}/N = 0.02$. The noncondensate oscillation shows the same pattern as in Fig. 12.7, but the oscillation of the condensate is now more complex. The dramatic variation of the amplitude is an indication of the coupling of the condensate to the oscillating mean-field potential of the thermal cloud. This coupling can re-excite the condensate mode and leads to an intermittent revival of the mode amplitude. The frequency of the oscillation, however, is still quite close to that of the condensate scissors mode. Since the condensate density is much larger than that of the thermal cloud where the two overlap, the condensate mean field is still dominating the dynamics.

The interaction between the condensate and thermal cloud in the course

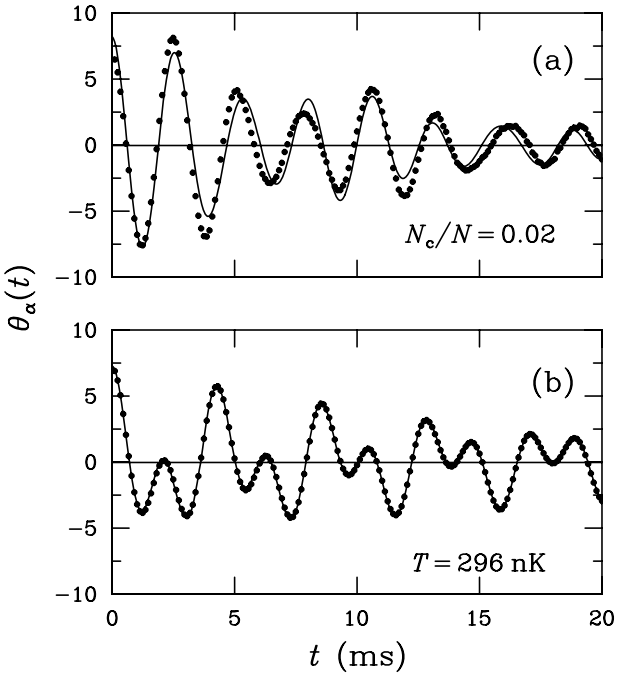


Fig. 12.9. As in Fig. 12.7, but at a higher temperature, $T = 296$ nK (from Jackson and Zaremba, 2002a).

of their oscillations is very clearly revealed in density contour plots of the two components. This is shown in Fig. 12.10 for a sequence of equally spaced time steps spanning approximately half a period of the condensate oscillation. During this sequence, the condensate is rotating counterclockwise while the thermal cloud is rotating in the opposite direction. There are clear signs of the thermal cloud's interaction with the repulsive mean-field of the condensate. As the condensate moves, it tends to push the thermal cloud out of its way, much like a boat moving through water. This strong perturbation of the thermal cloud is the source of the excitations leading to the Landau damping of the condensate scissors mode.

In order to make a comparison with the experimental data of Maragò *et al.* (2001), simulations of the kind shown in Figs. 12.7–12.9 were performed over a range of temperatures up to $T_{\rm BEC}$.[4] The frequencies and damping rates of the various scissors modes were obtained using three-mode fits to the numerical data. The results are shown along with the measured values

[4] The use of evaporative cooling in the experiments means that the total number of atoms varies from $N \simeq 2 \times 10^4$ at low T to $N \simeq 10^5$ close to $T_{\rm BEC}$. This variation was taken into account in the simulations.

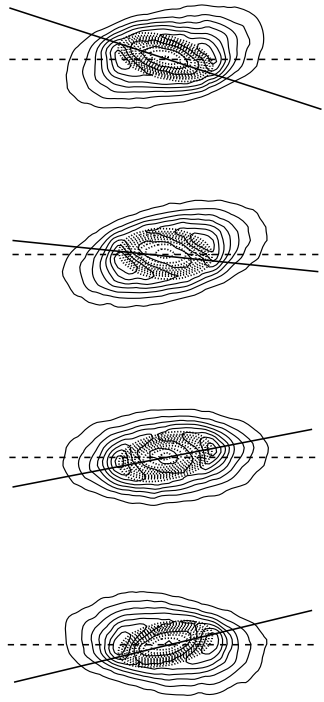


Fig. 12.10. Density contour plots of the condensate (dotted lines) and thermal cloud (solid lines) for a sequence of equally spaced time intervals spanning approximately one half-period of the scissors mode oscillation. The earliest time is at the top of the figure. In each image, the broken horizontal line denotes the symmetry axis of the trapping potential and the straight solid line indicates the major axis of the condensate (from Jackson and Zaremba, 2002a).

in Fig. 12.11. The values of the frequency and damping rate of the condensate scissors are found to be in very good agreement with experiment for temperatures below about $0.8T_{\mathrm{BEC}}$. Part of the discrepancy above this temperature is probably due to the fact that the condensate is strongly coupled to the thermal cloud in this temperature range, where as Fig. 12.9 shows, the condensate oscillation is quite irregular. The larger error bars for T approaching T_{BEC} presumably reflect a similar kind of behaviour in the experiments.

The downward trend of the condensate mode frequency with increasing temperature correlates with the increasing size of the thermal cloud. There are two possible explanations for this. If, as the density contours in Fig. 12.10

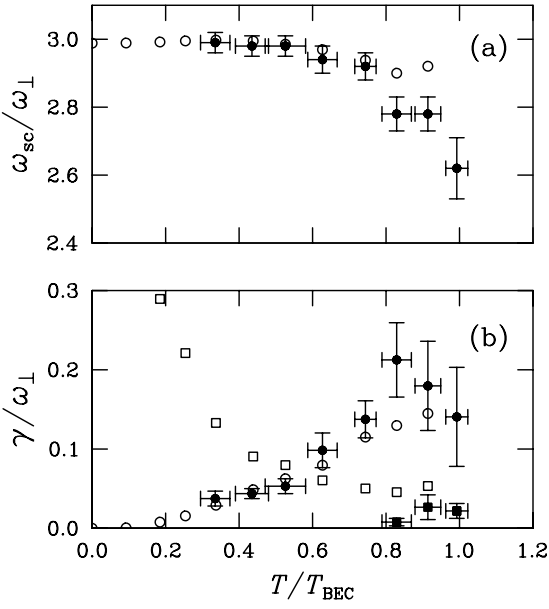


Fig. 12.11. Simulation results for (a) the frequency and (b) the damping rate of the scissors modes for the variable total number of atoms occurring in the experiments of Maragò *et al.* (2001). The open circles (theory) and solid circles (experiment) are for the condensate mode. The open squares in (b) show the calculated average damping rate of the two thermal cloud modes, while the solid squares are the corresponding experimental values (from Jackson and Zaremba, 2001).

suggest, the condensate were to drag part of the thermal cloud along with it, it would effectively have a larger inertia. Alternatively, the thermal cloud could be providing a counter-torque that effectively reduces the "spring constant" of the anisotropic trap. Either effect would lead to a lower mode frequency, as seen experimentally and calculated theoretically. As for the noncondensate, the scissors mode frequencies are found to be within a few per cent of the noninteracting gas values ($\omega_\pm = \omega_z \pm \omega_\perp$) over the whole temperature range.

The damping of the condensate mode increases monotonically with temperature; this is a consequence of Landau damping (see also Chapter 13) and the effects of C_{12} and C_{22} collisions. Simulations performed without collisions yielded damping rates that were as much as 50% lower than the values shown in Fig. 12.11. This shows that collisions play an important role in the simulations, either indirectly through the equilibrating effect of C_{22} collisions or directly through C_{12} collisions (see the discussion of the radial breathing mode in Section 12.3). The average damping rate of the

noncondensate modes, also shown in Fig. 12.11(b), *decreases* with increasing temperature. At the higher temperatures, the damping is purely a C_{22} effect, and is within a factor 2 of the experimental values. The increased damping rate at low temperatures is presumably associated with the fact that the condensate mean field is disrupting the noncondensate scissors mode oscillation.

As discussed in Chapter 10, the existence of a single condensate scissors mode is a consequence of the irrotational nature of the condensate velocity field. In contrast, the velocity field of the two thermal cloud modes has both irrotational and solenoidal components. This distinction becomes apparent when one considers the moment of inertia I. In Section 10.3, we reviewed the relation between the moment of inertia I of a trapped Bose superfluid gas and the Fourier transform $\chi_Q(\omega)$ of the quadrupole response function:

$$\chi_Q(\omega) = \frac{i}{\hbar}\int_0^\infty dt\, e^{i\omega t}\left\langle [\hat{Q}(t), \hat{Q}(0)]\right\rangle_0, \tag{12.8}$$

where $\hat{Q}$ is the quadrupole moment operator given in (10.96). One may verify that (12.8) agrees with the expression in terms of exact eigenstates given in (10.97). The function (12.8) determines the linear response of the system to a time-dependent perturbation that couples to $\hat{Q}$. The sudden rotation of the trap through a small angle θ_0 is described by the perturbation in (10.100), where the strength is $\lambda = m(\omega_z^2 - \omega_\perp^2)\theta_0$. This perturbation induces the time-dependent quadrupole moment $Q(t)$, which is the quantity calculated in our simulations. As discussed in Section 10.3, the imaginary part of $\chi_Q(\omega)$ and the Fourier transform of $Q(t)$ are related by (10.102). A three-mode fit to the simulation data for the induced quadrupole moment yields the results for $\chi_Q''(\omega)$ plotted in Fig. 12.12, expressed as a sum of three Lorentzian spectral densities. At low temperatures, the condensate mode is dominant and $\chi_Q''(\omega)$ exhibits a single peak. However, as the temperature increases, the strength of the condensate mode diminishes and the spectral density is found to be dominated by the two thermal cloud modes at $\omega_\pm = |\omega_\perp \pm \omega_z|$.

Zambelli and Stringari (2001) showed that I can be related to a frequency integral (10.98) involving the imaginary part $\chi_Q''(\omega)$ of the quadrupole response function. Combining this result for I and the expression for the rigid body value $I_{\rm rb}$ given by (10.106), one finds

$$\frac{I}{I_{\rm rb}} = (\omega_z^2 - \omega_\perp^2)\frac{\int_{-\infty}^{\infty} d\omega\, \chi_Q''(\omega)/\omega^3}{\int_{-\infty}^{\infty} d\omega\, \chi_Q''(\omega)\omega}. \tag{12.9}$$

However, a Lorentzian fit to the simulation data shown in Fig. 12.12 cannot

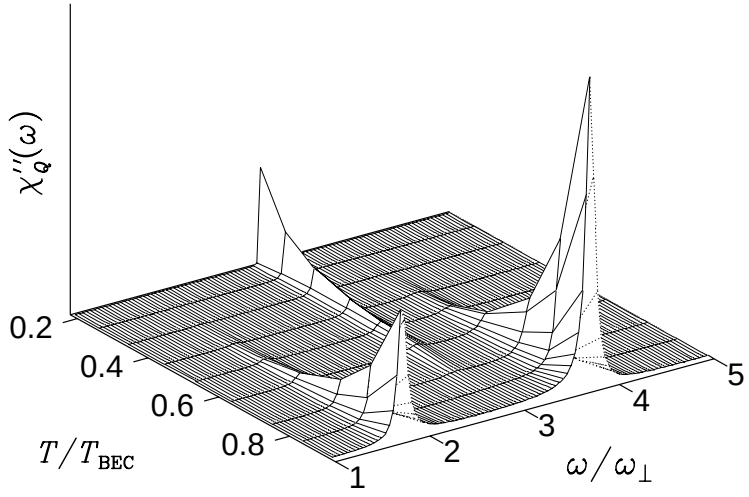


Fig. 12.12. Simulation results for the imaginary part of the quadrupole response function, $\chi''_Q(\omega)$, as a function of frequency and temperature. The parameters are the same as those used in Fig. 12.11 (from Jackson and Zaremba, 2001).

be used to evaluate the denominator in (12.9), which relates to $I_{\rm rb}$, since the first frequency moment of a Lorentzian is undefined. The problem arises because a decaying exponential does not capture the correct small-time behaviour of $Q(t)$; it would result in a spectral density that falls off at least as rapidly as ω^{-3}. To overcome this problem, Jackson and Zaremba (2002a) replaced the Lorentzian line shapes by delta functions with a spectral weight equal to the original Lorentzian peak. This procedure leads to the moment of inertia plotted in Fig. 12.13 as a function of temperature. At high temperatures, I approaches the value $I_{\rm rb}$ given by (10.107). As $T \to 0$, we see that the moment of inertia decreases in magnitude and approaches the irrotational value $I = \epsilon^2 I_{\rm rb}$, where $\epsilon = \langle x^2 - z^2 \rangle_{\rm c0} / \langle x^2 + z^2 \rangle_{\rm c0}$ describes the deformation of the condensate in equilibrium.

In the case of a *noninteracting* Bose gas at finite T, Stringari (1996b) showed that

$$I = \epsilon^2 \, I_{\rm c,rb} + I_{\rm n,rb}. \tag{12.10}$$

This gives the moment of inertia I as the sum of *separate* contributions from the Bose condensate and the thermal cloud. Somewhat surprisingly, this expression was also found in Chapter 10, using the moment method to include the HF mean-field coupling between the condensate and thermal cloud. The only difference between (12.10) for a noninteracting Bose gas and (10.111) is that in the latter the static density profiles $n_{\rm c0}(\mathbf{r})$ and $\tilde{n}_0(\mathbf{r})$ are

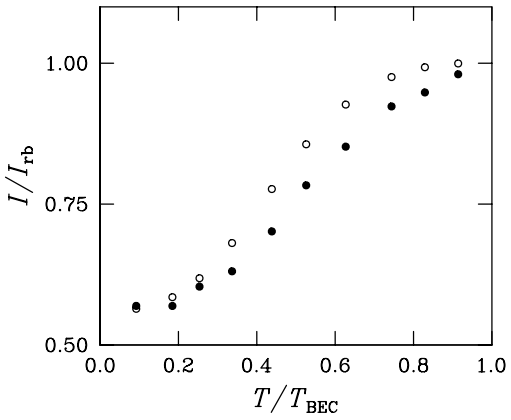


Fig. 12.13. The moment of inertia I of a trapped gas (normalized to the rigid-body moment of inertia) as a function of temperature. The solid circles are the results of (12.9) while the open circles were obtained using (12.10) (from Jackson and Zaremba, 2002a).

those of an *interacting* Bose-condensed gas (within the TF approximation). To check this expression in the present case, we calculated $I_{\rm c,rb}$ and $I_{\rm n,rb}$ in (10.111) using the equilibrium density profiles. The result for the moment of inertia (relative to $I_{\rm rb}$) is shown in Fig. 12.13, where one can compare it with that calculated from the expression (12.9) using simulation data for $\chi''_Q(\omega)$. The moments of inertia I in both cases increase monotonically with increasing temperature and show the transition from superfluid behaviour at low temperatures to normal fluid behaviour as one approaches $T_{\rm BEC}$. However, there is a significant difference at intermediate temperatures. Since the numerical simulations provide the most accurate treatment of the ZNG coupled dynamics, the difference[5] between (12.9) and (12.10) indicates that the moment method does not give a quantitatively correct treatment of the temperature dependence of the quadrupole response in the superfluid phase.

12.5 Quadrupole collective modes

In this section, we consider the quadrupole modes in axisymmetric traps. These modes were first studied experimentally by Jin *et al.* (1997) over a range of temperatures up to $T_{\rm BEC}$. This early experiment generated a great

[5] The origin of this difference merits further study. Jackson and Zaremba (2002a) fitted their simulation results for $\chi''_Q(\omega)$ to a sum of three delta functions in evaluating I from (12.9). However this procedure is not necessary since the denominator can be computed directly from the total density profile $n_0(\mathbf{r})$ using (10.107).

deal of interest because the frequency of one mode exhibited a strong unexpected temperature dependence. We now know that the observed behaviour is a result of a crossover from a condensate-like mode at low temperatures to a noncondensate-like mode at higher temperatures. The first to suggest this explanation were Bijlsma and Stoof (1999), on the basis of a mean-field analysis, which was later extended to include collisions by Khawaja and Stoof (2000b). Simulations by Jackson and Zaremba (2002c) based on the ZNG theory improved on these early calculations and were able to explain in detail the observed anomalous temperature dependence.

To place the experimental results in context, it is useful to consider first the condensate modes at $T = 0$. The quadrupole mode frequencies can be estimated in the TF approximation (Stringari, 1996b) and are found to depend on the trap geometry. For an isotropic trap, the modes can be labelled by an angular momentum index l and an azimuthal index m, but only the m index survives in axisymmetric traps. A trap anisotropy couples the lowest monopole ($l = 0$, $m = 0$) and quadrupole ($l = 2$, $m = 0$) modes of an isotropic trap, giving rise to two $m = 0$ modes with frequencies (Stringari, 1996b)

$$\omega_{\pm}^2 = \omega_{\perp}^2 \left(2 + \tfrac{3}{2}\lambda^2 \pm \tfrac{1}{2}\sqrt{9\lambda^4 - 16\lambda^2 + 16}\right), \tag{12.11}$$

where $\lambda \equiv \omega_z/\omega_\perp$ is the trap anisotropy parameter. The lower-frequency mode ω_+ corresponds to the $m = 0$ mode of interest in the experiments. With decreasing anisotropy, this mode evolves continuously into the $l = 2$, $m = 0$ quadrupole mode and thus takes on a quadrupolar character. The other condensate mode studied by Jin *et al.* (1997) was a true quadrupole mode with $m = 2$ and frequency $\omega_{m=2} = \sqrt{2}\omega_\perp$. For the experimental trap parameter values of $\omega_\perp/2\pi = 129$ Hz and $\lambda = \sqrt{8}$, the $m = 0$ mode has frequency $\omega_{m=0} \simeq 1.8\omega_\perp$, which is already close to the $\lambda \to \infty$ limiting value $\sqrt{10/3}\,\omega_\perp$.

Above $T_{\rm BEC}$, the thermal cloud can undergo breathing-like oscillations, one of which has frequency $\omega = 2\omega_\perp$. These are not true collective modes but rather correspond to a coherent motion of the atoms that is a special property of harmonic confinement. As we shall see, the close proximity of the $m = 0$ condensate mode to the $\omega = 2\omega_\perp$ thermal cloud mode is a key factor in the explanation of the temperature dependences of the mode frequencies observed by Jin *et al.* (1997).

In the experiments, the modes were excited by a parametric modulation of the harmonic trap frequencies. For example, the $m = 0$ mode was excited

by modulating the radial trap frequency according to

$$\omega_\perp^2(t) = \omega_\perp^2(1 + \epsilon \sin \Omega t), \tag{12.12}$$

where the modulation frequency Ω is an adjustable parameter. It is clear that this modulation affects both the condensate and thermal cloud, and so both components will be set into motion. Since the two components are strongly coupled by mean-field interactions, a consistent theoretical description must include the full dynamics of both components.

Simulations that faithfully follow the experimental excitation scheme will be described below. However, we consider first a procedure that can selectively couple to the individual components, by imposing velocity fields of the form $\mathbf{v}_\alpha(\mathbf{r}) = A_{\alpha 0}(x\hat{\mathbf{i}} + y\hat{\mathbf{j}}) + A_{\alpha 2}(x\hat{\mathbf{i}} - y\hat{\mathbf{j}})$ on the two components ($\alpha =$ c, n). The amplitudes $A_{\alpha 0}$ and $A_{\alpha 2}$ can be specified arbitrarily and excite the $m = 0$ and $m = 2$ modes, respectively. Although these are not exact mode-velocity fields, they do predominantly excite the modes of experimental interest. In the case of the condensate, the velocity is imposed by multiplying the equilibrium condensate wavefunction by an appropriate phase factor. For the thermal cloud, we simply add the position-dependent $\mathbf{v}_\mathrm{n}$ to the initial velocity of each test particle. Since the $m = 0$ and $m = 2$ modes are decoupled from each other in the linear response regime[6] owing to their different symmetries, both modes can be excited and studied in a single simulation. Information about each can then be extracted by evaluating the moments $Q_{\alpha 0} = \langle x^2 + y^2 \rangle_\alpha$ and $Q_{\alpha 2} = \langle x^2 - y^2 \rangle_\alpha$ that are specifically sensitive to the $m = 0$ and $m = 2$ modes, respectively. Again, α stands for either c or n.

In the first simulation to be discussed, *only* the condensate is excited initially ($A_{\mathrm{c}0} \neq 0$, $A_{\mathrm{c}2} \neq 0$, $A_{\mathrm{n}0} = A_{\mathrm{n}2} = 0$). Although the condensate mean field can excite the thermal cloud, one finds that the Q_n moments remain small during the subsequent time evolution. The condensate moments $Q_{\mathrm{c}\{0,2\}}(t)$, however, exhibit well-defined oscillations whose frequencies and damping rates can be extracted by means of a single-mode fit to the data. Except for the highest temperatures the fits are very good, indicating that a single mode of each symmetry is excited. The open squares in Fig. 12.14 show the frequencies as a function of the reduced temperature $T' = T/T_\mathrm{BEC}$. Both the $m = 0$ and $m = 2$ modes show a marked downward frequency shift with increasing temperature. This behaviour was also found in earlier calculations based on a variety of theoretical approaches (Hutchinson *et al.*,

[6] To remain in the linear regime, the amplitudes $A_{\alpha m}$ must be sufficiently small. This can be checked by making sure that the mode frequencies and damping rates are amplitude independent.

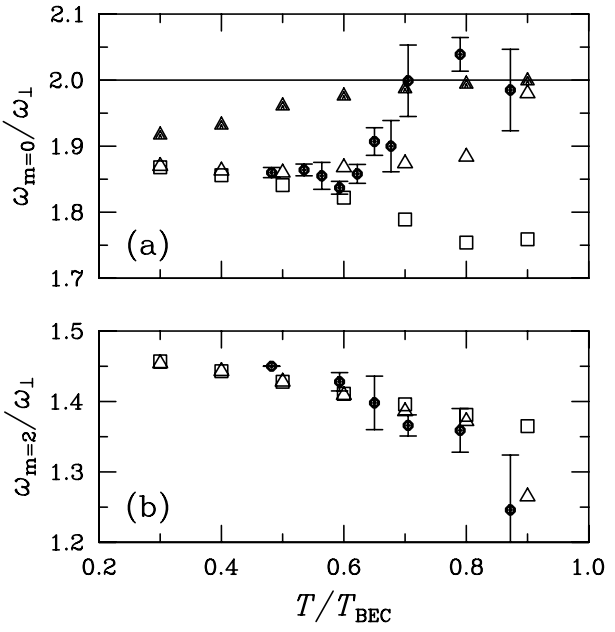


Fig. 12.14. Frequencies of (a) the $m = 0$ and (b) the $m = 2$ mode as functions of the reduced temperature $T' = T/T_{\rm BEC}$. The experimental data of Jin *et al.* (1997) is plotted with error bars, while our frequency results are shown for the condensate (open symbols) and noncondensate (solid symbols). The results of different initial thermal cloud conditions are represented by squares ($A_{{\rm n},m} = 0$) and triangles ($A_{{\rm n},m} \neq 0$). In both cases, condensate atoms are excited, $A_{{\rm c},m} \neq 0$ (from Jackson and Zaremba, 2002c).

1998; Bijlsma and Stoof, 1999; Reidl *et al.*, 2000; Duine and Stoof, 2001). However, none of these studies included the full thermal cloud dynamics. As in these earlier treatments, the agreement between theory and experiment is much better for the $m = 2$ mode than for the $m = 0$ mode. In the latter case, a strong deviation from experiment sets in above the reduced temperature $T' \simeq 0.6$. This is the anomalous behaviour mentioned at the beginning of this section.

We next investigate the influence of the thermal cloud dynamics by exciting both components *simultaneously*. In these simulations, the velocity fields imposed on the condensate and thermal cloud are identical. The triangles in Fig. 12.14 show the results of a single-mode fit to the condensate and noncondensate data. At high temperatures the thermal cloud frequency is close to the ideal gas value $\omega = 2\omega_\perp$ and then decreases as the temperature is lowered. The $m = 2$ condensate frequency shown in Fig. 12.14(b),

is largely independent of the initial conditions (as discussed below, it is also unchanged for a harmonic modulation of the trap). At the highest reduced temperature, $T' = 0.9$, the condensate frequency is dependent on the initial conditions of the thermal cloud owing to the strong mean-field coupling between the two components. In contrast, the $m = 0$ condensate frequency is generally higher than those found by exciting the condensate alone (compare the open triangles and the open squares in Fig. 12.14(a)) and approaches the noncondensate value $2\omega_\perp$ just below the transition.

More detailed information can be extracted from the simulation data by fitting two damped sinusoids to the $m = 0$ condensate data. This reveals the presence of two distinct frequency components at high temperatures, one close to that found for the condensate-only excitation, and the other close to the noncondensate value. This suggests that there are two well-defined normal modes of oscillation. The relative amplitude of the two modes is sensitive to the initial conditions and determines the form of the condensate oscillation. By using a single-mode fit to analyze the data, a frequency intermediate between the two mode frequencies is obtained, as shown by the open triangles in Fig. 12.14(a). In the case of the $m = 2$ modes, the condensate and noncondensate frequencies are sufficiently separated for the coupling between the two components to be relatively weak. In this case, one can truly speak of distinct "condensate" and "noncondensate" modes. But in general these two modes will each involve oscillations of both components.

The temperature dependence of the $m = 0$ mode, shown by the open triangles in Fig. 12.14(a), resembles the experimental behaviour (Jin *et al.*, 1997), but there are obvious discrepancies. To investigate their source, we now turn to simulations that more faithfully reproduce the harmonic excitation scheme employed emperimentally. These consist of a modulation of the radial trap frequency according to (12.12) over a time interval $\omega_\perp t = 30$, followed by a period of evolution in the original trap potential. This procedure excites $m = 0$ type oscillations in both the condensate and the thermal cloud; typical time-dependent plots of $Q_{\mathrm{c}0}(t)$ and $Q_{\mathrm{n}0}(t)$ are shown in Fig. 12.15. In the case of the condensate, we see a rather complex oscillation pattern. The revival of the amplitude is a consequence of the mean-field interaction between the condensate and thermal cloud.

In the experiments, a single-mode fit was used to extract mode frequencies from the data in a time interval of approximately $\omega_\perp t = 15$ following the excitation. We therefore analyzed our data within the observation window indicated by the vertical lines in Fig. 12.15. The results for the condensate frequency depend strongly on the drive frequency Ω, which affects the

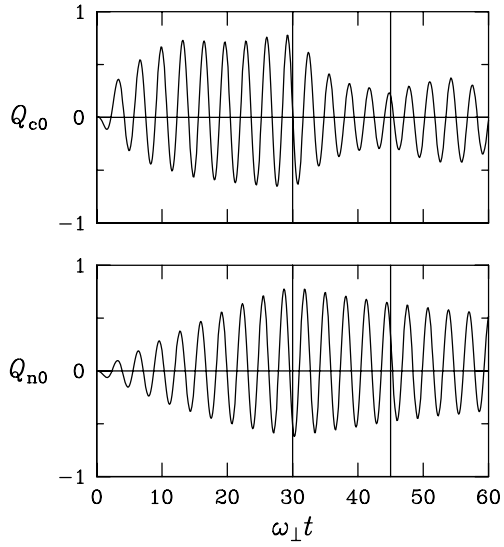


Fig. 12.15. Plot of the condensate and noncondensate $m = 0$ quadrupole moments Q_{c0} and Q_{n0} (in arbitrary units) as a function of time, at the reduced temperature $T/T_{BEC} = 0.8$. Up to $\omega_\perp t = 30$ (the first vertical line), the system is driven by a trap modulation of frequency $\Omega = 1.95\omega_\perp$ and amplitude $\epsilon = 0.02$. The subsequent evolution takes place in a static trap. The two vertical lines indicate the approximate range of the corresponding experimental measurements (from Jackson and Zaremba, 2002c).

relative amplitude of the condensate and the thermal cloud oscillations.[7] Figure 12.16 summarizes our results by showing frequencies from single-mode fits at each T for drive frequencies in the range $[1.75,\ 2.00]\,\omega_\perp$. At low T, the noncondensed component is small and has a minimal effect. As a result, all the frequencies are close to the condensate mode results shown by the open squares in Fig. 12.14. In contrast, above the reduced temperature $T' \sim 0.6$ there is a significant spread in the frequencies extracted from the fits. Taking $T' = 0.8$ as an example, the condensate mode is excited when $\Omega \simeq 1.75\omega_\perp$, while the frequency is close to that of the thermal cloud mode when $\Omega \simeq 2\omega_\perp$.

It is clear from the simulation data that there are two distinct branches above a reduced temperature $T' \sim 0.7$. The upper branch, which is obtained for a certain range of drive frequencies, is clearly the most relevant for the experiment. Although a record of the experimental drive frequencies is not

[7] The simulations described here includes the variation in the number of atoms N with temperature, as measured by Jin *et al.* (1997).

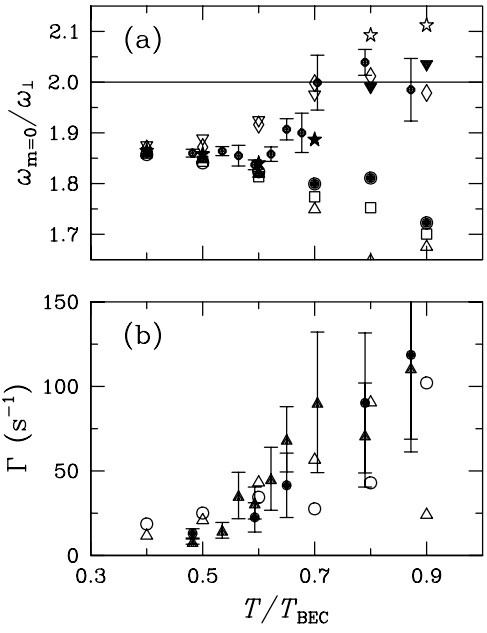


Fig. 12.16. (a) Frequency of the $m = 0$ mode as a function of temperature; the experimental results (Jin *et al.*, 1997) are plotted with error bars. We display simulation results for six drive frequencies: $\Omega = 1.75\omega_\perp$ (circles), $\Omega = 1.80\omega_\perp$ (squares), $\Omega = 1.85\omega_\perp$ (triangles), $\Omega = 1.90\omega_\perp$ (stars), $\Omega = 1.95\omega_\perp$ (diamonds), and $\Omega = 2.00\omega_\perp$ (inverted triangles). For each temperature value, the solid symbols represent the drive frequencies that produced the two largest condensate responses. (b) Damping rate vs. $T/T_{\rm BEC}$; the simulation results (open symbols) are compared with experiment (solid symbols), for the $m = 0$ (triangles) and $m = 2$ (circles) modes (from Jackson and Zaremba, 2002c).

available,[8] it is known that they were adjusted to maximize the amplitude of the condensate oscillation. Using this criterion, the drive frequencies that give a large response in the simulations are indicated by the solid points in Fig. 12.16(a). This still leaves some ambiguity as to which of the two branches the experiments would have followed. We conjecture that the experimental procedure resulted in a crossover from one branch to the other. Nevertheless, simulations indicate that the lower branch could be observed if the system were driven at the appropriate frequency.

We conclude with a few remarks about the damping rates shown in Fig. 12.16(b). For temperatures below $T' \sim 0.7$, the damping rates are in reasonably good agreement with experiment. However, at higher tempera-

[8] Eric Cornell, private communication.

tures there is considerable scatter in the theoretical results owing to their sensitivity to both the form of the drive and the time scale of the fit. This theoretical uncertainty may be related to the large error bars in the experimental data in this temperature range. As for the other examples we have considered, the damping is mainly of the Landau type, although C_{12} collisions also play a role. Within at most a factor 2, these mechanisms account for the experimental observations.

12.6 Transverse breathing mode

As our final example, we consider the transverse breathing mode in a cigar-shaped trap as studied by the Ecole Normale Supérieure (ENS) group in Paris (Chevy *et al.*, 2002). For a pure condensate at $T = 0$, this mode corresponds to the ω_+ frequency in (12.11), which approaches $2\omega_\perp$ in the $\lambda \to 0$ limit. It is associated with a breathing-like oscillation in the radial direction. For the experimental parameters $\omega_\perp/2\pi = 182.6$ Hz and $\lambda = 0.0646$, the TF transverse-breathing-mode frequency is $\omega_+ \simeq 2.00052\,\omega_\perp$. The frequency ω_- of the other mode approaches $\sqrt{5/2}\,\omega_z$ in the $\lambda \to 0$ limit, and corresponds to a breathing-like oscillation predominantly along the axial direction (the z-direction).

The thermal cloud dynamics above $T_{\rm BEC}$ is given by (3.42) with $n_{\rm c} = 0$ and $C_{12} = 0$. Since the thermal cloud mean-field term $2g\tilde{n}$ in (3.16) has a very small effect on the dynamics in the normal state, it can also be neglected and the distribution function is then a solution of the kinetic equation

$$\frac{\partial}{\partial t} f(\mathbf{p},\mathbf{r},t) + \frac{\mathbf{p}}{m}\cdot\nabla_{\mathbf{r}} f(\mathbf{p},\mathbf{r},t) - \nabla_{\mathbf{r}} V_{\rm trap}(\mathbf{r},t)\cdot\nabla_{\mathbf{p}} f(\mathbf{p},\mathbf{r},t) = C_{22}[f]\,. \tag{12.13}$$

The lowest-energy collective modes for the thermal gas can be found from (12.13) using the moment technique method discussed in Chapter 10. Guéry-Odelin *et al.* (1999) used this procedure to study the modes of a classical gas, but the results for a noncondensed Bose gas are essentially the same. The frequency and damping of the modes depend upon the mean collision rate τ^{-1} relative to the mode frequency ω. In the collisionless regime ($\omega\tau \to \infty$) the frequencies of the transverse breathing mode ($m = 0$) and the quadrupole mode ($m = 2$) are both $\omega = 2\omega_\perp$, and the damping is zero. In the opposite, collisional hydrodynamic, limit ($\omega\tau \to 0$), the modes are still undamped, but the frequencies of the modes are $\omega_{m=0} = \sqrt{10/3}\,\omega_\perp$ and $\omega_{m=2} = \sqrt{2}\,\omega_\perp$ (where the $m = 0$ frequency is specifically for the cigar-shaped limit $\lambda \ll 1$) (Griffin *et al.*, 1997; Guéry-Odelin *et al.*, 1999). In general, the modes will be damped through collisions and will have frequen-

cies between the collisionless and hydrodynamic limits. In the experiment of Chevy *et al.* (2002) the system is in the near-collisionless regime, so that both thermal cloud modes were only weakly damped, with frequency $\omega \simeq 2\omega_\perp$.

We now discuss the results of simulations based on the coupled ZNG equations. The experiments of Chevy *et al.* (2002) were performed over a range of temperatures where the condensate coexists with a thermal cloud. The main conclusion of these experiments is that the transverse breathing mode has a frequency that is almost independent of temperature and a damping rate that is much smaller than that of the $m = 2$ quadrupole mode.[9] As will become clear in the following, the observed behaviour of the breathing mode is a consequence of the near degeneracy of the condensate and thermal cloud mode frequencies, and of the way in which the two components are initially excited.

The experimental scheme used to excite the system was to suddenly change the radial trap frequency and then to reset it to its original value after some short time Δt. This stepped ("top hat") excitation can be represented by

$$\omega'_\perp(t) = \omega_\perp\{1 + \alpha[\Theta(t) - \Theta(t - \Delta t)]\}, \tag{12.14}$$

where $\Theta(x)$ is the Heaviside step function. The simulations were performed using this excitation procedure with the experimental parameters $\alpha = 0.26$, and $\omega_\perp \Delta t = 0.172$. It is clear that varying the trapping frequency affects both the condensate and thermal cloud and thus both will be excited.

We show in Fig. 12.17(a) the result of a simulation at $T = 125\,\mathrm{nK}$ (whereas the experimentally measured critical temperature was $T_{\mathrm{BEC}} \simeq 290\,\mathrm{nK}$). One sees that the normalized radial moments of both components oscillate in phase with approximately equal amplitudes. In addition, the oscillation frequency is very close to $2\omega_\perp$ and the damping rate is very small. As discussed earlier, the frequency is close to that expected for the condensate by itself and for the thermal cloud in the collisionless regime. This in-phase oscillation of the two components at a common frequency is reinforced by the mean-field interactions. The explanation of the small damping rate, however, is not so obvious. By comparison with other modes that we have studied, one might have expected Landau damping to be important. However, the present situation is unique in that the two components are oscillating in phase, and as a result, their *relative* motion is minimal. It is therefore reasonable to expect the Landau damping to be suppressed.

This can be checked by means of a simulation in which only the condensate

[9] The damping rate was also much smaller than that typically found for the quadrupole mode in other experiments (Jin *et al.*, 1997; Stamper-Kurn *et al.*, 1998; Maragò *et al.*, 2001).

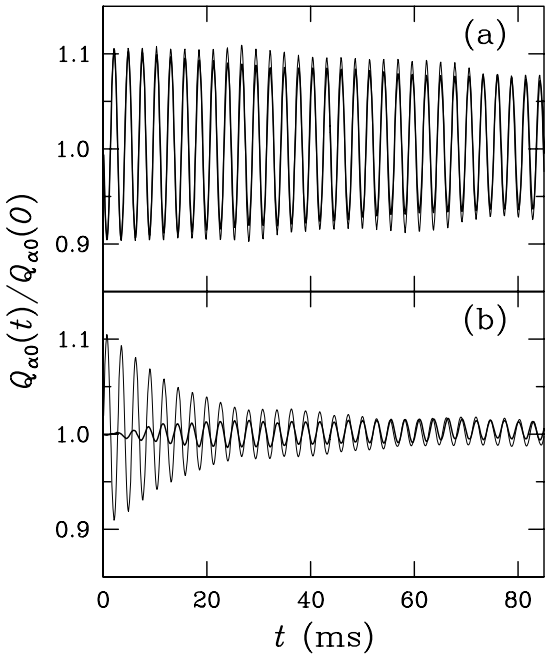


Fig. 12.17. Time-dependent radial moments for the condensate, $Q_{c0}(t)$ (thinner line) and thermal cloud, $Q_{n0}(t)$ (thicker line), divided by the respective initial values $Q_{c0}(0)$ and $Q_{n0}(0)$. The figures are for a temperature $T = 125\,\mathrm{nK}$ and show the result of (a) exciting the system using the "top hat" perturbation scheme employed experimentally, (12.14), and (b) exciting the condensate only by initially imposing a velocity field $\mathbf{v}_c \propto x\hat{\mathbf{i}} + y\hat{\mathbf{j}}$ (from Jackson and Zaremba, 2003a).

is excited. To do so, we imposed a radial velocity field $\mathbf{v}_c \propto x\hat{\mathbf{i}} + y\hat{\mathbf{j}}$ on the condensate but left the thermal cloud in its original equilibrium state. The results of this simulation are shown in Fig. 12.17(b). The condensate oscillation is seen to damp quite rapidly as the thermal cloud is set into motion by the dynamic mean field of the condensate. Once the oscillation has built up to the point where the two components are oscillating together, the subsequent damping is comparable to that seen in Fig. 12.17(a). The much larger damping rate at early times is thus clearly associated with Landau damping, which is operative as long as the two components are moving relative to each other. Further evidence supporting this explanation will be given at the end of this section.

This leaves unanswered the question regarding the source of the damping in Fig. 12.17(a). To answer this question, we performed a simulation in which collisions were artificially switched off. The only coupling between

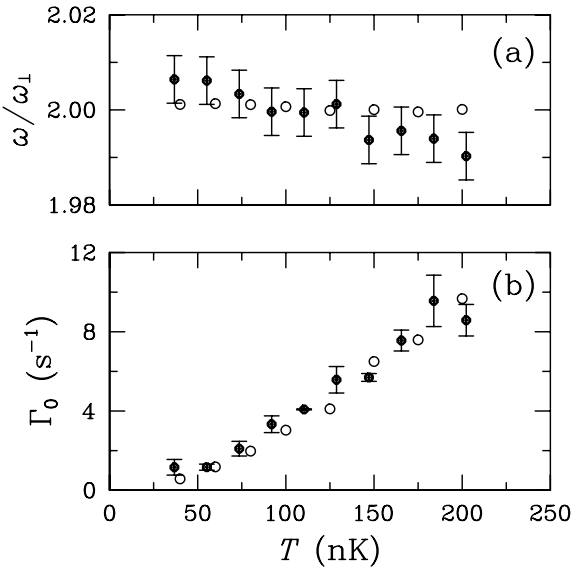


Fig. 12.18. (a) Frequency and (b) damping rate of the condensate breathing mode. The open circles give the simulation results while the solid circles give the experimental data (Chevy *et al.*, 2002). The simulation parameters and excitation scheme were chosen to correspond to the experimental conditions (from Jackson and Zaremba, 2002d).

the two components was then the mean-field interaction. The striking result found was that the damping is at least an order of magnitude smaller than for the collisional result in Fig. 12.17(a) (Jackson and Zaremba, 2002d). This residual damping is presumably Landau damping, but it clearly makes a negligible contribution once collisions are fully included. The transverse breathing mode is thus unique in that collisional damping is dominant even for the Bose-condensed gas.

To compare with experiment, data of the kind shown in Fig. 12.17(a) were analyzed using a single-mode fit to extract the frequency and damping rate. The results of this analysis are shown in Fig. 12.18 as functions of temperature. The experimental frequencies show a slightly stronger variation with temperature than predicted by the theory, but this is a 1% effect and the theoretical frequencies lie mainly within the experimental uncertainties. However, there is excellent agreement between theory and experiment for the damping rate. As discussed above, this damping rate must be due to C_{12} and C_{22} collisional processes and *not* to Landau damping. The ability of the

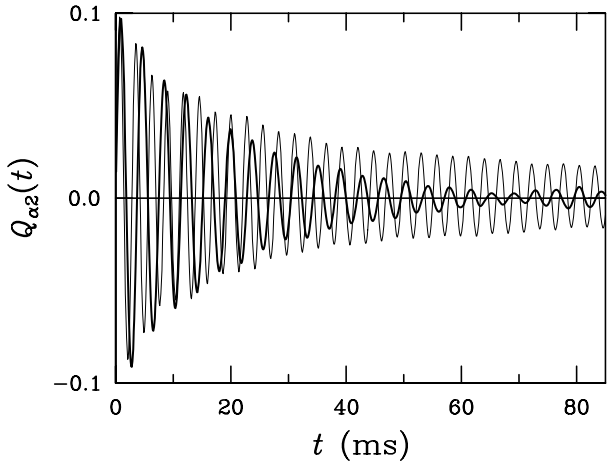


Fig. 12.19. Time-dependent quadrupole moments $\langle x^2 - y^2 \rangle$ for the condensate (thicker line) and thermal cloud (thinner line); the moments have been scaled so that they overlay one another. The plots are for a temperature $T = 125\,\mathrm{nK}$ and show the result of initially imposing a velocity field of the form $\mathbf{v}_{\{\mathrm{c,n}\}} \propto x\hat{\mathbf{i}} - y\hat{\mathbf{j}}$ on both components (from Jackson and Zaremba, 2003a).

ZNG theory to discriminate between these different damping mechanisms confirms that the theory incorporates the important physics.

Before leaving this topic, we provide some further evidence supporting our explanation of the small damping rate of the transverse breathing mode. Our work suggests that quite different results should be found when there is significant relative motion of the condensate and thermal cloud, as in the $m = 2$ quadrupole mode. Simulations of this mode were performed by imposing identical velocity fields of the form $\mathbf{v}_{\{\mathrm{c,n}\}} \propto x\hat{\mathbf{i}} - y\hat{\mathbf{j}}$ on the condensate and thermal cloud components. An example of the subsequent oscillations of the condensate and thermal cloud is shown in Fig. 12.19. One sees that the frequencies are indeed quite different in this case. The condensate oscillates with a frequency approximately equal to $\sqrt{2}\omega_\perp$, while the thermal cloud oscillates at a frequency close to $2\omega_\perp$. This nondegeneracy of the two modes implies that the two components are continually moving relative to each other. A fit to the condensate data yields a damping rate an order of magnitude *larger* than that observed for the transverse breathing mode.

Another way of lifting the degeneracy of the condensate and thermal cloud modes is to increase the anisotropy parameter λ. Specifically, we considered a trap whose frequencies were chosen to yield a geometric mean that was the

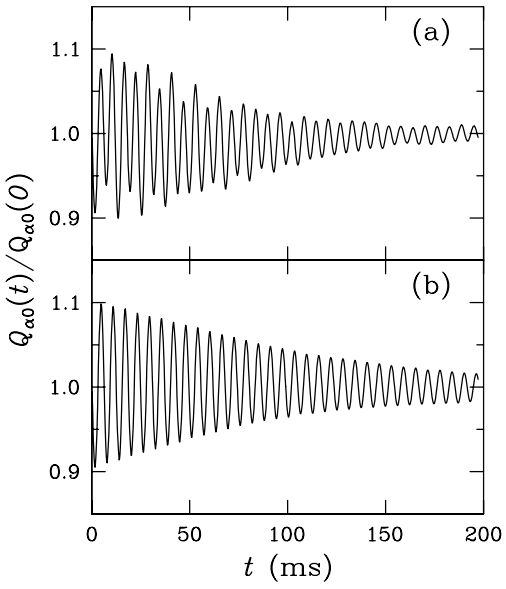


Fig. 12.20. Time-dependent radial moments for (a) the condensate and (b) the thermal cloud, divided by the corresponding values at $t = 0$, for a trap with anisotropy parameter $\lambda = 0.75$ and $\bar{\omega}/2\pi = 73.3$ Hz. The simulation was done at $T = 125\,\mathrm{nK}$, using the stepped excitation (12.14) (from Jackson and Zaremba, 2003a).

same as in the ENS experiment ($\bar{\omega}/2\pi = \lambda^{1/3}\omega_\perp/2\pi = 73.3$ Hz), but with $\lambda = 0.75$. Figure 12.20(b) shows the time dependence of the thermal cloud moment Q_n, which exhibits a damped oscillation at essentially one frequency. A single-mode fit to the data yields a frequency ratio $\omega/\omega_\perp \simeq 2.007$ and a damping rate $\Gamma/\omega_\perp \simeq 0.019$. The latter is approximately four times larger than the damping found for the ENS geometry. The behaviour of Q_c, Fig. 12.20(a), is more complex owing to the fact that both $\omega_\pm$ modes are excited in this case. A combination of two damped sinusoids provided a very good fit to the time dependence shown, yielding the following fit parameters: $\omega_-/\omega_\perp \simeq 1.12$, $\gamma_-/\omega_\perp \simeq 0.037$; $\omega_+/\omega_\perp \simeq 2.03$, $\gamma_+/\omega_\perp \simeq 0.024$. The frequencies are quite close to the values given by (12.11) for $\lambda = 0.75$. More importantly, we see that the damping of the ω_+ mode is approximately an order of magnitude larger than that of the transverse breathing mode from which it evolves. Thus, even a relatively small difference in the frequencies of the condensate and thermal cloud modes is sufficient to enhance significantly the Landau damping rate. This emphasizes the very special nature of the transverse breathing mode in a cigar-shaped trap studied by Chevy *et al.* (2002).

13

Landau damping in trapped Bose-condensed gases

With very few exceptions (such as the centre-of-mass dipole mode), collective oscillations in trapped superfluid Bose gases are damped. In the "collisionless" region the damping is second order in the interaction strength. There are three possible components. One is Beliaev damping, which is due to the decay of a single excitation into two excitations; this can occur even at $T = 0$. In addition, there is Landau damping, which is due to a collective mode scattering from thermally excited excitations. This process only occurs at finite temperatures but quickly becomes the dominant damping mechanism as the temperature increases. Both Landau and Beliaev damping arise naturally from the imaginary part of the Beliaev second-order self-energies, as given in (5.40) in the case of a *uniform* Bose gas. Finally there is the damping that arises from the C_{22} and C_{12} collision processes; this is discussed in Chapters 8, 12 and 19.

In Chapter 12 we calculated the damping of various condensate modes at finite temperatures using direct numerical simulations of the ZNG equations. These numerical results were generally in very good agreement with the available experimental data. From a theoretical perspective, one advantage of the simulations is that the Landau damping contribution can be isolated simply by setting the C_{12} and C_{22} collision terms to zero.

After providing an introduction to Landau damping in uniform Bose gases in Section 13.1, we present in Section 13.2 a detailed discussion of Landau damping based on a general formula in terms of Bogoliubov–Popov excitations. This discussion makes it clear that the Landau damping of condensate oscillations arises from the interaction with a thermal cloud of excitations. Starting from this formula, we can compare the results of various approximations. Specifically, in Section 13.3 we evaluate the Landau damping using either the Bogoliubov–Popov or the simpler Hartree–Fock approximation for the excitations. These results can also be compared with the semiclassical

treatment of the HF quantum states, thereby relating this to the ZNG approach. We find that all these approximations are in good agreement with each other, which justifies the semiclassical treatment of the excitations used in the ZNG approach. Sections 13.2 and 13.3 are mainly based on the work of Jackson and Zaremba (2003b).

In the region where strong collisions produce local equilibrium, the dynamics of a Bose-condensed gas are governed by the Landau two-fluid hydrodynamic equations. As will be discussed in Chapters 17–19, the normal-mode solutions of the linearized two-fluid equations exhibit hydrodynamic damping which is associated with transport coefficients. The physics behind this kind of damping in the collision-dominated hydrodynamic region is quite different from that of Landau and Beliaev damping, which arise in the weakly collisional region. In this regard, we emphasize that the Landau damping considered in this chapter has no relevance in the hydrodynamic region to be discussed in Chapters 17–19.

13.1 Landau damping in a uniform Bose gas

Landau damping has been mentioned already several times in Sections 5.3, 6.6 and 7.3 and was referred to extensively in Chapter 12 on numerical simulations. Particularly in the latter, it was emphasized that this damping is associated with the excitation of the thermal atoms by the dynamic mean field of the condensate. The essential physics involved in this process is illustrated by considering the problem of a *uniform normal* Bose gas in which the interaction between atoms is treated at a mean-field level. We begin this section by considering this simpler situation and then make contact with the Landau damping arising in the case of a uniform Bose-condensed gas. In subsequent sections of this chapter we discuss the analogous results for the Landau damping in a trapped Bose-condensed gas.

We consider a uniform normal Bose gas of density n_0 in which the particles interact via a contact potential of strength g. The collisionless dynamics of such a gas can be described by the kinetic equation (3.42) but with the collisional integrals on the r.h.s. and the condensate mean field in U set to zero. To drive the system out of equilibrium, an additional *weak* external potential $\delta U(\mathbf{r}, t)$ is applied, so that the total potential acting on the particles is $U(\mathbf{r}, t) = 2gn(\mathbf{r}, t) + \delta U(\mathbf{r}, t)$.

To determine the response of the gas to lowest order in the perturbation, we expand the distribution function $f(\mathbf{p}, \mathbf{r}, t)$ as

$$f(\mathbf{p}, \mathbf{r}, t) = f^0(\mathbf{p}) + \delta f(\mathbf{p}, \mathbf{r}, t), \tag{13.1}$$

where $f^0(\mathbf{p})$ is the equilibrium Bose distribution, which we note is independent of position for a uniform Bose gas. Expanding (3.42) to first order in the deviations δU and δf, we obtain the linearized kinetic equation

$$\frac{\partial \delta f}{\partial t} + \frac{\mathbf{p}}{m} \cdot \nabla_{\mathbf{r}} \delta f - \nabla_{\mathbf{r}} [2g\delta n(\mathbf{r}, t)] + \delta U(\mathbf{r}, t)] \cdot \nabla_{\mathbf{p}} f^0(\mathbf{p}) = 0. \tag{13.2}$$

We now assume that $\delta U(\mathbf{r}, t) = \delta U_{\mathbf{q}\omega} e^{i(\mathbf{q}\cdot\mathbf{r} - \omega t)}$, i.e. a wave-like perturbation. Writing the deviation as $\delta f(\mathbf{p}, \mathbf{r}, t) = \delta f_{\mathbf{q}\omega}(\mathbf{p}) e^{i(\mathbf{q}\cdot\mathbf{r} - \omega t)}$, we find that $\delta f_{\mathbf{q}\omega}(\mathbf{p})$ satisfies

$$\left(\frac{\mathbf{q} \cdot \mathbf{p}}{m} - \omega \right) \delta f_{\mathbf{q}\omega}(\mathbf{p}) - (2g\delta n_{\mathbf{q}\omega} + \delta U_{\mathbf{q}\omega})\, \mathbf{q} \cdot \nabla_{\mathbf{p}} f^0(\varepsilon_p) = 0, \tag{13.3}$$

where

$$\delta n_{\mathbf{q}\omega} = \int \frac{d\mathbf{p}}{(2\pi\hbar)^3} \delta f_{\mathbf{q}\omega}(\mathbf{p}), \tag{13.4}$$

and

$$f^0(\tilde{\varepsilon}_p) \equiv \frac{1}{e^{\beta(\tilde{\varepsilon}_p - \tilde{\mu}_0)} - 1}. \tag{13.5}$$

Here, $\tilde{\varepsilon}_p = p^2/2m + 2gn_0$ is the HF energy of an atom. Since the gas is assumed to be normal, the chemical potential $\tilde{\mu}_0$ takes a value less than the HF mean field $2gn_0$.

Solving (13.3) for $\delta f_{\mathbf{q}\omega}(\mathbf{p})$ and integrating the result over $\mathbf{p}$, one obtains a self-consistent equation for $\delta n_{\mathbf{q}\omega}$:

$$\delta n_{\mathbf{q}\omega} = -\int \frac{d\mathbf{p}}{(2\pi\hbar)^3} \frac{\mathbf{p} \cdot \mathbf{q}}{m} \frac{\partial f^0(\tilde{\varepsilon}_p)}{\partial \tilde{\varepsilon}_p} \frac{(2g\delta n_{\mathbf{q}\omega} + \delta U_{\mathbf{q}\omega})}{\omega - \mathbf{p} \cdot \mathbf{q}/m}. \tag{13.6}$$

This equation has the solution

$$\delta n_{\mathbf{q}\omega} = \chi_{\tilde{n}\tilde{n}}(\mathbf{q}, \omega) \delta U_{\mathbf{q}\omega}, \tag{13.7}$$

which defines the interacting density response function

$$\chi_{\tilde{n}\tilde{n}}(\mathbf{q}, \omega) = \frac{\chi^0_{\tilde{n}}(\mathbf{q}, \omega)}{1 - 2g\chi^0_{\tilde{n}}(\mathbf{q}, \omega)} \tag{13.8}$$

in terms of the noninteracting density response function

$$\chi^0_{\tilde{n}}(\mathbf{q}, \omega) = -\int \frac{d\mathbf{p}}{(2\pi\hbar)^3} \frac{\mathbf{p} \cdot \mathbf{q}}{m} \frac{\partial f_0(\tilde{\varepsilon}_p)}{\partial \tilde{\varepsilon}_p} \frac{1}{\omega + i0^+ - \mathbf{p} \cdot \mathbf{q}/m}. \tag{13.9}$$

For the uniform system under consideration, these response functions depend only on the magnitude of the wavevector $\mathbf{q}$. In addition, a small imaginary part has been added to the frequency ω to yield the causal response function.

One notes that (13.9) is the low-$\mathbf{q}$ limit of the more general expression for the density response function in (5.42). In the literature on the many body theory of Bose gases, the expression $\chi_{\tilde{n}\tilde{n}}(\mathbf{q},\omega)$ given in (13.8) is referred to as the generalized random phase approximation (GRPA) since it includes both Hartree and exchange interactions. This is the origin of the factor 2 in the denominator of (13.8) (for a more detailed discussion, see Section 5.4).

The preceding analysis demonstrates how the collisionless kinetic equation can be used to obtain the response of a uniform normal Bose gas to an external perturbation in terms of the density response function $\chi_{\tilde{n}\tilde{n}}(\mathbf{q},\omega)$. The poles of this function indicate a resonant response of the system and correspond to the collective density fluctuations of the interacting gas. In this GRPA theory, the poles occur when

$$1 - 2g\chi^0_{\tilde{n}}(\mathbf{q},\omega) = 0, \tag{13.10}$$

which is satisfied for some complex-valued frequency $\Omega(\mathbf{q}) = \omega(\mathbf{q}) - i\Gamma(\mathbf{q})$. This collective mode is referred to as zero sound, a terminology first introduced by Landau (1957) for the analogous mode in an interacting Fermi gas. The imaginary part of $\Omega(\mathbf{q})$ determines the damping of the mode and has come to be known as Landau damping. It is clear that the width of the mode is related to

$$\mathrm{Im}\chi^0_{\tilde{n}}(\mathbf{q},\omega(\mathbf{q})) = 2\pi \int \frac{d\mathbf{p}}{(2\pi\hbar)^3}[f^0(\tilde{\varepsilon}_{\mathbf{p}}) - f^0(\tilde{\varepsilon}_{\mathbf{p}+\hbar\mathbf{q}})] \times \delta(\hbar\omega(\mathbf{q}) - (\tilde{\varepsilon}_{\mathbf{p}+\hbar\mathbf{q}} - \tilde{\varepsilon}_{\mathbf{p}})). \tag{13.11}$$

Here we have made use of the more general result for $\chi^0_{\tilde{n}}(\mathbf{q},\omega)$ given by (5.42). As the energy-conserving delta function in (13.11) makes clear, Landau damping arises when the zero sound mode frequency overlaps with the allowed excitation spectrum of the noninteracting HF gas, i.e. whenever

$$\hbar\omega(\mathbf{q}) = \tilde{\varepsilon}_{\mathbf{p}+\hbar\mathbf{q}} - \tilde{\varepsilon}_{\mathbf{p}} \tag{13.12}$$

is satisfied for some momentum $\mathbf{p}$.

In the foregoing discussion, the single-particle excitations are induced by the dynamic mean field $2g\delta\tilde{n}(\mathbf{r},t)$ of the interacting normal gas. These excitations, however, can be induced equally well by an external perturbation. This is essentially what happens for the Bose-condensed gas, in which the role of the external perturbation is played by the dynamic mean field $2g\delta n_{\mathrm{c}}(\mathbf{r},t)$ of the condensate. This mean-field potential induces a density fluctuation $\delta\tilde{n}(\mathbf{r},t)$ in the thermal component that is given, in the frequency

domain, by (see (5.45))

$$\delta\tilde{n}(\mathbf{r},\omega) = \int d\mathbf{r}' \chi^0_{\tilde{n}}(\mathbf{r},\mathbf{r}',\omega) 2g\delta n_{\rm c}(\mathbf{r}',\omega). \tag{13.13}$$

Here $\chi^0_{\tilde{n}}$ is the density response function (5.42) for a noninteracting thermal gas of atoms, with HF spectrum $\tilde{\varepsilon}_p = p^2/2m + 2g(n_{\rm c0} + \tilde{n}_0)$ and with a thermal cloud chemical potential $\tilde{\mu}_0$ equal to $\varepsilon_{\rm c0}$ in (3.48). In obtaining (13.13), we have neglected on the r.h.s. the weak HF field $2g\delta\tilde{n}$ due to the noncondensate atoms themselves since, at low temperatures, this is negligible compared with the much larger mean field arising from the high-density condensate.

Within the ZNG theory, the condensate collective modes are given by solutions of the GGP equation (3.21). In Section 8.1, we ignored the fluctuations of the thermal cloud and concentrated on the damping associated with the source term R in (3.21). This term was evaluated to first order in $\delta n_{\rm c}$, assuming that all the thermal atoms were in thermal equilibrium. In the following, we continue to use this approximation for R, but now include the thermal cloud mean field $\delta\tilde{n}(\mathbf{r},t)$ by means of (13.13). Solving the linearized version of equation (3.21) for $\delta n_{\rm c}$, where both $\delta\tilde{n}$ and R are calculated to first order in $\delta n_{\rm c}$ as described above, the frequency of the (damped) condensate mode is the solution of (compare (5.72))

$$\omega^2 = c_0^2 q^2 \left[1 + 4g\chi^0_{\tilde{n}}(\mathbf{q},\omega)\right] - \frac{i\omega}{\tau'} \ . \tag{13.14}$$

Here $c_0 = \sqrt{gn_{\rm c0}/m}$ is the Bogoliubov phonon velocity (see (2.30) and below). The collision time τ' is defined in (8.22) and describes the effect of the C_{12} collisions arising in the absence of diffusive equilibrium between the condensate and the static thermal cloud.

Neglecting the small shift in the phonon energy arising from the real part of $\chi^0_{\tilde{n}}$ and setting $\omega = c_0 q$ on the r.h.s. of (13.14), one finds that

$$\Omega(q) = c_0 q \left[1 - i4g\,\mathrm{Im}\,\chi^0_{\tilde{n}}(\mathbf{q}, c_0 q) - \frac{i}{\tau' c_0 q}\right]^{1/2} \tag{13.15}$$

or

$$\Omega(q) \simeq c_0 q - i\left[\Gamma_{\rm L}(q) + \frac{1}{2\tau'}\right], \tag{13.16}$$

where the Landau damping contribution is given by

$$\Gamma_{\rm L}(q) = 2gc_0 q\,\mathrm{Im}\,\chi^0_{\tilde{n}}(\mathbf{q}, c_0 q). \tag{13.17}$$

Using (13.9) for the thermal atom response function, we have

$$\mathrm{Im}\,\chi^0_{\tilde{n}}(\mathbf{q},\omega) = -\pi \int \frac{d\mathbf{p}}{(2\pi\hbar)^3}\frac{\mathbf{q}\cdot\mathbf{p}}{m}\frac{\partial f^0(\tilde{\varepsilon}_p)}{\partial \tilde{\varepsilon}_p}\delta\left(\omega - \frac{\mathbf{q}\cdot\mathbf{p}}{m}\right). \tag{13.18}$$

Carrying out the angular integration first, we obtain

$$\begin{aligned}\mathrm{Im}\,\chi^0_{\tilde{n}}(\mathbf{q},\omega) &= -\frac{m\omega}{4\pi\hbar^3 q}\int_{m\omega/q}^{\infty} dp\, p\frac{\partial f^0(\tilde{\varepsilon}_p)}{\partial \tilde{\varepsilon}_p} \\ &= -\frac{m\omega}{4\pi\hbar^3 q}\int_{\bar{\varepsilon}(\omega/q)}^{\infty} d\varepsilon \frac{\partial f^0(\varepsilon)}{\partial \varepsilon},\end{aligned} \tag{13.19}$$

where $\bar{\varepsilon}(\omega/q) \equiv \frac{1}{2}m\,(\omega/q)^2 + 2g(n_{\mathrm{c}0} + \tilde{n}_0)$. Setting $\omega = c_0 q$, we find

$$\mathrm{Im}\,\chi^0_{\tilde{n}\tilde{n}}(\mathbf{q},\omega = c_0 q) = \frac{m^2 c_0}{4\pi\hbar^3} f^0(\bar{\varepsilon}), \tag{13.20}$$

with $\bar{\varepsilon} = \frac{1}{2}mc_0^2 + 2g(n_{\mathrm{c}0} + \tilde{n}_0)$. Using the relation $\tilde{\mu}_0 = \mu_{\mathrm{c}0} = g(n_{\mathrm{c}0} + 2\tilde{n}_0)$, valid in equilibrium, the equilibrium Bose distribution in (13.20) takes the form

$$f^0(\bar{\varepsilon}) = \frac{1}{e^{\beta(\bar{\varepsilon}-\tilde{\mu}_0)} - 1} = \frac{1}{e^{\beta(\frac{3}{2}gn_{\mathrm{c}0})} - 1}. \tag{13.21}$$

At temperatures sufficiently high that $k_{\mathrm{B}}T \gg gn_{\mathrm{c}0}$, (13.20) then reduces to

$$\lim_{q\to 0}\mathrm{Im}\,\chi^0_{\tilde{n}}(\mathbf{q},\omega = c_0 q) = \frac{1}{3}\frac{mk_{\mathrm{B}}T}{2\pi\hbar^3 c_0}, \tag{13.22}$$

where the factor 1/3 can be traced to the exchange term in the single-particle HF self-energy.

Using the preceding results based on the GRPA, we find that the Landau damping in (13.17) is given by

$$\Gamma_{\mathrm{L}}(q) = \frac{gmk_{\mathrm{B}}Tq}{3\pi\hbar^3} = \frac{4}{3}\frac{ak_{\mathrm{B}}Tq}{\hbar}, \tag{13.23}$$

where a is the s-wave scattering length. The *exact* result for Γ_{L} obtained from a careful evaluation of the Beliaev single-particle self-energies at finite T (see (8.28)) is the same as (13.23), apart from a slightly smaller numerical coefficient (namely, 4/3 is replaced by $3\pi/8$). This difference is due to our neglect of the fluctuations associated with the anomalous pair density $\tilde{m}(\mathbf{r},t)$. As discussed in Section 5.3, such terms are present in the exact equation of motion for $\Phi(\mathbf{r},t)$ given in (3.8) but are neglected in deriving (3.21). The original RPA calculation (given by time-dependent *Hartree* theory) carried out by Szépfalusy and Kondor (1974) ignored all exchange terms. Then the factor 1/3 would be missing in (13.23), as would the factor 4 appearing

in (13.14) (see also (5.72)). Thus the RPA result for the Landau damping found by Szépfalusy and Kondor was $\Gamma_{\rm L} = ak_{\rm B}Tq/\hbar$, without the 4/3 factor in (13.23) that arises when exchange is included. It seems fortuitous that quite different approximations for Landau damping give results very close to the exact result (8.28) based on the Beliaev self-energies.

13.2 Landau damping in a trapped Bose gas

In this section, we review the calculation of Landau damping in a trapped Bose gas at finite temperatures, treating the gas of thermal excitations in various approximations. These are based on a derivation of Landau damping using second-order perturbation theory and will then be compared with the results of direct numerical simulations of the ZNG equations.

13.2.1 Bogoliubov approximation

As discussed in Chapters 2 and 3, the Bogoliubov Hamiltonian can be diagonalized by means of the Bogoliubov transformation (recall that $\hat{\psi} = \Phi_0 + \tilde{\psi}$)

$$\tilde{\psi}(\mathbf{r}) = \sum_i \left[u_i(\mathbf{r})\alpha_i - v_i^*(\mathbf{r})\alpha_i^\dagger \right]. \tag{13.24}$$

The quasiparticle amplitudes u_i and v_i satisfy the Bogoliubov equations

$$\begin{aligned} \hat{L}u_i - gn_{\rm c0}v_i &= E_i u_i \\ \hat{L}v_i - gn_{\rm c0}u_i &= -E_i v_i \,, \end{aligned} \tag{13.25}$$

where we introduce the operator $\hat{L} \equiv -\hbar^2\nabla^2/2m + V_{\rm trap} + 2gn_{\rm c0} - \mu$. The orthonormality of the quasiparticle amplitudes is specified by the relation

$$\int d\mathbf{r}\ [u_i^*(\mathbf{r})u_j(\mathbf{r}) - v_i^*(\mathbf{r})v_j(\mathbf{r})] = \delta_{ij}\,. \tag{13.26}$$

Apart from a constant, the Bogoliubov Hamiltonian is given by (Fetter and Walecka, 1971)

$$\hat{H}_{\rm B} = \sum_i E_i \alpha_i^\dagger \alpha_i \,, \tag{13.27}$$

where the excitation energies E_i correspond to the condensate collective modes. In thermal equilibrium, these modes are populated according to the Bose distribution.

The cubic terms in the expansion of $\hat{H}$ (see Chapter 3)

$$V^{(3)} = g \int d\mathbf{r}\, \Phi_0 \left(\tilde{\psi}^\dagger \tilde{\psi} \tilde{\psi} + \tilde{\psi}^\dagger \tilde{\psi}^\dagger \tilde{\psi} \right), \tag{13.28}$$

couple the Bogoliubov excitations and lead to a mechanism for their decay. In particular, a condensate mode that has been excited, and is therefore no longer in equilibrium with the other thermal excitations, will undergo Landau damping. To represent this situation, we consider a nonequilibrium state in which one particular mode occupation $n_{\rm osc}$ is large compared with its equilibrium value. The total energy in this mode ($E_{\rm osc} \equiv \hbar\omega_{\rm osc} n_{\rm osc}$) then decays in time as $n_{\rm osc}$ relaxes towards its equilibrium value. The transition rate from the initial nonequilibrium state to any other state can be determined by second-order perturbation theory (Fermi's golden rule) and the average rate of change of the energy in this mode is found to be given by (Pitaevskii and Stringari, 1997)

$$\frac{\dot{E}_{\rm osc}}{E_{\rm osc}} = -\frac{2\pi}{\hbar}\sum_{ij}|A_{ij}|^2(f_i^0 - f_j^0)\delta(E_j - E_i - \hbar\omega_{\rm osc}), \tag{13.29}$$

where the transition matrix element is

$$A_{ij} = 2g\int d\mathbf{r}\,\Phi_0\left[\left(u_i u_j^* - v_i u_j^* + v_i v_j^*\right)u_{\rm osc} - \left(u_i u_j^* - u_i v_j^* + v_i v_j^*\right)v_{\rm osc}\right]. \tag{13.30}$$

The damping rate $\Gamma_{\rm L}$ is defined according to $2\Gamma_{\rm L} = -\dot{E}_{\rm osc}/E_{\rm osc}$ (which implies that the amplitude of the mode decays as $e^{-\Gamma_{\rm L}t}$). Using (13.29), we see that

$$\Gamma_{\rm L} = \frac{\pi}{\hbar}\sum_{ij}|A_{ij}|^2(f_i^0 - f_j^0)\delta(E_j - E_i - \hbar\omega_{\rm osc})\,. \tag{13.31}$$

This expression forms the basis of the calculation of Landau damping preformed by Guilleumas and Pitaevskii (2000, 2003). In their notation,

$$\frac{\Gamma_{\rm L}}{\omega_{\rm osc}} = \sum_{ij}\gamma_{ij}\delta(\omega_{ij} - \omega_{\rm osc}), \tag{13.32}$$

where

$$\gamma_{ij} \equiv \frac{\pi}{\hbar^2\omega_{\rm osc}}|A_{ij}|^2(f_i^0 - f_j^0). \tag{13.33}$$

Here γ_{ij} is the "damping strength" associated with the transition for which the frequency difference is $\omega_{ij} = (E_j - E_i)/\hbar$.

The calculation of the damping then consists of solving the Bogoliubov equations (13.25) for $u_{\rm osc}$, $v_{\rm osc}$ and $\omega_{\rm osc}$ for the condensate mode of interest, together with the corresponding quantities for the thermally populated excitations. The matrix elements A_{ij} in (13.30) can then be evaluated for transitions between each pair of excitations, which in turn yields γ_{ij}. The results of such a calculation will be discussed in Section 13.3.

The expression for Landau damping in (13.31) is the net result of "three-phonon" transitions in which a condensate excitation annihilates a thermal excitation to produce a more energetic thermal excitation, together with the inverse processes. It is of interest to note that (13.31) is equivalent to the imaginary part of the second-order Beliaev self-energies reviewed in Section 5.3. Furthermore, it is identical to the expression (7.45) that emerges from the linearized C_{12} collision integral for Bogoliubov excitations, as discussed in Section 7.3.

13.2.2 Quantum Hartree–Fock approximation

We next evaluate the Landau damping rate (13.31) using the Hartree–Fock approximation for the thermal excitations. This treatment can be further divided into a *quantum* HF and a *semiclassical* HF approximation (the latter being the basis of the ZNG equations).

The quantum HF approximation (Jackson and Zaremba, 2003b) corresponds to setting the Bogoliubov amplitude v_i (but not $v_{\rm osc}$) to zero in the Bogoliubov equations (13.25) and in the transition matrix element A_{ij} in (13.30). The quantum nature of the HF approximation lies in the fact that the wavefunction $u_i(\mathbf{r})$ is determined by the Schrödinger equation $\hat{L}\,u_i(\mathbf{r}) = E_i u_i(\mathbf{r})$ rather than the full Bogoliubov equations (13.25). In this approximation, the transition matrix element (13.30) reduces to

$$A_{ij}^{\rm HF} = 2g \int d\mathbf{r}\, \Phi_0(\mathbf{r}) u_i(\mathbf{r}) u_j^*(\mathbf{r}) [u_{\rm osc}(\mathbf{r}) - v_{\rm osc}(\mathbf{r})]. \tag{13.34}$$

It is instructive to give an alternative derivation of the expression for $\Gamma_{\rm L}$ in (13.31) within the quantum HF approximation described above. This derivation reveals more clearly that Landau damping is associated with the work done on the thermal cloud by the time-dependent mean field of the oscillating condensate.

Within the quantum HF approximation, the excitations of the condensate and thermal cloud are treated in different ways. The condensate excitation evolves dynamically according to a time-dependent GP equation while the thermal cloud excitations respond to this condensate mean field. This picture is quite distinct from the Bogoliubov approach, in which the "condensate mode" is distinguished from the "thermal excitations" only by the assumed different occupations of the respective modes. Common to both pictures, however, is the *absence* of a thermal cloud mean field acting back on the condensate. Such effects are included in the ZNG equations.

For small deviations from equilibrium, the time-dependent condensate

wavefunction is given by

$$\Phi(\mathbf{r},t) = \Phi_0(\mathbf{r}) + \delta\Phi(\mathbf{r},t)\,, \tag{13.35}$$

where the fluctuation $\delta\Phi$ is obtained from the linearized GP equation. The fluctuating part of the condensate density is then

$$\delta n_{\rm c}(\mathbf{r},t) = \Phi_0(\mathbf{r})\left[\delta\Phi(\mathbf{r},t) + \delta\Phi^*(\mathbf{r},t)\right], \tag{13.36}$$

which gives rise to a time-dependent condensate mean field $2g\delta n_{\rm c}(\mathbf{r},t)$ acting on the thermal cloud. Thus the thermal cloud experiences a perturbation given by

$$H'(t) = \int d\mathbf{r}\, 2g\delta n_{\rm c}(\mathbf{r},t)\tilde{n}(\mathbf{r})\,, \tag{13.37}$$

where $\tilde{n}(\mathbf{r})$ represents the density operator of the thermal atom component. The linearized GP equation has solutions of the form

$$\delta\Phi(\mathbf{r},t) = u_{\rm osc}(\mathbf{r})e^{-i\omega_{\rm osc}t} - v^*_{\rm osc}(\mathbf{r})e^{i\omega_{\rm osc}t}\,. \tag{13.38}$$

Here $u_{\rm osc}$ and $v_{\rm osc}$ are, apart from normalization, the Bogoliubov amplitudes introduced in subsection 13.2.1 and $\omega_{\rm osc}$ is the frequency of the condensate mode of interest. In terms of this wavefunction, the condensate density fluctuation in (13.36) is given by

$$\delta n_{\rm c}(\mathbf{r},t) = \delta n_-(\mathbf{r})e^{-i\omega_{\rm osc}t} + \text{c.c.} \tag{13.39}$$

where

$$\delta n_-(\mathbf{r}) \equiv \Phi_0(\mathbf{r})[u_{\rm osc}(\mathbf{r}) - v_{\rm osc}(\mathbf{r})]. \tag{13.40}$$

With the form (13.39) of the condensate density fluctuation, the perturbation given in (13.37) has a harmonic time dependence and can be treated by means of standard time-dependent perturbation theory. One thus finds that the time-averaged rate of change of the thermal cloud energy is given (to lowest order in the perturbation) by the expression

$$\overline{\frac{dE_{\rm th}}{dt}} = 2\omega_{\rm osc}(2g)^2 \int d\mathbf{r} \int d\mathbf{r}'\, \delta n^*_-(\mathbf{r})\chi''_{\tilde{n}\tilde{n}}(\mathbf{r},\mathbf{r}',\omega_{\rm osc})\delta n_-(\mathbf{r}'), \tag{13.41}$$

where $\chi''_{\tilde{n}\tilde{n}}(\mathbf{r},\mathbf{r}',\omega)$ is the imaginary part of the time Fourier transform of the thermal cloud density response function

$$\chi_{\tilde{n}\tilde{n}}(\mathbf{r},\mathbf{r}',t-t') = \frac{i}{\hbar}\theta(t-t')\langle[\tilde{n}(\mathbf{r},t),\tilde{n}(\mathbf{r}',t')]\rangle_0\,. \tag{13.42}$$

The angular brackets denote the equilibrium expectation value. Treating

the thermal atoms as independent HF excitations, the imaginary part of (13.42) is given by the spectral density

$$\chi''_{\tilde{n}\tilde{n}}(\mathbf{r},\mathbf{r}',\omega) = \pi \sum_{ij} u_i^*(\mathbf{r}) u_j(\mathbf{r}) u_j^*(\mathbf{r}') u_i(\mathbf{r}') \left(f_i^0 - f_j^0\right) \delta(\varepsilon_j - \varepsilon_i - \hbar\omega). \quad (13.43)$$

Here $u_i(\mathbf{r})$ is the HF eigenstate (obtained from (13.25) with $v_i(\mathbf{r}) = 0$) with energy $E_i = \varepsilon_i$, and f_i^0 is the thermal Bose distribution. Substituting (13.43) into (13.41), we find

$$\overline{\frac{dE_{\rm th}}{dt}} = 2\pi\omega_{\rm osc} \sum_{ij} |A_{ij}^{\rm HF}|^2 \left(f_i^0 - f_j^0\right) \delta(\varepsilon_j - \varepsilon_i - \hbar\omega_{\rm osc}) \quad (13.44)$$

with

$$A_{ij}^{\rm HF} = 2g \int d\mathbf{r}\, u_i(\mathbf{r}) u_j^*(\mathbf{r}) \delta n_-(\mathbf{r})\,. \quad (13.45)$$

Using (13.40), we see that this matrix element is identical to (13.34).

The rate of change of the condensate mode energy $\dot{E}_{\rm osc}$ is, of course, the negative of the rate of change of the thermal energy given by (13.44). To extract the damping rate we must divide this by the energy of the mode, which is determined by the amplitude of the condensate density fluctuation. If u and v are normalized according to (13.26), this energy is just $\hbar\omega_{\rm osc}$. We thus arrive at the formula (13.31), in which the thermal excitations are approximated by HF quantum states.

13.2.3 Semiclassical HF approximation and ZNG

In this section, we discuss Landau damping using the semiclassical (sc) Hartree–Fock approximation. Formally, this involves taking the $\hbar \to 0$ limit of the response function (13.42). As shown by Jackson and Zaremba (2003b), the Landau damping rate in this limit is given by

$$\frac{\Gamma_{\rm L}^{\rm sc}}{\omega_{\rm osc}} = \frac{2g^2}{\hbar k_{\rm B} T} \tilde{f}(\omega_{\rm osc}), \quad (13.46)$$

where $\tilde{f}(\omega)$ is the Fourier transform of

$$f(t) = \int d\mathbf{r} \int d\mathbf{r}' \delta n_-^*(\mathbf{r}) G_{\rm s}(\mathbf{r},\mathbf{r}',t) \delta n_-(\mathbf{r}'), \quad (13.47)$$

with the self-diffusion function defined as

$$G_{\rm s}(\mathbf{r},\mathbf{r}',t) = \tilde{N} \langle \delta(\mathbf{r} - \mathbf{r}_1(t)) \delta(\mathbf{r}' - \mathbf{r}_1(0)) \rangle_0. \quad (13.48)$$

The self-diffusion function describes the way in which the position of a representative thermal atom is correlated at two different times, and can be

evaluated by averaging over all possible classical trajectories of the atom. Examples of the evaluation of (13.47) and (13.48) are given in Jackson and Zaremba (2003b).

To the extent that C_{22} and C_{12} collisions are ignored, the semiclassical Landau damping rate (13.46) underlies the results of the numerical simulations of the ZNG equations of the kind discussed in Chapters 11 and 12. As an example, Fig. 12.6 gives the damping of a monopole mode *with* and *without* collisions. The latter is an example of the Landau damping we have been considering. However, it is of interest to note that these results for the damping of the mode were obtained by observing the dynamics of the condensate itself, as displayed in Fig. 12.5. Although the physical mechanism for the damping is the excitation of the thermal cloud by the dynamic mean field of the condensate, its effect on the dynamics of the condensate appears only through the mean field that *the thermal cloud exerts on the condensate*. It is therefore clear that there is a subtle and intricate interplay between the dynamics of the condensate and that of the thermal cloud in the ZNG simulations. Both must be described accurately to obtain a consistent description of the mode damping.

13.3 Numerical results for Landau damping

In this section, we compare the numerical results for the Landau damping rate $\Gamma_{\rm L}$ based on the various approximations discussed in Section 13.2. First, we evaluate the expression in (13.31) by treating the thermal component as a gas of Bogoliubov excitations.

The specific system we consider is a gas of ^{87}Rb atoms, with scattering length $a = 5.82 \times 10^{-9}$ m, in an isotropic harmonic trap with frequency $\omega_0/2\pi = 187\,\text{Hz}$. Following the approach of Guilleumas and Pitaevskii (2000), the results are conveniently displayed as a histogram showing the damping strengths γ_{ij} defined in (13.33) versus the excitation frequency $\omega_{ij} = (E_j - E_i)/\hbar$. The spectrum in Fig. 13.1 features several high peaks, which correspond to transitions between low-lying excitations with large overlaps, i.e. large A_{ij} matrix elements, together with large population factors $f_i^0 - f_j^0$. Such a peak or "resonance" is not expected to play a significant role in the Landau damping process unless it happens to lie very close to the mode frequency $\omega_{\rm osc}$. We will ignore these resonant transitions when calculating the Landau damping, and instead focus on the small-amplitude quasicontinuous "background".

Following Guilleumas and Pitaevskii (2000), each delta function in the

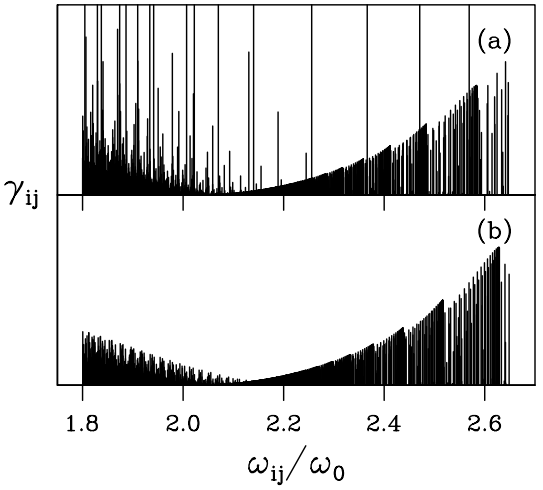


Fig. 13.1. Histogram showing the results of Landau damping calculations for $N_c = 2.5\times10^5$ ^{87}Rb atoms at a temperature such that $k_BT/\mu = 1.5$, where $\mu = 29.9\hbar\omega_0$ is the TF chemical potential. The height of the bars represents the damping strength γ_{ij} of each transition, with the frequency ω_{ij} plotted on the horizontal axis: (a) data for the Bogoliubov thermal excitations and (b) the corresponding quantum HF calculation. Note that in (a) the vertical scale has been expanded to display more clearly the "background" spectrum, which is of most interest. As a result, the highest peaks extend far beyond the vertical range. The mode oscillation frequency for this radial breathing mode is $\omega_{osc} = 2.235\omega_0$ (from Jackson and Zaremba, 2003b).

background spectrum is replaced by a Lorentzian

$$\frac{\Delta}{2\pi[(\omega_{ij}-\omega_{osc})^2+\frac{1}{4}\Delta^2]}. \tag{13.49}$$

In Jackson and Zaremba (2003b), it was shown that the dependence of the damping rate Γ_L on Δ is weak when Δ/ω_0 lies between 0.05 and 0.20. The results for Γ_L as a function of temperature are shown in Fig. 13.2, for different numbers of condensate atoms. There is a rapid increase in Γ_L at low temperatures, followed by a nearly linear temperature dependence at the higher temperatures. The results for $N_c = 5\times10^4$ are identical to those obtained by Guilleumas and Pitaevskii (2000).

We have repeated this calculation for quantum HF thermal excitations following the method discussed in subsection 13.2.2. The spectral density in this case is shown in Fig. 13.1(b). The main difference between the two plots is the absence in (b) of the strong resonances shown in (a), for which the Bogoliubov excitation spectrum was used. However, the background spectrum

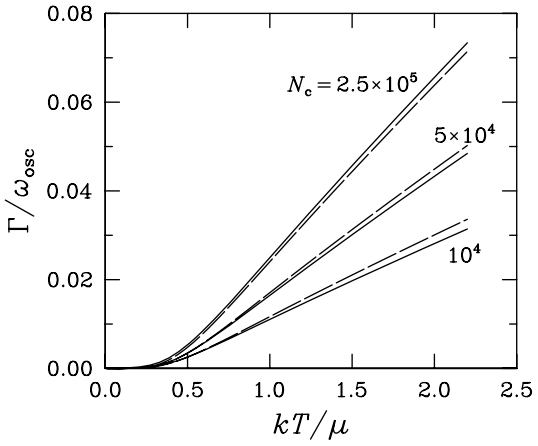


Fig. 13.2. The Landau damping rate, in units of the condensate frequency $\omega_{\rm osc}$, as a function of $k_{\rm B}T/\mu$, where μ is the Thomas–Fermi chemical potential. Results for three different values of the number of condensate atoms $N_{\rm c}$ are shown using the Bogoliubov (solid lines) and quantum HF (broken lines) approximations (from Jackson and Zaremba, 2003b).

is remarkably similar, especially in the vicinity of the condensate mode frequency $\omega_{\rm osc}$. Insight into this is provided by comparing the Bogoliubov and quantum HF excitation frequencies. As evident from Fig. 12.14 of Pitaevskii and Stringari (2003) (see also Fig. 4 of Jackson and Zaremba, 2003b), these two spectra are different at low energies E and angular momenta l, but converge at high E and l. This demonstrates that the excitations take on a single-particle character (You *et al.*, 1997; Dalfovo *et al.*, 1997) in this region. One finds that these single-particle-like excitations are responsible for the majority of the background transitions in Fig. 13.1. This explains the similarity between the HF and Bogoliubov damping rates, as can be seen from Fig. 13.2. The excellent agreement with the Bogoliubov results for a wide range of condensate sizes is in contrast with situation for the *uniform* Bose gas (Pitaevskii and Stringari, 1997), where one finds a significant difference between the results based on these two approximations for the excitation spectra. The reason for this is that the Bogoliubov spectrum for a uniform gas only approaches the single-particle HF form at temperatures much larger than the chemical potential. In a trapped gas, surface excitations with high multipolarities are important even at low temperatures. We conclude that retaining the "collective nature" of the excitations has little effect on the value of the Landau damping in trapped Bose gases.

It is also of interest to compare the Bogoliubov results with those of our

semiclassical simulations based on the ZNG theory (as discussed in subection 13.2.3). We plot the damping rates for both calculations as a function of the number of condensate atoms in Fig. 13.3, for three different temperatures. The behaviour is very similar for the two types of calculation. A comparison of the results in Fig. 13.3 shows that the agreement persists over a wide range of condensate number, although some systematic differences do exist. The simulation rates at low N_c tend to be lower than the Bogoliubov damping rate, while at high N_c there is a tendency for them to be slightly larger. These small differences are presumably due to the different approximations used in the two sets of calculations. These include (a) the use of HF as opposed to Bogoliubov excitations in the simulations, (b) the use of the semiclassical approximation for the thermal cloud dynamics and (c) the inclusion of the thermal cloud mean field in the self-consistent calculations of both the equilibrium and dynamical properties. Concerning (a), we saw in Fig. 13.2 that there is little difference between the Bogoliubov and quantum HF results (over a wide range of temperatures). We therefore do not expect the use of Bogoliubov, as opposed to HF, quasiparticles to change appreciably the results of our semiclassical simulation based on the ZNG equations.

In order to gain more insight into these results, we performed another numerical simulation which conforms more closely to the semiclassical HF approximation considered in Section 13.2.2. As in the full simulations in Chapter 11, we evolve the condensate mode using the time-dependent GP equation but now we do *not* include the mean field of the thermal cloud acting on the condensate. The condensate thus oscillates with a fixed amplitude, giving rise to a harmonic perturbation of the thermal cloud that itself starts off as an equilibrium distribution and evolves classically in time in the presence of the dynamic mean field of the condensate mode. We then calculate the thermal cloud energy $E_{\rm th}(t) = \sum_i \left[p_i^2(t)/2m + V_{\rm th}(\mathbf{r}_i(t))\right]$, where $V_{\rm th}(\mathbf{r})$ is the equilibrium effective potential acting on the thermal cloud. As a result of the work done by the condensate mode on the thermal cloud, the latter experiences an increase in energy with time that we can identify with the time-averaged rate of energy transfer in (13.44). These results agree quite well with the Bogoliubov damping results and the full simulations shown in Fig. 13.3.

In summary, in the present chapter we have studied the Landau damping of condensate modes in isotropic traps and have compared several methods for calculating this damping. Calculations based on Fermi's Golden Rule, involving sums over transitions between excited states, show excellent agreement between a Bogoliubov treatment of the excitations and the quantum

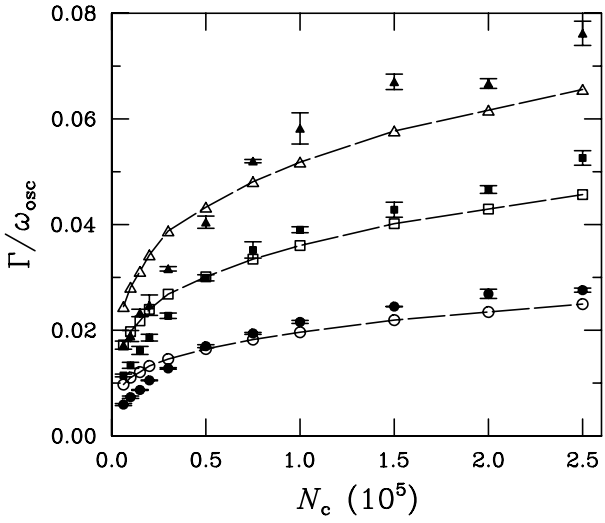


Fig. 13.3. The Landau damping rate, in units of mode frequency $\omega_{\rm osc}$, vs. the number of condensate atoms $N_{\rm c}$, for an isotropic trap. The results from a Bogoliubov calculation (open symbols) are compared with those of a semiclassical simulation (solid symbols) for three temperatures T such that $k_{\rm B}T/\mu = 1$ (circles), 1.5 (squares) and 2 (triangles). The broken lines through the Bogoliubov results serve as guides to the eye. Each simulation data point was calculated by taking the mean of the damping rates extracted from oscillations in the three directions; the error bars correspond to the standard deviation (from Jackson and Zaremba, 2003b).

Hartree–Fock approximation, over a wide range of temperatures and numbers of atoms. These results demonstrate that Landau damping in *trapped* Bose-condensed gases can be thought of as an essentially "single-particle" phenomenon in the sense that the collective nature of the excitations is not very significant. This is consistent with the fact that the Bogoliubov and HF eigenstates differ only at a quantitative level in the energy region of importance to Landau damping (see Fig. 4 of Jackson and Zaremba, 2003b). We have also calculated the Landau damping using two complementary methods involving the N-body simulations discussed in Chapters 11 and 12. In the first method, the condensate wavefunction is evolved using the time-dependent GP equation and its interaction with the thermal cloud leads to a decay of the condensate oscillation amplitude. In the second method, the condensate oscillates with a fixed amplitude and the rate of increase of the thermal cloud energy is determined. Both simulations give very similar

results, and in turn agree with the Fermi Golden Rule calculation based on (13.31) over a wide range of temperatures and condensate sizes.

In this chapter, we have limited ourselves to Landau damping in isotropic traps in order to focus on the essential physical differences between the various approximations. We refer to Jackson and Zaremba (2003b) and Guilleumas and Pitaevskii (2000, 2003) for a detailed discussion of Landau damping in anisotropic traps.

14

Landau's theory of superfluidity

In this chapter, we review the famous Landau theory of superfluidity at finite temperatures. This theory is based on coupled hydrodynamic equations for the superfluid and normal fluid components. Landau's two-fluid description is only valid when collisions among the thermal excitations making up the normal fluid are strong enough to produce local hydrodynamic equilibrium. These two-fluid equations were originally developed for liquid ^{4}He but are thought to be generic in form, describing the collision-dominated hydrodynamic region of all Bose superfluids. In this chapter, we will consider the solutions of the two-fluid equations mainly for a *uniform* superfluid. We discuss the existence of second sound (involving the out-of-phase motion of the superfluid and normal fluid components) as a characteristic feature of a Bose superfluid at finite temperatures.

This chapter gives background material needed for Chapters 15–19. In Chapter 15, we will show that, in the appropriate limit, the Landau two-fluid equations can be derived from the ZNG coupled equations given in Chapter 3 for a trapped dilute Bose-condensed gas. In Chapters 17–19, we extend this discussion and derive the Landau–Khalatnikov two-fluid equations, which include hydrodynamic damping associated with various transport coefficients. Useful reviews of the two-fluid equations in the context of dilute spatially uniform Bose-condensed gases are given by Pethick and Smith (2008, Chapter 10) and Pitaevskii and Stringari (2003, Chapter 6).

14.1 History of two-fluid equations

The original discovery of superfluidity in liquid ^{4}He was dramatically announced with the publication of the famous back-to-back papers of Kapitza (1938) in Moscow and Allen and Misener (1938) in Cambridge. These and subsequent experiments in the next few years showed that superfluid ^{4}He

could exhibit very bizarre hydrodynamic behaviour compared to classical liquids. This led to the development of a two-fluid theory of the hydrodynamic behaviour of liquid ^{4}He by Landau (1941).

An earlier but less complete version of Landau's hydrodynamic equations was developed by Tisza (1940) in the period 1938–40. Tisza's work was stimulated by the pioneering work of London (1938a,b) who argued that superfluidity was associated with some sort of BEC analogous to that in a noninteracting Bose gas. London's classic monograph on superfluidity in liquid ^{4}He, written in the early 1950s (London, 1954), gives a vivid picture of the approach of London and Tisza, based on the central role of BEC, in contrast with the phonon–roton theory of Landau. It was only around 1960 that London's ideas about the role of BEC were shown to provide a microscopic basis for Landau's very successful theory. For further discussion of this early history, see Griffin (1999a) and Balibar (2007).

In the first experimental studies, superfluidity (a term introduced by Kapitza, 1938) was associated entirely with the *relative* motion of the normal fluid and superfluid components under a variety of conditions, as discussed in Wilks (1967). The main point was that while the normal fluid exhibited the finite viscosity and thermal conductivity typical of an ordinary fluid, the superfluid component (which exhibited irrotational flow) did not. More recently, the aspects of superfluidity which have been emphasized are those more directly related to the fact that the superfluid velocity is associated with the gradient of the phase of the macroscopic wavefunction $\Phi(\mathbf{r}, t)$. While this aspect is more fundamental, it is still crucial to understand why superfluidity persists even in the presence of a dissipative normal fluid. This question is naturally addressed in the region where two-fluid hydrodynamics is valid.

In essence, Landau (1941) developed his generic two-fluid hydrodynamics by generalizing the standard theory of classical hydrodynamics to include equations of motion for a new "superfluid" degree of freedom. We recall that the equations of classical fluid dynamics were developed well before one knew about the existence of atoms. Since the work of Boltzmann and Maxwell in the 1880s, we know that the "coarse-grained" hydrodynamic description of a fluid in terms of a few quantities such as the local density $n(\mathbf{r}, t)$ and the local velocity $\mathbf{v}(\mathbf{r}, t)$ is only valid when the collisions between atoms are strong enough to produce local equilibrium (see for example Huang, 1987). As a result, the equations of collisional hydrodynamics only describe low-frequency phenomena, where the period of a collective oscillation is much longer than the relaxation time needed for the atoms to reach local equilibrium.

In his 1941 paper, Landau did not connect the superfluid component with

the motion of a "Bose condensate". Indeed, he rejected the efforts of London (1938a,b) and Tisza (1938, 1940) to use a Bose-condensed gas as a simple model to obtain some insight into the strange behaviour of superfluid ^{4}He. However, since the period 1957–65, Landau's superfluid degree of freedom has been understood microscopically in terms of the Bose order parameter $\Phi(\mathbf{r},t)$. In particular, the superfluid velocity field $\mathbf{v}_{\rm s}(\mathbf{r},t)$ is related to the gradient of the phase of $\Phi(\mathbf{r},t)$, as given by (2.20). A clear elementary discussion of the connection between $\Phi(\mathbf{r},t)$ and superfluidity in Bose fluids is given in the classic monograph by Nozières and Pines (1990). Probably the definitive account that formulates the various "levels of theory" for Bose superfluids is the classic paper by Hohenberg and Martin (1965), in which the central unifying role of $\Phi(\mathbf{r},t)$ is shown. Of special importance is their classification of various approximations for the collisionless domain.

Landau's original formulation of his two-fluid hydrodynamic equations was phenomenological in that the superfluid component was introduced without giving it an underlying *microscopic* basis. As mentioned above, Landau (1941) did not refer to the Bose statistics of the ^{4}He atoms. Indeed Landau's paper does not even mention atoms! The first microscopic derivation of the Landau hydrodynamic equations starting from the existence of the macroscopic order parameter $\Phi(\mathbf{r},t)$ was given by Bogoliubov (1970). This derivation was an extension of Bogoliubov's earlier work on deriving hydrodynamic equations for classical liquids without going through the intermediate stage of using Boltzmann-like kinetic equations.[1]

In Chapter 15, we will show that the generalized GP equation for the condensate atoms (given by (3.17)–(3.19)) and the kinetic equation for the noncondensate atoms (given by (3.42) with the collision integrals C_{22} and C_{12} in (3.40) and (3.41), respectively) can be used to derive the Landau two-fluid equations. These two-fluid hydrodynamic equations only emerge if the collisions between atoms are strong enough to drive the system to *local equilibrium*. Physically this requires that the relaxation time $\tau_{\rm R}$ needed to reach the local equilibrium state must be much less than the period of the collective mode that one is studying, i.e. $\omega\tau_{\rm R} \ll 1$. All the various relaxation times τ_{rmR} in a trapped Bose-condensed gas at finite temperatures will be derived in Chapter 18. As noted earlier, the two-fluid hydrodynamic region is the opposite extreme of the *collisionless* region discussed in Chapters 11–

[1] While Bogoliubov's derivation was quite general, buried in his complex analysis is the assumption that the normal fluid and the superfluid are in "local equilibrium" with each other. This assumption is not necessarily valid in the case of dilute Bose gases. In this sense, the two-fluid hydrodynamics of superfluid Bose gases, somewhat surprisingly, can be more complex than that of superfluid ^{4}He; an additional relaxational mode can appear (see Section 15.3).

13, in which collisions can be treated as a *weak perturbation* on the collective dynamics determined by the time-dependent mean fields.

14.2 First and second sound

We begin by writing down the Landau two-fluid equations. The first two equations are familiar from ordinary fluid dynamics. These are the continuity equation

$$\frac{\partial n}{\partial t} + \nabla \cdot \mathbf{j} = 0 \tag{14.1}$$

and the Euler equation

$$m\frac{\partial \mathbf{j}}{\partial t} = -\nabla P - n\nabla V_{\rm trap}, \tag{14.2}$$

where we have included an external trapping potential. Basic to Landau's approach is the existence of a new superfluid degree of freedom. Thus the total *mass* density and *mass* current are given by the sum of two components

$$mn \equiv \rho = \rho_{\rm s} + \rho_{\rm n}, \tag{14.3}$$

$$m\mathbf{j} \equiv \rho_{\rm s}\mathbf{v}_{\rm s} + \rho_{\rm n}\mathbf{v}_{\rm n}. \tag{14.4}$$

The new superfluid component was argued by Landau to exhibit only pure potential (irrotational) flow and to carry no entropy. Thus Landau's additional two hydrodynamic equations were

$$m\frac{\partial \mathbf{v}_{\rm s}}{\partial t} = -\nabla\mu, \tag{14.5}$$

$$\frac{\partial s}{\partial t} + \nabla \cdot (s\mathbf{v}_{\rm n}) = 0\ , \tag{14.6}$$

where s is the local entropy density. The two-fluid equations in the standard Landau form do not include *separate* continuity equations for the superfluid $\rho_{\rm s}(\mathbf{r}, t)$ and normal fluid $\rho_{\rm n}(\mathbf{r}, t)$ mass densities. Finally, it is assumed that these two components are always in *local equilibrium* with each other and hence that the fluctuations are related by the thermodynamic identity

$$n_0\delta\mu = -s_0\delta T + \delta P, \tag{14.7}$$

where n_0 is the equilibrium total density and s_0 is the equilibrium entropy per unit volume.

In their classic account of Bose superfluids, Nozières and Pines (1990) gave an extensive discussion of various formal definitions of the superfluid density and their equivalence. In particular, the normal fluid density $\rho_{\rm n}$

most naturally arises as the linear response of a Bose superfluid to a transverse perturbation (as in the rotating bucket experiment). Linear response theory shows that the superfluid density $\rho_s = \rho - \rho_n$ is given by the difference between the longitudinal and transverse parts of the velocity–velocity correlation function (in the long-wavelength, low-frequency, limit). These results are summarized in Section 6.1 of Griffin (1993). As discussed by Nozières and Pines (1990) (see also Hohenberg and Martin, 1965; Baym, 1969), these formal definitions of ρ_s can be shown to agree with the quantity that appears in the Landau two-fluid equations given above. Moreover, such definitions also make clear the close connection between a finite value of ρ_s and Bose broken symmetry (and the related Bose condensate density n_c). A key relation of this type is the Josephson relation, which is valid at all temperatures (for further discussion, see Gavoret and Noziéres, 1964; Section 6.1 of Griffin, 1993; Holzmann and Baym, 2007).

When the superfluid component vanishes ($\rho_s = 0, \mathbf{v}_s = 0$ and hence $\rho_n = \rho$), the three equations (14.1), (14.2) and (14.6) describe the hydrodynamics of a normal fluid in the absence of any dissipation (i.e. an "ideal" fluid). An excellent review of these equations for a normal fluid and their implications is given in Chapter 1 of *Fluid Mechanics* by Landau and Lifshitz (1959).

For later reference, it is useful to write the conservation equation (14.6) for the entropy in a different way. In terms of the local entropy per unit mass, $\bar{s}(\mathbf{r}, t) = s/\rho$, this equation can be written in the form

$$\frac{\partial \bar{s}}{\partial t} = \frac{\bar{s}}{\rho} \nabla \cdot \rho_s (\mathbf{v}_s - \mathbf{v}_n) - \mathbf{v}_n \cdot \nabla \bar{s}. \tag{14.8}$$

We have used (14.1), (14.3) and (14.4) in obtaining this result. In a normal liquid, where $\rho_s = 0$, the first term vanishes and (14.8) is then referred to as the adiabatic equation. In isentropic processes, the local entropy per unit mass $\bar{s}(\mathbf{r}, t)$ is independent of time and position. In a superfluid, (14.8) shows that an isentropic oscillation requires that $\mathbf{v}_s = \mathbf{v}_n$.

We now consider the *linearized* Landau equations, that is, we expand the local variables to first order in small fluctuations around the static equilibrium values. The local equilibrium entropy density is $s(\mathbf{r}, t) = s_0 + \delta s$, the local equilibrium kinetic pressure is $P(\mathbf{r}, t) = P_0 + \delta P$ and the local chemical potential is $\mu(\mathbf{r}, t) = \mu_0 + \delta\mu(\mathbf{r}, t)$. In this chapter, we restrict ourselves to a *uniform* superfluid, where $\rho_{s0},\ \rho_{n0},\ s_0$ and other thermodynamic quantities are all position independent. This will allow us to see how the linearized Landau equations lead to the existence of two kinds of hydrodynamic oscillation, first and second sound (Landau, 1941). We follow the approach given in Chapter 7 of Nozières and Pines (1990). Useful discussions of the two-

fluid equations in dilute Bose gases are given in Pethick and Smith (2008, p. 300) and in Pitaevskii and Stringari (2003, p. 75).

Multiplying (14.5) by $n_{\rm s0}$ gives (using (14.7))

$$\begin{aligned}\rho_{\rm s0}\frac{\partial\delta\mathbf{v}_{\rm s}}{\partial t} &= -n_{\rm s0}\nabla\delta\mu \\ &= -\frac{n_{\rm s0}}{n_0}\nabla(-s_0\delta T+\delta P) \\ &= -\frac{n_{\rm s0}}{n_0}\nabla\delta P+\frac{n_{\rm s0}}{n_0}s_0\nabla\delta T. \end{aligned} \tag{14.9}$$

Using this in (14.2) gives

$$\begin{aligned}-\nabla\delta P &= \rho_{\rm s0}\frac{\partial\delta\mathbf{v}_{\rm s}}{\partial t}+\rho_{\rm n0}\frac{\partial\delta\mathbf{v}_{\rm n}}{\partial t} \\ &= -\frac{n_{\rm s0}}{n_0}\nabla\delta P+\frac{n_{\rm s0}}{n_0}s_0\nabla\delta T+\rho_{\rm n0}\frac{\partial\mathbf{v}_{\rm n}}{\partial t} \end{aligned} \tag{14.10}$$

or

$$\rho_{\rm n0}\frac{\partial\delta\mathbf{v}_{\rm n}}{\partial t} = -\frac{n_{\rm n0}}{n_0}\nabla\delta P-\frac{n_{\rm s0}}{n_0}s_0\nabla\delta T. \tag{14.11}$$

Combining (14.1) and (14.2) gives

$$\frac{\partial^2\delta\rho}{\partial t^2} = -m\nabla\cdot\frac{\partial\delta\mathbf{j}}{\partial t} = \nabla^2\delta P. \tag{14.12}$$

Finally, (14.6) gives (using (14.11))

$$\begin{aligned}\frac{\partial^2\delta s}{\partial t^2} &= -s_0\nabla\cdot\frac{\partial\delta\mathbf{v}_{\rm n}}{\partial t} \\ &= \frac{s_0}{\rho_0}\nabla^2\delta P+\frac{s_0^2}{\rho_0}\left(\frac{\rho_{\rm s0}}{\rho_{\rm n0}}\right)\nabla^2\delta T \\ &= \frac{s_0}{\rho_0}\frac{\partial^2\delta\rho}{\partial t^2}+\frac{s_0^2}{\rho_0}\left(\frac{\rho_{\rm s0}}{\rho_{\rm n0}}\right)\nabla^2\delta T. \end{aligned} \tag{14.13}$$

The last equation gives the local entropy fluctuations in terms of the local mass density fluctuation $\delta\rho$ and temperature fluctuation δT. It can be rewritten in terms of the local entropy per unit mass $\bar{s}(\mathbf{r},t)$ defined earlier. Using

$$\delta\bar{s} = -\frac{s_0}{\rho_0^2}\delta\rho+\frac{1}{\rho_0}\delta s, \tag{14.14}$$

(14.13) takes on the simpler form

$$\frac{\partial^2\delta\bar{s}}{\partial t^2} = \bar{s}_0^2\left(\frac{\rho_{\rm s0}}{\rho_{\rm n0}}\right)\nabla^2\delta T\ . \tag{14.15}$$

Expanding δP and δT in terms of the $\delta\rho$ and $\delta\bar{s}$ fluctuations, we obtain

$$\begin{aligned}\delta P &= \left(\frac{\partial P}{\partial \rho}\right)_{\bar{s}} \delta\rho + \left(\frac{\partial P}{\partial \bar{s}}\right)_{\rho} \delta\bar{s}, \\ \delta T &= \left(\frac{\partial T}{\partial \rho}\right)_{\bar{s}} \delta\rho + \left(\frac{\partial T}{\partial \bar{s}}\right)_{\rho} \delta\bar{s}\,.\end{aligned} \tag{14.16}$$

Thus we see that (14.15) and (14.12) reduce to two coupled scalar equations for the time-dependent fluctuations $\delta\rho$ and $\delta\bar{s}$. These equations are easily solved for a uniform Bose superfluid. Inserting the normal mode plane-wave solutions appropriate to this case,

$$\delta\rho, \delta\bar{s} \propto e^{i(\mathbf{q}\cdot\mathbf{r}-\omega t)}, \tag{14.17}$$

one finds that these coupled algebraic equations have two solutions, $\omega^2 = u^2 q^2$, where u^2 is the solution of the quadratic equation

$$u^4 - u^2\left[\left(\frac{\partial P}{\partial \rho}\right)_{\bar{s}} + \frac{\rho_{\mathrm{s0}}}{\rho_{\mathrm{n0}}}\frac{T\bar{s}_0^2}{\bar{c}_v}\right] + \frac{\rho_{\mathrm{s0}}}{\rho_{\mathrm{n0}}}\left(\frac{T\bar{s}_0^2}{\bar{c}_v}\right)\left(\frac{\partial P}{\partial \rho}\right)_T = 0. \tag{14.18}$$

Here $\bar{c}_v = T\left(\partial\bar{s}/\partial T\right)_\rho$ is the equilibrium specific heat per unit mass. In writing down the coefficients in (14.18), we have used standard thermodynamic identities. One also has the exact thermodynamic identity (see Section 16 of Landau and Lifshitz, 1959)

$$\left(\frac{\partial P}{\partial \rho}\right)_{\bar{s}} = \left(\frac{\partial P}{\partial \rho}\right)_T + \frac{T}{\bar{c}_v}\left[\frac{1}{\rho_0}\left(\frac{\partial P}{\partial T}\right)_\rho\right]^2 = \frac{\bar{c}_p}{\bar{c}_v}\left(\frac{\partial P}{\partial \rho}\right)_T, \tag{14.19}$$

where $\bar{c}_p = T\left(\partial\bar{s}/\partial T\right)_P$ is the specific heat per unit mass at constant pressure.

The coefficients in (14.18) may appear daunting, but they involve only *equilibrium* thermodynamic quantities, which can be calculated for any given superfluid. We emphasize that (14.18) is valid *both* for a uniform Bose-condensed gas and for superfluid ^{4}He, assuming that the superfluid and normal fluid are in local equilibrium with each other. However, the detailed characteristics and behaviour of the two phonon modes (first and second sound) are quite different in a Bose gas and in superfluid ^{4}He, owing to the different kinds of thermal excitations involved in these two systems.

A key feature of superfluid ^{4}He is that (typically of any liquid) one finds that the temperature and pressure fluctuations are essentially uncoupled, which means that in (14.19) we have $\left(\partial P/\partial T\right)_\rho \simeq 0$. In this case, the adiabatic and isothermal compressibilities are equal $\left(\partial P/\partial\rho\right)_{\bar{s}} = \left(\partial P/\partial\rho\right)_T$, and hence (14.18) reduces to

$$u^4 - u^2(A+B) + AB = 0. \tag{14.20}$$

The two solutions for u^2 are given by

$$u_1^2 = A \equiv \left.\frac{\partial P}{\partial \rho}\right|_{\bar{s}} \quad \text{(first sound)} \tag{14.21}$$

$$u_2^2 = B \equiv \frac{\rho_{\rm s0}}{\rho_0}\left(\frac{T\bar{s}_0^2}{\bar{c}_v}\right) \quad \text{(second sound)}. \tag{14.22}$$

Working out the associated motions, one finds that the first sound mode ($\omega = u_1 k$) involves *in-phase* motion of the superfluid and normal fluid components, with $\delta\mathbf{v}_{\rm n} = \delta\mathbf{v}_{\rm s}$. One can also show that first sound is essentially an isothermal pressure wave ($\delta T = 0$). In contrast, the second sound mode ($\omega = u_2 k$) involves *out-of-phase* motion of the two components, with $\delta\mathbf{j} \simeq 0$ so that

$$\rho_{\rm n0}\,\delta\mathbf{v}_{\rm n} = -\rho_{\rm s0}\,\delta\mathbf{v}_{\rm s}. \tag{14.23}$$

This corresponds to an almost pure "temperature wave", with $\delta P = 0$.

The successful detection in 1946 of a second sound mode was of tremendous significance in low-temperature physics (see Noziéres and Pines, 1990; Wilks, 1967). The good agreement of the measured second sound velocity u_2 with the calculated values based on the Landau expression (14.22), obtained using Landau's postulated phonon–roton quasiparticle spectrum, vindicated *both* aspects of the Landau theory of superfluid ^{4}He. This agreement between theory and experiment was evidence for both the correctness of the two-fluid hydrodynamic equations and also the assumed dispersion relation of the phonon–roton thermal excitations.

In contrast, in a gas the pressure and temperature fluctuations are strongly coupled. Hence $(\partial P/\partial T)_\rho$ in (14.19) plays a significant role and cannot be neglected as in the case of superfluid ^{4}He. This means that the ratio of specific heats $\bar{c}_p/\bar{c}_v$ is quite different from unity; in a classical monatomic gas, one has $\bar{c}_p/\bar{c}_v = 5/3$.

For a dilute weakly interacting Bose gas, one can calculate all the thermodynamic derivatives in (14.18) using the ZNG model based on the single-particle HF spectrum given in (3.16). In this case, one finds that the *normal* fluid density is given by $\rho_{\rm n0} = m\tilde{n}_0$, where $\tilde{n}_0$ is the density of the thermally excited atoms. The *superfluid* density is given by $\rho_{\rm s0} = mn_{\rm c0}$, where $n_{\rm c0}$ is the density of atoms in the Bose condensate. In contrast, in superfluid ^{4}He the magnitude of the superfluid density is very different from the underlying condensate density. In liquid ^{4}He, the condensate fraction is only about 0.1 at $T = 0$ (Sokol, 1995).

To first order in the interaction strength g, one finds that in a dilute

Bose-condensed gas the first sound and second sound velocities are

$$u_1^2 = \frac{5}{3}\frac{k_{\rm B}T}{m}\frac{g_{5/2}(1)}{g_{3/2}(1)} + \frac{2g\tilde{n}_0}{m} \tag{14.24}$$

$$u_2^2 = \frac{gn_{\rm c0}}{m}, \tag{14.25}$$

where the $g_n(z)$ are the usual Bose–Einstein functions, which depend on the fugacity z of the gas. We will derive these results (and higher-order corrections) at the end of Section 15.3. It can be shown using the explicit solutions to be discussed in Chapter 15 that the first sound mode (14.24) in a Bose gas is largely an oscillation of the thermal cloud (the normal fluid), while the second sound mode (14.25) is largely an oscillation of the condensate (the superfluid). In addition, both involve density fluctuations and thus will have significant weight in the dynamic structure factor related to the density response function, as we discuss in Section 14.3 below. The fact that first and second sound in Bose gases are quite different from the analogous modes in superfluid ^{4}He should be remembered when reading standard texts about superfluid ^{4}He. Results equivalent to (14.24) and (14.25) were given by Lee and Yang (1959); Gay and Griffin (1985); Griffin and Zaremba (1997); Shenoy and Ho (1998).

Comparing (14.25) with (2.30), we see that in a uniform Bose-condensed gas the *hydrodynamic* second sound mode at finite T smoothly extrapolates to the Bogoliubov phonon mode in the *collisionless* region. The fact that the first sound velocity u_1 given by (14.24) does not depend explicitly on the interaction strength g, to lowest order, is typical of ordinary sound waves in a classical gas. However, one must remember that strong interactions (collisions) play a crucial indirect role in enforcing dynamic local equilibrium, which leads to collisional hydrodynamics.

The non-dissipative Landau two-fluid equations given in (14.1)–(14.6) play a key role in Chapters 15–19. Their structure is quite subtle, as was realized in the 1950s when these equations were written down in terms of a Lagrangian formalism. This approach is developed in Chapter 16 as a useful way of finding variational solutions of the two-fluid equations in trapped gases.

14.3 Dynamic structure factor in the two-fluid region

The non-dissipative two-fluid equations for a Bose superfluid discussed in Section 14.2 can be used to calculate the density response function in the two-fluid hydrodynamic region. In this section, we will calculate the den-

sity response function by directly solving these two-fluid equations in the presence of a small *time-dependent* external potential $\delta U(\mathbf{r},t)$ for a uniform superfluid ($V_{\rm trap}=0$). Equations (14.2) and (14.5) then become

$$m\frac{\partial \mathbf{j}}{\partial t} = -\nabla P - n\nabla \delta U, \tag{14.26}$$

$$m\frac{\partial \mathbf{v}_{\rm s}}{\partial t} = -\nabla(\mu + \delta U). \tag{14.27}$$

To solve the two-fluid equations, we repeat the calculations given in (14.9)–(14.15). In place of (14.12), we find (n_0 is the total equilibrium density)

$$\frac{\partial^2 \rho}{\partial t^2} = \nabla^2 P + n_0 \nabla^2 \delta U, \tag{14.28}$$

while the equation of motion for $\bar{s}$ is unchanged from (14.15). Using (14.16), we then obtain coupled equations for the linearized fluctuations $\delta\rho$ and $\delta\bar{s}$:

$$\begin{aligned} \frac{\partial^2 \delta\rho}{\partial t^2} &= \left(\frac{\partial P}{\partial \rho}\right)_{\bar{s}} \nabla^2\delta\rho + \left(\frac{\partial P}{\partial \bar{s}}\right)_{\rho} \nabla^2 \delta\bar{s} + n_0\nabla^2\delta U, \\ \frac{\partial^2 \delta\bar{s}}{\partial t^2} &= \bar{s}_0^2 \frac{\rho_{\rm s0}}{\rho_{\rm n0}} \left[\left(\frac{\partial T}{\partial \rho}\right)_{\bar{s}} \nabla^2\delta\rho + \left(\frac{\partial T}{\partial \bar{s}}\right)_{\rho} \nabla^2 \delta\bar{s} \right]. \end{aligned} \tag{14.29}$$

We consider an external potential which excites modes of frequency ω and wavevector $\mathbf{q}$, namely

$$\delta U(\mathbf{r},t) = \delta U_{\mathbf{q},\omega} e^{i(\mathbf{q}\cdot\mathbf{r}-\omega t)}. \tag{14.30}$$

The plane-wave solutions of (14.29),

$$\delta\rho(\mathbf{r},t) = \delta\rho_{\mathbf{q},\omega} e^{i(\mathbf{q}\cdot\mathbf{r}-\omega t)}, \quad \delta\bar{s}(\mathbf{r},t) = \delta\bar{s}_{\mathbf{q},\omega} e^{i(\mathbf{q}\cdot\mathbf{r}-\omega t)}, \tag{14.31}$$

give two coupled algebraic equations,

$$\omega^2 \delta\rho_{\mathbf{q},\omega} = q^2 \left(\frac{\partial P}{\partial \rho}\right)_{\bar{s}} \delta\rho_{\mathbf{q},\omega} + q^2 \left(\frac{\partial P}{\partial \bar{s}}\right)_{\rho} \delta\bar{s}_{\mathbf{q},\omega} + q^2 n_0 \delta U_{\mathbf{q},\omega}, \tag{14.32}$$

$$\omega^2 \delta\bar{s}_{\mathbf{q},\omega} = \frac{\rho_{\rm s0}}{\rho_{\rm n0}} \bar{s}_0^2 \left[q^2 \left(\frac{\partial T}{\partial \rho}\right)_{\bar{s}} \delta\rho_{\mathbf{q},\omega} + q^2 \left(\frac{\partial T}{\partial \bar{s}}\right)_{\rho} \delta\bar{s}_{\mathbf{q},\omega} \right]. \tag{14.33}$$

The solution of these equations is

$$\delta\rho_{\mathbf{q},\omega} = n_0 q^2 \frac{\omega^2 - \bar{s}_0^2 \dfrac{\rho_{\rm s0}}{\rho_{\rm n0}} \left(\dfrac{\partial T}{\partial \bar{s}}\right)_{\rho} q^2}{(\omega^2 - u_1^2 q^2)(\omega^2 - u_2^2 q^2)} \delta U_{\mathbf{q},\omega}\,, \tag{14.34}$$

$$\delta\bar{s}_{\mathbf{q},\omega} = n_0 q^4 \frac{\bar{s}_0^2 \dfrac{\rho_{\rm s0}}{\rho_{\rm n0}} \left(\dfrac{\partial T}{\partial \bar{s}}\right)_{\bar{s}}}{(\omega^2 - u_1^2 q^2)(\omega^2 - u_2^2 q^2)} \delta U_{\mathbf{q},\omega}\,, \tag{14.35}$$

where u_1 and u_2 are the first and second sound velocities given by the solutions of the quadratic equation (14.18).

The solution in (14.34) can always be written in terms of the density response function, defined as

$$\delta n_{\mathbf{q},\omega} = \chi_{nn}(\mathbf{q},\omega)\delta U_{\mathbf{q},\omega}. \tag{14.36}$$

We conclude that the density response function for a uniform Bose superfluid described by the non-dissipative Landau two-fluid equations is

$$\chi_{nn}(\mathbf{q},\omega) = \frac{n_0\, q^2}{m}\frac{\omega^2 - v^2q^2}{(\omega^2 - u_1^2q^2)(\omega^2 - u_2^2q^2)}, \tag{14.37}$$

where we have introduced a new velocity v defined as

$$v^2 \equiv \bar{s}_0^2\frac{\rho_{\rm s0}}{\rho_{\rm n0}}\left(\frac{\partial T}{\partial \bar{s}}\right)_\rho = T\frac{s_0^2}{\bar{c}_v}\frac{\rho_{\rm s0}}{\rho_{\rm n0}}. \tag{14.38}$$

We note that the first and second sound velocities given by the solutions of (14.18) satisfy the relations

$$\begin{aligned} u_1^2 + u_2^2 &= v^2 + \left(\frac{\partial P}{\partial \rho}\right)_{\bar{s}}, \\ u_1^2u_2^2 &= v^2\left(\frac{\partial P}{\partial \rho}\right)_T. \end{aligned} \tag{14.39}$$

The two-fluid density response function (14.37) was first derived for superfluid ^{4}He by Ginzburg (1943) and by Hohenberg and Martin (1964). It was first used in weakly interacting superfluid Bose gases by Gay and Griffin (1985). The generalization of (14.37) to include hydrodynamic damping is given in Hohenberg and Martin (1965). An expression analogous to (14.37) for $\chi_{nn}(\mathbf{q},\omega)$ in the two-fluid region for *trapped* Bose gases was derived by Taylor *et al.* (2007) . This extension is briefly discussed in Section 16.4.

We note that if $(\partial P/\partial\rho)_{\bar{s}} \simeq (\partial P/\partial\rho)_T$, as in superfluid ^{4}He, then the second sound velocity $u_2 \simeq v$, where v is defined in (14.38). In this case, the density response function in (14.37) is completely dominated by the first sound pole,

$$\chi_{nn}(\mathbf{q},\omega) \simeq \frac{n_0q^2}{m}\frac{1}{\omega^2 - u_1^2q^2}. \tag{14.40}$$

The situation is quite different in superfluid Bose-condensed gases at finite temperatures, where the thermal cloud excitations are particle-like excitations (i.e. the atoms move in a self-consistent HF field). In this case, one finds that the second sound velocity u_2 is quite different from the velocity

v. This means that the second sound pole makes a significant contribution to the density response function in (14.37). Explicit calculations of the dynamic structure factor $S(\mathbf{q},\omega)$ showing both the first and second sound resonances are given in Gay and Griffin (1985). In Bose gases, second sound should not be viewed as a pure temperature wave, as it is in superfluid ^{4}He.

The dynamic structure factor defined in (4.85) is directly related to the imaginary part of the density response function (see for example Chapter 2 of Griffin, 1993):

$$S(\mathbf{q},\omega) = -\frac{1}{\pi n_0}(f^0(\omega)+1)\,\mathrm{Im}\,\chi_{nn}(\mathbf{q},\omega+i0^+). \tag{14.41}$$

Using the two-fluid hydrodynamic response function (14.37), we find

$$S(\mathbf{q},\omega) = (f^0(\omega)+1)\frac{q^2}{m}\left[Z_1\delta(\omega^2-u_1^2q^2) + (1-Z_1)\delta(\omega^2-u_2^2q^2)\right], \tag{14.42}$$

where the relative weight of the first sound mode is given by

$$Z_1 = \frac{u_1^2-v^2}{u_1^2-u_2^2}. \tag{14.43}$$

One may verify that the hydrodynamic expression (14.42) exactly satisfies the f-sum rule

$$\int_{-\infty}^{\infty} d\omega\,\omega\,S(\mathbf{q},\omega) = \frac{q^2}{2m}, \tag{14.44}$$

and the compressibility sum rule

$$\begin{aligned}\int_{-\infty}^{\infty} d\omega\frac{S(\mathbf{q},\omega)}{\omega} &= \frac{1}{2m}\left(\frac{Z_1}{u_1^2}+\frac{1-Z_1}{u_2^2}\right)\\ &= \frac{1}{2m}\frac{v^2}{u_1^2u_2^2}\\ &= \frac{1}{2mv_{\rm I}^2}\ .\end{aligned} \tag{14.45}$$

In the last line, we have used the second relation in (14.39), which brings in the *isothermal* sound velocity $v_{\rm I}^2 \equiv (\partial P/\partial\rho)_T$. In addition, in the derivation of (14.44) and (14.45) we have used the identity

$$\delta(\omega^2-\omega_1^2) = \frac{1}{2\omega_1}\left[\delta(\omega-\omega_1)-\delta(\omega+\omega_1)\right], \tag{14.46}$$

and the relation satisfied by the Bose distribution,

$$-f^0(-\omega) = f^0(\omega)+1. \tag{14.47}$$

For the derivation and further discussion of the exact frequency sum rules

in (14.44) and (14.45), see for example Chapter 2 of Griffin (1993) and Chapter 7 of Pitaevskii and Stringari (2003).

We note that if we assume a two-pole expression such as (14.42) for $S(\mathbf{q},\omega)$, then the two sum rules (14.44) and (14.45) by themselves can be used to verify that the second sound mode has weight $Z_2 = 1 - Z_1$, with Z_1 given by (14.43). This use of frequency sum rules to determine the relative weights of first and second sound was pioneered by Nozières and Pines (1990). As discussed in standard texts (see for example Griffin, 1993, p. 31), the compressibility sum rule (14.45) is equivalent to the statement that

$$\lim_{\omega\to 0} \chi_{nn}(\mathbf{q},\omega) = -\frac{n_0}{m v_1^2} \ . \tag{14.48}$$

Clearly the two-fluid result for the density response in (14.37) satisfies this isothermal compressibility sum rule. We have derived the hydrodynamic density response function in (14.37) starting from the Landau two-fluid equations. However, it can also be viewed as the direct consequence of assuming that the first and second sound modes "exhaust" the spectral weight of the two exact frequency sum rules given above.

The density response function can be directly measured in superfluid Bose gases using two-photon Bragg scattering (for a review see Ozeri, 2005). This experimental probe involves the coherent absorption and stimulated emission of photons from two different lasers, leading to an energy ($\hbar\omega$) and momentum ($\hbar q$) transfer. This coherent process involving stimulated emission leads to a scattering cross section proportional to

$$S(\mathbf{q},\omega) + S(\mathbf{q},-\omega) = -\frac{1}{\pi n_0}\mathrm{Im}\chi_{nn}(\mathbf{q},\omega), \tag{14.49}$$

where $S(\mathbf{q},\omega)$ was defined in (14.41). Thus, in contrast with inelastic Brillouin light scattering, which is proportional to $S(\mathbf{q},\omega)$, the "detailed balance factor" $f^0(\omega)+1$ in (14.41) is absent in the two-photon Bragg (Raman) scattering cross section. For further discussion, see Zambelli *et al.* (2000) and Pitaevskii and Stringari (2003, p. 197).

15

Two-fluid hydrodynamics in a dilute Bose gas

In Chapters 11–13, we gave a detailed discussion of the dynamics of a trapped Bose gas at finite temperatures in a region where the collisions described by the C_{12} and C_{22} terms in the kinetic equation (3.42) do not play the central role. In this "collisionless" region, the dominant interaction effects are associated with the self-consistent fields which both the condensate and noncondensate atoms feel. Thus the dynamics can be understood to a first approximation by neglecting the C_{12} and C_{22} collision integrals in the kinetic equation and, at the next stage, treating them as a weak perturbation on the collisionless dynamics.

In the rest of this book (Chapters 15–19), we turn to the study of the coupled ZNG equations in the opposite limit, where the C_{12} and C_{22} collision integrals completely determine the dynamics of the thermal cloud. Specifically, the collisions lead to the thermal cloud being in local hydrodynamic equilibrium, and hence this regime is described by the equations of collisional hydrodynamics. Its characteristic feature is that the nonequilibrium behaviour of the thermal cloud atoms can be completely described in terms of a few differential equations involving coarse-grained variables that are dependent on position and time, analogous to the condensate variables $n_c(\mathbf{r}, t)$ and $\mathbf{v}_c(\mathbf{r}, t)$. In the present chapter, assuming that the thermal cloud distribution function $f(\mathbf{p}, \mathbf{r}, t)$ is given by the Bose distribution describing partial local equilibrium, (15.16), we show how the ZNG coupled equations lead precisely to Landau's two-fluid equations, reviewed in Chapter 14. This equivalence is not obvious, mainly because Landau's equations are expressed in terms of thermodynamic variables, which are not used in a more microscopic analysis such as that used in the ZNG approach.

The two-fluid regime was difficult to reach in the first wave of experiments on the properties of trapped Bose-condensed gases after the discovery of BEC in such gases in 1995. The reason was that the typical collisional

cross section and the achievable atomic densities were not sufficient to reach the region described by collisional two-fluid hydrodynamics. An exception was the pioneering experiment by Stamper-Kurn *et al.* (1998), which as later theoretical work showed, was well within the two-fluid region (Nikuni and Griffin, 2001b). New experimental developments seem promising for the study of the two-fluid collisional dynamics in trapped gases. A basic difficulty in trapped atomic Bose gases has been that if one works with high enough densities to obtain short collision times, the increase in three-body collisions leads to the destruction of the condensate. One recent suggestion has been to produce a condensate first, by using a very shallow trap (with low trapping frequencies), and then rapidly to compress the gas in order to enter the collisional hydrodynamic regime more deeply (van der Stam *et al.*, 2007) before significant three-body losses occur.

Possibly the most promising way of producing local equilibrium conditions in the future is to work with a strongly interacting gas of bosonic molecules formed in Fermi gases (for a review, see Giorgini *et al.*, 2007). We recall that in a trapped two-component Fermi gas, one can use a Feshbach resonance to adjust the magnitude and sign of the s-wave scattering length a between Fermi atoms in two different atomic hyperfine states. This allows one to study the collective modes across the BCS–BEC crossover region in great detail, with extremely large values of $|a|$. One also expects that close to the unitarity limit ($|a| \to \infty$) of trapped two-component Fermi superfluids, one is dealing with a region where the Landau two-fluid equations are the appropriate description of the dynamics at finite temperatures. The BEC side of the BCS–BEC crossover is well described as a strongly interacting Bose gas of very stable molecules (Petrov *et al.*, 2004, 2005). This molecular Bose condensate may prove to be the perfect example of a strongly interacting Bose gas needed to achieve local hydrodynamic equilibrium, and hence allow us to study the analogue of second sound in superfluid ^{4}He.

Our derivation in this chapter of the hydrodynamic two-fluid equations is built on the assumption of a partial local hydrodynamic equilibrium form for the thermal cloud distribution function $f(\mathbf{p}, \mathbf{r}, t)$. A more systematic derivation based on the Chapman–Enskog method of solving kinetic equations is developed in Chapters 17 and 18. This extends the two-fluid equations derived in this chapter by the inclusion of all transport coefficients associated with hydrodynamic damping.

The derivation of the Landau two-fluid hydrodynamics in this chapter (and its extension to include hydrodynamic damping in Chapter 17) is quite subtle and technically quite involved. This complexity arises even though our discussion is based on the coupled equations of the ZNG theory, in

which the thermal excitations are described in terms of a simple Hartree–Fock semiclassical spectrum. Of course, the final two-fluid equations are much more general than the simple microscopic model on which ZNG is based. However, considerable insight is gained from the explicit derivation of the Landau equations, as we hope this chapter will show.

15.1 Equations of motion for local equilibrium

As discussed in Chapter 3, the equation of motion for the condensate variables within the ZNG approximation is given by the generalized GP equation (3.21). In terms of the condensate density $n_c(\mathbf{r},t)$ and the velocity $\mathbf{v}_c(\mathbf{r},t)$ (the gradient of the phase), this GGP equation for the condensate is equivalent to the equations

$$\frac{\partial n_c}{\partial t} + \nabla\cdot(n_c\mathbf{v}_c) = -\Gamma_{12}[f,\Phi], \tag{15.1}$$

$$m\left(\frac{\partial}{\partial t} + \mathbf{v}_c\cdot\nabla\right)\mathbf{v}_c(\mathbf{r},t) = -\nabla\mu_c. \tag{15.2}$$

Here the nonequilibrium chemical potential is

$$\mu_c(\mathbf{r},t) \equiv -\frac{\hbar^2\nabla^2\sqrt{n_c(\mathbf{r},t)}}{2m\sqrt{n_c(\mathbf{r},t)}} + V_{\rm trap}(\mathbf{r}) + gn_c(\mathbf{r},t) + 2g\tilde{n}(\mathbf{r},t) \tag{15.3}$$

and the source term Γ_{12} is the result of C_{12} collisions between atoms in the condensate and noncondensate.

These equations describing the condensate are always hydrodynamic in form, in that the condensate is completely described by the two local variables n_c and $\mathbf{v}_c$. However, they do not form a *closed* set of equations since they depend on the nonequilibrium thermal cloud distribution function $f(\mathbf{p},\mathbf{r},t)$, through the source function Γ_{12} and the thermal cloud density $\tilde{n}$. Nevertheless, by taking moments $\int d\mathbf{p}\,\mathbf{p}^n (n=0,1,2)$ of the kinetic equation one can derive a set of equations for local variables that describe a general nonequilibrium state of the thermal cloud. In the limit of strong collisions, these equations can be reduced to a closed set of hydrodynamic equations for the thermal cloud coupled to the two condensate degrees of freedom, n_c and $\mathbf{v}_c$. This procedure is a natural generalization of how one derives the hydrodynamic equations in a classical gas, which is discussed in standard texts (see for example Ferziger and Kaper, 1972; Huang, 1987).

Taking moments (with respect to the momentum $\mathbf{p}$) of the kinetic equation (3.42) involves moments of the collision integrals C_{22} and C_{12} in (3.40) and (3.41). One can verify the following exact results (see Chapter 3 for the

definitions of the various quantities):

$$\begin{aligned} \int d\mathbf{p}\ \mathbf{p}C_{22} = 0, \qquad & \int d\mathbf{p}\ (\mathbf{p} - m\mathbf{v}_{\rm c})C_{12} = 0, \\ \int d\mathbf{p}\ \tilde{\varepsilon}_p C_{22} = 0, \qquad & \int d\mathbf{p}\ (\tilde{\varepsilon}_p - \varepsilon_{\rm c})C_{12} = 0. \end{aligned} \tag{15.4}$$

These collisional invariants show that both the C_{12} and C_{22} collision processes conserve energy and momentum. In the case of $1 \rightleftharpoons 2$ collisions, the momentum of the noncondensate atoms is conserved in the local rest frame of the condensate. In addition, the energy conservation condition takes into account explicitly the different mean-field energies of the condensate and noncondensate atoms. In addition to the exact relations (15.4), we have

$$\int d\mathbf{p}\ C_{22} = 0\,, \tag{15.5}$$

which is simply a statement that the number of noncondensate atoms is conserved in C_{22} collisions. In contrast, as already noted many times in earlier chapters, C_{12} does *not* conserve the number of noncondensate atoms and hence

$$\Gamma_{12}[f,\Phi] \equiv \int \frac{d\mathbf{p}}{(2\pi\hbar)^3}\ C_{12} \neq 0. \tag{15.6}$$

Using this in (15.1), one sees that C_{12} collisions and hence Γ_{12} will play a central role in all problems that involve the relaxation (or growth) of the condensate component.

Taking (15.4)–(15.6) into account, the exact moment equations derived from the kinetic equation (3.42) can be written (after some algebra) in the form

$$\frac{\partial \tilde{n}}{\partial t} + \nabla\cdot(\tilde{n}\mathbf{v}_{\rm n}) = \Gamma_{12}[f,\Phi], \tag{15.7}$$

$$m\tilde{n}\left(\frac{\partial}{\partial t} + \mathbf{v}_{\rm n}\cdot\nabla\right) v_{{\rm n}\mu} = -\frac{\partial P_{\mu\nu}}{\partial x_\nu} - \tilde{n}\frac{\partial U}{\partial x_\mu} - m(v_{{\rm n}\mu} - v_{c\mu})\Gamma_{12}[f,\Phi], \tag{15.8}$$

$$\begin{aligned} \frac{\partial \tilde{\epsilon}}{\partial t} + \nabla\cdot(\tilde{\epsilon}\mathbf{v}_{\rm n}) = & -\nabla\cdot\mathbf{Q} - D_{\mu\nu}P_{\mu\nu} \\ & + \left[\tfrac{1}{2}m(\mathbf{v}_{\rm n} - \mathbf{v}_{\rm c})^2 + \mu_{\rm c} - U\right]\Gamma_{12}[f,\Phi], \end{aligned} \tag{15.9}$$

Here the Greek subscripts μ and ν denote Cartesian components. The noncondensate local density and velocity are defined by

$$\tilde{n}(\mathbf{r},t) \equiv \int \frac{d\mathbf{p}}{(2\pi\hbar)^3} f(\mathbf{p},\mathbf{r},t) \tag{15.10}$$

$$\mathbf{v}_{\rm n}(\mathbf{r},t) \equiv \int \frac{d\mathbf{p}}{(2\pi\hbar)^3}\frac{\mathbf{p}}{m}\frac{f(\mathbf{p},\mathbf{r},t)}{\tilde{n}(\mathbf{r},t)}. \tag{15.11}$$

In addition, in the moment equations we have introduced the following new local hydrodynamic variables,

$$P_{\mu\nu}(\mathbf{r},t) \equiv m\int \frac{d\mathbf{p}}{(2\pi\hbar)^3}\left(\frac{p_\mu}{m}-v_{\mathrm{n}\mu}\right)\left(\frac{p_\nu}{m}-v_{\mathrm{n}\nu}\right) f(\mathbf{p},\mathbf{r},t), \quad (15.12)$$

$$\mathbf{Q}(\mathbf{r},t) \equiv \int \frac{d\mathbf{p}}{(2\pi\hbar)^3}\frac{1}{2m}(\mathbf{p}-m\mathbf{v}_{\mathrm{n}})^2\left(\frac{\mathbf{p}}{m}-\mathbf{v}_{\mathrm{n}}\right) f(\mathbf{p},\mathbf{r},t), \quad (15.13)$$

$$\tilde{\epsilon}(\mathbf{r},t) \equiv \int \frac{d\mathbf{p}}{(2\pi\hbar)^3}\frac{1}{2m}(\mathbf{p}-m\mathbf{v}_{\mathrm{n}})^2 f(\mathbf{p},\mathbf{r},t). \quad (15.14)$$

The symmetric rate-of-strain tensor appearing in (15.9) is defined by

$$D_{\mu\nu}(\mathbf{r},t) \equiv \frac{1}{2}\left(\frac{\partial v_{\mathrm{n}\mu}}{\partial x_\nu}+\frac{\partial v_{\mathrm{n}\nu}}{\partial x_\mu}\right). \quad (15.15)$$

We note that $\sum_\nu D_{\nu\nu} = \nabla\cdot\mathbf{v}_{\mathrm{n}}$. It is clear from the addition of (15.1) and (15.7) that the Γ_{12} terms cancel out, ensuring that the total number of atoms is conserved. The derivation of the results (15.7)–(15.9) involves manipulations identical to those used for classical gases. In the case of a normal Bose gas (where $C_{12} = 0$), a nice derivation of these equations is given on pp. 52–9 of Kadanoff and Baym (1962). We call attention to the fact that only C_{12} appears in these moment equations. This follows from the conservation of the number, momentum and energy of the noncondensate atoms in C_{22} collisions when one uses an HF spectrum.

The set of equations (15.7)–(15.9) is formally exact, but they are obviously not closed since the number of local variables ($\tilde{n}, n_{\mathrm{c}}, \mathbf{v}_{\mathrm{n}}, \mathbf{v}_{\mathrm{c}}$ as well as the stress tensor $P_{\mu\nu}$, the kinetic energy density $\tilde{\epsilon}$, the heat current $\mathbf{Q}$ and Γ_{12}) exceeds the total number (nine) of coupled scalar equations. To proceed we must specify the conditions under which the dynamics of the trapped gas is to be determined. The range of possibilities includes two extreme cases: (i) the collisionless regime in which we can, as a first approximation set $C_{12} = C_{22} = 0$ (in this regime the collisionless kinetic equation for the distribution function for noncondensate atoms must be solved explicitly to determine the various physical quantities of interest); (ii) a regime in which the collision rates are so high that a condition of *complete local equilibrium* is established. The most general situation will not fall neatly into either of these two extreme limits and will require a detailed solution of the kinetic equation (3.42) together with the quantum hydrodynamic equations for the condensate given by (15.1)–(15.3). Chapters 11 and 12 give a detailed discussion of this kind, for the case in which the collision rates are not too large but still have a significant effect on the dynamics.

Of particular interest is the transition from the partial local equilibrium

to complete local equilibrium conditions, in which the collisions C_{12} and C_{22} play a crucial role. An analysis of this transition will clarify the way in which the Landau two-fluid equations emerge within the context of our microscopic model. This turns out to be a rather subtle problem, and was first clarified in Zaremba *et al.* (1999).

We now consider the local hydrodynamic regime, where (15.7)–(15.9) for the noncondensate and (15.1)–(15.3) for the condensate can be shown to form a closed set of equations. When the collision rate among the thermal cloud atoms is high, the collision integral C_{22} drives the distribution function $f(\mathbf{p}, \mathbf{r}, t)$ towards the *partial local equilibrium* Bose distribution defined by

$$\tilde{f}(\mathbf{p}, \mathbf{r}, t) = \frac{1}{e^{\beta[(\mathbf{p}-m\mathbf{v}_{\rm n})^2/2m+U-\tilde{\mu}]} - 1}. \tag{15.16}$$

Here the temperature parameter β, the local fluid velocity $\mathbf{v}_{\rm n}$, the chemical potential $\tilde{\mu}$ and the mean field U are all functions of $\mathbf{r}$ and t. This partial local equilibrium expression for $\tilde{f}$ is determined by the requirement that it must satisfy

$$C_{22}[\tilde{f}, \Phi] = 0\,. \tag{15.17}$$

The expression (15.16) satisfies (15.17) for any value of $\tilde{\mu}(\mathbf{r}, t)$. It makes use only of the energy and momentum conservation factors in the definition of C_{22} in (3.40).

It is important to appreciate that the local chemical potential $\tilde{\mu}$ for the thermal cloud that appears in (15.16) is distinct from the local condensate chemical potential $\mu_{\rm c}$ defined in (15.3). How these two chemical potentials acquire a common value in the limit of *complete* or *diffusive* local equilibrium requires a careful analysis of the effect of the C_{12} collisions. Using the same kind of calculation as that in Section 3.3 for the thermal equilibrium Bose distribution f^0, one finds that $\tilde{f}$ in (15.16) leads to the following expression for $C_{12}[\tilde{f}, \Phi]$ in (3.41):

$$\begin{aligned} C_{12}[\tilde{f}, \Phi] = {} & \frac{2g^2 n_{\rm c}}{(2\pi)^2\hbar^4}\left[1 - e^{-\beta(\tilde{\mu}-\frac{1}{2}m(\mathbf{v}_{\rm n}-\mathbf{v}_{\rm c})^2-\mu_{\rm c})}\right] \\ & \times \int d\mathbf{p}_1 \int d\mathbf{p}_2 \int d\mathbf{p}_3\, \delta(m\mathbf{v}_{\rm c} + \mathbf{p}_1 - \mathbf{p}_2 - \mathbf{p}_3)\delta(\varepsilon_{\rm c} + \tilde{\varepsilon}_1 - \tilde{\varepsilon}_2 - \tilde{\varepsilon}_3) \\ & \times [\delta(\mathbf{p} - \mathbf{p}_1) - \delta(\mathbf{p} - \mathbf{p}_2) - \delta(\mathbf{p} - \mathbf{p}_3)](1 + \tilde{f}_1)\tilde{f}_2\tilde{f}_3. \end{aligned} \tag{15.18}$$

The factor in square brackets in (15.18) was discussed after (3.53). The important point is that $C_{12}[\tilde{f}, \Phi]$ is, in general, nonvanishing. This simply reflects the fact that atoms in the noncondensate need not be in *diffusive* local equilibrium with the condensate atoms ($\tilde{\mu} \neq \mu_{\rm c}$). The square bracket in

(15.18) will vanish (i.e. $C_{12}[\tilde{f}, \Phi] = 0$) only if the two chemical potentials μ_c and $\tilde{\mu}$ satisfy the condition (3.56), i.e. if there is *complete* local equilibrium between the two components.

Assuming that the collision rate among noncondensate atoms is sufficiently high, the partial local equilibrium distribution in (15.16) provides the appropriate zeroth-order solution to the kinetic equation (3.42). Of course, even in the absence of the C_{12} collision term, $\tilde{f}$ is not an exact solution of the kinetic equation. A systematic procedure based on the Chapman–Enskog method (see Chapters 17 and 18) starts from an expansion around the distribution function for *complete* local equilibrium. Such calculations were first carried out by Kirkpatrick and Dorfman (1983, 1985a) for a uniform Bose-condensed gas. In the present chapter we will use the expression $\tilde{f}$ in (15.16). This makes the physics transparent and will show in a very clear fashion how the Landau two-fluid equations emerge. The analysis in this chapter will set the stage for the more systematic approach given in Chapter 17.

Using (15.16) to evaluate the quantities in (15.12)–(15.14), it is convenient to integrate over $\mathbf{p}' \equiv \mathbf{p} - m\mathbf{v}_n$ in place of $\mathbf{p}$, since $\tilde{f}(\mathbf{p}, \mathbf{r}, t)$ is an even function of $\mathbf{p}'$. Since the integrand in (15.13) is odd in $\mathbf{p}'$, the heat current $\mathbf{Q}(\mathbf{r}, t)$ vanishes. For the same reason, the off-diagonal terms of $P_{\mu\nu}(\mathbf{r}, t)$ vanish and the diagonal components reduce to

$$P_{\mu\mu}(\mathbf{r}, t) \equiv \tilde{P}(\mathbf{r}, t) = \int \frac{d\mathbf{p}}{(2\pi\hbar)^3} \frac{p^2}{3m} \, \tilde{f}(\mathbf{p}, \mathbf{r}, t)\Big|_{\mathbf{v}_n = 0} . \tag{15.19}$$

The quantity $\tilde{P}(\mathbf{r}, t)$ clearly corresponds to the local kinetic pressure for a HF thermal gas of atoms. As noted in (15.19), $\tilde{P}(\mathbf{r}, t)$ has the same value as it does if $\mathbf{v}_n = 0$ in $\tilde{f}$. In addition, we note that, using $\tilde{f}$, the kinetic energy density defined in (15.14) is related to $\tilde{P}(\mathbf{r}, t)$ by

$$\tilde{\epsilon}(\mathbf{r}, t) = \tfrac{3}{2}\tilde{P}(\mathbf{r}, t) , \tag{15.20}$$

the same relation as far a uniform noninteracting Bose gas.

Since the momentum dependence of the integrand in (15.19) is the same as for a static equilibrium Bose distribution, the integrations can be carried out explicitly to give the expression

$$\tilde{P}(\mathbf{r}, t) = \frac{1}{\beta\Lambda^3} g_{5/2}(z) , \tag{15.21}$$

where the local thermodynamic variables β and z are again functions of $\mathbf{r}$ and t. The Bose–Einstein functions are defined as $g_n(z) \equiv \sum_{l=1}^{\infty} z^l / l^n$, the

local equilibrium gas fugacity is

$$z(\mathbf{r},t) \equiv e^{\beta(\mathbf{r},t)[\tilde{\mu}(\mathbf{r},t)-U(\mathbf{r},t)]} \tag{15.22}$$

and the local thermal de Broglie wavelength is

$$\Lambda(\mathbf{r},t) \equiv \left(\frac{2\pi\hbar^2}{mk_{\rm B}T(\mathbf{r},t)}\right)^{1/2}. \tag{15.23}$$

Finally, we note that the noncondensate density $\tilde{n}$ associated with $\tilde{f}$ is given in terms of the usual Bose–Einstein function:

$$\tilde{n}(\mathbf{r},t) = \int \frac{d\mathbf{p}}{(2\pi\hbar)^3}\, \tilde{f}(\mathbf{p},\mathbf{r},t)\Big|_{\mathbf{v}_{\rm n}=0} = \frac{1}{\Lambda^3} g_{3/2}(z)\,. \tag{15.24}$$

As with $\tilde{P}(\mathbf{r},t)$, $\tilde{n}$ has the same value as it would if $\mathbf{v}_{\rm n} = 0$ in $\tilde{f}$. The similarity of these expressions for $\tilde{P}$ and $\tilde{n}$ to those of a uniform noninteracting Bose gas, discussed in all textbooks on statistical mechanics, is a direct consequence of our treating the thermal atoms within a semiclassical Hartree–Fock approximation.

To summarize, using the partial local equilibrium approximation $\tilde{f}$ for the thermal cloud distribution function f, the hydrodynamic equations (15.7)–(15.9) for the noncondensate atoms simplify to

$$\frac{\partial\tilde{n}}{\partial t} + \nabla\cdot(\tilde{n}\mathbf{v}_{\rm n}) = \Gamma_{12}[\tilde{f},\Phi], \tag{15.25}$$

$$m\tilde{n}\left(\frac{\partial}{\partial t} + \mathbf{v}_{\rm n}\cdot\nabla\right)\mathbf{v}_{\rm n} = -\nabla\tilde{P} - \tilde{n}\nabla U - m(\mathbf{v}_{\rm n}-\mathbf{v}_{\rm c})\Gamma_{12}[\tilde{f},\Phi], \tag{15.26}$$

$$\frac{\partial\tilde{P}}{\partial t} + \nabla\cdot(\tilde{P}\mathbf{v}_{\rm n}) = -\tfrac{2}{3}\tilde{P}\nabla\cdot\mathbf{v}_{\rm n} + \tfrac{2}{3}\left[\tfrac{1}{2}m(\mathbf{v}_{\rm n}-\mathbf{v}_{\rm c})^2 + \mu_{\rm c} - U\right]\Gamma_{12}[\tilde{f},\Phi]. \tag{15.27}$$

The source term is given by (see (3.24) and (15.18))

$$\begin{aligned}\Gamma_{12}[\tilde{f},\Phi] &\equiv \int \frac{d\mathbf{p}}{(2\pi\hbar)^3} C_{12}[\tilde{f},\Phi] \\ &= -\frac{2g^2 n_{\rm c}}{(2\pi)^5\hbar^7}\left(1 - e^{-\beta[\tilde{\mu}-\mu_{\rm c}-\frac{1}{2}m(\mathbf{v}_{\rm n}-\mathbf{v}_{\rm c})^2]}\right)\int d\mathbf{p}_1\int d\mathbf{p}_2\int d\mathbf{p}_3 \\ &\quad\times\delta(m\mathbf{v}_{\rm c}+\mathbf{p}_1-\mathbf{p}_2-\mathbf{p}_3)\delta(\varepsilon_{\rm c}+\tilde{\varepsilon}_1-\tilde{\varepsilon}_2-\tilde{\varepsilon}_3) \\ &\quad\times(1+\tilde{f}_1)\tilde{f}_2\tilde{f}_3\,.\end{aligned} \tag{15.28}$$

It is clear that the $\Gamma_{12}[\tilde{f},\Phi]$ terms will play a crucial role in the solution of these hydrodynamic equations for the thermal cloud. These terms, which describe the transfer of atoms between the condensate and noncondensate,

are responsible for bringing these two components into complete (or diffusive) local equilibrium.

We shall concentrate on solutions involving small-amplitude oscillations of the condensate and noncondensate around static (i.e. absolute) thermal equilibrium. For the noncondensate atoms, absolute equilibrium is described by a Bose distribution at a uniform temperature T_0 and chemical potential $\tilde{\mu}_0$ (all *static* equilibrium quantities are indicated by a subscript 0). The equilibrium density and pressure are then given by the usual expressions, $\tilde{n}_0(\mathbf{r}) = g_{3/2}(z_0)/\Lambda_0^3$ and $\tilde{P}_0(\mathbf{r}) = g_{5/2}(z_0)/\beta_0\Lambda_0^3$, respectively. These are of the same form as (15.24) and (15.21), but here all thermodynamic quantities take on their equilibrium values. The equilibrium fugacity is $z_0(\mathbf{r}) = e^{\beta_0[\tilde{\mu}_0 - U_0(\mathbf{r})]}$, with

$$U_0(\mathbf{r}) = V_{\rm trap}(\mathbf{r}) + 2g[\tilde{n}_0(\mathbf{r}) + n_{\rm c0}(\mathbf{r})]. \tag{15.29}$$

It is clear that the condensate equilibrium density $n_{\rm c0}(\mathbf{r})$ is involved in the determination of the noncondensate equilibrium variables $\tilde{n}_0(\mathbf{r})$, $\tilde{P}_0(\mathbf{r})$ and $\tilde{\mu}_0$. We refer to Section 3.3 for a further discussion of complete thermal equilibrium, where $\mu_{\rm c0} = \tilde{\mu}_0$ (assuming that $\mathbf{v}_{\rm c0} = 0$ and $\mathbf{v}_{\rm n0} = 0$).

In this section we have derived a set of equations for the local density $\tilde{n}(\mathbf{r}, t)$, velocity $\mathbf{v}_{\rm n}(\mathbf{r}, t)$ and kinetic pressure $\tilde{P}(\mathbf{r}, t)$ of the noncondensate thermal atoms. These local variables are coupled to the condensate local variables, $n_{\rm c}(\mathbf{r}, t)$ and $\mathbf{v}_{\rm c}(\mathbf{r}, t)$, which are the solutions of (15.1) – (15.3). To conclude the section, we write down *linearized* versions of the equations derived above, which are valid close to static equilibrium. The linearized versions of the noncondensate equations (15.25)–(15.27) are

$$\frac{\partial \delta\tilde{n}}{\partial t} = -\nabla\cdot(\tilde{n}_0\delta\mathbf{v}_{\rm n}) + \delta\Gamma_{12}, \tag{15.30}$$

$$m\tilde{n}_0\frac{\partial \delta\mathbf{v}_{\rm n}}{\partial t} = -\nabla\delta\tilde{P} - \delta\tilde{n}\nabla U_0 - 2g\tilde{n}_0\nabla(\delta\tilde{n} + \delta n_{\rm c}), \tag{15.31}$$

$$\frac{\partial \delta\tilde{P}}{\partial t} = -\tfrac{5}{3}\nabla\cdot(\tilde{P}_0\delta\mathbf{v}_{\rm n}) + \tfrac{2}{3}\delta\mathbf{v}_{\rm n}\cdot\nabla\tilde{P}_0 + \tfrac{2}{3}(\mu_{\rm c0} - U_0)\delta\Gamma_{12}\,, \tag{15.32}$$

where we restrict ourselves to the equilibrium case $\mathbf{v}_{\rm n0} = 0$ and $\mathbf{v}_{\rm c0} = 0$. We have used the fact that in thermal equilibrium (where $\mu_{\rm c0} = \tilde{\mu}_0$), $\Gamma_{12}[f^0, \Phi]$ vanishes. We observe that $\delta\Gamma_{12}$ does not appear explicitly in (15.31) as it contributes only a second-order correction in g. These noncondensate equations must be solved in conjunction with the linearized equations for

the condensate fluctuations, which are given by

$$\frac{\partial \delta n_{\rm c}}{\partial t} = -\nabla \cdot (n_{\rm c0} \delta \mathbf{v}_{\rm c}) - \delta \Gamma_{12}, \tag{15.33}$$

$$m \frac{\partial \delta \mathbf{v}_{\rm c}}{\partial t} = -\nabla \delta \mu_{\rm c} \,. \tag{15.34}$$

Within the dynamic Thomas–Fermi (TF) approximation, i.e. ignoring the nonlocal quantum pressure terms in $\mu_{\rm c}(\mathbf{r}, t)$, the fluctuations in the local chemical potential (15.3) are given by

$$\delta \mu_{\rm c}(\mathbf{r}, t) = g \delta n_{\rm c}(\mathbf{r}, t) + 2g \delta \tilde{n}(\mathbf{r}, t) \,. \tag{15.35}$$

In our microscopic derivation we make use of the Thomas–Fermi approximation, which effectively treats the system as locally homogeneous. If this approximation were not made, the Landau two-fluid equations would not be obtained.

We also need an expression for $\delta \Gamma_{12}$, which is the linearized form of the expression (15.28). A straightforward calculation gives

$$\delta \Gamma_{12}[\tilde{f}, \Phi] = -\frac{\beta_0 n_{\rm c0}}{\tau_{12}} \delta \mu_{\rm diff}. \tag{15.36}$$

The difference between the local chemical potentials of the condensate and noncondensate is

$$\mu_{\rm diff} \equiv \tilde{\mu} - \mu_{\rm c}. \tag{15.37}$$

The variable $\mu_{\rm diff}(\mathbf{r}, t)$ will play an important role in our subsequent discussion. In complete thermal equilibrium, of course, $\mu_{\rm diff} = 0$. The equilibrium C_{12} collision time τ_{12} in (15.36) is defined as

$$\begin{aligned} \frac{1}{\tau_{12}^0} \equiv \frac{2g^2}{(2\pi)^5 \hbar^7} \int d\mathbf{p}_1 \int d\mathbf{p}_2 \int d\mathbf{p}_3 \, \delta(\mathbf{p}_1 - \mathbf{p}_2 - \mathbf{p}_3) \\ \times \delta(\mu_{\rm c0} + \tilde{\varepsilon}_1^0 - \tilde{\varepsilon}_2^0 - \tilde{\varepsilon}_3^0)(1 + f_1^0) f_2^0 f_3^0; \end{aligned} \tag{15.38}$$

here f_i^0 is the static Bose distribution function for complete equilibrium between the condensate and thermal cloud ($\mu_{\rm c0} = \tilde{\mu}_0$). This τ_{12}^0 collision time also appeared in earlier chapters, whenever Γ_{12} was calculated under the assumption that the thermal cloud atoms were in static thermal equilibrium.

15.2 Equivalence to the Landau two-fluid equations

In this section, we prove that the linearized coupled equations for the condensate and noncondensate components given at the end of Section 15.1 can

be reduced to the standard Landau two-fluid equations in the non-dissipative limit, generalized to include an external trapping potential. This derivation is of interest for several reasons. It shows how the source term $\Gamma_{12}[f, \Phi]$ associated with the C_{12} collisions plays a crucial role in ensuring that we obtain the correct Landau two-fluid equations. It also shows that our approximate ZNG equations in the strong collision region lead to the Landau two-fluid equations and local hydrodynamic equilibrium.

General derivations of the Landau two-fluid equations given in the literature start from the existence of the Beliaev Bose order parameter $\langle\hat{\psi}(\mathbf{r})\rangle$, but without making use of a kinetic equation. Leggett (2006) gave a derivation of the Landau two-fluid equations without invoking a Bose order parameter. His discussion is based on the Onsager–Penrose requirement for the existence of a condensate, that the single-particle density matrix has one eigenvalue which is macroscopic in size and proportional to the total number of atoms N (see (1.8) and related discussion). All such formal derivations, however, assume that both the normal and superfluid components can be described as being in *local hydrodynamic* equilibrium, without any discussion of how this state might be reached.

We believe that our microscopic derivation of the Landau two-fluid hydrodynamic equations, which starts from the microscopic ZNG equations for the coupled condensate and noncondensate components, helps one to understand several subtle features of Bose superfluids in the collisional hydrodynamic region. As with the pioneering derivation of Bogoliubov (1970), we will show how the existence of the Bose symmetry-breaking order parameter introduced by Beliaev (1958a) allows one to give a solid foundation to the Landau two-fluid equations for a dilute Bose-condensed gas and to state the precise conditions for such equations to be valid. We recall that the equations of the Landau two-fluid collisional hydrodynamics reviewed in Chapter 14 are valid for all Bose superfluids. As will be discussed in Section 16.4, the two-fluid equations also describe the dynamics of Fermi gas superfluids over the whole BCS–BEC crossover if the s-wave scattering length is large enough (i.e. near unitarity) to ensure local equilibrium.

For later comparison, it is convenient to write down the linearized version of the Landau two-fluid hydrodynamic equations discussed in Section 14.2:

$$\frac{\partial \delta n}{\partial t} + \nabla \cdot \delta \mathbf{j} = 0, \tag{15.39}$$

$$m\frac{\partial \delta \mathbf{j}}{\partial t} = -\nabla \delta P - \delta n \nabla V_{\mathrm{trap}}, \tag{15.40}$$

$$m\frac{\partial \delta \mathbf{v}_{\mathrm{s}}}{\partial t} = -\nabla \delta \mu, \tag{15.41}$$

$$\frac{\partial \delta s}{\partial t} + \nabla \cdot (s_0 \delta \mathbf{v}_{\rm n}) = 0\,, \tag{15.42}$$

where (see (14.3) and (14.4))

$$\begin{aligned} m\delta \mathbf{j}(\mathbf{r},t) &= \rho_{\rm s0}(\mathbf{r})\delta \mathbf{v}_{\rm s}(\mathbf{r},t) + \rho_{\rm n0}(\mathbf{r})\delta \mathbf{v}_{\rm n}(\mathbf{r},t), \\ m\delta n(\mathbf{r},t) &= \delta\rho_{\rm s}(\mathbf{r},t) + \delta\rho_{\rm n}(\mathbf{r},t). \end{aligned} \tag{15.43}$$

We have here distinguished the superfluid variables $\rho_{\rm s}$ and $\mathbf{v}_{\rm s}$ from the corresponding variables for the condensate. A similar distinction applies to the normal fluid variables as opposed to the noncondensate variables. However, in our simple microscopic approximation for a dilute Bose gas on which the ZNG equations are based, the condensate and noncondensate mass densities turn out to be equal to $\rho_{\rm s}$ and $\rho_{\rm n}$, respectively. The reason is that our approximation treats the thermal cloud excitations as atoms moving in a self-consistent Hartree–Fock field. We recall that in Landau's two-fluid theory (see Chapter 6 of Pitaevskii and Stringari, 2003), the normal fluid density is given by the formula

$$\rho_{\rm n0}(\mathbf{r}) = \int \frac{d\mathbf{p}}{(2\pi\hbar)^3}\frac{p^2}{3}\left[-\frac{\partial f^0(E_p)}{\partial E_p}\right]. \tag{15.44}$$

With a semiclassical HF approximation for E_p, the momentum integration can be done (using integration by parts) and one finds that

$$\rho_{\rm n0}(\mathbf{r}) = m\tilde{n}_0(\mathbf{r}), \tag{15.45}$$

where $\tilde{n}_0(\mathbf{r})$ is given by (15.24). This exact correspondence is no longer valid outside the HF approximation. For example, in the Hartree–Fock–Popov (HFP) approximation (see Chapter 5), there is a finite depletion (reduction) of the condensate even at zero temperature. In this case, the superfluid density $\rho_{\rm s}$ would no longer reduce precisely to the condensate mass density $mn_{\rm c}$.

The linearized versions of the ZNG continuity equations are given in (15.30) and (15.33). Adding these two equations, we obtain the continuity equation (15.39), which is one of Landau's two-fluid equations. Combining the two velocity equations (15.31) and (15.34), we obtain

$$\begin{aligned} \frac{\partial \delta \mathbf{j}}{\partial t} = &-\nabla \delta\tilde{P} - 2g(n_{\rm c0}+\tilde{n}_0)\nabla\delta\tilde{n} - g(n_{\rm c0}+2\tilde{n}_0)\nabla\delta n_{\rm c} \\ &- 2g\nabla(n_{\rm c0}+\tilde{n}_0)\delta\tilde{n} - \nabla V_{\rm trap}\delta\tilde{n}\,. \end{aligned} \tag{15.46}$$

One can prove that (15.46) is equivalent to the second Landau equation,

(15.40), as follows. The total local pressure is related to $\tilde{P}$ and $\tilde{n}$ by (see Lee and Yang, 1958; Zaremba *et al.*, 1999)

$$\begin{aligned} P &= \tilde{P} + \tfrac{1}{2}g(n^2 + 2n\tilde{n} - \tilde{n}^2) \\ &= \tilde{P} + \tfrac{1}{2}g(n_{\rm c}^2 + 4\tilde{n}n_{\rm c} + 2\tilde{n}^2), \end{aligned} \tag{15.47}$$

within the microscopic approximation on which the ZNG equations are based, i.e. the Hartree–Fock energy spectrum for the thermal atoms. To prove this, we note that (15.47) gives

$$\delta P = \delta\tilde{P} + 2gn_0\delta\tilde{n} + g(n_{\rm c0} + 2\tilde{n}_0)\delta n_{\rm c}, \tag{15.48}$$

where the total density $n_0 = n_{\rm c0} + \tilde{n}_0$, and hence we have

$$\begin{aligned} \nabla\delta P = \nabla\delta\tilde{P} &+ 2gn_0\nabla\delta\tilde{n} + g(n_{\rm c0} + 2\tilde{n}_0)\nabla\delta n_{\rm c} \\ &+ 2g\delta\tilde{n}\nabla n_0 + g\delta n_{\rm c}(\nabla n_{\rm c0} + 2\nabla\tilde{n}_0). \end{aligned} \tag{15.49}$$

Making use of the TF result for the condensate chemical potential,

$$\mu_{\rm c0} = gn_{\rm c0}(\mathbf{r}) + 2g\tilde{n}_{\rm c0}(\mathbf{r}) + V_{\rm trap}(\mathbf{r}), \tag{15.50}$$

we can rewrite the last term in (15.49) as $-\delta n_{\rm c}\nabla V_{\rm trap}$. Inserting the resulting expression for $\nabla\delta P$ into the Landau equation (15.40), this can be shown to reduce to the ZNG result in (15.46).

In order to make contact with the two remaining Landau equations (15.41) and (15.42) we must introduce the appropriate thermodynamic variables used by Landau which correspond to the chemical potential μ, the local temperature T and the local entropy s. Within the HF approximation for the noncondensate, the *total* energy density ϵ is given by (compare (15.47))

$$\begin{aligned} \epsilon &= \tilde{\epsilon} + \tfrac{1}{2}g\left\langle\hat{\psi}^\dagger\hat{\psi}^\dagger\hat{\psi}\hat{\psi}\right\rangle + nV_{\rm trap} \\ &= \tilde{\epsilon} + \tfrac{1}{2}g\left(n^2 + 2n\tilde{n} - \tilde{n}^2\right) + nV_{\rm trap}\,, \end{aligned} \tag{15.51}$$

where $\tilde{\epsilon}$ is defined by (15.20) and (15.21). For the local equilibrium entropy, we use the definition

$$\begin{aligned} s &\equiv k_{\rm B}\int\frac{d\mathbf{p}}{(2\pi\hbar)^3}\left[(1+\tilde{f})\ln(1+\tilde{f}) - \tilde{f}\ln\tilde{f}\right] \\ &= \frac{1}{T}\left[\tfrac{5}{2}\tilde{P} - \tilde{n}(\tilde{\mu} - U)\right]. \end{aligned} \tag{15.52}$$

The same result gives the entropy of a uniform ideal gas above $T_{\rm BEC}$ with fugacity $z = e^{\beta(\tilde{\mu}-U)}$. In static thermal equilibrium, (15.52) reduces to

$$T_0 s_0 = \tfrac{5}{2}\tilde{P}_0 + gn_{\rm c0}\tilde{n}_0\,, \tag{15.53}$$

where we have used $\tilde{\mu}_0 = \mu_{\rm c0}$ and $U_0 = V_{\rm trap} + 2gn_0$.

From the expressions for P, ϵ and s for local hydrodynamic equilibrium, given by (15.47), (15.51) and (15.52), respectively (all of which depend on $\mathbf{r}$ and t), we obtain the relation

$$\epsilon + P - Ts = \mu_{\rm c} n_{\rm c} + \tilde{\mu}\tilde{n} = \mu_{\rm c} n + \mu_{\rm diff}\tilde{n}\,, \tag{15.54}$$

where $\mu_{\rm diff}$ is the chemical potential difference defined in (15.37). As noted earlier, this variable $\mu_{\rm diff}$ plays a crucial role in describing the approach to diffusive local equilibrium (where $\mu_{\rm diff} = 0$). If $\mu_{\rm diff}$ is zero then (15.54) corresponds precisely to the usual thermodynamic relation involving these local variables, $\mu_{\rm c}$ playing the role of the equilibrium chemical potential.

Having defined these various thermodynamic functions, valid for partial local equilibrium, we now consider their local variations from absolute equilibrium (recalling that the equilibrium properties are often functions of position in a trapped gas). Taking the variation of the kinetic pressure (15.21) gives

$$\begin{aligned}\delta\tilde{P} &= \delta\left(\frac{1}{\beta\Lambda^3}\right) g_{5/2}(z_0) + \frac{1}{\beta_0\Lambda_0^3}\frac{\partial g_{5/2}(z_0)}{\partial z_0}\delta z \\ &= \frac{5}{2}\frac{\tilde{P}_0}{T_0}\delta T + \frac{\tilde{n}_0}{\beta_0 z_0}\delta z. \end{aligned} \tag{15.55}$$

Inserting the result for the variation in the fugacity $z(\mathbf{r}, t)$,

$$\delta z = \tilde{n}_0\left[\delta(\tilde{\mu} - 2gn) + \frac{gn_{\rm c0}}{T}\delta T\right]\beta_0 z_0, \tag{15.56}$$

into (15.55) one obtains

$$\delta\tilde{P} = s_0\delta T + \tilde{n}_0(\delta\tilde{\mu} - 2g\delta n). \tag{15.57}$$

In deriving this expression, we have made use of (15.53) and the Bose identity $z g_n'(z) = g_{n-1}(z)$, where $g_n(z)$ is a Bose–Einstein function as defined below (15.21). Combining (15.48) and (15.55), we find a thermodynamic relation for partial local equilibrium,

$$\delta P = s_0\delta T + n_0\delta\mu_{\rm c} + \tilde{n}_0\delta\mu_{\rm diff}\,. \tag{15.58}$$

The reason for the appearance of the last term is that the noncondensate has a chemical potential different from that of the condensate (i.e. $\mu_{\rm diff} \neq 0$).

Similarly, using (15.52) and (15.55), the entropy fluctuation δs is given by

$$T_0\delta s = \tfrac{3}{2}\delta\tilde{P} + gn_{\rm c0}\delta\tilde{n}. \tag{15.59}$$

This expression leads directly to the linearized form of the Landau entropy

conservation equation (14.6). Taking the time derivative of (15.59) and making use of the noncondensate density and pressure equations (15.30) and (15.32), we find that

$$\begin{aligned}\frac{\partial \delta s}{\partial t} &= \frac{3}{2T_0}\frac{\partial \delta \tilde{P}}{\partial t} + \frac{g n_{\mathrm{c}0}}{T_0}\frac{\partial \delta \tilde{n}}{\partial t} \\ &= -\nabla \cdot (s_0 \delta \mathbf{v}_{\mathrm{n}}).\end{aligned} \tag{15.60}$$

This is precisely the Landau entropy conservation equation (15.42). We note that the source terms $\delta\Gamma_{12}$ on the right-hand sides of (15.30) and (15.32) cancel out, so that a strict conservation law for the nonequilibrium local entropy defined in (15.52) is obtained.

Finally, a comparison of (15.34) and the Landau equation (15.41) shows that these two equations are equivalent if $\delta \mathbf{v}_{\mathrm{s}}$ is identified with $\delta \mathbf{v}_{\mathrm{c}}$ and $\delta\mu$ with $\delta\mu_{\mathrm{c}}$. In summary, we have derived a set of four hydrodynamic equations which are precisely of the form of the Landau two-fluid equations. At first sight, it might therefore appear that our starting equations are in fact equivalent to the Landau equations (15.39)–(15.42), but we have not proved this yet. The appearance of μ_{diff} in (15.54) implies that μ_{c} is *not* related to the other thermodynamic variables in the same way that the equilibrium chemical potential μ would be. As a result, $\delta\mu_{\mathrm{c}}$ and δs are not related directly to the thermodynamic fluctuations δT and δP as in (14.7). The latter relation was assumed in the derivation of the first and second sound velocities in Section 14.2.

To understand this difference more clearly, we can use (15.53) to define the equilibrium entropy function, writing $Ts^{\mathrm{eq}} \equiv \frac{5}{2}\tilde{P} + g n_{\mathrm{c}} \tilde{n}$. The notation s^{eq} is used here to distinguish this quantity from the local entropy s defined in (15.52). The variation of the equilibrium entropy s^{eq} can be expressed in terms of the thermodynamic fluctuations δT and δP via equilibrium thermodynamic derivatives. Considering a quasistatic change in the thermodynamic state of the system, (15.53) leads to the relation

$$T_0 \delta s^{\mathrm{eq}} = \tfrac{5}{2}\delta\tilde{P} + g n_{\mathrm{c}0}\delta\tilde{n} + g\tilde{n}_0 \delta n_{\mathrm{c}} - s_0 \delta T \tag{15.61}$$

and using (15.58) to eliminate δT, we find

$$T_0 \delta s^{\mathrm{eq}} = \tfrac{3}{2}\delta\tilde{P} + g n_{\mathrm{c}0}\delta\tilde{n} + \tilde{n}_0 \delta\mu_{\mathrm{diff}}. \tag{15.62}$$

This result clearly differs from the result for δs given in (15.59). The additional last term in (15.62) shows that the variation of the equilibrium entropy is not simply given by the equilibrium variation when $\delta\mu_{\mathrm{diff}}$ is finite.

In terms of $\delta s^{\rm eq}$, the entropy equation analogous to (15.42) takes the form

$$\frac{\partial \delta s^{\rm eq}}{\partial t} = -\nabla \cdot (s_0 \delta \mathbf{v}_{\rm n}) + \tilde{n}_0 \frac{\partial \delta \mu_{\rm diff}}{\partial t} \,. \tag{15.63}$$

The last term in (15.63) can be interpreted as the rate of production of entropy associated with the equilibration of the condensate and noncondensate chemical potentials. Its appearance emphasizes that the thermodynamic fluctuations δP and δT are coupled to fluctuations in $\delta \mu_{\rm diff}$, which are directly related to $\delta \Gamma_{12}$ by (15.36). In Chapter 17, we show that this kind of extra term always arises when one includes relaxation processes, for which the entropy is not conserved.

To make the final connection between the two-fluid equations derived above and the standard Landau two-fluid equations, we have to obtain an equation of motion for $\delta \mu_{\rm diff}(\mathbf{r}, t)$. Noting that $\tilde{\mu} - 2gn = \mu_{\rm diff} - g n_{\rm c}$, we see that (15.54) can be rewritten in the form

$$\delta \tilde{P} = s_0 \delta T + \tilde{n}_0 (\delta \mu_{\rm diff} - g \delta n_{\rm c}) \,. \tag{15.64}$$

Similarly, the variation of (15.24) yields the equation

$$\delta \tilde{n} = \left(\tfrac{3}{2} \tilde{n}_0 + \tilde{\gamma}_0 n_{\rm c0} \right) \frac{\delta T}{T_0} + \frac{\tilde{\gamma}_0}{g} (\delta \mu_{\rm diff} - g \delta n_{\rm c}) \,, \tag{15.65}$$

where we have introduced the dimensionless quantity

$$\tilde{\gamma}_0 \equiv \beta_0 g \int \frac{d\mathbf{p}}{(2\pi\hbar)^3} f^0 (1 + f^0) = \frac{\beta_0 g}{\Lambda_0^3} g_{1/2}(z_0) \,. \tag{15.66}$$

This quantity $\tilde{\gamma}_0$ can be related to thermodynamic derivatives by noting that (15.65) implies that an equilibrium variation at constant temperature gives $\delta \tilde{n}|_T = (\tilde{\gamma}_0/g) \delta(\tilde{\mu} - 2gn)|_T = -\tilde{\gamma}_0 (\delta n - \delta \tilde{n})|_T$. We thus obtain

$$\tilde{\gamma}_0 = \frac{(\partial \tilde{n}/\partial n)_T}{(\partial \tilde{n}/\partial n)_T - 1} . \tag{15.67}$$

In a similar manner, one can show that the coefficient of δT in (15.65) can be reduced to

$$\left(\tfrac{3}{2} \tilde{n}_0 + \tilde{\gamma}_0 n_{\rm c0} \right) \frac{1}{T_0} = (1 - \tilde{\gamma}_0) \left(\frac{\partial \tilde{n}}{\partial T} \right)_{\rm n} . \tag{15.68}$$

Equations (15.64) and (15.65) show that the fluctuations in δT and $\delta \mu_{\rm diff}$ can each be expressed in terms of $\delta n_{\rm c}$, $\delta \tilde{n}$ and $\delta \tilde{P}$. Eliminating $\delta \mu_{\rm diff}$ from these equations and making use of (15.30) and (15.32), one obtains an equation of motion for δT in terms of $\delta \Gamma_{12}$ and $\delta \mathbf{v}_{\rm n}$,

$$\frac{1}{T_0} \frac{\partial \delta T}{\partial t} = -\tfrac{2}{3} \nabla \cdot \delta \mathbf{v}_{\rm n} - g \left(\frac{\tilde{n}_0 + \frac{2}{3} \tilde{\gamma}_0 n_{\rm c0}}{\frac{5}{2} \tilde{\gamma}_0 \tilde{P}_0 - \frac{3}{2} g \tilde{n}_0^2} \right) \delta \Gamma_{12} \,. \tag{15.69}$$

Alternatively, eliminating δT from (15.64) and (15.65), we find

$$\frac{\partial \delta\mu_{\rm diff}}{\partial t} = \nabla U_0 \cdot \delta\mathbf{v}_{\rm n} + \tfrac{2}{3} g n_{\rm c0} \nabla\cdot\delta\mathbf{v}_{\rm n} - g\nabla\cdot(n_{\rm c0}\delta\mathbf{v}_{\rm c}) + g\left(\frac{\frac{5}{2}\tilde{P}_0 + 2g\tilde{n}_0 n_{\rm c0} + \frac{2}{3}\tilde{\gamma}_0 g n_{\rm c0}^2}{\frac{5}{2}\tilde{\gamma}_0\tilde{P}_0 - \frac{3}{2}g\tilde{n}_0^2} - 1\right)\delta\Gamma_{12}\,. \tag{15.70}$$

Making use of the expression (15.36), relating $\delta\Gamma_{12}$ to $\delta\mu_{\rm diff}$, we can rewrite (15.70) in the more transparent form

$$\frac{\partial \delta\mu_{\rm diff}}{\partial t} = \nabla U_0 \cdot \delta\mathbf{v}_{\rm n} + \frac{2}{3} g n_{\rm c0} \nabla\cdot\delta\mathbf{v}_{\rm n} - g\nabla\cdot(n_{\rm c0}\delta\mathbf{v}_{\rm c}) - \frac{\delta\mu_{\rm diff}}{\tau_\mu}\,, \tag{15.71}$$

where the relaxation time τ_μ associated with the chemical potential difference is defined via

$$\frac{1}{\tau_\mu} \equiv \frac{\beta_0 g n_{\rm c0}}{\tau_{12}}\left(\frac{\frac{5}{2}\tilde{P}_0 + 2g\tilde{n}_0 n_{\rm c0} + \frac{2}{3}\tilde{\gamma}_0 g n_{\rm c0}^2}{\frac{5}{2}\tilde{\gamma}_0\tilde{P}_0 - \frac{3}{2}g\tilde{n}_0^2} - 1\right) \equiv \frac{\beta_0 g n_{\rm c0}}{\tau_{12}\sigma_{\rm H}}\,. \tag{15.72}$$

Although the relaxation time τ_μ is proportional to the equilibrium collision time τ_{12} defined in (15.38), it has quite a different magnitude and temperature dependence, especially in the neighbourhood of $T_{\rm BEC}$, as a result of the factors $n_{\rm c0}$ and $\sigma_{\rm H}$ appearing in (15.72). The hydrodynamic factor $\sigma_{\rm H}$ defined in (15.72) plays an important role in the subsequent analysis.

This relaxation time τ_μ due to C_{12} collisions describes the rate of equilibration of the condensate and noncondensate chemical potentials; this equilibration is brought about by collisions between the two components. For a uniform gas with spatially homogeneous fluctuations, the solution of (15.71) is simply

$$\delta\mu_{\rm diff}(t) = \delta\mu_{\rm diff}(0)e^{-t/\tau_\mu}. \tag{15.73}$$

This shows that any initial difference in chemical potentials decays on a time scale set by τ_μ. Similar considerations will also apply to a trapped (nonuniform) gas. Thus, for fluctuations that occur on a time scale longer than the relaxation time τ_μ, we expect that $\delta\mu_{\rm diff}$ can effectively be set to zero (i.e. $\tilde{\mu} = \mu_{\rm c}$). It then follows from (15.54) that we recover the standard equilibrium relationships among all the thermodynamic variables we have defined. In summary, we have proved that, in the $\omega\tau_\mu \to 0$ limit, our hydrodynamic equations for a collective mode of frequency ω derived from the ZNG microscopic theory reproduce precisely the results given by the Landau two-fluid equations (15.39)–(15.42).

15.3 First and second sound in a Bose-condensed gas

To demonstrate the reduction of our equations to the Landau equations in the limit $\omega\tau_\mu \to 0$ more explicitly, in this section we will consider the hydrodynamic sound modes in a uniform Bose gas (Nikuni *et al.*, 1999). We take into account the fact that, according to (15.71), fluctuations in $\delta\mu_{\rm diff}$ are coupled to the fluctuations of the two velocities $\delta\mathbf{v}_{\rm c}$ and $\delta\mathbf{v}_{\rm n}$. It is convenient to introduce velocity potentials according to $\delta\mathbf{v}_{\rm c} \equiv \nabla\phi_{\rm c}$ and $\delta\mathbf{v}_{\rm n} \equiv \nabla\phi_{\rm n}$. In terms of these new variables, equations (15.33)–(15.35) for the condensate and equations (15.30)–(15.32) for the noncondensate can be combined to give

$$m\frac{\partial^2\phi_{\rm c}}{\partial t^2} = gn_{\rm c0}\nabla^2\phi_{\rm c} + 2g\tilde{n}_0\nabla^2\phi_{\rm n} + \frac{\sigma_{\rm H}}{\tau_\mu}\delta\mu_{\rm diff}, \tag{15.74}$$

$$m\frac{\partial^2\phi_{\rm n}}{\partial t^2} = \left(\frac{5\tilde{P}_0}{3\tilde{n}_0} + 2g\tilde{n}_0\right)\nabla^2\phi_{\rm n} + 2gn_{\rm c0}\nabla^2\phi_{\rm c} - \frac{2\sigma_{\rm H}}{3\tau_\mu}\delta\mu_{\rm diff}. \tag{15.75}$$

Here $\delta\Gamma_{12}$ has been expressed in terms of $\delta\mu_{\rm diff}$ using (15.36). These equations for $\phi_{\rm c}$, $\phi_{\rm n}$ and $\delta\mu_{\rm diff}$, in conjunction with (15.71), constitute a *complete* set describing the hydrodynamic modes of the gas. We look for solutions having a plane-wave form $\phi_{\rm c,n}(\mathbf{r},t) = \phi_{\rm c,n}e^{i(\mathbf{q}\cdot\mathbf{r}-\omega t)}$. In this case, (15.71) reduces to

$$\delta\mu_{\rm diff} = \frac{gn_{\rm c0}\tau_\mu}{1-i\omega\tau_\mu}\left(\phi_{\rm c} - \tfrac{2}{3}\phi_{\rm n}\right)q^2\,. \tag{15.76}$$

Substituting this result into (15.74) and (15.75), we are left with two coupled equations for the superfluid and normal fluid velocity potentials:

$$m\omega^2\phi_{\rm c} = gn_{\rm c0}\left(1 - \frac{\sigma_{\rm H}}{1-i\omega\tau_\mu}\right)q^2\phi_{\rm c} + 2g\tilde{n}_0\left(1 + \frac{\sigma_{\rm H}}{3(1-i\omega\tau_\mu)}\frac{n_{\rm c0}}{\tilde{n}_0}\right)q^2\phi_{\rm n} \tag{15.77}$$

and

$$m\omega^2\phi_{\rm n} = \left[\frac{5\tilde{P}_0}{3\tilde{n}_0} + 2g\tilde{n}_0\left(1 - \frac{2\sigma_{\rm H}}{9(1-i\omega\tau_\mu)}\frac{n_{\rm c0}^2}{\tilde{n}_0^2}\right)\right]q^2\phi_{\rm n} + 2gn_{\rm c0}\left(1 + \frac{\sigma_{\rm H}}{3(1-i\omega\tau_\mu)}\frac{n_{\rm c0}}{\tilde{n}_0}\right)q^2\phi_{\rm c}\,, \tag{15.78}$$

where the thermodynamic factor $\sigma_{\rm H}$ was defined by (15.72).

Taking the limit $\omega\tau_\mu \to 0$ of these coupled equations, we obtain

$$m\omega^2\phi_{\rm c} = gn_{\rm c0}(1-\sigma_{\rm H})q^2\phi_{\rm c} + 2g\tilde{n}_0\left(1 + \frac{\sigma_{\rm H}n_{\rm c0}}{3\tilde{n}_0}\right)q^2\phi_{\rm n}, \tag{15.79}$$

$$m\omega^2\phi_{\rm n} = \left[\frac{5\tilde{P}_0}{3\tilde{n}_0} + 2g\tilde{n}_0\left(1 - \frac{2\sigma_{\rm H}n_{\rm c0}^2}{9\tilde{n}_0^2}\right)\right]q^2\phi_{\rm n} + 2gn_{\rm c0}\left(1 + \frac{\sigma_{\rm H}n_{\rm c0}}{3\tilde{n}_0}\right)q^2\phi_{\rm c}\,, \tag{15.80}$$

with $\delta\mu_{\rm diff} \to gn_{\rm c0}\tau_\mu(\phi_{\rm c} - \frac{2}{3}\phi_{\rm n})q^2$. We see from these coupled equations that while $\delta\mu_{\rm diff}$ vanishes in the $\omega\tau_\mu \to 0$ limit, $\delta\Gamma_{12}$ remains finite even when $\omega\tau_\mu \to 0$ because it is proportional to $\delta\mu_{\rm diff}/\tau_\mu$. Thus, the $\delta\Gamma_{12}$ terms in equations (15.30) and (15.33) are still present in the Landau limit (defined by $\omega\tau_\mu \ll 1$). This simply reflects the fact that the establishment of diffusive local equilibrium requires a continual local readjustment of the number of atoms in the condensate and noncondensate through C_{12} collisions. In other words, the presence of strong collisional coupling between the two components, which ensures that $\delta\mu_{\rm diff} = 0$, also implies that the condensate and noncondensate densities are not separately conserved. Explicitly, in the Landau limit, we have in (15.30) and (15.33)

$$\delta\Gamma_{12}(\mathbf{q},\omega) = -\sigma_{\rm H}n_{\rm c0}\left(\phi_{\rm c} - \tfrac{2}{3}\phi_{\rm n}\right)q^2. \tag{15.81}$$

One sees from (15.79)–(15.81) that the thermodynamic factor $\sigma_{\rm H}$ plays an important role. In physical terms, one may view $\sigma_{\rm H}$ as defined in (15.72) as a renormalization factor that describes the change from the variables used in (15.30)–(15.32) for the thermal cloud in the ZNG equations to the thermodynamic variables used in Landau's equations (15.39)–(15.42).

Although this is not immediately apparent, equations (15.79) and (15.80) yield first and second sound velocities in precise agreement with those determined by the usual Landau two-fluid equations. This equivalence, of course, already follows from the fact that, as proved above, our microscopic collisional hydrodynamic equations are equivalent to the Landau two-fluid equations in the limit $\omega\tau_\mu \ll 1$.

We have seen that the diffusive equilibration of the superfluid and normal fluid (by the relaxation time τ_μ in (15.71) and (15.72)) plays a key role in arriving at the standard Landau two-fluid equations, even though these are valid only in the limit $\omega\tau_\mu \to 0$. In Chapter 17, we will extend the derivation of the non-dissipative two-fluid equations given in this section and include the hydrodynamic damping associated with the transport coefficients. We shall find that the relaxation time τ_μ is, in fact, related to what are called the coefficients of second viscosity in a superfluid. However, the physics of this relaxation time τ_μ is brought out most clearly by treating it in isolation from other transport processes, as we have done in this section. A few more

remarks about the role of τ_μ may be useful, since this new collision time is a special feature of Bose-condensed gases.

Another version of the two-fluid hydrodynamics was also discussed in Zaremba *et al.* (1999), one in which the collisional transfer of atoms between the condensate and thermal cloud was ignored. This corresponds to setting the source term $\Gamma_{12} = 0$ or, equivalently, taking the limit $\omega\tau_\mu \gg 1$. This version of two-fluid hydrodynamics was first discussed by Zaremba *et al.* (1998). It will be referred to as the "ZGN" limit, to contrast it with the Landau hydrodynamics that emerges from the ZNG equations. This ZGN limit was used to calculate the temperature dependence of the frequencies of first and second sound in a weakly interacting uniform Bose gas by Griffin and Zaremba (1997). Zaremba *et al.* (1999) later calculated the first and second sound velocities as a function of temperature in the two limits $\omega\tau_\mu \ll 1$ and $\omega\tau_\mu \gg 1$ and found that the results were in very close agreement.

This similarity of these two theories only arises in the limit of weak interactions. In the $\omega\tau_\mu \gg 1$ limit, the terms involving $\sigma_{\rm H}$ in (15.79) and (15.80) would not be present. In the Landau limit, $\omega\tau_\mu \ll 1$, these terms can also be neglected in the limit of weak interactions. Recalling the definition of $\sigma_{\rm H}$ from (15.72),

$$\sigma_{\rm H} \equiv \frac{\frac{5}{2}\tilde{\gamma}_0\tilde{P}_0 - \frac{3}{2}g\tilde{n}_0^2}{\frac{5}{2}\tilde{P}_0(1-\tilde{\gamma}_0) + 2gn_{\rm c0}\tilde{n}_0 + \frac{3}{2}g\tilde{n}_0^2 + \frac{2}{3}\tilde{\gamma}_0 g n_{\rm c0}^2}, \tag{15.82}$$

and that $\tilde{\gamma}_0$ in (15.66) is of order g, we see that to first order in g the expression $\sigma_{\rm H}$ reduces to

$$\sigma_{\rm H} \simeq \tilde{\gamma}_0 - \tfrac{3}{5}g\frac{\tilde{n}_0^2}{\tilde{P}_0} + O(g^2). \tag{15.83}$$

Solving (15.79) and (15.80), the squares of the first and second sound velocities $u^2 \equiv \omega^2/q^2$ are the solutions of the quadratic equation

$$\left[u^2 - \frac{gn_{\rm c0}}{m}(1-\sigma_{\rm H})\right]\left[u^2 - \frac{5\tilde{P}_0}{3\,m\tilde{n}_0} - \frac{2g\tilde{n}_0}{m}\left(1 - \frac{2\sigma_{\rm H}n_{\rm c0}^2}{9\tilde{n}_0^2}\right)\right] - \frac{4g^2\tilde{n}_0 n_{\rm c0}}{m^2}\left(1 + \frac{\sigma_{\rm H}n_{\rm c0}}{3\tilde{n}_0}\right)^2 = 0. \tag{15.84}$$

The terms involving $\sigma_{\rm H}$ clearly give rise to corrections to u^2 that are of order g^2. This explains why, for weak interactions, the first and second velocities are found to be essentially the same in the ZNG Landau limit ($\omega\tau_\mu \ll 1$) and the ZGN limit ($\omega\tau_\mu \gg 1$). In both cases, working to first order in g, the first and second sound velocities reduce to the results quoted in (14.24)

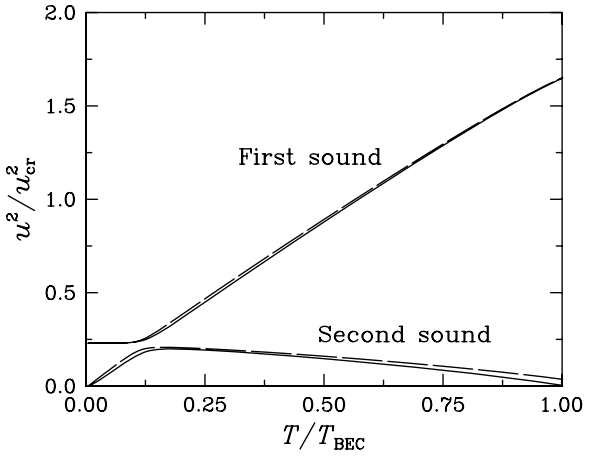


Fig. 15.1. Squares of the first and second sound velocities (normalized by the first sound velocity of an ideal gas at $T = T_{\rm BEC}$). The solid lines are for the Landau two-fluid hydrodynamics while the broken lines are for the ZGN hydrodynamics ($\omega\tau_\mu \gg 1$) described in the text. These results are for $gn/k_{\rm B}T_{\rm BEC} = 0.2$ (from Zaremba *et al.*, 1999).

and (14.25). Solving (15.84) for the velocities to order g^2, we find[1]

$$u_1^2 = \frac{5}{3}\frac{\tilde{P}_0}{m\tilde{n}_0} + \frac{2g\tilde{n}_0}{m}\left(1 - \frac{2}{9}\frac{n_{\rm c0}^2}{\tilde{n}_0^2}\sigma_{\rm H} + \frac{3m\tilde{n}_0}{5\tilde{P}_0}\frac{2gn_{\rm c0}}{m}\right), \tag{15.85}$$

$$u_2^2 = \frac{gn_{\rm c0}}{m}\left(1 - \sigma_{\rm H} - \frac{3m\tilde{n}_0}{5\tilde{P}_0}\frac{4g\tilde{n}_0}{m}\right), \tag{15.86}$$

where $\sigma_{\rm H}$ is approximated by (15.83) to order g.

To illustrate these results, the first and second sound velocities u_i^2 are plotted in Fig. 15.1 as a function of the temperature for a weakly interacting uniform Bose gas. As expected, the difference between the two limits ($\omega\tau_\mu \ll 1$ and $\omega\tau_\mu \gg 1$) is very small. The results in Fig. 15.1 also show that the first and second sound modes hybridize at low temperatures. This hybridization was originally discussed in the pioneering work of Lee and Yang (1959) on collective modes in a uniform Bose superfluid gas at finite temperatures. The two-fluid equations derived in the present chapter involve coefficients based on the ZNG approximation, which treats the thermal cloud in a high-temperature Hartree–Fock approximation ($gn_0 \ll k_{\rm B}T$). Thus we cannot

[1] The formulas for u_1^2 and u_2^2 given in Appendix C of Nikuni and Griffin (2001a) omit the last two terms in the bracketed expressions (of order g^2) in (15.85) and (15.86). The reason is that it was assumed there that the condensate and noncondensate velocity potentials $\phi_{\rm c}$ and $\phi_{\rm n}$ were uncoupled, an assumption that fact is only correct to first order in g.

expect our results for the first and second velocities to be quantitatively accurate at the low temperatures where the hybridization of the first and second sound modes occurs (see Fig. 15.1). The calculations of Lee and Yang (1959) are more complete since these authors considered both the low- and high-temperature limits in a Bose-condensed gas. In particular, in the low-temperature limit (where the thermal excitations are phonon-like rather than particle-like), the lower branch has a finite velocity at $T = 0$ (rather than going to zero as shown in Fig. 15.1). In spite of this deficiency at low temperatures, the results in Fig. 15.1 based on the coupled equations (15.77) and (15.78) still capture the hybridization that occurs when $u_1^2 \sim u_2^2$.

The preceding results are for the case of two-fluid hydrodynamic modes in a *uniform* Bose gas. However, they suggest that in trapped gases the Landau two-fluid equations will give results similar to those obtained from the ZGN two-fluid equations. Variational solutions of the ZGN equations describing the monopole and quadrupole hydrodynamic modes in a trapped Bose gas are discussed in Section VIIB of Zaremba *et al.* (1999). These solutions show that the hybridization of the in-phase and out-of-phase modes shown in Fig. 15.1 also occurs for the modes in trapped gases. We defer further discussion to the end of Section 16.3, where we discuss variational solutions of the Landau two-fluid equations for a trapped superfluid Bose gas.

The hydrodynamic modes are undamped in both the ZGN limit ($\omega\tau_\mu \to \infty$) and the ZNG Landau limit ($\omega\tau_\mu \to 0$). At intermediate values of $\omega\tau_\mu$, however, the damping arising from equilibration of the condensate and noncondensate can be quite significant. To obtain the hydrodynamic mode frequencies and damping, we set $\omega = \Omega - i\Gamma$ and solve the three hydrodynamic equations given by (15.74)–(15.76). These equations yield five mode frequencies: four modes corresponding to damped first and second sound ($\Omega = \pm u_{1,2} q$) and a purely imaginary (nonpropagating) relaxational mode. In the $\omega\tau_\mu \to 0$ limit, this relaxational mode has frequency $\omega \simeq -i/\tau_\mu$, consistent with (15.73).

In Fig. 15.2, we show the damping of the second sound mode in a uniform gas as a function of $\Omega\tau_\mu$, for $T/T_{\rm BEC} = 0.9$ and an interaction strength defined by $gn/k_{\rm B}T_{\rm BEC} = 0.2$. The relative damping Γ/Ω of the first sound mode is barely visible on the scale of Fig. 15.2, and therefore it is not displayed. It can be seen that the damping of the second sound mode peaks strongly at $\Omega\tau_\mu \simeq 1$, which marks the transition between the ZGN and the ZNG (Landau) regimes. If we neglect the coupling of the condensate to the noncondensate velocity potentials $\phi_{\rm c}$ and $\phi_{\rm n}$ in (15.77), the second sound

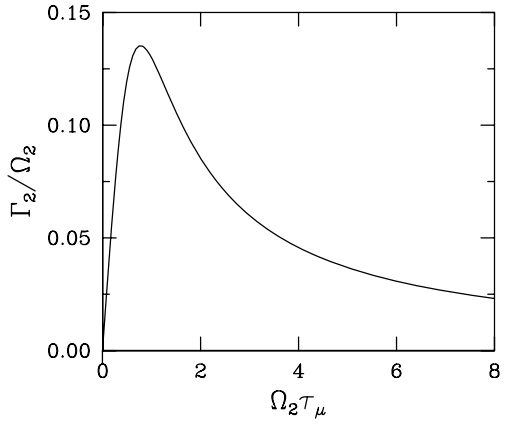


Fig. 15.2. Damping of the second sound mode in a uniform gas due to the relaxation of the condensate and noncondensate chemical potentials at a temperature such that $T/T_{\rm BEC} = 0.9$ (from Zaremba *et al.*, 1999).

frequency is given by

$$\omega^2 = \Omega_2^2 \left(1 - \frac{\sigma_{\rm H}}{1 - i\omega\tau_\mu} \right), \tag{15.87}$$

where $\Omega_2^2 \equiv (gn_{\rm c0}/m)q^2$. Setting $\omega = \Omega_2 - i\Gamma_2$ in (15.87), we find that the second sound damping is

$$\Gamma_2 = \frac{1}{2} \frac{\sigma_{\rm H}\Omega_2^2\tau_\mu}{1 + \Omega_2^2\tau_\mu^2}. \tag{15.88}$$

This formula gives an excellent fit to the numerical results in Fig. 15.2. The damping of the second sound mode is especially large since it involves an out-of-phase oscillation of the condensate and noncondensate density fluctuations. Such an out-of-phase oscillation would be expected to have the largest imbalance of local chemical potentials and thus the largest rate of transfer of atoms between the condensate and noncondensate. This damping mechanism always arises when the equilibration of the condensate and noncondensate is incomplete. Its analogue for a trapped Bose gas is discussed at the end of Section 19.5 (see (19.59)).

The damping involving the relaxation time τ_μ that has been discussed in this section is shown in Chapter 18 to be associated with the second viscosity transport coefficients arising in superfluids. In addition, there is hydrodynamic damping from the thermal conductivity and shear viscosity transport processes. These are neglected in this chapter, but are included in Chapters 18 and 19.

15.4 Hydrodynamic modes in a trapped normal Bose gas

Above the superfluid transition, the superfluid component is absent and the two-fluid equations described in Sections 15.2 and 15.3 reduce to the ordinary non-dissipative hydrodynamic equations for a trapped Bose gas. In discussing this, we follow the analysis of Griffin, Wu and Stringari (1997), which will be referred to as GWS. The linearized hydrodynamic equations (15.30)–(15.32) reduce to

$$\frac{\partial \delta n}{\partial t} = -\nabla \cdot [n_0(\mathbf{r})\delta \mathbf{v}], \tag{15.89}$$

$$mn_0(\mathbf{r})\frac{\partial \delta \mathbf{v}}{\partial t} = -\nabla \delta P - \delta n \nabla U_0 - 2gn_{\rm c}(\mathbf{r})\nabla \delta n, \tag{15.90}$$

$$\frac{\partial \delta P}{\partial t} = -\tfrac{5}{3}\nabla \cdot [P_0(\mathbf{r})\delta \mathbf{v}] + \tfrac{2}{3}\delta \mathbf{v} \cdot \nabla P_0(\mathbf{r}). \tag{15.91}$$

Note that since $n_{\rm c} = 0$, the source term Γ_{12} due to C_{12} collisions also vanishes. The tilde denoting the thermal gas component, used in previous sections, is no longer needed since now there is no condensate component. Here $U_0(\mathbf{r})$ is the effective (HF) static self-consistent trapping potential:

$$U_0(\mathbf{r}) = V_{\rm trap}(\mathbf{r}) + 2gn_0(\mathbf{r}). \tag{15.92}$$

In thermal equilibrium, the static local density and pressure are related to $U_0(\mathbf{r})$, which must satisfy the condition

$$\nabla P_0(\mathbf{r}) + n_{\rm c}(\mathbf{r})\nabla U_0(\mathbf{r}) = 0. \tag{15.93}$$

The local kinetic pressure and density are given by

$$n_0(\mathbf{r}) = \frac{1}{\Lambda_0^3} g_{3/2}(z_0), \quad P_0(\mathbf{r}) = \frac{1}{\beta_0 \Lambda_0^3} g_{5/2}(z_0). \tag{15.94}$$

One may easily verify that $\nabla n_0(\mathbf{r})$ is proportional to $\nabla U_0(\mathbf{r})$. In contrast with traditional discussions of the dynamics of normal fluids, the above hydrodynamic equations include the Hartree–Fock mean field contributions.

The above equations for a trapped Bose gas can be combined to give a closed equation for the velocity fluctuations $\delta \mathbf{v}(\mathbf{r}, t)$. Taking the time derivative of (15.90) and using (15.89) and (15.91) to eliminate the variables δP and δn, one finds

$$\begin{aligned} mn_0(\mathbf{r})\frac{\partial^2 \delta \mathbf{v}}{\partial t^2} = {} & \tfrac{5}{3}\nabla \{\nabla \cdot [P_0(\mathbf{r})\delta \mathbf{v}]\} - \tfrac{2}{3}\nabla [\nabla P_0(\mathbf{r}) \cdot \delta \mathbf{v}] \\ & + \nabla \cdot (n_0 \delta \mathbf{v})\nabla U_0 + 2gn_0 \nabla \nabla \cdot (n_0 \delta \mathbf{v}). \end{aligned} \tag{15.95}$$

Using (15.93), one can finally reduce (15.95) to (Nikuni and Griffin, 1998b)

$$m\frac{\partial^2\delta\mathbf{v}}{\partial t^2} = \frac{5}{3}\frac{P_0(\mathbf{r})}{n_0(\mathbf{r})}\nabla(\nabla\cdot\delta\mathbf{v}) - \nabla(\nabla U_0\cdot\delta\mathbf{v}) - \frac{2}{3}\nabla U_0(\nabla\cdot\delta\mathbf{v}) + 2gn_0(\mathbf{r})\nabla\left[\nabla\cdot(n_0(\mathbf{r})\delta\mathbf{v})\right]. \quad (15.96)$$

This is a vector equation, giving three coupled equations for the three components of $\delta\mathbf{v}(\mathbf{r},t)$. In deriving this, we have made use of the fact that $\nabla n_0(\nabla U_0\cdot\delta\mathbf{v}) = \nabla U_0(\nabla n_0\cdot\delta\mathbf{v})$. If one omits the HF mean fields ($U_0 = V_{\rm trap}$), (15.96) reduces to

$$m\frac{\partial^2\delta\mathbf{v}}{\partial t^2} = \frac{5}{3}\frac{P_0(\mathbf{r})}{n_0(\mathbf{r})}\nabla(\nabla\cdot\delta\mathbf{v}) - \nabla(\nabla V_{\rm trap}\cdot\delta\mathbf{v}) - \frac{2}{3}\nabla V_{\rm trap}(\nabla\cdot\delta\mathbf{v}). \quad (15.97)$$

This equation of motion is derived in an alternative way by Pethick and Smith (2008, p. 331).

In the paper GWS, (15.97) was used to derive some exact solutions for the hydrodynamic modes in a trap. We recall that, at $T = 0$, one can also obtain exact solutions of the time-dependent Gross–Pitaevskii equation for the condensate fluctuations (see Chapter 2). Thus one has exact solutions of the hydrodynamic equations both at $T = 0$ and for T above $T_{\rm BEC}$. These are very useful in dealing with coupled oscillations of the condensate and the thermal cloud at finite temperatures, for the following reason. One finds that, at intermediate temperatures, to a first approximation the condensate motion is similar to that at $T = 0$ (except that the condensate is thermally depleted). Similarly, the motion of the thermal cloud below $T_{\rm BEC}$ can be approximated by the exact solutions above $T_{\rm BEC}$, as long as we take into account that some fraction of the atoms is now in the condensate. These simple approximations are used in Section 16.3 as variational solutions to describe the two-fluid hydrodynamic modes. For this reason, we now review some of the exact solutions of the hydrodynamic equation (15.97) for a trapped gas above $T_{\rm BEC}$. For more details, see GWS, Kagan *et al.* (1997) and Pethick and Smith (2008).

An interesting class of normal mode solutions $\delta\mathbf{v}(\mathbf{r},t) = \delta\mathbf{v}_\omega e^{-i\omega t}$ of the wave equation (15.97) are those which are divergence free ($\nabla\cdot\delta\mathbf{v} = 0$). In this case, (15.97) reduces to

$$-m\omega^2\delta\mathbf{v}_\omega(\mathbf{r}) = -\nabla[\delta\mathbf{v}_\omega(\mathbf{r})\cdot\nabla V_{\rm trap}]. \quad (15.98)$$

From (15.98), one sees these solutions are also irrotational ($\nabla\times\delta\mathbf{v} = 0$). They do not depend on the equation of state of the trapped gas (which only enters in the first term on the r.h.s. of (15.97)) and thus are the same for

both classical and degenerate Bose gases. In GWS it is shown, using (15.94), (15.89) and (15.91), that the local hydrodynamic temperature satisfies the equation

$$\frac{\partial \delta T(\mathbf{r},t)}{\partial t} = -\tfrac{2}{3} T_0 \nabla \cdot \delta \mathbf{v}(\mathbf{r},t). \tag{15.99}$$

Using this, we see that the solutions of (15.98) are also isothermal, with $\delta T = 0$. It is interesting to note that the zero divergence solutions above $T_{\rm BEC}$ given by (15.98) are formally identical in form to the normal mode solutions of a pure condensate at $T = 0$, as given by the Stringari wave equation (2.28). These surface modes have the dispersion relation $\omega = \sqrt{l}\omega_0$ for $l = 1, 2, \ldots$ for an isotropic trap (see Section 2.1). This similarity between the surfaces modes at $T = 0$ and $T > T_{\rm BEC}$ is to be expected since the equation of state does not enter into the equation of motion (15.98). Problem 11.3 of Pethick and Smith (2008) gives further insight into this similarity.

The "breathing" monopole oscillation of (15.97) corresponds to a velocity spectrum $\delta \mathbf{v}_\omega(\mathbf{r}) \sim \mathbf{r}$, with $\nabla \cdot \delta \mathbf{v}_\omega(\mathbf{r}) =$ constant, for an isotropic trap $V_{\rm trap}(\mathbf{r}) = \frac{1}{2} m\omega_0^2 r^2$. This mode has frequency $\omega = 2\omega_0$, the same as a noninteracting trapped gas but quite different from the $\omega = \sqrt{5}\omega_0$ frequency of the condensate monopole mode at $T = 0$ discussed in Section 2.1. As noted in GWS, this is related to the fact that the monopole oscillation $\omega = 2\omega_0$ in a classical gas trapped in a harmonic potential trap is an exact solution of the full Boltzmann equation.

This unusual undamped mode in a harmonic trap was originally discovered by Boltzmann in 1872 (see Boltzmann, 1909) and has been extensively discussed in the mathematical literature on classical kinetic equations (see for example Uhlenbeck and Ford, 1963). Boltzmann unexpectedly found this undamped oscillation in an attempt to prove that his kinetic equation would always predict that a nonequilibrium (time-dependent) state of a classical gas would relax to thermal equilibrium as a result of dissipation due to collisions. The existence of such an undamped oscillation seemed to be in contradiction with the expected universal dissipative relaxation. Such "pathological" modes are now accessible to experiments using ultracold trapped gases.

Most Bose gas experiments involve a cigar-shaped trap with axial symmetry

$$V_{\rm trap}(\mathbf{r}) = \tfrac{1}{2} m(\omega_\perp^2 \rho^2 + \omega_z^2 z^2), \tag{15.100}$$

where the radial component is $\rho^2 \equiv x^2 + y^2$. As discussed in GWS, one can show that the exact solutions of (15.97) for such a trapping geometry

include

$$\begin{aligned} \delta n_\omega(\mathbf{r}) &\sim s^l\, e^{\pm il\phi} n_0(\mathbf{r}), \quad \omega^2_{l,m=\pm 1} = l\omega_\perp^2 \\ \delta n_\omega(\mathbf{r}) &\sim z s^{l-1}\, e^{\pm i(l-1)\phi} n_0(\mathbf{r}), \quad \omega^2_{l,m=\pm(l-1)} = (l-1)\omega_\perp^2 + \omega_z^2. \end{aligned} \tag{15.101}$$

In addition, coupled monopole–quadrupole surface mode solutions exist, corresponding to $\delta\mathbf{v}_\omega(\mathbf{r}) = \nabla\chi$ where $\chi(\mathbf{r}) = \alpha\rho^2 + \beta z^2$. This ansatz gives $\delta\mathbf{v}_\omega = 2(\alpha x, \alpha y, \beta z)$ and one finds the following coupled equations for the parameters α and β:

$$\begin{aligned} \left(\omega^2 - \tfrac{10}{3}\omega_\perp^2\right)\alpha - \tfrac{2}{3}\omega_\perp^2\beta &= 0 \\ -\tfrac{4}{3}\omega_z^2\alpha + \left(\omega^2 - \tfrac{8}{3}\omega_y^2\right)\beta &= 0. \end{aligned} \tag{15.102}$$

These solutions are characterized by a velocity field for which we have $\nabla\cdot\delta\mathbf{v}_\omega =$ constant. The two solutions of these equations are the coupled monopole-quadrupole oscillation frequencies given in (10.20),

$$\omega^2 = \tfrac{1}{3}\left[5\omega_\perp^2 + 4\omega_z^2 \pm (25\omega_\perp^4 + 16\omega_z^4 - 32\omega_\perp^2\omega_z^2)^{1/2}\right]. \tag{15.103}$$

A useful graph of these two normal mode frequencies is shown in Fig. 11.1 of Pethick and Smith (2008). These surface mode solutions are used in other chapters of the present book. Pethick and Smith also discuss the transverse scissors mode solutions for a general anisotropic harmonic trap, such as $\delta\mathbf{v}_\omega(\mathbf{r}) \sim \nabla(xy)$. Such scissors mode solutions are discussed in detail in Chapters 10 and 12, in both the normal and superfluid phases of a trapped Bose gas.

An analogue of (15.97) is also valid for the collisional hydrodynamic oscillations of a trapped normal Fermi gas. Analytical solutions for this case are discussed in Bruun and Clark (1999).

16

Variational formulation of the Landau two-fluid equations

In the collisional region at finite temperatures, the collective modes of superfluids are described by the Landau two-fluid hydrodynamic equations reviewed in Chapter 14. In the case of trapped Bose gases, these are coupled differential equations with position-dependent coefficients associated with the local thermodynamic functions. Building on the approach initiated by Zaremba *et al.* (1999) for trapped atomic Bose gases, in this chapter we develop an alternative *variational* formulation of two-fluid hydrodynamics. This is based on the work of Zilsel[1] (1950), originally developed to deal with superfluid ^{4}He. Assuming a simple variational ansatz for the superfluid and normal fluid velocity fields, this approach reduces the problem of finding the hydrodynamic collective mode frequencies to solving coupled algebraic equations for a few variational parameters. These equations contain constants that involve spatial integrals over various equilibrium thermodynamic derivatives. Such a variational approach is both simpler and more physical than a direct attempt to solve the Landau two-fluid equations numerically.

This chapter is mainly based on Taylor and Griffin (2005), Taylor (2008) and Zilsel (1950). In it, we discuss the normal modes of the non-dissipative Landau two-fluid equations for a trapped superfluid. In Section 16.3, we illustrate this formalism by deriving expressions for the frequencies of the dipole and breathing modes of a trapped Bose superfluid. In Chapters 17 and 18, we discuss an extended version of the two-fluid equations that includes hydrodynamic damping. The hydrodynamic damping of the collective modes is calculated in Chapter 19 using a generalized version of the variational approach developed in this chapter.

[1] Zilsel was a research associate of Fritz London. Zilsel's work is also summarized in the classic monograph by London (1954, p. 126ff).

16.1 Zilsel's variational formulation

Since two-fluid hydrodynamics only describes a system in local equilibrium, all thermodynamic quantities are functions of position and time. Even in static equilibrium, in the presence of a trapping potential, most thermodynamic quantities will be position dependent. This dependence will be, for the most part, left implicit.

Our variational principle will use a Lagrangian density of the form

$$\mathcal{L} = T - U, \tag{16.1}$$

where T is the kinetic energy density of the fluid and U is the internal energy density. Thus, we need to formulate the thermodynamics in terms of the internal energy density $U(\rho, \rho_{\rm n}, s)$, rather than the total energy $E = T + U$. The following discussion is based on Khalatnikov (1965) and the results will be referred to as the Landau–Khalatnikov (LK) two-fluid thermodynamics. For a recent summary, see also Section 8.6 of Mazenko (2003). For a trapped two-fluid system, the total energy density E is the sum of the kinetic and potential energy densities,

$$E = \tfrac{1}{2}\rho_{\rm s}\mathbf{v}_{\rm s}^2 + \tfrac{1}{2}\rho_{\rm n}\mathbf{v}_{\rm n}^2 + U + \rho U_{\rm trap}, \tag{16.2}$$

where $\rho_{\rm n}$ and $\rho_{\rm s}$ are the normal fluid and superfluid mass densities, $\mathbf{v}_{\rm n}$ and $\mathbf{v}_{\rm s}$ are the normal fluid and superfluid velocities and $\rho = \rho_{\rm s} + \rho_{\rm n}$ is the total mass density. In this section, the harmonic trap potential will be denoted by

$$U_{\rm trap}(\mathbf{r}) \equiv \tfrac{1}{2}(\omega_x^2 x^2 + \omega_y^2 y^2 + \omega_z^2 z^2) = \frac{V_{\rm trap}(\mathbf{r})}{m}, \tag{16.3}$$

because the *mass* density ρ is used in discussions of two-fluid hydrodynamics rather than the number density n. For the same reason, in this chapter the chemical potential $\mu_{\rm L}$ is written as μ/m, i.e. the chemical potential per unit mass.

Following LK, the total energy density $E_0(\mathbf{r}, t)$ as measured in a frame of reference moving with the *local* superfluid velocity is related to E in the lab frame of reference by (see for example Mazenko, 2003)

$$E_0 = E - \tfrac{1}{2}\rho\mathbf{v}_{\rm s}^2 - \mathbf{v}_{\rm s}\cdot\rho_{\rm n}(\mathbf{v}_{\rm n} - \mathbf{v}_{\rm s}). \tag{16.4}$$

The LK thermodynamic identity for local equilibrium is given by

$$dE_0 = Tds + \mu_{\rm L}d\rho + (\mathbf{v}_{\rm n} - \mathbf{v}_{\rm s})\cdot d\left[\rho_{\rm n}\left(\mathbf{v}_{\rm n} - \mathbf{v}_{\rm s}\right)\right], \tag{16.5}$$

where $\mu_{\rm L}$ is the Landau chemical potential defined above. Using (16.2) and

(16.4) in (16.5), we obtain the following thermodynamic identity for the internal energy density

$$dU = (\mu_{\rm L} - U_{\rm trap})d\rho + Tds + \tfrac{1}{2}(\mathbf{v}_{\rm n} - \mathbf{v}_{\rm s})^2 d\rho_{\rm n}. \tag{16.6}$$

In similar fashion, the LK expression for the pressure $P = \partial(E_0 V)/\partial V$ that includes the effects of a trapping potential can be rewritten in terms of the internal energy density to give

$$P = -U - \rho U_{\rm trap} + Ts + \mu_{\rm L}\rho + \tfrac{1}{2}\rho_{\rm n}(\mathbf{v}_{\rm n} - \mathbf{v}_{\rm s})^2. \tag{16.7}$$

Equations (16.6) and (16.7) can be combined to give

$$\rho d\mu_{\rm L} = dP - sdT - \rho_{\rm n}(\mathbf{v}_{\rm n} - \mathbf{v}_{\rm s}) \cdot d(\mathbf{v}_{\rm n} - \mathbf{v}_{\rm s}). \tag{16.8}$$

The three thermodynamic identities given by (16.6), (16.7) and (16.8) define all the thermodynamic properties that we will require in developing our variational approach. We note that (16.6) implies that the chemical potential per unit mass $\mu_{\rm L}$ is given by

$$\mu_{\rm L} = \left(\frac{\partial U}{\partial \rho}\right)_{s,\rho_{\rm n}} + U_{\rm trap}. \tag{16.9}$$

One can obtain the two-fluid equations of motion by taking the variation of the action defined by

$$A = \int d\mathbf{r} \int dt \left[\tfrac{1}{2}(\rho - \rho_{\rm n})\mathbf{v}_{\rm s}^2 + \tfrac{1}{2}\rho_{\rm n}\mathbf{v}_{\rm n}^2 - U(\rho, \rho_{\rm n}, s) - \rho U_{\rm trap}\right]. \tag{16.10}$$

Zilsel (1950) first used (16.10) to derive Landau's non-dissipative two-fluid equations in a uniform superfluid. In taking the variation of the action in (16.10) the five variables $\mathbf{v}_{\rm n}, \mathbf{v}_{\rm s}, \rho, \rho_{\rm n}$ and s are treated as independent, so that, for instance,

$$\frac{\delta U}{\delta \rho} \equiv \left(\frac{\partial U}{\partial \rho}\right)_{s,\rho_{\rm n}}. \tag{16.11}$$

Two important conservation laws in Landau's hydrodynamics not incorporated into the action defined in (16.10) are the conservation of mass and entropy,

$$\frac{\partial \rho}{\partial t} + \nabla \cdot [(\rho - \rho_{\rm n})\mathbf{v}_{\rm s} + \rho_{\rm n}\mathbf{v}_{\rm n}] = 0, \tag{16.12}$$

$$\frac{\partial s}{\partial t} + \nabla \cdot (s\mathbf{v}_{\rm n}) = 0. \tag{16.13}$$

The action integral in (16.10) is varied subject to the constraints (16.12) and (16.13).

An additional constraint, not employed by Zilsel, describes the conservation of circulation in the normal fluid (see Landau and Lifshitz, 1959, p. 14); this allows for the possibility of vorticity. This constraint takes the form (see Geurst, 1980, for discussion and references)

$$\frac{\partial s\eta}{\partial t} + \nabla \cdot (s\eta \mathbf{v}_{\mathrm{n}}) = 0, \tag{16.14}$$

where the new function η depends on $\mathbf{r}$ and t. Even though the inclusion of this constraint has no effect on the final form of the two-fluid equations, it eliminates the restrictions on the normal fluid velocity field $\mathbf{v}_{\mathrm{n}}$ that are present in Zilsel's original formulation. Since the conservation-of-circulation constraint does *not* affect the form of the equations resulting from the variational principle, for simplicity we will omit this term in our analysis. However, if one were interested in collective oscillations of the normal fluid with nonzero circulation, the conservation law (16.14) would have to be included in the action. This extension is discussed in Section III and Appendix A of Taylor and Griffin (2005).

Following the approach pioneered by Eckart (1938), the mass and entropy conservation laws (16.12) and (16.13) can be incorporated into the variational principle by introducing Lagrange multipliers ϕ and α (both dependent on $\mathbf{r}$ and t). In place of (16.10), the action is now given by

$$\begin{aligned} A = \int d\mathbf{r} \int dt \Big\{ & \tfrac{1}{2}(\rho - \rho_{\mathrm{n}})\mathbf{v}_{\mathrm{s}}^2 + \tfrac{1}{2}\rho_{\mathrm{n}}\mathbf{v}_{\mathrm{n}}^2 - U(\rho, \rho_{\mathrm{n}}, s) - \rho U_{\mathrm{trap}} \\ & + \phi \left[\frac{\partial \rho}{\partial t} + \nabla \cdot [(\rho - \rho_{\mathrm{n}})\mathbf{v}_{\mathrm{s}} + \rho_{\mathrm{n}}\mathbf{v}_{\mathrm{n}}] \right] + \alpha \left[\frac{\partial s}{\partial t} + \nabla \cdot (s\mathbf{v}_{\mathrm{n}}) \right] \Big\}. \end{aligned} \tag{16.15}$$

It is understood that the time and spatial integrations in (16.15) are performed between two fixed points, at which the fluctuations of the variables vanish.

We take the variation of the action in (16.15) with respect to $\rho, s, \mathbf{v}_{\mathrm{s}}$ and $\mathbf{v}_{\mathrm{n}}$ and set these to zero. Making use of the thermodynamic identities implied by (16.6) and using integration by parts to deal with the last two terms in (16.15), we obtain

$$\frac{\delta A}{\delta \rho} = \tfrac{1}{2}\mathbf{v}_{\mathrm{s}}^2 - \frac{\partial \phi}{\partial t} - \mathbf{v}_{\mathrm{s}} \cdot \nabla \phi - \mu_{\mathrm{L}} = 0, \tag{16.16}$$

$$\frac{\delta A}{\delta s} = -T - \frac{\partial \alpha}{\partial t} - \mathbf{v}_{\mathrm{n}} \cdot \nabla \alpha = 0, \tag{16.17}$$

$$\frac{\delta A}{\delta \mathbf{v}_{\mathrm{s}}} = (\rho - \rho_{\mathrm{n}})(\mathbf{v}_{\mathrm{s}} - \nabla \phi) = 0, \tag{16.18}$$

$$\frac{\delta A}{\delta \mathbf{v}_{\rm n}} = \rho_{\rm n}\left(\mathbf{v}_{\rm n} - \nabla\phi\right) - s\nabla\alpha = 0. \tag{16.19}$$

Taking the variation of the action with respect to $\rho_{\rm n}$ and using (16.18), one recovers a thermodynamic identity already known from (16.6):

$$\left(\frac{\partial U}{\partial \rho_{\rm n}}\right)_{s,\rho} = \tfrac{1}{2}(\mathbf{v}_{\rm n} - \mathbf{v}_{\rm s})^2. \tag{16.20}$$

Taking the variation of the action A with respect to ϕ and α, we recover the two conservation laws of two-fluid hydrodynamics given by (16.12) and (16.13).

Equations (16.16)–(16.19) can be rearranged to yield useful expressions. From (16.18), we obtain

$$\mathbf{v}_{\rm s} = \nabla\phi, \tag{16.21}$$

which ensures that the superfluid velocity is *irrotational*. Using this, (16.19) can be written as

$$\nabla\alpha = \frac{\rho_{\rm n}}{s}\left(\mathbf{v}_{\rm n} - \mathbf{v}_{\rm s}\right). \tag{16.22}$$

Combining (16.22) with (16.17) gives

$$\frac{\partial\alpha}{\partial t} = -T - \frac{\rho_{\rm n}}{s}\mathbf{v}_{\rm n}\cdot\left(\mathbf{v}_{\rm n} - \mathbf{v}_{\rm s}\right). \tag{16.23}$$

Finally, using (16.22) in (16.16), we obtain

$$\frac{\partial\phi}{\partial t} = -\left(\mu_{\rm L} + \tfrac{1}{2}\mathbf{v}_{\rm s}^2\right). \tag{16.24}$$

These results can now be used to derive the Landau two-fluid equation for $\mathbf{v}_{\rm s}$.

Taking the time derivative of (16.21) and the gradient of (16.24), the superfluid velocity satisfies the equation of motion

$$\frac{\partial\mathbf{v}_{\rm s}}{\partial t} = -\nabla\left(\mu_{\rm L} + \tfrac{1}{2}\mathbf{v}_{\rm s}^2\right). \tag{16.25}$$

We can write this equation for $\mathbf{v}_{\rm s}$ in another way. Making use of the thermodynamic derivatives in (16.9) and (16.20), one obtains

$$\begin{aligned}\nabla U &= \left(\frac{\partial U}{\partial\rho}\right)_{\rho_{\rm n},s}\nabla\rho + \left(\frac{\partial U}{\partial\rho_{\rm n}}\right)_{\rho,s}\nabla\rho_{\rm n} + \left(\frac{\partial U}{\partial s}\right)_{\rho,\rho_{\rm n}}\nabla s \\ &= (\mu_{\rm L} - U_{\rm trap})\nabla\rho + \tfrac{1}{2}\left(\mathbf{v}_{\rm n} - \mathbf{v}_{\rm s}\right)^2\nabla\rho_{\rm n} + T\nabla s.\end{aligned} \tag{16.26}$$

Taking the gradient of (16.7) and using (16.26), one obtains

$$-\nabla\mu_{\rm L} = -\frac{1}{\rho}\nabla P - \nabla U_{\rm trap} + \frac{s}{\rho}\nabla T + \frac{1}{2}\frac{\rho_{\rm n}}{\rho}\nabla\left(\mathbf{v}_{\rm n}-\mathbf{v}_{\rm s}\right)^2. \quad (16.27)$$

Inserting (16.27) into (16.25), the equation of motion for $\mathbf{v}_{\rm s}$ can be rewritten as

$$\left(\frac{\partial}{\partial t} + \mathbf{v}_{\rm s}\cdot\nabla\right)\mathbf{v}_{\rm s} = \frac{s}{\rho}\nabla T - \frac{1}{\rho}\nabla P - \nabla U_{\rm trap} + \frac{1}{2}\frac{\rho_{\rm n}}{\rho}\nabla\left(\mathbf{v}_{\rm n}-\mathbf{v}_{\rm s}\right)^2. \quad (16.28)$$

We have used the vector identity $\mathbf{v}\cdot\nabla\mathbf{v} = \frac{1}{2}\nabla(\mathbf{v}^2) - \mathbf{v}\times(\nabla\times\mathbf{v})$ and the fact that the superfluid velocity $\mathbf{v}_{\rm s}$ is irrotational ($\nabla\times\mathbf{v}_{\rm s} = 0$).

To determine an equation analogous to (16.28) for the velocity of the normal fluid, we take the time-derivative of (16.22) and the gradient of (16.23). Then, using (16.12), (16.13) and (16.22), one can show (after a lengthy calculation) that the normal fluid velocity satisfies

$$\left(\frac{\partial}{\partial t} + \mathbf{v}_{\rm n}\cdot\nabla\right)\mathbf{v}_{\rm n} = -\frac{\rho_{\rm s}}{\rho_{\rm n}}\frac{s}{\rho}\nabla T - \frac{1}{\rho}\nabla P - \nabla U_{\rm trap} - \frac{1}{2}\frac{\rho_{\rm s}}{\rho}\nabla\left(\mathbf{v}_{\rm n}-\mathbf{v}_{\rm s}\right)^2 - \frac{\Gamma}{\rho_{\rm n}}\left(\mathbf{v}_{\rm n}-\mathbf{v}_{\rm s}\right), \quad (16.29)$$

where the "source function" Γ in the last term is defined by

$$\Gamma \equiv \frac{\partial\rho_{\rm n}}{\partial t} + \nabla\cdot(\rho_{\rm n}\mathbf{v}_{\rm n}). \quad (16.30)$$

We note that Γ does not appear in the equation of motion for $\mathbf{v}_{\rm s}$ given by (16.28). However, from the continuity equation (16.12 and (16.30), the superfluid density satisfies

$$\frac{\partial\rho_{\rm s}}{\partial t} + \nabla\cdot(\rho_{\rm s}\mathbf{v}_{\rm s}) = -\Gamma. \quad (16.31)$$

We can combine equations (16.28)–(16.31) to obtain an equation of motion for the total current, $m\mathbf{j} = \rho_{\rm s}\mathbf{v}_{\rm s} + \rho_{\rm n}\mathbf{v}_{\rm n}$. The source term Γ cancels out and we are left with

$$m\frac{\partial\delta\mathbf{j}}{\partial t} = -\nabla P - \rho\nabla U_{\rm trap} - \rho_{\rm s}\mathbf{v}_{\rm s}\cdot\nabla\mathbf{v}_{\rm s} - \mathbf{v}_{\rm s}\nabla\cdot(\rho_{\rm s}\mathbf{v}_{\rm s}) - \rho_{\rm n}\mathbf{v}_{\rm n}\cdot\nabla\mathbf{v}_{\rm n} - \mathbf{v}_{\rm n}\nabla\cdot(\rho_{\rm n}\mathbf{v}_{\rm n}). \quad (16.32)$$

This equation is usually written in component form (Khalatnikov, 1965) as

$$m\frac{\partial j_i}{\partial t} = -\frac{\partial}{\partial x_j}\left(P\delta_{ij} + \rho_{\rm s}v_{{\rm s}i}v_{{\rm s}j} + \rho_{\rm n}v_{{\rm n}i}v_{{\rm n}j}\right) - \rho\frac{\partial U_{\rm trap}}{\partial x_i}, \quad (16.33)$$

where summation over the repeated index j on the r.h.s. is implied.

To summarize the discussion so far, the variational minimum of the action defined in (16.15) has been shown to give the equations (16.12), (16.13), (16.25) and (16.33). These are precisely the Landau two-fluid equations in the non-dissipative limit, generalized to include a static external potential. This variational formulation of the two-fluid equations will be used in Section 16.2.

Zilsel (1950) first derived (16.29) with the inclusion of Γ and argued that it had been omitted in the original two-fluid equations of Landau (1941). The reader may refer to the discussion of these results by London (1954, p. 126). Zilsel's conclusion did not take into account that Landau had discussed only equations of motion for the superfluid velocity and the *total* current density. If one works backwards from these two equations of Landau to derive a separate equation for the normal fluid velocity, one finds (16.29) with the source term Γ. Thus, the Landau two-fluid equations implicitly include the possibility of a finite value of Γ. In the later literature, when an equation such as (16.29) was used in the context of superfluid ^{4}He, Γ was usually set to zero. In fact, the appearance of Γ in the continuity equations (16.29) and (16.30) is consistent with the ZNG equations for a Bose-condensed gas (where $\rho_{\rm s} = mn_{\rm c}$ and $\rho_{\rm n} = m\tilde{n}$). In Section 17.2, we will show that in the Landau limit the source term in (15.1) is given by

$$\Gamma_{12}(\mathbf{r},t) = \sigma_{\rm H}\Big\{\nabla\cdot[n_{\rm c}(\mathbf{v}_{\rm c}-\mathbf{v}_{\rm n})] + \tfrac{1}{3}n_{\rm c}\nabla\cdot\mathbf{v}_{\rm n}\Big\}\,. \tag{16.34}$$

As discussed in Section 15.3, including such a source term Γ_{12} in the continuity equations (16.30) and (16.31) is vital in establishing the precise equivalence of the ZNG equations (in the limit $\omega\tau_\mu \to 0$) and the standard Landau two-fluid equations. Thus, Γ_{12} as given in (16.34) is an explicit microscopic expression for the source term Γ ($= m\Gamma_{12}$) first noted by Zilsel (1950) in his discussion of the two-fluid equations.

One can show explicitly that the above results do reproduce the Landau equations given in Section 14.1. Assuming that $\mathbf{v}_{\rm s0} = \mathbf{v}_{\rm n0} = 0$, the linearized version of equation (16.28) for the superfluid velocity reduces to

$$\begin{aligned}\frac{\partial\delta\mathbf{v}_{\rm s}}{\partial t} &= -\frac{1}{\rho_0}\nabla\delta P + \frac{s_0}{\rho_0}\nabla\delta T + \frac{\nabla P_0}{\rho_0^2}\delta\rho \\ &= -\nabla\delta\mu_{\rm L}.\end{aligned} \tag{16.35}$$

In static equilibrium in a harmonic trap, one has the relation (we recall that

$\nabla T_0 = 0$ since in equilibrium the temperature is constant)

$$\nabla \mu_{\rm L0} \equiv 0 = \frac{\nabla P_0}{\rho_0} + \nabla U_{\rm trap}. \tag{16.36}$$

Equation (16.35) reproduces in a linearized version the Landau equation (14.5). Linearizing the equation (16.29) for the normal fluid velocity $\mathbf{v}_{\rm n}$ and using (16.36), we find that it reduces to

$$\frac{\partial \delta \mathbf{v}_{\rm n}}{\partial t} = -\frac{1}{\rho_0}\nabla \delta P - \frac{\rho_{\rm s0}}{\rho_{\rm n0}}\frac{s_0}{\rho_0}\nabla \delta T + \frac{\nabla P_0}{\rho_0^2}\delta\rho. \tag{16.37}$$

Here we have made use of the fact that Γ in (16.29) vanishes in equilibrium (see Sections 3.3 and 8.1). We note that (16.35) and (16.37) are the same as (14.9) and (14.11) apart from the term $(\nabla P_0/\rho_0^2)\delta\rho = -(\nabla U_{\rm trap}/\rho_0)\delta\rho$ which describes the effect of the trapping potential; this was not included in Chapter 14.

The early work of Zilsel and others on variational forms of Landau's two-fluid hydrodynamics in superfluid ^{4}He was motivated by the desire to put such equations on a more secure basis. Today, there is no question that Landau's equations are the correct description of the collisional hydrodynamics of a Bose superfluid. We are interested in a variational formulation for a quite different reason, namely as a practical way of determining the normal mode solutions in a trapped nonuniform superfluid Bose gas.

16.2 The action integral for two-fluid hydrodynamics

In Section 16.1, we showed that the variation of the action integral (16.15) with respect to ρ, S, $\mathbf{v}_{\rm s}$, $\mathbf{v}_{\rm n}$, ϕ and α leads to the non-dissipative Landau two-fluid hydrodynamic equations (Zilsel, 1950). To determine the low-energy collective modes given by the solutions of the linearized hydrodynamic equations, we could expand the action about the equilibrium values of these variables. In discussing the collective modes, however, it is convenient to introduce displacement fields for the two velocity fields $\mathbf{v}_{\rm s}$ and $\mathbf{v}_{\rm n}$. This allows one to incorporate the two-fluid conservation laws (16.12) and (16.13) directly into expressions for $\delta\rho$ and δs, thereby eliminating the need for the Lagrange multipliers used in (16.15). In terms of the two displacement fields

$$\mathbf{v}_{\rm s}(\mathbf{r},t) \equiv \frac{\partial \mathbf{u}_{\rm s}(\mathbf{r},t)}{\partial t}, \quad \mathbf{v}_{\rm n}(\mathbf{r},t) \equiv \frac{\partial \mathbf{u}_{\rm n}(\mathbf{r},t)}{\partial t}, \tag{16.38}$$

the linearized continuity and entropy conservation equations (16.12) and (16.13) can be expressed as

$$\delta\rho(\mathbf{r},t) = -\nabla\cdot\left[\rho_{\rm s0}(\mathbf{r})\mathbf{u}_{\rm s}(\mathbf{r},t) + \rho_{\rm n0}(\mathbf{r})\mathbf{u}_{\rm n}(\mathbf{r},t)\right] \tag{16.39}$$

and

$$\delta s(\mathbf{r},t) = -\nabla\cdot\left[s_0(\mathbf{r})\mathbf{u}_{\rm n}(\mathbf{r},t)\right]. \tag{16.40}$$

We will use this simpler approach in the following discussion.

Assuming that $\mathbf{v}_{\rm n0} = \mathbf{v}_{\rm s0} = 0$ in equilibrium, the action integral in (16.10) to second order in the fluctuations δs and $\delta\rho$ is given by

$$\begin{aligned} A^{(2)} = \int d\mathbf{r}\int dt\Bigg[&\frac{1}{2}\rho_{\rm s0}\mathbf{v}_{\rm s}^2 + \frac{1}{2}\rho_{\rm n0}\mathbf{v}_{\rm n}^2 \;-\; \frac{1}{2}\left(\frac{\partial^2 U}{\partial\rho^2}\right)_{s,\rho_{\rm n}}(\delta\rho)^2 \\ &- \left(\frac{\partial^2 U}{\partial s\partial\rho}\right)_{\rho_{\rm n}}\delta s\delta\rho \;-\; \frac{1}{2}\left(\frac{\partial^2 U}{\partial s^2}\right)_{\rho,\rho_{\rm n}}(\delta s)^2\Bigg], \end{aligned} \tag{16.41}$$

subject to the constraints in (16.39) and (16.40) that relate $\delta\rho$ and δs to $\mathbf{u}_{\rm n}$ and $\mathbf{u}_{\rm s}$. Using the thermodynamic identity (16.6), the coefficients involving second derivatives in (16.41) can be simplified as follows:

$$\begin{aligned} \left(\frac{\partial^2 U}{\partial\rho^2}\right)_{s,\rho_{\rm n}} &= \left(\frac{\partial\mu_{\rm L}}{\partial\rho}\right)_{s,\rho_{\rm n}}, \\ \left(\frac{\partial^2 U}{\partial s\partial\rho}\right)_{\rho_{\rm n}} &= \left(\frac{\partial\mu_{\rm L}}{\partial s}\right)_{\rho,\rho_{\rm n}} = \left(\frac{\partial T}{\partial\rho}\right)_{s,\rho_{\rm n}}, \\ \left(\frac{\partial^2 U}{\partial s^2}\right)_{\rho,\rho_{\rm n}} &= \left(\frac{\partial T}{\partial s}\right)_{\rho,\rho_{\rm n}}. \end{aligned} \tag{16.42}$$

In the second line, we have used a well-known Maxwell relation. There is no contribution in (16.41) from fluctuations in $\rho_{\rm n}$ since $(\partial U/\partial\rho_{\rm n})_{s,\rho} = 0$, which follows from (16.20) and the fact that we are assuming $\mathbf{v}_{\rm s0} = \mathbf{v}_{\rm n0} = 0$. Using (16.42) and the constraints (16.39) and (16.40), the action integral to second order in the displacement fields $\mathbf{u}_{\rm n}$ and $\mathbf{u}_{\rm s}$ is given by

$$\begin{aligned} A^{(2)} = \int d\mathbf{r}\int dt\Bigg\{&\frac{1}{2}\rho_{\rm s0}\dot{\mathbf{u}}_{\rm s}^2 + \frac{1}{2}\rho_{\rm n0}\dot{\mathbf{u}}_{\rm n}^2 - \frac{1}{2}\left(\frac{\partial\mu_{\rm L}}{\partial\rho}\right)_{s,\rho_{\rm n}}\left[\nabla\cdot(\rho_{\rm s0}\mathbf{u}_{\rm s} + \rho_{\rm n0}\mathbf{u}_{\rm n})\right]^2 \\ &- \left(\frac{\partial T}{\partial\rho}\right)_{s,\rho_{\rm n}}\nabla\cdot(s_0\mathbf{u}_{\rm n})\left[\nabla\cdot(\rho_{\rm s0}\mathbf{u}_{\rm s} + \rho_{\rm n0}\mathbf{u}_{\rm n})\right] \\ &- \frac{1}{2}\left(\frac{\partial T}{\partial s}\right)_{\rho,\rho_{\rm n}}\left[\nabla\cdot(s_0\mathbf{u}_{\rm n})\right]^2\Bigg\}. \end{aligned} \tag{16.43}$$

There is no term in the action that is linear in the fluctuations, since they correspond to a variational minimum of A. It should be remembered that in evaluating the various local thermodynamic derivatives in (16.43), $\rho_{\rm n}$ is *always* fixed even though we leave this implicit.

The variational principle has now been reduced to taking the variation of the quadratic action given by (16.43) with respect to $\mathbf{u}_{\rm n}$ and $\mathbf{u}_{\rm s}$. The linearized hydrodynamic equations (and hence the low-energy collective modes of the system) are thus completely determined by the variational equations

$$\frac{\delta A^{(2)}}{\delta \mathbf{u}_{\rm s}(\mathbf{r},t)} = 0, \quad \frac{\delta A^{(2)}}{\delta \mathbf{u}_{\rm n}(\mathbf{r},t)} = 0. \tag{16.44}$$

Solutions of (16.44) corresponding to collective modes of frequency ω are given by

$$\mathbf{u}_{\rm s}(\mathbf{r},t) = \mathbf{u}_{\rm s}(\mathbf{r}) \cos\,\omega t, \quad \mathbf{u}_{\rm n}(\mathbf{r},t) = \mathbf{u}_{\rm n}(\mathbf{r}) \cos\,\omega t. \tag{16.45}$$

Substituting (16.45) into (16.43) and performing the time-integration, we obtain the Lagrangian (apart from an irrelevant constant factor)

$$L^{(2)} \equiv K[\mathbf{u}_{\rm s}, \mathbf{u}_{\rm n}]\omega^2 - U[\mathbf{u}_{\rm s}, \mathbf{u}_{\rm n}], \tag{16.46}$$

where

$$K[\mathbf{u}_{\rm s}, \mathbf{u}_{\rm n}] \equiv \frac{1}{2}\int d\mathbf{r}\left\{\rho_{\rm s0}(\mathbf{r})\mathbf{u}_{\rm s}^2 + \rho_{\rm n0}\mathbf{u}_{\rm n}^2\right\} \tag{16.47}$$

and

$$\begin{aligned} U[\mathbf{u}_{\rm s}, \mathbf{u}_{\rm n}] \equiv \frac{1}{2}\int d\mathbf{r}\ \Bigg\{ & \left(\frac{\partial \mu_{\rm L}}{\partial \rho}\right)_s \left[\nabla\cdot\left(\rho_{\rm s0}\mathbf{u}_{\rm s} + \rho_{\rm n0}\mathbf{u}_{\rm n}\right)\right]^2 \\ & + 2\left(\frac{\partial T}{\partial \rho}\right)_s \nabla\cdot(s_0\mathbf{u}_{\rm n})\left[\nabla\cdot\left(\rho_{\rm s0}\mathbf{u}_{\rm s} + \rho_{\rm n0}\mathbf{u}_{\rm n}\right)\right] \\ & + \left(\frac{\partial T}{\partial s}\right)_\rho \left[\nabla\cdot(s_0\mathbf{u}_{\rm n})\right]^2\Bigg\}. \end{aligned} \tag{16.48}$$

With (16.45), the Lagrangian variational equations (16.44) reduce to

$$\frac{\delta L^{(2)}}{\delta \mathbf{u}_{\rm s}(\mathbf{r})} = 0, \quad \frac{\delta L^{(2)}}{\delta \mathbf{u}_{\rm n}(\mathbf{r})} = 0. \tag{16.49}$$

Our motivation for developing this variational formulation is that we can use it to obtain approximate solutions for the collective mode oscillations.

We use a simplified Rayleigh–Ritz method and make an ansatz for the displacement fields of the form

$$\begin{aligned} \mathbf{u}_s(\mathbf{r}) &= (A_{\mathrm{s}1} f_1(\mathbf{r}),\ A_{\mathrm{s}2} f_2(\mathbf{r}),\ A_{\mathrm{s}3} f_3(\mathbf{r})), \\ \mathbf{u}_n(\mathbf{r}) &= (A_{\mathrm{n}1} g_1(\mathbf{r}),\ A_{\mathrm{n}2} g_2(\mathbf{r}),\ A_{\mathrm{n}3} g_3(\mathbf{r})), \end{aligned} \tag{16.50}$$

where the six coefficients $A_{\mathrm{s}i}$ and $A_{\mathrm{n}i}$ are variational parameters. The functions $f_i(\mathbf{r})$ and $g_i(\mathbf{r})$ are specific forms inspired by known exact solutions for $\mathbf{v}_{\mathrm{s}}$ at $T = 0$ (given in Section 2.1) and for $\mathbf{v}_{\mathrm{n}}$ at $T > T_{\mathrm{BEC}}$ (given in Section 15.4). Substituting this ansatz into (16.46), we obtain the six variational equations

$$\frac{\delta L^{(2)}}{\delta A_{\mathrm{s}i}} = 0, \quad \frac{\delta L^{(2)}}{\delta A_{\mathrm{n}i}} = 0. \tag{16.51}$$

In practice, symmetry usually allows one to reduce the number of equations. Such variational equations can be proved to give a rigorous *upper bound* for the collective mode frequencies.

Using the variational ansatz in (16.50), the Lagrangian given by (16.46)–(16.48) describes the dynamics in a form analogous to that of a pair of coupled harmonic oscillators, $A_{\mathrm{s}i}$ and $A_{\mathrm{n}i}$ representing the displacements of the two oscillators from equilibrium. The effective spring constants are determined by the equilibrium thermodynamic quantities of the system. This is a useful picture to keep in mind when envisioning the low-energy dynamics of the superfluid and normal components. It immediately implies, for instance, the existence of *in-phase* as well as *out-of-phase* oscillation modes of the two fluids.

16.3 Hydrodynamic modes in a trapped gas

To illustrate the variational formalism developed in the preceding section for two-fluid hydrodynamics in a trapped Bose gas, we will first consider the hydrodynamic modes in a *uniform* gas. In this case, $\rho_{\mathrm{s}0}, \rho_{\mathrm{n}0}$ and all thermodynamic derivatives appearing in (16.47) and (16.48) are independent of position. We take the displacement amplitudes to be along the x-axis:

$$\begin{aligned} \mathbf{u}_{\mathrm{s}}(\mathbf{r}) &= \mathcal{N} A_{\mathrm{s}} \cos qx\, \hat{\mathbf{i}}, \\ \mathbf{u}_{\mathrm{n}}(\mathbf{r}) &= \mathcal{N} A_{\mathrm{n}} \cos qx\, \hat{\mathbf{i}}, \end{aligned} \tag{16.52}$$

where A_{s} and A_{n} are the variational parameters defined in (16.50). The normalization constant $\mathcal{N}$ is chosen so that $\int d\mathbf{r}\ \mathbf{u}_{\mathrm{s}}^2 = A_{\mathrm{s}}^2$; $\mathcal{N}$ is independent

of $A_{\rm s}$ and $A_{\rm n}$. Inserting (16.52) into (16.47) and (16.48), these reduce to

$$K[A_{\rm s}, A_{\rm n}] = \frac{1}{2}\rho_{\rm s0}A_{\rm s}^2 + \frac{1}{2}\rho_{\rm n0}A_{\rm n}^2 \tag{16.53}$$

and

$$U[A_{\rm s}, A_{\rm n}] = q^2\left\{\frac{A_{\rm s}^2}{2}\left[\rho_{\rm s0}^2\left(\frac{\partial\mu_{\rm L}}{\partial\rho}\right)_s\right] + A_{\rm s}A_{\rm n}\left[\rho_{\rm s0}\rho_{\rm n0}\left(\frac{\partial\mu_{\rm L}}{\partial\rho}\right)_s + s_0\rho_{\rm s0}\left(\frac{\partial T}{\partial\rho}\right)_s\right]\right.$$
$$\left. + \frac{A_{\rm n}^2}{2}\left[\rho_{\rm n0}^2\left(\frac{\partial\mu_{\rm L}}{\partial\rho}\right)_s + 2s_0\rho_{\rm n0}\left(\frac{\partial T}{\partial\rho}\right)_s + s_0^2\left(\frac{\partial T}{\partial s}\right)_\rho\right]\right\}. \tag{16.54}$$

Using these expressions for K and U in (16.46), we can use the variational equations (16.51) to obtain a quadratic equation for $u^2 \equiv (\omega/q)^2$:

$$u^4 - u^2\left[\rho_0\left(\frac{\partial\mu_{\rm L}}{\partial\rho}\right)_s + 2s_0\left(\frac{\partial T}{\partial\rho}\right)_s + \frac{s_0^2}{\rho_{\rm n0}}\left(\frac{\partial T}{\partial s}\right)_\rho\right]$$
$$+\frac{\rho_{\rm s0}}{\rho_{\rm n0}}s_0^2\left[\left(\frac{\partial T}{\partial s}\right)_\rho\left(\frac{\partial\mu_{\rm L}}{\partial\rho}\right)_s - \left(\frac{\partial T}{\partial\rho}\right)_s^2\right] = 0. \tag{16.55}$$

In the two-fluid literature, one usually works with the entropy density $\bar{s} \equiv s/\rho$ rather than the entropy s per unit volume (Khalatnikov, 1965). Using thermodynamic identities, one can show that (16.55) is equivalent to (14.18). The two solutions give the sound velocities u_1 and u_2 corresponding to first and second sound, respectively. (See Section 14.2 for further discussion).

As a first application of our variational approach for a trapped gas, we consider the case $T = 0$. In this case only the superfluid component is present, and we then have zero entropy ($s_0 = 0$) and $\rho_{\rm n0} = 0$ (hence $\rho_{\rm s0} = \rho_0$). The Lagrangian given by (16.46)–(16.48) reduces to

$$L^{(2)}[\mathbf{u}_{\rm s}] = \frac{1}{2}\int d\mathbf{r}\,\left\{\rho_0\mathbf{u}_{\rm s}^2\omega^2 - \left(\frac{\partial\mu_{\rm L}}{\partial\rho}\right)\left[\nabla\cdot(\rho_0\mathbf{u}_{\rm s})\right]^2\right\}. \tag{16.56}$$

From $\delta L^{(2)}/\delta\mathbf{u}_{\rm s} = 0$ one obtains an equation of motion for the superfluid displacement field $\mathbf{u}_{\rm s}$:

$$\omega^2\mathbf{u}_{\rm s} = -\nabla\left[\left(\frac{\partial\mu_{\rm L}}{\partial\rho}\right)\nabla\cdot(\rho_0\mathbf{u}_{\rm s})\right]. \tag{16.57}$$

Using the $T = 0$ linearized continuity equation $\delta\rho = -\nabla\cdot(\rho_0\mathbf{u}_{\rm s})$ (see (16.39)), we obtain

$$\omega^2\mathbf{u}_{\rm s} = \nabla\left[\left(\frac{\partial\mu_{\rm L}}{\partial\rho}\right)\delta\rho\right]. \tag{16.58}$$

Multiplying both sides of this expression by ρ_0 and taking the divergence, we obtain a closed equation for the density fluctuations $\delta\rho$:

$$\omega^2\delta\rho = -\nabla\cdot\left\{\rho_0\nabla\left[\left(\frac{\partial\mu_{\rm L}}{\partial\rho}\right)\delta\rho\right]\right\}. \tag{16.59}$$

This equation is the basis of the $T = 0$ quantum hydrodynamic theory originally developed by Pitaevskii and Stringari (1998). Since it involves the correct chemical potential, this equation can include corrections to the formally similar GP equation (2.28).

The wave equation (16.59) describes the low-energy collective modes of *both* atomic Bose and two-component Fermi superfluid gases at $T = 0$. The only difference between the two quantum gases lies in the expression for the chemical potential $\mu_{\rm L}(\rho)$.

16.3.1 Dipole modes

Next we turn to a discussion of two-fluid hydrodynamic modes in a *trapped* superfluid gas, using our variational formalism. The simplest example is the dipole mode. This mode is characterized by displacements of the centre of mass of the condensate and the thermal cloud in a harmonic trap. We thus introduce the following ansatz for the displacement fields along the x-axis (the same results are found if we consider the y- and z-displacements):

$$\mathbf{u}_{\rm s} = A_{\rm s}\,\hat{\mathbf{i}}, \quad \mathbf{u}_{\rm n} = A_{\rm n}\,\hat{\mathbf{i}}. \tag{16.60}$$

Here $A_{\rm s}$ and $A_{\rm n}$ are the displacements of the two components from the trap centre and $\hat{\mathbf{x}}$ is a unit vector along the x direction. Substituting this variational ansatz into (16.47) and (16.48), we find

$$K[A_{\rm s}, A_{\rm n}] = \tfrac{1}{2}M_{\rm s}A_{\rm s}^2 + \tfrac{1}{2}M_{\rm n}A_{\rm n}^2 \tag{16.61}$$

and

$$U[A_{\rm s}, A_{\rm n}] = \tfrac{1}{2}k_{\rm s}A_{\rm s}^2 + \tfrac{1}{2}k_{\rm n}A_{\rm n}^2 + \tfrac{1}{2}k_{\rm sn}\left(A_{\rm s} - A_{\rm n}\right)^2, \tag{16.62}$$

where $M_{\rm s}$ and $M_{\rm n}$ are the *total* masses of the superfluid and normal components:

$$M_{\rm s} \equiv \int d\mathbf{r}\ \rho_{\rm s0}(\mathbf{r}), \quad M_{\rm n} \equiv \int d\mathbf{r}\rho_{\rm n0}(\mathbf{r}). \tag{16.63}$$

The "spring constants" $k_{\rm s}, k_{\rm n}$ and $k_{\rm sn}$ appearing in (16.62) are defined by

$$k_{\rm s} \equiv \int d\mathbf{r}\ \left[\left(\frac{\partial\mu_{\rm L}}{\partial\rho}\right)_s\frac{\partial\rho_0}{\partial x} + \left(\frac{\partial\mu_{\rm L}}{\partial s}\right)_\rho\frac{\partial s_0}{\partial x}\right]\frac{\partial\rho_{\rm s0}}{\partial x}, \tag{16.64}$$

$$k_{\rm n} \equiv \int d\mathbf{r} \left\{ \left[\left(\frac{\partial \mu_{\rm L}}{\partial \rho} \right)_s \frac{\partial \rho_0}{\partial x} + \left(\frac{\partial \mu_{\rm L}}{\partial s} \right)_\rho \frac{\partial s_0}{\partial x} \right] \frac{\partial \rho_{\rm n0}}{\partial x} + \left[\left(\frac{\partial T}{\partial \rho} \right)_s \frac{\partial \rho_0}{\partial x} + \left(\frac{\partial T}{\partial s} \right)_\rho \frac{\partial s_0}{\partial x} \right] \frac{\partial s_0}{\partial x} \right\} \tag{16.65}$$

and

$$k_{\rm sn} \equiv - \int d\mathbf{r} \left[\left(\frac{\partial \mu_{\rm L}}{\partial \rho} \right)_s \frac{\partial \rho_{\rm n0}}{\partial x} + \left(\frac{\partial T}{\partial \rho} \right)_s \frac{\partial s_0}{\partial x} \right] \frac{\partial \rho_{\rm s0}}{\partial x}. \tag{16.66}$$

The variational equations in (16.51) give the following secular equation for the dipole hydrodynamic modes:

$$\begin{pmatrix} M_{\rm s}\omega^2 - k_{\rm s} - k_{\rm sn} & k_{\rm sn} \\ k_{\rm sn} & M_{\rm n}\omega^2 - k_{\rm n} - k_{\rm sn} \end{pmatrix} \begin{pmatrix} A_{\rm s} \\ A_{\rm n} \end{pmatrix} = 0. \tag{16.67}$$

Applying some thermodynamic identities, the spring constants $k_{\rm s}$ and $k_{\rm n}$ defined in (16.64) and (16.65) can be simplified considerably. Equation (16.6) gives useful expressions for the gradients of various equilibrium thermodynamic quantities. From the definition $T = (\partial U/\partial s)_\rho$, we find

$$\begin{aligned} \nabla T_0 &= \left(\frac{\partial^2 U}{\partial s \partial \rho} \right) \nabla \rho_0 + \left(\frac{\partial^2 U}{\partial s^2} \right)_\rho \nabla s_0 \\ &= \left(\frac{\partial T}{\partial \rho} \right)_s \nabla \rho_0 + \left(\frac{\partial T}{\partial s} \right)_\rho \nabla s_0. \end{aligned} \tag{16.68}$$

Since $\nabla T_0 = 0$, (16.68) gives us the useful relation

$$\left(\frac{\partial T}{\partial \rho} \right)_s \frac{\partial \rho_0}{\partial x} + \left(\frac{\partial T}{\partial s} \right)_\rho \frac{\partial s_0}{\partial x} = 0. \tag{16.69}$$

Taking the gradient of (16.9), we find that in static equilibrium

$$\nabla \mu_{\rm L0} = \left(\frac{\partial \mu_{\rm L}}{\partial \rho} \right)_s \nabla \rho_0 + \left(\frac{\partial \mu_{\rm L}}{\partial s} \right)_\rho \nabla s_0 + \nabla U_{\rm trap}. \tag{16.70}$$

For a harmonic trap, noting that $\nabla \mu_{\rm L0} = 0$, (16.70) gives another useful expression

$$\left(\frac{\partial \mu_{\rm L}}{\partial \rho} \right)_s \frac{\partial \rho_0}{\partial x} + \left(\frac{\partial \mu_{\rm L}}{\partial s} \right)_\rho \frac{\partial s_0}{\partial x} = - \frac{\partial}{\partial x} U_{\rm trap} = -\omega_x^2 x; \tag{16.71}$$

here ω_x is the harmonic trap frequency along the x-axis.

Substituting (16.71) and (16.69) into (16.64) and (16.65) and integrating

by parts, one finds that the expressions for the spring constants $k_{\rm s}$ and $k_{\rm n}$ simplify to

$$k_{\rm s} = \omega_x^2 M_{\rm s}, \quad k_{\rm n} = \omega_x^2 M_{\rm n}. \tag{16.72}$$

Using these values, the solutions of the two coupled equations in (16.67) are given by

$$\left[M_{\rm s}M_{\rm n}\left(\omega^2 - \omega_x^2\right) - k_{\rm sn}\left(M_{\rm s} + M_{\rm n}\right)\right]\left(\omega^2 - \omega_x^2\right) = 0. \tag{16.73}$$

This gives an *in-phase* dipole mode at the trap frequency

$$\omega = \omega_x \tag{16.74}$$

and an *out-of-phase* dipole mode at frequency

$$\omega^2 = \omega_x^2 + \frac{k_{\rm sn}}{\mu}. \tag{16.75}$$

The reduced mass of the superfluid and normal fluid components is $\mu = M_{\rm s}M_{\rm n}/(M_{\rm s} + M_{\rm n})$. As noted earlier, the same dipole modes arise if we consider displacements along the y- and z- axes in a harmonic trap.

The in-phase mode given by (16.74) is the expected Kohn mode in a harmonic trap and corresponds to the solution $A_{\rm s} = A_{\rm n}$. This mode is a *rigid* in-phase oscillation of both the superfluid and normal fluid static distributions and as a result, the interactions have no effect on the mode frequency. We recall that (see Section VI of Zaremba *et al.*, 1999) the two-fluid equations derived in Section 15.2 exhibit an exact solution described by

$$\begin{aligned} n_{\rm c}(\mathbf{r}, t) &= n_{\rm c0}(\mathbf{r} - \mathbf{r}_0(t)) \\ \tilde{n}(\mathbf{r}, t) &= \tilde{n}_0(\mathbf{r} - \mathbf{r}_0(t)). \end{aligned} \tag{16.76}$$

The solution at $T = 0$ is given by (2.31). The centre-of-mass displacement $\mathbf{r}_0(t) = (x_0, y_0, z_0)$ executes simple harmonic motion at the harmonic trap frequencies; for example,

$$\frac{\partial^2 x_0(t)}{\partial t^2} = -\omega_x^2 x_0(t). \tag{16.77}$$

The Kohn mode (16.76) valid at finite T corresponds to a rigid in-phase oscillating displacement of the static density profiles of *both* the condensate and thermal cloud. It can be shown that in a harmonic trap this mode exists under all conditions and thus it must be a solution of the Landau two-fluid equations.

The frequency of the mode given by (16.75) does depend on the interactions, since these determine the thermodynamic functions appearing in the

spring constant $k_{\rm sn}$ defined in (16.66). Using (16.72) and (16.75) in (16.67), one can show that this mode corresponds to the solution[2] $M_{\rm s}A_{\rm s}+M_{\rm n}A_{\rm n}=0$. This is precisely the analogue in a trapped Bose gas of second sound. The displacements of the superfluid and normal fluid have opposite signs, producing an out-of-phase oscillation of the two components.

16.3.2 Breathing modes

As a trial solution for the velocity displacements of the superfluid and normal components involved in a breathing mode, we take

$$\mathbf{u}_{\rm s}=(A_{\rm s1}x,A_{\rm s2}y,A_{\rm s3}z),\quad \mathbf{u}_{\rm n}=(A_{\rm n1}x,A_{\rm n2}y,A_{\rm n3}z). \tag{16.78}$$

These forms are consistent with the exact solutions at $T=0$ (where $\mathbf{u}_{\rm n}\equiv 0$) and at $T>T_{\rm BEC}$ (where $\mathbf{u}_{\rm s}\equiv 0$). Substituting (16.78) into (16.47) and (16.48), we find

$$K[A_{\rm s},A_{\rm n}]=\tfrac{1}{2}\textstyle\sum_i\left[\tilde{M}_{{\rm s}i}A_{{\rm s}i}^2+\tilde{M}_{{\rm n}i}A_{{\rm n}i}^2\right] \tag{16.79}$$

and

$$U[A_{\rm s},A_{\rm n}]=\tfrac{1}{2}\textstyle\sum_{ij}\left[k_{{\rm s},ij}A_{{\rm s}i}A_{{\rm s}j}+k_{{\rm n},ij}A_{{\rm n}i}A_{{\rm n}j}+2k_{{\rm sn},ij}A_{{\rm s}i}A_{{\rm n}j}\right], \tag{16.80}$$

where the *weighted* masses $\tilde{M}_i$ are now defined by (compare with (16.63) for the dipole mode)

$$\tilde{M}_{{\rm s}i}\equiv\int d\mathbf{r}\;\rho_{\rm s0}(\mathbf{r})x_i^2,\quad \tilde{M}_{{\rm n}i}\equiv\int d\mathbf{r}\;\rho_{\rm n0}(\mathbf{r})x_i^2. \tag{16.81}$$

The constants $k_{{\rm s},ij},k_{{\rm n},ij}$, and $k_{{\rm sn},ij}$ in (16.80) (which can be thought of as effective spring constants) are now defined by

$$k_{{\rm s},ij}\equiv\int d\mathbf{r}\;\left(\frac{\partial\mu_{\rm L}}{\partial\rho}\right)_s\frac{\partial(\rho_{\rm s0}x_i)}{\partial x_i}\frac{\partial(\rho_{\rm s0}x_j)}{\partial x_j}=k_{{\rm s},ji}, \tag{16.82}$$

$$\begin{aligned}k_{{\rm n},ij}\equiv\int d\mathbf{r}\Bigg[&\left(\frac{\partial\mu_{\rm L}}{\partial\rho}\right)_s\frac{\partial(\rho_{\rm n0}x_i)}{\partial x_i}\frac{\partial(\rho_{\rm n0}x_j)}{\partial x_j}\\&+2\left(\frac{\partial T}{\partial\rho}\right)_s\frac{\partial(\rho_{\rm n0}x_i)}{\partial x_i}\frac{\partial(s_0x_j)}{\partial x_j}\\&+\left(\frac{\partial T}{\partial s}\right)_\rho\frac{\partial(s_0x_i)}{\partial x_i}\frac{\partial(s_0x_j)}{\partial x_j}\Bigg],\end{aligned} \tag{16.83}$$

[2] This out-of-phase dipole mode was first discussed by Zaremba *et al.* (1998) using a modified form of two-fluid hydrodynamics that involves neglecting the source term Γ_{12} (see the discussion at the end of Section 15.3). This gives an expression for the spring constant $k_{\rm sn}$ that is slightly different from (16.66).

and

$$k_{\mathrm{sn},ij} \equiv \int d\mathbf{r} \left[\left(\frac{\partial \mu_{\mathrm{L}}}{\partial \rho} \right)_s \frac{\partial(\rho_{\mathrm{s}0} x_i)}{\partial x_i} \frac{\partial(\rho_{\mathrm{n}0} x_j)}{\partial x_j} + \left(\frac{\partial T}{\partial \rho} \right)_s \frac{\partial(\rho_{\mathrm{s}0} x_i)}{\partial x_i} \frac{\partial(s_0 x_j)}{\partial x_j} \right]. \tag{16.84}$$

In general, we have six variational parameters (one for each Cartesian component of the two displacement fields), and hence the collective mode frequencies are found from the six coupled algebraic equations given by (16.51), namely

$$\begin{aligned} \tilde{M}_{\mathrm{s}i} \omega^2 A_{\mathrm{s}i} &= \frac{1}{2} \sum_j \Big[(k_{\mathrm{s},ij} + k_{\mathrm{s},ji}) A_{\mathrm{s}j} + 2 k_{\mathrm{sn},ij} A_{\mathrm{n}j} \Big], \\ \tilde{M}_{\mathrm{n}i} \omega^2 A_{\mathrm{n}i} &= \frac{1}{2} \sum_j \Big[(k_{\mathrm{n},ij} + k_{\mathrm{n},ji}) A_{\mathrm{n}j} + 2 k_{\mathrm{sn},ji} A_{\mathrm{s}j} \Big]. \end{aligned} \tag{16.85}$$

In experiments performed on trapped superfluid gases, one usually has an axisymmetric trap, so that $\omega_1 = \omega_2 \equiv \omega_\perp$, $\omega_3 = \omega_z$. In this case, the modes of interest are the *radial* and *axial* breathing modes, with $A_{\mathrm{s}1} = A_{\mathrm{s}2}$ and $A_{\mathrm{n}1} = A_{\mathrm{n}2}$. For an isotropic trap, the variational equations (16.85) simplify to

$$\begin{aligned} \tilde{M}_{\mathrm{s}} \omega^2 A_{\mathrm{s}} &= A_{\mathrm{s}} \sum_{ij} k_{\mathrm{s},ij} + A_{\mathrm{n}} \sum_{ij} k_{\mathrm{sn},ij}, \\ \tilde{M}_{\mathrm{n}} \omega^2 A_{\mathrm{n}} &= A_{\mathrm{n}} \textstyle\sum_{ij} k_{\mathrm{n},ij} + A_{\mathrm{s}} \sum_{ij} k_{\mathrm{sn},ij}, \end{aligned} \tag{16.86}$$

where we define (see (16.81))

$$\tilde{M}_{\mathrm{s}} \equiv \int d\mathbf{r}\ \rho_{\mathrm{s}0}(r) r^2, \quad \tilde{M}_{\mathrm{n}} \equiv \int d\mathbf{r}\ \rho_{\mathrm{n}0}(r) r^2. \tag{16.87}$$

Since the normal fluid vanishes at $T = 0$, (16.85) reduces to a single equation for the superfluid component:

$$\tilde{M}_{\mathrm{s}i} \omega^2 A_{\mathrm{s}i} = \sum_j k_{\mathrm{s},ij} A_{\mathrm{s}j}. \tag{16.88}$$

At $T = 0$, the spring constants $k_{\mathrm{n},ij}$ and $k_{\mathrm{sn}_i j}$ are zero, and $k_{\mathrm{s},ij}$ in (16.82) reduces to

$$\int d\mathbf{r} \left(\frac{\partial \mu_{\mathrm{L}}}{\partial \rho} \right) \frac{\partial(\rho_0 x_i)}{\partial x_i} \frac{\partial(\rho_0 x_j)}{\partial x_j}. \tag{16.89}$$

To proceed we must know the equation of state, or equivalently the chemical

potential $\mu_{\rm L}(\rho)$, as a function of the mass density ρ. Assuming a *polytropic* equation of state, defined by

$$\mu_{\rm L}(\rho) \propto \rho^{\gamma}, \tag{16.90}$$

one can show after some calculation that (16.89) reduces to the remarkably simple result

$$k_{{\rm s},ij} = \tilde{M}_i \left(2\omega_i^2 \delta_{ij} + \gamma \omega_i^2\right). \tag{16.91}$$

Here $\tilde{M}_i$ is now the weighted mass as defined in (16.81) but now with respect to the total density profile $\rho_0(\mathbf{r})$. For further details, we refer to Taylor and Griffin (2005). Using (16.89), our variational equation in (16.88) reduces to

$$\omega^2 A_{{\rm s}i} = 2\omega_i^2 A_{{\rm s}i} + \gamma \omega_i^2 \sum_j A_{{\rm s}j}. \tag{16.92}$$

The breathing modes of trapped superfluid Fermi gases close to unitarity have been studied extensively at $T = 0$ (see Astrakharchik *et al.* (2005), where further references are given). As we noted earlier in this section, the collective modes are identical at $T = 0$ for Bose and Fermi superfluids. Equation (16.92) is precisely equivalent to Eq. (3) derived in Astrakharchik *et al.* (2005).

In order to illustrate the qualitative features of the two-fluid hydrodynamic modes in a trapped Bose gas, we conclude this section with some numerical results for the breathing modes given by Zaremba *et al.* (1999). These calculations are based on a variational solution of the ZGN version of the two-fluid equations. The resulting equations are identical to (16.85) but with slightly different expressions for the spring constants $k_{\rm s}$, $k_{\rm n}$, and $k_{\rm sn}$. However, as discussed at the end of Section 15.3, the difference between the ZGN two-fluid equations and the correct Landau equations is expected to be small for weak interactions. We will consider the breathing modes in an isotropic trap (also referred to as $l = 0$ monopole modes). The variational displacement fields given in (16.78) reduce to

$$\mathbf{u}_{\rm s} = A_{\rm s}\mathbf{r}, \quad \mathbf{u}_{\rm n} = A_{\rm n}\mathbf{r}, \tag{16.93}$$

where $A_{\rm s}$ and $A_{\rm n}$ are given by the coupled equations in (16.86). There are two solutions corresponding to in-phase and out-of-phase oscillations of the superfluid and normal components. We note that within the present approximation the superfluid is identified with the condensate.

In Fig. 16.1, we give the results for $\omega_0/2\pi = 200$ Hz. The calculations were performed for 5000 ^{87}Rb atoms with an s-wave scattering length $a = 58$ Å. The results shown are similar to those for other values of N, from a

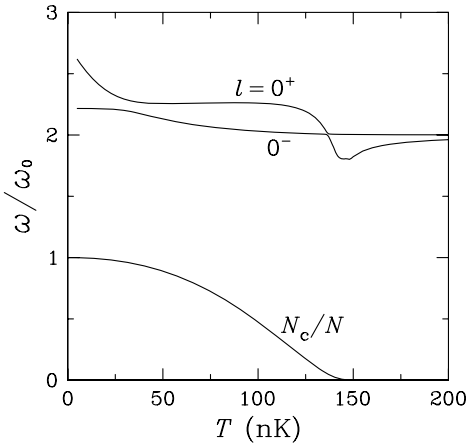


Fig. 16.1. Variational calculation of the breathing mode frequencies in the two-fluid region as a function of temperature. Calculations are for 5000 ^{87}Rb atoms in an isotropic trap. The lower curve gives the condensate fraction as a function of the temperature (from Zaremba *et al.* 1999).

lower limit of 2000 to an upper limit of 20 000. The behaviour shown in Fig. 16.1 should persist to arbitrarily large N, where a collisional hydrodynamic description would be valid. While the numerical calculations we discuss here are only for illustrative purposes, they should capture correctly the behaviour of the two-fluid breathing modes predicted by the Landau two-fluid equations.

In the lower part of Fig. 16.1 we show the condensate fraction as a function of temperature (Zaremba *et al.*, 1999), with transition temperature $T_{\rm BEC} \simeq$ 149 nK. In Fig. 11.5, we showed the condensate and noncondensate radial densities used in Fig. 16.1, at a temperature $T = 100$ nK (at which the total numbers of condensate and noncondensate atoms are approximately equal). The depletion in the noncondensate density at the centre of the trap due to the repulsive interaction with the condensate in the overlap region is clearly evident in Fig. 11.5. The static TF approximation was not used in calculating the density profiles in Fig. 11.5.

In the upper part of Fig. 16.1, the two radial breathing-mode frequencies are shown as a function of temperature. As a first approximation, these results can be viewed as a superposition of the $T = 0$ condensate breathing mode discussed in Chapter 2 (Stringari, 1996a) and the noncondensate mode above $T_{\rm BEC}$ discussed in Section 15.4. For both modes, one has to take into account the varying condensate and noncondensate fractions as a function

of temperature. Hutchinson *et al.* (1997) calculated the condensate mode frequencies at finite temperatures in the static thermal cloud approximation discussed in Chapter 8. This kind of calculation involves the static Hartree–Fock–Popov (HFP) approximation (see Section 5.2).

At $T = 0$, the condensate breathing mode (denoted as 0^-) starts at $\omega \approx 2.25\omega_0$, which is close to the TF limit $\sqrt{5}\omega_0$ discussed in Chapter 2. Apart from hybridization effects (to be discussed below), this mode basically follows the HFP behaviour with increasing temperature (taking into account the *decreasing* condensate fraction N_c/N as we approach $T_{\rm BEC}$). The *noncondensate* breathing mode (denoted as 0^+) has a frequency close to that of the breathing mode at $\omega = 2\omega_0$ found above $T_{\rm BEC}$. For temperatures below $T_{\rm BEC}$, the noncondensate density fluctuations couple with the condensate fluctuations and these modes show an increasing frequency shift with decreasing T. In addition, hybridization of the modes occurs where the uncoupled modes cross. The $\omega = 2\omega_0$ mode in Fig. 16.1 hybridizes with the condensate mode just below $T_{\rm BEC}$, and once again at around 35 nK. As discussed in Section 15.3, this kind of hybridization is already exhibited by first and second sound modes in a uniform Bose gas (see Fig. 15.1).

Shenoy and Ho (1998) gave results for the hydrodynamic modes obtained by solving the Landau two-fluid hydrodynamic equations directly. The temperature dependence of their noncondensate breathing mode is in agreement with the frequencies shown in Fig. 16.1. However, the condensate breathing mode does not approach the expected GP frequency, $2.25\omega_0$, as the temperature decreases. For further discussion of these differences, we refer to Zaremba *et al.* (1999).

The hybridization mentioned above is most clearly shown in Fig. 16.2, which gives the amplitudes of the condensate and noncondensate fluctuations associated with each breathing mode. Starting at low temperatures, the mode denoted by 0^+ is essentially a noncondensate oscillation ($|A_{\rm n+}| \simeq 1$) with a very small condensate amplitude ($|A_{\rm s+}| \simeq 0$). After passing through the hybridization point near $T = 35$ nK (see Fig. 16.1), the 0^+ mode changes over to a condensate oscillation ($|A_{\rm s+}| \simeq 1$) with only a small out-of-phase noncondensate amplitude ($|A_{\rm n+}| \simeq 0$). At the second hybridization point, near $T = 136$ nK, the amplitude of the noncondensate component grows dramatically as the mode switches over to become a noncondensate mode with frequency $2\omega_0$. In the temperature range between $T = 136$ nK and $T_{\rm BEC} \simeq 149$ nK, the amplitude of the condensate component is still appreciable. However, the condensate has a minimal effect on the

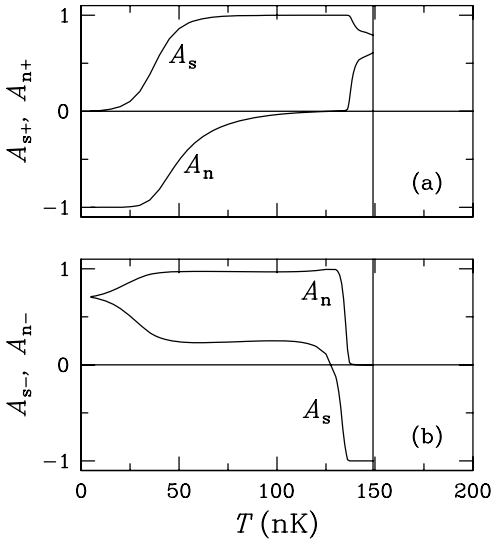


Fig. 16.2. Amplitude of the condensate (A_s) and noncondensate (A_n) components in the breathing modes as a function of temperature. The frequencies of the 0^+ and 0^- breathing modes are shown in Fig. 16.1. The vertical line denotes the BEC transition temperature, 149 nK (from Zaremba *et al.*, 1999).

noncondensate oscillation frequency since the condensate fraction N_c/N is quite small in the temperature region close to T_{BEC}.

The amplitude for the lower frequency breathing mode (0^-) is also shown in Fig. 16.2. Below the hybridization temperature at $T = 136$ nK, the 0^- mode is seen to be largely an oscillation of the noncondensate component ($|A_{n-}| \simeq 1$). Above the hybridization point at $T \simeq 136$ nK, this mode becomes essentially a condensate oscillation ($|A_{s-}| \simeq 1$), with only a very small noncondensate amplitude. A mode of this type is to be expected since the noncondensate is relatively massive and as a result, the condensate simply oscillates in the presence of a more or less static noncondensate. In this respect, the mode in this region is clearly equivalent to the mode discussed by Hutchinson *et al.* (1997), for which the collective dynamics of the noncondensate was ignored from the beginning (i.e. the static HFP approximation was used).

The results shown in Figs. 16.1 and 16.2 are only approximate, but they do indicate the complex behaviour of the coupled condensate and noncondensate dynamics involved in the two-fluid hydrodynamic region in trapped Bose gases. In this chapter, we have illustrated the variational approach

using the simplest ansatz for the two velocity fields. The results can of course be improved by using more variational parameters (see Zaremba *et al.*, 1999).

16.4 Two-fluid modes in the BCS–BEC crossover at unitarity

The variational equations for the two-fluid hydrodynamic modes given in Section 16.3 have also been used to discuss Fermi superfluids in the BCS–BEC crossover (for a review see Giorgini *et al.*, 2008). At unitarity, the s-wave scattering length a between Fermi atoms in two different hyperfine states becomes infinite. The effective variational spring constants defined in Section 16.3 can then be simplified by making use of the universal properties that thermodynamic functions exhibit at unitarity (Ho, 2004). Taylor *et al.* (2008) have reported explicit results for the temperature dependence of the dipole and breathing frequencies (the analogue of those discussed in Section 16.3) in a Fermi superfluid at unitarity, working within the local density approximation (LDA).

In addition, Taylor *et al.* (2007) have extended the variational formalism developed in this chapter to derive an expression for the density response function $\chi_{nn}(\mathbf{q}, \omega)$ in the Landau two-fluid hydrodynamic region in trapped superfluid gases. The dynamic structure factor $S(\mathbf{q}, \omega)$ was calculated for the dipole and breathing two-fluid modes within this variational formulation, giving the relative weights of the hybridized superfluid and normal fluid modes as functions of the temperature and the momentum transfer $\mathbf{q}$. These results are a natural generalization of the expression $S(\mathbf{q}, \omega)$ for a *uniform* Bose superfluid given by (14.42) and discussed in Chapter 14. Two-photon Bragg scattering (for a review see Ozeri *et al.*, 2005) is a promising way of studying the illusive out-of-phase hydrodynamic modes in trapped superfluid gases. For further discussion, see Taylor (2008).

The same variational formalism can be used to discuss the BEC limit, in which all the Fermi atoms have formed bound states in the open channel. These long-lived bosonic molecules are strongly interacting and can form a molecular Bose condensate. If the collisions are strong enough to produce local hydrodynamic equilibrium, the dynamics will be described by the generic Landau two-fluid equations. The dynamics of molecular Bose condensates using Fermi gases with a Feshbach resonance promises to be a very rich topic in ultracold atom physics in the future.

17

The Landau–Khalatnikov two-fluid equations

In Chapter 15, we showed that in the limit of short collision times the coupled equations of motion for the condensate and noncondensate atoms lead to Landau's non-dissipative two-fluid hydrodynamics. However the approach used in Chapter 15 was not based on a small expansion parameter, in contrast with the more systematic Chapman–Enskog procedure used to derive hydrodynamic damping in the kinetic theory of classical gases. In the present chapter, we generalize the procedure of Chapter 15 to trapped Bose-condensed gases, in order to derive two-fluid hydrodynamic equations that include dissipation due to transport processes. We solve the kinetic equation by expanding the nonequilibrium single-particle distribution function $f(\mathbf{p}, \mathbf{r}, t)$ around the distribution function $f^{(0)}(\mathbf{p}, \mathbf{r}, t)$ that describes *complete* local equilibrium between the condensate and the noncondensate components. All hydrodynamic damping effects are included by taking into account deviations from the local equilibrium distribution function $f^{(0)}$. Our discussion for a trapped Bose gas is a natural extension of the pioneering work of Kirkpatrick and Dorfman (1983, 1985a) for a uniform Bose-condensed gas. This chapter is mainly based on their work as well as on Nikuni and Griffin (2001a,b).

We will prove that, with appropriate definitions of various thermodynamic variables, our two-fluid hydrodynamic equations including damping have precisely the *structure* of those first derived by Landau and Khalatnikov for superfluid ^{4}He. In particular, the damping associated with the collisional exchange of atoms between the condensate and noncondensate components, which is discussed at length in Chapter 15, is now expressed in terms of frequency-dependent second viscosity coefficients. This special type of damping is a characteristic signature of a dilute Bose superfluid and

exists in addition to the hydrodynamic damping associated with the shear viscosity and thermal conductivity of the normal fluid.

The discussion that follows involves somewhat complicated calculations, as well as some subtle conceptual issues. However, the main result is very simple. Starting from a microscopic theory based on the ZNG equations of Chapter 3, we are able to derive the full two-fluid equations, including damping from all the transport coefficients that arise in the hydrodynamics of a superfluid at finite temperatures. The results emphasize the rich physics that can be studied in trapped Bose gases when one is in the collisional hydrodynamic two-fluid region.

Pethick and Smith (2008, pp. 334) discuss the calculation of collisional relaxational effects in a trapped normal Bose gas. This gives a useful introduction to some key ideas and techniques that we will develop in this and the next two chapters.

17.1 The Chapman–Enskog solution of the kinetic equation

Section 1.2 of Ferziger and Kaper (1972) gives a brief history of some of the main developments that finally led to the Chapman–Enskog method of solving the Boltzmann equation in the case of a classical gas. Very early on, it was realized that the kinetic equation first derived by Boltzmann[1] in 1872 (in connection with his attempt to prove that gases would always evolve to equilibrium) could be solved analytically only for one specific kind of interatomic potential (that describing so-called Maxwell molecules). In 1912, the mathematician Hilbert analyzed the structure of the Boltzmann equation using the theory of integral equations and proved the existence and uniqueness of solutions. In particular, Hilbert's analysis related various momentum averages involving $f(\mathbf{p}, \mathbf{r}, t)$ to the physical variables $n(\mathbf{r}, t)$, $T(\mathbf{r}, t)$ and $\mathbf{v}(\mathbf{r}, t)$ that arise in hydrodynamic equations. Building on Hilbert's work, Enskog (1917) developed a systematic procedure for solving the Boltzmann equation using a series of successive approximations. Chapman had independently obtained similar results via a different approach a few years earlier. The Chapman–Enskog approach clarified the physics in that it showed how successive approximations to the solution of a kinetic equation could be used to give a microscopic basis for the Euler hydrodynamic equations (zeroth-order, i.e. no damping) and the Navier–Stokes hydrodynamic equations (first-order, i.e. including damping from transport coefficients). These successive ap-

[1] For an English translation of Boltzmann's monumental paper published in 1872, see Brush (1966, p. 88).

proximations emphasized the key role of the local density, temperature and velocity in parameterizing solutions of the underlying Boltzmann equations for the single-particle distribution function $f(\mathbf{p}, \mathbf{r}, t)$.

As we will show, the Chapman–Enskog approach can be naturally extended to deal with the kinetic equation describing a gas of thermal atoms coupled to a Bose-condensate degree of freedom, as given in Chapter 3. Following the standard Chapman–Enskog approach (Ferziger and Kaper, 1972), we introduce a small expansion parameter α and rewrite the kinetic equation (3.42) for the distribution of $f(\mathbf{p}, \mathbf{r}, t)$

$$\frac{\partial f(\mathbf{p}, \mathbf{r}, t)}{\partial t} + \frac{\mathbf{p}}{m} \cdot \nabla_{\mathbf{r}} f(\mathbf{p}, \mathbf{r}, t) - \nabla_{\mathbf{r}} U \cdot \nabla_{\mathbf{p}} f(\mathbf{p}, \mathbf{r}, t) = \frac{1}{\alpha} \left(C_{12}[f] + C_{22}[f] \right). \tag{17.1}$$

The expansion parameter α will be eventually taken to be unity, but it allows one to develop a perturbative solution of (17.1). In order to solve this quantum kinetic equation, we formally expand the distribution function $f(\mathbf{p}, \mathbf{r}, t)$ in powers of α:

$$f = f^{(0)} + \alpha f^{(1)} + \cdots. \tag{17.2}$$

Using the expansion (17.2), we can also expand the various hydrodynamic variables in the equations (15.7)–(15.9):

$$\begin{aligned} \tilde{n} &= \tilde{n}^{(0)} + \alpha \tilde{n}^{(1)} + \cdots, \quad P_{\mu\nu} = P_{\mu\nu}^{(0)} + \alpha P_{\mu\nu}^{(1)} + \cdots, \\ \mathbf{Q} &= \mathbf{Q}^{(0)} + \alpha \mathbf{Q}^{(1)} + \cdots, \quad \tilde{\epsilon} = \tilde{\epsilon}^{(0)} + \alpha \tilde{\epsilon}^{(1)} + \cdots. \end{aligned} \tag{17.3}$$

The superscript (0) denotes the *diffusive local equilibrium* solution (see below) which is determined by the vanishing of both collision integrals in (17.1) (formally, $\alpha \to 0$).

We now redefine the source function Γ_{12} of earlier chapters and also expand it in powers of α:

$$\Gamma_{12} \equiv \frac{1}{\alpha} \int \frac{d\mathbf{p}}{(2\pi\hbar)^3} C_{12}[f, \Phi] = \frac{1}{\alpha} \left(\Gamma_{12}^{(0)} + \alpha \Gamma_{12}^{(1)} + \alpha^2 \Gamma_{12}^{(2)} + \cdots \right). \tag{17.4}$$

We also expand the condensate wavefunction in powers of α:

$$\Phi = \Phi^{(0)} + \alpha \Phi^{(1)} + \cdots. \tag{17.5}$$

In these expansions, however, a crucial point is that we are assuming that the total local density $n \equiv n_{\rm c} + \tilde{n}$ is *not* altered by the higher-order correction

terms $f^{(i)}$ $(i \geq 1)$ in (17.2). That is, although we have

$$\begin{aligned} n_{\rm c} &= n_{\rm c}^{(0)} + \alpha n_{\rm c}^{(1)} + \cdots, \\ \tilde{n} &= \tilde{n}^{(0)} + \alpha \tilde{n}^{(1)} + \cdots, \end{aligned} \tag{17.6}$$

the total density is always given by the local equilibrium result ($\alpha = 0$)

$$n = n_{\rm c}^{(0)} + \tilde{n}^{(0)} \ . \tag{17.7}$$

We also assume that nonlocal correction terms $f^{(i)}, f^{(2)}, \ldots$ make no contribution to the noncondensate velocity fields $\mathbf{v}_{\rm n}$ or to the phase θ of the condensate wavefunction. Thus the condensate velocity $\mathbf{v}_{\rm c}$ is not affected by the deviations of f from the local equilibrium value of $f^{(0)}$. The condensate chemical potential (15.3), within the Thomas–Fermi approximation in which we neglect the quantum pressure, is given by the expansion

$$\begin{aligned} \mu_{\rm c}(\mathbf{r},t) &= V_{\rm trap}(\mathbf{r}) + g[n(\mathbf{r},t) + \tilde{n}(\mathbf{r},t)] \\ &= \mu_{\rm c}^{(0)}(\mathbf{r},t) + \alpha \mu_{\rm c}^{(1)}(\mathbf{r},t) + \cdots . \end{aligned} \tag{17.8}$$

Taking (17.6) and (17.7) into account, this implies that

$$\mu_{\rm c}^{(0)} \equiv V_{\rm trap} + g(n + \tilde{n}^{(0)}), \quad \mu_{\rm c}^{(1)} = g\tilde{n}^{(1)}. \tag{17.9}$$

Using the expansion (17.2) in the kinetic equation (17.1), we find that the lowest-order ($\alpha = 0$) solution $f^{(0)}$ is determined by the condition

$$C_{12}[f^{(0)}, \Phi^{(0)}] + C_{22}[f^{(0)}] = 0. \tag{17.10}$$

The *unique* solution of (17.10) is given by the diffusive local equilibrium Bose distribution function, namely

$$f^{(0)}(\mathbf{p},\mathbf{r},t) = \frac{1}{e^{\beta(\mathbf{r},t)[(\mathbf{p}-m\mathbf{v}_{\rm n}(\mathbf{r},t))^2/2m + U(\mathbf{r},t) - \tilde{\mu}^{(0)}(\mathbf{r},t)]} - 1} \ . \tag{17.11}$$

Here the local equilibrium noncondensate chemical potential $\tilde{\mu}^{(0)}$ is determined by the condition $C_{12}[f^{(0)}, \Phi^{(0)}] = 0$, which gives (see Chapters 8 and 15)

$$\tilde{\mu}^{(0)}(\mathbf{r},t) = \mu_{\rm c}^{(0)} + \tfrac{1}{2}m(\mathbf{v}_{\rm n} - \mathbf{v}_{\rm c})^2. \tag{17.12}$$

Using (17.9), this result is equivalent to

$$\tilde{\mu}^{(0)}(\mathbf{r},t) = V_{\rm trap} + gn + g\tilde{n}^{(0)} + \tfrac{1}{2}m(\mathbf{v}_{\rm n} - \mathbf{v}_{\rm c})^2, \tag{17.13}$$

in conjunction with

$$\tilde{n}^{(0)}(\mathbf{r},t) = \int \frac{d\mathbf{p}}{(2\pi\hbar)^3} f^{(0)}(\mathbf{p},\mathbf{r},t) = \frac{1}{\Lambda^3} g_{3/2}(z^{(0)}) \ . \tag{17.14}$$

Here $z^{(0)}(\mathbf{r},t) \equiv e^{\beta[\tilde{\mu}^{(0)}(\mathbf{r},t)-U(\mathbf{r},t)]}$ is the local fugacity in the case of diffusive local equilibrium.

It is important for the reader to note that the diffusive equilibrium condition in (17.10) does not just imply that the distribution function is given by $f^{(0)}$ in (17.11); it also requires that the noncondensate chemical potential $\tilde{\mu}^{(0)}$ satisfies the relation (17.12). One may verify that $f^{(0)}$ satisfies $C_{22}[f^{(0)}] = 0$ for arbitrary values of $\tilde{\mu}^{(0)}$. In contrast, the requirement (17.10) is *only* satisfied if the local chemical potential of the thermal cloud $\tilde{\mu}^{(0)}(\mathbf{r},t)$ is given by (17.12), the condensate and noncondensate densities being determined self-consistently. Of course, it immediately follows that, since $C_{12}[f^{(0)},\Phi^{(0)}] = 0$, we have $\Gamma_{12}^{(0)} \equiv \Gamma_{12}[f^{(0)},\Phi^{(0)}] = 0$. This last result means that (17.4) reduces to

$$\Gamma_{12} = \Gamma_{12}^{(1)} + \alpha\Gamma_{12}^{(2)} + \cdots . \tag{17.15}$$

We note that the first-order term $\Gamma_{12}^{(1)}$ is determined by the deviation $f^{(1)}$ from the local equilibrium distribution $f^{(0)}$ in (17.11). Using $f^{(0)}$ to evaluate the moment in (15.13), we find that the heat current $\mathbf{Q}^{(0)}(\mathbf{r},t)$ vanishes. Using $f^{(0)}$ in the pressure tensor in (15.12) gives

$$\begin{aligned} P_{\mu\nu}^{(0)}(\mathbf{r},t) &= \delta_{\mu\nu}\tilde{P}^{(0)}(\mathbf{r},t) \equiv \delta_{\mu\nu}\int \frac{d\mathbf{p}}{(2\pi)^3}\frac{(\mathbf{p}-m\mathbf{v}_{\rm n})^2}{3m} f^{(0)}(\mathbf{p},\mathbf{r},t) \\ &= \delta_{\mu\nu}\frac{1}{\beta\Lambda^3} g_{5/2}(z^{(0)}). \end{aligned} \tag{17.16}$$

Finally, the local kinetic energy density in (15.14) in local equilibrium is found to be $\tilde{\epsilon}^{(0)}(\mathbf{r},t) = \frac{3}{2}\tilde{P}^{(0)}(\mathbf{r},t)$, where $\tilde{P}^{(0)}$ is defined in (17.16).

To summarize the above results, the lowest-order hydrodynamic equations for the noncondensate variables are given by (see also Section 15.1)

$$\frac{\partial\tilde{n}}{\partial t} + \nabla\cdot(\tilde{n}\mathbf{v}_{\rm n}) = \Gamma_{12}^{(1)} \ , \tag{17.17}$$

$$m\tilde{n}\left(\frac{\partial}{\partial t} + \mathbf{v}_{\rm n}\cdot\nabla\right)\mathbf{v}_{\rm n} = -\nabla\tilde{P} - \tilde{n}\nabla U - m(\mathbf{v}_{\rm n}-\mathbf{v}_{\rm c})\Gamma_{12}^{(1)} \ , \tag{17.18}$$

$$\frac{\partial\tilde{P}}{\partial t} + \nabla\cdot(\tilde{P}\mathbf{v}_{\rm n}) = -\tfrac{2}{3}\tilde{P}\nabla\cdot\mathbf{v}_{\rm n} + \tfrac{2}{3}\left[\tfrac{1}{2}m(\mathbf{v}_{\rm n}-\mathbf{v}_{\rm c})^2 + \mu_{\rm c} - U\right]\Gamma_{12}^{(1)} \ , \tag{17.19}$$

where $\tilde{n} = \tilde{n}^{(0)}$, $\tilde{P} = \tilde{P}^{(0)}$ and $\mu_{\rm c} = \mu_{\rm c}^{(0)}$ are given by (17.14), (17.16) and (17.12), respectively. These coupled equations describe diffusive local hydrodynamic equilibrium, as defined in (17.11) and (17.13). They involve the source term $\Gamma_{12}^{(1)}$ from (17.15), which depends on the contribution from

the next order correction, $f^{(1)}$. Later, we shall derive an explicit expression for Γ_{12} when we include the effect of deviations from the local equilibrium distribution, which are associated with various transport processes. We will show from (17.68) that the lowest-order contribution, which enters (17.17)–(17.19), is given by

$$\Gamma_{12}^{(1)}(\mathbf{r},t) = \sigma_{\rm H}\left\{\nabla\cdot[n_{\rm c}(\mathbf{v}_{\rm c}-\mathbf{v}_{\rm n})] + \tfrac{1}{3}n_{\rm c}\nabla\cdot\mathbf{v}_{\rm n}\right\}, \tag{17.20}$$

where the thermodynamic function $\sigma_{\rm H}$ is defined in (15.72).

It is useful to compare the present analysis based on the expansion in (17.2) around the diffusive local equilibrium distribution $f^{(0)}(\mathbf{p},\mathbf{r},t)$ (as given by (17.11), (17.13) and (17.14)) with the discussion in Section 15.1 based on the partial local equilibrium distribution $\tilde{f}(\mathbf{p},\mathbf{r},t)$ in (15.16). The key difference is that in $\tilde{f}$ the thermal cloud chemical potential $\tilde{\mu}(\mathbf{r},t)$ is *not* constrained to satisfy (17.12). That is, (15.16) describes partial local equilibrium, in which the thermal cloud atoms are in thermal equilibrium with themselves, but may not be in diffusive equilibrium with the condensate. The partial local equilibrium distribution $\tilde{f}$ thus involves corrections to $f^{(0)}$ as defined in (17.11). Chapter 15 is based on deriving the collisional hydrodynamic equations using the approximation $f(\mathbf{p},\mathbf{r},t)\simeq\tilde{f}(\mathbf{p},\mathbf{r},t)$.

The difference of $\tilde{f}$ from $f^{(0)}(\mathbf{p},\mathbf{r},t)$ corresponds to a new kind of relaxation process, discussed in Section 15.2, characterized by the relaxation time τ_μ defined in (15.72). We showed that in the "Landau" limit $\omega\tau_\mu\to 0$ the non-dissipative two-fluid Landau equations emerge from the coupled ZNG equations of Chapter 3. Obtaining this Landau limit from the ZNG equations is quite subtle. In the limit $\omega\tau_\mu\to 0$, we shall find that

$$\delta\mu_{\rm diff} = -\tau_\mu g\left[\nabla\cdot n_{\rm c0}(\delta\mathbf{v}_{\rm c}-\delta\mathbf{v}_{\rm n}) + \tfrac{1}{3}n_{\rm c0}\nabla\cdot\delta\mathbf{v}_{\rm n}\right], \tag{17.21}$$

where (see (15.72))

$$\frac{1}{\tau_\mu} = \frac{\beta_0 g n_{\rm c0}}{\tau_{12}\sigma_{\rm H}}. \tag{17.22}$$

However, since

$$\Gamma_{12}^{(1)} = \delta\Gamma_{12} = -\frac{\delta\mu_{\rm diff}}{\tau_\mu}\frac{\sigma_{\rm H}}{g}, \tag{17.23}$$

in the limit $\tau_\mu\to 0$, one finds that $\delta\mu_{\rm diff}\to 0$ in such a way that $\delta\Gamma_{12}$ in (17.23) remains finite, namely

$$\delta\Gamma_{12} = \sigma_{\rm H}\left[\nabla\cdot n_{\rm c0}(\delta\mathbf{v}_{\rm c}-\delta\mathbf{v}_{\rm n}) + \tfrac{1}{3}n_{\rm c0}\nabla\cdot\delta\mathbf{v}_{\rm n}\right]. \tag{17.24}$$

The result is equivalent to (15.81) for a uniform gas. This shows explicitly

how the relaxation time τ_μ drops out of the result for $\delta\Gamma_{12}$ in the "Landau limit" ($\omega\tau_\mu \to 0$). As discussed in Chapter 15, the contribution in (17.24) plays a key role in ensuring that the ZNG coupled equations of motion for the condensate and noncondensate reproduce precisely the Landau two-fluid hydrodynamic equations.

17.2 Deviation from local equilibrium

We now consider the first-order Chapman–Enskog deviation $f^{(1)}$ from the diffusive local equilibrium distribution function $f^{(0)}(\mathbf{p},\mathbf{r},t)$ in (17.11). The deviation $f^{(1)}$ will give rise to additional dissipative terms in the hydrodynamic equations. In determining these dissipative terms, we restrict ourselves to terms that are of first order in the two velocity fields $\mathbf{v}_{\rm n}$ and $\mathbf{v}_{\rm c}$. Following standard practice, we write the first correction term in (17.2) in the form

$$f^{(1)}(\mathbf{p},\mathbf{r},t) \equiv f^{(0)}(\mathbf{p},\mathbf{r},t)[1+f^{(0)}(\mathbf{p},\mathbf{r},t)]\psi(\mathbf{p},\mathbf{r},t) \tag{17.25}$$

and work with the function $\psi(\mathbf{p},\mathbf{r},t)$. Setting $f = f^{(0)} + \alpha f^{(1)}$, the C_{22} and C_{12} collision terms in (3.40) and (3.41) reduce to

$$\begin{aligned}\frac{1}{\alpha}C_{22}[f,\Phi] \simeq C_{22}[f^{(1)}] \simeq \frac{2g^2}{(2\pi)^5\hbar^7}\int d\mathbf{p}_2\int d\mathbf{p}_3\int d\mathbf{p}_4 \\ \times \delta(\mathbf{p}+\mathbf{p}_2-\mathbf{p}_3-\mathbf{p}_4)\delta(\tilde{\varepsilon}_{p_1}+\tilde{\varepsilon}_{p_2}-\tilde{\varepsilon}_{p_3}-\tilde{\varepsilon}_{p_4}) \\ \times f^{(0)}f_2^{(0)}\left(1+f_3^{(0)}\right)\left(1+f_4^{(0)}\right)(\psi_3+\psi_4-\psi_2-\psi) \\ \equiv \hat{L}_{22}[\psi],\end{aligned} \tag{17.26}$$

$$\begin{aligned}\frac{1}{\alpha}C_{12}[f,\Phi] \simeq \frac{2g^2 n_{\rm c}}{(2\pi)^2\hbar^4}\int d\mathbf{p}_1\int d\mathbf{p}_2\int d\mathbf{p}_3 \\ \times \delta(m\mathbf{v}_{\rm c}+\mathbf{p}_1-\mathbf{p}_2-\mathbf{p}_3)\delta(\varepsilon_{\rm c}^{(0)}+\tilde{\varepsilon}_{p_1}-\tilde{\varepsilon}_{p_2}-\tilde{\varepsilon}_{p_3}) \\ \times [\delta(\mathbf{p}-\mathbf{p}_1)-\delta(\mathbf{p}-\mathbf{p}_2)-\delta(\mathbf{p}-\mathbf{p}_3)] \\ \times (1+f_1^{(0)})f_2^{(0)}f_3^{(0)}(-\beta\mu_{\rm c}^{(1)}+\psi_2+\psi_3-\psi_1) \\ \equiv -\beta g\tilde{n}^{(1)}\hat{L}_{12}[1]+\hat{L}_{12}[\psi],\end{aligned} \tag{17.27}$$

where $\varepsilon_{\rm c}^{(0)} = \mu_{\rm c}^{(0)} + \frac{1}{2}mv_{\rm c}^2$, with $\mu_{\rm c}^{(0)}$ and $\mu_{\rm c}^{(1)}$ given in (17.9). Here the ψ_i stand for $\psi(\mathbf{p}_i,\mathbf{r},t)$. The linearized $\hat{L}_{12}$ operator in (17.27) is given by

$$\hat{L}_{12}[\psi] \equiv \frac{2g^2 n_{\rm c}}{(2\pi)^2\hbar^4}\int d\mathbf{p}_1 \int d\mathbf{p}_2 \int d\mathbf{p}_3 \times \delta(m\mathbf{v}_{\rm c} + \mathbf{p}_1 - \mathbf{p}_2 - \mathbf{p}_3)\delta(\varepsilon_{\rm c}^{(0)} + \tilde{\varepsilon}_{p_1} - \tilde{\varepsilon}_{p_2} - \tilde{\varepsilon}_{p_3}) \times [\delta(\mathbf{p} - \mathbf{p}_1) - \delta(\mathbf{p} - \mathbf{p}_2) - \delta(\mathbf{p} - \mathbf{p}_3)] \times (1 + f_1^{(0)}) f_2^{(0)} f_3^{(0)} (\psi_2 + \psi_3 - \psi_1). \tag{17.28}$$

One may verify that the linearized collision integrals (17.26) and (17.27) satisfy the relations

$$\begin{aligned} \hat{L}_{22}[a\psi(\mathbf{p}) + b] &= a\hat{L}_{22}[\psi(\mathbf{p})] \\ \hat{L}_{12}[a\psi(\mathbf{p}) + b] &= a\hat{L}_{12}[\psi(\mathbf{p})] + b\hat{L}_{12}[1], \end{aligned} \tag{17.29}$$

where a and b are independent of momentum but may depend on position.

Using (17.26)–(17.29) and expanding the kinetic equation (17.1) to first order in α, we find that the function $\psi(\mathbf{p}, \mathbf{r}, t)$ defined in (17.25) is determined by the equation

$$\frac{\partial^0 f^{(0)}(\mathbf{p}, \mathbf{r}, t)}{\partial t} + \frac{\mathbf{p}}{m} \cdot \nabla_{\mathbf{r}} f^{(0)}(\mathbf{p}, \mathbf{r}, t) - \nabla_{\mathbf{r}} U \cdot \nabla_{\mathbf{p}} f^{(0)}(\mathbf{p}, \mathbf{r}, t) = -\beta g \tilde{n}^{(1)} \hat{L}_{12}[1] + \hat{L}_{12}[\psi] + \hat{L}_{22}[\psi]. \tag{17.30}$$

The notation $\partial^0/\partial t$ introduced in (17.30) means that we use the lowest-order local equilibrium hydrodynamic equations (17.17)–(17.19) in evaluating the *time* derivatives of $\mathbf{v}_{\rm n}$, $\tilde{\mu}$, T and U in the local distribution function $f^{(0)}$. Making use of the expression (17.11) for $f^{(0)}$, the l.h.s. of (17.30) can be reduced to

$$\begin{aligned} &\left[\frac{\partial^0}{\partial t} + \frac{\mathbf{p}}{m} \cdot \nabla_{\mathbf{r}} - \nabla_{\mathbf{r}} U(\mathbf{r}, t) \cdot \nabla_{\mathbf{p}}\right] f^{(0)}(\mathbf{p}, \mathbf{r}, t) \\ &= \left[\frac{1}{z}\left(\frac{\partial^0}{\partial t} + \frac{\mathbf{p}}{m} \cdot \nabla\right) z + \frac{mu^2}{2k_{\rm B}T^2}\left(\frac{\partial^0}{\partial t} + \frac{\mathbf{p}}{m} \cdot \nabla\right) T \right. \\ &\quad \left. + \frac{m\mathbf{u}}{k_{\rm B}T} \cdot \left(\frac{\partial^0}{\partial t} + \frac{\mathbf{p}}{m} \cdot \nabla\right) \mathbf{v}_{\rm n} + \frac{\nabla U(\mathbf{r}, t)}{k_{\rm B}T} \cdot \mathbf{u}\right] f^{(0)}(1 + f^{(0)}). \end{aligned} \tag{17.31}$$

The thermal velocity $\mathbf{u}$ is defined by $m\mathbf{u} \equiv \mathbf{p} - m\mathbf{v}_{\rm n}$, and $z \equiv z^{(0)}$ is the local equilibrium fugacity given below (17.14).

Using the expressions for the local noncondensate density $\tilde{n}^{(0)}$ in (17.14)

and the kinetic pressure $\tilde{P}^{(0)}$ in (17.16), one finds

$$\begin{aligned}\frac{\partial^0 \tilde{n}^{(0)}}{\partial t} &= \frac{3\tilde{n}^{(0)}}{2T}\frac{\partial^0 T}{\partial t} + \frac{\gamma^{(0)} k_{\rm B} T}{z}\frac{\partial^0 z}{\partial t}, \\ \frac{\partial^0 \tilde{P}^{(0)}}{\partial t} &= \frac{5\tilde{P}^{(0)}}{2T}\frac{\partial^0 T}{\partial t} + \frac{\tilde{n}^{(0)} k_{\rm B} T}{z}\frac{\partial^0 z}{\partial t}.\end{aligned} \tag{17.32}$$

Here we have defined $\gamma^{(0)}(\mathbf{r},t) \equiv (\beta_0/\Lambda_0^3) g_{1/2}(z^{(0)}(\mathbf{r},t)) \equiv \tilde{\gamma}_0/g$, where $\tilde{\gamma}_0$ was introduced earlier in (15.66). Combining these equations with (17.17) and (17.19), one can show that they reduce to

$$\begin{aligned}\frac{\partial^0 T}{\partial t} &= -\frac{2}{3}T(\nabla\cdot\mathbf{v}_{\rm n}) - \mathbf{v}_{\rm n}\cdot\nabla T + \frac{2T}{3\tilde{n}^{(0)}}\sigma_1\Gamma_{12}, \\ \frac{\partial^0 z}{\partial t} &= -\mathbf{v}_{\rm n}\cdot\nabla z + \sigma_2 z\frac{\Gamma_{12}}{\tilde{n}^{(0)}}.\end{aligned} \tag{17.33}$$

The dimensionless local thermodynamic functions σ_1 and σ_2 appearing in (17.33) are defined as

$$\begin{aligned}\sigma_1(\mathbf{r},t) &\equiv \frac{\gamma^{(0)}\tilde{n}^{(0)}\left(\tilde{\mu}^{(0)} - U\right) - \frac{3}{2}[\tilde{n}^{(0)}]^2}{\frac{5}{2}\tilde{P}^{(0)}\gamma^{(0)} - \frac{3}{2}[\tilde{n}^{(0)}]^2}, \\ \sigma_2(\mathbf{r},t) &\equiv \beta\frac{\frac{5}{2}\tilde{P}^{(0)}\tilde{n}^{(0)} - [\tilde{n}^{(0)}]^2\left(\tilde{\mu}^{(0)} - U\right)}{\frac{5}{2}\tilde{P}^{(0)}\gamma^{(0)} - \frac{3}{2}[\tilde{n}^{(0)}]^2}.\end{aligned} \tag{17.34}$$

An equation for $\partial^0\mathbf{v}_{\rm n}/\partial t$ analogous to the equations of motion (17.33) is given by (17.18).

Combining all these results, one finds that (17.31) becomes

$$\begin{aligned}&\left(\frac{\partial^0}{\partial t} + \frac{\mathbf{p}}{m}\cdot\nabla_{\mathbf{r}} - \nabla_{\mathbf{r}}U\cdot\nabla_{\mathbf{p}}\right)f^{(0)} \\ &= \left\{\frac{1}{T}\mathbf{u}\cdot\nabla T\left(\frac{mu^2}{2k_{\rm B}T} - \frac{5\tilde{P}^{(0)}}{2\tilde{n}^{(0)}k_{\rm B}T}\right) + \frac{m}{k_{\rm B}T}\left[\mathbf{u}\cdot(\mathbf{u}\cdot\nabla)\mathbf{v}_{\rm n} - \tfrac{1}{3}u^2\nabla\cdot\mathbf{v}_{\rm n}\right]\right. \\ &\quad \left. + \left[\sigma_2 + \frac{mu^2}{3k_{\rm B}T}\sigma_1 + \frac{m}{k_{\rm B}T}\mathbf{u}\cdot(\mathbf{v}_{\rm c} - \mathbf{v}_{\rm n})\right]\frac{\Gamma_{12}}{\tilde{n}^{(0)}}\right\}f^{(0)}(1+f^{(0)}),\end{aligned} \tag{17.35}$$

where we recall that $\mathbf{u} \equiv \mathbf{p}/m - \mathbf{v}_{\rm n}$. In calculating the dissipative terms, we will only consider terms to *first order* in the velocity fields $\mathbf{v}_{\rm n}$ and $\mathbf{v}_{\rm c}$. Since Γ_{12} as given by (17.24) is proportional to $\mathbf{v}_{\rm n}$ and $\mathbf{v}_{\rm c}$ (see (17.68) below), we can neglect the last term, which is proportional to $\mathbf{v}_{\rm c} - \mathbf{v}_{\rm n}$ in (17.35). Using this linearized version of (17.35) in the l.h.s. in the equation of motion

(17.30), we obtain

$$\left\{\frac{\mathbf{u}\cdot\nabla T}{T}\left[\frac{mu^2}{2k_{\rm B}T}-\frac{5g_{5/2}(z)}{2g_{3/2}(z)}\right]+\frac{m}{k_{\rm B}T}D_{\mu\nu}\left(u_\mu u_\nu-\tfrac{1}{3}\delta_{\mu\nu}u^2\right)\right.$$
$$\left.+\left(\sigma_2+\frac{mu^2}{3k_{\rm B}T}\sigma_1\right)\frac{\Gamma_{12}^{(1)}}{\tilde{n}^{(0)}}\right\}f^{(0)}(1+f^{(0)})+\beta g\tilde{n}^{(1)}\hat{L}_{12}[1]$$
$$=\hat{L}_{12}[\psi]+\hat{L}_{22}[\psi]\equiv\hat{L}[\psi], \tag{17.36}$$

where $D_{\mu\nu}$ is defined in (15.15). This linearized integral equation determines the deviation function ψ defined in (17.25) and hence the first-order correction $f^{(1)}(\mathbf{p},\mathbf{r},t)$ to the diffusive local equilibrium distribution $f^{(0)}(\mathbf{p},\mathbf{r},t)$. The rest of this chapter is based on this key equation for $\psi(\mathbf{p},\mathbf{r},t)$, which determines $f^{(1)}$ in (17.25).

The linearized collision operators $\hat{L}_{22}$ and $\hat{L}_{12}$ defined in (17.26) and (17.28) can be shown to satisfy the following conditions:

$$\begin{aligned}&\hat{L}_{12}[\mathbf{p}-m\mathbf{v}_{\rm c}]=0,\quad \hat{L}_{12}[\tilde{\varepsilon}_p-\varepsilon_{\rm c}^{(0)}]=0,\\ &\hat{L}_{22}[1]=0,\quad \hat{L}_{22}[\mathbf{p}]=0,\quad \hat{L}_{22}[\tilde{\varepsilon}_p]=0.\end{aligned} \tag{17.37}$$

In the derivation of these collisional invariants, we make crucial use of the energy and momentum conservation factors in $\hat{L}_{12}$ and $\hat{L}_{22}$. In order to find a *unique* solution of (17.36) for ψ, we impose the additional constraints:

$$\int d\mathbf{p}\ \mathbf{u}\,f^{(0)}(1+f^{(0)})\psi=0,$$
$$\int d\mathbf{p}\Big(\tfrac{1}{2}mu^2+U-\tilde{\mu}^{(0)}\Big)f^{(0)}(1+f^{(0)})\psi=\frac{1}{\beta}\int d\mathbf{p}\ln\Big(\frac{1+f^{(0)}}{f^{(0)}}\Big)f^{(1)}=0. \tag{17.38}$$

Physically, the first constraint in (17.38) means that the deviation $f^{(1)}$ from local equilibrium makes no contribution to the local velocity field $\mathbf{v}_{\rm n}$ as defined in (15.11). As we discuss in more detail in Section 17.3, the second constraint in (17.38) means that the total energy density and the local entropy density are also not affected by the deviation $f^{(1)}$. This means that these quantities have the same value as when we approximate f by $f^{(0)}$. This is a key feature of the Chapman–Enskog solutions and gives a unique correspondence between successive approximations of the Boltzmann equation and the position- and time-dependent hydrodynamic variables describing the diffusive local equilibrium solution $f^{(0)}$.

Next we turn to the solution of the linear integral equation for ψ given

by (17.36). The most general solution has the following form (Uehling and Uhlenbeck, 1933; Kirkpatrick and Dorfman, 1985a)

$$\psi(\mathbf{p},\mathbf{r},t)=\frac{\nabla T\cdot\mathbf{u}}{T}A(u)+D_{\mu\nu}\left(u_\mu u_\nu-\tfrac{1}{3}u^2\delta_{\mu\nu}\right)B(u)+\Gamma_{12}^{(1)}D(u). \tag{17.39}$$

The last term is not present in a normal Bose gas, since Γ_{12} vanishes. Here the dependence of the functions on $(\mathbf{r},t)$ is left implicit, and u_μ is a component of the thermal velocity $\mathbf{u}$ defined below (17.35). The functions $A(u), B(u)$ and $D(u)$ in (17.39) are given by the solutions to the following three linear integral equations:

$$\mathbf{u}\left[\frac{mu^2}{2k_{\rm B}T}-\frac{5g_{5/2}(z)}{2g_{3/2}(z)}\right]f^{(0)}(1+f^{(0)})=\hat{L}[\mathbf{u}A(u)], \tag{17.40}$$

$$\frac{m}{k_{\rm B}T}\left(u_\mu u_\nu-\tfrac{1}{3}\delta_{\mu\nu}u^2\right)f^{(0)}(1+f^{(0)})=\hat{L}[(u_\mu u_\nu-\tfrac{1}{3}\delta_{\mu\nu}u^2)B(u)], \tag{17.41}$$

$$\left(\sigma_2+\frac{mu^2}{3k_{\rm B}T}\sigma_1\right)\frac{1}{\tilde{n}^{(0)}}f^{(0)}(1+f^{(0)})=-\frac{\beta g\tilde{n}^{(1)}}{\Gamma_{12}^{(1)}}\hat{L}_{12}[1]+\hat{L}[D(u)], \tag{17.42}$$

where $\hat{L}\equiv\hat{L}_{12}+\hat{L}_{22}$. The linearized collision integrals $\hat{L}_{22}$ and $\hat{L}_{12}$ are defined in (17.26) and (17.28), respectively. For the constraints (17.38) to be satisfied, we require that the solutions of (17.40) and (17.42) satisfy the conditions

$$\begin{aligned}\int\frac{d\mathbf{p}}{(2\pi\hbar)^3}f^{(0)}(1+f^{(0)})u^2A(u)=0,\\ \int\frac{d\mathbf{p}}{(2\pi\hbar)^3}f^{(0)}(1+f^{(0)})\left(\tfrac{1}{2}mu^2+U-\tilde{\mu}^{(0)}\right)D(u)=0.\end{aligned} \tag{17.43}$$

In terms of the solution for ψ given by (17.39), one finds that the corrections due to $f^{(1)}$ in (17.25) to the various hydrodynamic variables defined in (17.3) are given by

$$\begin{aligned}\tilde{n}^{(1)}&=\left[\int\frac{d\mathbf{p}}{(2\pi\hbar)^3}f^{(0)}(1+f^{(0)})D(u)\right]\Gamma_{12}^{(1)}(\mathbf{r},t)\equiv-\tau\Gamma_{12}^{(1)},\\ P_{\mu\nu}^{(1)}&=\delta_{\mu\nu}\tilde{P}^{(1)}-2\eta\left(D_{\mu\nu}-\tfrac{1}{3}\delta_{\mu\nu}\,\mathrm{Tr}\,D\right),\\ \mathbf{Q}^{(1)}&=-\kappa\nabla T,\\ \tilde{\epsilon}^{(1)}&\equiv\tfrac{3}{2}\tilde{P}^{(1)},\end{aligned} \tag{17.44}$$

where we have defined

$$\tilde{P}^{(1)} \equiv \tau\tfrac{2}{3}(U - \tilde{\mu}^{(0)})\Gamma_{12}^{(1)} \simeq \tau\tfrac{2}{3}gn_{\rm c}^{(0)}\Gamma_{12}^{(1)}. \tag{17.45}$$

We note that $\tilde{n}$ and $\tilde{P}$ are both altered by an amount proportional to $\Gamma_{12}^{(1)}$.

The transport coefficients η and κ that appear in (17.44) are associated with the functions $A(u)$ and $B(u)$ in (17.39), namely

$$\begin{aligned}\eta &\equiv -\frac{m}{15}\int \frac{d\mathbf{p}}{(2\pi\hbar)^3}u^4 B(u) f^{(0)}(1+f^{(0)}),\\ \kappa &\equiv -\frac{m}{6T}\int \frac{d\mathbf{p}}{(2\pi\hbar)^3}u^4 A(u) f^{(0)}(1+f^{(0)}).\end{aligned} \tag{17.46}$$

Later we shall see that η corresponds to the shear viscosity and κ to the thermal conductivity of the thermal cloud. The relaxation time τ *defined* in (17.44), namely

$$\tau \equiv -\int \frac{d\mathbf{p}}{(2\pi\hbar)^3} f^{(0)}(1+f^{(0)})D(u), \tag{17.47}$$

plays a crucial role in the subsequent analysis. We will find that τ is the relaxation time associated with another kind of transport coefficient, called the second viscosity. Using the expression in the first line of (17.44), we can eliminate $\Gamma_{12}^{(1)}$ from the integral equation (17.42) for $D(u)$ to give

$$\left(\sigma_2 + \frac{mu^2}{3k_{\rm B}T}\sigma_1\right)\frac{1}{\tilde{n}^{(0)}} f^{(0)}(1+f^{(0)}) - \tau\beta g\hat{L}_{12}[1] = \hat{L}[D(u)]. \tag{17.48}$$

In Chapter 18, we will solve the three linear integral equations (17.40)–(17.42) and use the results to find explicit expressions for η, κ and τ. We show in Chapter 18 that τ in (17.47) is equal to the relaxation time τ_μ defined in (17.22)[2]. In the discussion in subsequent paragraphs, we use τ_μ rather than τ. The physical meaning of the relaxation time τ_μ can be clearly seen by writing the first equation of (17.44) in the form[3]

$$\Gamma_{12}^{(1)} = -\frac{\tilde{n}^{(1)}}{\tau_\mu} = -\frac{\tilde{n} - \tilde{n}^{(0)}}{\tau_\mu}, \tag{17.49}$$

where we have used (17.3) for $\tilde{n}^{(1)}$ (with $\alpha = 1$).

We now summarize what we have achieved by this lengthy analysis of the

[2] We caution the reader that μ here refers to the chemical potential and not to a component of a vector or tensor.

[3] This kind of relaxation term in the two-fluid hydrodynamic equations was introduced in a pioneering paper by Miyake and Yamada (1976) discussing liquid ^{4}He near the superfluid transition (their equations involved a phenomenological relaxation time equivalent to τ_μ).

first correction $f^{(1)}$ to the diffusive local equilibrium single-particle distribution function $f^{(0)}$. We have derived the following hydrodynamic equations for the noncondensate, which include the normal fluid transport coefficients (we set the expansion parameter α to unity):

$$\frac{\partial \tilde{n}}{\partial t} + \nabla \cdot (\tilde{n}\mathbf{v}_{\rm n}) = \Gamma_{12}, \tag{17.50}$$

$$\begin{aligned} m\tilde{n}\left(\frac{\partial}{\partial t} + \mathbf{v}_{\rm n}\cdot\nabla\right) v_{{\rm n}\mu} + \frac{\partial \tilde{P}}{\partial x_\mu} + \tilde{n}\frac{\partial U}{\partial x_\mu} \\ = -m(v_{{\rm n}\mu} - v_{{\rm c}\mu})\Gamma_{12} + \frac{\partial}{\partial x_\nu}\left[2\eta\left(D_{\mu\nu} - \tfrac{1}{3}\delta_{\mu\nu}\,\mathrm{Tr}\,D\right)\right], \end{aligned} \tag{17.51}$$

$$\begin{aligned} \frac{\partial \tilde{\epsilon}}{\partial t} + \nabla\cdot(\tilde{\epsilon}\mathbf{v}_{\rm n}) + (\nabla\cdot\mathbf{v}_{\rm n})\tilde{P} \\ = \left[\tfrac{1}{2}m(\mathbf{v}_{\rm n} - \mathbf{v}_{\rm c})^2 + \mu_{\rm c} - U\right]\Gamma_{12} \\ + 2\eta\left(D_{\mu\nu} - \tfrac{1}{3}\delta_{\mu\nu}\,\mathrm{Tr}\,D\right)^2 + \nabla\cdot(\kappa\nabla T), \end{aligned} \tag{17.52}$$

where $\tilde{n}$ and $\tilde{P}$ now include the effect of $f^{(1)}$ (using the results in (17.49) and (17.45)):

$$\begin{aligned} \tilde{n} &= \tilde{n}^{(0)} + \tilde{n}^{(1)} = \tilde{n}^{(0)} - \tau_\mu \Gamma_{12}^{(1)}, \\ \tilde{P} &= \tilde{P}^{(0)} + \tilde{P}^{(1)} = \tilde{P}^{(0)} + \tfrac{2}{3}\tau_\mu g n_{\rm c}^{(0)}\Gamma_{12}^{(1)}, \\ \tilde{\epsilon} &= \tfrac{3}{2}\tilde{P}. \end{aligned} \tag{17.53}$$

The lowest-order (complete or diffusive local equilibrium) values of $\tilde{n}^{(0)}$ and $\tilde{P}^{(0)}$ are given by (17.14) and (17.16), respectively.

The equivalent quantum hydrodynamic equations for the Bose condensate are given by (15.33) and (15.34), with condensate chemical potential

$$\begin{aligned} \mu_{\rm c}(\mathbf{r},t) &= \mu_{\rm c}^{(0)} + g\tilde{n}^{(1)} \\ &= \mu_{\rm c}^{(0)} - g\tau_\mu\Gamma_{12}^{(1)}. \end{aligned} \tag{17.54}$$

In the last step we have used the expression in the first line of (17.44).

We turn to the question of determining an explicit expression for the source function Γ_{12} that appears in the preceding equations. This requires some lengthy (but straightforward) calculations, and the reader could skip to the final result (17.68). As noted below (17.35), we are only considering terms to first order in the velocity fields. Using (17.53) in (17.50), we obtain

$$\frac{\partial \tilde{n}}{\partial t} = -\nabla\cdot(\tilde{n}\mathbf{v}_{\rm n}) + \Gamma_{12} = \frac{\partial \tilde{n}^{(0)}}{\partial t} - \tau_\mu\frac{\partial \Gamma_{12}}{\partial t}. \tag{17.55}$$

Using the expression for $\tilde{n}^{(0)}$ in (17.14), one finds

$$\frac{\partial \tilde{n}^{(0)}}{\partial t} = \left(\tfrac{3}{2}\tilde{n}^{(0)} + g\gamma^{(0)} n_{\rm c}\right) \frac{1}{T}\frac{\partial T}{\partial t} + g\gamma^{(0)}\left(\frac{\partial \tilde{n}^{(0)}}{\partial t} - \frac{\partial n}{\partial t}\right), \tag{17.56}$$

where $\gamma^{(0)}$ is defined below (17.32). For brevity, we denote $\gamma^{(0)}$ by γ in the subsequent discussion. In (17.56), all the quantities on the l.h.s take their local equilibrium values. For simplicity of notation, we omit the superscript (0), except in $\tilde{n}^{(0)}$. Using the continuity equation for the total density n, (17.56) can be reduced to

$$\frac{\partial \tilde{n}^{(0)}}{\partial t} = \frac{1}{1-g\gamma}\left[\left(\tfrac{3}{2}\tilde{n} + g\gamma n_{\rm c}\right)\frac{1}{T}\frac{\partial T}{\partial t} + g\nabla\cdot(\tilde{n}\mathbf{v}_{\rm n} + n_{\rm c}\mathbf{v}_{\rm c})\right]. \tag{17.57}$$

Inserting (17.57) into (17.55), we find

$$\tau_\mu \frac{\partial \Gamma_{12}}{\partial t} + \Gamma_{12} = \frac{1}{1-g\gamma}\left[\left(\tfrac{3}{2}\tilde{n} + g\gamma n_{\rm c}\right)\frac{1}{T}\frac{\partial T}{\partial t} + \nabla\cdot\tilde{n}\mathbf{v}_{\rm n} + g\gamma\nabla\cdot(n_{\rm c}\mathbf{v}_{\rm c})\right]. \tag{17.58}$$

We can transform (17.52) into an equation for $\tilde{P}$ using $\tilde{\epsilon} = \frac{3}{2}\tilde{P}$:

$$\frac{\partial \tilde{P}}{\partial t} = \frac{\partial \tilde{P}^{(0)}}{\partial t} + \tfrac{2}{3}\tau_\mu g n_{\rm c}\frac{\partial \Gamma_{12}}{\partial t} = -\nabla\tilde{P}\cdot\mathbf{v}_{\rm n} - \tfrac{5}{3}\tilde{P}\nabla\cdot\mathbf{v}_{\rm n} - \tfrac{2}{3}gn_{\rm c}\Gamma_{12} + \tfrac{2}{3}\nabla\cdot(\kappa\nabla T). \tag{17.59}$$

Taking the time derivative of $\tilde{P}^{(0)}$ in (17.16) gives

$$\frac{\partial \tilde{P}^{(0)}}{\partial t} = \left(\tfrac{5}{2}\tilde{P} + g\tilde{n}n_{\rm c}\right)\frac{1}{T}\frac{\partial T}{\partial t} + g\tilde{n}\frac{\partial \tilde{n}}{\partial t} + g\tilde{n}\nabla\cdot(\tilde{n}\mathbf{v}_{\rm n} + n_{\rm c}\mathbf{v}_{\rm c}). \tag{17.60}$$

Substituting this into (17.59), we obtain

$$-\left(\tfrac{2}{3}gn_{\rm c} + g\tilde{n}\right)\left(\tau_\mu\frac{\partial\Gamma_{12}}{\partial t} + \Gamma_{12}\right) = \left(\tfrac{5}{2}\tilde{P} + g\tilde{n}n_{\rm c}\right)\frac{1}{T}\frac{\partial T}{\partial t} + g\tilde{n}\nabla\cdot(n_{\rm c}\mathbf{v}_{\rm c}) + \nabla\tilde{P}\cdot\mathbf{v}_{\rm n} + \tfrac{5}{3}\tilde{P}\nabla\cdot\mathbf{v}_{\rm n} - \tfrac{2}{3}\nabla\cdot(\kappa\nabla T). \tag{17.61}$$

The next step involves eliminating the derivative $\partial T/\partial t$ from (17.58) and (17.61). To do this, we multiply (17.58) by $(1-g\gamma)(\frac{5}{2}\tilde{P} + g\tilde{n}n_{\rm c})$ and (17.61) by $(\frac{3}{2}\tilde{n} + gn_{\rm c}\gamma)$. Subracting the two equations then leads to

$$\begin{aligned}
&\left[(1-g\gamma)\left(\tfrac{5}{2}\tilde{P}+g\tilde{n}n_{\rm c}\right)+\left(\tfrac{3}{2}\tilde{n}+g\gamma n_{\rm c}\right)\left(\tfrac{2}{3}gn_{\rm c}+g\tilde{n}\right)\right]\left(\tau_\mu\frac{\partial\Gamma_{12}}{\partial t}+\Gamma_{12}\right)\\
&=\left(\tfrac{5}{2}\tilde{P}+g\tilde{n}n_{\rm c}\right)\nabla\cdot(\tilde{n}\mathbf{v}_{\rm n})-\left(\tfrac{3}{2}\tilde{n}+g\gamma n_{\rm c}\right)\left(\nabla\tilde{P}\cdot\mathbf{v}_{\rm n}+\tfrac{5}{3}\tilde{P}\nabla\cdot\mathbf{v}_{\rm n}\right)\\
&\quad+\left(\tfrac{5}{2}\tilde{P}+g\tilde{n}n_{\rm c}\right)g\gamma\nabla\cdot(n_{\rm c}\mathbf{v}_{\rm c})-\left(\tfrac{3}{2}\tilde{n}+g\gamma n_{\rm c}\right)g\tilde{n}\nabla\cdot(n_{\rm c}\mathbf{v}_{\rm c})\\
&\quad+\left(\tfrac{3}{2}\tilde{n}+g\gamma n_{\rm c}\right)\tfrac{2}{3}\nabla\cdot(\kappa\nabla T).
\end{aligned}\tag{17.62}$$

The l.h.s of (17.62) can be reduced to

$$\left[\tfrac{5}{2}\tilde{P}+2g\tilde{n}n_{\rm c}+\tfrac{2}{3}g^2n_{\rm c}^2\gamma-g\left(\tfrac{5}{2}\tilde{P}\gamma-\tfrac{3}{2}\tilde{n}^2\right)\right]\left(\tau_\mu\frac{\partial\Gamma_{12}}{\partial t}+\Gamma_{12}\right).\tag{17.63}$$

The first two terms on the r.h.s of (17.62) can be rewritten as

$$\begin{aligned}
&\left[\left(\tfrac{5}{2}\tilde{P}+g\tilde{n}n_{\rm c}\right)\nabla\tilde{n}-\left(\tfrac{3}{2}\tilde{n}+g\gamma n_{\rm c}\right)\nabla\tilde{P}\right]\cdot\mathbf{v}_{\rm n}\\
&+\left[\left(\tfrac{5}{2}\tilde{P}\tilde{n}+g\tilde{n}^2n_{\rm c}\right)-\left(\tfrac{5}{2}\tilde{P}\tilde{n}+\tfrac{5}{3}\tilde{P}g\gamma n_{\rm c}\right)\right]\nabla\cdot\mathbf{v}_{\rm n}.
\end{aligned}\tag{17.64}$$

From the local equilibrium expressions for $\tilde{n}$ and $\tilde{P}$, we obtain

$$\begin{aligned}
\nabla\tilde{n}&=\left[\tfrac{3}{2}\tilde{n}-\gamma(\tilde{\mu}-U)\right]\frac{1}{T}\nabla T-g\gamma\nabla n_{\rm c},\\
\nabla\tilde{P}&=\left[\tfrac{5}{2}\tilde{P}-\tilde{n}(\tilde{\mu}-U)\right]\frac{1}{T}\nabla T-g\tilde{n}\nabla n_{\rm c}.
\end{aligned}\tag{17.65}$$

Using these expressions in (17.64), the latter reduces to

$$-\left(\tfrac{5}{2}\tilde{P}\gamma-\tfrac{3}{2}\tilde{n}^2\right)\left[g\nabla\cdot(n_{\rm c}\mathbf{v}_{\rm n})-\tfrac{1}{3}gn_{\rm c}\nabla\cdot\mathbf{v}_{\rm n}\right].\tag{17.66}$$

Combining the above equations, we obtain

$$\begin{aligned}
&\left[\tfrac{5}{2}\tilde{P}+2g\tilde{n}n_{\rm c}+\tfrac{2}{3}g^2n_{\rm c}^2\gamma-g\left(\tfrac{5}{2}\tilde{P}\gamma-\tfrac{3}{2}\tilde{n}^2\right)\right]\left(\tau_\mu\frac{\partial\Gamma_{12}}{\partial t}+\Gamma_{12}\right)\\
&=g\left(\tfrac{5}{2}\tilde{P}\gamma-\tfrac{3}{2}\tilde{n}^2\right)\left(\nabla\cdot[n_{\rm c}(\mathbf{v}_{\rm c}-\mathbf{v}_{\rm n})]+\tfrac{1}{3}n_{\rm c}\nabla\cdot\mathbf{v}_{\rm n}\right)\\
&\quad+\left(\tfrac{3}{2}\tilde{n}+g\gamma n_{\rm c}\right)\tfrac{2}{3}\nabla\cdot(\kappa\nabla T).
\end{aligned}\tag{17.67}$$

We can now eliminate $\partial T/\partial t$ in (17.61), to obtain the following relatively simple equation of motion for Γ_{12}:

$$\tau_\mu\frac{\partial\Gamma_{12}}{\partial t}+\Gamma_{12}=\sigma_{\rm H}\left\{\nabla\cdot[n_{\rm c}(\mathbf{v}_{\rm c}-\mathbf{v}_{\rm n})]+\frac{1}{3}n_{\rm c}\nabla\cdot\mathbf{v}_{\rm n}\right\}-\frac{2\sigma_{\rm H}\sigma_1}{3g\tilde{n}}\nabla\cdot(\kappa\nabla T).\tag{17.68}$$

If we keep the expansion parameter α and expand Γ_{12} as in (17.15), namely as $\Gamma_{12}=\Gamma_{12}^{(1)}+\alpha\Gamma_{12}^{(2)}$ (recall that $\Gamma_{12}^{(0)}\equiv 0$), we find that $\Gamma_{12}^{(1)}$ is given by the

expression already quoted in (17.20). The second-order correction is given by

$$\Gamma_{12}^{(2)} = -\tau_\mu \frac{\partial}{\partial t}\Gamma_{12}^{(1)} - \frac{2\sigma_{\rm H}\sigma_1}{3g\tilde{n}}\nabla\cdot(\kappa\nabla T), \tag{17.69}$$

where the thermodynamic functions $\sigma_{\rm H}$ and σ_1 are defined in (15.82) and (17.34), respectively. We have accomplished our goal of finding explicit expressions for $\Gamma_{12}^{(1)}$ and $\Gamma_{12}^{(2)}$.

In closing this section, we will discuss the relation between the analysis given in this section and the ZNG results reviewed in Section 15.2. A careful analysis is needed to obtain a deeper understanding of the physics and the choice of the various hydrodynamic variables. In this chapter, we started with the diffusive local equilibrium distribution $f^{(0)}$ given by (17.11). We then included the deviation ψ from local equilibrium, given by (17.25) and (17.39). We showed that the deviation term proportional to $D(u)$ in (17.39) gives rise to corrections to the local thermodynamic quantities $\tilde{n}$, $\tilde{P}$ and $\tilde{\epsilon}$. The interesting fact is that the type of contribution associated with $D(u)$ in $f^{(1)}$ is already contained in the *partial* local equilibrium distribution function $\tilde{f}$ given in (15.16), which was the basis of the analysis in Sections 15.1 and 15.2. If we linearize the distribution function $\tilde{f}$ around *static* equilibrium (denoted by f^0), using $f \simeq f^0 + \delta f$, one finds to lowest order that

$$\delta f = \beta_0 f^0(1+f^0)\Big[\frac{\delta\tilde{T}}{T_0}\Big(\frac{p^2}{2m} + U_0 - \mu_{\rm c0}\Big) + \mathbf{p}\cdot\mathbf{v}_{\rm n} - 2g\delta n + \delta\tilde{\mu}\Big]. \tag{17.70}$$

In the discussion based on (17.39), in contrast, one finds that

$$\begin{aligned}\delta f = {}& \beta_0 f^0(1+f^0)\Big[\frac{\delta T}{T_0}\Big(\frac{p^2}{2m} + U_0 - \mu_{\rm c0}\Big) + \mathbf{p}\cdot\mathbf{v}_{\rm n} - 2g\delta n + \delta\mu_{\rm c}^{(0)}\Big] \\ & + f^0(1+f^0)D(u)\delta\Gamma_{12},\end{aligned} \tag{17.71}$$

where $\delta\Gamma_{12} \equiv \Gamma_{12}^{(1)}$. The terms in (17.39) associated with the functions $A(u)$ and $B(u)$ are not relevant here. The first term in the square brackets in (17.71) represents the difference between $f^{(0)}$ and f^0. The second term is the Chapman–Enskog deviation $f^{(1)}$. Using the explicit solution for $D(u)$ given by (18.33) and (18.34), we find that (17.71) can be written as

$$\begin{aligned}\delta f = \beta_0 f^0(1+f^0)\Big[&\Big(\frac{\delta T}{T_0} - \frac{2\sigma_1\tau_\mu}{3\tilde{n}_0}\delta\Gamma_{12}\Big)\Big(\frac{p^2}{2m} + U_0 - \mu_{\rm c0}\Big) + \mathbf{p}\cdot\mathbf{v}_{\rm n} \\ & - 2g\delta n + \delta\mu_{\rm c}^{(0)} - g(\sigma_{\rm H}^{-1}+1)\tau_\mu\delta\Gamma_{12}\Big].\end{aligned} \tag{17.72}$$

In the above discussion, we have distinguished the local temperature associated with the partial equilibrium deviation in (17.70) ($\tilde{T}(\mathbf{r},t) = T_0 + \delta\tilde{T}$) from that associated with the Chapman–Enskog result in (17.71) and (17.72) ($T(\mathbf{r},t) = T_0 + \delta T$). Comparing the two expressions for δf, we conclude that

$$\delta\tilde{T} \equiv \delta T - T_0 \frac{2\sigma_1\tau_\mu}{3\tilde{n}_0}\delta\Gamma_{12}, \tag{17.73}$$

and

$$\delta\tilde{\mu} \equiv \delta\mu_{\rm c}^{(0)} - g(\sigma_{\rm H}^{-1}+1)\tau_\mu\delta\Gamma_{12}. \tag{17.74}$$

The results (17.73) and (17.74) mean that the fluctuations in the local temperature and chemical potential of the thermal cloud have *different* values in the partial equilibrium solution ($\tilde{f}$) and Chapman–Enskog solution ($f = f^{(0)} + f^{(1)}$).

With the expression $\delta\mu_{\rm c}(\mathbf{r},t) = \delta\mu_{\rm c}^{(0)} - g\tau_\mu\delta\Gamma_{12}$ from (17.54) and $\delta\tilde{\mu}$ from (17.74), we obtain

$$\delta\mu_{\rm diff}(\mathbf{r},t) \equiv \delta\tilde{\mu} - \delta\mu_{\rm c} = -g\sigma_{\rm H}^{-1}\tau_\mu\delta\Gamma_{12} = -\frac{\tau_{12}}{\beta n_{\rm c0}}\delta\Gamma_{12}. \tag{17.75}$$

In the last line, we have used the hydrodynamic renormalization parameter $\sigma_{\rm H}$ defined in (15.72). The relation between $\delta\mu_{\rm diff}(\mathbf{r},t)$ and $\delta\Gamma_{12}(\mathbf{r},t)$ in (17.75) is precisely that given by the ZNG result (15.36) (see also (17.23)). The physical significance of the renormalized thermodynamic variables given by (17.73) and (17.74) will be discussed at the end of Section 17.3.

17.3 Equivalence to Landau–Khalatnikov two-fluid equations

In this section, we prove that our hydrodynamic equations for the condensate and equations (17.50)–(17.52) for the thermal cloud can be written precisely in the form of the Landau–Khalatnikov (LK) two-fluid equations. The standard form of the complete LK two-fluid equations involving dissipative terms is (Khalatnikov, 1965):

$$\frac{\partial n}{\partial t} + \nabla\cdot\mathbf{j} = 0, \tag{17.76}$$

$$\begin{aligned} &m\frac{\partial j_\mu}{\partial t} + \frac{\partial}{\partial x_\nu}\Big(\delta_{\mu\nu}P + m\tilde{n}v_{{\rm n}\mu}v_{{\rm n}\nu} + mn_{\rm c}v_{{\rm c}\mu}v_{{\rm c}\nu}\Big) + n\frac{\partial V_{\rm trap}}{\partial x_\mu} \\ &= \frac{\partial}{\partial x_\nu}\left[2\eta\left(D_{\mu\nu} - \tfrac{1}{3}\delta_{\mu\nu}\,{\rm Tr}\,D\right) + \delta_{\mu\nu}\left(\zeta_1\nabla\cdot[mn_{\rm c}\left(\mathbf{v}_{\rm c}-\mathbf{v}_{\rm n}\right)] + \zeta_2\nabla\cdot\mathbf{v}_{\rm n}\right)\right], \end{aligned} \tag{17.77}$$

$$\frac{\partial \mathbf{v}_{\rm c}}{\partial t} = -\nabla\Big(\frac{\mu}{m} + \tfrac{1}{2}v_{\rm c}^2 - \zeta_3 \nabla\cdot[mn_{\rm c}(\mathbf{v}_{\rm c}-\mathbf{v}_{\rm n})] - \zeta_4 \nabla\cdot\mathbf{v}_{\rm n}\Big), \quad (17.78)$$

$$\frac{\partial s}{\partial t} + \nabla\cdot\left(s\mathbf{v}_{\rm n} - \frac{\kappa\nabla T}{T}\right) = \frac{R_s}{T}. \quad (17.79)$$

Here, repeated Greek subscripts are summed over and the strain tensor $D_{\mu\nu}$ is defined in (15.15). The total current is given by $\mathbf{j} = n_{\rm c}\mathbf{v}_{\rm c} + \tilde{n}\mathbf{v}_{\rm n}$ and the dissipative function R_s in (17.79) describes the entropy production rate produced by the various transport processes:

$$R_s = \zeta_2(\nabla\cdot\mathbf{v}_{\rm n})^2 + 2\zeta_1(\nabla\cdot\mathbf{v}_{\rm n})\nabla\cdot[mn_{\rm c}(\mathbf{v}_{\rm c}-\mathbf{v}_{\rm n})] + \zeta_3\Big\{\nabla\cdot[mn_{\rm c}(\mathbf{v}_{\rm c}-\mathbf{v}_{\rm n})]\Big\}^2$$
$$+ 2\eta\left(D_{\mu\nu} - \tfrac{1}{3}\delta_{\mu\nu}\,\mathrm{Tr}\,D\right)^2 + \frac{\kappa}{T}(\nabla T)^2. \quad (17.80)$$

As can be seen, these equations involve the thermal conductivity κ, the shear viscosity η and four second viscosity coefficients ζ_i. In a normal fluid, the equations of fluid dynamics (see Landau and Lifshitz, 1959, p. 48) only involve one second viscosity coefficient, namely ζ_2. In this case the second viscosity coefficient is usually called the bulk viscosity.

As discussed earlier, the normal fluid and superfluid densities that appear in the standard Landau two-fluid theory can be identified with the corresponding noncondensate and condensate densities, within the context of the ZNG finite-temperature model based on a Hartree–Fock approximation for the single-particle excitation spectrum. We have made use of this correspondence explicitly in writing the LK hydrodynamic equations given above.

The thermodynamic functions that appear in these LK two-fluid equations satisfy the following superfluid local thermodynamic relations (see also Section 16.1):

$$\begin{aligned} P + \epsilon &= \mu n + sT + m\tilde{n}(\mathbf{v}_{\rm n}-\mathbf{v}_{\rm c})^2, \\ dP &= n\,d\mu + s\,dT - m\tilde{n}(\mathbf{v}_{\rm n}-\mathbf{v}_{\rm c})\cdot d(\mathbf{v}_{\rm n}-\mathbf{v}_{\rm c}), \\ d\epsilon &= \mu\,dn + T\,ds + (\mathbf{v}_{\rm n}-\mathbf{v}_{\rm c})\cdot d[m\tilde{n}(\mathbf{v}_{\rm n}-\mathbf{v}_{\rm c})]. \end{aligned} \quad (17.81)$$

The various local thermodynamic functions that appear in the LK two-fluid equations must be carefully defined to ensure that they satisfy these relations. The local entropy is defined by (see also (15.52))

$$s = k_{\rm B}\int \frac{d\mathbf{p}}{(2\pi\hbar)^3}\left[(1+f)\ln(1+f) - f\ln f\right]. \quad (17.82)$$

Using $f = f^{(0)} + f^{(1)}$ and working to first order in $f^{(1)}$, one finds that this local entropy reduces to

$$s = k_{\rm B} \int \frac{d\mathbf{p}}{(2\pi\hbar)^3} \biggl[(1+f^{(0)}) \ln(1+f^{(0)}) - f^{(0)} \ln f^{(0)} + \ln\biggl(\frac{1+f^{(0)}}{f^{(0)}}\biggr) f^{(1)} \biggr]. \tag{17.83}$$

Because of the constraint on $f^{(1)}$ given in (17.38), the last term in (17.83) makes no contribution to the local entropy. One is left with

$$\begin{aligned} s &= \frac{1}{T} \left[\tfrac{5}{2} \tilde{P}^{(0)} - \tilde{n}^{(0)} (\tilde{\mu}^{(0)} - U) \right] \\ &\simeq \frac{1}{T} \left[\tfrac{5}{2} \tilde{P}^{(0)} - \tilde{n}^{(0)} (\mu_{\rm c}^{(0)} - U) - \tfrac{1}{2} m \tilde{n} (\mathbf{v}_{\rm n} - \mathbf{v}_{\rm c})^2 \right], \end{aligned} \tag{17.84}$$

where we have used the expression for $\tilde{\mu}^{(0)}$ in (17.12) and also the fact that $\tilde{n}^{(0)} = \tilde{n} + O(v_{\rm n}, v_{\rm c})$.

The local energy density ϵ in the LK superfluid thermodynamic equations (17.81) is defined in the *local* rest frame, where $\mathbf{v}_{\rm c} = 0$. In the context of the ZNG microscopic model, ϵ is given by

$$\epsilon = \tilde{\epsilon} + n V_{\rm trap} + \tfrac{1}{2} g \left(n^2 + 2 n \tilde{n} - \tilde{n}^2 \right) + \tfrac{1}{2} m \tilde{n} (\mathbf{v}_{\rm n} - \mathbf{v}_{\rm c})^2, \tag{17.85}$$

where $\tilde{\epsilon}$ is defined in (17.53). Using (17.53) for $\tilde{n}$ in (17.85), one finds that the first-order corrections from $\Gamma_{12}^{(1)}$ cancel out, leaving

$$\epsilon = \tilde{\epsilon}^{(0)} + n V_{\rm trap} + \tfrac{1}{2} g \left[n^2 + 2 n \tilde{n}^{(0)} - (\tilde{n}^{(0)})^2 \right] + \tfrac{1}{2} m \tilde{n} (\mathbf{v}_{\rm n} - \mathbf{v}_{\rm c})^2, \tag{17.86}$$

where $\tilde{\epsilon}^{(0)} = \frac{3}{2} \tilde{P}^{(0)}$. From (17.83) and (17.86), we conclude that both the local entropy density s and the local energy density ϵ are entirely determined by the diffusive local equilibrium distribution function $f^{(0)}$ alone. They are *not* affected by the deviation $f^{(1)}$ from $f^{(0)}$.

In contrast, as we now show, the definition of the local pressure and chemical potential must be carefully chosen to ensure that they satisfy the LK relations in (17.81). We first define the *nonequilibrium* pressure P' as

$$P' \equiv \tilde{P} + \tfrac{1}{2} g \left(n^2 + 2 n \tilde{n} - \tilde{n}^2 \right). \tag{17.87}$$

Using the result for $\tilde{P}$ in (17.53) and working to first order in $\Gamma_{12}^{(1)}$, (17.87) reduces to

$$P' = P - \tfrac{1}{3} \tau_\mu g n_{\rm c}^{(0)} \Gamma_{12}^{(1)}, \tag{17.88}$$

where we have defined the (diffusive) local equilibrium pressure P (see also (15.47))

$$P \equiv \tilde{P}^{(0)} + \tfrac{1}{2} g \left[n^2 + 2 n \tilde{n}^{(0)} - (\tilde{n}^{(0)})^2 \right]. \tag{17.89}$$

A crucial observation is that the LK thermodynamic relations (17.81) would not be satisfied if in these LK equations we identified P' in (17.88) with the pressure and $\mu_{\rm c}$ in (17.54) with the chemical potential μ. Spurious terms would appear, associated with the corrections proportional to $\Gamma_{12}^{(1)}$.

We should recall from Section 15.2 that in deriving the Landau equations from the ZNG two-fluid equations we did define the total pressure as the P' given in (17.87) and used $\mu = \mu_{\rm c}$. However, this identification also gave rise to extra terms proportional to $\delta\mu_{\rm diff}$ in the thermodynamic relation (15.54). This is why the precise equivalence between the ZNG hydrodynamics and the Landau theory, proved in Section 15.2, is restricted to the limit $\omega\tau_\mu \to 0$, i.e. $\delta\mu_{\rm diff} \to 0$. In contrast, if the pressure P is defined as in (17.89) and μ is defined as $\mu_{\rm c}^{(0)}$, one can show that the LK relations (17.81) are satisfied. Therefore we conclude that the local equilibrium pressure P defined in (17.89) and the local equilibrium chemical potential $\mu_{\rm c}^{(0)}$ given by (17.9) are, in fact, the *correct* variables to use in the Landau–Khalatnikov equations. We show later that the corrections to the total pressure in (17.88) and to the chemical potential in (17.54) that are proportional to $\Gamma_{12}^{(1)}$ actually give rise to additional *damping* terms associated with the four second viscosity coefficients ζ_i that appear in the LK equations (17.77)–(17.79).

We now proceed to derive the LK two-fluid equations from the preceding microscopic theory, one by one. Our continuity equations for $n_{\rm c}$ and $\tilde{n}$ are given by (15.1) and (17.50). Adding them, we obtain the continuity equation for the total density, (17.76); the Γ_{12} terms cancel out. To derive the LK generalized Euler equation (17.77) for the total current $\mathbf{j}$, we combine our two continuity equations for $\tilde{n}$ and $n_{\rm c}$ and the two velocity equations (17.51) and (15.34) to give

$$m\frac{\partial j_\mu}{\partial t} + \frac{\partial}{\partial x_\nu}(\delta_{\mu\nu}P' + m\tilde{n}v_{{\rm n}\mu}v_{{\rm n}\nu} + mn_{\rm c}v_{{\rm c}\mu}v_{{\rm c}\nu}) + n\frac{\partial V_{\rm trap}}{\partial x_\mu} = \frac{\partial}{\partial x_\nu}\left[2\eta\left(D_{\mu\nu} - \tfrac{1}{3}\delta_{\mu\nu}\,\mathrm{Tr}\,D\right)\right]. \tag{17.90}$$

Using (17.88), P' can be eliminated from (17.90) to give (see footnote 2)

$$m\frac{\partial j_\mu}{\partial t} + \frac{\partial}{\partial x_\nu}(\delta_{\mu\nu}P + m\tilde{n}v_{{\rm n}\mu}v_{{\rm n}\nu} + mn_{\rm c}v_{{\rm c}\mu}v_{{\rm c}\nu}) + n\frac{\partial V_{\rm trap}}{\partial x_\mu} = \frac{\partial}{\partial x_\nu}\left[2\eta\left(D_{\mu\nu} - \tfrac{1}{3}\delta_{\mu\nu}\,\mathrm{Tr}\,D\right) + \tfrac{1}{3}\delta_{\mu\nu}\tau_\mu g n_{\rm c}\Gamma_{12}^{(1)}\right]. \tag{17.91}$$

To consistently include the damping due to the first order correction term in the Chapman–Enskog expansion, we must use the expression for $\Gamma_{12}^{(1)}$ given in (17.20). We then find that (17.91) is identical with the LK equation

(17.77), the second viscosity coefficients ζ_1 and ζ_2 being given by

$$\zeta_1 = \frac{gn_c}{3m}\tau_\mu\sigma_H, \quad \zeta_2 = \frac{gn_c^2}{9}\tau_\mu\sigma_H. \tag{17.92}$$

Similarly, using (17.54) for μ_c and (17.20) for $\Gamma_{12}^{(1)}$ in the equation for the condensate velocity (15.33), we obtain an equation of motion for $\mathbf{v}_c$ precisely in the LK form (17.78). By directly comparing the two equations, one finds explicit expressions for the second viscosity coefficients ζ_3 and ζ_4 in (17.78), namely[4]

$$\zeta_3 = \frac{g}{m^2}\tau_\mu\sigma_H, \quad \zeta_4 = \frac{gn_c}{3m}\tau_\mu\sigma_H. \tag{17.93}$$

The second viscosity coefficients in the LK two-fluid equations (17.77) and (17.78) may be viewed as the direct result of the correction terms in the pressure P' in (17.88) and to the chemical potential μ_c in (17.54), which are both proportional to $\Gamma_{12}^{(1)}$ given in (17.20). The microscopic derivation of the LK two-fluid equations given above thus allows a deeper understanding of the renormalized thermodynamic variables such as the temperature (17.73), pressure (17.88), and chemical potential (17.54), which arise when we include the effect of the term $\tau_\mu\Gamma_{12}^{(1)}$ in the two-fluid equations. It makes clear that the correct local thermodynamic variables must be carefully chosen when we generalize the Landau limit ($\omega\tau_\mu \to 0$) discussed in Section 15.2 and include hydrodynamic damping due to transport processes.

Finally, we derive the LK equation (17.79) for the local entropy. Using (17.81) we obtain

$$T\frac{\partial s}{\partial t} = \frac{\partial \epsilon}{\partial t} - \mu\frac{\partial n}{\partial t} - m(\mathbf{v}_n - \mathbf{v}_c)\frac{\partial}{\partial t}\left[\tilde{n}(\mathbf{v}_n - \mathbf{v}_c)\right]. \tag{17.94}$$

Using the expression for the local energy density ϵ given by (17.85), we can show that (17.94) reduces to

$$\begin{aligned}\frac{\partial s}{\partial t} &= \frac{\partial \tilde{\epsilon}}{\partial t} + \left[V_{\rm trap} - \mu + g(n+\tilde{n})\right]\frac{\partial n}{\partial t} + gn_c\frac{\partial \tilde{n}}{\partial t} - \frac{1}{2}\frac{\partial n}{\partial t}m(\mathbf{v}_n - \mathbf{v}_c)^2 \\ &= \frac{\partial \tilde{\epsilon}}{\partial t} + g\tilde{n}^{(1)}\frac{\partial n}{\partial t} + gn_c\frac{\partial \tilde{n}}{\partial t}.\end{aligned} \tag{17.95}$$

We have neglected the last term in the first line of (17.95), since it is third order in the local velocities. Using our two-fluid equations (17.50)–(17.52),

[4] We note that the results (17.92) and (17.93) satisfy the Onsager reciprocal relation, $\zeta_1 = \zeta_4$. This latter equality can be proved to hold quite generally, as shown by equation (4.28) of Hohenberg and Martin (1965).

we find that (17.95) reduces to the desired LK form (17.79), the entropy production rate R_s being identified with the following expression:

$$R_s = \tau_\mu g \Gamma_{12} \left\{ g \nabla \cdot [n_{\rm c}(\mathbf{v}_{\rm c} - \mathbf{v}_{\rm n})] + \tfrac{1}{3} n_{\rm c} \nabla \cdot \mathbf{v}_{\rm n} \right\} + 2\eta \left(D_{\mu\nu} - \tfrac{1}{3} \delta_{\mu\nu} \, {\rm Tr}\, D \right)^2 + \frac{\kappa}{T} (\nabla T)^2. \tag{17.96}$$

Using (17.20) for $\Gamma_{12}^{(1)}$ and the explicit expressions for ζ_i given in (17.92) and (17.93), it can be seen that R_s in (17.96) does indeed reduce to the Landau–Khalatnikov expression (17.80).

This completes our microscopic derivation of the Landau–Khalatnikov two-fluid equations, which included the hydrodynamic damping from transport processes. The derivation started from the ZNG microscopic equations and used the Chapman–Enskog technique to solve the kinetic equation for the thermal cloud atoms. In the next section, we summarize some of the essential physics involved in this somewhat complex derivation and relate it to the simpler discussion in Chapter 15.

17.4 The C_{12} collisions and the second viscosity coefficients

Starting from the ZNG equations in Chapter 3, we have solved the kinetic equation for the thermal cloud atoms using the Chapman–Enskog expansion. Including the first-order correction $f^{(1)}(\mathbf{p}, \mathbf{r}, t)$ to the diffusive local equilibrium distribution function $f^{(0)}(\mathbf{p}, \mathbf{r}, t)$ in Section 17.3, we reduced our coupled equations to a form precisely identical to the well-known phenomenological Landau–Khalatnikov two-fluid equations for a superfluid. We included fully the hydrodynamic damping associated with the shear viscosity, thermal conductivity and the four second viscosity coefficients. The most interesting and novel aspect of our results, in comparison with an analogous development of the hydrodynamic equations for a normal Bose gas (Nikuni and Griffin, 1998a), is the extremely subtle role of the source function Γ_{12} in the kinetic equation and in the GGP equation. In particular, we found that when we included the lowest-order correction to the the distribution function $f^{(0)}(\mathbf{p}, \mathbf{r}, t)$ that describes diffusive local equilibrium, we obtained corrections to various hydrodynamic variables that were proportional to $\Gamma_{12}^{(1)}$, which arises from C_{12} collisions. This required great care in defining local thermodynamic variables that are consistent with the LK dissipative two-fluid equations. These difficulties do not arise in the dissipative hydrodynamics of Bose gases above $T_{\rm BEC}$ or in the non-dissipative two-fluid equations in the Landau limit discussed in Chapter 15.

An analogous derivation of the Landau–Khalatnikov equations for a *uniform* Bose gas was first given by Kirkpatrick and Dorfman (KD) (1985a). However, at finite temperatures, where the dominant thermal excitations are particle-like Hartree–Fock excitations, KD did not obtain the second viscosity coefficients ζ_i given above. The reason for this difference is that KD neglected the source term Γ_{12} associated with the deviation from local equilibrium produced by the C_{12} collisions. In Section 17.3, we showed that the second viscosity coefficients are directly related to this Γ_{12} source term, which describes the exchange of atoms between the condensate and noncondensate components due to C_{12} collisions. For this reason, these second viscosity coefficients ζ_i are a very special feature of the two-fluid hydrodynamics of a Bose-condensed gas.

In the derivation of the second viscosity coefficients (17.92) and (17.93), we used the lowest-order correction, $\Gamma_{12} = \Gamma_{12}^{(1)}$, as defined in (17.4) and (17.15). This restricts the validity of our results to the region $\omega\tau_\mu \ll 1$ when we consider collective fluctuations with frequency ω. However, our discussion can be easily extended to deal with small but finite values of $\omega\tau_\mu$, in which case (17.20) is replaced by

$$\Gamma_{12}(\omega) = \frac{\sigma_{\rm H}}{1 - i\omega\tau_\mu}\left\{\nabla\cdot[n_{\rm c}(\mathbf{v}_{\rm c} - \mathbf{v}_{\rm n})] + \tfrac{1}{3}n_{\rm c}\nabla\cdot\mathbf{v}_{\rm n}\right\}. \tag{17.97}$$

Using this expression, we can still write our two-fluid equations in the Landau–Khalatnikov form, but now with *frequency-dependent* second viscosity coefficients:

$$\zeta_i(\omega) = \frac{\zeta_i}{1 - i\omega\tau_\mu}. \tag{17.98}$$

Everything else in the derivation given in Section 17.3 goes through. A similar extension to frequency-dependent transport coefficients, corresponding to the thermal conductivity κ and the shear viscosity η, is discussed in more detail in Section 19.3 and in Appendix C.

The expression (17.98) for the frequency-dependent second viscosity coefficients has the form expected derived from general considerations (see Landau and Lifshitz, 1959, p. 384). In a normal gas, the bulk viscosity (or, equivalently, the second viscosity ζ_2) is usually associated with compression and expansion processes. It also arises when a gas is coupled to an *internal* relaxation process (for example, the transfer of energy from the translational degrees of freedom of molecules to their vibrational degrees of freedom). If the relaxation time of this internal process is denoted by $\tau_{\rm R}$, the frequency-dependent second viscosity coefficient is given by $\zeta(\omega) = \zeta_0/(1 - i\omega\tau_{\rm R})$,

where $\zeta_0 \propto \tau_R$. In the case of a Bose-condensed gas at finite temperatures, the analogue of this is the collisional transfer of atoms between the non-condensate and condensate components. In Section 15.2, we showed that the characteristic relaxation time for the equilibration between these two components is given by τ_μ. Thus it is quite natural that we find τ_μ determines the second viscosity coefficients, as seen in (17.92) and (17.93). Above T_{BEC} (when $n_c = 0$), the four second viscosity coefficients ζ_i all vanish, as expected in a dilute single-component Bose gas in the normal phase.

In our derivation of the LK two-fluid equations in Section 17.3, we included to lowest order the dissipation associated with the deviations from diffusive local equilibrium. These deviations are proportional to the various transport coefficients. In terms of the transport relaxation times τ_i which will be evaluated in Chapter 18, this means that the LK hydrodynamic equations only include corrections to first order in $\omega\tau_i$ (where ω is the frequency of the hydrodynamic mode under consideration).

In Chapter 15, which was based on the partial local equilibrium distribution function $\tilde{f}$, we derived equations of motion which included an equation for $\delta\mu_{diff}$, (15.71). We showed that this could lead to a new relaxational mode $\omega \simeq -i/\tau_\mu$, where the diffusive relaxation time τ_μ describes how fast the chemical potentials of the condensate and thermal cloud equilibrate. If we expand (15.77) and (15.78) to first order in $\omega\tau_\mu$, coupling to this relaxational mode gives rise to damping of the first sound and second sound modes. In the language of transport coefficients, this is hydrodynamic damping due to the second viscosity coefficients.

We also discussed in Chapter 15 the two-fluid equations in the two limits $\omega\tau_\mu \ll 1$ (the Landau limit) and $\omega\tau_\mu \gg 1$. It is only in the latter case that the relaxational mode (related to the equilibration of the condensate and thermal cloud) would be observable as an additional relaxational mode. An interesting question which deserves more study is whether there is a region where the thermal cloud is internally in local thermal equilibrium ($\omega\tau_\eta \ll 1$ and $\omega\tau_\kappa \ll 1$) but is not in diffusive equilibrium with the condensate (i.e. $\omega\tau_\mu \gg 1$). For further discussion, we refer to Section 18.2 and Nikuni *et al.* (1999, 2000).

18

Transport coefficients and relaxation times

In Chapter 17, we derived two-fluid hydrodynamic equations that include damping related to transport coefficients. Our entire analysis was based on the coupled ZNG equations for the condensate and in the thermal cloud. These involved a generalized GP equation for the condensate and a kinetic equation for the thermal atoms. A crucial role is played by the C_{12} collision term in the kinetic equation, which describes the interactions between atoms in the condensate and in the thermal cloud.

Our analysis of the deviation from the diffusive local equilibrium solution of the kinetic equation was based on the Chapman–Enskog approach, extensively developed for classical gases and first applied to Bose-condensed gases by Kirkpatrick and Dorfman (1983, 1985a). This approach required a careful treatment of the novel feature relating to the C_{12} collisions both in the kinetic equation describing the thermal atoms and also in the source term Γ_{12} in the generalized GP equation for the condensate. Using the Chapman–Enskog approach to solve the kinetic equation for a trapped Bose gas, we obtained explicit expressions for the function $\psi(\mathbf{p}, \mathbf{r}, t)$ that describes the deviation from diffusive local equilibrium, as defined by (17.25) and (17.39). This deviation can be related to various transport coefficients, as discussed in Chapter 17.

These transport coefficients are determined by the solutions of the three integral equations (17.40)–(17.42) for the three contributions to the deviation function $\psi(\mathbf{p}, \mathbf{r}, t)$ in (17.39). In Section 18.1, we will solve these integral equations and obtain explicit expressions for the thermal conductivity κ, the shear viscosity η and the four second viscosity coefficients ζ_i. Most of the manipulations and arguments are natural generalizations of the Chapman–Enskog derivation of the transport coefficients in a classical gas (see Ferziger and Kaper, 1972). However, this kind of calculation may not

be well known to most readers interested in ultracold quantum gases and so we will give a detailed account in Section 18.1 and in Appendix B. A reader more interested in the final results and how they determine the damping of hydrodynamic modes (the subject of Chapter 19) can go straight to Section 18.2.

These transport coefficients can be used to define various transport relaxation times in a precise manner. The relaxation times play a crucial role in the nonequilibrium behaviour of trapped Bose-condensed gases. Some explicit model calculations are given in Section 18.2.

The transport coefficients are associated with the dissipative response of the thermal cloud. However, the condensate density is always much larger than that of the thermal cloud in the centre of the trap, even at temperatures just below $T_{\rm BEC}$. As a result, the C_{12} collisions between the condensate and noncondensate atoms make a larger contribution to all the transport coefficients than the C_{22} collisions between the thermal atoms.

18.1 Transport coefficients in trapped Bose gases

Linearized integral equations for the functions $A(u)$, $B(u)$ and $D(u)$ from (17.39) are given in (17.40)–(17.42). Equation (17.39) determines the deviation ψ from *local diffusive* equilibrium, as defined in (17.25). Once we have explicit expressions for these corrections, we can use them to determine the transport coefficients η and κ from the integrals in (17.46). We will follow the standard Chapman–Enskog method, as reviewed (for example) in the monograph by Ferziger and Kaper (1972). In this approach, one solves the linearized kinetic equation by expanding the function ψ in a basis set of polynomial functions. These polynomial functions are chosen to satisfy constraints such as (17.38) that the function ψ given in (17.39) itself must satisfy. In a classical gas, one uses Sonine polynomials but one can also define analogous polynomials for a degenerate uniform Bose gas.

This procedure was first applied to a uniform degenerate quantum gas by Uehling and Uhlenbeck (1933). With this Chapman–Enskog method, we will calculate the transport coefficients in a trapped Bose-condensed gas using the lowest-order polynomial approximation. In classical gases, this is known to give very accurate results for the transport coefficients. We follow the pioneering work of Kirkpatrick and Dorfman (1983, 1985a) on a uniform Bose-condensed gas. Their analysis was extended to a trapped Bose gas above $T_{\rm BEC}$ by Nikuni and Griffin (1998a) and to the superfluid phase by Nikuni and Griffin (2001a).

In a classical gas, all transport coefficients involve the same Maxwell–Boltzmann elastic collision time (see for example Huang, 1987) given by

$$\frac{1}{\tau_{\rm MB}(\mathbf{r})} = \sqrt{2}\tilde{n}_0(\mathbf{r})\sigma\bar{v}. \tag{18.1}$$

This formula is often used as an estimate of the relaxation rate in trapped Bose-condensed gases. In particular, the condition $\omega\tau_{\rm MB} = 1$ is used to estimate the "crossover" between collisionless modes ($\omega\tau_{\rm MB} \gg 1$) and the collision-dominated hydrodynamic modes ($\omega\tau_{\rm MB} \ll 1$). In (18.1), $\tilde{n}_0(\mathbf{r})$ is the local density of the thermal atoms, $\sigma = 8\pi a^2$ is the atomic cross section for bosons in the s-wave approximation and $\bar{v} = \sqrt{8k_{\rm B}T/\pi m}$ is the mean value of the velocity distribution in a classical gas (this is very close to the value in a noninteracting Bose gas). However, for a Bose-condensed gas, we will show in this chapter that the collision time in (18.1) is not relevant.

18.1.1 Thermal conductivity

In evaluating the thermal conductivity κ, it is convenient to rewrite the expression for κ in (17.46) as follows:[1]

$$\begin{aligned}\kappa &= -\frac{1}{3}k_{\rm B}\int \frac{d\mathbf{p}}{(2\pi\hbar)^3}\mathbf{u}A(u)\cdot\mathbf{u}\left[\frac{mu^2}{2k_{\rm B}T} - \frac{5g_{5/2}(z_0)}{2g_{3/2}(z_0)}\right]f^0(1+f^0)\\ &= -\frac{1}{3}k_{\rm B}\int \frac{d\mathbf{p}}{(2\pi\hbar)^3}\mathbf{u}A(u)\cdot\hat{L}[\mathbf{u}A(u)].\end{aligned} \tag{18.2}$$

where we recall that $m\mathbf{u} \equiv \mathbf{p} - m\mathbf{v}_{\rm n}$. The addition of the second term in the square brackets in the first line of (18.2) makes no contribution when one uses the first constraint in (17.38) in conjunction with the fact that ψ is proportional to $\mathbf{u}$ (see (17.39)). However, this form is useful since we can then employ the integral equation (17.40) to obtain the result given in the second line. The reader should note that the static equilibrium distribution function f^0 will be used to evaluate all transport coefficients in this section.

To solve the linear integral equation (17.40) for $A(u)$, we introduce a simple ansatz of the form (Uehling and Uhlenbeck, 1933; Kirkpatrick and Dorfman, 1985a; Nikuni and Griffin, 1998a)

$$A(u) = -A\left[\frac{mu^2}{2k_{\rm B}T} - \frac{5g_{5/2}(z_0)}{2g_{3/2}(z_0)}\right], \tag{18.3}$$

where A is a constant to be determined. This is the lowest-order polynomial

[1] For simplicity, we denote the equilibrium temperature by T rather than T_0.

function that satisfies the constraint on $A(u)$ given in the first equation in (17.43). Using this ansatz for $A(u)$ in (18.2) gives

$$\kappa = \frac{1}{3} k_{\mathrm{B}} A \int \frac{d\mathbf{p}}{(2\pi\hbar)^3} u^2 \left[\frac{mu^2}{2k_{\mathrm{B}}T} - \frac{5g_{5/2}(z_0)}{2g_{3/2}(z_0)} \right]^2 f^0(1+f^0). \tag{18.4}$$

This momentum integral can be evaluated in terms of the Bose–Einstein functions $g_n(z_0)$ and is given by equation (B.9) in Appendix B. Our final expression for κ is

$$\kappa = \frac{5}{2} A \frac{\tilde{n}_0 k_{\mathrm{B}}^2 T}{m} \left[\frac{7g_{7/2}(z_0)}{2g_{3/2}(z_0)} - \frac{5g_{5/2}^2(z_0)}{2g_{3/2}^2(z_0)} \right]. \tag{18.5}$$

We still need to determine the constant A in (18.3). Inserting the expression (18.3) for $A(u)$ into the integral equation (17.40), multiplying by $\mathbf{u}[mu^2/2k_{\mathrm{B}}T - 5g_{5/2}(z_0)/2g_{3/2}(z_0)]$ and finally integrating over $\mathbf{p}$, we find that A is given by the expression

$$A = -\frac{\displaystyle\int \frac{d\mathbf{p}}{(2\pi\hbar)^3} u^2 \left[\frac{mu^2}{2k_{\mathrm{B}}T} - \frac{5g_{5/2}(z_0)}{2g_{3/2}(z_0)} \right]^2 f^0(1+f^0)}{\displaystyle\int \frac{d\mathbf{p}}{(2\pi\hbar)^3} \left[\frac{mu^2}{2k_{\mathrm{B}}T} - \frac{5g_{5/2}(z_0)}{2g_{3/2}(z_0)} \right] \mathbf{u} \cdot \hat{L} \left[\mathbf{u} \left(\frac{mu^2}{2k_{\mathrm{B}}T} - \frac{5g_{5/2}(z_0)}{2g_{3/2}(z_0)} \right) \right]}. \tag{18.6}$$

The integral in the numerator of this expression for A is given by (B.9). The denominator involves the collision integral $\hat{L}$ and is much more complex. Changing variables as in (B.3), this denominator can be written in the form

$$D_\kappa = \left(\frac{2k_{\mathrm{B}}T}{m} \right)^{5/2} \frac{m^3}{(2\pi)^3} \int d\mathbf{s} \left(s^2 - \frac{5g_{5/2}(z_0)}{2g_{3/2}(z_0)} \right) \mathbf{s} \cdot \hat{L} \left[\mathbf{s} \left(s^2 - \frac{5g_{5/2}(z_0)}{2g_{3/2}(z_0)} \right) \right]. \tag{18.7}$$

As shown in (B.13), one can write

$$\hat{L}[\psi] = \frac{8m(k_{\mathrm{B}}T)^2 a^2}{\pi^3 \hbar^3} \left(\hat{L}'_{22}[\psi] + n_{\mathrm{c}0} \Lambda^3 \hat{L}'_{12}[\psi] \right), \tag{18.8}$$

where $\hat{L}'_{22}$ and $\hat{L}'_{11}$ are dimensionless collision operators defined in (B.14) and (B.15). After some work, one can reduce (18.7) to

$$D_\kappa = -m^3 \left(\frac{2k_{\mathrm{B}}T}{m} \right)^{5/2} \frac{8a^2 m (k_{\mathrm{B}}T)^2}{\pi^3} \frac{1}{(2\pi)^3} \left(I_{22}^\kappa + n_{\mathrm{c}0} \Lambda^3 I_{12}^\kappa \right), \tag{18.9}$$

where (see Appendix B for the details)

$$I_{22}^{\kappa} = -\int d\mathbf{s}\, s^2\mathbf{s}\cdot\hat{L}'_{22}(s^2\mathbf{s}),$$
$$I_{12}^{\kappa} = -\pi^{3/2}\int d\mathbf{s}\, s^2\mathbf{s}\cdot\hat{L}'_{12}(s^2\mathbf{s}). \tag{18.10}$$

These dimensionless integrals depend on the condensate density $n_{c0}(\mathbf{r})$ only through the fugacity $z_0 = e^{-\beta g n_{c0}(\mathbf{r})}$. Using them in (18.6), we obtain

$$A = \frac{15}{4}\frac{\pi^{9/2}\hbar^3}{8a^2m(k_BT)^2}\frac{\left[\frac{7}{2}g_{7/2}(z_0) - \frac{5}{2}\frac{g_{5/2}^2(z_0)}{g_{3/2}(z_0)}\right]}{I_{22}^{\kappa} + n_{c0}\Lambda^3 I_{12}^{\kappa}}. \tag{18.11}$$

Our final expression for the thermal conductivity κ in (18.5) can be written as

$$\kappa = \frac{75k_B}{64a^2}\left(\frac{\pi}{\sqrt{2}}\right)^3\left(\frac{k_BT}{m}\right)^{1/2}\frac{\left[\frac{7}{2}g_{7/2}(z_0) - \frac{5g_{5/2}^2(z_0)}{2g_{3/2}(z_0)}\right]^2}{I_{22}^{\kappa}(z_0) + \Lambda_0^3 n_{c0} I_{12}^{\kappa}(z_0)} \tag{18.12}$$

In Appendix B, we give expressions for the integrals (18.10) that are useful for numerical evaluation (see (B.36) and (B.40)).

The physical significance of the constant A in (18.11) is made clear by introducing a "relaxation time" approximation for the collision integral on the r.h.s. of (17.36):

$$\hat{L}[\psi] = -f^0(1+f^0)\frac{\psi}{\tau_\kappa}. \tag{18.13}$$

On inserting the first term in (17.39), (18.13) gives

$$\hat{L}[\mathbf{u}A(u)] = -f^0(1+f^0)\frac{\mathbf{u}A(u)}{\tau_\kappa}. \tag{18.14}$$

Using this in the integral equation (17.40) that determines the function $A(u)$, one can show that (17.40) reduces to $A = \tau_\kappa$. Thus the constant A in our ansatz (18.3) may be identified as the relaxation time τ_κ associated with the thermal conductivity.

In terms of τ_κ, the expression for κ in (18.5) is

$$\kappa = \tau_\kappa\frac{5\tilde{n}_0k_B^2T}{2m}\left[\frac{7g_{7/2}(z_0)}{2g_{3/2}(z_0)} - \frac{5g_{5/2}^2(z_0)}{2g_{3/2}^2(z_0)}\right]. \tag{18.15}$$

Using (18.11), the inverse relaxation time is naturally expressed as the sum

of contributions from C_{12} and C_{22} collisions:

$$\frac{1}{\tau_\kappa} = \frac{1}{\tau_{\kappa,12}} + \frac{1}{\tau_{\kappa,22}}, \tag{18.16}$$

where

$$\begin{aligned} \tau_{\kappa,12} &= \tau_{\rm MB} \frac{15\sqrt{\pi}}{8} \frac{\left[\frac{7}{2} g_{7/2}(z_0) g_{3/2}(z_0) - \frac{5}{2} g_{5/2}^2(z_0)\right]}{n_{\rm c0} \Lambda_0^3 I_{12}^\kappa(z_0)}, \\ \tau_{\kappa,22} &= \tau_{\rm MB} \frac{15\sqrt{\pi}}{8} \frac{\left[\frac{7}{2} g_{7/2}(z_0) g_{3/2}(z_0) - \frac{5}{2} g_{5/2}^2(z_0)\right]}{I_{22}^\kappa(z_0)}; \end{aligned} \tag{18.17}$$

in the above expressions we have introduced the Maxwell–Boltzmann relaxation time $\tau_{\rm MB}$ defined in (18.1).

For a trapped Bose gas *above* $T_{\rm BEC}$, where the term involving $I_{12}^\kappa(z_0)$ vanishes, the expression for κ in (18.12) reduces to the result obtained by Nikuni and Griffin (1998a). In the classical gas limit, the fugacity z_0 is much less than 1 and the function F_{22} in (B.37) can be expanded in powers of z_0. To order z_0^3, one can compute $I_{22}^\kappa(z)$ analytically to obtain

$$I_{22}^\kappa(z_0) = \left(\frac{2\pi}{\sqrt{2}}\right)^3 \sqrt{\pi} z_0^2 \Big(1 + \frac{9}{16}\sqrt{\frac{3}{2}} z_0 + \cdots\Big). \tag{18.18}$$

Using this expression for $I_{22}^\kappa(z_0)$ in (18.12), the thermal conductivity reduces to

$$\kappa = \frac{1}{8}\left(\frac{75}{64}\right) \frac{k_{\rm B}}{a^2} \left(\frac{k_{\rm B} T}{\pi m}\right)^{1/2} (1 - 0.07 z_0). \tag{18.19}$$

In the classical-gas limit the fugacity is $z_0 = n_0 \Lambda_0^3$, which is clearly proportional to the static equilibrium density profile $n_0(\mathbf{r})$ of the trapped gas. In obtaining the correction term in (18.19) for a degenerate Bose gas, we have made use of the expansion of the Bose–Einstein functions for small z (Pathria, 1972):

$$g_n(z) = z + \frac{z^2}{2^n} + \cdots. \tag{18.20}$$

The numerical coefficient of the first (density-independent) term in (18.19) agrees with the well-known Chapman–Enskog result for the thermal conductivity in a classical gas of hard spheres of radius a. The extra overall factor 1/8 in (18.19) is due to the fact that we have used the correct quantum mechanical scattering cross section $\sigma = 8\pi a^2$ for bosons, rather than the classical expression πa^2.

18.1.2 Shear viscosity

The shear viscosity η was defined in (17.46). We follow the same procedure as in our preceding discussion of the thermal conductivity. To solve the integral equation (17.41), one finds that the simplest consistent ansatz (Uehling and Uhlenbeck, 1933; Kirkpatrick and Dorfman, 1985a; Nikuni and Griffin, 1998a) is to take $B(u)$ to be a constant, $B(u) \equiv B$. Carrying out the momentum integral for η in (17.46) using the results in Appendix B, one finds that

$$\begin{aligned}\eta &= -B\frac{(k_{\rm B}T)^2}{m}\tilde{n}_0\frac{g_{5/2}(z_0)}{g_{3/2}(z_0)}\\ &= -B\frac{k_{\rm B}T}{m}\tilde{P}_0(\mathbf{r}).\end{aligned} \tag{18.21}$$

The local density $\tilde{n}_0$ and pressure $\tilde{P}_0$ are defined in (17.14) and (17.16).

The constant B can be determined by multiplying (17.41) by the traceless tensor $\overline{u_\mu u_\nu} \equiv u_\mu u_\nu - \delta_{\mu\nu}u^2/3$ and integrating over $\mathbf{p}$, which gives

$$B = \frac{\dfrac{m}{k_{\rm B}T}\displaystyle\int \frac{d\mathbf{p}}{(2\pi\hbar)^3}\overline{u_\mu u_\nu}^2 f^0(1+f^0)}{\displaystyle\int \frac{d\mathbf{p}}{(2\pi\hbar)^3}\overline{u_\mu u_\nu}\hat{L}\left[\overline{u_\mu u_\nu}\right]}. \tag{18.22}$$

Recalling that repeated Greek indices are summed over, one finds that $(\overline{u_\mu u_\nu})^2 = \frac{2}{3}u^4$. Writing this expression for B as N/D_η, the numerator N is easily evaluated to obtain[2]

$$N = \frac{5}{16}m^3\left(\frac{m}{k_{\rm B}T}\right)\left(\frac{2k_{\rm B}T}{m}\right)^{3/2}\frac{1}{\pi^{3/2}}\,g_{5/2}(z_0). \tag{18.23}$$

The denominator in (18.22) is given by

$$D_\eta = -\frac{m^3}{(2\pi)^3}\left(\frac{2k_{\rm B}T}{m}\right)^{7/2}\frac{8m(k_{\rm B}T)^2a^2}{\pi^3}\left(I_{22}^\eta + n_{\rm c0}\Lambda^3 I_{12}^\eta\right), \tag{18.24}$$

where the dimensionless integrals I_{22}^η and I_{12}^η (which depend on position only through the fugacity z_0) are defined as

$$\begin{aligned}I_{22}^\eta &\equiv -\int d\mathbf{s}\, s_\mu s_\nu \hat{L}'_{22}\left[s_\mu s_\nu\right],\\ I_{12}^\eta &\equiv -\pi^{3/2}\int d\mathbf{s}\, s_\mu s_\nu \hat{L}'_{12}\left[s_\mu s_\nu\right].\end{aligned} \tag{18.25}$$

The detailed derivation of (18.24) is discussed in Appendix B, where the

[2] We leave these intermediate expressions in an unsimplified form to guide the reader in their derivation.

integrals I_{22}^{η} and I_{12}^{η} are also given in a form convenient for numerical evaluation (see (B.43) and (B.44)).

As in the case of the thermal conductivity discussed earlier, we can understand the physical meaning of $B(\mathbf{u}) = B$ as follows. Let us introduce a relaxation-time approximation for the collision integral on the r.h.s. of (17.36) using $\psi = \overline{u_\mu u_\nu}$, which corresponds to the second term in (17.39):

$$\begin{aligned}\hat{L}\left[\overline{u_\mu u_\nu}\right] &= -\frac{f - f^0}{\tau_\eta} \\ &= -\overline{u_\mu u_\nu}\frac{f^0(1+f^0)}{\tau_\eta}.\end{aligned} \tag{18.26}$$

Using (18.26) the momentum integrals in the denominator and numerator of (18.22) can be shown to be equal and hence to cancel out. One then obtains a simple expression relating B and the relaxation time τ_η, namely

$$B = -\frac{m}{k_{\mathrm{B}}T}\tau_\eta. \tag{18.27}$$

Now using (18.23) and (18.24) to calculate $B = N/D_\eta$, one finds that the relaxation time associated with the shear viscosity is given by

$$\tau_\eta = \frac{5\pi^{3/4}\hbar^3 g_{5/2}(z_0)}{16m(k_{\mathrm{B}}T)^2 a^2 (I_{22}^{\eta} + n_{\mathrm{c}0}\Lambda^3 I_{12}^{\eta})}. \tag{18.28}$$

Combining (18.27), (18.28) and (18.21), one can show that the shear viscosity reduces to

$$\begin{aligned}\eta &= \tau_\eta \tilde{P}_0 \\ &= \frac{5\pi^3}{32\sqrt{2}a^2}(mk_{\mathrm{B}}T)^{1/2}\frac{g_{5/2}^2(z_0)}{I_{22}^{\eta}(z_0) + \Lambda_0^3 n_{\mathrm{c}0} I_{12}^{\eta}(z_0)}.\end{aligned} \tag{18.29}$$

We note that the inverse of the shear viscosity relaxation time τ_η (18.28) can be usefully written as the sum of two contributions

$$\frac{1}{\tau_\eta} \equiv \frac{1}{\tau_{\eta,12}} + \frac{1}{\tau_{\eta,22}}, \tag{18.30}$$

where

$$\begin{aligned}\tau_{\eta,12} &= \tau_{\mathrm{MB}}\frac{5\sqrt{2}\pi^{7/2}}{2}\left[\frac{g_{5/2}(z_0)g_{3/2}(z_0)}{n_{\mathrm{c}0}\Lambda_0^3 I_{12}^{\eta}(z_0)}\right], \\ \tau_{\eta,22} &= \tau_{\mathrm{MB}}\frac{5\sqrt{2}\pi^{7/2}}{2}\left[\frac{g_{5/2}(z_0)g_{3/2}(z_0)}{I_{22}^{\eta}(z_0)}\right].\end{aligned} \tag{18.31}$$

These relaxation times will be numerically evaluated and discussed in Section 18.2.

Above $T_{\rm BEC}$, the viscosity (18.29) reduces to the result of Nikuni and Griffin (1998a) for a degenerate (but normal) Bose gas. In a classical Maxwell–Boltzmann gas, the analytical result for the viscosity analogous to that for the thermal conductivity (18.19) is

$$\eta = \frac{1}{8}\left(\frac{5}{16}\right)\frac{m}{a^2}\left(\frac{k_{\rm B}T}{\pi m}\right)^{1/2}(1-0.33z_0)\ . \tag{18.32}$$

The comments following (18.19) concerning the thermal conductivity κ also apply to the shear viscosity in a normal Bose gas.

18.1.3 The second viscosity coefficients

To determine an expression for the relaxation time τ as defined in (17.47), we can use the simplest ansatz for the solution of (17.42), namely

$$D(u) = D\Big(\sigma_2 + \frac{mu^2}{3k_{\rm B}T}\sigma_1\Big), \tag{18.33}$$

where $\sigma_1(\mathbf{r},t)$ and $\sigma_2(\mathbf{r},t)$ are defined in (17.34). A lengthy but straightforward calculation shows that (18.33) does satisfy the constraint in the second line of (17.43). Using the ansatz (18.33) in the expression for τ in (17.47), we can express τ in terms of the constant D as

$$\tau = -D\int\frac{d\mathbf{p}}{(2\pi\hbar)^3}f^0(1+f^0)\Big(\sigma_2+\frac{mu^2}{3k_{\rm B}T}\sigma_1\Big) = -\tilde{n}_0 D. \tag{18.34}$$

In the last step, we have used the integral results (B.9) and (B.11).

The constant D in (18.33) can be determined by integrating (17.42) over the momentum $\mathbf{p}$:

$$\begin{aligned} &D\int\frac{d\mathbf{p}}{(2\pi\hbar)^3}\hat{L}_{12}\left[\sigma_2+\frac{mu^2}{3k_{\rm B}T}\sigma_1\right] \\ &= \frac{1}{\tilde{n}^{(0)}}\int\frac{d\mathbf{p}}{(2\pi\hbar)^3}\left(\sigma_2+\frac{mu^2}{3k_{\rm B}T}\sigma_1\right)f^0(1+f^0) - \beta_0 g\tau\int\frac{d\mathbf{p}}{(2\pi\hbar)^3}\hat{L}_{12}[1]. \end{aligned} \tag{18.35}$$

Making use of the collisional invariants (17.37), one finds that $\hat{L}_{22}[D(\mathbf{u})] = 0$. As a result $\hat{L}_{22}$ does not appear in (18.35). We have used the expression for $\tilde{n}^{(1)}$ in the first line of (17.44) in writing down the last term on the r.h.s. of

(18.35). Now considering the l.h.s, one can show that

$$\hat{L}_{12}\left[\frac{mu^2}{2k_{\rm B}T}\right] = -\beta_0 g n_{\rm c0}\hat{L}_{12}[1]\ , \quad \int \frac{d\mathbf{p}}{(2\pi\hbar)^3}\hat{L}_{12}[1] = -\frac{n_{\rm c0}}{\tau_{12}^0}, \tag{18.36}$$

where the collision time τ_{12}^0 is defined in (15.38). Using the results (18.36), (18.35) can be solved to give

$$D = -\frac{\tau_{12}^0 + \beta_0 g n_{\rm c0}\tau}{n_{\rm c0}(\sigma_2 - \frac{2}{3}\sigma_1\beta_0 g n_{\rm c0})}. \tag{18.37}$$

Using this solution for D in (18.34), we find a self-consistent expression for τ:

$$\tau = \frac{\tilde{n}_0(\tau_{12}^0 + \beta_0 g n_{\rm c0}\tau)}{n_{\rm c0}(\sigma_2 - \frac{2}{3}\sigma_1\beta_0 g n_{\rm c0})}. \tag{18.38}$$

Solving (18.38) for τ, we finally obtain the result

$$\frac{1}{\tau} = \frac{1}{\tau_{12}^0}\left[\frac{n_{\rm c0}}{\tilde{n}_0}\left(\sigma_2 - \tfrac{2}{3}\sigma_1\beta_0 g n_{\rm c0}\right) - \beta_0 g n_{\rm c0}\right] = \frac{\beta_0 g n_{\rm c0}}{\sigma_{\rm H}\tau_{12}^0} \equiv \frac{1}{\tau_\mu}. \tag{18.39}$$

In this reduction, we have used the explicit expressions for σ_1 and σ_2 as well as the definition of $\sigma_{\rm H}$ in (15.82). These quantities involve only *static* thermodynamic functions for the trapped Bose gases. The result (18.39) shows that τ is precisely the relaxation time τ_μ introduced in the ZNG derivation of two-fluid hydrodynamics discussed in Section 15.2. This equivalence was used in Chapter 17.

We conclude that the Chapman–Enskog correction proportional to $D(u)$ in the last term in (17.39) describes the second viscosity coefficients given in (17.92) and (17.93). We can express all four second viscosity coefficients explicitly in terms of the collision time τ_{12}^0 associated with the collision integral C_{12} as follows:

$$\begin{aligned} \zeta_1 &= \frac{k_{\rm B}T}{3m}\sigma_{\rm H}^2\tau_{12}^0, \qquad \zeta_2 = \frac{n_{\rm c}k_{\rm B}T}{9}\sigma_{\rm H}^2\tau_{12}^0, \\ \zeta_3 &= \frac{k_{\rm B}T}{m^2 n_{\rm c}}\sigma_{\rm H}^2\tau_{12}^0, \quad \zeta_4 = \zeta_1. \end{aligned} \tag{18.40}$$

These second viscosity coefficients,[3] which enter the dissipative two-fluid

[3] As we noted in Section 17.4, the pioneering calculations of Kirkpatrick and Dorfman (1985a) concluded that the four second viscosity coefficients vanish at high temperatures. This was due to the neglect of the source term Γ_{12} in the GGP equation for the condensate.

equations for a superfluid Bose gas, depend only on the C_{12} collisions. In contrast, the shear viscosity η and the thermal conductivity κ depend on both the C_{12} and C_{22} collisions. We also note that the second viscosity coefficients that we have obtained above using the Chapman–Enskog procedure were already captured by a simpler approach originally used by ZNG, based on the partial local equilibrium distribution function $\tilde{f}$. This approach (developed in Section 15.2) gives a more transparent picture of the physics associated with the relaxation time τ_μ. However, it leaves out the hydrodynamic damping associated with the other transport processes.

18.2 Relaxation times for the approach to local equilibrium

In the present section, we will discuss the three characteristic relaxation times for a trapped Bose-condensed gas defined in Section 18.1; each is associated with a distinct relaxation process due to collisions between atoms. As in the case of a classical gas, these relaxation times associated with transport coefficients play a crucial role in understanding the dynamics of a Bose-condensed gas. For example, they determine how fast thermal equilibrium is reached when the gas is perturbed, the rate of evaporative cooling in the Bose-condensed phase and the characteristic frequency dividing the so-called collisionless region (discussed in Chapters 11–13) from the collision-dominated hydrodynamic region (described by the Landau two-fluid equations derived in Chapters 15 and 17).

We recall from the discussion in Chapter 17 that the transport coefficients describe the deviations of the noncondensate distribution function $f(\mathbf{p}, \mathbf{r}, t)$ from the diffusive local equilibrium form $f^{(0)}(\mathbf{p}, \mathbf{r}, t)$. Thus it is natural that they should determine the rate at which such deviations relax. In Section 18.1, we derived microscopic expressions for the various relaxation times in a trapped Bose-condensed gas. They are all functions of position since they depend on the local fugacity, which in turn depends on the local static condensate density $n_{c0}(\mathbf{r})$. These expressions are quite complex and involve various integrals that must be evaluated numerically. In the present section, we will illustrate the behaviour of these transport relaxation times as functions of both the position in the trap and also the temperature.

Thus in a trapped Bose-condensed gas, there are several distinct relaxation times associated with various transport processes. In the detailed solution of the Boltzmann equation for $f(\mathbf{p}, \mathbf{r}, t)$ for such a gas at finite temperatures, an important distinction is made between collisions between the thermal atoms (the C_{22} collisions) and collisions between the thermal and condensate atoms

(the C_{12} collisions). In the classical limit, the C_{22} collisions give rise to the relaxation time (18.1). The C_{12} collisions only appear at temperatures below the Bose–Einstein transition temperature $T_{\rm BEC}$ and are a unique feature of a Bose-condensed gas. Our detailed calculations in this chapter include both the C_{22} and C_{12} collisions. However, we will show that the various transport relaxation times for the thermal atoms are dominated by the C_{12} collisions in a trapped Bose gas because of the high condensate density. This is easily understood, since a key feature of trapped gases is that the condensate is strongly peaked (i.e. has a high density) at the centre of the trap while the thermal cloud (the noncondensate density profile) is spread out over a much larger spatial region. Even at temperatures very close to $T_{\rm BEC}$, the condensate density at the centre of the trap is still comparable with the noncondensate density.

Before discussing the numerical calculations carried out by Nikuni and Griffin (2001b), we briefly summarize some of the main features. One important finding is that the thermal conductivity relaxation time τ_κ given by (18.16) and (18.17) is very well approximated by a simple empirical formula,

$$\frac{1}{\tau_\kappa(\mathbf{r})} \simeq \sqrt{2} n_{\rm c0}(\mathbf{r}) \sigma \bar{v} \equiv \frac{1}{\tau_{\rm BE}(\mathbf{r})}, \tag{18.41}$$

provided that

$$n_{\rm c0}(\mathbf{r}) \gg \tilde{n}_0(\mathbf{r}). \tag{18.42}$$

In fact, numerical calculations show that all three characteristic relaxation times τ_μ, τ_κ and τ_η are of the same magnitude within a factor 2. Thus we may conclude that, to a good first approximation, the relaxation time $\tau_{\rm BE}(\mathbf{r})$ as defined in (18.41) can be used as a simple "universal" relaxation rate in trapped Bose-condensed gases, replacing the classical-gas expression $\tau_{\rm MB}$ in (18.1). One immediately sees that since the condition (18.42) holds in the centre of the trap even at temperatures just below $T_{\rm BEC}$, in this region one always has $\tau_{\rm BE}(\mathbf{r}) \ll \tau_{\rm MB}(\mathbf{r})$. For collective modes involving oscillations concentrated in the condensate region, we conclude that it is $\tau_{\rm BE}(\mathbf{r})$ that determines the "crossover frequency" between the collisionless and hydrodynamic frequency regions. In the thermal gas outside the strongly peaked condensate region and at temperatures above $T_{\rm BEC}$, the classical-gas collision time $\tau_{\rm MB}$ still gives a good estimate for transport relaxation times.

In (18.17), $\tau_{\rm BE}/\tau_{\kappa,12}$ and $\tau_{\rm MB}/\tau_{\kappa,22}$ are given as universal functions of the local fugacity $z_0(\mathbf{r})$. Analytical calculations show that $\tau_{\rm BE}/\tau_{\kappa,12} \simeq 1$ for $z_0 \leq 0.3$ and $\tau_{\rm MB}/\tau_{\kappa,22} \simeq 8/15$ for $0 \leq z_0 \leq 1$. Similar calculations based on (18.31) show that $\tau_{\rm BE}/\tau_{\eta,12}$ and $\tau_{\rm MB}/\tau_{\eta,22}$ are also universal functions

of z_0 but have a more complicated dependence, increasing as z_0 increases. These analytical results are confirmed by the numerical results plotted in Figs. 18.2 and 18.3. We emphasize that while $\tau_{\rm BE}$ in (18.41) gives a good estimate of $\tau_{\kappa,12}$ in (18.17), it is only an *empirical* formula.

We note that the transport relaxation rates calculated in Section 18.1 and in Appendix B involve integrands that suppress the weight of the low-energy collisions. As a result, there is no Bose enhancement from these processes and hence no divergence in either $1/\tau_\kappa, 1/\tau_\eta$ or $1/\tau_\mu$ when $z_0 = e^{-\beta g n_{\rm c0}(\mathbf{r})}$ approaches unity (at the edge of the condensate in a trap or at $T_{\rm BEC}$). This is in contrast with the collision times τ_{12} and τ_{22} defined in Nikuni *et al.* (1999), since $1/\tau_{12}$ and $1/\tau_{22}$ have an infrared divergence for $z_0 = 1$ (i.e. at $T = T_{\rm BEC}$) due to the contribution from low-momentum collisions. We also call attention to the quite different behaviour of the relaxation times related to the transport coefficients and the collision times $\bar{\tau}_{12}$ and $\bar{\tau}_{22}$ calculated in the numerical simulations in Chapter 11.

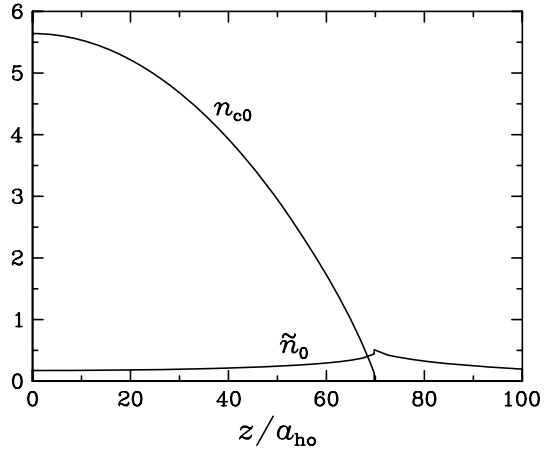


Fig. 18.1. The Thomas–Fermi density profiles ($\times 10^{-3} a_{\rm ho}^3$) along the z-axis of the condensate and the noncondensate for ^{23}Na atoms at $T = 1\,\mu$K $\simeq 0.7 T_{\rm BEC}$, with $N = 4 \times 10^7$ (from Nikuni and Griffin, 2001b).

As an illustration of the behaviour of these various transport relaxation times, Nikuni and Griffin (2001b) considered the pioneering MIT study of collective modes at finite temperatures. In the experiments of Stamper-Kurn *et al.* (1998), the trap frequencies were $\omega_x/2\pi = \omega_y/2\pi = 230\,$Hz and $\omega_z/2\pi = 18\,$Hz, the scattering length of ^{23}Na was $a = 27.5\,$Å and the total number of atoms was $N = 4 \times 10^7$. Both the $m = 0$ quadrupole oscillations and the out-of-phase dipole oscillations were studied. In each

case, the condensate and the thermal cloud oscillate along the z-axis with a collective mode frequency close to the axial trap frequency ω_z. It is therefore convenient to compare the various transport relaxation rates with the trap frequency ω_z.

In Fig. 18.1, we plot the spatial dependence of both the condensate and noncondensate density profiles along the axial z-direction at $T \simeq 0.7T_{\rm BEC}$. These calculations are based on the TF approximation. In Fig. 18.2, we plot the spatial dependence of the inverse relaxation times. We find that all the relaxation rates are much larger than the trap frequency ω_z, and thus the collective oscillations, with the frequency $\omega \sim \omega_z$, should be described by collisional hydrodynamics. In Fig. 18.2, the open circles show the relaxation time $\tau_{\rm BE}$ defined in (18.41). As mentioned above, this empirical relaxation time is seen to be in excellent agreement with the thermal relaxation time τ_κ determined from the microscopic expressions (18.16) and (18.17). For comparison, the broken line shows the classical-gas elastic collision time (18.1), which only includes the C_{22} collisions between the atoms in the thermal cloud. Clearly this classical-gas collision rate is much smaller than all the transport relaxation rates. This shows the dominant role of the C_{12} collisions over C_{22} collisions in a trapped Bose-condensed gas, as discussed earlier.

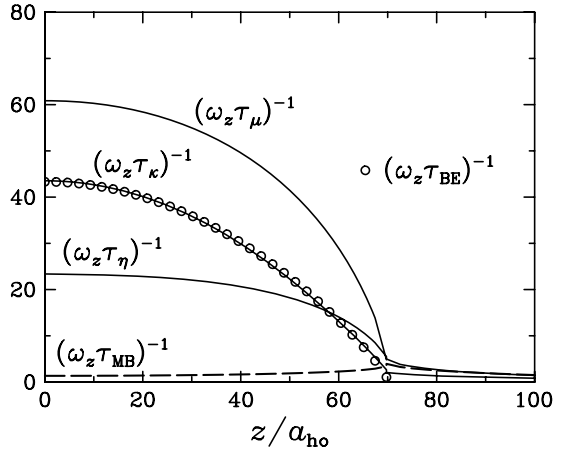


Fig. 18.2. The axial position dependence of the various (dimensionless) relaxation rates for ^{23}Na atoms, with $N = 4 \times 10^7$. The circles show the relaxation rate $1/\omega_z\tau_{\rm BE}$ calculated using (18.41) and the broken line is the classical collision rate $1/\omega_z\tau_{\rm MB}$ calculated using (18.1) (from Nikuni and Griffin, 2001b).

Outside the central condensate region τ_μ^{-1} vanishes, while τ_κ^{-1} and τ_η^{-1} only have contributions from the C_{22} collisions. One can show that they

are well approximated by the classical-gas results, namely $\tau_\kappa \simeq \frac{15}{8}\tau_{\rm MB}$ and $\tau_\eta \simeq \frac{5}{4}\tau_{\rm MB}$. In order to see the dependence on the total number of the atoms, in Fig. 18.3 we plot the relaxation times for a much smaller number of atoms, $N = 10^5$. In this case, all the relaxation rates are such that $\omega_z\tau \sim 1$ and thus the collective modes studied in such a gas would be in the transition region between the collisionless and collisional hydrodynamic domains. We recall that the collective modes in a trapped gas with a much larger number of atoms (as in Fig. 18.2) are predicted to be within the collisional hydrodynamic region.

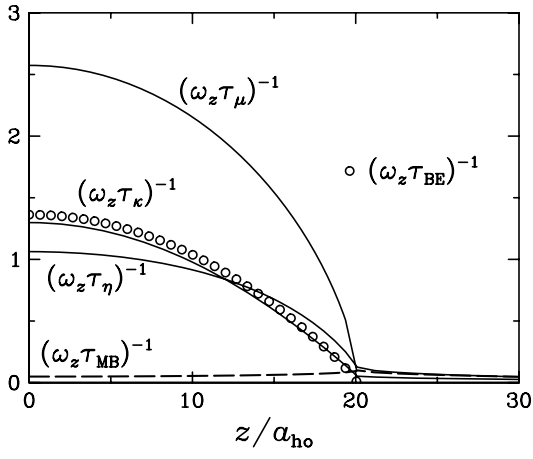


Fig. 18.3. The position dependence of the (dimensionless) relaxation rates, as in Fig. 18.2, for $N = 10^5$ at $T = 160\,n\mathrm{K} \simeq 0.8T_{\rm BEC}$ (from Nikuni and Griffin, 2001b).

We also note that the diffusive equilibrium relaxation time τ_μ is always slightly smaller than the other relaxation times (see Figs. 18.2 and 18.3). In principle one could have $\omega\tau_\mu \gg 1$ even in the collisional hydrodynamic region, where $\omega\tau_\kappa, \omega\tau_\eta \ll 1$. In a uniform Bose gas, this can occur in a fairly wide region near $T_{\rm BEC}$ where τ_μ becomes very large (see Nikuni *et al.*, 1999). In this region one expects to find a new relaxational mode (centred at zero frequency) in the superfluid phase. This relaxational mode, a unique feature of a Bose-condensed gas, is discussed in Section 15.3 and in Nikuni *et al.* (2000). However, the high condensate density in the centre of most traps (relative to the low density of the thermal cloud) precludes observation of this relaxational mode, except under special conditions. However, one might try to observe this new mode by using a quasi-one-dimensional trap such that the gas is almost uniform in the axial direction.

We can evaluate the peak value of the approximate relaxation rate in

(18.41) using the $T = 0$ Thomas–Fermi expression for the peak density $n_{c0}(0)$ at the centre of the trap. One finds the useful formula

$$\frac{1}{\bar{\omega}\tau_{\rm BE}(\mathbf{r}=0)} \simeq 6.47 \left(\frac{a}{a_{\rm ho}}\right)^{7/5} N^{17/30} \left(\frac{T}{T_{\rm BEC}}\right)^{1/2} \left(\frac{N_{\rm c}}{N}\right)^{2/5}, \qquad (18.43)$$

where the harmonic oscillator length $a_{\rm ho} \equiv (\hbar/m\bar{\omega})^{1/2}$ with $\bar{\omega} \equiv (\omega_x\omega_y\omega_z)^{1/3}$ is the average frequency of the anisotropic trap and $N_{\rm c}$ is the total number of the atoms in the condensate at temperature T. For a given value of N, the expression (18.43) can be used to estimate the temperature at which we pass from the collisionless to the hydrodynamic region. The crossover temperature is given by $\omega\tau_{\rm BE}(0) = 1$.

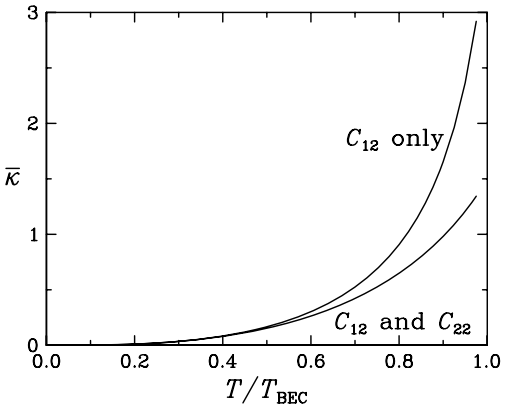


Fig. 18.4. The thermal conductivity $\bar{\kappa}$ in (18.44) in a uniform gas (the results are for $gn = 0.2k_{\rm B}T_{\rm BEC}$) as a function of temperature. The upper curve ignores the contribution from C_{22} collisions, i.e. I_{22}^{κ} is set to zero (from Nikuni and Griffin, 2001a).

To be in the two-fluid hydrodynamic region ($\omega\tau_{\rm BE} < 1$) one must work with $N \sim 10^6$ atoms or more when dealing with ^{87}Rb or ^{23}Na atoms. One can also make use of a highly anisotropic trap potential $\omega_x, \omega_y \gg \omega_z$ in order to produce a low-frequency collective mode, $\omega \sim \omega_z$, while keeping the high density due to tight confinement in the radial direction. However, the most promising way of achieving the condition of local hydrodynamic equilibrium is to work with a molecular Bose gas on the BEC side of a two-component Fermi superfluid, where a Feshbach resonance can be used to increase the s-wave scattering length (for further discussion see Section 16.4). The point to be emphasized is that if one is already close to the hydrodynamic domain above $T_{\rm BEC}$ as estimated using (18.1), one will enter deeply into the hydro-

dynamic region ($\omega\tau_{\rm BE} \ll 1$) once the Bose condensate forms. This is simply a result of the fact that in the highly localized condensate core region one always finds that $\tau_{\rm BE} \ll \tau_{\rm MB}$.

In trapped gases, one has a further complication in that while the thermal cloud in the condensate region may be in the collisional hydrodynamic region near the centre of the trap, eventually one always enters the collisionless region in the low density tail of the thermal cloud. This fact was not addressed in the derivation of the two-fluid hydrodynamic equations given in Chapters 15 and 17. While this has little effect on the frequency of a condensate mode, it may be important in the damping of other collective modes. In Chapter 19, we will discuss a new approach to deal with the inevitable collisionless region in the tail of the thermal cloud, based on using frequency-dependent transport coefficients.

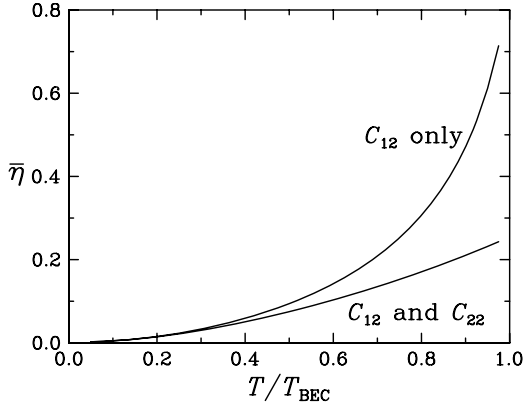


Fig. 18.5. The shear viscosity coefficient $\bar{\eta}$ in a uniform gas (for $gn = 0.2k_{\rm B}T_{\rm BEC}$) as a function of temperature. We also plot the results obtained when only C_{12} collisions are retained, i.e. setting $I^{\eta}_{22} = 0$ (from Nikuni and Griffin, 2001a).

As a concrete example, we calculate the transport coefficients (18.12) and (18.29) for a *uniform* Bose gas ($V_{\rm trap} = 0$). In Figs. 18.4 and 18.5, we plot the temperature dependence of the dimensionless transport coefficients $\bar{\kappa}$ and $\bar{\eta}$, defined by

$$\bar{\kappa} \equiv \frac{\kappa}{n v_{\rm cl}^2 \tau_0 k_{\rm B}}, \quad \bar{\eta} \equiv \frac{\eta}{n v_{\rm cl} m \tau_0}. \tag{18.44}$$

Here $v_{\rm cl} \equiv (5k_{\rm B}T_{\rm BEC}/3m)^{1/2}$ is the hydrodynamic sound velocity of a classical gas at $T = T_{\rm BEC}$ and $\tau_0^{-1} \equiv \sqrt{2}(8\pi a^2)n(8k_{\rm B}T_{\rm BEC}/\pi m)^{1/2}$ is the inverse of the classical-gas mean collision time (18.1) evaluated at $T = T_{\rm BEC}$. We also show the results obtained by keeping only the C_{12} collisions, i.e. setting

$I_{22}^{\kappa,\eta} = 0$. At lower temperatures $T \leq 0.5T_{\rm BEC}$, it can be seen that both κ and η are dominated by the C_{12} collisions. In Fig. 18.6, we plot the four second viscosity coefficients given in (18.40) for a uniform Bose gas, using the dimensionless expressions

$$\bar{\zeta}_1 \equiv \frac{\zeta_1}{v_{\rm cl}^2 \tau_0}, \quad \bar{\zeta}_2 \equiv \frac{\zeta_2 m}{n v_{\rm cl}^2 \tau_0}, \quad \bar{\zeta}_3 \equiv \frac{\zeta_2 n}{m v_{\rm cl}^2 \tau_0}. \tag{18.45}$$

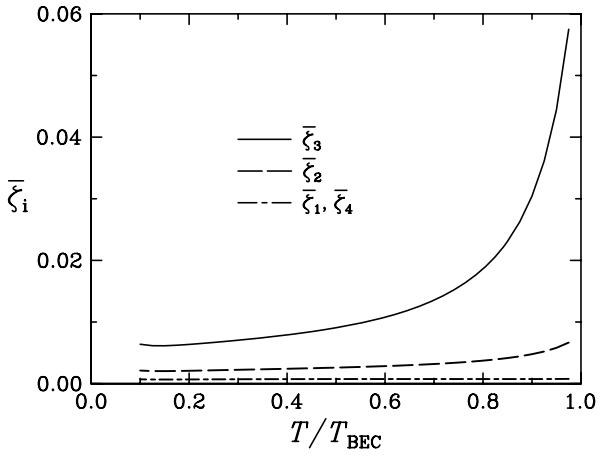


Fig. 18.6. The second viscosity coefficients $\bar{\zeta}_i$ defined in (18.45) for a uniform Bose-condensed gas (for $gn = 0.2k_{\rm B}T_{\rm BEC}$) as a function of temperature. These second viscosity coefficients vanish for $T > T_{\rm BEC}$ (from Nikuni and Griffin, 2001a).

We emphasize that the transport coefficients plotted in Figs. 18.4–18.6 are for a uniform Bose-condensed gas. The transport coefficients in a trapped gas are very different because of the increased contribution of the C_{12} collisions. This is due to the high condensate density relative to the thermal cloud.

18.3 Kinetic equations versus Kubo formulas

In many textbook treatments, rather than starting from a kinetic equation, the two-fluid equations are derived from general conservation laws and constitutive relations describing currents. In such discussions, transport coefficients appear as linear response functions. As shown in the classic papers of Kadanoff and Martin (1963) and Hohenberg and Martin (1965), one can invert such macroscopic two-fluid hydrodynamic equations to give expressions for various correlation functions which are valid in the low $\mathbf{q}$ and ω colli-

sional hydrodynamic region. The correlation functions exhibit poles that correspond to hydrodynamic collective modes, with damping proportional to the various transport coefficients.

The density response function in the two-fluid region with no dissipation was discussed in Section 14.3. The generalization of these results to include hydrodynamic damping is given in Hohenberg and Martin (1965). One can use them to express the transport coefficients explicitly in terms of current–current correlation functions evaluated in a thermal equilibrium ensemble (such expressions were first discussed by Kubo (1957) and are known as Kubo formulas). One can calculate transport coefficients in a superfluid Bose gas starting from these current correlation functions. This is an alternative to the Chapman–Enskog procedure that makes use of a kinetic equation.

Kirkpatrick and Dorfman (1985b) illustrated this Kubo formula approach by directly calculating the transport coefficients of a dilute weakly interacting uniform Bose gas starting from correlation functions. They reproduced the same results as those obtained using the Chapman–Enskog approach (Kirkpatrick and Dorfman, 1985a). The advantage of the correlation function approach is that in a strongly interacting gas (such as a Fermi superfluid near unitarity in the BCS–BEC crossover region), kinetic equations based on second-order binary collisions (with transition rates $\propto a^2$) are not obviously valid. In contrast, one may be able to use powerful non-perturbative techniques to calculate transport coefficients; see for example Son (2007) and Schaefer (2007) and references therein. For a recent discussion of the viscosity transport coefficient in a normal Fermi gas near unitarity, see Bruun and Smith (2007).

19

General theory of damping of hydrodynamic modes

In Chapter 17, we derived the Landau–Khalatnikov two-fluid hydrodynamics which describes the collision-dominated region of a trapped Bose condensate interacting with a thermal cloud. In this chapter, we use these equations to discuss the damping of the hydrodynamic collective modes in a trapped Bose gas at finite temperatures. We derive variational expressions based on these equations for both the frequency and the damping of collective modes. This extends the analysis in Chapter 16 in which a variational approach was developed to calculate the hydrodynamic two-fluid oscillation frequencies in the non-dissipative limit. A novel feature of our treatment is the introduction of *frequency-dependent* transport coefficients, which produce a natural cutoff eliminating the collisionless region in the low-density tail of the thermal cloud. Our expression for the damping in trapped superfluid Bose gases is a natural generalization of the approach used by Landau and Lifshitz (1959) for uniform classical fluids. This chapter is mainly based on Nikuni and Griffin (2004).

In Chapters 15 and 17, we derived a closed set of two-fluid hydrodynamic equations for a trapped Bose-condensed gas starting from the simplified microscopic model describing the coupled dynamics of the condensate and noncondensate atoms given in Chapter 3. These hydrodynamic two-fluid equations include dissipative terms associated with the shear viscosity, the thermal conductivity and the four second viscosity coefficients. Explicit formulas for these transport coefficients were derived in Section 18.1. Our goal in this chapter is to find a general expression for the damping of the two-fluid modes in terms of these transport coefficients. We emphasize that the damping of hydrodynamic two-fluid oscillations is completely different in nature from the Landau and Beliaev damping of oscillations in the collisionless region which is treated in Chapters 12 and 13.

In Section 19.1, we give a brief review of the two-fluid equations derived in Chapter 17. In Section 19.2, we derive a general formula for the damping, which depends only on knowing undamped normal mode solutions (such as those given by the variational approach in Chapter 16) and the transport coefficients (given in Chapter 18). This kind of expression is very convenient when working out the damping of hydrodynamic modes in *trapped* Bose gases, as first pointed out by Kavoulakis *et al.* (1998a). In Section 19.4, we will illustrate these results by calculating the damping of the monopole–quadrupole two-fluid modes in trapped Bose gases.

Appendix C gives some details of damping calculations for hydrodynamic modes based on the use of *frequency-dependent* transport coefficients. These allow a very natural way of dealing with the fact that, in a trapped Bose gas, the outer regions of the low-density thermal cloud will always be in the collisionless limit.

19.1 Review of coupled equations for hydrodynamic modes

The linearized hydrodynamic equations for the condensate are given by

$$\begin{aligned} \frac{\partial \delta n_{\rm c}}{\partial t} &= -\nabla \cdot (n_{\rm c0} \delta \mathbf{v}_{\rm c}) - \delta\Gamma_{12}, \\ m \frac{\partial \delta \mathbf{v}_{\rm c}}{\partial t} &= -g \nabla (\delta n_{\rm c} + 2\delta \tilde{n}). \end{aligned} \tag{19.1}$$

The hydrodynamic equations for the noncondensate atoms (the normal fluid) are given by (see footnote 2 in Chapter 17)

$$\begin{aligned} \frac{\partial \delta \tilde{n}}{\partial t} &= -\nabla \cdot (\tilde{n}_0 \delta \mathbf{v}_{\rm n}) + \delta\Gamma_{12}, \\ m\tilde{n}_0 \frac{\partial \delta v_{{\rm n}\mu}}{\partial t} &= -\frac{\partial \delta \tilde{P}}{\partial x_\mu} - \delta\tilde{n} \frac{\partial U_0}{\partial x_\mu} - 2g\tilde{n}_0 \frac{\partial \delta n}{\partial x_\mu} \\ &\quad + \frac{\partial}{\partial x_\nu} \left[2\eta \left(D_{\mu\nu} - \tfrac{1}{3}\delta_{\mu\nu} {\rm Tr}\, D \right) \right], \\ \frac{\partial \delta \tilde{P}}{\partial t} &= -\tfrac{5}{3} \nabla \cdot (\tilde{P}_0 \delta \mathbf{v}_{\rm n}) + \tfrac{2}{3} \delta \mathbf{v}_{\rm n} \cdot \nabla \tilde{P}_0 \\ &\quad + (\mu_{\rm c0} - U_0) \delta\Gamma_{12} + \tfrac{2}{3} \nabla \cdot (\kappa \nabla \delta T). \end{aligned} \tag{19.2}$$

These equations are the linearized versions of results (17.50) – (17.52). They involve the fluctuations of the source function and the temperature, $\delta\Gamma_{12}$ and δT. These fluctuations can also be written in terms of the condensate and noncondensate velocity fields, as discussed in Chapter 18. Defining

$\delta\Gamma_{12} = \delta\Gamma_{12}^{(1)} + \delta\Gamma_{12}^{(2)}$, one has

$$\begin{aligned}\delta\Gamma_{12}^{(1)} &= \sigma_{\rm H}\left\{\nabla\cdot[n_{\rm c0}(\delta\mathbf{v}_{\rm c}-\delta\mathbf{v}_{\rm n})]+\tfrac{1}{3}n_{\rm c0}\nabla\cdot\delta\mathbf{v}_{\rm n}\right\},\\ \delta\Gamma_{12}^{(2)} &= -\tau_\mu\frac{\partial}{\partial t}\delta\Gamma_{12}^{(1)}-\frac{2\sigma_{\rm H}\sigma_1}{3g\tilde{n}_0}\nabla\cdot(\kappa\nabla\delta T),\end{aligned} \tag{19.3}$$

and (as in Chapter 18, T denotes the equilibrium temperature)

$$\frac{\partial\delta T}{\partial t} = -\frac{2}{3}T\nabla\cdot\delta\mathbf{v}_{\rm n}+\frac{2T\sigma_1}{3\tilde{n}_0}\delta\Gamma_{12}^{(1)}. \tag{19.4}$$

These are the linearized versions of (17.20), (17.69) and the first equation in (17.33), respectively. The thermodynamic functions $\sigma_{\rm H}$ and σ_1 involve only the static local thermodynamic functions of the trapped gas and are defined in (15.82) and (17.34), respectively. In (19.3), τ_μ is the relaxation time given by (17.22) (see also (15.72)). It describes how fast the condensate and noncondensate atoms reach *diffusive* local equilibrium with each other as a result of C_{12} collisions. We remark that the expression for σ_1 in (17.34) can be reduced to

$$\sigma_1 = -\tfrac{3}{2}\tilde{n}_0 g\left(\frac{\tilde{n_0}+\frac{2}{3}\gamma^{(0)}gn_{\rm c0}}{\frac{5}{2}\tilde{P}_0 g\gamma^{(0)}-\frac{3}{2}g\tilde{n}_0^2}\right). \tag{19.5}$$

Using this expression, we see that (19.4) is equivalent to (15.69).

To derive a closed set of equations for the velocity fields $\mathbf{v}_{\rm c}$ and $\mathbf{v}_{\rm n}$, we take the time derivatives of (19.1) and (19.2). This gives

$$m\frac{\partial^2\delta\mathbf{v}_{\rm c}}{\partial t^2} = g\nabla[\nabla\cdot(n_{\rm c0}\delta\mathbf{v}_{\rm c})+2\nabla\cdot(\tilde{n}_0\delta\mathbf{v}_{\rm n})]-g\nabla\delta\Gamma_{12}, \tag{19.6}$$

and

$$\begin{aligned}m\tilde{n}_0\frac{\partial^2\delta v_{{\rm n}\mu}}{\partial t^2} &= \frac{5}{3}\frac{\partial}{\partial x_\mu}\left[\nabla\cdot(\tilde{P}_0\mathbf{v}_{\rm n})\right]-\frac{2}{3}\frac{\partial}{\partial x_\mu}(\delta\mathbf{v}_{\rm n}\cdot\nabla\tilde{P}_0)+\nabla\cdot(\tilde{n}_0\delta\mathbf{v}_{\rm n})\frac{\partial U_0}{\partial x_\mu}\\ &\quad+2g\tilde{n}_0\frac{\partial}{\partial x_\mu}\left[\nabla\cdot(n_{\rm c0}\delta\mathbf{v}_{\rm c})+\nabla\cdot(\tilde{n}_0\delta\mathbf{v}_{\rm n})\right]\\ &\quad-\frac{1}{3}\frac{\partial U_0}{\partial x_\mu}\delta\Gamma_{12}+\frac{2}{3}gn_{\rm c0}\frac{\partial\delta\Gamma_{12}}{\partial x_\mu}\\ &\quad+\frac{\partial}{\partial x_\nu}\left\{2\eta\left[\frac{\partial}{\partial t}D_{\mu\nu}-\frac{1}{3}\delta_{\mu\nu}\left({\rm Tr}\,\frac{\partial D}{\partial t}\right)\right]\right\}\\ &\quad-\frac{2}{3}\frac{\partial}{\partial x_\mu}\nabla\cdot(\kappa\nabla\delta T).\end{aligned} \tag{19.7}$$

We now consider normal mode solutions of the form

$$\mathbf{v}_{\rm n}(\mathbf{r},t)\equiv\mathbf{v}_{\rm n}(\mathbf{r})e^{-i\omega t},\qquad \mathbf{v}_{\rm c}(\mathbf{r},t)\equiv\mathbf{v}_{\rm c}(\mathbf{r})e^{-i\omega t}. \tag{19.8}$$

The coupled equations (19.6) and (19.7) reduce to

$$m\omega^2\mathbf{v}_{\rm c} = -g\nabla[\nabla\cdot(n_{\rm c0}\mathbf{v}_{\rm c})] - 2g\nabla[\nabla\cdot(\tilde{n}_0\mathbf{v}_{\rm n})] + g\nabla\delta\Gamma_{12,\omega}[\mathbf{v}_{\rm n},\mathbf{v}_{\rm c}] \quad (19.9)$$

and

$$\begin{aligned} m\tilde{n}_0\omega^2 v_{{\rm n}\mu} = &-\frac{5}{3}\frac{\partial}{\partial x_\mu}\Big[\nabla\cdot(\tilde{P}_0\mathbf{v}_{\rm n})\Big] + \frac{2}{3}\frac{\partial}{\partial x_\mu}(\mathbf{v}_{\rm n}\cdot\nabla\tilde{P}_0) - \nabla\cdot(\tilde{n}_0\mathbf{v}_{\rm n})\frac{\partial U_0}{\partial x_\mu} \\ &- 2g\tilde{n}_0\frac{\partial}{\partial x_\mu}\Big[\nabla\cdot(n_{\rm c0}\mathbf{v}_{\rm c}) + \nabla\cdot(\tilde{n}_0\mathbf{v}_{\rm n})\Big] \\ &+ \frac{1}{3}\frac{\partial U_0}{\partial x_\mu}\delta\Gamma_{12,\omega}[\mathbf{v}_{\rm n},\mathbf{v}_{\rm c}] - \frac{2}{3}gn_{\rm c0}\frac{\partial\delta\Gamma_{12,\omega}[\mathbf{v}_{\rm n},\mathbf{v}_{\rm c}]}{\partial x_\mu} \\ &+ i\omega\frac{\partial}{\partial x_\nu}\left\{2\eta\left[v_{{\rm n}\mu\nu} - \frac{1}{3}\delta_{\mu\nu}\left(\nabla\cdot\mathbf{v}_{\rm n}\right)\right]\right\} \\ &+ \frac{2}{3}\frac{\partial}{\partial x_\mu}\nabla\cdot(\kappa\nabla\delta T_\omega[\mathbf{v}_{\rm n},\mathbf{v}_{\rm c}]), \end{aligned} \quad (19.10)$$

where the temperature fluctuation is given by $\delta T \equiv \delta T_\omega[\mathbf{v}_{\rm n},\mathbf{v}_{\rm c}]e^{-i\omega t}$. The symmetric tensor $v_{{\rm n}\mu\nu}$ associated with the normal fluid velocity amplitude is defined by

$$v_{{\rm n}\mu\nu} \equiv \frac{1}{2}\left(\frac{\partial v_{{\rm n}\nu}}{\partial x_\mu} + \frac{\partial v_{{\rm n}\mu}}{\partial x_\nu}\right). \quad (19.11)$$

The source function $\delta\Gamma_{12}$ appearing in the above equations can be expressed in terms of of $\mathbf{v}_{\rm n}$ and $\mathbf{v}_{\rm c}$. Setting $\delta\Gamma_{12,\omega} = \delta\Gamma_{12,\omega}[\mathbf{v}_{\rm n},\mathbf{v}_{\rm c}]e^{-i\omega t}$, one finds that

$$\delta\Gamma_{12,\omega}[\mathbf{v}_{\rm n},\mathbf{v}_{\rm c}] = \delta\Gamma^{(1)}_{12,\omega}[\mathbf{v}_{\rm n},\mathbf{v}_{\rm c}] + \delta\Gamma^{(2)}_{12,\omega}[\mathbf{v}_{\rm n},\mathbf{v}_{\rm c}], \quad (19.12)$$

with (using (19.3))

$$\delta\Gamma^{(1)}_{12,\omega}[\mathbf{v}_{\rm n},\mathbf{v}_{\rm c}] = \sigma_{\rm H}\left\{\nabla\cdot[n_{\rm c0}(\mathbf{v}_{\rm c}-\mathbf{v}_{\rm n})] + \frac{1}{3}n_{\rm c0}\nabla\cdot\mathbf{v}_{\rm n}\right\}, \quad (19.13)$$

and (again using (19.3))

$$\delta\Gamma^{(2)}_{12,\omega} = i\omega\tau_\mu\delta\Gamma^{(1)}_{12,\omega}[\mathbf{v}_{\rm n},\mathbf{v}_{\rm c}] - \frac{2\sigma_{\rm H}\sigma_1}{3g\tilde{n}_0}\nabla\cdot(\kappa\delta T_\omega[\mathbf{v}_{\rm n},\mathbf{v}_{\rm c}]). \quad (19.14)$$

Similarly, (19.4) gives

$$\delta T_\omega[\mathbf{v}_{\rm n},\mathbf{v}_{\rm c}] = -\frac{1}{i\omega}\left[-\frac{2}{3}T(\nabla\cdot\mathbf{v}_{\rm n}) + \frac{2}{3}\frac{T}{\tilde{n}_0}\sigma_1\delta\Gamma^{(1)}_{12,\omega}[\mathbf{v}_{\rm n},\mathbf{v}_{\rm c}]\right]. \quad (19.15)$$

Using these results in (19.9) and (19.10), we obtain a closed set of equations for the local velocity amplitudes $\mathbf{v}_{\rm n}$ and $\mathbf{v}_{\rm c}$ defined in (19.8).

19.2 Normal mode frequencies

We first consider the undamped normal mode solutions of our collisional hydrodynamic equations, neglecting all hydrodynamic dissipation. Formally this means that we set η, κ, τ_μ to zero in the two-fluid hydrodynamic equations. As discussed in Chapter 15, this limit corresponds to the Landau two-fluid collisional hydrodynamics without dissipation. In this limit, the coupled equations (19.9) and (19.10) for $\mathbf{v}_{\rm c}$ and $\mathbf{v}_{\rm n}$ reduce to

$$m\omega^2\mathbf{v}_{\rm c} = -g\nabla[\nabla\cdot(n_{\rm c0}\mathbf{v}_{\rm c})] - 2g\nabla[\nabla\cdot(\tilde{n}_0\mathbf{v}_{\rm n})] + g\nabla\delta\Gamma_{12}^{(1)}[\mathbf{v}_{\rm n},\mathbf{v}_{\rm c}], \quad (19.16)$$

$$\begin{aligned} m\omega^2\mathbf{v}_{\rm n} = &-\tfrac{5}{3}\nabla[\nabla\cdot(\tilde{P}_0\mathbf{v}_{\rm n})] + \tfrac{2}{3}\nabla[\mathbf{v}_{\rm n}\cdot\nabla\tilde{P}_0] - \tfrac{2}{3}\nabla\left(gn_{\rm c0}\delta\Gamma_{12}^{(1)}[\mathbf{v}_{\rm n},\mathbf{v}_{\rm c}]\right) \\ &- \delta\Gamma_{12}^{(1)}[\mathbf{v}_{\rm n},\mathbf{v}_{\rm c}]\nabla U_0 - \nabla\cdot(\tilde{n}_0\mathbf{v}_{\rm n})\nabla U_0 \\ &- 2g\tilde{n}_0\nabla\left[\nabla\cdot(\tilde{n}_0\mathbf{v}_{\rm n}) + \nabla\cdot(n_{\rm c0}\mathbf{v}_{\rm c})\right]. \end{aligned} \quad (19.17)$$

The ω-dependence of $\delta\Gamma_{12,\omega}^{(1)}$ is left implicit. In general, the solutions of these coupled hydrodynamic equations are very complicated in the presence of a trapping potential. An alternative approach is to reformulate them so that their solutions are given in terms of a variational functional, as we did in Chapter 16 starting from the usual Landau form of the two-fluid equations. The present discussion closely follows the variational analysis originally developed[1] by Zaremba *et al.* (1999). The same formalism was also used by Nikuni and Griffin (1998b) to calculate the hydrodynamic mode frequencies for a trapped Bose gas above $T_{\rm BEC}$.

From the discussion in Chapters 15 and 17, we know that the two-fluid equations (19.16) and (19.17) are equivalent to the usual Landau two-fluid equations. The only difference is the choice of variables. Thus the present variational formulation will lead to the same two-fluid collective modes as the variational formulation given in Chapter 16.

It is convenient to introduce a *six-component* local velocity vector

$$\boldsymbol{v} = \begin{pmatrix} \mathbf{v}_{\rm n} \\ \mathbf{v}_{\rm c} \end{pmatrix}, \quad (19.18)$$

We can then write the coupled equations for $\mathbf{v}_{\rm n}$ and $\mathbf{v}_{\rm c}$ in (19.17) and (19.16) in the form of a 6×6 matrix equation

$$\boldsymbol{A}\boldsymbol{v} = \omega^2\boldsymbol{D}\boldsymbol{v}. \quad (19.19)$$

[1] To avoid confusion, the reader should note that Zaremba *et al.* (1999) actually discussed the variational solutions of what we call the ZGN version of two-fluid hydrodynamics, in which the effect of the C_{12} collisions is ignored (see Section 15.3 for more details).

The 6×6 matrix $\boldsymbol{A}$ has the following block structure:

$$\boldsymbol{A} = \begin{pmatrix} \mathbf{A}_{11}\ \mathbf{A}_{12} \\ \mathbf{A}_{21}\ \mathbf{A}_{22} \end{pmatrix}, \tag{19.20}$$

where the 3×3 matrix elements $\mathbf{A}_{ij}$ are defined by the vector components[2]

$$\begin{aligned}(\mathbf{A}_{11}\mathbf{v}_{\rm n})_\mu \equiv & -\frac{5}{3}\frac{\partial}{\partial x_\mu}[\nabla\cdot(\tilde{P}_0\mathbf{v}_{\rm n})] + \frac{2}{3}\frac{\partial}{\partial x_\mu}(\mathbf{v}_{\rm n}\cdot\nabla\tilde{P}_0) \\ & - \nabla\cdot(\tilde{n}_0\mathbf{v})\frac{\partial U_0}{\partial x_\mu} - 2g\tilde{n}_0\frac{\partial}{\partial x_\mu}[\nabla\cdot(\tilde{n}_0\mathbf{v}_{\rm n})] \\ & - \frac{\partial}{\partial x_\mu}\left(\frac{2}{3}gn_{\rm c0}\delta\Gamma_{12}^{(1)}[\mathbf{v}_{\rm n},0]\right) + \delta\Gamma_{12}^{(1)}[\mathbf{v}_{\rm n},0]\frac{\partial U_0}{\partial x_\mu}, \end{aligned} \tag{19.21}$$

$$\begin{aligned}(\mathbf{A}_{12}\mathbf{v}_{\rm c})_\mu \equiv & -2g\tilde{n}_0\frac{\partial}{\partial x_\mu}\nabla\cdot(n_{\rm c0}\mathbf{v}_{\rm c}) - \frac{\partial}{\partial x_\mu}\left(\frac{2}{3}gn_{\rm c0}\delta\Gamma_{12}^{(1)}[0,\mathbf{v}_{\rm c}]\right) \\ & + \delta\Gamma_{12}^{(1)}[0,\mathbf{v}_{\rm c}]\frac{\partial U_0}{\partial x_\mu}, \end{aligned} \tag{19.22}$$

$$(\mathbf{A}_{21}\mathbf{v}_{\rm n})_\mu \equiv -2gn_{\rm c0}\frac{\partial}{\partial x_\mu}\nabla\cdot(\tilde{n}_0\mathbf{v}_{\rm n}) + gn_{\rm c0}\frac{\partial}{\partial x_\mu}\delta\Gamma_{12}^{(1)}[\mathbf{v}_{\rm n},0], \tag{19.23}$$

$$(\mathbf{A}_{22}\mathbf{v}_{\rm c})_\mu \equiv -gn_{\rm c0}\frac{\partial}{\partial x_\mu}\nabla\cdot(n_{\rm c0}\mathbf{v}_{\rm c}) + gn_{\rm c0}\frac{\partial}{\partial x_\mu}\delta\Gamma_{12}^{(1)}[0,\mathbf{v}_{\rm c}]. \tag{19.24}$$

The 6×6 matrix $\boldsymbol{D}$ in (19.19) is block-diagonal ($\mathbf{D}_{12} = \mathbf{D}_{21} = 0$), with diagonal elements $\mathbf{D}_{11} = m\tilde{n}_0\mathbf{I}$, $\mathbf{D}_{22} = mn_{\rm c0}\mathbf{I}$, where $\mathbf{I}$ is the 3×3 unit matrix. We note that the matrix $\boldsymbol{A}$ has the important Hermitian property

$$\int d\mathbf{r}\,\boldsymbol{v}'\cdot(\boldsymbol{A}\boldsymbol{v}) = \int d\mathbf{r}\,\boldsymbol{v}\cdot(\boldsymbol{A}\boldsymbol{v}'), \tag{19.25}$$

where $\boldsymbol{v}' \equiv (\mathbf{v}'_{\rm n}, \mathbf{v}'_{\rm c})$.

The coupled equations (19.16) and (19.17) can be rewritten in terms of the variational functional

$$J[\mathbf{v}_{\rm n}, \mathbf{v}_{\rm c}] \equiv \frac{U[\mathbf{v}_{\rm n}, \mathbf{v}_{\rm c}]}{K[\mathbf{v}_{\rm n}, \mathbf{v}_{\rm c}]}, \tag{19.26}$$

where

$$U[\mathbf{v}_{\rm n}, \mathbf{v}_{\rm c}] \equiv \frac{1}{2}\int d\mathbf{r}\,\boldsymbol{v}\cdot(\boldsymbol{A}\boldsymbol{v}), \qquad K[\mathbf{v}_{\rm n}, \mathbf{v}_{\rm c}] \equiv \frac{1}{2}\int d\mathbf{r}\,\boldsymbol{v}\cdot(\boldsymbol{D}\boldsymbol{v}). \tag{19.27}$$

Using the Hermitian property (19.25) of $\boldsymbol{A}$ one can prove that the requirement that the functional J be stationary leads to the equations (19.16) and

[2] We note that 6×6 matrices as in (19.20) and six-component vectors as in (19.18) are given in bold italic; 3×3 matrices and three-component vectors are denoted as bold roman.

(19.17), ω^2 being identified with the stationary value of the functional J in (19.26). One can therefore calculate the normal mode frequencies ω^2 using a physically reasonable variational ansatz for $\mathbf{v}_{\rm n}$ and $\mathbf{v}_{\rm c}$ in the functional $J[\mathbf{v}_{\rm n}, \mathbf{v}_{\rm c}]$ and then minimizing the expression.

19.3 General expression for damping of hydrodynamic modes

In this section, we derive a general expression for the hydrodynamic damping of a collective mode due to transport processes. This kind of expression for hydrodynamic damping in a trapped Bose gas was first derived above $T_{\rm BEC}$ (and later generalized to below $T_{\rm BEC}$) by Nikuni and Griffin (1998a, 2004). A reader not interested in the derivation can go immediately to the final, relatively simple, result for the damping in (19.46).

In what follows, we retain the terms involving the transport coefficients κ, η and τ_μ in the two-fluid equations given in Section 19.1. Starting from the coupled equations (19.9) and (19.10) for $\mathbf{v}_{\rm c}$ and $\mathbf{v}_{\rm n}$, we can write them in matrix form:

$$\boldsymbol{A}\boldsymbol{v} + \boldsymbol{F}\boldsymbol{v} = \omega^2 \boldsymbol{D}\boldsymbol{v}. \tag{19.28}$$

This generalizes the non-dissipative result (19.19): the 6×6 matrix $\boldsymbol{F}$ represents the dissipative terms in the two-fluid equations and has a block structure made up of 3×3 matrices (see footnote 2)

$$\boldsymbol{F} = \begin{pmatrix} \mathbf{F}_{11} & \mathbf{F}_{12} \\ \mathbf{F}_{21} & \mathbf{F}_{22} \end{pmatrix}. \tag{19.29}$$

These 3×3 matrix elements $\mathbf{F}_{ij}$ are given by

$$\begin{aligned}
(\mathbf{F}_{11}\mathbf{v}_{\rm n})_\mu = -\frac{\partial}{\partial x_\mu} \Bigg\{ & \frac{2}{3} g n_{\rm c0} i\omega \tau_\mu \delta\Gamma_{12}^{(1)}[\mathbf{v}_{\rm n}, 0] \\
& - \left(\frac{2}{3} + \frac{4 n_{\rm c0} \sigma_{\rm H} \sigma_1}{9\tilde{n}_0} \right) \nabla \cdot (\kappa \nabla \delta T_\omega[\mathbf{v}_{\rm n}, 0]) \Bigg\} \\
& + \left\{ i\omega\tau_\mu \delta\Gamma_{12}^{(1)}[\mathbf{v}_{\rm n}, 0] - \frac{2\sigma_{\rm H}\sigma_1}{3g\tilde{n}_0} \nabla \cdot (\kappa \nabla \delta T_\omega[\mathbf{v}_{\rm n}, 0]) \right\} \frac{\partial U_0}{\partial x_\mu} \\
& + i\omega \frac{\partial}{\partial x_\nu} 2\eta \left(v_{{\rm n}\mu\nu} - \frac{1}{3}\delta_{\mu\nu} \nabla \cdot \mathbf{v}_{\rm n} \right),
\end{aligned}$$

$$
\begin{aligned}
(\mathbf{F}_{12}\mathbf{v}_{\rm c})_\mu = -\frac{\partial}{\partial x_\mu}\Bigg\{ & \frac{2}{3} g n_{\rm c0}(i\omega\tau_\mu)\delta\Gamma_{12}^{(1)}[0,\mathbf{v}_{\rm c}] \\
& - \left(\frac{2}{3} + \frac{4n_{\rm c0}\sigma_{\rm H}\sigma_1}{9\tilde{n}_0}\right)\nabla\cdot(\kappa\nabla\delta T_\omega[0,\mathbf{v}_{\rm c}])\Bigg\} \\
& + \left\{ i\omega\tau_\mu\delta\Gamma_{12}^{(1)}[0,\mathbf{v}_{\rm c}] - \frac{2\sigma_{\rm H}\sigma_1}{3g\tilde{n}_0}\nabla\cdot(\kappa\nabla\delta T_\omega[0,\mathbf{v}_{\rm c}])\right\}\frac{\partial U_0}{\partial x_\mu}, \\
(\mathbf{F}_{21}\mathbf{v}_{\rm n})_\mu = & \, g n_{\rm c0}\frac{\partial}{\partial x_\mu}\left\{ i\omega\tau_\mu\delta\Gamma_{12}^{(1)}[\mathbf{v}_{\rm n},0] - \frac{2\sigma_{\rm H}\sigma_1}{3g\tilde{n}_0}\nabla\cdot(\kappa\nabla\delta T_\omega[\mathbf{v}_{\rm n},0])\right\}, \\
(\mathbf{F}_{22}\mathbf{v}_{\rm c})_\mu = & \, g n_{\rm c0}\frac{\partial}{\partial x_\mu}\left\{ i\omega\tau_\mu\delta\Gamma_{12}^{(1)}[0,\mathbf{v}_{\rm c}] - \frac{2\sigma_{\rm H}\sigma_1}{3g\tilde{n}_0}\nabla\cdot(\kappa\nabla\delta T_\omega[0,\mathbf{v}_{\rm c}])\right\}.
\end{aligned}
\tag{19.30}
$$

In our subsequent analysis, we will treat $\boldsymbol{F}$ in (19.28) as a small perturbation to the equation (19.19).

To find a solution for $\mathbf{v}_{\rm n}, \mathbf{v}_{\rm c}$ including damping, we expand the six-dimensional vector $\boldsymbol{v}$ in terms of undamped normal mode solutions $\boldsymbol{v}_\alpha$ (α denotes the mode label, not a cartesian component):

$$
\boldsymbol{v} = \sum_\alpha C_\alpha \boldsymbol{v}_\alpha, \qquad \text{where} \qquad \boldsymbol{A}\boldsymbol{v}_\alpha = \omega_\alpha^2 \boldsymbol{D}\boldsymbol{v}_\alpha. \tag{19.31}
$$

From the Hermitian property of the operator $\boldsymbol{A}$, one can show that these normal mode solutions $\mathbf{v}_\alpha$ satisfy the orthonormality relation

$$
\begin{aligned}
\int d\mathbf{r}\, \boldsymbol{v}_\alpha\cdot(\boldsymbol{D}\boldsymbol{v}_\beta) &= \int d\mathbf{r}\Big(m n_{\rm c0}\mathbf{v}_{{\rm c}\alpha}\cdot\mathbf{v}_{{\rm c}\beta} + m\tilde{n}_0\mathbf{v}_{{\rm n}\alpha}\cdot\mathbf{v}_{{\rm n}\beta}\Big) \\
&= \delta_{\alpha\beta},
\end{aligned}
\tag{19.32}
$$

where $\boldsymbol{D}$ is defined below (19.24). We assume that the normal mode solutions are normalized. Making use of (19.32), we obtain a linear equation for the coefficients C_α in (19.31) :

$$
(\omega^2 - \omega_\alpha^2)C_\alpha = \int d\mathbf{r}\, \boldsymbol{v}_\alpha\cdot\sum_{\alpha'} C_{\alpha'}\boldsymbol{F}\boldsymbol{v}_{\alpha'} \equiv \sum_{\alpha'} V_{\alpha\alpha'}C_{\alpha'}, \tag{19.33}
$$

where the matrix element $V_{\alpha\alpha'}$ is defined by

$$
V_{\alpha\alpha'} \equiv \int d\mathbf{r}\, \boldsymbol{v}_\alpha\cdot\boldsymbol{F}\boldsymbol{v}_{\alpha'}. \tag{19.34}
$$

We note that $V_{\alpha\alpha'}$ depends on the mode frequency ω.

Expanding the mode frequency to first order in the perturbation $\boldsymbol{F}$ as

$\omega = \omega_\alpha + \Delta\omega_\alpha$, we obtain

$$\Delta\omega_\alpha = \frac{1}{2\omega_\alpha} V_{\alpha\alpha}(\omega_\alpha), \tag{19.35}$$

where

$$V_{\alpha\alpha}(\omega_\alpha) \equiv -i\omega_\alpha \int d\mathbf{r} \Bigg\{ \frac{g\tau_\mu}{\sigma_{\rm H}} \left(\delta\Gamma_{12}^{(1)}[\mathbf{v}_{{\rm n}\alpha}, \mathbf{v}_{{\rm c}\alpha}] \right)^2 + \frac{\kappa}{T} |\nabla \delta T_{\omega_\alpha}[\mathbf{v}_{{\rm n}\alpha}, \mathbf{v}_{{\rm c}\alpha}]|^2 + \frac{\eta}{2} \left(\frac{\partial v_{{\rm n}\alpha\nu}}{\partial x_\mu} + \frac{\partial v_{{\rm n}\alpha\mu}}{\partial x_\nu} - \frac{2}{3}\delta_{\mu\nu} \nabla \cdot \mathbf{v}_{{\rm n}\alpha} \right)^2 \Bigg\}. \tag{19.36}$$

Since the last term in (19.36) involves a square, it is implicit that the Cartesian component indices μ and ν are summed over. The detailed calculations leading to (19.36) are given in Appendix D. Using (19.36), we conclude that $\Delta\omega_\alpha = -i\Gamma_\alpha$, where the damping rate Γ_α of the normalized mode α is given by

$$\Gamma_\alpha = \frac{1}{2} \int d\mathbf{r} \Bigg[\frac{g\tau_\mu}{\sigma_{\rm H}} \left(\delta\Gamma_{12}^{(1)}[\mathbf{v}_{{\rm n}\alpha}, \mathbf{v}_{{\rm c}\alpha}] \right)^2 + \frac{\kappa}{T} |\nabla \delta T_{\omega_\alpha}[\mathbf{v}_{{\rm n}\alpha}, \mathbf{v}_{{\rm c}\alpha}]|^2 + \frac{\eta}{2} \left(\frac{\partial v_{{\rm n}\alpha\nu}}{\partial x_\mu} + \frac{\partial v_{{\rm n}\alpha\mu}}{\partial x_\nu} - \frac{2}{3}\delta_{\mu\nu} \nabla \cdot \mathbf{v}_{{\rm n}\alpha} \right)^2 \Bigg]. \tag{19.37}$$

Expressing the first term in the damping rate (19.37) in terms of the second viscosity coefficients in (17.92) and (17.93), we obtain

$$\Gamma_\alpha = \int d\mathbf{r} \Bigg\{ \zeta_2 (\nabla \cdot \mathbf{v}_{{\rm n}\alpha})^2 + 2\zeta_1 (\nabla \cdot \mathbf{v}_{{\rm n}\alpha}) \nabla \cdot [m n_{{\rm c}0} (\mathbf{v}_{{\rm c}\alpha} - \mathbf{v}_{{\rm n}\alpha})] + \zeta_3 \left\{ \nabla \cdot [m n_{{\rm c}0} (\mathbf{v}_{{\rm c}\alpha} - \mathbf{v}_{{\rm n}\alpha})] \right\}^2 + \frac{\kappa}{T} |\nabla \delta T_{\omega_\alpha}[\mathbf{v}_{{\rm n}\alpha}, \mathbf{v}_{{\rm c}\alpha}]|^2 + \frac{\eta}{2} \left(\frac{\partial v_{{\rm n}\alpha\nu}}{\partial x_\mu} + \frac{\partial v_{{\rm n}\alpha\mu}}{\partial x_\nu} - \frac{2}{3}\delta_{\mu\nu} \nabla \cdot \mathbf{v}_{{\rm n}\alpha} \right)^2 \Bigg\} \times \left[2 \int d\mathbf{r}\, m (n_{{\rm c}0} \mathbf{v}_{{\rm c}\alpha}^2 + \tilde{n}_0 \mathbf{v}_{{\rm n}\alpha}^2) \right]^{-1}. \tag{19.38}$$

The damping expression (19.38) is written for normal mode solutions which are not normalized, i.e. do not satisy (19.32). As a result, the denominator in (19.38) contains the additional normalization factor

$$\int d\mathbf{r}\, m \left(n_{{\rm c}0} \mathbf{v}_{{\rm c}\alpha}^2 + \tilde{n}_0 \mathbf{v}_{{\rm n}\alpha}^2 \right). \tag{19.39}$$

The formula (19.38) for the damping rate Γ_α in can be understood physically in terms of entropy production (Landau and Lifshitz, 1959). The local entropy production rate $R_s(\mathbf{r}, t)$ in the two-fluid equations (Khalatnikov,

1965) is given by (17.80). Assuming a normal mode oscillation of the form

$$\mathbf{v}_{\rm n}(\mathbf{r},t)=\mathbf{v}_{{\rm n}\alpha}(\mathbf{r})\cos\omega_\alpha t,\qquad \mathbf{v}_{\rm c}(\mathbf{r},t)=\mathbf{v}_{{\rm c}\alpha}(\mathbf{r})\cos\omega_\alpha t, \tag{19.40}$$

the time *average* of the total entropy production rate can be calculated and we obtain

$$\begin{aligned}\langle\dot{R}_s\rangle &\equiv \frac{\omega_\alpha}{2\pi}\int_0^{2\pi/\omega_\alpha} dt\int d\mathbf{r}\, R_s(\mathbf{r},t)\\ &=\frac{1}{2}\int d\mathbf{r}\Big[\zeta_2(\nabla\cdot\mathbf{v}_{{\rm n}\alpha})^2+2\zeta_1(\nabla\cdot\mathbf{v}_{{\rm n}\alpha})\nabla\cdot[mn_{\rm c0}(\mathbf{v}_{{\rm c}\alpha}-\mathbf{v}_{{\rm n}\alpha})]\\ &\qquad+\zeta_3\left\{\nabla\cdot[mn_{\rm c0}(\mathbf{v}_{{\rm c}\alpha}-\mathbf{v}_{{\rm n}\alpha})]\right\}^2+\frac{\kappa}{T}\left|\nabla\delta T_{\omega_\alpha}[\mathbf{v}_{{\rm n}\alpha},\mathbf{v}_{{\rm c}\alpha}]\right|^2\\ &\qquad+\frac{\eta}{2}\left(\frac{\partial v_{{\rm n}\alpha\nu}}{\partial x_\mu}+\frac{\partial v_{{\rm n}\alpha\mu}}{\partial x_\nu}-\frac{2}{3}\delta_{\mu\nu}\nabla\cdot\mathbf{v}_{{\rm n}\alpha}\right)^2\Big].\end{aligned} \tag{19.41}$$

Furthermore, the total mechanical energy of the normal mode is given by

$$\langle E_{\rm mech}\rangle=\frac{1}{2}\int d\mathbf{r}(mn_{\rm c0}\mathbf{v}_{{\rm c}\alpha}^2+m\tilde{n}_0\mathbf{v}_{{\rm n}\alpha}^2). \tag{19.42}$$

Using (19.41) and (19.42), we see that an expression for the damping rate calculated using

$$\Gamma_\alpha=\frac{\langle\dot{R}_s\rangle}{2\langle E_{\rm mech}\rangle} \tag{19.43}$$

reduces exactly to (19.38).

The equivalence of (19.38) and (19.43) shows that our expression for hydrodynamic damping in the two-fluid region is a natural generalization of the approach first given in the classic work by Landau and Lifshitz (1959) for classical fluids. This approach was first used by Kavoulakis *et al.* (1998a) to calculate damping in trapped Bose gases above $T_{\rm BEC}$. The Landau–Lifshitz damping formula (19.43) was used by Wilks (1967) to discuss the damping of first and second sound in superfluid ^{4}He.

So far we have not dealt with the fact that in a trapped Bose gas, the decreasing density in the tail of the thermal cloud always leads to the breakdown of the collisional hydrodynamic description (which requires that $\omega\tau\ll 1$). As pointed out by Kavoulakis *et al.* (1998a), this causes trouble in using (19.38) to evaluate the damping of modes in a trapped Bose gas. This problem can be handled in a physically motivated (but *ad hoc*) manner, by introducing a spatial cutoff in the integral. Following Nikuni and Griffin (2004), we will use here a more microscopic procedure. As we discussed in Section 17.4, the fact that the condensate and noncondensate atoms are not in complete local equilibrium can be taken into account by introducing

frequency-dependent second viscosity coefficients

$$\zeta_i(\omega) = \frac{\zeta_i}{1 - i\omega\tau_\mu}. \tag{19.44}$$

Similarly, one can also introduce frequency dependence into the shear viscosity and the thermal conductivity:

$$\kappa(\omega) = \frac{\kappa}{1 - i\omega\tau_\kappa}, \quad \eta(\omega) = \frac{\eta}{1 - i\omega\tau_\eta}. \tag{19.45}$$

In Appendix C, we give details of the derivation of such frequency-dependent transport coefficients starting from the kinetic equation for the thermal cloud atoms.

Replacing the transport coefficients in (19.38) by $\kappa(\omega_\alpha), \eta(\omega_\alpha)$ and $\zeta_i(\omega_\alpha)$, and taking the real part, we find that the hydrodynamic damping rate of a collective mode of frequency ω_α in a trapped Bose gas is now given by

$$\begin{aligned}\Gamma_\alpha = \int d\mathbf{r} \Bigg[& \frac{1}{1 + (\omega_\alpha\tau_\mu)^2} \frac{g\tau_\mu}{\sigma_{\rm H}} \left(\delta\Gamma_{12}^{(1)}[\mathbf{v}_{{\rm n}\alpha}, \mathbf{v}_{{\rm c}\alpha}] \right)^2 \\ & + \frac{1}{1 + (\omega_\alpha\tau_\kappa)^2} \frac{\kappa}{T} \left| \nabla \delta T_{\omega_\alpha}[\mathbf{v}_{{\rm n}\alpha}, \mathbf{v}_{{\rm c}\alpha}] \right|^2 \\ & + \frac{1}{1 + (\omega_\alpha\tau_\eta)^2} \frac{\eta}{2} \left(\frac{\partial v_{{\rm n}\alpha\nu}}{\partial x_\mu} + \frac{\partial v_{{\rm n}\alpha\mu}}{\partial x_\nu} - \frac{2}{3}\delta_{\mu\nu} \nabla \cdot \mathbf{v}_{{\rm n}\alpha} \right)^2 \Bigg] \\ & \times \left[2 \int d\mathbf{r}\, m(n_{\rm c0}\mathbf{v}_{{\rm c}\alpha}^2 + \tilde{n}_0 \mathbf{v}_{{\rm n}\alpha}^2) \right]^{-1}. \end{aligned} \tag{19.46}$$

The fluctuations $\delta T_{\omega_\alpha}[\mathbf{v}_{{\rm n}\alpha}, \mathbf{v}_{{\rm c}\alpha}]$ and $\delta\Gamma_{12}^{(1)}[\mathbf{v}_{{\rm n}\alpha}, \mathbf{v}_{{\rm c}\alpha}]$ are given explicitly in (19.13) and (19.15) in terms of the velocities $\mathbf{v}_{{\rm n}\alpha}$ and $\mathbf{v}_{{\rm c}\alpha}$. The formula (19.46) allows one to calculate the hydrodynamic damping due to various transport processes in a trapped Bose gas (both above and below $T_{\rm BEC}$) in a well-defined manner. The frequency-dependent transport coefficients in these expressions automatically yield the factors $1/[1 + (\omega_\alpha\tau_i)]^2$, $i = \mu, \kappa, \eta$. These effectively introduce a spatial cutoff for $\omega_\alpha\tau_i > 1$, i.e. when hydrodynamic local equilibrium breaks down and we enter the collisionless regime. Thus the collisionless region in the low density tail of the thermal cloud in trapped gases is "eliminated" in a natural manner.

19.4 Hydrodynamic damping in a normal Bose gas

To illustrate the general theory of hydrodynamic damping derived in Sections 19.2 and 19.3, we first consider collective modes in a trapped non-condensed Bose gas *above* $T_{\rm BEC}$. In particular, we consider the $m = 0$

monopole–quadrupole collective mode in an axisymmetric trap ($\omega_x = \omega_y \equiv \omega_\perp \neq \omega_z$). This type of collective mode above $T_{\rm BEC}$ was first observed in a pioneering MIT experiment by Stamper-Kurn *et al.* (1998). The $m = 0$ mode in trapped gases above $T_{\rm BEC}$ was also measured by Buggle *et al.* (2005). Data from the latter paper is shown in Fig. 10.2.

The collisional hydrodynamic modes in a trapped Bose gas with $m = 0$ symmetry were discussed by Griffin *et al.* (1997); see also Section 15.4 of the present book. Above $T_{\rm BEC}$, the Hartree-Fock mean field is negligible and thus the equilibrium density is simply given by $n_0(\mathbf{r}) = g_{3/2}(z_0(\mathbf{r}))/\Lambda^3$ with $z_0(\mathbf{r}) = e^{\beta[\mu_0 - V_{\rm trap}(\mathbf{r})]}$. The chemical potential μ_0 is determined as a function of the temperature through $N = (k_{\rm B}T/\hbar\bar{\omega})^3 g_3(z_0)$, where $\bar{\omega} \equiv (\omega_x\omega_y\omega_z)^{1/3}$. The two normal mode frequencies are temperature independent and are given by (15.103). These two solutions will be denoted as $\Omega_\pm$. The normal fluid velocity field amplitude is given by

$$\mathbf{v}_{\rm n} = (ax, ay, bz). \tag{19.47}$$

Using (15.102), the coefficients $a_\pm$ and $b_\pm$ can be shown to satisfy the following relations:

$$\frac{a_\pm}{b_\pm} = \left(\frac{3\Omega_\pm^2}{4\omega_z^2} - 2\right), \qquad \frac{b_\pm}{a_\pm} = \left(\frac{3\Omega_\pm^2}{2\omega_\perp^2} - 5\right). \tag{19.48}$$

Kavoulakis *et al.* (1998a) discussed the hydrodynamic damping of this $m = 0$ mode using the Landau–Lifshitz formula (19.43), with a spatial cutoff to remove the collisionless regime in the low-density tail of the thermal cloud. In contrast, we have calculated the damping rate from (19.46), which eliminates the collisionless regime in the low-density tail through the use of the frequency-dependent transport coefficients. In the normal phase above $T_{\rm BEC}$, there is no contribution from the second viscosity transport coefficients. Moreover, for the $m = 0$ mode above $T_{\rm BEC}$ one can show that the temperature is uniform and hence that $\nabla T = 0$ (Griffin *et al.*, 1997; Kavoulakis *et al.*, 1998a). Thus the thermal conductivity also makes no contribution to the hydrodynamic damping of the $m = 0$ mode. We conclude that only the shear viscosity η in (19.46) contributes to the damping of these monopole–quadrupole normal modes. We will limit our discussion to the low-frequency mode Ω_-. For a cigar-shaped trap $\omega_\perp \gg \omega_z$, we have $\Omega_- \simeq \sqrt{\frac{12}{5}}\omega_z$. Using (19.47) and (19.48), the damping of this Ω_- mode is

given by (Kavoulakis *et al.*, 1998a)

$$\Gamma_- = \frac{\frac{2}{3}(a_- - b_-)^2 \int d\mathbf{r} \frac{\eta(\mathbf{r})}{1 + [\Omega_- \tau_\eta(\mathbf{r})]^2}}{\int d\mathbf{r}\ [a_-^2(x^2+y^2) + b_-^2 z^2] m\tilde{n}_0(\mathbf{r})}. \tag{19.49}$$

The expression (19.49) for the viscous damping Γ_- can be written in a more transparent form. First of all, one can rewrite the denominator of (19.49) making use of the fact that the equilibrium density profile given by (17.14) is a function of $V_{\rm trap}(\mathbf{r})$; hence

$$\Gamma_- = \frac{\frac{2}{3}(a_- - b_-)^2 \int d\mathbf{r} \frac{\eta(\mathbf{r})}{1 + [\Omega_- \tau_\eta(\mathbf{r})]^2}}{\left(\frac{2a_-^2}{\omega_\perp^2} + \frac{b_-^2}{\omega_z^2}\right) \frac{m}{3} \int d\mathbf{r} [\omega_\perp^2 (x^2+y^2) + \omega_z^2 z^2] \tilde{n}_0(\mathbf{r})}. \tag{19.50}$$

In (19.50), the factor involving the coefficients a_- and b_- can be written in terms of the undamped frequencies Ω_- and Ω_+ as

$$\frac{\frac{2}{3}(a_-^2 - b_-^2)}{\left(\frac{2a_-^2}{\omega_\perp^2} + \frac{b_-^2}{\omega_z^2}\right)} = \frac{\frac{2}{3}\left(\frac{a_-}{b_-} - 1\right)\left(1 - \frac{b_-}{a_-}\right)}{\left(\frac{2a_-}{\omega_\perp^2 b_-} + \frac{b_-}{\omega_z^2 a_-}\right)} = \frac{3(\Omega_-^2 - 4\omega_z^2)(\Omega_-^2 - 4\omega_\perp^2)}{4(5\omega_\perp^2 + 4\omega_z^2 - 3\Omega_-^2)}$$

$$= \frac{(\Omega_-^2 - 4\omega_z^2)(\Omega_-^2 - 4\omega_\perp^2)}{2(\Omega_+^2 - \Omega_-^2)}. \tag{19.51}$$

We have used (19.48) and (15.103) to arrive at the final expression (19.51). Using this in (19.49), we obtain the following expression for the hydrodynamic damping rate due to the shear viscosity

$$\Gamma_- = \frac{\tilde{\tau}}{2(\Omega_+^2 - \Omega_-^2)} (\Omega_-^2 - 4\omega_z^2)(\Omega_-^2 - 4\omega_\perp^2), \tag{19.52}$$

where we have defined a new relaxation time $\tilde{\tau}$ as

$$\tilde{\tau} \equiv \frac{\int d\mathbf{r} \frac{\tau_\eta(\mathbf{r}) \tilde{P}_0(\mathbf{r})}{1 + (\Omega_- \tau_\eta)^2}}{\frac{m}{3} \int d\mathbf{r} [\omega_\perp^2 (x^2+y^2) + \omega_z^2 z^2] \tilde{n}_0(\mathbf{r})}. \tag{19.53}$$

Here we have used the result for the viscosity given in (18.29), namely $\eta(\mathbf{r}) = \tau_\eta(\mathbf{r}) \tilde{P}_0(\mathbf{r})$. Using the equilibrium relation given in (15.93), we can

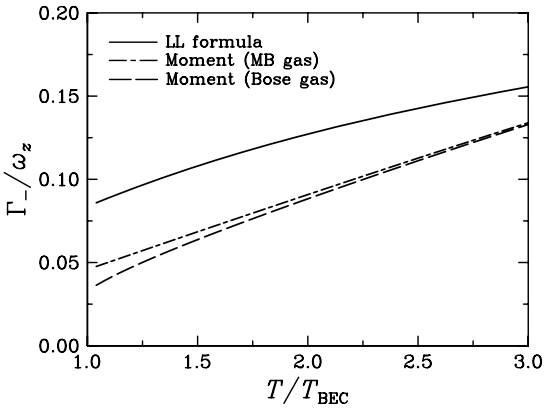


Fig. 19.1. Temperature dependence of (solid line) the hydrodynamic damping rate of the $m = 0$ mode in a degenerate Bose gas above $T_{\rm BEC}$, calculated from (19.52) based on the Landau–Lifshitz formula, for the experimental parameters used by Buggle *et al.* (2005). The moment method result (given in Section 10.1) is shown by the broken line. For comparison, the moment result for a Maxwell–Boltzmann (MB) gas is also plotted (the broken-and-dotted line).

also rewrite the denominator of (19.53) in terms of $\tilde{P}_0(\mathbf{r})$, and hence the expression for $\tilde{\tau}$ finally reduces to

$$\tilde{\tau} = \frac{\int d\mathbf{r} \dfrac{\tau_\eta \tilde{P}_0(\mathbf{r})}{1 + (\Omega_- \tau_\eta)^2}}{\int d\mathbf{r} \tilde{P}_0(\mathbf{r})}. \tag{19.54}$$

It is useful to comment on the relation between the present calculation of hydrodynamic damping and the moment method developed by Guéry-Odelin *et al.* (1999) for the Boltzmann equation describing a classical gas. As reviewed in Section 10.1, in the moment method for the monopole–quadrupole mode, collisions are characterized by a single parameter, the quadrupole relaxation time τ defined in (10.14). In the collisional hydrodynamic limit $\omega_z \tau \ll 1$, the moment method reproduces the hydrodynamic frequency (Ω_- given by (10.20) or (15.103)). The damping of this mode has the same form as that given by (19.52) except that τ replaces $\tilde{\tau}$. Both $\tilde{\tau}$ and τ are related to the same position-dependent viscous relaxation time $\tau_\eta(\mathbf{r})$, but involve slightly different spatial averages (compare (19.54) and (10.14)).

For illustration, we will evaluate the damping Γ_- given by (19.52) for the low-frequency solution Ω_-. We use the trap parameters from Buggle *et al.* (2005) for ^{87}Rb: the trap frequencies are $\omega_\perp/2\pi = 474$Hz, $\omega_z/2\pi = 16.8$Hz,

and the total number of atoms is $N = 10^7$.[3] Their experimental data are shown in Fig. 10.2. In the temperature region $T_{\rm BEC} < T < 3T_{\rm BEC}$, our calculations show that $\omega_z \tau_\eta(\mathbf{r} = 0) \ll 1$. Thus the dominant contribution in (19.53) arises from the low-density tail of the cloud where $\omega_z \tau_\eta(\mathbf{r}) \sim 1$. In Fig. 19.1, we plot the temperature dependence of the damping rate. For comparison, we also plot the damping calculated using the moment method technique reviewed in Chapter 10. The moment results underestimate the damping by about a factor 2 compared to our generalized Landau–Lifshitz result.

19.5 Hydrodynamic damping in a superfluid Bose gas

In this section, we use the expression (19.46) to calculate the damping of the $m = 0$ collective mode in the *superfluid* phase. Below $T_{\rm BEC}$, the gas in general exhibits coupled oscillations of the condensate (superfluid) and noncondensate (normal fluid) components. At temperatures slightly below $T_{\rm BEC}$, one has (a) a "condensate mode", in the sense that it is mainly the condensate component that oscillates, and (b) a "noncondensate mode", in which it is mainly the noncondensate component that oscillates. As one might expect, the frequency of the condensate mode is very close to that of a *pure* condensate mode at $T = 0$. Similarly the frequency of the noncondensate mode is close to that of the *pure* noncondensate mode above $T_{\rm BEC}$. These frequencies will be slightly shifted, of course, owing to coupling between the two components. However, in the calculations in this section, we will neglect these relatively small frequency shifts and concentrate on calculating the *damping* of these two modes. Thus we neglect the condensate component in the noncondensate mode and also the noncondensate component in the condensate mode oscillation.

We first consider the damping of the noncondensate mode of monopole–quadrupole symmetry. In (19.46), we use $\mathbf{v}_{\rm n}$ as given by (19.47) with the coefficients in (19.48). On the basis of the discussion given in the preceding paragraph, we set $\mathbf{v}_{\rm c} = 0$ as a first approximation. For simplicity, we assume that $\nabla T = 0$ also holds *below* $T_{\rm BEC}$ (i.e. the mode is isothermal). We then find that the damping of the low-frequency solution in (15.103) consists of two contributions: $\Gamma = \Gamma_1 + \Gamma_2$. The first contribution Γ_1 is from the shear

[3] Nikuni and Griffin (2004) originally used the trap parameters for metastable triplet He* atoms (Pereira dos Santos *et al.*, 2001), with an estimated s-wave scattering length $a \sim 160$ Å. Later high-precision work by Moal *et al.* (2006) showed that $a = 75$ Å, which means that the oscillations studied by Pereira dos Santos *et al.* (2001) were not in the collisional hydrodynamic region.

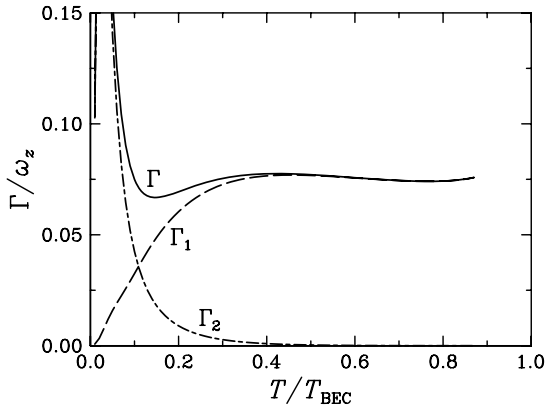


Fig. 19.2. Temperature dependence of the damping rate of the low-frequency monopole–quadrupole mode in the superfluid phase. The broken and broken-and-dotted lines give the noncondensate contributions from the shear viscosity (Γ_1) and from the second viscosity coefficients (Γ_2). The system parameters are the same as in Fig. 19.1, where analogous results are given for temperatures above $T_{\rm BEC}$ (from Nikuni and Griffin, 2004).

viscosity that we calculated in Section 19.3 (see (19.52)). The second contribution Γ_2 is due to the second viscosity coefficients in (18.40) associated with the relaxation time τ_μ and is given by

$$\Gamma_2 = \frac{\int d\mathbf{r} \frac{1}{1+(\Omega_-\tau_\mu)^2} \frac{g\tau_\mu}{\sigma_{\rm H}} \left(\delta\Gamma_{12}^{(1)}[\mathbf{v}_{\rm n},0]\right)^2}{2m\int d\mathbf{r}\, \tilde{n}_0 \mathbf{v}_{\rm n}^2}. \tag{19.55}$$

As noted above, the frequency Ω_- of this mode in the superfluid phase is well approximated by the solution Ω_- given in (15.103) for $T > T_{\rm BEC}$. In Fig. 19.2 we plot the noncondensate mode damping below $T_{\rm BEC}$. The contribution from the shear viscosity is seen to be dominant for $T > 0.3T_{\rm BEC}$. The contribution from the second viscosity coefficients only takes over at very low temperatures (where our microscopic model has, of course, decreasing validity).

Finally, we consider the condensate mode below the superfluid transition temperature. The pure condensate mode frequencies at $T = 0$ were given by Stringari (1996a):

$$\Omega_\pm^2 = 2\omega_\perp^2 + \tfrac{3}{2}\omega_z^2 \pm \tfrac{1}{2}\sqrt{16\omega_\perp^4 + 9\omega_z^4 - 16\omega_\perp^2\omega_z^2}. \tag{19.56}$$

The associated superfluid velocity field is given by

$$\mathbf{v}_{\rm c} = (ax, ay, bz), \tag{19.57}$$

in which the coefficients satisfy

$$\frac{a_\pm}{b_\pm} = \left(\frac{\Omega_\pm^2}{2\omega_z^2} - \frac{3}{2}\right), \quad \frac{b_\pm}{a_\pm} = \left(\frac{\Omega_\pm^2}{\omega_\perp^2} - 4\right) a_\pm. \tag{19.58}$$

As before, the coefficients corresponding to the two solutions of (19.56) are labeled by $\pm$. As a first approximation, we use this condensate mode solution for $\mathbf{v}_{\rm c}$ in (19.46) and set the normal fluid velocity $\mathbf{v}_{\rm n}$ to zero. We also ignore any temperature fluctuation δT. Within these approximations, our expression for the hydrodynamic damping of the low-frequency condensate mode Ω_- reduces to

$$\Gamma_- = \frac{\displaystyle\int d\mathbf{r} \frac{g\tau_\mu \sigma_{\rm H}}{1 + (\Omega_- \tau_\mu)^2} [\nabla \cdot (n_{\rm c0} \mathbf{v}_{\rm c})]^2}{\displaystyle 2m \int d\mathbf{r} n_{\rm c0} \mathbf{v}_{\rm c}^2}. \tag{19.59}$$

Here Ω_- is the lower frequency given by (19.56) and can be approximated as $\Omega_- \simeq \sqrt{\frac{5}{2}}\omega_z$ for $\omega_\perp \gg \omega_z$.

This damping is entirely due to the fact that the condensate is not in diffusive local equilibrium with the static thermal cloud. One may verify that in a uniform gas the expression (19.59) reduces to the simple formula given in (15.88) for the damping of second sound (see also Fig. 15.2). The damping rate given by (19.59) formally reduces to the expression obtained by Nikuni (2002) in the collisionless limit $\Omega_- \tau_\mu \gg 1$. In this limit, the damping also agrees with the result derived by Williams and Griffin (2000), where it was assumed that the thermal cloud always remains in *static* thermal equilibrium (see the discussion in Chapter 8). This agreement makes sense since the simplifying assumptions that we made above (i.e. that $\mathbf{v}_{\rm n} = 0$ and $\nabla T = 0$) are equivalent to assuming that the thermal cloud is always in thermal equilibrium. The damping of the condensate mode given by (19.59) is extremely small, simply because we are always in the extreme collisional hydrodynamic limit $\omega_z \tau_\mu \ll 1$, where hydrodynamic damping disappears.

This completes our calculation of the hydrodynamic damping of the monopole–quadrupole mode in a trapped Bose gas using the general formula (19.46). For simplicity, we have assumed that the modes involve motion of either the superfluid component or the normal fluid component. An improved calculation based on (19.46) that would take into account the simultaneous oscillation of both components is clearly possible. The expression (19.46) is based on a very general approach to hydrodynamic damping and should be useful in future studies of collective modes in trapped gases in the two-fluid regime.

Appendix A
Monte Carlo calculation of collision rates

In this appendix we calculate the collision probabilities needed to perform a dynamical simulation. As indicated in Chapter 11, this is done by means of a Monte Carlo sampling of the multidimensional integrals defining the collision rates in (11.32), (11.40) and (11.42). A general discussion of this numerical technique can be found in Press *et al.* (1992).

A.1 Monte Carlo evaluation of $\Gamma_{22}^{\rm out}$

The evaluation of the $\Gamma_{22}^{\rm out}$ collision rate in (11.32) involves integration over all possible incoming momenta $\mathbf{p}_1$ and $\mathbf{p}_2$ and all scattering angles Ω. To set up this calculation, we first consider the simpler problem of performing the one-dimensional integral

$$I = \int_0^{x_{\rm max}} dx\, f(x)\,, \tag{A.1}$$

where $f(x)$ is a positive definite function with maximum value $f_{\rm max}$ in the range of interest. If random points (x_i, y_i) are distributed uniformly in the x-f plane with $0 < x_i < x_{\rm max}$ and $0 < y_i < f_{\rm max}$, the integral can be estimated as

$$I \simeq x_{\rm max} f_{\rm max} F\,, \tag{A.2}$$

where F is the fraction of points that lie below the $f(x)$ curve. This fraction is determined simply by counting the number of times $N_{\rm a}$ that $y_i < f(x_i)$ in a sample of $N_{\rm b}$ trials, that is,

$$F = \frac{N_{\rm a}}{N_{\rm b}}\,. \tag{A.3}$$

Moreover, $P_i \equiv f(x_i)/f_{\rm max}$ is the probability that the random variable y_i, uniformly distributed between 0 and $f_{\rm max}$, satisfies $y_i < f(x_i)$. The expected

fraction of successes in N_b trials is therefore

$$F' \simeq \frac{1}{N_\mathrm{b}} \sum_{i=1}^{N_\mathrm{b}} P_i \,. \tag{A.4}$$

If this probabilistic estimate is substituted into (A.2), we obtain

$$I \simeq \frac{x_\mathrm{max}}{N_\mathrm{b}} \sum_{i=1}^{N_\mathrm{b}} f(x_i) \,. \tag{A.5}$$

We can recognize this as a simple approximation to the integral, since $x_\mathrm{max}/N_\mathrm{b}$ is the mean step size Δx when the x_i are uniformly distributed on $(0, x_\mathrm{max})$. The fact that $F'/F \to 1$ as $N_\mathrm{b} \to \infty$ confirms that the result (A.2) is indeed an estimate of the integral of interest.

A byproduct of the sampling procedure described above is that it generates a discrete approximation to the distribution $f(x)$. To see this, we consider a small interval Δx centred on x_0. The expected number of sampling points falling in this interval is $(\Delta x/x_\mathrm{max})N_\mathrm{b}$. Of these, the fraction retained is $f(x_0)/f_\mathrm{max}$. Thus the number of points that are retained in any given interval Δx is proportional to the value of $f(x)$ in this interval, that is, these points are distributed along the x axis according to the distribution function $f(x)$. Since the normalization of f is I, the discrete approximant to the distribution is

$$f(x) \simeq \frac{I}{N_\mathrm{a}} \sum_{i=1}^{N_\mathrm{a}} \delta(x - x_i) \,. \tag{A.6}$$

This result can now be used to evaluate the more general integral

$$J = \int_0^{x_\mathrm{max}} dx\, f(x) g(x) \,. \tag{A.7}$$

Using (A.6), we have

$$J \simeq \frac{I}{N_\mathrm{a}} \sum_{i=1}^{N_\mathrm{a}} g(x_i) = x_\mathrm{max} f_\mathrm{max} \frac{1}{N_\mathrm{b}} {\sum_i}' g(x_i) \,, \tag{A.8}$$

where the prime on the summation restricts the sum to the N_a points that have been retained.

We now apply these results to the calculation of the local collision rate (11.32), rewriting it as (suppressing for convenience the space and time

arguments)[1]

$$\Gamma_{22}^{\rm out} = \int \frac{d\mathbf{p}_1}{(2\pi\hbar)^3} \int \frac{d\mathbf{p}_2}{(2\pi\hbar)^3} f(\mathbf{p}_1) f(\mathbf{p}_2) g(\mathbf{p}_1, \mathbf{p}_2) \equiv \int d\boldsymbol{p}\, w(\boldsymbol{p}) g(\boldsymbol{p})\,, \quad \text{(A.9)}$$

where $\boldsymbol{p}$ is a point in six-dimensional momentum space and the factor $w(\boldsymbol{p}) \equiv f(\mathbf{p}_1) f(\mathbf{p}_2)/(2\pi\hbar)^6$ is considered as a weight function. We denote the maximum value of $w(\boldsymbol{p})$ by $w_{\max}$ and define the domain on which the integrand is nonzero by $[-p_{\max}/2,\, p_{\max}/2]$ for each momentum component. Choosing a point $\boldsymbol{p}_i$ at random in the hypervolume $(p_{\max})^6$ and a random number R_i uniformly distributed on $[0, w_{\max}]$, the point $\boldsymbol{p}_i$ is accepted if $R_i < w(\boldsymbol{p}_i)$. In this way, we generate a set of $\boldsymbol{p}_i$ points that are distributed in phase space according to $w(\boldsymbol{p})$. A straightforward extension of (A.8) then gives

$$\Gamma_{22}^{\rm out} \simeq (p_{\max})^6 w_{\max} \frac{1}{N_{\rm b}} {\sum_i}' g(\boldsymbol{p}_i)\,, \quad \text{(A.10)}$$

where $N_{\rm b}$ is the number of random points $\boldsymbol{p}_i$ sampled and the prime on the summation indicates that only those points for which $R_i < w(\boldsymbol{p}_i)$ are included. There are of course statistical fluctuations in the resulting estimate of the integral but these decrease in relative magnitude as $1/\sqrt{N_{\rm b}}$ as $N_{\rm b} \to \infty$. Our result for $\Gamma_{22}^{\rm out}$ is essentially an average of $g(\mathbf{p}_1, \mathbf{p}_2)$ over pairs of momenta $(\mathbf{p}_1,\, \mathbf{p}_2)$, each of which is distributed according to $f(\mathbf{p}, \mathbf{r}, t)$.

For the special case $g \equiv 1$, the integral (A.9) is simply $\tilde{n}(\mathbf{r}, t)^2$. The Monte Carlo estimate of this quantity is thus

$$\tilde{n}^2 \simeq (p_{\max})^6 w_{\max} \frac{N_{\rm a}}{N_{\rm b}}\,, \quad \text{(A.11)}$$

where $N_{\rm a}$ is the total number of points in the sample that are accepted. With this identification, (A.10) can be written as

$$\Gamma_{22}^{\rm out} \simeq \tilde{n}^2 \frac{1}{N_{\rm a}} {\sum_i}' g(\boldsymbol{p}_i)\,. \quad \text{(A.12)}$$

The sample of the $N_{\rm a}$ accepted points consists of $N_{\rm a}$ $\mathbf{p}_1$-values and $N_{\rm a}$ $\mathbf{p}_2$-values, each of which is distributed according to $f(\mathbf{p}, \mathbf{r}, t)$, as mentioned above. To relate this sample to the N-body simulation, we consider a particular spatial cell of volume $\Delta\mathbf{r}$ which contains $N_{\rm cell}$ test particles. If this set is to be representative of the local density, we must have

$$\tilde{n}(\mathbf{r}, t) = \gamma \frac{N_{\rm cell}}{\Delta\mathbf{r}}\,, \quad \text{(A.13)}$$

[1] The reader should note that roman bold symbols refer to three-dimensional vectors, while italic bold symbols refer to six-dimensional vectors.

where $\gamma = \tilde{N}/\tilde{N}_{\rm tp}$ is the ratio of the actual number $\tilde{N}$ of thermal atoms in the system and the number of test particles $\tilde{N}_{\rm tp}$ chosen in the simulation. Effectively each test particle has a weight γ when sums over particles are performed.

The required average of $g(\mathbf{p}_1, \mathbf{p}_2)$ in (A.9) is generated as follows. We select successive pairs of particles at random from the set of $N_{\rm cell}$ particles in the cell. We then have

$$\Gamma_{22}^{\rm out} \simeq \tilde{n}^2 \frac{2}{N_{\rm cell}} \sum_{(ij)} g(\mathbf{p}_i, \mathbf{p}_j), \tag{A.14}$$

where the sum extends over the $N_{\rm cell}/2$ pairs ij of particles selected from the set. Using (A.13), we have

$$\Gamma_{22}^{\rm out} \Delta\mathbf{r} = 2\gamma\tilde{n} \sum_{(ij)} g(\mathbf{p}_i, \mathbf{p}_j). \tag{A.15}$$

The average number of collisions occurring in $\Delta\mathbf{r}$ in a time interval Δt is thus given by

$$\frac{1}{2}\Gamma_{22}^{\rm out} \Delta\mathbf{r}\Delta t \simeq \gamma\tilde{n} \sum_{(ij)} g(\mathbf{p}_i, \mathbf{p}_j)\Delta t. \tag{A.16}$$

This expression implies that the probability that the pair ij suffers a collision in the time interval Δt is

$$P_{ij}^{22} = \gamma\tilde{n} g(\mathbf{p}_i, \mathbf{p}_j)\Delta t\,. \tag{A.17}$$

The appearance of the factor γ again accounts for the fact the the number of test particles in the simulation is not the same as the actual number of particles in the system. Inserting the explicit form of $g(\mathbf{p}_1, \mathbf{p}_2)$ for the C_{22} collision rate, we have

$$P_{ij}^{22} = \gamma\tilde{n}(\mathbf{r})\sigma|\mathbf{v}_i - \mathbf{v}_j| \int \frac{{\rm d}\Omega}{4\pi}(1 + f_3)(1 + f_4)\Delta t, \tag{A.18}$$

which is the expression given in (11.36).

A.2 Monte Carlo evaluation of $\Gamma_{12}^{\rm out}$

We next consider the C_{12} collision probabilities to be used in the simulations. Following (A.9), we write (11.40) as

$$\Gamma_{12}^{\rm out} = \int d\mathbf{p}_2\, w(\mathbf{p}_2) g(\mathbf{p}_2)\,, \tag{A.19}$$

where $w(\mathbf{p}_2) = f(\mathbf{p}_2)/(2\pi\hbar)^3$ and $g(\mathbf{p}_2)$ is the remaining part of the integrand. In contrast to (A.9), the momenta here are three-dimensional. Carrying out the integration using a Monte Carlo sampling technique, we have

$$\Gamma_{12}^{\rm out} \simeq (p_{\rm max})^3 w_{\rm max} \frac{1}{N_{mb}} {\sum_i}' g(\mathbf{p}_i)\,. \tag{A.20}$$

With $g = 1$ the integral is simply $\tilde{n}$, and we therefore have

$$\Gamma_{12}^{\rm out} \simeq \frac{\tilde{n}}{N_{\rm a}} {\sum_i}' g(\mathbf{p}_i)\,, \tag{A.21}$$

where $N_{\rm a}$ is the number of points accepted in the sample. In other words, $\Gamma_{12}^{\rm out}$ is simply the average of $g(\mathbf{p})$ over a sample of momenta $\mathbf{p}_i$ distributed according to $w(\mathbf{p})$.

In a simulation, the average defined in (A.21) is replaced by the cell average $\sum_{i=1}^{N_{\rm cell}} g(\mathbf{p}_i)/N_{\rm cell}$, where $N_{\rm cell}$ is the number of test particles in a cell of volume $\Delta\mathbf{r}$. Using (A.13), we thus find

$$\Gamma_{12}^{\rm out} \Delta\mathbf{r} \simeq \gamma \sum_{i=1}^{N_{\rm cell}} g(\mathbf{p}_i)\,. \tag{A.22}$$

This implies that the probability that a test particle in the cell will suffer a collision of this kind in the time interval Δt is

$$P_i^{\rm out} = \gamma n_{\rm c} \sigma v_r^{\rm out} \int \frac{d\Omega}{4\pi} (1 + f_3)(1 + f_4) \Delta t\,. \tag{A.23}$$

Following the same reasoning for $\Gamma_{12}^{\rm in}$, the probability that an atom in the simulation undergoes an "in" collision is

$$P_i^{\rm in} = \gamma \frac{g n_{\rm c}}{\pi v_r^{\rm in}} \int d\mathbf{v}_\perp f_4 (1 + f_3) \Delta t\,. \tag{A.24}$$

This completes the derivation of (11.48) and (11.49) for the C_{12} collisions.

Appendix B

Evaluation of transport coefficients: technical details

In this appendix, we give further details of the calculations that lead to the results given in Section 18.1 for the transport coefficients and transport relaxation times in a trapped Bose superfluid gas.

B.1 Integrals involving the Bose–Einstein functions $g_n(z)$

In the determination in Chapter 18 of explicit expressions for the transport coefficients, various momentum integrals were involved. Here we give a few more details about evaluating these integrals in terms of the Bose–Einstein functions $g_n(z_0)$.

The thermal conductivity κ in (18.4) involves the integral

$$I_1 \equiv \int \frac{d\mathbf{p}}{(2\pi\hbar)^3} u^2 \left[\frac{mu^2}{2k_{\rm B}T} - \frac{5}{2}\frac{g_{5/2}(z_0)}{g_{3/2}(z_0)}\right]^2 f^0(1+f^0), \tag{B.1}$$

where f^0 is the *static* equilibrium Bose distribution

$$\begin{aligned} f^0(\mathbf{p},\mathbf{r}) &= \frac{1}{e^{\beta_0[p^2/2m+U_0(\mathbf{r})-\mu_{\rm c0}(\mathbf{r})]}-1} \\ &= \frac{1}{e^{\beta_0[p^2/2m+gn_{\rm c0}(\mathbf{r})]}-1} \end{aligned} \tag{B.2}$$

In our linearized theory, we can set $\mathbf{p} = m\mathbf{u}$ (ignoring $\mathbf{v}_{\rm n}$). Changing variables to

$$\mathbf{s} = \mathbf{u}\left(\frac{m}{2k_{\rm B}T}\right)^{1/2} = \mathbf{p}\left(\frac{1}{2mk_{\rm B}T}\right)^{1/2}, \tag{B.3}$$

one finds that

$$I_1 = \left(\frac{2k_{\rm B}T}{m}\right)^{5/2} \frac{m^3}{2\pi\hbar^2} \int ds\, s^2 \left[s^2 - \frac{5}{2}\frac{g_{5/2}(z_0)}{g_{3/2}(z_0)}\right]^2 f^0(1+f^0). \tag{B.4}$$

In terms of the variable $x \equiv s^2$, we have

$$I_1 = \left(\frac{2k_{\rm B}T}{m}\right)^{5/2} \frac{m^3}{4\pi\hbar^2} \int_0^\infty dx\, x^{3/2} \left[x - \frac{5}{2}\frac{g_{5/2}}{g_{3/2}}\right]^2 f^0(x)[1 + f^0(x)], \quad \text{(B.5)}$$

where

$$f^0(x) = \left(z_0^{-1}e^x - 1\right)^{-1}, \text{ with } z_0 = e^{\beta_0(\mu_{\rm c0} - U_0)} = e^{-\beta_0 g n_{\rm c0}}. \quad \text{(B.6)}$$

The integral involved in (B.5) is of the type

$$\begin{aligned} \int dx\, x^{n-1} f^0(1 + f^0) &= z_0 \frac{\partial}{\partial z_0}\left(\int dx\, x^{n-1} f^0\right) \\ &= z_0 \frac{\partial}{\partial z_0}[\Gamma(n) g_n(z_0)] \\ &= \Gamma(n) g_{n-1}(z_0). \end{aligned} \quad \text{(B.7)}$$

Here $g_n(z)$ is a Bose–Einstein (BE) integral (see Appendices B and D in Pathria, 1972) and, for n an odd multiple of $\frac{1}{2}$, the gamma function is given by

$$\begin{aligned} \Gamma(n) &\equiv \int_0^\infty dx\, x^{n-1} e^{-x} \\ &= (n-1)(n-2) \times \ldots \times \frac{1}{2}\sqrt{\pi}. \end{aligned} \quad \text{(B.8)}$$

Using these BE functions, one finds that (B.5) reduces to

$$\begin{aligned} I_1 &= \frac{15m^3}{32\pi^{3/2}} \left(\frac{2k_{\rm B}T}{m}\right)^{5/2} \left[\frac{7}{2} g_{7/2}(z_0) - \frac{5g_{5/2}^2(z_0)}{2g_{3/2}(z_0)}\right] \\ &= \frac{15\tilde{n}_0 k_{\rm B} T}{2m} \left[\frac{7g_{7/2}(z_0)}{2g_{3/2}(z_0)} - \frac{5g_{5/2}^2(z_0)}{2g_{3/2}^2(z_0)}\right]. \end{aligned} \quad \text{(B.9)}$$

This expression is used to obtain (18.5) from (18.4).

In calculating the shear viscosity (18.21), the numerator N of the constant B in (18.22) involves the integral (repeated indices are summed over)

$$\begin{aligned} I_2 &= \int \frac{d\mathbf{p}}{(2\pi\hbar)^3} \left(u_\nu u_\mu - \tfrac{1}{3}\delta_{\mu\nu} u^2\right)^2 f^0(1 + f^0) \\ &= \int \frac{d\mathbf{p}}{(2\pi\hbar)^3} \tfrac{2}{3} u^4 f^0(1 + f^0). \end{aligned} \quad \text{(B.10)}$$

Carrying out the same kind of calculation as that given above for the integral I_1 defined in (B.1), using the BE integrals in (B.7), we find that

$$I_2 = \frac{5}{16}\left(\frac{m}{k_{\rm B}T}\right) \frac{m^3}{\pi^{3/2}} \left(\frac{2k_{\rm B}T}{m}\right)^{5/2} g_{7/2}(z_0). \quad \text{(B.11)}$$

Here we have used $\Gamma(n=\frac{7}{2})=\frac{15}{8}\sqrt{\pi}$, as given by (B.8). For completeness we also give two other integrals used in Chapter 18:

$$\begin{aligned}\int \frac{d\mathbf{p}}{(2\pi\hbar)^3} f^0(1+f^0) &= \frac{g_{1/2}(z_0)}{\Lambda_0^3},\\ \int \frac{d\mathbf{p}}{(2\pi\hbar)^3} u^2 f^0(1+f^0) &= \frac{3k_{\rm B}T}{m}\frac{g_{3/2}(z_0)}{\Lambda_0^3}.\end{aligned} \tag{B.12}$$

B.2 Reduction of collision integrals in transport coefficients

For the calculation of the thermal conductivity κ, we need to calculate the constant A given in (18.6). This involves the integral D in (18.7).

It is convenient to introduce the dimensionless velocity defined in (B.3). With this new variable, we can rewrite the linearized collision operator $\hat{L}$ in (17.36) as

$$\hat{L}[\psi] = \frac{8m(k_{\rm B}T_0)^2a^2}{\pi^3\hbar^3}\left(\hat{L}'_{22}[\psi] + n_{\rm c0}\Lambda_0^3\hat{L}'_{12}[\psi]\right). \tag{B.13}$$

Here the *dimensionless* collision operators $\hat{L}'_{22}$ and $\hat{L}'_{12}$ are defined by

$$\begin{aligned}\hat{L}'_{22}[\psi] \equiv \int d\mathbf{s}_2 \int d\mathbf{s}_3 \int d\mathbf{s}_4\, \delta(\mathbf{s}+\mathbf{s}_2-\mathbf{s}_3-\mathbf{s}_4)\delta(s^2+s_2^2-s_3^2-s_4^2)\\ \times f_1^0 f_2^0(1+f_3^0)(1+f_4^0)(\psi_3+\psi_4-\psi_2-\psi),\end{aligned} \tag{B.14}$$

$$\begin{aligned}\hat{L}'_{12}[\psi] \equiv \pi^{3/2}\int d\mathbf{s}_1 \int d\mathbf{s}_2 \int d\mathbf{s}_3\, \delta(\mathbf{s}-\mathbf{s}_2-\mathbf{s}_3)\delta(s_1^2-\beta_0 g n_{\rm c0}-s_2^2-s_3^2)\\ \times[\delta(\mathbf{s}-\mathbf{s}_1)-\delta(\mathbf{s}-\mathbf{s}_2)-\delta(\mathbf{s}-\mathbf{s}_3)]\\ \times(1+f_1^0)f_2^0 f_3^0(\psi_2+\psi_3-\psi_1),\end{aligned} \tag{B.15}$$

where $f_i^0 = (z_0^{-1}e^{s_i^2}-1)^{-1}$.

The linearized collision integral $L'_{22}[\psi]$ defined in (B.14) has the following general properties. First, owing to the momentum and energy conserving delta functions, this linearized collision integral satisfies the condition

$$\hat{L}'_{22}[\mathbf{s}] = 0, \qquad \hat{L}'_{22}[s^2] = 0. \tag{B.16}$$

Second, we assume that ψ and ϕ are functions of the variable $\mathbf{s}$ and consider the integral

$$\begin{aligned}\int d\mathbf{s}\,\phi\hat{L}'_{22}[\psi] = \int d\mathbf{s}_1 \int d\mathbf{s}_2 \int d\mathbf{s}_3 \int d\mathbf{s}_4\, \delta(\mathbf{s}_1+\mathbf{s}_2-\mathbf{s}_3-\mathbf{s}_4)\\ \times\delta(s_1^2+s_2^2-s_3^2-s_4^2)\\ \times\phi_1 f_1^0 f_2^0(1+f_3^0)(1+f_4^0)\end{aligned}$$

$$\times (\psi_3 + \psi_4 - \psi_2 - \psi_1), \tag{B.17}$$

where $\phi_i \equiv \phi(\mathbf{s}_i)$. In (B.17), one can exchange the variables $\mathbf{s}_i$ to write the integral in several different ways:

$$\begin{aligned}
\int d\mathbf{s}\, \phi \hat{L}'_{22}[\psi] &= \int d\mathbf{s}_1 \int d\mathbf{s}_2 \int d\mathbf{s}_3 \int d\mathbf{s}_4 \\
&\quad \times \delta(\mathbf{s}_1 + \mathbf{s}_2 - \mathbf{s}_3 - \mathbf{s}_4)\delta(s_1^2 + s_2^2 - s_3^2 - s_4^2) \\
&\quad \times \phi_2 f_1^0 f_2^0 (1 + f_3^0)(1 + f_4^0) \\
&\quad \times (\psi_3 + \psi_4 - \psi_2 - \psi_1) \quad (1 \leftrightarrow 2) \\
&= \int d\mathbf{s}_1 \int d\mathbf{s}_2 \int d\mathbf{s}_3 \int d\mathbf{s}_4 \\
&\quad \times \delta(\mathbf{s}_1 + \mathbf{s}_2 - \mathbf{s}_3 - \mathbf{s}_4)\delta(s_1^2 + s_2^2 - s_3^2 - s_4^2) \\
&\quad \times \phi_3 f_3^0 f_4^0 (1 + f_1^0)(1 + f_2^0) \\
&\quad \times (\psi_1 + \psi_2 - \psi_3 - \psi_4) \quad (1 \leftrightarrow 3,\ 2 \leftrightarrow 4) \\
&= \int d\mathbf{s}_1 \int d\mathbf{s}_2 \int d\mathbf{s}_3 \int d\mathbf{s}_4 \delta(\mathbf{s}_1 + \mathbf{s}_2 - \mathbf{s}_3 - \mathbf{s}_4) \\
&\quad \times \delta(s_1^2 + s_2^2 - s_3^2 - s_4^2) \\
&\quad \times \phi_4 f_3^0 f_4^0 (1 + f_1^0)(1 + f_{20}) \\
&\quad \times (\psi_1 + \psi_2 - \psi_3 - \psi_4) \quad (1 \leftrightarrow 4,\ 2 \leftrightarrow 3).
\end{aligned} \tag{B.18}$$

Using the equilibrium relation

$$f_1^0 f_2^0 (1 + f_3^0)(1 + f_4^0) = f_3^0 f_4^0 (1 + f_1^0)(1 + f_2^0), \tag{B.19}$$

these results can be combined to give

$$\begin{aligned}
\int d\mathbf{s}\, \phi \hat{L}'_{22}[\psi] &= \frac{1}{4} \int d\mathbf{s}_1 \int d\mathbf{s}_2 \int d\mathbf{s}_3 \int d\mathbf{s}_4\, \delta(\mathbf{s}_1 + \mathbf{s}_2 - \mathbf{s}_3 - \mathbf{s}_4) \\
&\quad \times \delta(s_1^2 + s_2^2 - s_3^2 - s_4^2)(\phi_1 + \phi_2 - \phi_3 - \phi_4) \\
&\quad \times f_1^0 f_2^0 (1 + f_3^0)(1 + f_4^0)(\psi_3 + \psi_4 - \psi_2 - \psi_1).
\end{aligned} \tag{B.20}$$

By the same sort of analysis, one can show that $\int d\mathbf{s}\, \psi \hat{L}'_{22}[\phi]$ is also given by the r.h.s. of (B.20). We thus have proved that the linearized collision integral $\hat{L}'_{22}$ has the important Hermitian property

$$\int d\mathbf{s}\, \phi \hat{L}'_{22}[\psi] = \int d\mathbf{s}\, \psi \hat{L}'_{22}[\phi]. \tag{B.21}$$

This is a very useful feature of the linearized collision integral related to C_{22} collisions and is well known in the case of classical gases (Ferziger and Kaper, 1972).

The Hermitian property (B.21) is very useful. Let us consider an integral

of the kind in the denominator of (18.6), which is given in (18.7):

$$I_A = \int d\mathbf{s} \left[s^2 - \frac{5g_{5/2}(z)}{2g_{3/2}(z)} \right] \mathbf{s} \cdot \hat{L}'_{22} \left[\left(s^2 - \frac{5g_{5/2}(z)}{2g_{3/2}(z)} \right) \mathbf{s} \right]. \tag{B.22}$$

Using (B.16), one finds that (B.22) reduces to

$$I_A = \int d\mathbf{s} \left[s^2 - \frac{5g_{5/2}(z)}{2g_{3/2}(z)} \right] \mathbf{s} \cdot \hat{L}'_{22}[s^2 \mathbf{s}]. \tag{B.23}$$

Using (B.21) again, we find that (B.23) can be re-written in the form

$$I_A = \int d\mathbf{s}\, s^2 \mathbf{s} \cdot \hat{L}'_{22} \left[\left(s^2 - \frac{5g_{5/2}(z)}{2g_{3/2}(z)} \right) \mathbf{s} \right]. \tag{B.24}$$

Using (B.16) again, we find that (B.24) reduces to

$$I_A = \int d\mathbf{s}\, s^2 \mathbf{s} \cdot \hat{L}'_{22}[s^2 \mathbf{s}]. \tag{B.25}$$

All the preceding discussion of the $\hat{L}'_{22}$ linearized collision integral can be extended to the $\hat{L}'_{12}$ collision integral defined in (B.15). One can thus show that $\hat{L}'_{12}$ satisfies the collision invariants

$$\hat{L}'_{12}[\mathbf{p} - m\mathbf{v}_{\rm c}] = 0, \qquad \hat{L}'_{12}[\tilde{\varepsilon}_p - \varepsilon^0_{\rm c}] = 0, \tag{B.26}$$

where $\tilde{\varepsilon}_p \equiv p^2/2m + U$ and $\varepsilon^0_{\rm c} = V_{\rm trap} + gn_{\rm c0} + 2g\tilde{n} = U - gn_{\rm c0}$. In our linearized theory, we can set $\mathbf{v}_{\rm c} = 0$ in (B.26). Using the variable $\mathbf{s}$ defined in (B.3), one finds that $\hat{L}'_{12}[\mathbf{s}] = 0$. In addition, $\hat{L}'_{12}[\psi]$ satisfies a Hermitian condition analogous to (B.21):

$$\int d\mathbf{s}_1 \phi(\mathbf{s}_1) \hat{L}'_{12}[\psi(\mathbf{s}_1)] = \int d\mathbf{s}_1 \psi(\mathbf{s}_1) \hat{L}'_{12}[\phi(\mathbf{s}_1)]. \tag{B.27}$$

We will now sketch the proof of this important property.

Writing out the three terms in (B.15) corresponding to the second line of the equation, one obtains

$$\begin{aligned} \int d\mathbf{s}_1 \phi(\mathbf{s}_1) \hat{L}'_{12}[\psi(\mathbf{s}_1)] = \int d\mathbf{s}_1 \int d\mathbf{s}_2 \int d\mathbf{s}_3\, \delta(\mathbf{s}_1 - \mathbf{s}_2 - \mathbf{s}_3) \\ \times \delta(s_1^2 - s_2^2 - s_3^2 - \beta_0 g n_{\rm c0})(\psi_2 + \psi_3 - \psi_1) \\ \times (1 + f_1^0) f_2^0 f_3^0 [\phi(\mathbf{s}_1) - 2\phi(\mathbf{s}_2)]. \end{aligned} \tag{B.28}$$

Interchanging the dummy variables $\mathbf{s}_1$ and $\mathbf{s}_3$ in (B.28) and adding the resulting two equal expressions, one can replace $\phi_1 - 2\phi_2$ in (B.28) by

$$\tfrac{1}{2} (\phi_1 - 2\phi_2 + \phi_1 - 2\phi_3) = -(\phi_2 + \phi_3 - \phi_1). \tag{B.29}$$

The next step is to calculate the r.h.s. of (B.27) using the same tricks as in the case of $\hat{L}'_{22}$ discussed above. One finds that

$$\int d\mathbf{s}_1 \psi(\mathbf{s}_1) \hat{L}'_{12}[\phi(\mathbf{s}_1)] = \int d\mathbf{s}_1 \int d\mathbf{s}_2 \int d\mathbf{s}_3 \, \delta(\mathbf{s}_1 - \mathbf{s}_2 - \mathbf{s}_3) \times \delta(s_1^2 - s_2^2 - s_3^2 - \beta_0 g n_{c0})(\psi_2 + \psi_3 - \psi_1) \times (1 + f_1^0) f_2^0 f_3^0 (\phi_2 + \phi_3 - \phi_1), \tag{B.30}$$

which is clearly equal to the integral in (B.28). This proves the result in (B.27).

From the Hermiticity relation (B.27) and the fact that $\hat{L}'_{12}[\mathbf{s}] = 0$, one may easily verify that if the integral I_A in (B.22) involved $\hat{L}'_{12}$ instead of $\hat{L}'_{22}$ the reduction to (B.25) would still hold. This simplification of integrals involving $\hat{L}' \equiv \hat{L}'_{12} + \hat{L}'_{22}$ was used in Section 18.1 in evaluating the thermal conductivity of a trapped Bose-condensed gas. Specifically, these results were used in the derivation of the expressions (18.9) and (18.12).

The shear viscosity η in (18.21) requires evaluation of the constant B in (18.22). The denominator of this expression involves an integral of the type

$$I_B \equiv \int d\mathbf{s} \left(s_\mu s_\nu - \tfrac{1}{3} \delta_{\mu\nu} s^2 \right) \hat{L}'_{22} \left[s_\mu s_\nu - \tfrac{1}{3} \delta_{\mu\nu} s^2 \right]. \tag{B.31}$$

Using the fact that $L_{22}[\psi]$ is a linear operator and the conditions (B.16), one finds that

$$\hat{L}'_{22} \left[s_\mu s_\nu - \tfrac{1}{3} \delta_{\mu\nu} s^2 \right] = \hat{L}'_{22}[s_\mu s_\nu] - \tfrac{1}{3} \delta_{\mu\nu} \hat{L}'_{22}[s^2] = \hat{L}'_{22}[s_\mu s_\nu], \tag{B.32}$$

and hence the integral (B.31) reduces to

$$I_B = \int d\mathbf{s} \left(s_\mu s_\nu - \tfrac{1}{3} \delta_{\mu\nu} s^2 \right) \hat{L}'_{22}[s_\mu s_\nu]. \tag{B.33}$$

Employing the Hermitian property of $\hat{L}'_{22}$ in (B.21), one can rewrite (B.33) in the equivalent form

$$I_B = \int d\mathbf{s} \, s_\mu s_\nu \hat{L}'_{22} \left[s_\mu s_\nu - \tfrac{1}{3} \delta_{\mu\nu} s^2 \right]. \tag{B.34}$$

Again using (B.16), this result for I_B simplifies to

$$I_B = \int d\mathbf{s} \, s_\mu s_\nu \hat{L}'_{22}[s_\mu s_\nu]. \tag{B.35}$$

Equation (B.35) enables us to express the denominator of (18.22) in the form given in (18.24); the dimensionless integrals I^η_{22} and I^η_{12} in this equation are defined in (18.25).

We conclude this appendix with expressions for the dimensionless integrals

in (18.10) and (18.25) suitable for numerical calculations (Nikuni and Griffin, 1998a). Starting with I_{22}^{κ}, one can reduce this to

$$I_{22}^{\kappa}(z_0) = \int_0^\infty ds_0 \int_0^\infty ds_r \int_{-1}^1 dy \int_{-1}^1 dy' F_{22}(s_0, s_r, y, y'; z_0) \times s_0^4 s_r^7 (y^2 + y'^2 - 2y^2 y'^2), \tag{B.36}$$

where the function F_{22} is defined as

$$F_{22}(s_0, s_r, y, y'; z_0) \equiv \frac{z_0^2 e^{-s_0^2 - s_r^2}}{(1 - z_0 e^{-s_1^2})(1 - z_0 e^{-s_2^2})(1 - z_0 e^{-s_3^2})(1 - z_0 e^{-s_4^2})}. \tag{B.37}$$

We have defined the new variables $y = \cos\theta, y' = \cos\theta'$, and have expressed all the s_i^2 variables in terms of s_0, s_r, y and y':

$$\begin{aligned} s_1^2 &= \tfrac{1}{2}(s_0^2 + 2s_0 s_r y + s_r^2), \quad & s_2^2 &= \tfrac{1}{2}(s_0^2 - 2s_0 s_r y + s_r^2), \\ s_3^2 &= \tfrac{1}{2}(s_0^2 + 2s_0 s_r y' + s_r^2), \quad & s_4^2 &= \tfrac{1}{2}(s_0^2 - 2s_0 s_r y' + s_r^2). \end{aligned} \tag{B.38}$$

To derive a similar expression for I_{12}^{κ} in (18.10) suitable for numerical calculations, we introduce the variable transformation

$$\mathbf{s}_2 = \tfrac{1}{2}(\mathbf{s}_0 + \mathbf{s}_r), \qquad \mathbf{s}_3 = \tfrac{1}{2}(\mathbf{s}_0 - \mathbf{s}_4). \tag{B.39}$$

We then express $\mathbf{s}_r$ in polar coordinates s_r, θ, ϕ, where θ is the azimuthal angle with respect to the vector $\mathbf{s}_0$, i.e. $\mathbf{s}_r \cdot \mathbf{s}_0 = s_r s_0 \cos\theta$. With these new variables, one can reduce the expression (18.10) for I_{12}^{κ} to

$$I_{12}^{\kappa}(z_0) = 8\pi^{7/2} \int_0^\infty ds_r \int_{-1}^1 dy\, F_{12}(s_r, y; z_0) s_r^2 (s_r^2 + \beta_0 g n_{\mathrm{c}0})^{3/2} \times \left[s_r^2 (x_r^2 + 3\beta_0 g n_{\mathrm{c}0})(1 - y^2) + \tfrac{9}{4}(\beta_0 g n_{\mathrm{c}0})^2 \right], \tag{B.40}$$

where $y = \cos\theta$. Here we have introduced the function F_{12} defined as

$$F_{12}(s_r, y; z_0) \equiv \frac{z_0 e^{-s_1^2}}{(1 - z_0 e^{-s_1^2})(1 - z_0 e^{-s_2^2})(1 - z_0 e^{-s_3^2})}. \tag{B.41}$$

All the variables s_i^2 in (B.41) are expressed as functions of s_r and y:

$$\begin{aligned} s_1^2 &= 2(s_r^2 + \beta_0 g n_{\mathrm{c}0}), \\ s_2^2 &= s_r^2 + s_r y \sqrt{s_r^2 + \beta_0 g n_{\mathrm{c}0}} + \tfrac{1}{2}\beta_0 g n_{\mathrm{c}0} \\ s_3^2 &= s_r^2 - x_r y \sqrt{s_r^2 + \beta_0 g n_{\mathrm{c}0}} + \tfrac{1}{2}\beta_0 g n_{\mathrm{c}0}. \end{aligned} \tag{B.42}$$

Finally, we consider the dimensionless integrals defined in (18.25), which enter the expression (18.28) for the shear viscosity. After some calculation,

one finds that these integrals can be written in forms that are useful for numerical calculations:

$$\begin{aligned} I_{22}^{\eta} &\equiv -\int d\mathbf{s}\, s_\mu s_\nu \hat{L}'_{22}\,[s_\mu s_\nu] \\ &= \frac{\pi^3}{\sqrt{2}} \int_0^\infty ds_0 \int_0^\infty ds_r \int_{-1}^1 dy \int_{-1}^1 dy' F_{22}(s_0, s_r, y, y'; z_0) \\ &\quad \times s_0^2 s_r^7 (1 + y^2 + y'^2 - 3y^2 y'^2), \end{aligned} \tag{B.43}$$

and

$$\begin{aligned} I_{12}^{\eta} &\equiv -\pi^{3/2} \int d\mathbf{s}\, s_\mu s_\nu \hat{L}'_{12}\,[s_\mu s_\nu] \\ &= 8\pi^{7/2} \int_0^\infty ds_r \int_{-1}^1 dy\, F_{12}(s_r, y; z_0) s_r^2 \sqrt{s_r^2 + \beta_0 g n_{\mathrm{c}0}} \\ &\quad \times \left[s_r^2 (s_r^2 + \beta_0 g n_{\mathrm{c}0})(1 - y^2) + \tfrac{1}{3} (\beta_0 g n_{\mathrm{c}0})^2 \right]. \end{aligned} \tag{B.44}$$

These expressions involve the functions F_{22} and F_{12} defined in (B.37) and (B.41), respectively.

Appendix C
Frequency-dependent transport coefficients

In this appendix, we give some details on the frequency-dependent coefficients of the shear viscosity and thermal conductivity, starting from the Zaremba–Nikuni–Griffin (ZNG) kinetic equation for the noncondensate atoms.

As in the Chapman–Enskog procedure described in Chapter 18, we insert the local equilibrium distribution function $f^{(0)}$ into the left-hand side of the kinetic equation, noting that

$$f^{(0)}(\mathbf{p},\mathbf{r},t) = \frac{1}{z^{-1}(\mathbf{r},t)e^{\beta(\mathbf{r},t)[\mathbf{p}-m\mathbf{v}_{\rm n}(\mathbf{r},t)]^2/2m}-1} \tag{C.1}$$

and $z(\mathbf{r},t) = e^{\beta(\mathbf{r},t)[\tilde{\mu}(\mathbf{r},t)-U(\mathbf{r},t)]}$ is the local fugacity. As shown in Appendix A of Nikuni and Griffin (2001a), the kinetic equation is then given by

$$\frac{\partial f}{\partial t} + \left[\frac{1}{z}\frac{\mathbf{p}}{m}\cdot\nabla z + \frac{(\mathbf{p}-m\mathbf{v}_{\rm n})^2}{2mk_{\rm B}T^2}\frac{\mathbf{p}}{m}\cdot\nabla T + \frac{\mathbf{p}-m\mathbf{v}_{\rm n}}{k_{\rm B}T}\cdot\left(\frac{\mathbf{p}}{m}\cdot\nabla\right)\mathbf{v}_{\rm n} + \frac{\nabla U}{mk_{\rm B}T}\cdot(\mathbf{p}-m\mathbf{v}_{\rm n})\right] f^{(0)}(1+f^{(0)}) = C_{12}+C_{22}. \tag{C.2}$$

In contrast with the analysis in Sections 17.1 and 17.2, we now keep the first term on the l.h.s., the time derivative of f. Since we are interested in small-amplitude collective oscillations, we will work to first order in the fluctuations around static thermal equilibrium (which is described by f^0).

We first consider the shear viscosity transport coefficient, which is associated with the anisotropic pressure tensor. In a linearized theory, this is given by

$$P'_{\mu\nu} \equiv P_{\mu\nu} - \delta_{\mu\nu}\tilde{P} = \int \frac{d\mathbf{p}}{(2\pi\hbar)^3}\frac{1}{m}\left(p_\mu p_\nu - \tfrac{1}{3}\delta_{\mu\nu}p^2\right) f(\mathbf{p},\mathbf{r},t). \tag{C.3}$$

The equation of motion for $P'_{\mu\nu}$ can be obtained by taking moments of (C.2)

and linearizing it around static thermal equilibrium. To save space, we introduce the abbreviation

$$\overline{p_\mu p_\nu} \equiv p_\mu p_\nu - \tfrac{1}{3}\delta_{\mu\nu}p^2. \tag{C.4}$$

We then find that the contribution of the second term on the l.h.s. of (C.2) to this moment is

$$\begin{aligned}\int \frac{d\mathbf{p}}{(2\pi\hbar)^3}\frac{1}{m}\overline{p_\mu p_\nu}\frac{\mathbf{p}}{k_\mathrm{B}T}\cdot\left(\frac{\mathbf{p}}{m}\cdot\nabla\right)\mathbf{v}_\mathrm{n}f^0(1+f^0)\\ = \tilde{P}_0\left(\frac{\partial v_{n\nu}}{\partial x_\mu}+\frac{\partial v_{n\mu}}{\partial x_\nu}-\tfrac{2}{3}\delta_{\mu\nu}\nabla\cdot\mathbf{v}_\mathrm{n}\right).\end{aligned} \tag{C.5}$$

One thus obtains

$$\frac{\partial P'_{\mu\nu}}{\partial t}+\tilde{P}_0\left(\frac{\partial v_{n\nu}}{\partial x_\mu}+\frac{\partial v_{n\mu}}{\partial x_\nu}-\tfrac{2}{3}\delta_{\mu\nu}\nabla\cdot\mathbf{v}_\mathrm{n}\right)=\left\langle\frac{1}{m}\overline{p_\mu p_\nu}\right\rangle_\mathrm{coll}, \tag{C.6}$$

where we define

$$\left\langle\frac{1}{m}\overline{p_\mu p_\nu}\right\rangle_\mathrm{coll}\equiv\int\frac{d\mathbf{p}}{(2\pi\hbar)^3}\frac{1}{m}\overline{p_\mu p_\nu}(C_{12}+C_{22}). \tag{C.7}$$

The collisional contribution on the right-hand side of (C.6) arises from deviation $f^{(1)}$ of the distribution function f from the local equilibrium solution (C.1). Following the Chapman–Enskog procedure, we use the ansatz $f = f^0 + \delta f$, where

$$\delta f=\sum_{\mu\nu}B_{\mu\nu}\overline{p_\mu p_\nu}f^0(1+f^0), \tag{C.8}$$

where $B_{\mu\nu}$ is some momentum-independent symmetric tensor. The relation between $B_{\mu\nu}$ and $P'_{\mu\nu}$ can be found by using (C.8) in (C.3) and carrying out the momentum integral, which gives

$$\begin{aligned}P'_{\mu\nu}&=\sum_{\mu'\nu'}B_{\mu'\nu'}\int\frac{d\mathbf{p}}{(2\pi\hbar)^3}\frac{1}{m}\overline{p_\mu p_\nu}\;\overline{p_{\mu'}p_{\nu'}}f^0(1+f^0)\\ &=\tfrac{1}{5}\left(B_{\mu\nu}-\tfrac{1}{3}\delta_{\mu\nu}\operatorname{Tr}B\right)\sum_{\mu'\nu'}\int\frac{d\mathbf{p}}{(2\pi\hbar)^3}\frac{1}{m}\left(\overline{p_{\mu'}p_{\nu'}}\right)^2\\ &=2mk_\mathrm{B}T\tilde{P}_0\left(B_{\mu\nu}-\tfrac{1}{3}\delta_{\mu\nu}\operatorname{Tr}B\right).\end{aligned} \tag{C.9}$$

Using (C.8) in (C.7), we find

$$\begin{aligned}\left\langle\frac{1}{m}\overline{p_\mu p_\nu}\right\rangle_\mathrm{coll}&=\sum_{\mu'\nu'}B_{\mu'\nu'}\int\frac{d\mathbf{p}}{(2\pi\hbar)^3}\frac{1}{m}\overline{p_\mu p_\nu}\hat{L}\left[\overline{p_{\mu'}p_{\nu'}}\right]\\ =\tfrac{1}{5}\left(B_{\mu\nu}-\tfrac{1}{3}\delta_{\mu\nu}\operatorname{Tr}B\right)&\sum_{\mu'\nu'}\int\frac{d\mathbf{p}}{(2\pi\hbar)^3}\frac{1}{m}\overline{p_{\mu'}p_{\nu'}}\hat{L}\left[\overline{p_{\mu'}p_{\nu'}}\right],\end{aligned} \tag{C.10}$$

where $\hat{L}$ is the linearized collision operator as defined by (17.26), (17.28) and (17.36). Combining (C.9) and (C.10) and using the definition of the viscous relaxation time τ_η in (18.28), we find that the collision term reduces to

$$\left\langle \frac{1}{m}\overline{p_\mu p_\nu} \right\rangle_{\rm coll} = -\frac{P'_{\mu\nu}}{\tau_\eta}. \tag{C.11}$$

Assuming the harmonic time dependence $P'_{\mu\nu} \propto e^{-i\omega t}$, we finally obtain

$$\begin{aligned} P'_{\mu\nu} &= -\frac{2\tau_\eta \tilde{P}_0}{1-i\omega\tau_\eta}\left(D_{\mu\nu} - \tfrac{1}{3}\delta_{\mu\nu}\,\mathrm{Tr}\,D\right) \\ &= -2\eta(\omega)\left(D_{\mu\nu} - \tfrac{1}{3}\delta_{\mu\nu}\,\mathrm{Tr}\,D\right), \end{aligned} \tag{C.12}$$

where the frequency-dependent viscosity coefficient $\eta(\omega)$ is naturally defined as

$$\eta(\omega) \equiv \frac{\tau_\eta \tilde{P}_0}{1-i\omega\tau_\eta} = \frac{\eta}{1-i\omega\tau_\eta}. \tag{C.13}$$

The frequency-dependent *thermal conductivity* can also be obtained in the same manner. The linearized heat current defined by (15.13) is given (to first order in $\mathbf{v}_{\rm n}$) by

$$\begin{aligned} \mathbf{Q}(\mathbf{r},t) &= \int \frac{d\mathbf{p}}{(2\pi\hbar)^3}\frac{p^2}{2m}\frac{\mathbf{p}}{m} f(\mathbf{p},\mathbf{r},t) - \frac{5}{2}\mathbf{v}_{\rm n}\tilde{P}_0(\mathbf{r}) \\ &= \int \frac{d\mathbf{p}}{(2\pi\hbar)^3}\left[\frac{p^2}{2m} - \frac{5}{2}k_{\rm B}T\frac{g_{5/2}(z_0)}{g_{3/2}(z_0)}\right]\frac{\mathbf{p}}{m} f(\mathbf{p},\mathbf{r},t). \end{aligned} \tag{C.14}$$

In the second line, the term proportional to $\mathbf{v}_{\rm n}$ has been rewritten using the definition of $\mathbf{v}_{\rm n}$ in (15.11) and the thermal equilibrium values of $\tilde{n}_0$ and $\tilde{P}_0$ given by (15.24) and (15.21), respectively. Taking the moment of the kinetic equation (C.2), one finds that the corresponding contribution to $\mathbf{Q}$ is given by

$$\begin{aligned} &\int \frac{d\mathbf{p}}{(2\pi\hbar)^3}\left[\frac{p^2}{2m} - \frac{5}{2}k_{\rm B}T\frac{g_{5/2}(z_0)}{g_{3/2}(z_0)}\right]\frac{\mathbf{p}}{m}\frac{p^2}{2mk_{\rm B}T^2}\left(\frac{\mathbf{p}}{m}\cdot\nabla\delta T\right) f^0(1+f^0) \\ &\qquad = \frac{5}{2}\frac{\tilde{n}_0 k_{\rm B}^2 T}{m}\left\{\frac{7g_{7/2}(z_0)}{2g_{3/2}(z_0)} - \frac{5}{2}\left[\frac{g_{5/2}(z_0)}{g_{3/2}(z_0)}\right]^2\right\}\nabla\delta T. \end{aligned} \tag{C.15}$$

One thus obtains

$$\frac{\partial \mathbf{Q}}{\partial t} + \frac{5}{2}\frac{\tilde{n}_0 k_{\mathrm{B}}^2 T}{m}\left\{\frac{7g_{7/2}(z_0)}{2g_{3/2}(z_0)} - \frac{5}{2}\left[\frac{g_{5/2}(z_0)}{g_{3/2}(z_0)}\right]^2\right\}\nabla\delta T$$
$$= \int \frac{d\mathbf{p}}{(2\pi\hbar)^3}\left[\frac{p^2}{2m} - \frac{5}{2}k_{\mathrm{B}}T\frac{g_{5/2}(z_0)}{g_{3/2}(z_0)}\right]\frac{\mathbf{p}}{m}(C_{12}+C_{22}) = -\frac{\mathbf{Q}}{\tau_\kappa}. \quad \text{(C.16)}$$

Here τ_κ is the thermal relaxation time defined in (18.13). Assuming a harmonic time dependence of the heat current, $\mathbf{Q} \propto e^{-i\omega t}$, we obtain

$$\mathbf{Q} = -\frac{\kappa}{1 - i\omega\tau_\kappa}\nabla\delta T \equiv -\kappa(\omega)\nabla\delta T. \quad \text{(C.17)}$$

The frequency-dependent thermal conductivity is then defined naturally by

$$\kappa(\omega) \equiv \frac{\kappa}{1 - i\omega\tau_\kappa}, \quad \text{(C.18)}$$

where κ is given by (18.15).

Appendix D
Derivation of hydrodynamic damping formula

In this appendix, we give a detailed derivation of (19.36) and (19.37). This leads to a Landau–Lifshitz expression for the damping of two-fluid a hydrodynamic mode labeled by α. It is convenient to divide $V_{\alpha\alpha}$ in (19.34) into four terms:

$$V_{\alpha\alpha} = \int d\mathbf{r}\, \boldsymbol{v}_\alpha \cdot \boldsymbol{F}\boldsymbol{v}_\alpha = V_\alpha^{11} + V_\alpha^{12} + V_\alpha^{21} + V_\alpha^{22}, \tag{D.1}$$

where

$$\begin{aligned} V_{\alpha\alpha}^{11} &= \textstyle\int d\mathbf{r}\, v_{\mathrm{n}\alpha} \cdot (\mathbf{F}_{11}\mathbf{v}_{\mathrm{n}}), \quad V_{\alpha\alpha}^{12} = \int d\mathbf{r}\, v_{\mathrm{n}\alpha} \cdot (\mathbf{F}_{12}\mathbf{v}_{\mathrm{c}}), \\ V_{\alpha\alpha}^{21} &= \textstyle\int d\mathbf{r}\, v_{\mathrm{c}\alpha} \cdot (\mathbf{F}_{21}\mathbf{v}_{\mathrm{n}}), \quad V_{\alpha\alpha}^{22} = \int d\mathbf{r}\, v_{\mathrm{c}\alpha} \cdot (\mathbf{F}_{22}\mathbf{v}_{\mathrm{c}}). \end{aligned} \tag{D.2}$$

From (19.30), we obtain

$$\begin{aligned} V_{\alpha\alpha}^{11} = \int d\mathbf{r} \Big[& \nabla \cdot \mathbf{v}_{\mathrm{n}\alpha} \Big\{ \frac{2}{3} g n_{\mathrm{c}0} i\omega\tau_\mu\, \delta\Gamma_{12}^{(1)}[\mathbf{v}_{\mathrm{n}\alpha}, 0] \\ & - \Big(\frac{2}{3} + \frac{4 n_{\mathrm{c}0}\sigma_{\mathrm{H}}\sigma_1}{9\tilde{n}_0} \Big) \nabla \cdot (\kappa \nabla \delta T_\omega[\mathbf{v}_{\mathrm{n}\alpha}, 0]) \Big\} \\ & + \Big[i\omega\tau_\mu\, \delta\Gamma_{12}^{(1)}[\mathbf{v}_{\mathrm{n}\alpha}, 0] - \frac{2\sigma_{\mathrm{H}}\sigma_1}{3g\tilde{n}_0} \nabla \cdot (\kappa \nabla \delta T_\omega[\mathbf{v}_{\mathrm{n}\alpha}, 0]) \Big] \mathbf{v}_{\mathrm{n}\alpha} \cdot \nabla U_0 \\ & - i\omega \frac{\partial v_{\mathrm{n}\mu\alpha}}{\partial x_\nu} \eta \Big(\frac{\partial v_{\mathrm{n}\mu\alpha}}{\partial x_\nu} + \frac{\partial v_{\mathrm{n}\nu\alpha}}{\partial x_\mu} - \frac{2}{3}\delta_{\mu\nu} \nabla \cdot \mathbf{v}_{\mathrm{n}\alpha} \Big) \Big], \end{aligned} \tag{D.3}$$

$$\begin{aligned} V_{\alpha\alpha}^{12} = \int d\mathbf{r} \Big(& \nabla \cdot \mathbf{v}_{\mathrm{n}\alpha} \Big[\frac{2}{3} g n_{\mathrm{c}0} i\omega\tau_\mu\, \delta\Gamma_{12}^{(1)}[0, \mathbf{v}_{\mathrm{c}\alpha}] \\ & - \Big(\frac{2}{3} + \frac{4 n_{\mathrm{c}0}\sigma_{\mathrm{H}}\sigma_1}{9\tilde{n}_0} \Big) \nabla \cdot (\kappa \nabla \delta T_\omega[0, \mathbf{v}_{\mathrm{c}\alpha}]) \Big] \\ & + \Big[i\omega\tau_\mu \delta\Gamma_{12}^{(1)}[0, \mathbf{v}_{\mathrm{c}\alpha}] - \frac{2\sigma_{\mathrm{H}}\sigma_1}{3g\tilde{n}_0} \nabla \cdot (\kappa \nabla \delta T_\omega[0, \mathbf{v}_{\mathrm{c}\alpha}]) \Big] \mathbf{v}_{\mathrm{n}\alpha} \cdot \nabla U_0 \Big), \end{aligned} \tag{D.4}$$

$$V_{\alpha\alpha}^{21} - \int d\mathbf{r}\, g\nabla\cdot(n_{\mathrm{c}0}\mathbf{v}_{\mathrm{c}\alpha})\left[i\omega\tau_\mu\delta\Gamma_{12}^{(1)}[\mathbf{v}_{\mathrm{n}\alpha},0] - \frac{2\sigma_{\mathrm{H}}\sigma_1}{3g\tilde{n}_0}\nabla\cdot(\kappa\nabla\delta T_\omega[\mathbf{v}_{\mathrm{n}\alpha},0])\right], \tag{D.5}$$

$$V_{\alpha\alpha}^{22} = -\int d\mathbf{r}\, g\nabla\cdot(n_{\mathrm{c}0}\mathbf{v}_{\mathrm{c}\alpha})\left[i\omega\tau_\mu\delta\Gamma_{12}^{(1)}[0,\mathbf{v}_{\mathrm{c}\alpha}] - \frac{2\sigma_{\mathrm{H}}\sigma_1}{3g\tilde{n}_0}\nabla\cdot(\kappa\nabla\delta T_\omega[0,\mathbf{v}_{\mathrm{c}\alpha}])\right]. \tag{D.6}$$

Adding (D.3) and (D.4), we obtain

$$\begin{aligned} V_{\alpha\alpha}^{11} + V_{\alpha\alpha}^{12} = \int d\mathbf{r}\Big[\nabla\cdot\mathbf{v}_{\mathrm{n}\alpha}\Big\{&\frac{2}{3}gn_{\mathrm{c}0}i\omega\tau_\mu\,\delta\Gamma_{12}^{(1)}[\mathbf{v}_{\mathrm{n}\alpha},\mathbf{v}_{\mathrm{c}\alpha}] \\ &- \left(\frac{2}{3} + \frac{4n_{\mathrm{c}0}\sigma_{\mathrm{H}}\sigma_1}{9\tilde{n}_0}\right)\nabla\cdot(\kappa\nabla\delta T_\omega[\mathbf{v}_{\mathrm{n}\alpha},\mathbf{v}_{\mathrm{c}\alpha}])\Big\} \\ &+ \Big\{i\omega\tau_\mu\,\delta\Gamma_{12}^{(1)}[\mathbf{v}_{\mathrm{n}\alpha},\mathbf{v}_{\mathrm{c}\alpha}] \\ &- \frac{2\sigma_{\mathrm{H}}\sigma_1}{3g\tilde{n}_0}\nabla\cdot(\kappa\nabla\delta T_\omega[\mathbf{v}_{\mathrm{n}\alpha},\mathbf{v}_{\mathrm{c}\alpha}])\Big\}\mathbf{v}_{\mathrm{n}\alpha}\cdot\nabla U_0 \\ &- i\omega\frac{\partial v_{\mathrm{n}\mu\alpha}}{\partial x_\nu}\eta\left(\frac{\partial v_{\mathrm{n}\mu\alpha}}{\partial x_\nu} + \frac{\partial v_{\mathrm{n}\nu\alpha}}{\partial x_\mu} - \frac{2}{3}\delta_{\mu\nu}\nabla\cdot\mathbf{v}_{\mathrm{n}\alpha}\right)\Big]. \end{aligned} \tag{D.7}$$

Adding (D.5) and (D.6), we obtain

$$\begin{aligned} V_{\alpha\alpha}^{21} + V_{\alpha\alpha}^{22} = -\int d\mathbf{r} g\nabla\cdot(n_{\mathrm{c}0}\mathbf{v}_{\mathrm{c}\alpha})\Big[&i\omega\tau_\mu\delta\Gamma_{12}^{(1)}[\mathbf{v}_{\mathrm{n}\alpha},\mathbf{v}_{\mathrm{c}\alpha}] \\ &- \frac{2\sigma_{\mathrm{H}}\sigma_1}{3g\tilde{n}_0}\nabla\cdot(\kappa\nabla\delta T_\omega[\mathbf{v}_{\mathrm{n}\alpha},\mathbf{v}_{\mathrm{c}\alpha}])\Big]. \end{aligned} \tag{D.8}$$

Adding (D.7) and (D.8) and collecting together terms involving $\delta\Gamma_{12}^{(1)}, \kappa$ and η, we obtain

$$V_{\alpha\alpha} = V_{\alpha\alpha}^{(1)} + V_{\alpha\alpha}^{(2)} + V_{\alpha\alpha}^{(3)}, \tag{D.9}$$

where

$$\begin{aligned} V_{\alpha\alpha}^{(1)} = &\int d\mathbf{r}\, i\omega\tau_\mu\delta\Gamma_{12}^{(1)}[\mathbf{v}_{\mathrm{n}\alpha},\mathbf{v}_{\mathrm{c}\alpha}] \\ &\times\left[\frac{2}{3}gn_{\mathrm{c}0}\nabla\cdot\mathbf{v}_{\mathrm{n}\alpha} + \mathbf{v}_{\mathrm{n}\alpha}\cdot\nabla U_0 - g\nabla\cdot(n_{\mathrm{c}0}\mathbf{v}_{\mathrm{c}\alpha})\right], \end{aligned} \tag{D.10}$$

$$\begin{aligned} V_{\alpha\alpha}^{(2)} = &\int d\mathbf{r}\nabla\cdot(\kappa\nabla\delta T_\omega[\mathbf{v}_{\mathrm{n}\alpha},\mathbf{v}_{\mathrm{c}\alpha}])\Big\{-\frac{2}{3}\nabla\cdot\mathbf{v}_{\mathrm{n}\alpha} \\ &+\frac{2\sigma_1\sigma_{\mathrm{H}}}{3\tilde{n}_0}\Big[\nabla\cdot(n_{\mathrm{c}0}\mathbf{v}_{\mathrm{c}\alpha}) - \frac{1}{g}\mathbf{v}_{\mathrm{n}\alpha}\cdot\nabla U_0 - \frac{2}{3}n_{\mathrm{c}0}\nabla\cdot\mathbf{v}_{\mathrm{n}\alpha}\Big]\Big\}, \end{aligned} \tag{D.11}$$

$$V_{\alpha\alpha}^{(3)} = -i\omega\eta\int d\mathbf{r}\,\frac{\partial v_{\mathrm{n}\mu\alpha}}{\partial x_\nu}\left(\frac{\partial v_{\mathrm{n}\mu\alpha}}{\partial x_\nu} + \frac{\partial v_{\mathrm{n}\nu\alpha}}{\partial x_\mu} - \frac{2}{3}\delta_{\mu\nu}\nabla\cdot\mathbf{v}_{\mathrm{n}\alpha}\right). \tag{D.12}$$

Using $\nabla U_0 = g\nabla n_{c0}$ and (19.13), we find that (D.10) simplifies to

$$V_{\alpha\alpha}^{(1)} = -i\omega \int d\mathbf{r}\, \frac{g\tau_\mu}{\sigma_H} \left(\delta\Gamma_{12}^{(1)}[\mathbf{v}_{n\alpha}, \mathbf{v}_{c\alpha}]\right)^2 . \tag{D.13}$$

Similarly, one can rewrite (D.11), using (19.13) and (19.15) and integrating by parts, as

$$V_{\alpha\alpha}^{(2)} = -i\omega \int d\mathbf{r}\, \frac{\kappa}{T} \left|\nabla \delta T_\omega[\mathbf{v}_{n\alpha}, \mathbf{v}_{c\alpha}]\right|^2 . \tag{D.14}$$

Here we have used the fact that $\delta T_{\omega\alpha}$ is, from the definition in (19.15), purely imaginary. Finally, (D.12) can be written as

$$V_{\alpha\alpha}^{(3)} = -i\omega \frac{\eta}{2} \int d\mathbf{r} \left(\frac{\partial v_{n\mu\alpha}}{\partial x_\nu} + \frac{\partial v_{n\nu\alpha}}{\partial x_\mu} - \frac{2}{3}\delta_{\mu\nu} \nabla \cdot \mathbf{v}_{n\alpha} \right)^2 , \tag{D.15}$$

where summation over the repeated indices μ and ν is implicit. We have made use of the fact

$$\sum_{\mu\nu} \delta_{\mu\nu} \left(\frac{\partial v_{n\mu\alpha}}{\partial x_\nu} + \frac{\partial v_{n\nu\alpha}}{\partial x_\mu} - \frac{2}{3}\delta_{\mu\nu} \nabla \cdot \mathbf{v}_{n\alpha} \right) = 0. \tag{D.16}$$

Adding (D.11)–(D.13), we obtain the result in (19.36), namely

$$\begin{aligned} V_{\alpha\alpha} = -i\omega \int d\mathbf{r} \Bigg[& \frac{g\tau_\mu}{\sigma_H} \left(\delta\Gamma_{12}^{(1)}[\mathbf{v}_{n\alpha}, \mathbf{v}_{c\alpha}]\right)^2 + \frac{\kappa}{T} |\nabla \delta T_\omega[\mathbf{v}_{n\alpha}, \mathbf{v}_{c\alpha}]|^2 \\ & + \frac{\eta}{2} \left(\frac{\partial v_{n\alpha\nu}}{\partial x_\mu} + \frac{\partial v_{n\alpha\mu}}{\partial x_\nu} - \frac{2}{3}\delta_{\mu\nu} \nabla \cdot \mathbf{v}_{n\alpha} \right)^2 \Bigg]. \end{aligned} \tag{D.17}$$

Using (D.17) in the expression (19.35), we obtain the final expression given in (19.37). As shown in (19.38) and the subsequent discussion, this expression is a generalized version of the Landau–Lifshitz expression for hydrodynamic damping, applicable to two-fluid modes in trapped superfluid gases.

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